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TA-20	06 FEB 2024	38.0.0	TA-64	20 NOV 2023	37.0.0
TA-21	06 FEB 2024	38.0.0	TA-65	20 NOV 2023	37.0.0
TA-22	06 FEB 2024	38.0.0	TA-66	20 NOV 2023	37.0.0
TA-23	06 FEB 2024	38.0.0	TA-67	20 NOV 2023	37.0.0
TA-24	06 FEB 2024	38.0.0	TA-68	20 NOV 2023	37.0.0
TA-25	06 FEB 2024	38.0.0	TA-69	20 NOV 2023	37.0.0
TA-26	06 FEB 2024	38.0.0	TA-70	20 NOV 2023	37.0.0
TA-27	06 FEB 2024	38.0.0	TA-71	20 NOV 2023	37.0.0
TA-28	06 FEB 2024	38.0.0	TA-72	20 NOV 2023	37.0.0
TA-29	06 FEB 2024	38.0.0	TA-73	13 OCT 2025	42.0.0
TA-30	06 FEB 2024	38.0.0	TA-74	13 OCT 2025	42.0.0
TA-31	20 NOV 2023	37.0.0	TA-75	13 OCT 2025	42.0.0
TA-32	20 NOV 2023	37.0.0	TA-76	13 OCT 2025	42.0.0
TA-33	20 NOV 2023	37.0.0	TA-77	13 OCT 2025	42.0.0
TA-34	20 NOV 2023	37.0.0	TA-78	15 JUL 2025	41.0.0
TA-35	20 NOV 2023	37.0.0	TA-79	15 JUL 2025	41.0.0
TA-36	20 NOV 2023	37.0.0	TA-80	15 JUL 2025	41.0.0
TA-37	20 NOV 2023	37.0.0	TA-81	15 JUL 2025	41.0.0
TA-38	20 NOV 2023	37.0.0	TA-82	15 JUL 2025	41.0.0
TA-39	20 NOV 2023	37.0.0	TA-83	15 JUL 2025	41.0.0
TA-40	20 NOV 2023	37.0.0	TA-84	15 JUL 2025	41.0.0
TA-41	20 NOV 2023	37.0.0	TA-85	15 JUL 2025	41.0.0
TA-42	20 NOV 2023	37.0.0	TA-86	15 JUL 2025	41.0.0
TA-43	20 NOV 2023	37.0.0	TA-87	15 JUL 2025	41.0.0
TA-44	20 NOV 2023	37.0.0	TA-88	15 JUL 2025	41.0.0
TA-45	20 NOV 2023	37.0.0	TA-89	15 JUL 2025	41.0.0
TA-46	20 NOV 2023	37.0.0	TA-90	15 JUL 2025	41.0.0
TA-47	20 NOV 2023	37.0.0	TA-91	15 JUL 2025	41.0.0
TA-48	20 NOV 2023	37.0.0	TA-92	15 JUL 2025	41.0.0
TA-49	20 NOV 2023	37.0.0	TA-93	08 JUN 2026	45.0.0
TA-50	20 NOV 2023	37.0.0	TA-94	08 JUN 2026	45.0.0

Highlights

ROR Record of Revisions	ROR
New revision.	
Added table regarding the authority's prior approval of the manual.	
0.2 System of Amendment and Revision	0.2
Added new role - Revision Lead.	
0.3.2 Identification of Changes	0.3.2
Edited paragraph on revision markings.	
0.3.3 Revision Process Responsibilities	0.3.3
Updated section and reviewed table with responsibilities.	
1.2.1 Deputies to Nominated Persons	1.2.1
Changed Security Manager Deputy from Vacant to N/A.	
1.4.2 Specific Responsibilities	1.4.2
Added item on Crew Alertness Risk Assessment (CARA).	
1.5.1 Second-in-Command (SIC)	1.5.1
Added item on Crew Alertness Risk Assessment (CARA).	
1.5.2 Cabin Crew	1.5.2
Added item on Crew Alertness Risk Assessment (CARA).	
Updated paragraph on pre-flight checks.	
2.1.3.4 Post-flight Analysis of Documents	2.1.3.4
Updated section.	
2.1.3.6 Record-keeping	2.1.3.6
Added paragraph on Skybook.	
4.1.12.1 In-flight Service Representative (ISR)	4.1.12.1
Added paragraph on Crew Alertness Risk Assessment (CARA).	
6.6.3 Critical Incident Stress Debriefing for Crew Members	6.6.3
Added CISM contact number.	
7.1.2 Crew Member Responsibility	7.1.2
Updated section. Added content on CARA.	
7.1.3 Definitions	7.1.3
Added CARA definition.	
7.2.2 Briefing on Tour	7.2.2
Updated section, added subsections.	
7.2.5.1 Reduced Scheduled Show-times	7.2.5.1
Corrected OM-A reference.	
7.2.5.3 Increased Showtime Due to Aerodrome Logistics	7.2.5.3
Updated title and section.	
7.2.6 Scheduled Minimum Turn Times	7.2.6
Added note on IJET limitations.	
7.2.6.5 Increased Scheduled Show / Turn Time Due to Remote Operations	7.2.6.5
Added exclusion to the Caribbean.	
7.2.10.1.2 Daily FDP (Augmented flight Crew / In-flight Rest)	7.2.10.1.2
Updated paragraph on planned block times.	
7.2.10.1.2.1 Augmented Flight Crew on Revenue Flights	7.2.10.1.2.1

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Updated procedures' list.

[7.2.10.1.2.2 Augmented Flight Crew on Non-revenue Flights](#) [7.2.10.1.2.2](#)

Updated paragraph on cabin crew rest.

[7.2.11 Maximum Scheduled FDPs Encroaching on the WOCLs in Seven Consecutive Calendar Days](#) [7.2.11](#)

Updated footnote on consecutive WOCL FDPs.

[7.2.19 Crew Positioning](#) [7.2.19](#)

Added paragraph on positioning by Hire Car, Chauffered Car or Taxi.

[7.2.21.1 Rest Conditions](#) [7.2.21.1](#)

Updated paragraph on company mobile phone.

[7.2.21.6 Transportation Times to and from Rest Locations \(Travelling Time\)](#) [7.2.21.6](#)

Added notes on transportation time.

[7.2.21.6.1 Availability of suitable accommodation at the rest location](#) [7.2.21.6.1](#)

Added new section.

[8.1.8.2 Methods, Procedures, and Responsibility for Preparation and Acceptance of the Mass and Centre of Gravity Calculations](#) [8.1.8.2](#)

Updated item on change in passenger manifest, masses, or baggage.

Added item on acceptance of the Mass and Balance documentation by hand signature or equivalent.

[8.1.8.5 Last Minute Changes Procedure](#) [8.1.8.5](#)

Updated paragraph on flight crew being unable to access the revised Mass and Balance.

[8.1.10.1 General](#) [8.1.10.1](#)

Updated section.

[8.1.12 Documents, Forms, and Additional Information to be Carried](#) [8.1.12](#)

Added mention to E.

Added mention to E.

Added mention to E.

Added mention to E.

Added row with OFP Annex.

[8.2.2.5.3 Live Animals](#) [8.2.2.5.3](#)

Removed previous item 11 on Chief Pilot approval.

[8.3.7.4.1 Protecting the Final Reserve](#) [8.3.7.4.1](#)

Added paragraphs on FRF.

[8.3.10.1 Cabin Crew Seating](#) [8.3.10.1](#)

Updated paragraph.

Updated list with order of preference.

[8.3.15.8 Cabin Secure Checks](#) [8.3.15.8](#)

Updated paragraph on Window Blinds.

[8.3.15.10.1 PED Types](#) [8.3.15.10.1](#)

Updated section.

[8.3.15.10.2 Procedures for PEDs](#) [8.3.15.10.2](#)

Updated section. Added subsections.

[8.3.15.10.3 Use of PEDs During Flights](#) [8.3.15.10.3](#)

Updated Table 1 - Use of type of PEDs by phase of flight.

Updated Use of PEDs during LVO approach operations section.

[8.3.15.10.4 Exceptions to PEDs Rules](#) [8.3.15.10.4](#)

Reworded item. Removed previous item e.

8.3.15.10.5 PEDs Not Accessible During Flight	8.3.15.10.5
Updated section.	
8.3.15.10.6 Use of PEDs in the Flight Crew Compartment	8.3.15.10.6
Updated section.	
8.3.15.10.7 PED Reportable Events	8.3.15.10.7
Added mention to the PED Policy.	
8.3.16.1.1 Prior to Take-off	8.3.16.1.1
Updated item adding information on power banks.	
8.3.19.3 Monitoring and Crosschecking	8.3.19.3
Added mention to ground path. Added mention to verbally stating intention.	
Added mention to ground path.	
Added content on escalation path.	
8.3.24 Taxiing	8.3.24
Added item on conditional clearances.	
Updated item.	
Added item on Strobe / Anti-collision or Landing lights.	
Updated item on ATC.	
Added item on landing lights.	
Added item on increasing workload.	
8.3.24.1 Taxi and Runway Entry - Crew Coordination	8.3.24.1
Added new section.	
8.3.24.10 Runway Status Lights (RWSL)	8.3.24.10
Added new section.	
8.3.25 Line-up	8.3.25
Added paragraph.	
8.3.27.15 Fatigue Management (Controlled Rest on FCC)	8.3.27.15
Updated section. Added subsections.	
8.3.28 Approach and Landing	8.3.28
Removed reference.	
Added paragraph.	
Updated paragraph.	
Added paragraph.	
8.3.28.21 Land and Hold Short Operations (LAHSO)	8.3.28.21
Added section.	
8.4.2.1 Low Visibility Take-off Procedure	8.4.2.1
Updated 400 to 550 m.	
8.18.1 Introduction and System Description	8.18.1
Added content on Electronic Flight Folder (EFF).	
Added row for Electronic Flight Folder. Updated layout.	
8.18.2.2 Portable EFB	8.18.2.2
Updated content on Portable EFB Connectivity to Aircraft Systems.	
8.18.3.2.1 Flight Crew Portable EFB	8.18.3.2.1
Updated Wi-Fi item.	
Updated Bluetooth item.	
Updated Cellular data item.	
Updated item on EFB databases / contents.	
Removed mention to Jepp Mobile.	
Added content on Electronic Flight Folder.	

Added item on Flight progress.

Added item on Skybook.

[8.18.3.2.2 Cabin Crew Portable EFB](#)

[8.18.3.2.2](#)

Updated Wi-Fi item.

Updated Bluetooth item.

Updated Cellular Data item.

Updated item on EFB databases / contents.

[8.18.4 Contingency or Non-normal Procedure](#)

[8.18.4](#)

Added row on EFF.

Added row on EFF.

Added row on EFF.

Added row on EFF.

Added row on EFF.

Updated row on SUSPECTED WI-FI or BLUETOOTH (TPED) INTERFERENCE.

[8.18.6.2 Portable EFB](#)

[8.18.6.2](#)

Added row on Skybook.

[8.18.7.2 Portable EFB](#)

[8.18.7.2](#)

Added row on Skybook.

[Appx2 APPENDIX 2 TO CHAPTER 8 -Approvals by Aeroplane Type](#)

[Appx2](#)

Updated to enable RNP-1 Navigation specification on the Falcon fleet.

Updated to enable RNP-1 Navigation specification on the Falcon fleet.

[9.1.3.1 Airworthiness and Operational Items](#)

[9.1.3.1](#)

Added mention to Note 1.

Added mention to Note 1. Added mention to power banks.

Added Note 1.

[9.2 Provisions for Dangerous Goods Carried by Passengers or Crew](#)

[9.2](#)

Added mention to power banks.

Added item.

Added item on power banks. Removed previous item i. on spare batteries.

Added note.

[10.1.4 NTA Security Organisation](#)

[10.1.4](#)

Updated Accountable Manager contact information.

Updated Head of Security Title.

Updated organisation chart.

[10.2.2.1 Access to Security Restricted Areas and Other Areas](#)

[10.2.2.1](#)

Updated paragraph on conditions.

[10.2.2.2 Requirements for Crew Identification Cards and Airport Identification Cards](#)

[10.2.2.2](#)

Updated paragraphs.

Updated paragraph on misuse of airport identification cards.

[10.2.2.3 Requirements for Vehicle Passes](#)

[10.2.2.3](#)

Updated list of requirements.

[10.2.2.5 Recognised Identification Cards](#)

[10.2.2.5](#)

Updated ID cards and badges.

[10.3.1 Searching and Checking the Aeroplane](#)

[10.3.1](#)

Updated paragraph on critical part.

Updated paragraph on critical part.

[10.3.2 Protection of Aeroplanes](#)

[10.3.2](#)

Removed item on access aids.

[10.3.3.1 Security Seals](#)

[10.3.3.1](#)

Updated paragraphs.

Removed placements for DA-2EASy.

Added CL650 information.

[10.3.3.2 Specific Aeroplane Threat](#)

[10.3.3.2](#)

Changed 'Security department' to 'Head of Security'.

[10.4.2.1 Screening of Passengers](#)

[10.4.2.1](#)

Updated paragraph and item list on screening.

Added paragraph on WTMD equipment alarms.

Added paragraph on HHMD equipment.

Added paragraph on explosive detection dogs, ETD equipment, SMD equipment and SED equipment.

[10.4.2.2 Screening of Cabin Baggage](#)

[10.4.2.2](#)

Updated and added content on cabin baggage screening.

Added content on Screening of liquids, aerosols, and gels (LAGs).

[10.4.5.1 Restrain Kits](#)

[10.4.5.1](#)

Added section.

[10.4.6 Prohibited Articles](#)

[10.4.6](#)

Added Example of information on prohibited articles available to the passenger.

[10.4.8 Identification of Passengers](#)

[10.4.8](#)

Updated section.

[10.5.0 General Provisions](#)

[10.5.0](#)

Removed content on an article listed in paragraphs (a) – (e) of Attachment 4-C to the Implementing Regulation being permitted to remain within accompanied hold baggage.

[10.5.1.1 Accompanied Hold Baggage](#)

[10.5.1.1](#)

Added new paragraph.

Added new paragraph.

Added new paragraph.

[10.5.4 Prohibited Articles](#)

[10.5.4](#)

Added content on an article listed in paragraphs (a) - (e) of Attachment 4-C to the Implementing Regulation being permitted to remain within accompanied hold baggage.

[10.5.4.1 Attachment 5-B - List of Prohibited Articles](#)

[10.5.4.1](#)

Added section.

[10.8.2.1 Screening](#)

[10.8.2.1](#)

Added item on EVD equipment.

Added paragraph.

[10.8.3 NTA Cleaning, Stores, and Supplies](#)

[10.8.3](#)

Updated paragraph on placement of Security Seals.

[10.11.3 Recruitment](#)

[10.11.3](#)

Updated section.

[10.11.4 Training](#)

[10.11.4](#)

Updated section.

[10.12.1 Transport of Weapons and Munitions of War](#)

[10.12.1](#)

Updated paragraph on weapons of war and munitions of war.

[10.12.3 Bodyguards](#)

[10.12.3](#)

Updated section.

[10.13.1.1 Aeroplane Hijack](#)

[10.13.1.1](#)

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.
 Changed 'Security Manager' to 'Head of Security'.

[10.13.2 Bomb Threat](#)

[10.13.2](#)

Changed 'Security Manager' to 'Head of Security'.

[10.13.2.1.1 Receipt of the Bomb Threat Call](#)

[10.13.2.1.1](#)

Updated Bomb Threat Checklist.

Updated Bomb Threat Checklist.

[10.13.2.1.2 Report of the Bomb Threat Call](#)

[10.13.2.1.2](#)

Changed 'Security Manager' to 'Head of Security'.

[10.13.2.1.3 Threat Assessment](#)

[10.13.2.1.3](#)

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.

[10.13.2.1.4 Actions to be Taken](#)

[10.13.2.1.4](#)

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.

[10.13.3 Discovery of a Suspect or Prohibited Article](#)

[10.13.3](#)

Changed 'Security Manager' to 'Head of Security'.

[10.13.7 Intoxicated Passengers](#)

[10.13.7](#)

Updated section.

[10.14.2 Responsibilities](#)

[10.14.2](#)

Changed 'Security Manager' to 'Head of Security'.

[10.14.3.1 Reportable Incidents](#)

[10.14.3.1](#)

Changed 'Security Manager' to 'Head of Security'.

[10.14.3.2 Reporting Methods](#)

[10.14.3.2](#)

Changed 'Security Manager' to 'Head of Security'.

[10.14.3.3 Crime on Board](#)

[10.14.3.3](#)

Changed 'Security Manager' to 'Head of Security'.

[10.15 Security Quality Assurance](#)

[10.15](#)

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.

[10.15.1 Purpose](#)

[10.15.1](#)

Changed 'Security Manager' to 'Head of Security'.

[10.15.2 Security Quality Assurance Applicability](#)

[10.15.2](#)

Changed 'Security Manager' to 'Head of Security'.

Changed 'Security Manager' to 'Head of Security'.

[TA 3 Meters to Feet Conversion Table](#)

[TA 3](#)

Updated table.

[TA 34 Runway Status Lights \(RWSL\)](#)

[TA 34](#)

Added new Technical Annex.

Abbreviations

ABBREVIATION	DESCRIPTION
A-CDM	Airport Collaborative Decision Making
AAD	Altitude Alerting Device
AAL	Above Aerodrome Level
ABP	Able Body Passenger
ACARS	Airborne Communications Addressing & Reporting System
ACAS	Airborne Collision Avoidance System
ACC	Area Control Centre
ACSD	Aviation Child Safety Device
AD	Altitude Deviation
ADF	Automatic Direction Finder
ADO	Assistant Director of Operations
ADS-C	Automatic Dependent Surveillance - Contract
AEA	Association of European Airlines
AFI	Africa-Indian Ocean Region
AFIS	Aerodrome Flight Information Service
AFM	Aeroplane Flight Manual
AFTN	Aeronautical Fixed Telecommunication Network
AGL	Above Ground Level
AHCC	Altitude Height Conversation Chart
AHQ	Advanced Handling Qualification
AIP	Aeronautical Information Publication
AIR OPS	Commission Regulation (EU) NO 965/2012 and Amendments Thereof Pursuant to the Regulation (EC) NO 216/2008 of the European Parliament and of the Council
AIREP	Air-Report
AIS	Aeronautical Information Service
AM	Accountable Manager
AMC	Acceptable Means of Compliance
AMCC	Aircraft Maintenance Control Centre
AME	Aviation Medical Examiner
AMSL	Above Mean Sea Level
ANAC	Autoridade Nacional de Aviação Civil
ANS	Air Navigation Service
ANSP	Air Navigation Service Provider
AOC	Air Operator Certificate
AOG	Aircraft On Ground
AOM	Aeroplane Operating Manual
APU	Auxiliary Power Unit
APV	Approach with Vertical Guidance
ASD	Accelerate-Stop Distance
ASE	Altimetry System Error
ASIA	Asia Region
AT	Autothrottle
ATA	Actual Time of Arrival
ATC	Air Traffic Control

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ABBREVIATION	DESCRIPTION
ATD	Actual Time of Departure
ATFCM	Air Traffic Flow and Capacity Management
ATIS	Automatic Terminal Information Service
ATL	Aircraft Technical Log
ATM	Air Traffic Management
ATO	Approved Training Organisation
ATS	Air Traffic Service
ATSU	Air Traffic Service Unit
BAC	Blood Alcohol Concentration
BALS	Basic Approach Light System
BECMG	Becoming
BEM	Basic Empty Mass
BKN	Broken
CAA	Civil Aviation Authority
CAME	Continuing Airworthiness Management Exposition
CAMO	Continuing Airworthiness Management Organisation
CAR	Caribbean Region
CARA	Crew Alertness Risk Assessment
CB	Circuit Breaker
CBT	Computer-Based Training
CCATT	Cabin Crew Attestation
CCM	Cabin Crew Member
CDFA	Continuous Descent Final Approach
CDL	Configuration Deviation List
CFMU	Central Flow Management Unit
CG	Centre of Gravity
CISM	Critical Incident Stress Manager
CMV	Converted Meteorological Visibility
CP	NJE Chief Pilot
CPDLC	Controller Pilot Data Link Communications
CRD	Child Restraint Device
CRM	Crew Resource Management
CRP	Cruise Relief Pilot
CSSF	Cold Soaked Surface Frost
CTOT	Calculated Take-Off Time
CVR	Cockpit Voice Recorder
DA	Decision Altitude
DA/H	Decision Altitude/ Height
DGOR	Dangerous Goods Occurrence Report
DGR	Dangerous Goods Regulation
DH	Descent Height
DME	Distance-Measuring Equipment
DMS	Document Management System
DO	Director of Operations
DOA	Design Organisation Approval
DOD	Department Of Defense

ABBREVIATION	DESCRIPTION
DOI	Dry Operating Index
DOM	Dry Operating Mass
DOT	Director Of Training
DP	Duty Period
DS	Dust Storm
DT	Director of Training
DTM	Duty Training Manager
EASA	European Union Aviation Safety Agency
EDP	Electronic Data Processing
EFB	Electronic Flight Bag
EGNOS	European Geostationary Navigation Overlay Service
EGPWS	Enhanced Ground Proximity Warning System
EOBT	Advanced Estimated Off-Block Time
ERA	En-Route Alternate
ERP	Emergency Response Plan
ETA	Estimated Time of Arrival
ETD	Estimated Time of Departure
ETOPS	Extended Range Two-Engined Operations
EU	European Union
EUR	European Region
EUR-RODEX	European Regional OPMET Data Exchange
EVS	Enhanced Vision System
FAA	Federal Aviation Administration
FAF	Final Approach Fix
FAS	Final Approach Segment
FATO	Final Approach and Take-Off Area
FBO	Fixed-Base Operator
FCC	Flight Crew Compartment
FCM	Flight Crew Member
FCOM	Flight Crew Operating Manual
FDM	Flight Data Monitoring
FDP	Flight Duty Period
FDR	Flight Data Recorder
FFS	Full Flight Simulator
FIC	Flight Information Centre
FL	Flight Level
FLDTA	Factored Landing Distance at Time of Arrival
FM	From
FMA	Flight Mode Annunciator
FMC	Flight Management Computer
FMS	Flight Management System
FOD	Foreign Object Damage
FOLD	Factored Operational Landing Distance
FOSM	Flight Operations Support Manager
FPFM	Flight Planning and Fuel Management
FRMS	Fatigue Risk Management System

ABBREVIATION	DESCRIPTION
FSB	Fasten Seat Belt
FSTD	Flight Simulator Training Device
FTOM	Flight Test Operations Manual
GDPR	General Data Protection Regulation
GHS	Globally Harmonized System of classification and labelling of chemicals
GIDS	Ground Ice Detection System
GNSS	Global Navigation Satellite System
GPIAAF	Gabinete de Prevenção e Investigação de Acidentes com Aeronaves e de Acidentes Ferroviários
GPS	Global Positioning System
GPU	Ground Power Unit
GPWS	Ground Proximity Warning System
GRF	Global Reporting Format
GRT	Ground Refresher Training
HF	High Frequency
HF	Human Factor
HOFO	Head Of Flight Operations
HOP	Head of Operations Planning
HUD	Head Up Display
IALS	Intermediate Approach Light System
IAP	Instrument Approach Procedure
ICAO	International Civil Aviation Organisation
IFR	Instrument Flight Rules
ILS	Instrument Landing System
IMC	Instrument Meteorological Conditions
INS	Inertial Navigation System
IRS	Inertial Reference System
ISA	International Standard Atmosphere
ISO	International Organization for Standardization
ISR	In-flight Service Representative
KIAS	Knots Indicated Air Speed
LAHSO	Land and Hold Short Operations
LDA	Landing Distance Available
LDG	Landing
LDTA	Landing Distance at Time of Arrival
LHS	Left Hand Seat
LNAV	Lateral Navigation
LOC	Localiser
LPV	Localizer Performance with Vertical guidance
LRC	Long Range Cruise
LTC	Line Training Captain
LVO	Low Visibility Operations
LVP	Low Visibility Procedures
LVTO	Low Visibility Take-Off
MAC	Manual handling Assessment Chart

ABBREVIATION	DESCRIPTION
MCC	Maintenance Control Centre
MCF	Maintenance Check Flight
MCTOM	Maximum Certified Take-Off Mass
MDA	Minimum Descent Altitude
MDA/H	Minimum Descent Altitude / Height
MDH	Minimum Descent Height
MEA	Minimum En-route Altitude
MEL	Minimum Equipment List
MET	Meteorological
METAR	Aviation Routine Meteorological Report
MID	Middle East Region + India / Pakistan
MLM	Maximum Landing Mass
MLS	Microwave Landing System
MMEL	Master Minimum Equipment List
MNPS	Minimum Navigation Performance Specification
MOCA	Minimum Obstacle Clearance Altitude
MOPSC	Maximum Operational Passenger Seating Configuration
MORA	Minimum Off-Route Altitude
MSA	Minimum Sector Altitude
MSL	Mean Sea Level
MTOM	Maximum Take-Off Mass
MVA	Minimum Vectoring Altitude
MZFM	Maximum Zero Fuel Mass
NALS	No Approach Light System
NAM	North American Region
NAT	North Atlantic Region
NAT HLA	North Atlantic High Level Airspace (formerly NAT MNPSA)
NAV	Navigation Equipment
NAVAID	Navigation Aid
NJCSP	NetJets Crew Support Procedure
NJE	NetJets Europe
NJML	NetJets Management Limited
NJMP	NetJets Maintenance Procedures
NM	Nautical Miles
NOTAM	Notice To Airmen
NPA	Non-Precision Approach
NS	No Smoking
NSSM	NetJets Service Standards Manual
NTA	NetJets Transportes Aéreos, S.A.
NTO	No-Technical Objection
OAT	Outside Air Temperature
OCA	Oceanic Control Area
OCC	Operations Control Centre
OCH	Obstacle Clearance Height
OCL	Obstruction Clearance Limit
OEI	One Engine Inoperative

ABBREVIATION	DESCRIPTION
OEM	Original Equipment Manufacturer
OFP	Operational Flight Plan
OHMP	Occupational Health Medical Practitioner
OM	Operations Manual
OM-A	Operations Manual Part A
OM-B	Operations Manual Part B
OM-C	Operations Manual Part C
OM-D	Operations Manual Part D
OML	Operational Multi-Pilot Limitation
OPC	Operator Proficiency Check
OVC	Overcast
PA	Pressure Altitude
PAPI	Precision Approach Path Indicator
PAR	Precision Approach Radar
PART-CAT	Annex IV (Commercial Air Transport) to the Regulation (EU) 965/2012
PART-CC	Annex V (Cabin Crew) to the Regulation (EU) No. 1178/2011
PART-FCL	Annex I (Flight Crew Licence) to the Regulation (EU) No. 1178/2011
PART-MED	Annex IV (Medical Requirements for Aircrew) to the Regulation (EU) No. 1178/2011
PART-SPA	Annex V (Operations Requiring Specific Approvals) to the Regulation (EU) 965/2012
PBC	Performance-Based Communication
PBCS	Performance-Based Communication and Surveillance
PBE	Protective Breathing Equipment
PBN	Performance Based Navigation
PBS	Performance-Based Surveillance
PDCS	Performance Data Computation System
PDSC	Pre-Departure Service Check
PED	Portable Electronic Devices
PF	Pilot Flying
PFD	Primary Flight Display
PIC	Pilot-In-Command
PL	Payload
PM	Pilot Monitoring
PNR	Point of No Return
POA	Production Organisation Approval
POH	Pilot Operating Handbook
PPR	Prior Permission Required
PRM	Passenger with Reduced Mobility
PROB	Probability
QC	Quality Check
QFE	Height above Airport Elevation
QNH	Atmospheric Pressure Adjusted to Mean Sea Level
QRH	Quick Reference Handbook
RA	Resolution Advisory (ACAS)
RAIM	Receiver Autonomous Integrity Monitoring

ABBREVIATION	DESCRIPTION
RCAM	Runway Condition Assessment Matrix
RCC	Runway Condition Code
RCF	Reduced Contingency Fuel
RCP	Required Communication Performance
RCR	Runway Condition Report
REA	Ready message
RFFS	Rescue and Fire Fighting Services
RHS	Right Hand Seat
RNAV	Area Navigation
RNP	Required Navigation Performance
RSAS	Runway Situational Awareness Systems
RSD	Runway Surface Descriptor
RSP	Required Surveillance Performance
RTF	Radiotelephone
RTO	Rejected Take-Off
RVR	Runway Visual Range
RVSM	Reduced Vertical Separation Minimum
RWSL	Runway Status Lights
RWY	Runway
RWYCC	Runway Condition Code
SAE	Society of Automotive Engineers
SAFA	Safety Assessment of Foreign Aircraft
SAFE	System for Aircrew Fatigue Evaluation
SAP	Stabilised Approach
SAT	Static Air Temperature
SBAS	Satellite Based Augmentation System
SBF	Seat Belt Fastened
SCCM	Senior Cabin Crew Member
SCP	Special Categories Passenger
SCT	Scattered
SELCAL	Selective Calling
SEP	Safety and Emergency Procedures
SERA	Standardised European Rules of the Air
SFI	Synthetic Flying Instructor
SIC	Second In Command
SID	Standard Instrument Departure
SIGMET	Significant Meteorological information
SITA	Société Internationale de Télécommunications Aéronautiques
SMS	Safety Management System
SNOWTAM	Snow NOTAM
SOP	Standard Operating Procedures
SPECI	Special Report Amending a METAR
SRA	Surveillance Radar Approach
SRM	Structure Repair Manual
SS	Sand Storm
SSR	Secondary Surveillance Radar

ABBREVIATION	DESCRIPTION
STAR	Standard Terminal Arrival Route
STC	Supplemental Type Certificate
STD	Scheduled Time of Departure
TA	Traffic Advisory
TAF	Terminal Aerodrome Forecast
TAS	True Airspeed
TAWS	Terrain Awareness and Warning System
TCAS	Traffic Collision Avoidance System
TCDS	Type Certification Data Sheet
TERPS	USA Standard for terminal instrument procedures
TFC	Trigger Flow Checklist
TL	Traffic Load
TOBT	Target Off-Block Time
TOC	Top Of Climb
TOD	Top Of Descent
TODA	Take-Off Distance Available
TODR	Take-Off Distance Required
TORA	Take-Off Run Available
TORR	Take-Off Run Required
TRE	Type Rating Examiner
TRI	Type Rating Instructor
TSAT	Target Start-up Approval Time
TTOT	Target Take-Off Time
TVE	Total Vertical Error
ULD	Unit Load Devices
UM	Unaccompanied Minor
UTC	Universal Time Coordinated
VAT	Indicated Airspeed at Threshold
VCi	Voice Contact Instruction
VFR	Visual Flight Rule
VHF	Very High Frequency
VIS	Visibility
VMC	Visual Meteorological Conditions
VMO	Maximum Operating Speed Limit
VNAV	Vertical Navigation
VOLMET	Meteorological Information for Aeroplane in Flight
VREF	Reference Speed
WAAS	Wide Area Augmentation System
WHO	World Health Organisation
WOCL	Window Of Circadian Low
ZFTT	Zero Flight Time Training

Record of Revisions

Revision Number	Issue date and Remark	Subject to authority's prior approval
Orig.	31 Aug 01	Yes
1	26 Jul 02	Yes
2	22 Aug 02	Yes
3	27 Jan 03	Yes
4	27 Oct 03	Yes
5	10 Jan 04	Yes
6	20 Feb 04	Yes
7	10 Mar 04	Yes
Reissue	17 Sep 04	Yes
1	10 Dec 04	Yes
2	14 Mar 05	Yes
3	24 Jun 05	Yes
4	22 Sep 05	Yes
5	27 Oct 05	Yes
6	02 may 06	Yes
7	06 Dec 06	Yes
TR01	07 Mar 07 (incorporated into Reissue - 30 May 07)	Yes
Reissue (Rev.8)	30 May 07	Yes
9	30 Nov 07	Yes
TR02	31 Jan 08 (incorporated into Rev.10 - 05 May 08)	Yes
10	05 May 08	Yes
TR03	14 Jul 08 (incorporated into Rev.11 - 05 Sep 08)	Yes
11	05 Sep 08	Yes
12	13 Oct 08	Yes
13	03 Nov 08	Yes
14	16 Mar 09	Yes
15	20 Apr 09	Yes
16	24 Sep 09	Yes
TR04	10 Dec 09 (incorporated into Rev.17 - 26 Apr 10)	Yes
17	26 Apr 10	Yes
TR05	27 Sep 10 (incorporated into Rev.18 - 22 Dec 10)	Yes

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Revision Number	Issue date and Remark	Subject to authority's prior approval
TR06	15 Oct 10 (incorporated into Rev.18 - 22 Dec 10)	Yes
18	22 Dec 10	Yes
19	21 Mar 11	Yes
TR07	14 Nov 11 (incorporated into Rev.10 - 12 Dec 11)	Yes
20	12 Dec 11	Yes
TR08	01 Jun 12 (incorporated into Rev.21 - 10 Apr 13)	Yes
TR09	02 Jul 12 (incorporated into Rev.21 - 10 Apr 13)	Yes
TR10	15 Feb 13 (incorporated into Rev.21 - 10 Apr 13)	Yes
21	10 Apr 13 (incorporated into Rev.21 - 10 Apr 13)	Yes
TR11	20 Sep 13 (incorporated into Rev.22 - 28 Dec 13)	Yes
22	28 Dec 13	Yes
TR12	07 Mar 14 (incorporated into Reissue - 10 Sep 14)	Yes
Reissue	26 Sep 14	Yes
TR13	12 Nov 14 (incorporated into Rev.24 - 13 Nov 15)	Yes
TR14	24 Dec 14 (incorporated into Rev.24 - 13 Nov 15)	Yes
TR15	28 Apr 15 (incorporated into Rev.24 - 13 Nov 15)	Yes
TR16	25 Jun 15 (incorporated into Rev.24 - 13 Nov 15)	Yes
TR17	06 Aug 15 (incorporated into Rev.24 - 13 Nov 15)	Yes
24	13 Nov 15	Yes
TR18	17 Mar 16 (incorporated into Rev.25 - 12 Dec 16)	Yes
TR19	29 Jun 16 (incorporated into Rev.25 - 12 Dec 16)	Yes
TR20	04 Aug 16 (incorporated into Rev.25 - 12 Dec 16)	Yes
TR21	22 Sep 16 (incorporated into Rev.25 - 12 Dec 16)	Yes
TR22	04 Oct 16 (incorporated into Rev.25 - 12 Dec 16)	Yes
25	12 Dec 16	Yes
TR23	17 Mar 17 (incorporated into Rev.26 - 08 Sep 17)	Yes
TR24	05 Sep 17 (incorporated into Rev.26 - 08 Sep 17)	Yes
26	08 Sep 17	Yes
TR25	15 Nov 17 (incorporated into Rev.27 - 04 Dec 17)	Yes
27	04 Dec 17	Yes
TR26	28 Mar 18 (incorporated into Rev.28 - 07 Feb 19)	Yes
TR27	26 Apr 18 (incorporated into Rev.28 - 07 Feb 19)	Yes

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Revision Number	Issue date and Remark	Subject to authority's prior approval
TR28	16 Aug 18 (incorporated into Rev.28 - 07 Feb 19)	Yes
TR29	21 Nov 18 (incorporated into Rev.28 - 07 Feb 19)	Yes
28	07 Feb 19	Yes
TR30	11 Apr 19 (incorporated into Rev.29 - 30 Jan 20)	Yes
TR31	09 Oct 19 (incorporated into Rev.29 - 30 Jan 20)	Yes
TR32	05 Dec 19 (incorporated into Rev.29 - 30 Jan 20)	Yes
TR33	29 Jan 20 (incorporated into Rev.29 - 30 Jan 20)	Yes
29	30 Jan 20	Yes
TR34	18 Mar 20 (incorporated into Rev.30 - 09 Set 20)	Yes
TR35	05 May 20 (incorporated into Rev.30 - 09 Set 20)	Yes
30	09 Set 20	Yes
TR36	13 Nov 20 (incorporated into Rev.31 - 12 Mar 21)	Yes
31	12 Mar 21	Yes
32	28 Jul 21	Yes
33	18 Apr 22	Yes
34	29 Jul 22	Yes
35	29 Jul 22	Yes
36	31 Mar 23	Yes
37.0.0	20 Nov 2023	No
38.0.0	06 Feb 2024	Yes
39.0.0	04 Sep 2024	Yes
40.0.0	09 Jan 2025	Yes
41.0.0	15 Jul 2025	No
42.0.0	13 Oct 2025	Yes
43.0.0	09 Feb 2026	No
44.0.0	07 Apr 2026	Yes
45.0.0	08 Jun 2026	Yes

Revisions subject to the authority's prior approval	See Letter of Approval.
Notification-only revisions (not subject to the authority's prior approval)	Internal review and acceptance were conducted by the Compliance Manager or Accountable Manager in accordance with ORO.GEN.130 (c) and Section 0.3 of the OM-A.
	Name:
	Signature:

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Compliance Statement

The Accountable Manager hereunder certifies that the contents of the Operations Manuals are in compliance with the requirements of AIR OPS - COMMISSION REGULATION (EU) No. 965/2012 pursuant to the Regulation (EU) 2018/1139 of the European Parliament and of the Council.

Name: Richard Weeks

Title: Accountable Manager

Richard
Weeks

Digitally signed by: Richard
Weeks
DN: CN = Richard Weeks
email = rmhweeks@ntasa.pt
C = PT O = ELT OU = COO
Date: 2024.02.12 12:01:08 Z

Signature:

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Chapter 0 - Administration and Control of Operations Manual

0.1 Introduction

The Operations Manual Part A is the NetJets Transportes Aéreos, S.A. OM-A accepted by the Autoridade Nacional de Aviação Civil (ANAC).

It refers to the Manufacturer's Flight and Operating Manuals (AFM and AOM), and the Manufacturer's normal, abnormal and emergency checklists.

The OM is primarily written in the ICAO language, English. Gender-neutral language is used at all times unless it reduces clarity or precision. When used in the OM, the following terms shall have the meanings indicated below:

- **'Shall'** or an action verb in imperative sense means its usage in a regulation or procedure or provision is mandatory.
- **'Should'** means that the application of a procedure or provision is recommended.
- **'May'** means that the application of a procedure or provision is optional.

Human factors principles shall be observed and used when developing the OM.

The OM and all associated manuals are and remain the property of NTA.

0.1.1 Rules and Regulations

The OM has been prepared per Regulation (EU) No 965/2012 (Air Operations), as amended, the applicable rules and regulations of Portugal, and the terms and conditions contained in the Air Operator's Certificate of NTA and associated operations specifications. In addition, it considers any relevant ICAO standards and procedures for Air Navigation Services.

The OM reflects the current and valid company policies, regulations, and procedures.

Deviations from regulations and acceptable means of compliance are only permissible if the state or competent authority has granted a derogation, exemption, or approved the implementation of Alternative Means of Compliance (AltMOC). These deviations will be documented in [Appendix 2 of Chapter 0](#).

0.1.2 Operations Manual Applicability

This Operations Manual (OM) contains regulations that shall be adhered to at all times by NTA crew members, operational personnel involved with safety of flight operations, as well as those persons not directly employed by NTA, who hold positions directly related to this company's operations.

In the event of willful or negligent disobedience to these rules and regulations, the personnel involved may become subject to disciplinary, legal, or penal action.

However, nothing contained in the OM shall prevent personnel from exercising their own best judgment in any situation for which the OM makes no provision or in an emergency.

If an NTA employee finds that the NTA OM is deficient in providing adequate information to perform a function properly and requires urgent revision, they shall contact the DO by email at lis-DO@netjets.com or by telephone to describe the deficiency and recommended changes. The DO in coordination with the relevant nominated persons analyses comments to determine whether a single or a systemic problem exists in a manual. Feedback will be provided to the reporter.

0.1.3 Operations Manual Description

This OM consists of four separate parts:

The **Operations Manual Part A (OM-A)** contains non-type related operational policies, instructions, and procedures required for safe operation. It details the duties and responsibilities of all flight and ground operations personnel, and their interrelationship to the operation as a whole. A complete copy of the OM-A shall be supplied to all flight crew members.

The **Operations Manual Part B (OM-B)** contains all type related instructions and procedures required for safe operation of the aeroplane. It comprises the manufacturer's aeroplane documentation, relevant checklists, and a description and instructions for the use of emergency equipment, and the actions to be taken in emergencies. It also contains other type-specific material, and NTA Standard Operating Procedures. A complete copy of the OM-B is supplied to all flight crew members.

The **Operations Manual Part C (OM-C)** comprises all the navigation and other data required for the NetJets area of operations. The Jeppesen Airway Manual, which is supplied to all flight crew members, contains all relevant information for departure, destination, alternate destination, emergency fields and the route. The Route Manual covers the different geographic regions of NetJets operations, and contains information regarding flight navigation, as well as the regulations and procedures of respective states / regions, or specific places, which may vary from or enhance the regulations herein. The Jeppesen Route Manual corresponds to the requirements for AIR OPS OM-C.

The **Operations Manual Part D (OM-D)** contains all training instructions and qualifications required for safe operation.

In addition to the OM, NetJets uses the:

- **Continuing Airworthiness Management Exposition (CAME)**, which contains all instructions and procedures to be followed by maintenance personnel to satisfy NetJets' maintenance quality requirements.
- **Safety Equipment and Procedures Manual**, which contains all emergency equipment carried on board of NTA aeroplanes and outlines the standard emergency and safety procedures applied by the company.

0.2 System of Amendment and Revision

The Accountable Manager is ultimately responsible for the contents of the OM. Under the duties and responsibilities detailed in this manual, this responsibility is delegated to the Director of Operations for the OM-A, OM-B, and OM-C, and to the Director of Training for the OM-D.

To effectively support the OM, functional responsibility for issuing and reviewing each chapter is further delegated to the appropriate positions within the NTA organisation, based on their areas of responsibility and expertise. The delegation is made in accordance with [0.3.3 Revision Process Workflow](#). In the context of the workflow necessary for the creation, revision, and approval of changes in the OM, the following responsibilities are assigned to each position:

Revision Lead

The Revision Lead coordinates and oversees the amendment process for the assigned part(s) of the OM. Their responsibilities include:

- tracking amendment cycles;
- coordinating with subject matter experts;
- drafting or revising draft text in response to regulatory or procedural changes;
- ensuring alignment with regulatory requirements (notwithstanding the responsibilities of the Compliance Manager and the Responsible Manager);
- ensuring consistency with safety, regulatory, and internal quality processes;
- coordinating with Document Control to ensure timely submission, publication, and distribution of amendments.

To maintain impartiality and avoid conflicts of interest, the Revision Lead should not hold formal approval responsibility for technical or operational content.

For OM-A and OM-C, the Revision Lead role is typically performed by a member of Operations Regulatory and Technical Support (ORTS).

Responsible Manager

- Submit proposals for changes to the OM section to the relevant nominated person or Department Director following the revision process;
- Ensure that the relevant section is kept up to date and compliant with with AIR OPS, the national rules and regulations of Portugal, the boundaries of the AOC of NTA, Company policies and procedures, and also that any amendments required by the Competent Authority are introduced;
- Ensure that the information taken from approved documents, and any amendment thereof, is correctly reflected in the OM;
- Run an Operational Risk Assessment if deemed necessary;
- Process feedback from the Reviewer(s) before contents are sent for approval.

Reviewer

- Provide an independent review of amendments prepared by the responsible manager, focusing on:
 - Appropriateness of the amendment in the context of the reviewer's operational area;
 - Accuracy and clarity of the amendment;
 - Compliance with the applicable requirements and regulations.

Nominated Person / Director

- Review and sign-off the revised content of the relevant OM section before it is submitted to the Director of Operations and Accountable Manager for final approval;
- Ensure all contents for the section are continuously revised and amended to comply with all legal and company regulations;
- Ensure that any prior approval from the authority is obtained before that amendment becoming effective, or in cases where an immediate revision is required in the interests of safety, that any required approval has been requested before the amendment becomes effective.

Director of Operations (DO) and Accountable Manager (AM)

- Approve all contents (after sign-off from the nominated person or Department Director);
- Submit all proposed amendments and revisions to the Authority in accordance with [0.3 Control of Amendments](#).

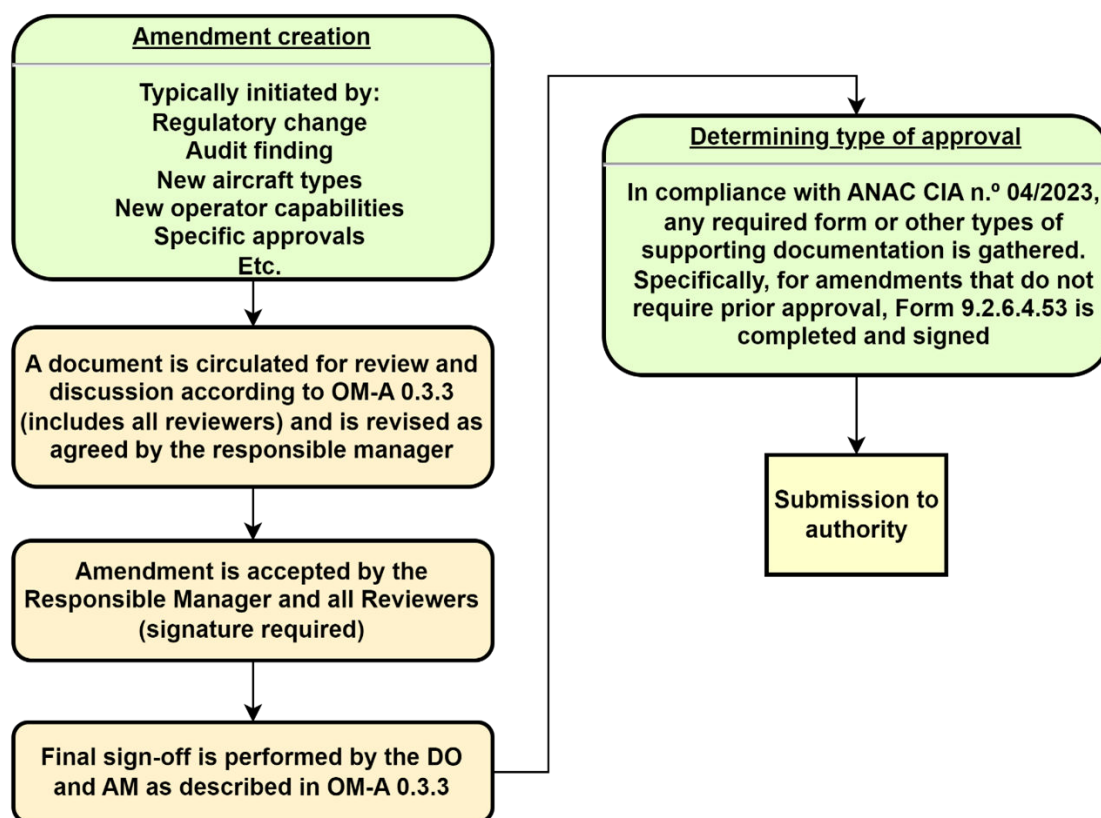
0.3 Control of Amendments

The OM will be updated to reflect any changes in internal structure, applicable regulations, procedures or policies in NTA, and to correct typos and / or to increase its clarity or readability. Also, changes required by the competent authority will be incorporated. The aim is to perform at least one amendment cycle per year and a review 6 months after the last cycle in the event that another cycle has not occurred.

All NTA employees are encouraged to submit suggestions for improvements to the OM, subject to the provisions of AIR OPS and the Portuguese legislation. All suggestions should be forwarded to Lis-OM@netjets.com or the applicable responsible manager.

Internal Control

All amendments, regardless of whether they are subject to the prior approval of the competent authority, must go through the complete revision process workflow depicted in [0.33 Revision Process Workflow](#), except those classified as minor changes (as listed below). This process is shown with a simplified flowchart below.



Examples of minor changes are:

1. Editorial issues, typos, etc. (without changing content);
2. Minor changes due to regulatory update (update of EASA documents references);
3. Edition for clarity improvement of the text, without changes to the original content or intent;
4. Glossary update;
5. Change of document templates, except the documents requiring direct ANAC approval;

6. Update of references.

Control by the Competent Authority

All changes, regardless of whether they require prior approval, shall be managed using the published internal validation and review process, approved by the competent authority. With this process, responsibility, accountability and traceability of the changes are guaranteed. Proposed amendments will be supplied to the Authority for approval in advance of the effective date.

a. Amendments Requiring Prior Approval

For any changes requiring prior approval, NetJets shall apply for and obtain an approval, in writing, issued by the Competent Authority. The amendments shall not become effective before approval has been issued.

The application shall be submitted at least 30 calendar days prior to the intended effective date to allow the Competent Authority to determine continued compliance with Regulation (EU) 2018/1139 and its Implementing Rules and to amend, if necessary, the operator certificate and related terms of approval attached to it. NTA shall provide all relevant documentation to the competent authority. This includes any risk assessment or management of change requested by the competent authority.

- Each time amendments requiring prior approval are submitted, the following documents must be included:
 - 'Requerimento para aprovação do OM' (9.2.6.4.46), which shall be updated from the previous submission.
 - The compliance checklist ('Lista de verificação de conformidade OM') corresponding to the manual being submitted (Forms 9.2.6.4.47 to 9.2.6.4.50).
 - In cases where specific approval is being requested (i.e., DG, LVOs, RVSM, ETOPS, PBN, MNPS, HHO, HEMS, HOFO, SET-IMC, PinS, VEMS, EFB), the applicable form for that approval, along with all required supporting documentation.

Appendix 1 to Chapter 0 - Items Requiring Prior Approval provides a non-exhaustive list of items which require prior approval; if any doubt exists as to whether a change requires approval it shall be assumed that it does and the Competent Authority shall be consulted to clarify.

NOTE

Exceptionally, when prior approval is required, but an immediate amendment or revision is required in the interest of safety, it may be published as a temporary change effective immediately, provided that any approval required has been applied for.

b. Amendments not requiring prior approval - Notification only

All amendments not requiring prior approval will be submitted to the Competent Authority with a minimum of 10 working days in advance of the intended effective date, accompanied by the form 9.2.6.4.53 (ANAC's Notificação De Alterações Sem Necessidade De Aprovação Prévia).

To qualify for the notification-only procedure, the amendment shall either:

- Not be related to the items included in the exhaustive list of items requiring prior approval (ANAC form 9.2.6.4.46);

- If related to the items included in the exhaustive list of items requiring prior approval:
 - the amendment must not have any impact on operational procedures or on safety; or,
 - be of an editorial nature, without impact on the intention of the text.
- Result from the update of an OEM manual or equivalent document.

Each time amendments following this procedure are submitted, the compliance checklist ('Lista de verificação de conformidade OM') corresponding to the manual being submitted (Forms 9.2.6.4.47 to 9.2.6.4.50) shall also be provided.

The competent authority shall assess the information supplied by NTA to confirm compliance with any relevant requirements. If compliance is not confirmed, the competent authority shall inform NTA. The amendment shall only take effect after the 10-day period has elapsed, provided there is confirmation that the submission email was received (automatic response emails serve as acknowledgement) and no formal objection has been raised by the competent authority.

Amendments following the notification-only procedure should preferably be submitted separately from those requiring prior approval.

NOTE

Exceptionally, when an immediate amendment or revision is required in the interest of safety, it may be published as a temporary change effective immediately, provided that the competent authority has been notified.

Release of amendments

Before issuing a new revision, the Operations Regulatory and Technical department shall issue an announcement (OM-Alert) to advise crews and other relevant personnel of any amendments to the OM, their effective date, and to provide any additional context if deemed necessary. The effective date should reflect the need to provide enough time for the affected personnel to study the changes, unless the revision needs to be effective immediately in the interest of safety.

Hand-written amendments are not permitted.

0.3.1 Temporary Revisions

If it is necessary to effect changes to the OM only for a limited time, an OM Alert / OM-Temporary Text will be issued rather than a revision being made to the applicable OM.

Temporary changes shall be valid for a maximum of 6 months, after which time they shall either be cancelled, incorporated into the applicable OM or re-issued.

0.3.2 Identification of Changes

This manual contains a '[Record of Revisions](#)' which will register all revisions (including Temporary Revisions) and their dates. The 'Record of Revisions' table will show in the 'Subject to authority's prior approval' column:

- 'YES' – for amendments requiring prior approval in accordance with [0.3, Control by the Competent Authority, par. \(a\)](#);
- 'N/A' – for amendments not requiring prior approval in accordance with [0.3, Control by the Competent Authority, par. \(b\)](#).

With each amendment, the revised pages of the 'List of Effective Pages' will be updated.

Each page of the manual is annotated with the required control information which includes:

- the title of the chapter;
- the number of the chapter;
- the page number within that chapter;
- the revision date for that page.

Contents of this manual that have been changed (added, revised or moved) are marked with a vertical sidebar on the left side of the text. A vertical sidebar without corresponding text indicates that content has been deleted. These markings apply solely to changes made in the current revision. Editorial changes (correction of typos and text formatting) will not be marked.

0.3.3 Revision Process Responsibilities

The following table defines the responsibilities for the creation, revision, review, declaration, and approval of amendments to the Operations Manual.

Element	Title	Responsible Manager	Reviewers	Operator Declaration ¹	Final Approval
		(Nominated Persons OM-A 1.2, otherwise noted)			
OM-A Ch0	Administration and Control of OM	ADO ²	Compliance	Compliance	AM
OM-A Ch1	Organisation and Responsibilities	AM	Safety / Compliance / Continuing Airworthiness / Crew Training / Ground Operations / Dangerous Goods / Flight Operations	Compliance	AM
OM-A Ch2	Operational Control and Supervision	ADO ²	Safety / Compliance / Continuing Airworthiness / Crew Training / Ground Operations / Dangerous Goods / Flight Operations	Compliance	AM
OM-A Ch3	Management System	Compliance	Safety	Compliance	AM
OM-A Ch4	Crew Composition	ADO ²	Crew Training / Flight Operations	Compliance	AM
OM-A Ch5	Qualification Requirements	Crew Training	Flight Operations	Compliance	AM
OM-A Ch6	Crew Health Precautions	Safety	Flight Operations	Compliance	AM
OM-A Ch7	Flight Time Limitations	ADO ²	Ground Operations / Flight Operations / Safety	Compliance	AM
OM-A Ch8	Operating Procedures	ADO ²	Crew Training / Flight Operations	Compliance	AM
OM-A Ch9	Dangerous Goods and Weapons	Dangerous Goods	Crew Training / Flight Operations	Compliance	AM
OM-A Ch10	Security	Security	Flight Operations	Compliance	AM
OM-A Ch11	Management of accidents, incidents, and occurrences	Safety	Flight Operations	Compliance	AM
OM-A Ch12	Rules of the Air	ADO ²	Crew Training / Flight Operations	Compliance	AM
OM-A Ch13	Leasing	ADO ²	Flight Operations	Compliance	AM
SEP Manual	As per SEP Manual				
OM-B	Type-specific operating matters	Fleet Program Manager ³	Crew Training / Flight Operations	Compliance	AM

Element	Title	Responsible Manager	Reviewers	Operator Declaration ¹	Final Approval
		(Nominated Persons OM-A 1.2, otherwise noted)			
OM-C	Route and Aerodrome Instructions and Information	ADO ²	Crew Training	Compliance	AM
OM-D	Training	Crew Training	Standards ⁴	Compliance	AM

¹ **Operator Declaration:** Required for all changes to the OM and confirms compliance with all applicable regulations. Normally signed by Compliance Monitoring.

² **ADO:** Assistant Director of Operations. Not a Nominated Person; deputisation done as laid in this section.

³ **Fleet Programme Managers:** Technical leads for each aircraft type, appointed by the NP Flight Operations. Not a Nominated Person; deputisation done as laid in this section.

⁴ **Standards:** Person under the delegation of the NP Crew Training. Not a Nominated Person; deputisation done as laid in this section.

Deputisation of the revision process responsibilities

When a Responsible Manager or Reviewer is absent, deputisation is performed in accordance with [OM-A 1.2.1](#). For positions that are not Nominated Persons, duties are undertaken by the individual's line manager during their absence.

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0.4 Distribution System

The OM is made available electronically in the DMS, on the Portable EFB, and on the internal network to all relevant persons who are required to be familiar with and comply with its content.

A Master Copy (Original) is kept in the Technical Library by Document Control.

The Competent Authority, ANAC, has been provided with a copy of the OM, and receives all amendments and revisions thereto.

Subcontractors shall be provided a copy if and as applicable.

Description	Aircraft	FCM	CCM	Ops Personnel	Crew Training Personnel
Operations Manual Part A	Yes	Yes	Yes	Yes	Yes
Safety Equipment Manual	Yes	Yes	Yes	Yes	Yes
Operations Manual Part B (fleet specific)	Yes	Yes	No	Yes	Yes
Operations Manual Part C	Yes	Yes	No	Yes	Yes
Operations Manual Part D	Yes	Yes	Yes	Yes	Yes

0.5 Glossary

This glossary contains terms and abbreviations that are used in this OM, Weight and Balance, and other operational and handling documents.

0.5.1 Abbreviations

For abbreviations used in air navigation / aeroplane operations, see **Jeppesen Route Manual Part GEN**.

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0.5.2 Terms and Definitions

These terms and definitions are proposed by the member states. Note that these definitions may change in the future, and that the definitive reference should be individual Parts of AIR OPS.

Adequate Aerodrome. An adequate aerodrome means an aerodrome that at the time of the expected time of use is available and can be operated, taking account of the applicable performance requirements and runway characteristics and is equipped with the necessary ancillary services such as air traffic services (ATS), sufficient lighting, communications, meteorological reports, navigation aids, and emergency services.

Aerodrome Operating Minima means the limits of usability of an aerodrome for:

- a. take-off, expressed in terms of runway visual range (RVR) and / or visibility and, if necessary, ceiling;
- b. landing in 2D instrument approach operations, expressed in terms of visibility and / or RVR, minimum descent altitude / height (MDA/H) and, if necessary, ceiling;
- c. landing in 3D instrument approach operations, expressed in terms of visibility and / or RVR and decision altitude / height (DA/H) as appropriate to the type and / or category of the operation.

Aeroplane means an engine-driven fixed-wing aeroplane heavier than air that is supported in flight by the dynamic reaction of the air against its wings.

Aeroplane Equipment means articles, other than stores and spare parts of a removable nature, for use on board an aeroplane during flight, including first-aid and survival equipment.

Aeroplane Flight Manual (AFM) is a manual associated with the certificate of airworthiness that contains information (limits, procedures, data, etc.) required to operate the aeroplane at the level of safety established by the aeroplane's certification basis.

Aeroplane Operating Manual (AOM) is part of the Operations Manual describing in detail the characteristics and operation of the aeroplane and its systems.

Aeroplane Variant, as used with respect to the licensing and operation of flight crew, means an aeroplane of the same basic certificated type which contains modifications not resulting in significant changes of handling and / or flight characteristics, or flight crew complement, but causing significant changes to equipment and / or procedures.

Air Operator Certificate (AOC) is a certificate authorising an operator to carry out specified commercial air transport operations.

Air Traffic Control Unit means, variously, area control centre, approach control office, or aerodrome control tower.

Air Traffic Incident means a serious occurrence involving air traffic such as:

1. Near collision;
2. Serious difficulty caused by:
 - faulty procedures or lack of compliance with applicable procedures;
 - failure of ground facilities.

Alternate Aerodrome means an adequate aerodrome to which an aircraft may proceed when it becomes either impossible or inadvisable to proceed to or land at the aerodrome of intended landing. It implies that all the necessary services and facilities are available, the aircraft performance requirements can be met, and which is operational at the expected time of use. It includes the following:

- take-off alternate aerodrome: an alternate aerodrome at which an aircraft would be able to land shortly after take-off, in cases where it is not possible to use the aerodrome of departure;
- en route alternate (ERA) aerodrome: an alternate aerodrome at which an aircraft would be able to land if a diversion becomes necessary while en route;
- fuel en route alternate (fuel ERA) aerodrome: means an ERA aerodrome that is required at the planning stage for use in the calculation of fuel;
- 'destination alternate aerodrome': an alternate aerodrome at which an aircraft would be able to land if it becomes either impossible or inadvisable to land at the aerodrome of intended landing.

Approach with Vertical Guidance (APV) Operation means an instrument approach which utilises lateral and vertical guidance, but does not meet the requirements established for precision approach and landing operations, with a DH not lower than 250 ft and a runway visual range (RVR) of not less than 600 m. These are coded in Jeppesen charts as RNAV (GNSS): LNAV/VNAV (APV Baro) or LPV (APV SBAS). See also Instrument approach procedure (IAP).

Approved by the Authority means documented by the Authority as suitable for the purpose intended.

Audit is a systematic, independent and documented process for obtaining evidence and evaluating it objectively to determine the extent to which requirements are complied with.

Auditee is the area or organisation that is being audited.

Auditor is an individual that satisfies NTA qualification and competence standards and has been formally approved to conduct an audit in a specific area.

Authority means the competent body responsible for the safety regulation of Civil Aviation.

Best practice is a strategy, process, approach, method, tool or technique that is generally recognised as being effective in helping to achieve operational objectives.

Cabin Crew Member is an appropriately qualified crew member, other than a flight crew or technical crew member, who is assigned to perform duties related to safety of passengers and flight during operations.

Calibrated Airspeed means indicated airspeed of an aeroplane, corrected for position and instrument error. Calibrated airspeed is equal to true airspeed in standard atmosphere at sea level.

Captain means a pilot meeting all the requirements of [5.2.1.2 Minimum Qualification Requirements](#) for Commanders. For the purpose of this manual, the term is used in the context of a specific role related to crew training.

Cargo in relation to Commercial Air Transportation, means any property, including animals and mail, carried by an aeroplane other than stores and accompanied baggage.

Ceiling means the height above the ground or water of the base of the lowest layer of cloud below 6 000 m (20 000 ft) covering more than half the sky.

Circling means the visual phase of a circling approach operation.

Circling Approach Operation means a Type A instrument approach operation to bring an aircraft into position for landing on a runway / final approach that is not suitably located for a straight-in approach.

Civil Aeroplane means any aeroplane on the civil register of a State, other than those which that State treats as being in the service of the State, either permanently or temporarily.

Cockpit Voice Recorder (CVR) means a crash-protected flight recorder that uses a combination of microphones and other audio and digital inputs to collect and record the aural environment of the flight crew compartment and communications to, from and between the flight crew members.

Commander means the pilot designated as being in command and charged with the safe conduct of the flight.

Commercial Air Transportation means the transportation by air of passengers, cargo, or mail for remuneration or hire.

Commercial Operations means any operation of an aeroplane, in return for remuneration or other valuable consideration, which is available to the public or, when not made available to the public, which is performed under a contract between an operator and a customer, where the latter has no control over the operator.

Competent Authority is the authority designated by the member state for the oversight over operators subject to certification obligation. For NTA this is the ANAC.

Conformity is the fulfillment of specifications contained in the requirements as being documented and / or implemented by the auditee.

Contaminated Runway means a runway of which a significant portion of its surface area (whether in isolated areas or not) within the length and width being used is covered by one or more of the substances listed under the runway surface condition descriptors.

Contingency Fuel means the fuel required to compensate for unforeseen factors that could have an influence on the fuel consumption to the destination aerodrome.

Continuous Descent Final Approach (CDFA) is a specific technique for flying the final-approach segment of a non-precision instrument approach procedure as a continuous descent, without level-off, from an altitude / height at or above the Final Approach Fix altitude / height to a point approximately 15 m (50 ft) above the landing runway threshold or the point where the flare manoeuvre should begin for the type of aeroplane flown.

Converted Meteorological Visibility (CMV) is a value (equivalent to an RVR) which is derived from the reported meteorological visibility, as converted in accordance with the requirements in [8.1.3.6.7](#).

Corrective Action is the action(s) taken by the auditee to address the non-conformity and eliminate its recurrence.

Crew Briefing means the information provided to the crew prior to duty or flight activities that defines and supports those activities. It consists primarily of messages in the Waypoint application on the company mobile device, which may include information from or point to other sources, such as feasibility notes, NTA Aerodrome Briefings, or OM-C Briefings. The purpose of the Crew Briefing is to ensure that the crew is aware of all relevant information related to their duty or flight activities, including any changes or updates.

Crew Member means a person assigned by an operator to perform duties on board an aeroplane.

Crew Memo is used to inform crews about issues which are not directly related to the operations but rather related to Crew Policies or any other topics under the responsibility of the Crew Support and Planning department.

Critical Phases of Flights means:

- a. Take-off run;
- b. Take-off flight path;
- c. Final approach, missed approach, and landing including the landing roll;
- d. Any other phase(s) of flight at the discretion of the Commander.

Cruise Relief Pilot (CRP) is a first officer who has successfully completed the Cruise Relief Pilot training (initial / recurrent) as detailed in the OM-D, making them qualified to act as Pilot Relieving the Commander.

Cruise Relief Co-pilot means a flight crew member who relieves the SIC (co-pilot) of their duties at the controls during the cruise phase of a flight in multi-pilot operations above FL 200. NTA only allows fully qualified pilots to serve as flight crew members and, consequently, in-flight reliefs. Therefore, any flight crew designated to the flight may relieve the SIC in the cruise phase.

Dangerous Goods means articles or substances which are capable of posing a risk to health, safety, property, or the environment and which are shown in the list of dangerous goods in the Technical Instructions, or which are classified according to those Instructions.

Decision Altitude / Height (DA/H) is a specified altitude / height in the precision approach or approach with vertical guidance at which a missed approach shall be initiated if the required visual reference to continue has not been established.

NOTE

Decision Altitude (DA), is referenced to mean sea level (MSL) and Decision Height (DH) is referenced to the runway threshold elevation.

'Dry Runway' means a runway whose surface is free of visible moisture and not contaminated within the area intended to be used.

EFB Administrator is a person appointed by the operator to be held responsible for the administration of the EFB system within the company. The EFB administrator is the primary link between the operator and the EFB system and software suppliers.

EFB Host Platform (equipment) - When considering an EFB system, the EFB host platform is the equipment (i.e. hardware) in which the computing capabilities and basic software (e.g. operating system, input / output software) reside.

EFB Software Application is the software installed on an EFB system that allows specific operational functionality.

EFB System comprises the hardware (including any battery, connectivity provision, I/O devices) and software (including databases) needed to support the intended EFB function(s).

Electronic Charts are electronic navigation charts displayed on an EFB system; includes both terminal and en-route charts.

Electronic Flight Bag (EFB) is an information system for crew members which allows storing, updating, delivering, displaying, and / or computing digital data to support flight operations or duties.

Electronic Terminal Charts are electronic navigation charts displayed on an EFB system; includes only terminal charts.

Enhanced Flight Vision System (EFVS) means an enhanced vision system (EVS) that also incorporates a flight guidance system and displays the image on a HUD or an equivalent display. It provides the flight crew with real-time display, or an enhanced display, of the external topography (natural and human-made) through the use of imaging sensors; an EFVS may allow for EFVS operation or operational credits if applicable requirements are met.

EFVS Operation means an operation in which visibility conditions require an EFVS to be used instead of natural vision in order to perform an approach or landing, identify the required visual references or conduct a roll-out.

Enhanced Vision System (EVS) is an electronic means of displaying a real-time image of the external scene through the use of imaging sensors.

Estimated Off-Block Time (EOBT) is the estimated time at which the aeroplane will commence movement associated with a departure.

Evidence is data or information found during an audit that is analysed by the auditor to determine fulfillment of the requirements.

Exemption is a formal authorisation issued by the competent Authority or Member State providing relief from part or all of the provisions of a regulation. The authorisation may or may not be conditional.

Factored Landing Distance at Time of Arrival (FLDTA) LDFA factored by the required 15% operational distance factor so as to comply with the requirement for the LDA to be a minimum of 115% of the LDR. This may also be referred to as 'Factored Operational Landing Distance (FOLD)' by some OEMs.

Fail-Operational Flight Control System means that a flight control system is fail-operational if, in the event of a failure below alert height, the approach, flare and landing can be completed by the remaining part of the automatic system, which becomes a fail-passive system. Hybrid systems are considered to be fail operational if, in the event of a failure below decision height, the landing can be completed either by the pilot using head up guidance, or by the landing system itself.

Fail-Passive Flight Control System means that a flight control system is fail-passive if, in the event of a failure, there is no significant out-of-trim condition or deviation of flight path or altitude, but the landing is not completed automatically. For a fail-passive automatic flight control system, the pilot assumes control of the aeroplane after a failure.

Final Approach is that part of an instrument approach procedure which commences at the specified final approach fix or point, or where such a fix or point is not specified:

- At the end of the last procedure turn, base turn or inbound turn of a racetrack procedure, if specified; or
- At the point of interception of the last track specified in the approach procedure; and ends at a point in the vicinity of an aerodrome from which:
 - A landing can be made; or
 - A missed approach procedure is initiated.

Final Approach Segment (FAS) means that segment of an instrument approach procedure (IAP) in which alignment and descent for landing are accomplished.

Final Reserve Fuel means an amount of fuel for all turbine powered (jet) aeroplanes, calculated to fly for 30 minutes at holding speed at 1500 feet AGL / ISA conditions, calculated with the expected landing weight at alternate (or destination if no alternate is required). For piston engine aeroplanes, the final reserve fuel shall be for 45 minutes at holding speed.

Final Take-off Speed means the speed of the aeroplane that exists at the end of the take-off path in the en-route configuration with one engine inoperative.

Finding is the non-fulfilment of specifications contained in requirements as determined by the auditor in terms of having been documented and / or implemented by the auditee.

First Aid Oxygen means the additional oxygen provided for the use of passengers, who do not satisfactorily recover following subjection to excessive cabin altitudes, during which they had been provided with supplemental oxygen.

Flight Crew Member means a licensed crew member charged with duties essential to the operation of an aeroplane during a flight duty period.

Flight Data Recorder (FDR) means a crash-protected flight recorder that uses a combination of data sources to collect and record parameters that reflect the state and performance of the aircraft.

Flight Plan can be one of the following:

1. **ATS Flight Plan:** Specified information to Air Traffic Service Units, relative to an intended flight or portion of a flight of an aeroplane.
2. **Current ATS Flight Plan:** The ATS Flight Plan, including changes, if any, brought about by subsequent clearances.
3. **Filed ATS Flight Plan:** The Flight Plan as filed with an ATS unit by the pilot or their designated representative, without any subsequent changes.
4. **Operational Flight Plan:** The operator's plan for the safe conduct of the flight, based on considerations of the aeroplane performance, other operating limitations, and relevant expected conditions on the route to be followed and at the aerodromes concerned.
5. **Repetitive ATS Flight Plan:** A Flight Plan related to a series of frequently recurring, regularly operated individual flights with identical basic features, submitted by an operator for retention and repetitive use by ATS units.

Flight Monitoring means the following functions performed by flight dispatch:

- operational monitoring of flight progression throughout all phases of the flight;
- communication of all available and relevant safety information between the operational-control personnel on the ground and the flight crew;
- critical assistance to the flight crew in the event of an in-flight emergency or security issue, or at the request of the flight crew.

Flight Recorder means any type of equipment installed on the aeroplane and capable of processing data which can be used to:

- Support the investigation of incidents and accidents;
- Monitor the operation of the aeroplane;

- Support maintenance activities.

The processed data can be stored on board or transmitted to an external system for remote storage and / or analysis.

Examples of Flight Recorders include Flight Data Recorder, Cockpit Voice Recorder, Quick Access Recorder, Terrain Awareness and Warning System, Central Maintenance Computer, Electronic Engine Controller, Airborne Reporting System. This list is not exhaustive.

Flight Time is the total time from the moment an aeroplane first moves under its own power for the purpose of taking off until the moment it comes to rest at the end of the flight for the purpose of unloading persons (including crew members or cargo).

Fractional Operations are Non-Commercial Operations performed by NTA without owners on board and Non-commercial Operations with owners on board where NTA performs the role of a fractional programme manager, in accordance with the procedures and limitations in the OM, except for the procedures and limitations in the Fractional Operations Manual.

Fuel Scheme means a set of policies, detailed in this operations manual, which are appropriate for NTA's operations and corresponding to NTA's capabilities to support them, as defined in Commission Regulation (EU) No 965/2012. It comprises of:

- a fuel / energy planning and in-flight re-planning policy;
- an aerodrome selection policy; and
- an in-flight fuel / energy management policy.

Go-around means a transition from an approach operation to a stabilised climb. This includes manoeuvres conducted at or above the MDA/H or DA/H, or below the DA/H (balked landing).

Handicapped Passenger is a person who is unable to move by themselves to the nearest emergency exit at floor level, in the event of an emergency evacuation.

Head-Up Display (HUD) is a display system which presents flight information into the pilot's forward external field of view and which does not significantly restrict the external view.

IFR Conditions means weather conditions below the minimum for flight under visual flight rules.

Inspection is an independent documented conformity evaluation by observation and judgement accompanied as appropriate by measurement, testing or gauging, in order to verify compliance with applicable requirements.

Installed EFB means that there are two possibilities for the hardware of EFB systems: portable and installed. An EFB host platform installed in the aeroplane and considered as an aeroplane part, covered, thus, by the aeroplane airworthiness approval.

Instrument Approach Operation means an approach and landing using instruments for navigation guidance based on an instrument approach procedure (IAP). There are two methods for executing instrument approach operations:

- a two-dimensional (2D) instrument approach operation, using lateral navigation guidance only;
- a three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance.

Instrument Approach Procedure (IAP) means a series of predetermined manoeuvres by reference to flight instruments with specified protection from obstacles from the initial approach fix or, where applicable, from the beginning of a defined arrival route to a point from which a landing can be completed and thereafter, if a landing is not completed, to a position at which holding or en-route obstacle clearance criteria apply. IAPs are classified as follows:

- i. non-precision approach (NPA) procedure, which means an IAP designed for 2D instrument approach operations Type A;
- ii. approach procedure with vertical guidance (APV) means a performance-based navigation (PBN) IAP designed for 3D instrument approach operations Type A;
- iii. precision approach (PA) procedure means an IAP based on navigation systems designed for 3D instrument approach operations Type A or B.

Instrument Flight Time means time during which a pilot is piloting an aeroplane solely by reference to instruments and without external reference points.

iPad is a line of tablet computers designed and marketed by Apple Inc., which runs Apple's iOS. The iPad is a PED (Portable Electronic Device).

Isolated Aerodrome means an aerodrome for which the alternate and final fuel reserve required to the nearest adequate destination alternate aerodrome is more than fuel to fly for 2 hours at normal cruise consumption above the destination aerodrome, including final reserve fuel.

Landing Distance at Time of Arrival (LDTA) means a landing distance that is achievable in normal operations based on landing performance data and associated procedures determined for the prevailing conditions at the time of landing (Note - this does not include any operational distance factor).

Large Aeroplane means an aeroplane of more than 5700 kg (12,500 pounds) maximum certificated take-off weight. The category 'Large Aeroplane' does not include the commuter aeroplane category.

Line Check means a check conducted by NTA and completed by the pilot or cabin crew member to demonstrate competence in carrying out normal line operations described in the operations manual.

Legacy Performance Data 'Dry', 'Wet' and 'Contaminated' data referenced to the surface contamination (e.g. compacted snow) i.e. not RCC data.

Low Visibility Procedures (LVP) means procedures applied at an aerodrome for the purpose of ensuring safe operations during lower than Standard Category I, other than Standard Category II, Category II and III approaches, and low visibility take-offs.

Low Visibility Operations (LVOs) means approach or take-off operations on a runway with a RVR below 550 m or with a decision height less than 200 ft.

Low Visibility Take-off (LVTO) means a take-off with an RVR less than 550 m, operated in compliance with the conditions described in this manual. LVTO operations in an RVR of less than 400 m require a specific approval.

Mach Number means the ratio of true air speed to the speed of sound.

Maintenance means any one or combination of overhaul, repair, inspection, replacement, modification, or defect rectification of an aeroplane / aeroplane component.

Master Minimum Equipment List (MMEL) is a list established for a particular aeroplane type by the manufacturer with the approval of the State of Manufacture containing items, one or more of which is permitted to be unserviceable at the commencement of a flight. The MMEL may be associated with special operating conditions, limitations, or procedures.

Maximum Certificated Take-off Mass means the maximum total weight of the aeroplane and its contents at which the aeroplane may take-off anywhere in the world, in the most favourable circumstances in accordance with the certificate of airworthiness in force in respect of the aeroplane.

Maximum Operational Passenger Seating Configuration (MOPSC) means the maximum passenger seating capacity of an individual aeroplane, excluding crew seats, established for operational purposes and specified in the operations manual. Taking as a baseline the maximum passenger seating configuration established during the certification process conducted for the type certificate (TC), supplemental type certificate (STC) or change to the TC or STC as relevant to the individual aeroplane, the MOPSC may establish an equal or lower number of seats, depending on the operational constraints.

Minimum Descent Altitude / Height (MDA/H) is a specified altitude / height in a 2D instrument approach operation or circling approach operations below which descent shall not be made without the required visual references.

Minimum Equipment List means a list which provides for the operation of the aeroplane, subject to specified conditions, with particular equipment inoperative, prepared by the operator in conformity with, or more restrictive than, the MMEL established for the aeroplane type.

Missed Approach Point (MAPt) is that point in an instrument approach procedure at or before which the prescribed missed approach procedure shall be initiated, in order to ensure that the minimum obstacle clearance is not infringed.

Mountainous Area (Terrain) is:

- a. an area of changing terrain profile where the changes of terrain elevation exceed 3000 ft (900 m) within a distance of 10 nm; or
- b. when so designated by the NTA.

Night means the hours between 30 minutes after sunset and 30 minutes before sunrise, except in a designated Mountainous area where night shall mean the hours between 15 minutes after sunset and 15 minutes before sunrise or if more restrictive such other period between sunset and sunrise, as may be specified by the state authority.

Non Intentional Transmitting PED (NIT-PED) are PEDs that do not have intended radio frequency (RF) transmission capabilities but may radiate non-intentional RF transmissions, sometimes referred to as spurious emissions.

Non-Precision Approach (NPA) Operation means an instrument approach with a minimum descent height (MDH), or DH when flying a CDFA technique, not lower than 250 ft and an RVR/CMV of not less than 750 m for aeroplanes. See also Instrument approach procedure (IAP).

Observation is a documented statement of non-compliance with a recommended practice or a recommendation for improvement.

Obstacle Clearance Altitude / Height OCA / OCH is the lowest altitude (OCA), or, alternatively, the lowest height above the relevant runway threshold or above the aerodrome elevation as applicable (OCH), used in establishing compliance with appropriate obstacle clearance criteria.

Obstacle Clearance Limit (OCL) is the height above aerodrome elevation below which the minimum prescribed vertical clearance cannot be maintained either on approach, or in the event of a missed approach.

OM-Alert is used to alert crews about changes to the Operations Manuals.

Operational Alert - OPS ALERT is used to alert crews about information which is time critical and not necessarily safety or security relevant.

Operational Control is the exercise of authority over the initiation, continuation, diversion, or termination of a flight in the interest of the safety of the flight.

Operational Credit means a credit for operations enabling lower aerodrome operating minima than would normally be established, based upon the performance of advanced aircraft systems utilising the available external infrastructure. Lower operating minima may include a lower DH/A or MDH/A, reduced visibility requirements or reduced ground facilities or a combination of these.

Operator Proficiency Check (OPC) means a check conducted by NTA and completed by the pilot or cabin crew member to demonstrate competence in carrying out normal, abnormal and emergency procedures.

Operator means any legal or natural person, operating or proposing to operate one or more aeroplane.

Operational Memo - OPS MEMO is used to inform crews about issues which are not related to changes to the Operations Manuals, and are not time, safety, or security critical.

Passenger means a person other than a crew member travelling or about to travel on an aeroplane.

Performance Class A Aeroplanes means multi-engined aeroplanes powered by turbopropeller engines with an MOPSC of more than nine or a maximum take-off mass exceeding 5 700 kg, and all multi-engined turbo-jet powered aeroplanes.

Performance-Based Communication (PBC) is communication based on performance specifications applied to the provision of air services.

Performance-Based Communication and Surveillance (PBCS) is a concept that optimises air traffic management by aligning communication (RCP) and surveillance (RSP) performance requirements. This ensures efficient and reliable data exchange between aircraft and control centres, which supports the reduction of separation minima between authorised aircraft.

Performance-Based Surveillance (PBS) is surveillance based on performance specifications applied to the provision of air traffic services.

Performance Data Computation System (PDCS) is the system used to combine aircraft performance data, runway / obstacle data, and then, for a given set of conditions, provide a limiting mass respecting NTA and regulatory requirements in a user-friendly format that can be used by Operations Planning and Control and Flight Crew.

Persons with Reduced Mobility (PRM) are understood to be any person whose mobility is reduced due to any physical disability, sensory or locomotory, permanent or temporary, intellectual disability or impairment, any other cause of disability, or age.

Pilot Flying (PF) means a pilot who is responsible for controlling and monitoring the aircraft's actual flight plan, configuration, automation mode, and system status. They will also conduct tasks as detailed in the task allocation SOPs or as directed by the Commander.

Pilot-In-Command means:

- i. In the context of a qualification / training, a pilot meeting all the requirements of [5.2.1.2 Minimum Qualification Requirements for Commanders](#).
- ii. In the context of an assignment within an aircraft crew, the pilot designated as being in command and charged with the safe conduct of the flight. For the purpose of this manual, the term 'commander' is also used to refer to this particular meaning.

Pilot Monitoring (PM) means a pilot who is observing the behaviour and action of the PF. They shall observe and compare the aircraft's actual flight path, configuration, automation mode, and system status, with those expected for the phase of flight and make timely and appropriate interventions when there is a significant deviation. They will also conduct tasks as detailed in the task allocation SOPs or as directed by the Commander.

Pilot Relieving The Commander means a suitably qualified pilot (either PIC or Cruise Relief Pilot qualified) to whom the Commander has delegated the safe conduct of the flight whilst the Commander is absent from the FCC during Augmented Crew operations in accordance with [Chapter 7 - Flight Duty Time Limitations](#).

Policy is the stated intentions and direction of an organisation.

Portable EFB means that there are two possibilities for the hardware of EFB systems: portable and installed. A portable EFB is a portable EFB host platform, used on the flight deck, which is not part of the certified aeroplane configuration.

Portable Electronic Devices (PEDs) are typically consumer electronic devices, which have functional capability for communications, entertainment, data processing, and / or utility. There are two basic categories of PEDs - those with and those without intentional transmitting capability; please refer to ED-130/RTCA DO-294().

Precision Approach means an instrument approach using Instrument Landing System, Microwave Landing System, or Precision Approach Radar for guidance in both azimuth and elevation. See also Instrument approach procedure (IAP).

Precision Approach and Landing is an instrument approach and landing using precision lateral and vertical guidance with minima as determined by the category of operation.

Preventive Action is the action(s) taken by the auditee to eliminate the cause of a potential non-conformity or a potential undesired situation.

Procedure is a detailed sequence of actions required to produce a specific result.

Process is a coordinated sequence of procedures, actions, and / or decisions designed to produce specific results.

Psychoactive substances mean alcohol, opioids, cannabinoids, sedatives and hypnotics, cocaine, other psychostimulants, hallucinogens, and volatile solvents, with the exception of caffeine and tobacco.

Quality is the degree to which a system consistently meets specified requirements, satisfies stated needs, or produces desired outcomes.

RCC FLDTA Data is the RCC performance data provided by the OEM to use for use during In-flight Assessment of the Landing Distance at Time of Arrival.

RCC Performance Data for Dispatch is the RCC Performance data provided by the OEM where the OEM has stated the data may be used at the time of dispatch (i.e. it is not limited to use during In-flight Assessment of the Landing Distance at Time of Arrival).

Reported RVR is the RVR communicated to the commander of an aeroplane by, or on behalf of, the person in charge of the aerodrome.

Required Communication Performance (RCP) specification is a set of requirements for air traffic services provision and associated ground equipment, aircraft capability, and operations needed to support performance-based communication.

Required Surveillance Performance (RSP) specification is a set of requirements for air traffic services provision and associated ground equipment, aircraft capability, and operations needed to support performance-based surveillance.

Required Visual Reference means that section of the visual aids of the approach area which shall be in view for sufficient time for the pilot to make an assessment of the position and rate of change of position, in relation to the desired flight path.

Requirement is a specification in legislation, regulation, adopted standard, or any internal / external manual that is considered to be a part of the operational practice.

RNAV (GNSS) Approach is a GNSS RNAV approach promulgated by a state and designed in accordance with PANS-OPS Criteria ICAO Doc 8168.

Root Cause is the most basic cause (or causes) that can reasonably be identified that management has control to fix and, when fixed, will prevent (or significantly reduce the likelihood of) the problem's recurrence.

Runway Condition Assessment Matrix (RCAM) means a matrix that allows the assessment of the runway condition code (RWYCC), using associated procedures, from a set of observed runway surface condition(s) and pilot report of braking action.

Runway Condition Code (RWYCC) means a number to be used in the runway condition report (RCR), that describes the effect of the runway surface condition on aeroplane deceleration performance and lateral control. The purpose of the runway condition code (RWYCC) is to permit an operational aeroplane landing performance calculation by the flight crew.

Runway Condition Report (RCR) means a comprehensive standardised report relating to the conditions of the runway surface and their effect on the aeroplane landing and take-off performance, described by means of runway conditions code.

Runway Surface Condition means a description of the condition of the runway surface used in the RCR which establishes the basis for the determination of the RWYCC for aeroplane performance purposes.

Runway Surface Condition Descriptor (RSD) means one of the following elements on the surface of the runway:

1. 'compacted snow': snow that has been compacted into a solid mass such that aeroplane tyres, at operating pressures and loadings, will run on the surface without significant further compaction or rutting of the surface;
2. 'dry snow': snow from which a snowball cannot readily be made;
3. 'frost': ice crystals formed from airborne moisture on a surface whose temperature is at or below freezing. Frost differs from ice in that the frost crystals grow independently and, therefore, have a more granular texture;
4. 'ice': water that has frozen or compacted snow that has transitioned into ice in cold and dry conditions;
5. 'slush': snow that is so water-saturated that water will drain from it when a handful is picked up or will splatter if stepped on forcefully;

6. 'standing water': water of depth greater than 3 mm (0.125 in);
7. 'Wet ice': ice with water on top of it or ice that is melting;
8. 'wet snow': snow that contains enough water to be able to make a well compacted, solid snowball, but water will not squeeze out.

Runway Visual Range (RVR) is the range over which the pilot of an aeroplane on the centreline of a runway can see the runway surface markings or the lights delineating the runway for identifying its centreline.

Safe Landing means, in the context of the fuel policy or fuel schemes, a landing at an adequate aerodrome with no less than the final reserve fuel remaining and in compliance with the applicable operational procedures and aerodrome operating minima.

Second-in-Command (SIC) means:

- i. In the context of a qualification / training, a pilot meeting all the requirements of [5.2.3.1 Minimum Qualification Requirements for Second-in-Command](#).
- ii. In the context of an assignment within an aircraft crew, a pilot serving in any piloting capacity other than as pilot-in-command or commander, but excluding a pilot who is on board the aeroplane for the sole purpose of receiving flight instruction for a licence or rating.^{1 2}

Slippery Wet Runway means a wet runway where the surface friction characteristics of a significant portion of the runway have been determined to be degraded.

Special Categories of Passengers (SCPs) are persons requiring special conditions, assistance, and / or devices when carried on a flight and include at least the following:

- a. PRMs;
- b. Infants and unaccompanied minors below the age of 12;
- c. Deportees, inadmissible passengers, or persons in custody.

Specially Prepared Winter Runway means a runway with a dry frozen surface of compacted snow or ice which has been treated with sand or grit or has been mechanically treated to improve runway friction.

Stabilised Approach (SAP) is an approach which is flown in a controlled and appropriate manner in terms of configuration energy and control of the flight path from a predetermined point or altitude / height down to a point 50 ft above the threshold or the point where the flare manoeuvre is initiated or higher.

State of Registry is the State on whose register the aeroplane is entered.

State of the Operator means the State in which the Operator has their principal place of business, or, where the operator has no principal place of business, the place where the operator is established or resides.

Supernumeraries are persons, in addition to the flight crew, that are not cabin crew member or passengers, but are on board because they either:

- are assigned to the flight by NTA and are necessary for the safety of operations; or
- have a relationship with the operator (e.g. employee, maintenance personnel); or

¹ A PIC-qualified pilot may be assigned to act as an SIC within an aircraft crew.

² When an augmented crew in accordance with Chapter 7 - Flight Duty Time Limitations is used, more than one FCM may be assigned as SIC on that flight.

- are a state- or operator-authorised inspector, auditor, or observer in the performance of their duties; or
- have certain (operator-required) knowledge and abilities gained through selection and mandatory training; or
- are assigned to a passenger flight by the operator to conduct certain customer service activities in the cabin; not designated to perform any safety duties.

Supplemental Oxygen means the additional oxygen required to protect each occupant against the adverse effects of excessive cabin altitude and to maintain acceptable physiological conditions.

Synthetic Flight Trainer means any one of the following three types of apparatus in which flight conditions are simulated on the ground:

1. A Flight Simulator which provides an accurate representation of the flight deck of a particular aeroplane type to the extent that the mechanical, electrical, electronic, etc., aeroplane systems, control functions, the normal environment of flight crew members, and the performance and flight characteristics of that type of aeroplane are realistically simulated.
2. A Flight Procedures Trainer which provides a realistic flight deck environment, and which simulates instrument responses, simple control functions of mechanical, electrical, electronic, etc., aeroplane systems, and the performance and flight characteristics of the aeroplane of a particular class.
3. A Basic Instrument Flight Trainer which is equipped with appropriate instruments and which simulates the flight deck environment of an aeroplane in flight in IMC.

System is a combination of interacting or interrelated elements within an organisation functioning in a coordinated manner to achieve the desired outcomes.

Take-off and Landing Data is a fleet specific form provided with the Operational Flight Plan for the flight crew members to record relevant take-off and landing data for the flight.

Take-off Safety Speed means a referenced airspeed obtained after lift-off at which the required one-engine-inoperative climb performance can be achieved.

Transmitting PED (T-PED) have intended radio frequency (RF) transmission capabilities.

Type A Instrument Approach Operation means an instrument approach operation with an MDH or a DH at or above 250 ft.

Type B Instrument Approach Operation means an operation with a DH below 250 ft. Type B instrument approach operations are categorised as follows:

- i. Category I (CAT I): a DH not lower than 200 ft and with either a visibility not less than 800 m or an RVR not less than 550 m, unless accepted by the Authority;
- ii. Category II (CAT II): a DH lower than 200 ft but not lower than 100 ft, and an RVR not less than 300 m;
- iii. Category III (CAT III): a DH lower than 100 ft or no DH, and an RVR less than 300 m or no RVR limitation.

'V1' means the maximum speed in the take-off at which the pilot must take the first action to stop the aeroplane within the accelerate-stop distance. V1 also means the minimum speed in the take-off, following a failure of the critical engine at VEF, at which the pilot can continue the take-off and achieve the required height above the take-off surface within the take-off distance.

'VEF' means the speed at which the critical engine is assumed to fail during take-off.

Vertical Navigation is a method of navigation which permits aeroplane operation on a vertical flight profile using altimetry sources, external flight path references, or a combination of these.

Viewable Stowage Device means a non-certified device that is attached to the flight crew member (e.g. with a kneeboard) or to an existing aircraft part (e.g. using suction cups), and is intended to hold charts or to hold low-mass portable electronic devices that are viewable by the flight crew members at their assigned duty stations.

Visual Approach means an approach when either part of all of an instrument approach procedure is not completed, and the approach is executed with visual reference to the terrain.

A visual approach shall not be conducted when the RVR, or factored equivalent, is less than 800 metres.

Weather-Permissible Aerodrome means an adequate aerodrome where, for the anticipated time of use, meteorological reports, or forecasts, or any combination thereof, indicate that the meteorological conditions will be at or above the required aerodrome operating minima, and the runway surface condition reports indicate that a safe landing will be possible.

Wet Runway means a runway of which the surface is covered by any visible dampness or water up to and including 3 mm deep within the area intended to be used.

Synonyms

Within the context of this manual, the following words all have the same meanings, and may be used interchangeably:

AERODROME	AIRPORT, AIRFIELD
AEROPLANE	AIRPLANE, AIRCRAFT
CAT I, II, III	CATEGORY 1, 2, 3
FLIGHT ATTENDANT	CABIN ATTENDANT, STEWARD(ESS), CABIN CREW MEMBER
LUGGAGE	BAGGAGE
NETJETS	NETJETS TRANSPORTES AÉREOS S.A., NETJETS EUROPE
PILOT-IN-COMMAND	PIC
PROGRAMS	PROGRAMMES
SECOND-IN-COMMAND	FIRST OFFICER, F/O, CO-PILOT, SIC

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0.6 Appendix 1 to Chapter 0 - Items Requiring Prior Approval**ORO.GEN.130**

Any change affecting:

- The scope of the certificate, the operations specifications or any of the elements of the management system, as required in ORO.GEN.200(a)(1) and (a)(2). Typical examples are listed below:
 1. the name of the operator;
 2. a change of legal entity;
 3. the operator's principal place of business;
 4. the operator's scope of activities;
 5. additional locations of the operator;
 6. the accountable manager referred to in ORO.GEN.210(a);
 7. reporting lines between the accountable manager and the nominated person;
 8. the operator's documentation, as required by Annex III of COMMISSION REGULATION (EU) No 965/2012;
 9. the facilities;
 10. Safety policy and safety procedures;
 11. Quality policy.
- Procedures for implementing changes with and without the need for prior approval from the Competent Authority.

Flights Duty Schemes / Flight Time Limitations

ORO.GEN.120 // ARO.GEN.120	Alternative means of compliance.
ORO.GEN.110(j)	Dangerous Goods training programmes for personnel as required by the technical instructions.
ORO.GEN.130(c)	Procedure for implementing changes without prior approval.
ORO.GEN.310	Use of aircraft listed on an AOC for non-commercial operations and specialised operations.
ORO.AOC.110	Leasing Agreements.
ORO.AOC.120(a)	Provision of initial cabin crew training courses.
ORO.AOC.120(b)	Issue of cabin crew training attestations.
ORO.MLR.105(b)	Minimum Equipment List.
ORO.MLR.105(f)	Rectification Interval Extensions.
ORO.FC.145(c)	Flight crew training and checking programmes including syllabi.

ORO.FC.240(a)	Operation on more than one type or variant.
ORO.FC.A.245(a)	Alternative Training and Qualification Programme.
ORO.FC.A.245(e)	ATQP extended validity dates.
ORO.CC.125	Cabin crew conversion training and checking programmes.
ORO.CC.130	Cabin crew differences training and checking programmes.
ORO.CC.140	Cabin crew recurrent training and checking programmes.
ORO.CC.145	Cabin crew refresher training and checking programmes.
ORO.CC.215(a)	Cabin crew training and checking programmes including syllabi.
ORO.CC.250(a)	Cabin crew operating on more than three aircraft types.
CAT.GEN.MPA.155(a)	Carriage of Weapons/Munitions of War.
CAT.OP.MPA.110	Changes in the method used to establish aerodrome operating minima
CAT.OP.MPA.115(a)	Use of non-Stabilised Approaches (SAp) for particular approaches to particular runways.
CAT.OP.MPA.115(b)	Use of non-CDFA technique for particular approaches to particular runways.
CAT.OP.MPA.140(a)(2)	Operations 180 minutes from an adequate aerodrome for non-ETOPS aircraft.
CAT.OP.MPA.145(b)	Method for establishing minimum flight altitudes.
CAT.OP.MPA.180	Fuel/energy schemes.
CAT.OP.MPA.182(d)	Use of isolated aerodromes
CAT.OP.MPA.200	Special refuelling or defuelling procedures.
CAT.OP.MPA.270(b)	Procedures for descending below minimum flight altitudes.
CAT.OP.MPA.320(d)	Use of a permanent lower landing mass for determining threshold speed.
CAT.POL.A.240(a)	Operations with increased bank angles.
CAT.POL.A.245(a)	Steep approach operations - Performance Class A.
CAT.POL.A.250(a)	Short landing operations - Performance Class A.
CAT.POL.A.345(a)	Steep approach operations - Performance Class B.

CAT.POL.A.350(a)	Short landing operations - Performance Class B.
SPA.PBN.100	Performance-based navigation operations.
SPA.MNPS.100	Operations with specified minimum navigation performance.
SPA.RVSM.100	Operations in airspace with reduced vertical separation minima.
SPA.LVO.100(a)	Take-off operations with visibility conditions of less than 400 m RVR.
SPA.LVO.100(b)	Instrument approach operations in low-visibility conditions.
SPA.LVO.100(c)	Operations with operational credits.
SPA.LVO.120	FCM low visibility training and checking syllabus.
SPA.ETOPS.100	Extended range operations with two-engined aeroplanes.
SPA.DG.100	Transport of Dangerous Goods.
CAT.POL.A.255	Reduced required landing distance operations.
CAT.POL.MAB.100(e)	Use of revised standard mass values for passengers and baggage.
CAT.POL.MAB.100 (f)	Standard masses for load items other than standard masses for passengers and checked baggage.
ORO.FTL.125(b)	Flight Duty Schemes / Flight Time Limitations.
ORO.MLR.105 (j)	Operating other than in accordance with the MEL, but within the constraints of the MMEL.
SPA.EFB.100	Use of type B EFB applications.
SPA.GEN.115	Changes to a specific approval.
SPO.GEN.005 // ORO.SPO.115	Performing Maintenance Check Flights.

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0.7 Appendix 2 to Chapter 0 - Approved Deviations

Type	Subject	Description	Validity
AltMOC	Carriage of Animals in excess of 8 Kg in the passenger compartment	NTA has been approved to implement an AltMoC to <i>CAT.OP.MPA.160 Stowage of baggage and cargo approved</i>, modifying <i>AMC2 CAT.OP.MPA.160 Stowage of baggage and cargo</i> so as to permit the carriage of pets weighing more than 8 Kg in the passenger compartment. Procedures and limitations pertaining to the application of this AltMoC are shown in OM-A 8.2.2.5.3 Live Animals and OM-B 1.5.11 Animal Seating Configuration.	Permanent
Derogation	Fight Time Limitation: total duty hours in any seven consecutive days	NTA holds a derogation to EU OPS Subpart Q - 1.1100 Flight and duty limitations / 1.1 Cumulative Duty Hours / (b) permitting the maximum total duty hours in any seven consecutive days to be extended from 60 to 70, provided that any hours in excess of 60 shall only be used for positioning duty to start an extended recovery rest period. Procedures and limitations pertaining to the application of this derogation are shown in OM-A 7.2.12 Accumulated Duty, Flight, and Working Time Limitations.	Permanent

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Chapter 1 - Organisation and Responsibilities

Scope of Operations

NTA provides unscheduled, worldwide air transport services to customers. The NTA's AOC, ATO, and FSTD qualify as complex organisations due to the nature and complexity of their activities, complying with all applicable regulatory requirements. To support its in-house operations, NTA contracts third parties to provide services. All third parties shall satisfy EASA, ANAC, IS-BAO and NTA requirements and standards, which are verified by the compliance monitoring programme.

AOC Operating Area

Under the terms of the AOC, the company is authorised to operate Worldwide with the exception of the Polar Regions (North of 78° N and South of 60° S). Operations to specific aerodromes in the Polar Regions may be approved for use by the Director of Operations and this will be reflected in the applicable Aerodrome Briefing.

Aeroplane Types And Airframes Operate

The aeroplane types and airframes operated are listed on the AOC.

Facilities

The offices where the business is directed from and where the day to day operational control and supervision activities occur, are located at Rua Calvet de Magalhães, 245 Bloco B, 2774-550, Paço de Arcos, Portugal.

The facilities are reviewed annually to ensure that they allow the performance and management of all planned tasks and activities in accordance with the applicable requirements. Remote facilities are also subject to review as part of the compliance monitoring programme to ensure that they are adequate for their applicable scope of work.

Information concerning individual persons within this chapter serves the purpose of informing personnel of the general functions and responsibilities of that person. Detailed information regarding their duties is stipulated in the respective description of duties, as recorded and updated by the Human Resources department or by individual departments heads.

Each individual person described here is responsible, in the event of their absence, for nominating a representative during such periods, and making this nomination known to all members of the company.

Urgent Operational Contact

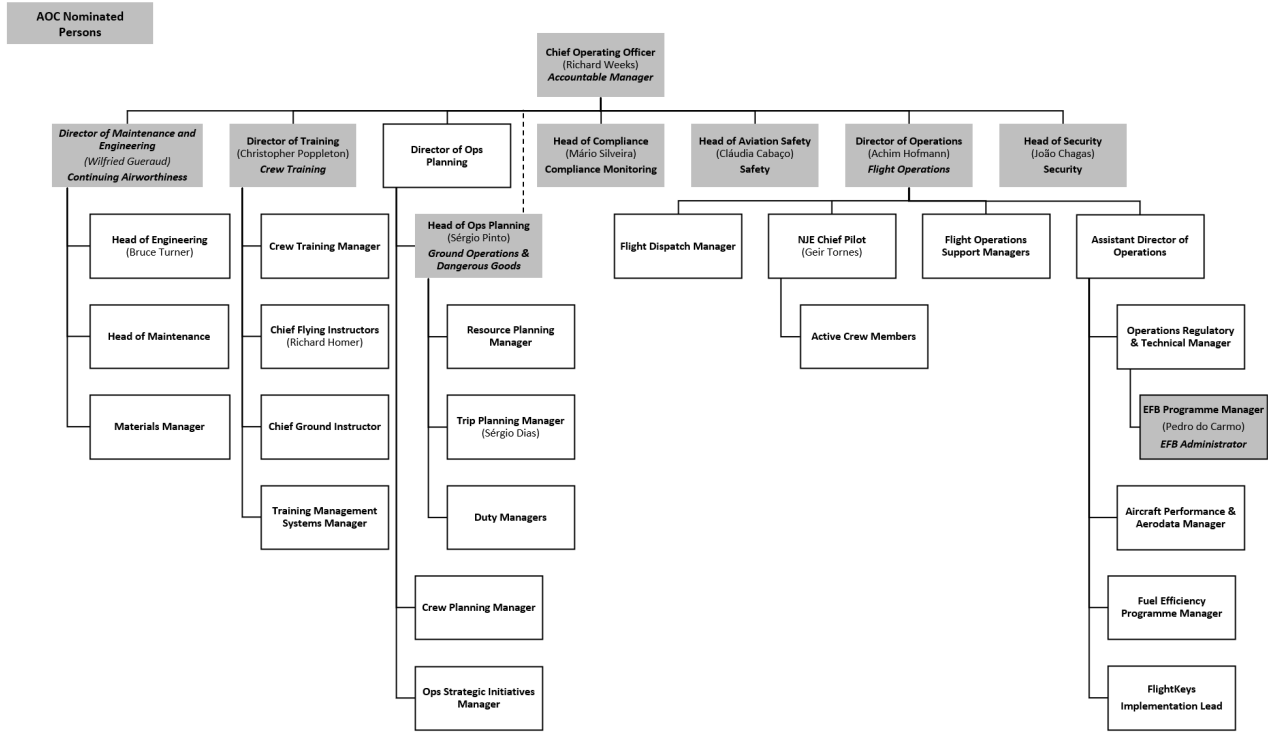
The point of contact for urgent operational issues is the Dispatch Hotline which goes directly to the Dispatch Supervisor, a position which is permanently manned. The Dispatch Supervisor is able to initiate contact / conference calls with all other departments and Nominated Persons as required and has the authority to activate the Emergency Response Plan.

Dispatch Hotline: +351 214 469 090

1.1 Organisational Structure

The following pages depict the organisational structure within the departmental descriptions of NTA.

Organisational Table - NTA



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1.2 Nominated Persons

The following personnel are nominated for the management and supervision of the operations in accordance with Commission Regulation (EU) No 965/2012, as amended or other applicable regulations:

Area / Responsibility	Name	Company Title
Accountable Manager	Richard Weeks	Chief Operation Officer
Flight Operations Manager ¹	Achim Hofmann	Director of Operations
Crew Training Manager ¹	Christopher Poppleton	Director of Training
Ground Operations Manager ¹	Sérgio Pinto	Head of Operations Planning
Continuing Airworthiness Manager ¹	Wilfried Gueraud	Director of Maintenance and Engineering
Compliance Monitoring Manager	Mário Silveira	Head of Compliance
Safety Manager	Cláudia Cabaço	Head of Aviation Safety
Security Manager	João Chagas	Head of Security
Dangerous Goods	Sérgio Pinto	Head of Operations Planning
EFB Administrator	Pedro do Carmo	EFB Programme Manager

¹ Persons nominated in accordance with ORO.AOC.135

1.2.1 Deputies to Nominated Persons

Continuity of supervision in the absence of a nominated person will be ensured. Therefore, the following deputies have been designated.

Area / Responsibility	Deputy Name	Company Title
Accountable Manager	Achim Hofmann	Director of Operations
Flight Operations Manager	Geir Tornes	NJE Chief Pilot
Crew Training Manager	Richard Homer	Chief Flying Instructor
Ground Operations Manager	Sérgio Dias	Trip Planning Manager
Continuing Airworthiness Manager	Bruce Turner Paulo Pestana	Head of Engineering CAMO Regulatory Manager
Compliance Monitoring Manager	Rita Santos	Senior Compliance Auditor
Safety Manager	André Sousa	Aviation Safety Lead
Security Manager	N/A	N/A
Dangerous Goods	Sérgio Dias	Trip Planning Manager
EFB Administrator	António Vilela	Avionics Technical Specialist

When an NP's absence is planned, they are required to notify the other NPs if it exceeds 24 hours. The notification shall clearly identify a replacement if their nominated deputy is unavailable. They shall also brief their deputies in advance on any noteworthy ongoing issues.

In the case of unexpected absences, the deputy(ies) shall immediately take over the respective NP duties and responsibilities. If the deputy is unavailable, the most senior department staff member shall assume the duties and responsibilities of the NP, which shall be communicated to the AM.

The delegation of duties and responsibilities is made to ensure business continuity only. Unless explicitly instructed and authorised by the absent NP, the delegation shall not include the authority to initiate or accept significant changes in company manuals and procedures or organisational changes.

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1.3 Responsibilities and Duties of Management

1.3.0 General

The key responsibilities of the NTA management team are:

1. Maintain the safety and efficiency of NTA operations, in particular within their areas of responsibility, following all the relevant regulations;
2. Decide on the acceptance of risk with respect to the safety and security of aircraft operations within their areas of responsibility in accordance with NTA risk assessment processes and risk tolerability;
3. Ensure compliance with all relevant regulations and standards within their areas of responsibility;
4. Take the necessary corrective actions resulting from compliance audit discrepancies in their areas of responsibility;

1.3.1 Chief Operating Officer

The Chief Operating Officer shall act as the Accountable Manager in accordance with ORO.GEN.210 and CAMO.A.305.

The Accountable Manager has corporate authority to ensure that all NTA operations activities can be financed and carried out according to the conditions and restrictions of the Air Operator Certificate (AOC) and in compliance with applicable regulations, standards, and NTA requirements.

The Accountable Manager has direct accountability for safety, is responsible for establishing and maintaining an effective management system, and chairs the safety and compliance review boards.

Further responsibilities are documented in the Management System Manual.

1.3.2 Director of Operations

The Director of Operations (DO) is the nominated person for Flight Operations according to Part-ORO (ORO.AOC.135) and a direct reporting subordinate of the Accountable Manager. The DO is the Deputy Accountable Manager and has an independent budget.

The DO is responsible for the overall management of NTA operations including:

1. The safe and efficient management of Flight Operations in accordance with all the required regulations;
2. The Command and Control of daily operations through the Operational Control Centre (OCC);
3. To direct, motivate, and develop the NJE Chief Pilot and Cabin and Fleet Program Manager teams in order to efficiently maximise the performance of the teams and ensure that the fleet goals are achieved. Those goals are derived from the Safety Management System, the NetJets Service mandate, and by collaboration with the Maintenance department to maximise aircraft availability;
4. The activities of the Operations Regulatory and Technical Support, to ensure the following:
 - Compliance with all relevant national and international regulations and the provisions of the company's AOC.

- Control of the usability of aerodromes in accordance with [Chapter 8 - Operating Procedures](#), determination of the restrictions for their use and respective operating minima;
 - Control of the usability of areas and airways to be utilised, and associated minimum flight altitudes.
 - Revision of contents of the Operation Manual Part A;
 - Monitoring of the regulatory equipment of all fleets in order to comply with the requirements of future airspace structures and / or the legal changes associated with them.
 - Filing of the required reports with ANAC or foreign authorities, and apply to them for necessary approvals (i.e. RVSM, ETOPS, NAT HLA, PART-SPA, etc.).
 - Keeping current on equipment developments and flight operational procedures;
 - To ensure Airport Runway and Airport / Aircraft Performance data is updated to ensure safe operation of Company aeroplanes.
5. To ensure compliance with the Management System within Flight Operations;
 6. To ensure that aeroplanes are operated in accordance with applicable requirements;
 7. To implement and control the operating standards of NTA fleets;
 8. To make the AFM, the AOM, and the relevant documentation available to crews and ensure that the aeroplane documentation is updated in accordance with the manufacturer's revisions or mandates by ANAC;
 9. To review the aircraft equipment of the fleet(s) under their responsibility and keep current on the development of new aircraft specific equipment and flight operational requirements in order to safeguard that the aeroplanes comply with any future or legal requirements and ensuring that NetJets crew members make best use of available equipment;
 10. To publish other necessary directives for crew, as outlined in [Chapter 2 - Operational Control and Supervision](#);
 11. To process Safety Reports (see [Chapter 11 - Handling, Notifying and Reporting Accidents, Serious Incidents and Occurrences](#)), and other reports, investigate flight operational irregularities, SAFA reports and recommend remedial actions;
 12. To call and manage hearings in the event of incidents or accidents, or, whenever deemed necessary, in case of irregularities or violations of legal provisions or internal directives;
 13. To provide guidelines to the Director of Training, in order to establish flight crew training requirements, check procedures, and training syllabi;
 14. The routine contacts with ANAC and authorities of other countries;
 15. The representation, as far as flight operations are concerned, of the company's interests in national and international bodies and institutions;
 16. Ensuring that any relevant contractor employed meets all company standards;
 17. The budget and planning within the Operations department.

1.3.3 NJE Chief Pilot

The NJE Chief Pilot reports to the DO and has the following duties and responsibilities:

- To assume the responsibilities of the DO in their absence;
- Manage the NTA daily flight operations of the fleets, ensuring the application of all relevant regulations within the scope of the AOC and enforcing company procedures and standards;
- Direct, motivate, and develop the Duty Managers, Flight Ops Support Managers and Dispatch teams, ensuring they have the necessary knowledge, information and resources;
- Guarantee alignment between the teams being supervised and maximise their performance for goals achievement;
- Ensure the revision of all manuals relevant for the operation, following the flight operations philosophy and policies, and ensuring that regulatory and technical requirements are continuously reviewed;
- Support all fleet specific projects and ensure tasks are coordinated in accordance with the respective procedures to ensure full compliance with regulations, NTA policies and procedures and departmental goals;
- To support the Safety department by implementing fleet specific measures with the aim of developing NetJets safety culture;
- As Line Manager of all cabin and cockpit crew members to direct and assist in delivering safe, effective and efficient flight operations;
- Analyse cases of serious breaching of workplace policies and apply mitigating actions in accordance with the applicable procedures;
- Publish global crew directives and procedures to ensure crew members have access to relevant and up to date information regarding the aircraft's daily operation;
- Support the Senior Manager, Crew Support and Labour Relations in the design, implementation and monitoring of all crew non-operational related policies and procedures;
- Deliver and plan effective and regular communications to all crew members to deploy essential information, actively and consistently engaging crew on a decentralised basis to improve employee relations and enhance office-line communication;

1.3.4 Head of Operations Planning

The Head of Operations Planning (HOP) is the nominated person for Ground Operations and is ultimately responsible to the accountable manager, as required by Part-ORO.

The Head of Operations Planning is responsible for the management and supervision of ground operations, including the responsibility of ensuring that NTA remains in compliance with the applicable ground operations requirements.

The HOP has an independent budget.

The HOP has the following duties and responsibilities:

1. To manage and control Ground Operations (Operations Planning) to ensure safe and efficient operations.
2. To ensure that all legal requirements and the provisions of the AOC relating to ground operations are strictly observed.
3. To ensure compliance with the Management System within Operations Planning.
4. To act as the company's representative to all authorities, as far as ground operations are concerned.
5. To ensure the safe, punctual, and economic operations of all company flights by establishing or contracting out for professional ground handling of company aeroplanes, passengers, their load, crew members at all aerodromes to which the company operates.
6. To ensure that operations are only conducted along routes and within areas that meet all Safety, Legal, Aircraft Performance and NTA requirements for which.
 - Ground facilities and services, including meteorological services, are adequate for the planned operation.
 - The performance of the aeroplane to be used is adequate to comply with minimum flight altitude requirements.
 - The equipment of the aeroplane to be used meets the requirements for the planned operation.
 - Current maps, charts, and other relevant navigational data are available.
 - The limitations of 8.5 Extended Range Operations with Two-Engined Aeroplanes (ETOPS) are met for all twin-engine aeroplanes.
7. To ensure the planning of the Company's operations and the checking of the suitability of area routes and aerodromes as defined by the Director of Operations.
8. To manage crew scheduling and to maintain adequate duty time records, in accordance with procedures and limitations that meet all Safety, Legal, and NTA requirements.
9. To ensure that the travelling of the crew members to their duty location, rest locations and back to the gateway and for the provision of a suitable rest location, is in accordance with procedures and limitations that meet all Safety, Legal and NTA requirements.

1.3.5 Director of Training

The Director of Training (DT) is the nominated person for Training according to Part-ORO (ORO.AOC.135) and a direct reporting subordinate of the Accountable Manager and has an independent budget.

The duties and responsibilities of the Director of Training are:

1. To plan, implement, and manage the Training department strategy, in liaison with the Accountable Manager, to ensure that training is provided according to business needs, in compliance with current legislation and NetJets policy, procedures, and standards;
2. To appoint Type Rating Examiners (TRE), Type Rating Instructors (TRI), and Line Trainers in cooperation with the NJE Chief Pilot, who will nominate suitable persons;
3. To arrange ground instruction for newly recruited personnel and refresher training when required;
4. To arrange initial and recurrent simulator training;
5. To arrange initial and recurrent flight training;
6. To be responsible for all training records, ensure that crew members checks are conducted in accordance with current legislation, and the Training Manual;
7. To manage and control the NetJets Training Manual (OM-D), approved by the Authority, to ensure that it contains details of all relevant training programmes, policies, procedures, requirements, includes standards for flight crew training and evaluation, and other guidance necessary to administer the NetJets training programme;
8. To ensure that the training programme consists of ground, simulator, and flight training to guarantee that crew members have sufficient knowledge, skills, and proficiency to competently perform their assigned duties;
9. To check and record the validity and currency of all crew member Training Records, and the validity dates of Crew Licences, ensuring that this information is passed on to the Operations Staff;
10. To compile and maintain current lists of all flight and cabin personnel, so that checks, simulator checks, training, first-aid instruction, and medical examinations are administered in due time for licensing and renewal of licences;
11. To maintain area, route, and airport qualification records for all pilots;
12. To ensure that all training providers and contractors have the required approval or acceptance of the Authority, meet the required performance standards of NetJets, and are periodically evaluated to enable the continuous improvement of the quality of the service provided and ensure compliance with the required qualification and performance standards;
13. To ensure that training facilities, devices, and course materials shall reasonably reflect the configuration of the applicable fleet. Where such facilities, devices, or course materials are not available, any differences must be highlighted and additional training must be provided and recorded for the company equipment;
14. To ensure compliance with and functioning of the Management System within the Training department;

15. To assist the DO with review of the Operations Manual;
16. To represent, as far as training is concerned, the company's interests in national and international bodies and institutions;
17. To ensure that there are sufficient number of qualified instructors, examiners, and support personnel to administer the training programmes efficiently, on time, in a standardised format, and in accordance with the requirements of the Competent Authority and NetJets;
18. To be directly responsible to the Authority for the full compliance of the ATO in line with the PART-FCL requirements;
19. To implement and manage a process for the purpose of achieving training programme improvements;
20. Ensure that NetJets TRI and TRE are approved and have been trained in accordance with the current legislation and NetJets requirements;
21. Responsible for the training analysis of Cat C Airports;
22. To monitor crew member currency;
23. To manage the budget and planning within the Training department;
24. To ensure compliance with the Management System within the Training department;
25. To provide strategic guidance to the NTA standards programme and ensure that it is effectively managed and resourced to competently perform its duties;
26. To organise flights to check the overall standards of their fleets and identify potential improvements in procedures and training.

1.3.6 Director of Maintenance and Engineering

The NTA nominated person for the Continuing Airworthiness is the Director of Maintenance and Engineering.

The Director of Maintenance and Engineering reports to the Accountable Manager and is responsible to ensure that all maintenance is carried out on time and to the approved standards.

In case of absence, the designated Director of Maintenance and Engineering Deputies will exercise the assigned duties for the nominated person for continuing airworthiness.

The Duties and Responsibilities of the Director of Maintenance and Engineering include but are not limited to:

1. Responsible for Part-M and Part-CAMO compliance;
2. Ensure, through the actions of the respective maintenance organisations, that all maintenance (scheduled and non-scheduled) is carried out on time, to an approved standard, and in accordance with the approved programme, such that the maintenance responsibility requirements prescribed in applicable regulations are satisfied, with minimum down time, maximizing aircraft availability in accordance with the annual budget and forecast;
3. Direct and manage the teams that provide the Service Level Agreements deliverables in the scope of the subcontracted tasks defined in the CAMO Subcontract Agreements to achieve the NJE service objectives and goals, ensuring compliance with NJE standards and requirements;
4. Ensure, through the actions of the respective maintenance organisations, that defects and damage affecting safe operation are rectified to an approved standard, considering the minimum equipment list and configuration deviation list for the aeroplane type;
5. Review Aircraft inspection requirements to develop and monitor efficient inspection programs to minimise aircraft scheduled downtime;
6. Ensure a continual review of relevant internal and external maintenance data to enable the formulation of recommended improvements of reliability to include pre-emptive maintenance to reduce unscheduled downtime;
7. Review internal and external maintenance-related processes;
8. Maintain close liaison with all OEMs to ensure supply chain-related disruptions are minimised and inventory management is developed and monitored, and economically maintained;
9. Maintain close liaison in collaboration with the procurement team of all MROs to ensure that external services are conducted economically and with minimal downtime;
10. Ensure that the department is properly staffed, managed, organised, trained, and directed to fulfil its responsibilities according to internal and EASA regulations, as well as direct, motivate, develop, and control the Maintenance team to optimise their performance and ensure that objectives are achieved;
11. Ensure the provision of a maintenance programme approved by ANAC for each aeroplane operated by the company, including the associated reliability programme, and obtain approval of all amendments to the programme;
12. Ensure that the Certificate of Airworthiness for each aircraft remains valid to keep the aircraft in service;

13. Ensure the provision of a Minimum Equipment List (MEL), approved by ANAC for each aeroplane operated by the company, and obtain approval of all amendments to the MEL;
14. Ensure, through the actions of NJE Flight Operations personnel and maintenance organisations, that pre-flight inspections are carried out in such a manner as to render the aeroplane fit for their intended flights and, where these tasks are contracted to other organisations, that their accomplishment is subject to NJE technical requirements;
15. Ensure liaison with the Director of Operations regarding technical matters, including the coordination and implementation of NJE policy regarding aeroplane maintenance to ensure the maximum benefits of operating experience are incorporated into all phases of maintenance;
16. Ensure that all aircraft maintenance records are maintained in accordance with the approved procedures and in compliance with the applicable regulations, thereby supporting the integrity of the documentation control and the residual value of the aircraft;
17. Ensure the integrity and validity of the Continuous Airworthiness Management Exposition (CAME) and NTA Maintenance Procedures;
18. Promote the Continuous Improvement Process within the teams to ensure a continued focus on Safety Reporting (incl. JUST Culture), Compliance and process efficiency;
19. The management and timely follow-up of both Internal and External Audit findings, ensuring the identified root causes and corrective actions continue to support the Safety and Compliance requirements of the CAMO.

1.3.7 CAMO Regulatory Manager

The description of the duties and responsibilities of the CAMO Regulatory Manager are described in the Continuing Airworthiness Management Exposition Manual (CAME).

In their absence, the Director of Maintenance will delegate overall responsibility of the Maintenance department and the associated functions and activities to the CAMO Regulatory Manager.

1.3.8 Head of Aviation Safety

The Head of Aviation Safety shall act as the Safety Manager in accordance with ORO.GEN.200(a)(1) and CAMO.A.305(a)(5).

The Head of Aviation Safety is the focal point for all safety matters and is responsible for developing, administering, and maintaining an effective safety management system.

The Head of Aviation Safety has direct access to the Accountable Manager.

Further responsibilities are documented in the Management System Manual.

1.3.9 Head of Standards

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1.3.10 EFB Administrator

The EFB Programme Manager is assigned with the role of EFB Administrator and reports directly to the Operations Regulatory and Technical Manager. Their duties and responsibilities are:

1. Administration of all EFB systems and procedures within NTA;
2. Providing technical assistance in case of EFB failures, and coordinating with the Maintenance, Flight Operations, and IT departments on EFB repair and spare parts;
3. Liaising with industry partners concerning all aspects of the EFB;
4. Coordinating with the IT department for any software security and control-related issues;
5. Coordinating with database suppliers on reliability, content, and the availability of updates and providing support to Flight Operations when database issues are detected;
6. Managing and monitoring all electronic subscription services (FMS, NAV, DB, Echarts, Emaps), including the distribution and loading processes;
7. Requesting the Compliance department to perform periodic EFB audits to internal and external departments, to ensure that company procedures are strictly adhered;
8. Maintaining and updating, if needed, any qualifications and training as required by the competent authorities;
9. Keeping current on the subject of data management with NetJets' external providers.

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1.3.11 Security Manager

The Security Manager is responsible for all Aviation Security matters and reports directly to the Director of Safety and Compliance for all aviation security matters. Their duties and responsibilities are:

1. Ensure that NTA's requirement to be a secure operator is achieved through sound company policy, timely advice, and security performance by the company staff and its business partners;
2. Maintain a security policy for the protection of aeroplanes, staff, and passengers against potential and / or known threats;
3. Ensure that the flow of information and interaction between the Security Department and other departments within NetJets is effective, accurate, and prompt to guarantee the highest possible levels of safety and security;
4. Provide security risk assessments to all destinations NTA aeroplanes and crews operate;
5. Ensure security briefings for NetJets crews operating into locations requiring such assessments are provided to crew members in a timely manner;
6. Develop and oversee NTA Security Programme and the Operations Manual [Chapter 10 - Security](#), in line with the relevant legislation and company policy;
7. Provide 24-hour response to incidents, including bomb threat assessments;
8. Maintain routine contacts with ANAC, Authorities of foreign countries, and International organisations;
9. Develop policies and counter-measures which are commensurate with the threat and risks and which are compliant with relevant Portuguese and host state legislation;
10. Ensure that staff receives security training appropriate to their responsibilities;
11. Manage NTA Security Quality Control programme and report directly to the Accountable Manager on the aviation security performance of the company;
12. Be the NTA primary point of contact on all security related matters with Portuguese and Foreign Authorities, Regulators, and International Organisations;
13. Represent, as far as Aviation Security is concerned, the company's interest in national and international bodies, institutions, and advisement groups;
14. Ensure that contracts with elements covering security work include detailed specifications of the tasks involved;
15. Ensure that adequate security arrangements are in place at the airports network where NTA operates;
16. Ensure that all service providers have implemented security measures to acceptable standards.

1.3.12 Head of Compliance

The Head of Compliance shall act as the Compliance Monitoring Manager in accordance with ORO.GEN.200(a)(6) and CAMO.A.305(a)(4).

The Head of Compliance is the focal point for all compliance matters and is responsible for developing, administering, and maintaining an effective compliance management system. The Head of Compliance has direct access to the Accountable Manager.

Further responsibilities are documented in the Management System Manual.

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1.4 Authority, Responsibilities, and Duties of the Commander

NTA will nominate a pilot to be the aeroplane Commander for each flight or series of flights.

The Commander reports to the NJE Chief Pilot.

1.4.1 General Responsibilities

The commander shall be responsible for the proper execution of their duties related to the safety of the aeroplane and its occupants in accordance with the procedures and instructions in the company operations manuals.

The commander shall take all reasonable steps to:

1. Maintain familiarity with relevant Portuguese State, EU, AIR OPS, ICAO air legislation, and agreed aviation practices and procedures;
2. Maintain familiarity with such provisions of the company Operations Manuals as are necessary to fulfil their function;
3. Read and acknowledge all DMS publications and communications requiring acknowledgment (compliance) prior to each flight duty;
4. Read all DMS publications and communications not requiring acknowledgment (compliance) prior to commencing a tour. If a crew member is on tour or at a training event, it can be read prior to the start of the next tour;
5. Ensure that DMS database is updated with the latest publications before each flight duty;
6. Ensure that when areas of improvement are identified within the operation, a crew report will be filed by one of the crew members by means of the appropriate form.

1.4.2 Specific Responsibilities

The Commander shall:

1. Ensure that the passengers are met by a crew member at the agreed meeting point before the flight and ensure that after the flight they are escorted to their next mode of transport and assisted as necessary by a crew member or NetJets representative;
2. Ensure that the passengers are assisted to board or disembark the aeroplane;
3. Ensure that a crew member assists in loading or unloading passenger luggage or cargo, as required;
4. Be responsible for the safety of all crew members, passengers, supernumeraries, and cargo on board, as soon as they arrive on board, until they leave the aeroplane at the end of the flight;
5. Be responsible for the operation and safety of the aeroplane from the moment the aeroplane is first ready to move for the purpose of taxiing prior to take-off until the moment it finally comes to rest at the end of the flight and the engine(s) used as primary propulsion units are shutdown;
6. Have authority to give all commands they deem necessary for the purpose of securing the safety of the aeroplane and of persons or property carried therein, and all persons carried in the aeroplane shall obey such commands;
7. Have authority to disembark any person, or any part of the cargo, which in their opinion may represent a potential hazard to the safety of the aeroplane or its occupants;

8. Not allow a person to be carried in an aeroplane who appears to be under the influence of alcohol or drugs, to the extent that the safety of the aeroplane or its occupants is likely to be endangered;
9. Have the right to refuse transportation of inadmissible passengers, deportees, or persons in custody, if their carriage poses any risk to the safety of the aeroplane or its occupants;
10. Ensure that all passengers are briefed in accordance with [8.3.16 Passenger and Supernumerary Briefing Procedures](#) to include the location of emergency exits and the location and use of relevant safety and emergency equipment, and ensure a sufficient quantity of passenger safety cards is available and stored in their designated location;
11. Ensure that all operational procedures and checklists are complied with in accordance with the Operations Manual and ensure compliance with all operating limitations defined by the OEM and established by ANAC;
12. Ensure that the aeroplane cabin interior and the aeroplane external surfaces are clean; When unable to clean the aeroplane exterior, an exterior cleaning should be requested at the next available opportunity;
13. Ensure that the fuel on board will be equal to or exceeds the minimum required fuel according to the OFP;
14. Ensure that the weather forecast and reports for the proposed operating area and flight duration indicate that the flight may be conducted without infringing Company operating minima. Will monitor weather information whilst en-route to include destination, destination alternate (if required) and, en-route alternates (if required);
15. Decide whether or not to accept an aeroplane with unserviceabilities allowed by the Configuration Deviation List or Minimum Equipment List, and already reported to the NTA Maintenance department.;
16. Take all reasonable steps to ensure that the aeroplane and any required equipment is serviceable, and be satisfied that the relevant emergency equipment remains easily accessible for immediate use;
17. Delay the flight and call the NJE Chief Pilot if the crew is of the opinion that the aeroplane is not in an airworthy condition.
18. Ensure that aeroplane refuelling is supervised, with particular attention being paid to:
 - The correct amount and grade of fuel;
 - Fuel water checks;
 - Fire safety precautions;
 - Checking filler caps for security and correct replacement after refuelling.
19. Take all reasonable steps to ensure that the aeroplane mass and balance is within the calculated limits for the operating conditions.
20. Confirm that the aeroplane's performance will enable it to safely complete the proposed flight. They shall also, shortly before take-off and landing, verify that the actual weather conditions are similar or conservative compared to the weather data used for the take-off performance calculations and the in-flight landing distance assessment.
21. Not permit any crew member to perform any activity during critical phases of flight, except those duties required for the safe operation of the aeroplane.

22. Take all reasonable steps to ensure that before take-off and before landing the flight, cabin crew, and passengers are properly secured in their allocated seats.

NOTE

Required cabin crew should be properly secured in their allocated seats during taxi except for the performance of essential safety-related duties.

23. Make any decisions necessary to start, delay, or cancel the flight when operational conditions dictate.
24. Immediately report any unforeseen changes to the planned flight (departure times, destination, delays, etc.) to Flight Dispatch.
25. Take all reasonable steps to ensure that whenever the aeroplane is taxiing, taking-off, or landing, or whenever they consider it advisable (i.e. in turbulent conditions), all passengers are properly secured in their seats and all cabin baggage is stowed in the approved stowage areas.
26. Ensure that the documents and Manuals in [Chapter 8 - Operating Procedures](#) are carried and will remain valid throughout the flight or series of flights.
27. Ensure that a thorough pre-flight inspection has been carried out according to the Aircraft Flight Manual. The inspection shall include:
- Pitot and static ports are not damaged or obstructed;
 - Flight Controls are not locked or disabled (as applicable, depending on aeroplane type);
 - Frost, snow, or ice is not present on critical surfaces;
 - Aeroplane structure or structural components are not damaged.
28. With regards to flight recorders (FDR and CVR) shall ensure that:
- Flight recorders are not disabled or switched off during flight; and
 - In the event an occurrence other than an accident or a serious incident that is subject to mandatory reporting, flight recorders' recordings are not intentionally erased; and
 - In the event of an accident or a serious incident, or if preservation of recordings of flight recorders is directed by the investigating authority:
 - flight recorders' recordings are not intentionally erased;
 - flight recorders are deactivated immediately after the flight is completed; and
 - precautionary measures to preserve the recordings of flight recorders are taken before leaving the flight crew compartment;
 - flight recorders are reactivated only with the agreement of the investigating authority, if involved, or the Safety department.
29. Maintain a high standard of discipline, conduct, and appearance as a senior representative of the Company;

30. Ensure that the aeroplane is certificated to fly in any expected icing conditions and that the external surfaces are clear of any deposit which might adversely affect the performance and / or controllability of the aeroplane, unless permitted by the Aeroplane Flight Manual.
31. Ensure that all Journey and Technical Logs are completed and ensure that for each flight a description of known or suspected defects that affect operation of the aeroplane are recorded in the Technical Logs. They shall also ensure that all required times and fuel figures are logged via the company mobile phone tool or, if no data communication is possible, via a telephone call to Flight Dispatch.
32. Ensure that all occurrences are reported as required in [Chapter 11- Handling, Notifying and Reporting Accidents, Incidents and Occurrences and Using the CVR Recording](#) of this Manual.
33. In an emergency situation that requires immediate decision and action, take any action they consider necessary under the circumstances. In such cases, they may deviate from rules and operational procedures and methods in the interest of safety.
34. Use their authority to apply greater safety margins, including aerodrome operating minima, if they deem it necessary.
35. Ensure that a continuous listening watch is maintained on the appropriate radio communication frequencies at all times whenever the flight crew is manning the aeroplane for the purpose of commencing and / or conducting a flight and when taxiing. This includes the monitoring of the emergency frequency 121.5 on the secondary VHF radio when not being used for other required and relevant communications.

The monitoring of the emergency frequency 121.5 during the take-off, landing, or any other phase of flight, where the monitoring is very distracting, is at the discretion of the commander.
36. For the aeroplane types that permit the flight crew to update the FMS navigation database and / or Jeppesen E-charts:
 - Download the respective FMS Nav Data Base and, where applicable, the Jeppesen E-charts to the NTA issued USB Memory stick in accordance with OM-B, or in the absence thereof, other interim published NTA procedures;
 - Be in possession of the NTA USB Memory stick containing the updated FMS Nav Data Base and, where applicable, Jeppesen E Charts.
37. Ensure that the cabin is inspected after a flight for any remaining passenger items;
38. Ensure that the aeroplane interior is cleaned after each flight and the aeroplane stock is sufficient for subsequent passenger flights;
39. Maintain currency and proficiency on their NetJets assigned aeroplane type as an active crew member;
40. Maintain up-to-date personal records, e.g. Part-MED medical certificate, Part-FCL, ANAC documents, passports, proficiency checks, and any other records applicable to the performance of flight duty;
41. Provide updated copies of these records to the Training department;
42. Ensure that manual amendments available at aeroplane lockers or placed on board of the aeroplane are incorporated in the manuals at the first available opportunity;

43. For aeroplane that normally carry cabin crew or ISRs, when a cabin crew or ISR is not carried, ensure that all on-board items that are left at the FBO for cleaning or storage (plates, cutlery, linen, chocolates, etc.) are taken back on board.
44. Ensure that the aeroplane's exterior / interior pre-flight inspection is conducted as soon as practicable at the beginning of a flight duty period and promptly inform the company of any defects which may result in a delay to the flight.
45. Ensure that the IJET feasibility notices for both the planned departure and destination aerodromes have been checked prior to departure and any operational restrictions contained therein complied with.
46. Ensure they have with them their assigned Portable EFB and that it has been updated with the latest FD-Pro X databases.
47. Identify any hazards within the planned operation. If a commander is unable to mitigate the identified hazards or has safety concerns about the planned operation, they shall call the FOSM team to discuss an appropriate course of action.
48. Ensure that a suspected communicable disease is reported promptly to air traffic control and the company in order to facilitate provision for the presence of any special medical personnel and equipment necessary for the management of public health risks on arrival.
49. Clearances received are safe with respect to terrain clearance during climb / descent and en route.
50. Compliance with the provisions of a clearance will not involve violation of other regulations governing the flight.
51. As soon as possible, report to the appropriate air traffic services (ATS) unit any hazardous weather or flight conditions encountered that are likely to affect the safety of other aircraft.
52. When notified that a Crew Alertness Risk Assessment (CARA) is required for a given duty period, complete that CARA no later than 30 minutes after reporting time for the applicable duty period.

1.5 Duties and Responsibilities of Crew Members other than the Commander

1.5.1 Second-in-Command (SIC)

Function

The Second-in-command is responsible to the Commander to assist with the safe and efficient conduct of the flight. In the event of the incapacitation of the Commander, the Second in Command will assume command.

The Second-in-command departmental Head to whom they are responsible when not reporting directly to a Commander is the NJE Chief Pilot.

General Responsibilities

The Second-in-command shall take all reasonable steps to:

1. Maintain familiarity with relevant Portuguese State, EU, AIR OPS, ICAO air legislation, and agreed aviation practices and procedures.
2. Maintain familiarity with such provisions of the Company Operations Manual as are necessary to fulfil their function:
 - Read and acknowledge all DMS publications and communications requiring acknowledgement (compliance) prior to each flight duty.
 - Read all DMS publications and communications not requiring acknowledgement (compliance) prior to commencing a tour. If a crew member is on tour or at a training event, it can be read prior to the start of the next tour.
 - Ensure that the DMS database is updated with the latest publications before each flight duty.
3. Assist the Commander, as requested, concerning administrative duties in relation to the flight.
4. Support the Commander in the maintenance of a proper standard of crew discipline, conduct, and personal appearance.
5. Maintain currency and proficiency on their NetJets assigned aeroplane type as an active crew member.
6. Maintain up-to-date personal records, e.g. PART-MED medical certificate, PART-FCL, documents, passports, proficiency checks, and any other records applicable to the performance of flight duty.
7. Provide updated copies of these records to the Training department.

Specific Responsibilities

It is the specific responsibility of the Second-in-command:

1. To properly execute their duties related to the safety of the aeroplane and its occupants in accordance with the procedures and instructions in the company operations manuals, including procedures, limitations, and performance relating to the specific aeroplane type, as well as additional duties allocated to them by the Commander.
2. To confirm the safe navigation of the aeroplane, maintaining a continuous and independent check upon both the geographical position of the aeroplane and its safe terrain clearance.

3. To volunteer such advice, information and assistance to the Commander, as may contribute favourably towards the safe and efficient conduct of the flight.
4. To seek and receive such information and / or explanations from the Commander, as may be necessary to enable the Second-in-command to fulfil their function.
5. To maintain a high personal standard of discipline, conduct, and appearance as a representative of the Company.
6. To support the Commander, by active example, in the development and maintenance of a high standard of professional expertise and morale amongst the crew.
7. To report to the Commander any incident that has endangered, or may have endangered, the safety of the operation, if not already reported by another crew member.
8. Ensure that all occurrences are reported as required in [Chapter 11 - Handling, Notifying and Reporting Accidents, Incidents and Occurrences and Using the CVR Recording](#) of this Manual;
9. To report to the Commander any fault, failure, malfunction, or defect which they believe may affect the airworthiness or safe operation of the aeroplane, including emergency systems.
10. For the aeroplane types that permit the flight crew to update the FMS navigation database and / or Jeppesen E Charts:
 - To download the FMS Nav Database and, where applicable, the Jeppesen E Charts to the NTA-issued USB Memory stick in accordance with OM-B, or, in the absence thereof, other interim published NTA procedures.
 - To be in possession of the NTA USB Memory stick containing the updated FMS Nav Database and, where applicable, Jeppesen E-charts.
11. Ensure they have their assigned Portable EFB with them and that it has been updated with the latest FD-Pro X databases.
12. To notify the Commander of any hazards or safety concerns they have identified within the planned operation.
13. When notified that a Crew Alertness Risk Assessment (CARA) is required for a given duty period, complete that CARA no later than 30 minutes after reporting time for the applicable duty period, and notify the Commander that the CARA has been completed.

1.5.2 Cabin Crew

Cabin crew on aeroplanes fulfil three important functions:

1. They ensure, under the overall authority and responsibility of the Commander, cabin safety ([Chapter 8 - Operating Procedures](#)) (i.e. the safety of the passengers and the security of cabin load and galleys).
2. Form the link between the passengers and the flight crew.
3. Provide an optimum level of comfort and service for the company's customers, namely the passengers, and by their courtesy, efficiency, thoughtfulness, and ready assistance, directly shape the company's image.

Cabin crew shall also:

1. Be responsible for the proper execution of their duties related to the safety of the aeroplane and its occupants in accordance with the procedures and instructions in the company operations manuals, including procedures and additional duties allocated to them by the Commander.
2. Report to the Commander any fault, failure, malfunction, or defect which they believe may affect the airworthiness or safe operation of the aeroplane, including emergency systems.
3. Report to the Commander any incident that has endangered, or may have endangered, the safety of the operation.
4. Comply with the NTA safety reporting requirements.
5. Maintain familiarity with relevant Portuguese State, EU, and ICAO air legislation, and agreed aviation practices and procedures.
6. Maintain familiarity with such provisions of the Company Operations Manual as are necessary to fulfil their function:
 - Read and acknowledge all DMS publications and communications requiring acknowledgment (compliance) prior to each flight duty.
 - Read all DMS publications and communications not requiring acknowledgment (compliance) prior to commencing a tour. If a crew member is on tour or at a training event, then it can be read prior to the start of the next tour.
 - Ensure that the DMS database is updated with the latest publications before each flight duty.
7. Notify the Commander of any hazards or safety concerns they have identified within the planned operation.
8. When notified that a Crew Alertness Risk Assessment (CARA) is required for a given duty period, complete that CARA no later than 30min after reporting time for the applicable duty period, and notify the Commander that the CARA has been completed.

Whenever more than one cabin crew member is assigned to a flight, a senior cabin crew member shall be designated in accordance with [4.1.7 Designation of a Senior Cabin Crew Member](#) to:

1. Direct, coordinate, and organise the functions and tasks of the cabin crew.
2. Be responsible to the Commander for the conduct and coordination of cabin safety, normal and emergency procedures specified in the Operations Manual, including for discontinuing non safety-related duties for safety or security purposes.

3. Ensure the flight crew is aware of any irregularities or malfunctions and, at the prescribed times, ensure reports of cabin secure or cabin not secure are made.

Cabin crew members must meet the qualification and recency requirements of [Chapter 5- Qualifications Requirements](#). Prior to flight, all cabin crew must participate in the Commander's cabin crew briefing ([Chapter 8- Operating Procedures](#)), receiving information and particulars pertinent to the individual flight.

Immediately after boarding the aircraft, cabin crew members shall carry out all required pre-flight checks, including inspection of safety and emergency equipment, security searches, and any other equipment inspections in accordance with NetJets procedures. Any deficiencies detected must be reported immediately to the Senior Cabin Crew Member or Commander for further corrective action.

In flight, they conduct the prescribed passenger services, as well as fulfilling the safety relevant tasks of [Chapter 8- Operating Procedures](#). In the event of an emergency situation, they act in accordance with company Standard Operating Procedures and Checklists, notifying the Commander immediately and requesting further instructions.

Between flights, cabin crew ensures proper cleaning of the aeroplane's cabin and ensure re-catering for the next flight. The Cabin Crew Member must ensure that all luggage is stowed in accordance with the safety standards in the cabin and that all baggage is handed over after the flight to the passengers, FBO, or Flight Crew. This includes coats and small belongings. The cabin must be properly searched after each flight to ensure that nothing is left behind and the Flight Crew notified.

After flight, cabin-related documentation shall be closed. Cabin crew shall ensure the orderly handover of the cabin to the next cabin crew or, where applicable, to the catering or cleaning personnel or to a maintenance contractor.

1.5.3 Line Training Captain, Supervisory Captain

Refer to the Training Manual (Part D of the Operations Manual).

1.5.4 Pilot Relieving the Commander

This paragraph is applicable to the Pilot Relieving The Commander in Augmented Crew operations when required by [Chapter 7 - Flight Duty Time Limitations](#).

1.5.4.1 Duties in the Event of an Emergency

The actions associated with the following guidelines may be applied as the actual emergency dictates and must be considered in conjunction with the applicable procedures and requirements specified in OM-A, OM-B, other applicable manuals and checklists, as relevant to the aircraft type concerned.

Rapid Depressurisation

- a. In the event of a rapid depressurisation, the Pilot Relieving The Commander shall ensure that the aircraft is operated in accordance with the OM-A, OM-B, other applicable manuals and checklists as relevant to the aircraft type concerned.
- b. The Pilot Relieving the Commander should not attempt to summon the Commander to the FCC, until a safe cabin altitude has been established.
- c. The Commander should remain on oxygen at their rest station until such time as the Pilot Relieving The Commander summons them to the FCC.
- d. When operating in airspace where specific procedures are prescribed for ensuring vertical and / or lateral separation in the event of an in-flight contingency (e.g. NAT HLA or RVSM airspace), the Pilot Relieving The Commander will be responsible for conducting any descent and diversion in accordance with these special procedures, as far as circumstances allow.

Engine Failure or Fire

- a. In the event of an engine failure or fire warning, the Pilot Relieving The Commander shall ensure that the aircraft is operated correctly and in accordance with the OM-A, OM-B, other applicable manuals and checklists as relevant to the aircraft type concerned, until the return of the Commander to the FCC.
- b. When operating in airspace where specific procedures are prescribed for ensuring vertical and / or lateral separation in the event of an in-flight contingency (e.g. NAT HLA or RVSM airspace), the Pilot Relieving The Commander will be responsible for conducting any descent and diversion in accordance with these special procedures, as far as circumstances allow.

Other Emergencies

Other emergencies or non-normal shall be actioned in accordance with the OM-A, OM-B, other applicable manuals and checklists as relevant to the aircraft type concerned. Whenever possible, the aircraft shall continue to be flown in accordance with the current clearance (i.e. track and flight level), pending the return of the Commander to the FCC.

1.5.4.2 Alerting the Commander

The Pilot Relieving The Commander, after successfully containing an emergency and ensuring the continued safe operation of the aircraft, shall alert the Commander as soon as practicable and summon them to the cockpit by any of the following means:

- a. Summon a cabin crew member to the cockpit and request the cabin crew to personally alert the Commander.
- b. If option 'a)' is not possible, use whatever other means considered appropriate to the situation which may include making a P.A. call 'COMMANDER TO THE FLIGHT DECK'.

Additionally the Pilot Relieving The Commander shall summon the Commander to the FCC in the event of any non-normal event that may impact the continued safe operation of the planned flight or for any other reason they consider it appropriate for the Commander to return to the FCC.

1.5.5 General Responsibilities

1.5.5.1 Crew Policy Manual

1. Crews will find their duties and responsibilities as employees of the company, as well as further company regulations and requirements in the Crew Policy Manual.
2. The current Crew Policy Manual is available to all crew members on the company portal.
3. All flight crew members are required to be familiar with the content of the Crew Policy Manual and abide by the policies and practices detailed therein.
4. In order to attain the highest levels of safety and service, all flight crews shall report all circumstances in which NTA levels of safety could be compromised and shall make recommendations for improvement to company systems and services.

Chapter 2 - Operational Control and Supervision

2.1 Supervision of the Operation by the Operator

NTA has implemented a method to ensure operational control and supervision of flight operations according to Part-ORO (ORO.GEN.110). This method is maintained and updated as required.

ORO.GEN.110 prescribes that:

- The operator shall ensure that all personnel are adequately instructed, have demonstrated their abilities in their particular duties, and are aware of their responsibilities and the relationship of such duties to the operation as a whole.
- Every flight shall be conducted in accordance with the provisions of the operations manual.

These requirements are based upon the overriding priority of safety. When conducting non-commercial operations under the scope of ORO.AOC.125 or ORO.GEN.310, specific, pre-identified provisions of the operations manuals may be replaced by procedures tailored to accommodate the unique requirements of those activities, namely Maintenance Check Flights, Training Flights under the NTA ATO and Fractional Operations. Operational control is always kept with NTA, and any alternative procedure is limited to the scope described in [section 8.7](#).

2.1.1 Licensing

All licence and qualification requirements of the ANAC shall apply.

Each licence entitles its holder to exercise their authority and execute their responsibilities based upon the licence only as long as it remains valid. It is therefore of the utmost importance that the licence holder for revalidation is in due time re-trained and re-examined by the company, or by approved third parties, as prescribed.

The Director of Training shall designate a section of their department as responsible for scheduling in due time licensed personnel for appropriate retraining, checks, and ascertaining that revalidation of each licence occurs before expiry. It must be stressed, however, that the final responsibility for retaining a licence's validity rests with its holder.

2.1.2 Competence of Operations Personnel

General

It is of the utmost importance, in the interest of safety and that of the company, to maintain a high level of competence, personnel proficiency, and skills.

Tools used for monitoring competence are:

1. (Crews) Check and training flights, route and line checks, simulator sessions;
2. (Ground personnel) Ad hoc assessment by the Ground Operations on random visits to contractors providing NTA with services;
3. Collecting and assessing accident and incident reports and customer complaints.

(In cases of high safety relevance, personnel who might otherwise become subject to disciplinary, legal, or penal action shall be encouraged to write anonymous reports).

For points 1 and 2 above, whenever a crew member does not achieve or maintain the required standards, the following procedure in the OM-D shall be applied: Procedures for Training, Checking, and Standards Not Achieved.

Tools used for maintaining and improving proficiency are:

1. The system for promulgating of additional operational instructions and information (see [2.2 System of Promulgation of Additional Operational Instructions and Information](#));
2. The Flight Safety programmes outlined in [Chapter 3 - Management System](#);
3. In individual cases, retraining as determined by supervisory staff.

Management staff shall convey to each individual the fact that their cooperation within the company team is essential, that their opinion is required, and that their ideas and suggestions are valuable. Apart from the esprit-de-corps this generates, individuals will be motivated not only to retain interest in their profession and in the company's welfare, but to remain proficient, skilled, and competent.

If doubt ever exists about an individual's relevant duties and responsibilities, or company procedures or policies, the next company superior should be informed.

Management attitudes and behaviour have a profound effect on staff, and the willingness by management to accept a lower standard will lead to the lower standard quickly becoming the norm. Staff shall never be tempted or pressured into lowering margins of safety.

2.1.2.1 Training

2.1.2.1.1 Training Captain

In the event of a Flight Crew Member demonstrating a deficiency in safety standards, the Line Training Captain / Supervisory Captain is authorised to recommend extraordinary Line Training, a Line Check, Simulator Training, and / or a Simulator Check, or if the situation warrants, even recommend removal of the Flight Crew Member from further active flight duty.

2.1.2.1.2 Operations Control

See [2.3 Operational Control and Supervision](#).

2.1.2.1.3 Dispatch

Flight Dispatch personnel must have demonstrated competence in:

- a. NTA manuals and their applicability;
- b. Air law, rules and regulations;
- c. Aeroplane data, performance, analysis and route;
- d. Weather analysis and weather hazards;
- e. NOTAMs and SNOWTAMS;
- f. Flight Planning and flight release procedures;
- g. Eurocontrol CFMU flight plan management and procedures;
- h. Dangerous Goods;
- i. Knowledge of Mass and Balance;
- j. Initiate, in the case of an Emergency, applicable procedures as described in the Emergency Response Plan.

2.1.2.1.4 Scheduling

Before a Scheduler is engaged in independent duty, they are required to successfully complete a six-month training and probation period, during which they are instructed in all legal and operational conditions for crew composition as well as duty time and flight time limitations. They are additionally supervised by their superiors during the initial compilation of duty rosters.

2.1.2.1.5 Ground Operations Personnel

The Ground Handling instructions contained in [8.2 Ground Handling Instructions](#) of this manual are the binding authority for all information and procedures in the field of aeroplane ground handling.

The instructions govern the responsibilities for aeroplane ground handling and describes the procedures for the handling of NTA flights.

The overall responsibility (editorial as well as professional) for the Ground Handling instructions lies with the Ground Operations nominated person.

All aeroplane handling agents are expected to comply with the standards and requirements detailed in the Ground Handling instructions.

2.1.2.1.6 Supervisory Staff

The number of supervisors appointed is related to the number of staff employed. The duties and responsibilities of each supervisor are clearly defined.

The supervision of all crew members shall be exercised by individuals possessing experience and personal qualities sufficient to ensure the attainment of standards specified in the operations manual.

2.1.2.1.7 Qualification for Employment

NTA has defined a high standard of requirements for employment. At all management levels, requirements have been defined for each subordinate position and applicants for these positions have to meet these requirements.

The rapidly evolving nature of flight operations in general, changes in procedures and electronic data processing, changes in medical assessments, and changes in legal provisions may consequently change such demands. Therefore, the department manager shall, in the course of routine meetings regularly held with supervisory staff from time to time, address these qualification requirements, discuss them respectively, have them re-evaluated, and, where necessary, have them changed.

NOTE

For qualification requirements of operations personnel, refer to [Chapter 5 - Qualifications Requirements](#).

2.1.2.2 Training, Examinations, and Licences

In a manner similar to that outlined in [2.1.1 Licensing](#), the department managers shall lay down the content and the extent of company training to be conducted for each position.

Whenever legal provisions or the special responsibilities of a position necessitate an examination (prior to which a holder may exercise their authority and execute responsibilities only under the supervision of a fully qualified person), the department manager defines the contents and extent of the examination and the minimum requirements for passing. For particulars, refer to [Chapter 5 - Qualifications Requirements](#).

Legal provisions may prescribe a licence (with a set expiry date and the requirement to renew it by undergoing defined checks / examinations) whenever the responsibilities of a position (i.e. flight crew member) necessitate the regular reassessment of the capabilities and the proficiency of a holder. Additionally, it may establish for defined positions the need for a company-specific licence. For particulars, refer to [Chapter 5 - Qualifications Requirements](#).

Similar to the provisions outlined in [2.1.1 Licensing](#) above, the department managers shall, in the course of specific meetings held with appropriate staff, regularly address all aspects of training, examinations, and licences.

It hardly needs to be mentioned that not only legal requirements and their changes must be observed meticulously, but that the state of the art must be monitored as well.

The NJE Chief Pilot shall closely cooperate with the crew training department. Type-specific aeroplane characteristics, experience gained by check, training, and line personnel, occurrence reports, statistics, trends, results of incident and accident investigations will all contribute to establishing and re-evaluating the requirements in training, examining, and licensing.

2.1.3 Control, Analysis, and Storage of Documents and Records

2.1.3.1 Documents to be Carried on Each Flight

The complete list of documents to be carried on each flight is available at [8.1.12 Documents, Forms, and Additional Information to be Carried](#).

2.1.3.2 Documents Available During Flight

The following documents will be available in the Operational Control Centre for the duration of each flight:

- a. A copy of the Flight Package (including - OFP, Weather and NOTAMs, and Mass and Balance documentation).
- b. Copies of the relevant Journey and Technical Logs.
- c. Notification of special loads and / or special categories of passengers.

2.1.3.3 Jeppesen Update Procedure

2.1.3.3.1 EFB Approved Fleet

Refer to [8.18.3 Normal Procedures](#) Updating of EFB Database and Content.

2.1.3.4 Post-flight Analysis of Documents

After completion of each sector or at the earliest opportunity, the crew shall ensure operational flight documentation is submitted and retained electronically via Skybook by:

1. Completing all required post-flight entries and performing Sector Sign-Off; and
2. Ensuring the information is successfully transmitted to the Skybook server. If a sign-off is performed when Skybook is offline, the crew shall transmit once connectivity is available.

The following documents are retained electronically, and no mailing is required:

Document type	Capture method
Operational Flight Plan (OFP); Flight Release package (incl. Mass & Balance record).	Automatically on Skybook and internal flight planning database
OFP Annex; Fuel documentation; Any other relevant operational documents e.g.: • deicing delivery notes; • operationally-relevant invoices (see Note); • documents related to oceanic crossings.	By photo (taken by crew) and uploaded to Skybook

All annotations made by the crew on electronic documents (in particular the OFP), including notes and markings, are automatically captured and retained within Skybook as part of the corresponding document, forming an integral element of the official operational record.

In normal circumstances, the OFP Annex and the NAT position reports and change of routing are the only flight documents used in physical paper form. After completion (and signature in the case of the OFP Annex), a photo of the documents must be uploaded to Skybook as part of the post-flight actions.

NOTE

Any supplier invoices or receipts paid by the crew must be submitted via the standard expenses process (do not email documents). Invoices or receipts provided to crews for products or services that are directly billed to NetJets must be captured by photo and uploaded to Skybook.

2.1.3.5 Document Storage Periods

The following information / documentation shall be stored in an acceptable form, accessible to ANAC, for at least the periods shown in the tables below.

The information used for the preparation and execution of a flight and personnel training records shall be preserved, even if NTA ceases to be the operator of that aeroplane or employer of that crew member, provided this is within the time-scales in this chapter.

A summary of training shall be maintained to show every crew member's completion of each stage of training and checking.

An annotation is made after the document name to state the format in which it is kept on record: hard copy (H) and / or electronic (E).

Information used for the preparation and execution of a flight:

Operational Flight Plan (E,H)	3 months
Technical Log (E)	36 months after the date of the last entry
Route specific NOTAMS / AIS briefing documentation if edited by NetJets (E,H)	3 months
Mass and Balance documentation (E,H)	3 months
Notification of special loads including Dangerous Goods (E)	3 months

Reports

Journey Log (E,H)	3 months
All reports for recording details of any occurrence, as prescribed in Part- ORO (ORO.GEN.160), or any event which the Commander deems necessary to report / record, with the exception of those listed in this table (E)	3 months
Discretion reports of extended flight duty periods, extended flight hours and reduced rest periods (E)	6 months
Air Traffic Incident Report, Bird Strike Report, Dangerous Goods Occurrence Report (E)	24 months
Accident Report (E)	Indefinitely

Flight and Cabin Crew Records

Flight duty, duty and rest period records (E)	2 years
Flight Crew licence and Cabin Crew attestation (E)	As long as the crew member is exercising the privileges of the licence or attestation for NTA and 5 years from the termination date ¹
Crew member training, checking and qualification (E)	5 years
Records on crew member recent experience (E)	15 months

Crew member route and aerodrome / task and area competence, as appropriate (E)	5 years
Dangerous Goods training, as appropriate (E)	5 years
Training / qualification records of other personnel for whom a training programme is required (E)	Last 2 training records

¹ Licence or attestation records, as applicable, will be held for 5 years from the termination date, except when the termination follows a Training Review Board. In such cases, and as per OM-D, the records will be retained for a period of 10 years from the termination date.

Records for other Operations Personnel

Initial Training (E)	Lifetime
Recurrent Training (E)	3 years and at least the last 2 training records
Dispatcher's Competency Check (E)	2 years and at least the last 2 training records
Flight deck familiarisation training (E)	2 years and at least the last 2 training records
Flight Dispatchers Licence (E)	5 years and at least the last 2 training records

Training / qualification records of other personnel for whom an approved training programme is required.

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Other Records

Records on cosmic and solar radiation dosage (E)	Until 12 months after the crew member has left the employ of the operator
Compliance Monitoring Programme Records (E)	5 Years
Dangerous Goods Transport Document (E)	3 months after completion of the flight
Dangerous Goods Acceptance Checklist (E)	3 months after completion of the flight
Documentation required to gain a specific approval under Part-SPA (E)	As long as the approval is valid

2.1.3.6 Record-keeping

To comply with the requirements for record-keeping laid in Part-ORO, NTA will ensure all records are accessible and secured, in both paper and electronic formats, for at least the entire period specified in the relevant subpart of the regulation.

RECORDS IN PHYSICAL FORMAT

Records required to be held in physical format for less than three months are stored in a locked, environmentally stable area of NTA's offices, effectively protecting them from theft, alteration, and damage during their retention period.

Records required to be held in physical format for more than three months are stored in a restricted area of NTA's offices. The records room is designed for long-term storage and to prevent document damage. It is equipped with environmental control, fire detection, and automatic fire extinguishing systems.

RECORDS IN ELECTRONIC FORMAT

- Electronic records are stored in systems with built-in redundancy in at least two separate facilities, ensuring reliability in case of a catastrophic failure;
- NTA's IT staff maintains on-premises storage systems. Off-premises storage systems have guaranteed high availability and data recovery protocols. All data is continuously backed up;
- Access to electronic records is restricted to authorised personnel based on their roles within NTA;
- Files with confidential or personal information will be encrypted while being transferred outside NTA's network;
- Hardware and software used to store records will be evaluated whenever new systems are implemented;
- Data will be migrated to new platforms to ensure continuity of record retention over the entire period required by regulations.

Unless specified in this OM or other approved internal procedure records will be held for at least five years.

NTA uses the Skybook service / application for the electronic distribution of flight records and the respective storage and management of crew annotations. Any electronic records in Skybook, including the Nav Log and digital sign-off, are uploaded to the application servers (managed by the platform provider) for long-term archival.

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2.1.3.7 Production and Inspection of Documents

All documents relating to flight operations and maintenance must be produced for inspection upon request by any person authorised by an authority.

The Commander shall, within a reasonable time of being requested to do so by a person authorised by an authority, produce to that person the documentation required to be carried on board.

If a crew member becomes a crew member for another operator, the crew member's records shall be made available to the new operator, provided this is within the timescales prescribed [2.1.3.5 Document Storage Periods](#).

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2.2 System of Promulgation of Additional Operational Instructions and Information

General

Information of an operational nature, which is supplementary to that contained in the Operations Manual, is either of a semi-permanent nature (it will normally be printed and distributed as a leaflet or as a booklet), or of an ad hoc nature with a more or less high degree of urgency necessitating expeditious means of distribution.

Due to its operational relevance, such information shall contain the date / time of its issue and state the beginning (date / time) and the end (where appropriate, otherwise the statement 'valid until further notice' should be included) of its applicability.

The following modes of publication / communication for promulgation are used by the company:

Print	(Leaflet, Circular, Booklet) and distribution by mail and company mail
Distribution	To a staff member's home address
Email	To a staff member's home address
Mailing	To a staff member's home address
Electronic Data Processing	Enabling all company departments to receive or distribute information where flights and / or operations personnel have access to a terminal.
SITA	Teletype messages
AFTN	Teletype messages
Facsimile	Faxed information
Telephone	Voice Communications
AFIS	To aeroplane in flight
HF telephone patch or SATCOM	To aeroplane in flight
Relay by ATS	To aeroplane in flight
Internet	Future use for publication of company manuals

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The system of promulgation is as follows:

Instruction / Information	Responsible By Means of	By Means of
Industrial agreements, including duty times	Flight Operations	EDP, email, Print
Country and aerodrome regulations	Flight Operations, Ground Operations	EDP, email, Print
List of authorised aerodromes	Flight Operations	EDP
List of Ground Handling Agents	Ground Operations	EDP, email, Print
List of Maintenance Agents	Maintenance	EDP, email, Print
List of Fuel Suppliers	Procurement, Ground Operations	EDP, email, Print
Crew Scheduling	Ground Operations	EDP, email, Print
Crew Rostering	Flight Operations	EDP, email, Print
Revocation of aerodrome authorisation or pilot's classification	Flight Operations	EDP, email, Print
Changes in Ground Handling information	Ground Operations	EDP, email, Print
Technical aeroplane type information	Maintenance Fleet Manager	EDP, email, Print
Changes in information pertaining to Maintenance	Maintenance	EDP, email, Print
Information on immigration, visa, health, customs, Gen Dec, passenger manifest requirements	Ground Operations	EDP, email, Print
Change of an aeroplane's routing or schedule	Ground Operations	All
Significant meteorological information and NOTAMs.	Ground Operations	Facsimile, AFIS, SITA, EDP, email, HF, SATCOM.
Changes of approach procedures, aerodrome operating minima, departure or missed approach procedures.	Flight Operations	Jeppesen Distribution, EDP, email, Print.
Runway length reductions, short term installation of obstacles.	Flight Operations, Ground Operations	Airport Performance Database, EDP, email, HF, SATCOM.
Other significant aeronautical information.	Flight Operations, Ground Operations	Print, Facsimile, SITA, EDP, email, AFIS, HF, SATCOM.

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2.3 Operational Control and Supervision

According to Part-ORO (ORO.GEN.110 (b)), every flight shall be conducted in accordance with the provisions of the operations manual (OM-A, B, C and D).

Flight Dispatch maintains overall control of NTA daily flight activities under a system of shared responsibility, including the initiation, continuation, diversion, or termination of each flight in the interest of safety.

For all NTA flights, Flight Dispatch shares authority with the Pilot-in-Command for all pre-flight decision functions, including planning, weather and NOTAM analysis, aeroplane performance, mass & balance, and flight release.

When exercising the responsibilities and authority mentioned above, both the Flight Dispatch team and the Pilot-in-Command must ensure that their decision-making is not influenced by actual or perceived commercial pressure from passengers or other departments to operate a flight in a specific manner or at all.

Supervision of the operation is accomplished by the Operations Control Center, which includes flight monitoring, providing in-flight assistance to crews, recording post-flight data, and any other task associated with the operational control of a flight. During the flight, the Pilot-in-Command has sole authority over safety.

Operating near or over conflict areas

Upon the recognition of airspace affected by an armed conflict or other types of security threats, the Security Department may decide, in coordination with all relevant internal stakeholders, to limit the use of such zones, either by prohibiting their use or subjecting every flight operating in that zone to prior authorisation. The decisions are made visible to internal stakeholders by marking and highlighting these zones in the planning and oversight tools used by OCC. Crew members can access the same information via the software tools in the portable EFB.

Director of Operations/NJE Chief Pilot supervision and approval of non-routine operations

To ensure an adequate level of supervision of non-routine activities, certain operations require DO/CP approval as specified in the OM-A paragraphs listed below. The provisions in this subsection are only to be applied where explicitly referenced in the OM-A text. The DO/CP may not approve any other deviations from the OM-A requirements.

- [5.2.5.5 MCF\(B\) Flight Crew Requirements](#)
- [5.4.1.2.6 Ad Hoc Route / Area Qualification](#)
- [7.2.5.1 Reduced Scheduled Show-Times](#)
- [7.2.6.1 Reduced Scheduled Turn Time](#)
- [8.1.3.8 Maximum Operating Winds](#)

Requests for DO/CP approval shall be submitted in writing using the *DO approval tool*, which collects relevant information for the type of approval required through pre-formatted questions. These include, but are not limited to, the applicable OM-A reference, any risk mitigation measures already planned, and other questions designed to provide sufficient details about the proposed operation to enable the DO/CP to assess the level of risk.¹

¹ The 'DO Approval Tool' is a web-based application featuring a pre-formatted request form tailored to the applicable type of approval, which shall be completed by the person submitting the approval request. Requests for approval, approvals, and denials should normally be processed using this 'DO Approval Tool'; however, in cases where this is impractical, email may also be used but must include the same information as the 'DO Approval Tool' and the emails must be retained for the required period.

If insufficient information is provided, the DO/CP shall request more information and/or conduct additional investigations into the proposal.

If, according to the DO/CP's assessment of the information provided, the DO/CP considers that the proposed operation will maintain an acceptable level of safety, then the DO/CP may approve the request using the response function of the *DO approval tool*.

If the DO/CP judges that the proposed operation will not maintain an acceptable level of safety, the DO/CP shall either deny the operation or implement sufficient additional risk mitigation measures to ensure an acceptable level of safety prior to approving the operation with this added risk mitigation. A record of all DO/CP approval requests and their outcome shall be maintained for two years.

2.3.1 Operations Control Centre - OCC

The OCC is responsible for the day to day management of company aeroplanes in the most safe, efficient and economic manner in accordance to company procedures with due regard to customer service.

The OCC is led by the Duty Manager and is comprised of:

- Duty Manager;
- Daily Scheduler;
- Ground Services;
- Flight Dispatcher.

Duty Manager monitors the execution of the daily operational programme, seeking the expert views of all members anticipating potential areas of operational inefficiency, or service disruption, in order to maintain high levels of service and safety. Initiate and approve the appropriate recovery plan needed to minimise the impact on NetJets customers. Ensure that operational activities on the day are performed in compliance with company policies, procedures, standards, and legal requirements. Ensures that Owner Services (generally Owner Services Current Day) is informed about issues that will require communication to the Owner, via the SOC Track or other communication channels as needed.

Daily Scheduler is responsible for scheduling solutions in terms of aeroplanes and crew, maintaining IJet updated in accordance with all on day changes, as directed in [2.3.4 Scheduling](#).

Ground Services Specialist is responsible for coordinating, re-arranging, and cancelling on the day airport slots and PPRs, or provide solutions for allocations and / or ETD 'frustrations'.

Responsible for ground services in respect to the company flight programme.

Flight Dispatcher is responsible for flight preparation in terms of flight planning, on the day weather analysis, flight release, in-flight assistance, flight briefing / debriefing, closely monitoring ATC scenarios likely to impact the daily operation, and liaising with flight dispatch personnel to mitigate delays and pro-active 'service keeping' measures.

2.3.2 Flight Dispatch

Duties and Responsibilities

- Responsible for all NetJets Transportes Aéreos, S.A. Flight Dispatch functions.
- Ensure compliance with AIR OPS, ICAO, company, and work-rule procedures and regulations.
- Prepare and modify operational flight plans, determine the most suitable routing for the flight in accordance with all relevant factors (e.g. meteorological conditions, aeroplane performance, airspace availability, flight schedule, NTA fuel policy, etc.) that affect the operation and, if necessary, cancel a flight so that a high level of safety can be assured within the bounds of regulation.
- File ATC flight plans and monitor ATC regulations in order to find the most efficient and economic route for NTA flights and ensure a high on-time performance.
- Ensure that all flights are released safely, in accordance with AIR OPS, ICAO, National Authority, and company rules and regulations.
- Monitor the daily flight activity, aeroplane / crew arrivals and departures, crew airline transfers, and all other duties and tasks associated with the daily operations of NTA falling within Operational Control.
- Proactively oversee NetJets programme to ensure safe, efficient, and economic operation of the company. Maintain responsibility for all NetJets Transportes Aéreos, S.A. dispatch operational functions.
- Be responsible to the Head of Operations Planning and Control for the suggestion and implementation of policies to improve performance, efficiency, and service to customers and crew.
- Immediately inform the OCC as well as other relevant Operations Planning and Control and OS teams of any unforeseen situation and / or changes that may affect the Customer's flight activity.
- Evaluate weather, aeroplane performance, and flight releases to assure that they fall within the operational parameters of company safety policy.
- Complete flight data logs.
- Coordinate with other departments to enhance communication and fine tune the delivery process to ensure customer satisfaction.
- Coordinate with NJE Chief Pilot and FPM team regarding any operational requirement in order to assure the highest levels of safety and service.
- Examine work for exactness, neatness, and conformance to policies, rules, and regulations.
- Try to ensure punctuality, accuracy, tact (customer-handling), detail of information, etc.
- Provide the paperwork (itinerary / briefing sheet) quality standards required for proper execution of flights.
- Assure that all operational problems in daily activity are followed and customers are properly informed via standard company channels.
- Update crew members on their daily flight activity and revise the required flight documentation (Flight Package) according to the new schedule.

- Ensure compliance with the company procedures on activating the ERP, and keep the different ERP teams briefed.

Flight Briefing includes a telephone call or messaging system, notifying the crew about their planned assigned flight activity, including forwarding all relevant flight documentation (Flight Package).

Flight Preparation is a process of planning flights, which includes preparing and modifying operational flight plans (OFP), determining the most suitable routing for the flight considering all relevant factors, such as meteorological conditions, NOTAMs, aeroplane performance, airspace availability, flight schedule and NTA fuel policy. It also includes determining the most suitable departure and arrival alternate(s) for the flight and filing the required ATC flight plan according to the respective OFP.

Flight Release is a process of shared responsibility between the Commander and the Dispatcher, achieved by direct communication (telephone or electronic means), which includes mutual approval of weather analyses, mass and balance, minimum legal fuel to be carried, and information about the respective ATC slot and airport slot.

In-flight Assistance is accomplished by providing all relevant assistance to the safe and efficient flight progression. The assistance shall be performed via SATCOM, AFIS, or other effective means of communication.

Flight Monitoring is accomplished by constant monitoring of flight progression and anticipating abnormal scenarios that can compromise safety and the normal progression of a flight. Flight monitoring shall be done using IJet, Flight Explorer, CFMU, AFIS, SATCOM, JetPlan, or other effective means of flight monitoring.

2.3.2.1 Flight Release Procedure

No NTA aeroplane shall depart without being previously released by a NTA Flight Dispatcher.

The Flight Release procedure is a process of shared responsibility between Operations Planning and Control personnel (Flight Dispatchers) and the Commander of the flight, where flight initiation is assessed and mutual agreement is achieved with regards to the flight plan data, weather conditions, mass and balance, total fuel on board, and ATC slot information. However, the Commander has the final decision with regards to fuel related issues.

NOTE

The Commander has sole responsibility for safety decisions relating to the flight when the aeroplane is airborne.

The Flight Release consists of messages sent using the Flight Dispatch Support system to the crew's company mobile device and vice versa.

No flight shall be released if the amount of fuel on board and any changes to passenger and baggage masses are not received and recorded.

Any email exchange related to flight releases shall be addressed to lis-flightrelease@netjets.com.

The Flight Release shall contain, or have attached to it, NOTAMs, actual weather reports, available weather forecasts, or a combination thereof, for the departure airport, destination airport, and alternate airports that are the latest available at the time the release is signed (electronically) by the Flight Dispatcher. It may include any additional available weather reports or forecasts that the PIC or the Flight Dispatcher considers necessary or desirable.

The Flight Release is a three step process designed to complete the Flight Release procedure.

Step 1

Every Flight Release will be initiated by Flight Dispatch with a Flight Release Ready message sent electronically, containing but not limited to: OFP, Mass and Balance form, meteorological conditions, NOTAMs, and performance, known as the flight package. The message shall be sent and should be available to the flight crew at the show time prior to the first flight of the FDP and, when possible, as early as 90 minutes prior to the schedule time of departure of the remaining flights. This flight package shall be used for the flight preparation;

Step 2

The Commander or the person delegated by the Commander shall reply to the Flight Release Ready Message, as follows:

Primary method: Using the Waypoint application on the company mobile device.

NOTE

Changes to the number or classification of passengers requires flight crew to advise electronically or via telephone.

Backup method: Via email using the RLS - Reply to Flight Release Ready Message set format (see RLS - Reply to Flight Release Ready Message).

RLS - Reply to Flight Release Ready message shall be submitted by email to lis-flightrelease@netjets.com. If no data communication is possible, the crew shall communicate the data to Flight Dispatch by telephone.

Step 3

The process is concluded when a **Flight Release** message is sent by Flight Dispatch and received by the flight crew. When the flight release message is received, the Commander shall ensure that the Flight Release data is crosschecked to ensure that it matches the data in the on-board documentation and any changes / differences are recorded on the on-board documentation (OFP, Mass and Balance data, etc).

NOTE

1. Step 2 and Step 3 can be repeated as many times as necessary, depending on the amount of changes occurring after the completion of Step 3.
2. Step 3 is normally accomplished electronically but may be done so via telephone. If Step 3 is accomplished by telephone, then the Flight Release message shall be sent electronically as soon as possible.

RLS - Reply to Flight Release Ready Message (via email)

The subject of the message shall contain the following fields as required (see Reply to Flight Release Ready Message Fields and Codes table):

RLS - Reply to Flight Release Ready Message Fields	
Subject:	RLS REG ADEP ADEST FUEL PAX BAG

The message can either be in upper or lower case and the field divider can be either a space or a dash.

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Table 2.3.2.1 A - Reply to Flight Release Ready message Fields and Codes

Field	Description and Codes																										
RLS	RLS - Code for Release: Always required																										
REG	Last 3 letters of the Aeroplane registration: Always required.																										
ADEP	ICAO identifier of the Departure aerodrome: Always required.																										
ADEST	ICAO identifier of the Destination aerodrome: Always required.																										
FUEL	TTL fuel on-board (lbs): Always required.																										
	NOTE																										
	The following codes shall be used and placed after the TTL fuel figure when the TTL fuel is more than the rounding up or down to the nearest two hundred lbs (200 lbs):																										
	<table> <tr> <th>Code</th><th>Reason</th></tr> <tr> <td>ATC</td><td>More than usual expected ATC delay</td></tr> <tr> <td>DISP</td><td>On request of Flight Dispatch</td></tr> <tr> <td>LESS</td><td>LESS</td></tr> <tr> <td>MEL</td><td>MEL reference nbr. With fuel penalty</td></tr> <tr> <td>OTHER</td><td>Specify the reason</td></tr> <tr> <td>PREV</td><td>Fuel already on board from previous leg</td></tr> <tr> <td>TAXI</td><td>De-icing procedures, taxi queues</td></tr> <tr> <td>TANK</td><td>Tankering more for economic reasons</td></tr> <tr> <td>WX-DEP</td><td>Forecasted weather in departure area</td></tr> <tr> <td>WX-RTE</td><td>Forecasted weather en route</td></tr> <tr> <td>WX-DEST</td><td>Forecasted weather in arrival area</td></tr> <tr> <td>WX-ALT</td><td>Forecasted weather in alternate area</td></tr> </table>	Code	Reason	ATC	More than usual expected ATC delay	DISP	On request of Flight Dispatch	LESS	LESS	MEL	MEL reference nbr. With fuel penalty	OTHER	Specify the reason	PREV	Fuel already on board from previous leg	TAXI	De-icing procedures, taxi queues	TANK	Tankering more for economic reasons	WX-DEP	Forecasted weather in departure area	WX-RTE	Forecasted weather en route	WX-DEST	Forecasted weather in arrival area	WX-ALT	Forecasted weather in alternate area
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LESS	LESS																										
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TANK	Tankering more for economic reasons																										
WX-DEP	Forecasted weather in departure area																										
WX-RTE	Forecasted weather en route																										
WX-DEST	Forecasted weather in arrival area																										
WX-ALT	Forecasted weather in alternate area																										

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Field	Description and Codes
PAX nW	Number and weight category of passengers: This is only required if the passenger number / category / mass data in the mass and balance documentation changes. Code is PAX followed by the total number of passengers on-board (nW) where: • n = number of pax for the specific type; • W = passenger weight category, possible values are: • M for adult male pax; • F for adult female pax; • X for other adults; • C for child. NOTE nW can be repeated for as many passenger types as needed.
BAG	Total baggage mass (lbs): Only required if there is a change to the baggage mass data in the mass and balance documentation. Code is BAG followed by the total amount of baggage mass being carried on-board.

Reply to Flight Release Ready message examples	
1.	RLS DKD LFPB LGAV 15000
2.	RLS DKD LFPB LGAV 16000 ATC
3.	RLS DKD LFPB LGAV 15000 PAX 4M 6F 2C
4.	RLS DKD LFPB LGAV 15000 BAG 464
5.	RLS GLA EGLL KTEB 25000 TANK PAX 1M 1F BAG 174
6.	RLS DXR EGGW LFPB 4000 PREV BAG 200

Explanation:

1. No change in fuel or change in fuel only within the rounding up or down to the nearest 200 lbs.
2. Change in fuel: new fuel TTL amount 16000 lbs reason being 'more than usual expected ATC delay'.
3. Change in pax: new breakdown, 4 adult male, 6 adult female, and 2 children.
4. Change in baggage mass: new total baggage mass of 464 lbs.
5. Change in fuel, pax and baggage: new fuel TTL amount 25000 lbs, reason being 'tankering more for economic reasons', new pax breakdown 1 adult male and 1 adult female, new total baggage mass of 174 lbs.

6. Change in fuel and baggage amount: new fuel TTL amount 4000 lbs, reason being 'fuel already on-board from previous leg' and new total baggage weight of 200 lbs.

For more complex changes, the flight crew shall contact Flight Dispatch by telephone to make the necessary changes.

2.3.2.1.1 Procedure Specifics

- a. A flight shall not depart unless a Flight Dispatcher specifically releases the flight and the Commander accepts the release.
- b. A Flight Dispatcher shall not release a flight unless they are thoroughly familiar with reported and forecast weather conditions on the route to be flown.
- c. A Flight Dispatcher shall not release a flight unless they are thoroughly familiar with all information available for the route to be flown and airports to be used, including all planned alternates.
- d. A Flight Dispatcher shall not release a flight when the combination of paragraph b. and c. does not allow the flight to be legally dispatched and, in that situation, alternate airports or routes must be found to comply with the legal requirements. Nevertheless, a flight may be released if weather conditions can be met as per Planning Minima for Destination Aerodromes, En Route Alternate (when applicable) and Destination Alternate Aerodromes.
- e. A Flight Dispatcher shall not change an original destination or alternate airport that is specified in the original flight release to another airport while the aeroplane is en-route, unless the other airport is authorised for that type of aeroplane and the appropriate requirements are met at the time of re-dispatch or amendment of the flight release, and the Commander agrees with the change.
- f. A Flight Dispatcher shall inform the Commander en-route if there is a change to the weather forecasts specified in the flight release that indicates that the destination or alternate aerodromes may be below the required minima at the time the aeroplane is scheduled to arrive. However, the flight may continue to a destination with weather below minimums if, in the opinion of the commander, it is preferable to hold over the destination waiting for weather improvement, subject to normal fuel policy. The flight release may be amended after departure, as per point i. below, to include any alternate airport that is within the fuel range of the aeroplane with the agreement of the Commander. In the above circumstances, the Flight Dispatcher shall use AFIS, SATCOM, HF, or any mean of communication available to contact the aeroplane in flight and pass the available information to the crew.
- g. NTA flight releases will always be run through IntelliJet system, except in the cases that the aeroplane data is not loaded or due to system failure.
- h. In the situations that the aeroplane data is not loaded, the fuel loaded in the aeroplane has to be cross-checked with the Operation Flight Plan (OFP). If the fuel load is at or above the minimum required fuel for that sector, the flight can be released.
- i. The original flight release cannot be changed. In the case of an in-flight re-planning, a Flight Dispatcher shall produce an amended flight release and record the amendment and the agreement of the Commander.
- j. A Flight Dispatcher must file CHG item 16 of the ATC flight plan with a change (CHG) message whenever the alternate(s) airport(s) changes prior to departure, due to weather, operational and technical reasons.
- k. When running the flight release, all passenger names are passed to the crew company mobile phone using IntelliJet. In case a deviation on the number and / or names of any of the passengers, the crew members will pass to Flight Dispatch the required information

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about the passenger(s) such as names, nationality, and passport number. All the information shall be immediately passed on to Owner Services, Ground Services, and Catering Logistics for proper update in IntelliJet and customs / immigration notification at the destination airport.

- I. At the time of the flight release, on revenue legs, the Flight Dispatcher is to assure that the Commander holds the security code, if any, for the flight. If they do not, the code is to be given.

For further information on the Flight Release procedure, refer to [Flight Dispatch Systems Manual, Chapter 3.6.](#)

2.3.3 International Trip Planning

International Trip Planning is responsible for obtaining and managing overflights and landing permits by liaising with the appropriate aeronautical authorities.

It is responsible for ensuring the following:

- Application for and continual management of overflight and landing permissions and traffic rights.
- Requests for flights to Europe, Eastern Europe, North Africa, and the Middle East.
- Preparing quotes for flights operating outside of the Primary Operations Area.
- Preparing a flight folder for all flights requiring overflight or landing permission, applying and updating permissions, and recording as necessary.
- Ensuring fuel supply is available at airports that are not NTA-contracted.

2.3.4 Scheduling

Scheduling is responsible for creating and refining optimum daily trip schedules for owner aeroplane, by managing crews and aeroplanes in compliance with the applicable regulations, in the most efficient and economic manner with the due regard to safety, crew productivity and availability, crew training, crew licensing, and aeroplane maintenance availability.

Additionally, the Scheduling staff maintains a daily record of the crew members that exceeded daily duty, crew rest reduction, non-standard shows, and the accepted or declined extended days on the Scheduling Report Form.

Duties and Responsibilities

Schedule Planner / Scheduler

NOTE

Schedule Planner and Scheduler have the same responsibilities with different degrees of scope.

- Produce and refine daily trip schedules for owner aeroplanes, utilising available crew and aeroplane resources in compliance with the applicable regulations, and company procedures to create optimal and cost effective trip schedules to fulfil owner's requirements.
- Liaise closely with Maintenance, Flight Operations, Training, and other Ops Planning departments regarding all resources required for flight activity; being aware of all required variables that can influence a flight schedule; define the most effective daily trip schedules maximising aeroplane utilisation and crew productivity.

Scheduling Supervisor

- Supervise and control the team of Schedule Planners in the assigned shift, assigning activities based on daily needs in order to efficiently allocate resources and maximise the performance of the team, guaranteeing cost effectiveness.
- Monitor the performance of the team to guarantee compliance with policies and procedures, consistency, and quality of service.
- Oversee the daily trip schedules for owner aeroplanes defined in the assigned shift to guarantee that trips are efficiently developed based on the maximum aeroplane utility, owner request, and crew availability.

2.3.5 Trip Review

Trip Review is responsible for flight feasibility studies in terms of aeroplane performance and range, aeroplane performance versus aerodromes, aerodrome analysis, and weather prognostics.

Duties and Responsibilities

Trip Review has the following main responsibilities:

- Monitor daily and future flight activity, researching and using the available resources, and anticipating unusual events that may have a negative impact on an Owner's flight itinerary and recommend alternative solutions.
- Verify and approve the Quality Check (QC) for new airports or those with specific restrictions.
- Evaluate aeroplane performance.
- Complete a Quality Check (QC) for new flight requests and request changes, including analysis of aeroplane performance and airport restrictions.

2.3.6 Operational Irregularities

2.3.6.1 Definition

Operational irregularities as referred to in the following section are all deviations from the planned schedule and may be:

- Early departure of a flight;
- Late departure of a flight;
- Rerouting of a flight;
- Diversion of a flight en-route;
- Cancellation of a flight.

2.3.6.2 Measures Taken

In case of irregularities, all necessary measures shall be taken and decided by the Operations Control Centre, be it for the present or further planning, after conferring with the departments concerned.

The above decisions must:

- Not influence the safety of the flight under any circumstances;
- Most effectively re-plan flight operations as close as possible to normal operations;
- Deliver passengers to their destinations with as little delay as possible;
- Reduce loss of revenue to a minimum, wherever possible, without ever impacting on customer service obligations.

The Operations Control Centre will confer with all concerned sections and, after final decisions as to the course of action have been made, shall pass all relevant information regarding aeroplane, crew, and passengers to the departments and individuals concerned.

2.3.7 Early Departures

2.3.7.1 Definition

An early departure of a flight is defined as a flight departing ahead of the scheduled departure time.

2.3.7.2 Policy

Since NTA operates ad hoc services based on the needs of its aeroplanes owners, provided that ATC, aerodrome, and all other factors permit, company aeroplanes may depart as soon as practicable after arrival of the passengers.

2.3.8 Late Departures

If a late departure cannot be avoided, it should be kept to a minimum.

The Commander may delay the departure in order to avoid adverse meteorological conditions at the departure aerodrome, en-route or at the destination aerodrome. Any decision for such a delay must be communicated to the Operations Control Centre immediately.

2.3.9 Rerouting of Flight

2.3.9.1 Definition

A change of the planned destination after departure.

2.3.9.2 Application

Reason for such an action could be:

- Passenger request;
- Operational or scheduling requirement to position an aeroplane at a different aerodrome;
- Weather conditions at the original planned destination.

This decision is made by the Operations Control Centre after conferring with the sections and individuals concerned.

2.3.10 Diversion of Flight En-route

2.3.10.1 Definition

A flight proceeding to an aerodrome other than the planned destination at departure.

2.3.10.2 Application

Reason for such an action may be:

- Operational (e.g. weather at destination);
- Technical, requiring a landing at the nearest suitable aerodrome;
- The medical condition of any person on board requiring an urgent landing;
- Meteorological conditions en-route.

In case of a diversion, the crew is required to report this fact to the Operations Control Centre by AFIS, if time permits, or by telephone immediately after landing, once any matters of an urgent nature have been addressed.

2.3.11 Cancellation of Flight

2.3.11.1 Definition

The cancellation of a previously planned flight before departure.

2.3.11.2 Application

Reasons for such action could be, for example:

- Aeroplane owner's decision not to proceed with journey;
- Technical or mechanical malfunction of the aeroplane;
- Extended delay;
- Meteorological conditions.

Before a flight is cancelled for technical reasons, the Operations Control Centre should confer with the Maintenance department and Vendor Operations department to find an alternative solution.

After a decision to cancel a flight has been made, Operations Control Centre shall forward the change to all concerned departments and sub-contractors associated with the flight.

2.4 Power of Authority - ANAC(Autoridade Nacional de Aviação Civil)

2.4.1 General

Any person authorised by ANAC is permitted at any time to board and fly in any NTA aeroplane operated in accordance with the AOC. They may enter and remain on the flight deck. However, the Commander may refuse access to the flight deck if, in their opinion, the safety of the aeroplane or its occupants would thereby be endangered.

Chapter 3 - Management System

The NTA management system is described in the NetJets Management System Manual (MSM), which includes:

- The NTA management system policies;
- The NTA processes for identifying safety hazards and for evaluating and managing the associated risks;
- The NTA compliance monitoring programme;
- The accountabilities and responsibilities for NTA operations;
- A description of all key NTA management system processes.

The NTA management system policies are reproduced below.

3.1 Safety Policy

Safety Accountability

Safety is the first of our core principles at NetJets. As Accountable Manager, I am legally accountable for the safety of NetJets' operations. The nominated persons are also accountable for the safety of operations within their functional area and are responsible for ensuring that NetJets remains compliant with the applicable regulations. I am committed to balancing the allocation of resources to reduce any safety risk to a point which is as low as reasonably practical. The senior management team and I shall provide the necessary human and financial resources to implement this Safety Policy.

Safety Objectives

NetJets shall carry out an annual safety review and set clear safety objectives. The safety objectives aim to promote a strong safety culture, where safety concerns are communicated and viewed as opportunities to improve the safety of our operations, the service we provide our customers and the work environment for our employees. We shall measure the effectiveness of any safety actions we implement, monitor our safety performance, and identify areas for improvement in our management system.

Following our core principles of safety and integrity, NetJets shall comply with all applicable legislation, meet all applicable standards, and work in accordance with the Company's documentation.

Manager Responsibility

We aim to lead our industry in safety practices and standards, including applying human factors principles throughout the operation. All NetJets' managers are responsible for the following:

- The safety of their functional areas;
- The protection of safety-critical processes and procedures;
- The management of the safety consequences of any internal or external changes and conducting an appropriate risk assessment;
- Ensuring that all externally supplied systems and services meet our safety standards.

Employee Responsibility

All NetJets employees are responsible for ensuring the safety and regulatory compliance of NetJets' operations. All employees are required to report any safety issue or event that may affect the safety of NetJets' customers, employees, or operations. Employees shall report any suspected non-compliance with regulatory requirements. NetJets provides a safety reporting system to enable any employee to report a safety event or issue, including the ability to report confidentially. All employees are responsible for supporting the compliance monitoring programme and any internal safety investigations.

Just Culture

The purpose of safety reporting, flight data monitoring and internal investigations is to improve safety, not to apportion blame to individuals; therefore, as an Accountable Manager, I require all managers to follow the 'Just Culture' principles and I guarantee that no disciplinary action will be taken as a result of reporting a safety issue / event or from a safety event identified by the flight data monitoring program unless there is clear evidence of wilful misconduct or gross negligence.

In supporting NetJets Just Culture, safety information, such as safety reports, flight data, or safety investigations, shall only be used for the improvement of the safety of NetJets' operations. Sharing of safety information for any other purpose is strictly prohibited.

Safety and Compliance Team

The Safety and Compliance department is responsible for the following:

- Ensuring that the management system is effective;
- Supporting all departments in identifying hazards and conducting appropriate risk assessments;
- Promoting the NetJets safety culture and communicating relevant safety information to employees;
- Ensuring that all employees are aware of their safety responsibilities and receive adequate training and support to fulfil those responsibilities.

3.2 Fatigue Policy

Safety is the first of our core principles at NetJets. As such, NetJets has a Fatigue Risk Management System, as part of its Management System, to manage fatigue-related risks within its operations. NetJets is committed to balancing the allocation of resources to reduce fatigue risk in all crew members, to an acceptable level.

The FRMS is a core part of the business and provides an opportunity to improve the safety and efficiency of the operation while maximising the well-being of crew members.

This is achieved through the shared responsibility and commitment of management, crew members and supporting staff.

In accordance with the NetJets core principles of safety and integrity, NetJets shall observe all applicable legal requirements and international standards; in addition, NetJets shall implement good industry practices based on scientific fatigue management principles.

NetJets shall carry out an annual fatigue review and set clear fatigue objectives. The Accountable Manager and the senior management team shall provide adequate resources to achieve NetJets' FRMS objectives. The objectives aim to promote a strong safety culture, where fatigue concerns are communicated and viewed as opportunities to reduce fatigue exposure in NetJets operations. NetJets shall measure the effectiveness of any implemented fatigue mitigating actions, monitor fatigue performance, and identify areas for improvement in the management of fatigue risk.

All NetJets employees are responsible for creating an environment that promotes open and honest reporting of fatigue related hazards and incidents, ensuring that the fatigue risks within their area(s) of responsibility are managed appropriately, providing all crew members with adequate opportunity for recovery sleep between duties. The nominated persons shall manage the fatigue consequences of any internal or external changes and conduct an appropriate risk assessment for changes in safety critical processes.

All NetJets crew members have a responsibility to ensure the safety of NetJets' operations; those responsibilities include, but are not limited to:

- Making appropriate use of their rest periods (between duty periods) to obtain sleep;
- Participate in fatigue risk management education and training programmes;
- Report fatigue-related hazards and incidents as described in NetJets Safety Reporting System;
- Comply with the Fatigue Risk Management Policy; and
- Inform Scheduling department immediately prior to or during work if they know or suspect they or another crew member are suffering from unacceptable levels of fatigue;
- Inform their supervisor if they have any doubt about their or another crew member's capability to accomplish their duties due to fatigue.

NetJets supports a 'Just Culture' and guarantees that no disciplinary action will be taken as a result of reporting fatigue, a fatigue related event, or a potential fatigue contributing factor, unless there is clear evidence of wilful misconduct or gross negligence.

The Safety department is responsible for ensuring that the FRMS is effective and for supporting all departments in identifying fatigue contributing factors and conducting appropriate risk assessments.

3.3 Compliance Monitoring Programme

The main purpose of the Compliance Monitoring Programme is to monitor the continued compliance with the relevant regulations and adopted standards, as well as to make continuous improvements to the overall level of safety of NTA operations. The programme covers all NTA approvals as an AOC Operator, CAMO, ATO and FSTD.

All NTA employees are responsible for the safety and compliance of the service provided to our customers and for supporting the continuous improvement of the management system.

All NTA operational personnel are responsible for complying with the applicable laws, regulations and procedures in all locations where operations are conducted. NTA employees shall ensure that when contracting or purchasing any part of NTA activities, the contracted or purchased service or product conforms to the applicable requirements.

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Chapter 4 - Crew Composition

4.1 Crew Composition

4.1.1 Aeroplane Type

NTA's fleet is composed of the following Aeroplane Types:

Manufacturer	TCDS Designator	Type Designator (ICAO Doc 8643)	Model / Marketing
Embraer	EMB-505	E55P	Phenom 300/300E
Cessna / Textron	560XLS ¹	C56X	Citation XLS
Cessna / Textron	680A*	C68A	Citation Latitude
Dassault	Falcon 2000EX	F2TH	Falcon 2000EX
Bombardier	BD-100-1A10	CL35	Challenger 350
Bombardier	CL-600-2B16	CL60	Challenger 650
Bombardier	BD-700-1A11	GL5T	Global 5000
Bombardier	BD-700-1A10	GLEX	Global 6000

¹ For historical reasons, and in the context of this manual, the prefix 'CE-' is used in the designator of these types (CE-560XLS and CE-680A).

4.1.2 Area and Type of Operation

NTA aeroplanes Area of Operations are all ICAO Air NAV Regions.

Type of Operation - Non Scheduled.

4.1.3 Phase of Flight

The crew composition in this chapter refers to all phases of flight.

4.1.4 Minimum Crew

4.1.4.1 Minimum Flight Crew

For any flight, the number and composition of the flight crew must never be less than the number required by the Aeroplane Operating Manual (AOM) or Aeroplane Flight Manual (AFM), as applicable. This minimum number may be increased by the terms of [Chapter 7 - Flight Duty Time Limitations](#). Any such increase then becomes the essential minimum for Company operations.

All flight crew must hold applicable and valid licences acceptable to the ANAC and be suitably qualified and competent to conduct the duties assigned to them.

Normally, for commercial flight operations, the minimum number of flight crew for NTA aeroplanes must be as given in the table below:

Aeroplane Type	Minimum Flight Crew	Minimum Cabin Crew
EMB-505	2	-
CE-560XLS	2	-
CE-680A	2	-
BD-100-1A10	2	-
DA-2EASy	2	1
CL-600-2B16	2	1
BD-700-1A10	2	1
BD-700-1A11	2	1

4.1.4.2 Minimum Cabin Crew

Regulatory-wise, NTA is not required to have a cabin crew assigned to its flights, in accordance with ORO.CC.100.

However, to enhance safety and customer service NTA will normally assign the number of cabin crew as prescribed in [4.1.4.1](#) for all flights.

All cabin crew must be suitably qualified in accordance with [5.3 Cabin Crew Qualification Requirements](#) and competent to conduct the duties assigned to them.

The cabin crew member's primary function is to perform duties in the interest of safety as specified in the Operations Manuals.

Commercial flight operations with less than the minimum cabin crew prescribed in [4.1.4.1](#) are permitted only when no other viable option exists.

Where more than one cabin crew is assigned to a flight, the senior cabin crew member will be designated according to [4.1.7 Designation of Senior Cabin Crew Member](#).

All designated cabin crew shall wear the company uniform.

4.1.5 Crew Member Experience, Recency, and Qualification

Under normal circumstances, the commander, Second-in-command, and third crew member (if required or directed by flight operations) will occupy only the seats to which they are qualified and assigned. For the commander, this is the left-hand seat and for the Second-in-command, the right-hand seat.

PIC-qualified crew members without right hand seat qualification occupying the right pilot seat shall, under normal conditions, not act as pilot flying during take-off and landing. No pilot shall exercise the duties or the responsibilities of the commander from the right hand seat unless they have successfully undergone the necessary additional training and checking (refer to OM-D, Training Manual).

Each cabin crew must have successfully passed initial and conversion or differences training and familiarisation flights, and must have a valid emergency training / proficiency certificate on the respective aeroplane type. No person may accept an assignment to duty knowing that they are not fully qualified and fit for duty.

If a cabin crew member has not, during the preceding six months, undertaken flying duties on an aeroplane type, they may not operate as one of the minimum number of cabin crew required until either:

- i. Completion of the refresher training specified in the Training Manual;
- ii. Two familiarisation sectors under supervision have been satisfactorily completed.

The Company will maintain a record of the training undertaken, the initial and recurrent checks completed, and the current qualification status for each cabin crew member as specified in [Chapter 2- Operational Control and Supervision](#).

4.1.6 Designation of Commander and In-flight Relief of a Flight Crew Member

For each flight, NTA will designate one pilot only to be the Commander.

When, by order of flight operations or in accordance with the provisions of [Chapter 7 - Flight Duty Time Limitations](#), extra flight crew members are scheduled for in-flight relief, the following rules shall be observed:

A flight crew member may be relieved in flight of their duties at the controls by another suitably qualified flight crew member.

a. Relief of the commander

The commander may delegate the conduct of the flight to:

1. another suitably qualified Commander; or
2. a suitably qualified Second-in-command, in accordance with [5.2.2 Pilot Relieving the Commander \(ORO.FC.A.201\)](#), for operations above FL200.

NOTE

The overall responsibility for the flight remains with the company designated Commander as per [4.2 Designation of the Commander](#) and cannot be transferred, except in the event of Commander's incapacitation.

If two suitably qualified Second-in-command, as per a.2, are crewed together, then the Commander shall designate which pilot shall be the pilot relieving the Commander.

Normally, the pilot that has been on the fleet the longest should be designated as the pilot relieving the Commander.

When the commander is relieved by a suitably qualified Second-in-command, the PF shall be the pilot in the right-hand seat.

b. Relief of the Second-in-command:

The Second-in-command may be relieved by another suitably qualified pilot.

- c. The company designated commander shall be at the controls for take-off and landing.
- d. The company designated commander will assign the seats and duties of PF / PM for the flight crew at the controls for take-off and landing. When doing so, the commander shall ensure that the respective flight crew are suitably qualified.

4.1.7 Designation of Senior Cabin Crew Member

When more than one cabin crew is assigned to a flight, NTA will designate one of them to be the senior cabin crew member, as follows:

A. Normal Line Operations

Subject to the commander's decision, the usual succession to act as senior cabin crew member will be as follows:

- i. Cabin Crew Line Trainer, if applicable;
- ii. The more senior cabin crew member (by company seniority) with at least one year of experience as CCM.

B. Training and Checking Events

During a training event, the SCCM shall be assigned in accordance with the procedure in OM-D, section 2.2.4 Senior Cabin Crew Member (SCCM).

The designation of the SCCM shall be covered during the First Flight of Tour briefing, ensuring that all crew members acknowledge that role.

If the SCCM becomes incapacitated or unfit for duty during a flight, the commander must be informed immediately. Acknowledging that most flights with more than one CCM are training events, it's important to determine if a reassignment is possible after an incapacitation. If no qualified and current CCM is available, the remainder of the flight shall be performed without a CCM.

4.1.8 Crewing of Inexperienced Flight Crew Members

Flight Operations has established the following procedures to prevent the crewing together of inexperienced flight crew members.

1. There shall not be more than one inexperienced flight crew member in any flight crew.
2. A flight crew member is considered inexperienced, following completion of a type rating or command course and the associated line training, until they are achieved either a further:
 - a. 100 flight hours or 30 sectors on the type within a consolidation period of 120 consecutive days; or
 - b. 150 flight hours and flown 20 sectors on the type (no time limit).
3. After acceptance by the ANAC, in the designated implementation period of a new aeroplane type, for the first 9 months, the above restrictions are not applicable.

NOTE

The 120 consecutive days specified in (a) above is a rolling 120-day period which does not necessarily commence immediately on completion of the conversion course.

Restrictions applicable to Inexperienced PICs Flying to Cat B and C Aerodromes: the following table depicts at which CAT C aerodromes inexperienced PICs can operate as commanders.

CAT C Aerodrome	Inexperienced PIC operation at aerodrome as commander
EDFE (Egelsbach)	No
EGLC (London City)	Yes (except BD100-1A10)
ENTC (Tromsø)	Yes
LESO (San Sebastian)	Yes
LFKC (Calvi)	No
LFLB (Chambery)	No
LFLP (Annecy)	No
LFTZ (La Mole)	No
LICG (Pantelleria)	No
LICR (Reggio Calabria)	No
LIMG (Albenga)	No
LIPB (Bolzano)	No
LOWI (Innsbruck)	No
LSGS (Sion)	No
LSZA (Lugano)	No
LSZS (Samedan)	No
VNKT (Kathmandu)	No
VQPR (Paro, Bhutan)	No

NOTE

1. Some aerodromes have additional experience (e.g. hours) requirements. In these cases, the most restrictive requirement will apply.
2. For those aerodromes which require 'In Flight Familiarisation' there are additional restrictions regarding the performance of a 'Familiarisation Flight' by an inexperienced PIC. These are depicted in [5.4.1.2.1](#).

For CAT B aerodromes, any Inexperienced FCM restrictions will be depicted in the respective NTAAerodrome briefing.

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4.1.9 Age Limit for Flight Crew

(FCL.065)

A FCM who has attained the age of 65 years shall not act as a pilot of an aircraft that is engaged in commercial air transport.

The FCM shall not be scheduled for any flights within 24 hours before their 65th birthday. Due to unforeseen circumstances, the FCM may actually operate an NTA aeroplane within the 24 hours before their 65th birthday if, at the time of take-off, the flight is expected to be completed before 00:00 (Lisbon time) of the day of their 65th birthday.

4.1.10 Flight Crew with an Operational Multi-pilot Limitation (OML)

A FCM with an OML on their medical certificate shall only operate an aeroplane in multi-pilot operations when the other flight crew member:

- a. is fully qualified on the relevant aeroplane type; and
- b. is not subject to an OML; and
- c. has not attained the age of 60 years.

NOTE

In case an FCM with an OML is part of an augmented flight crew, all other FCM in that crew shall meet the requirements of a), b) and c) above.

4.1.11 External Flight Crew

NTA and NJML flight crew shall not be part of a crew where any of the required minimum flight crew members is not an NTA or NJML flight crew member, except for training flights with an external TRI / TRE.

4.1.12 Personnel or Crew Members Other than Cabin Crew in the Passenger Compartment

Personnel or Crew Members Assigned to Specialist Duties

For operational reasons, it might be necessary to supplement the minimum required number of cabin crew by additional crew members assigned to specialist duties. These crew members, to whom the qualification and recency requirements for cabin crew are not applicable, include, but are not limited to, the following:

- a. Child escorts;
- b. Entertainers;
- c. Ground engineers;
- d. Interpreters;
- e. Medical personnel;
- f. Secretaries;
- g. Security staff.

The additional crew members that perform duties in the passenger's compartment shall:

- a. not be confused by the passengers with the operating cabin crew members;
- b. not occupy required cabin crew assigned stations;
- c. not impede the operating cabin crew members in their duties.

The commander shall ensure that the passengers are briefed that any additional crew members are not members of the operating cabin crew and are not responsible for any safety related tasks.

Additional crew members shall not wear a uniform in such a way that might lead the passengers to identify them as being part of the operating cabin crew.

4.1.12.1 In-flight Service Representative (ISR)

The ISR is part of the crew and is assigned the duty of in-flight service to the Owner / passenger(s). The ISR is not responsible for any safety-related tasks, including the safety briefing, which shall be provided by a Flight Crew Member. The ISR is considered an ABP and shall be briefed beforehand by the Flight Crew on the safety features of the aeroplane, including the operation of the Emergency and normal exits.

The ISR shall wear the NTA uniform. However, to ensure that the passengers are aware that they are not operating cabin crew members, the commander shall ensure that the passengers are briefed that they are not members of the operating cabin crew and that they are not responsible for any safety-related tasks.

When the ISR is notified that a Crew Alertness Risk Assessment (CARA) is required for a given duty period, they shall complete that CARA no later than 30 minutes after reporting time for the applicable duty period, and notify the Commander that the CARA has been completed.

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4.2 Designation of the Commander

For each flight, NTA will designate one pilot only to be the commander. The handling of an aeroplane or conduct of flight may be delegated by the commander to another suitably qualified pilot who they deem competent and fit for duty.

The commander is the flight crew member holding overall responsibility for the safety of the aeroplane, its crew, its passengers, and its load. They are the pilot-in-command unless they have been relieved in flight of their duties at the controls by another pilot.

NTA will nominate a pilot as commander only if they meet the minimum qualification and recency requirements laid down in [Chapter 5 - Qualifications Requirements](#).

No pilot may accept a nomination as commander unless, in addition to their licence qualifications and their training, they have the recent experience and knowledge required, and consider themselves to be in all respects competent and fit for the task.

Whenever two PIC-qualified crew members are crewed together in an aeroplane, the pilot who has been qualified as PIC longest within NTA shall be designated the commander of the flight by the Scheduling department, with possible exceptions for operational requirements (e.g. aerodrome and route qualification requirements), during training and evaluation flights, unless a notification to the contrary has been received from the crew. Any change in commander designation must be agreed by both pilots and communicated by email to the Scheduling department the day prior to flight(s). Due to the non-automated process, the commander designation shall not be changed by crew request during an FDP. The OFP, ATC flight plan, and IntelliJet need to reflect the agreed commander designation.

Any PIC-qualified pilot that actively holds a management position within the Flight Operations department or Training department and meets the requirements of [4.1.5 Crew Member Experience, Recency, and Qualification](#) will be designated the Commander.

Any change in command during flight shall be reported as soon as possible to NTA Operations Control Centre and to ATC, where legally required, and shall be entered in the Flight and Maintenance Reports.

4.2.1 Commander Assignment During Training and Evaluation Flights

The LTC, STC, SSTC, and / or TRI/TRE shall be designated Commander on training, instruction, and evaluation / check flights if they occupy the RHS or LHS.

On flights during which the LTC, STC, SSTC, and / or TRI / TRE is observing the conduct of the flight from the observer's (jump) seat, they may not be the designated commander. They may assume command of the flight from a pilot only if, in their judgment, the safety of the flight necessitates it.

4.3 Flight Crew Incapacitation

The symptoms and effects of incapacitation are described in the SEP Manual. The following procedures shall be followed to determine the succession of command in the event of the incapacitation of the commander.

4.3.1 Aeroplanes with a Flight Crew of Two

The Second-in-Command will assume command of the aircraft.

However, if there is a pilot on board who is qualified with NTA to act as pilot-in-command on the same aircraft type and is fit to fly, the Second-in-Command may exceptionally request that they assume the position of Commander and if they accept then they shall become the commander.

4.3.2 Aeroplanes with an Augmented Flight Crew

On those occasions when there is an additional pilot among the crew who holds a current licence with type rating and the qualification to act as pilot-in-command, the senior pilot will take command.

4.3.3 Flights with Commander under Supervision

On flights conducted with a commander under the supervision of a check or training pilot, the supervising commander will take over command.

4.3.4 Consideration as an Emergency

Any situation in which the commander or any member of the minimum flight crew is incapacitated and incapable of further duty is to be regarded as an emergency.

In this case, nothing shall prevent the remaining crew from asking for the assistance of any person on board the aeroplane, who they deem to be competent, to perform any necessary duties on the flight deck.

Any case of incapacitation of any member of the minimum flight crew must be reported immediately to ATC. This is necessary in order that suitable priority treatment of the flight can be claimed and the proper precautions taken by the controller. Furthermore, a full report of the circumstances must be made to NTA as soon as practicable.

4.4 Operation on More than One Type or Variant

4.4.1 Flight Crew

A flight crew member shall not operate more than 2 aeroplane types or variants for which a separate licence endorsement is required.

A flight crew member may only fly aeroplanes within one licence endorsement in any one flight duty period.

The minimum flight crew complement will be as specified in [4.1.4 Minimum Crew](#).

The following aeroplanes are considered as one type:

- C560XL(S): Citation C560XL, Citation C560XLS.
- BD-700-1A10 (Global 6000) and BD-700-1A11 (Global 5000).

4.4.2 Cabin Crew

A cabin crew member shall not operate on more than 3 aeroplane types.

They may operate on 4 types only if approved by the competent authority and if for at least two of the types:

- a. safety and emergency equipment and type-specific normal and emergency procedures are similar; and
- b. non-type-specific normal and emergency procedures are identical.

BD-700-1A10 (Global 6000) and BD-700-1A11 (Global 5000) are the same type hence considered a variant for the purpose of CCM training and qualification.

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Chapter 5 - Qualifications Requirements

This chapter contains the requirements for the licences, ratings, qualifications (i.e. for route and aerodromes), experience, courses, training, checking, and recency for operations personnel to conduct their duties.

Detailed training programmes and the minimum qualification, experience, and recency requirements for FCM, CCM, and personnel with training / supervisory roles are described in the OM-D.

5.1 Flight Crew Qualification Requirements

The minimum requirements for employment and promotion are specified here in accordance with EASA AIR OPS, EASA Aircrew Regulation, as applicable, and ANAC regulations.

All Training and checking programmes are specified in OM-D and are approved by ANAC. To operate company aeroplanes, flight crew must have at least the following:

- a. A licence and ratings issued or accepted in accordance with PART-FCL;
- b. A valid Class 1 Medical Certificate issued in accordance with PART-MED;
- c. A current Validation, if necessary for maintenance or delivery flights, issued by ANAC;
- d. A type rating for the type to be flown;
- e. A multi-pilot instrument rating for the relevant type to include PBN privileges;
- f. Minimum Level 4 English language endorsement on the licence;
- g. A valid European passport;
- h. Additional qualifications as required.

Flight Crew shall demonstrate their level of English language proficiency by method of assessment established and defined by their State of Licence Issue, in order to ensure effective communication during the performance of duties.

An FCM may operate as part of the operating crew of an NTA aeroplane up to and including the age of 64 (reference paragraph [4.1.9 Age Limit for Flight Crew](#)).

5.2 Flight Crew

5.2.1 Commander (PIC)

5.2.1.1 Nomination as Commander

(ORO.FC.105)

Before flight crew members may be appointed as Commander, they must meet the requirements under [5.2.1.2](#).

5.2.1.2 Minimum Qualification Requirements for Commanders

The minimum qualification requirements for a flight crew member to be qualified as PIC and act as a Commander of an NTA flight are:

- 3,000 hours total time in aeroplanes;
- Successful completion of the NTA command course;
- An Airline Transport Pilots Licence with appropriate type rating;
- A valid Instrument Rating to include PBN privileges for all operations;
- Applicable conversion and / or recurrent training and checks;
- Recency of experience according to [4.1.5 New Member Experience, Recency, and Qualification](#);
- Route / Area and aerodrome competence training according to [5.4.1.2](#);
- A current European passport and all required visas;
- Command of the English language;
- Additional qualifications, as required.

5.2.1.3 PIC's Qualification to Operate in Either Pilot's Seat

(ORO.FC.135)

PIC's whose duties require them to operate also in the right-hand seat and carry out the duties of a Second-in-command, or PICs required to conduct training or examining duties from the right-hand seat, must complete additional training and checking as specified in OM-D.

5.2.2 Pilot Relieving the Commander

5.2.2.1 Minimum Qualifications

- a. The Commander may delegate the conduct of the flight to a suitable NTA PIC- qualified pilot.
- b. The Commander may delegate the conduct of the flight to an NTA Second-in-Command for operations above FL200 providedt they have the following qualifications.
 - i. 1500 hours total time in aeroplanes;

- ii. An unrestricted, Airline Transport Pilots Licence with appropriate type rating;
- iii. A valid Instrument Rating to include PBN privileges for all operation;
- iv. Applicable conversion and / or recurrent training and checks;
- v. Recency of experience according to [4.1.5 Crew Member Experience, Recency, and Qualification](#);
- vi. Route and aerodrome competence training according to [5.4.1.2](#);
- vii. A current European passport and all required visas;
- viii. Command of the English language;
- ix. Applicable pilot relieving the commander training;
- x. At least 500 hours total flight time on the applicable aeroplane type and a minimum of 2 years on the applicable fleet after the initial line check;
- xi. Additional qualifications, as required.

5.2.3 Second-in-Command

5.2.3.1 Minimum Qualification Requirements for Second-in-Command

The minimum qualification requirements for an FCM to act as a Second-in-Command on an NTA flight are:

- 1500 hours total time in aeroplanes;
- PART-FCL Airline Transport Pilot Licence for Aeroplanes (ATPL - A) or Commercial Pilot Licence for Aeroplanes (CPL - A) including all prerequisites for an ATPL, except actual flying experience (hours) requirements;
- A valid Instrument Rating to include PBN Privileges for all operations;
- Applicable conversion and / or recurrent training and checks;
- Recency of experience according to [4.1.5 Crew Member Experience, Recency, and Qualification](#);
- Route and aerodrome competence training according to [5.4.1.2](#);
- A current European passport and all required visas;
- Minimum Level 4 English language endorsement on the licence;
- Additional qualifications, as required.

5.2.3.2 Pilot Relieving the Second-in-Command

A pilot relieving the Second-in-Command shall be either PIC-qualified or SIC-qualified in NTA.

5.2.4 Recent Experience

A pilot shall not operate an aeroplane in commercial air transport or carry passengers as Commander or Second-in-Command unless they have carried out, within the preceding 90 days, at least three take-offs, approaches, and three landings as pilot flying in an aeroplane

or an approved full flight simulator (FFS) representing that type or class. The 3 take-offs and landings shall be performed in a multi-pilot operation. The flight simulator must be acceptable to ANAC for take-off and landing.

In the case of commercial air transport, the 90-day period prescribed above may be extended up to a maximum of 120 days, as long as the pilot undertakes line flying under the supervision of a type rating instructor or examiner.

When the pilot does not comply with the above requirement, they shall complete a training flight (non-commercial) with a type rating instructor or examiner in the aeroplane or an FFS of the aeroplane type to be used, which shall include at least the above requirements before they can exercise their privileges.

5.2.5 Maintenance Check Flight (MCF)

(SPO.SPEC.MCF.100)

Maintenance Check Flights are divided into two types:

- a. Maintenance Check Flight Level (A) - Level A maintenance check flights are maintenance check flights for which the use of abnormal or emergency procedures as defined in the aircraft flight manual is expected. Level A maintenance check flights also include operations required to prove the functioning of a backup system or other safety devices.
- b. Maintenance Check Flight Level (B) - Any other check flight.

NOTE

High Speed Taxi Checks (those with a ground speed greater than 20 kts) are classified as MCF(B).

5.2.5.1 MCF(A) Flight Crew Selection

MCF(A) Flight Crew shall be selected and qualified in accordance with the *Maintenance Check Flight Manual*.

5.2.5.2 MCF(A) Flight Crew Pre-requisites

The PIC of an MCF(A) shall meet the requirements detailed in the Maintenance Check Flight Manual, including holding a current MCF(A) qualification.

The SIC of an MCF(A) shall not be an inexperienced FCM as described in paragraph [4.1.8 Crewing of Inexperienced Flight Crew Members](#) of this manual.

5.2.5.3 MCF(A) Flight Crew Training and Qualification

The PICs that are eligible to perform MCF(A) flights shall, in order to be qualified, successfully complete the relevant training as documented in OM-D.

5.2.5.4 MCF(A) Flight Crew Recency Requirements

The PIC shall not perform an MCF(A) unless they have carried out an MCF within the last 36 months.

Recency can be regained by performing an MCF(A) flight as an observer or after acting as PIC during an MCF(A) in a full flight simulator session.

5.2.5.5 MCF(B) Flight Crew Requirements

The PIC and SIC of an MCF(B) shall not be an inexperienced FCM as described in paragraph [4.1.8 Crewing of Inexperienced Flight Crew Members](#) of this manual unless exempted from this requirement by the DO/CP in accordance with [OM-A 2.3 Operational Control and Supervision](#).

There are no other specific requirements for MCF(B) Flight Crew.

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5.2.6 Line Flying Under Supervision

Following completion of flying training and checking as part of a conversion course, or as part of the command course when upgrading to PIC, a pilot must fly a minimum number of sectors and / or flying hours under the supervision of a Line Training Captain or a Supervisor Captain.

Detailed requirements for Line Flying Under Supervision are described in OM-D.

5.2.7 Operations on More than One Type or Variant

(ORO.FC.240)

5.2.7.1 Qualification Requirements

In addition to the minimum qualification requirements in [5.2.1 Commander \(PIC\)](#) and [5.2.3 Second-in-Command](#), a flight crew member must meet the following requirements before exercising the privileges of a second licence endorsement:

- a. Flight crew members must have completed two consecutive OPCs and must have 500 hours in the relevant crew position in commercial air transport operations with NTA.

In the case of a pilot having experience with NTA and exercising the privileges of 2 licence endorsements, and then being promoted to PIC on one of the types, the required minimum experience as commander is 6 months and 300 hours, and the pilot must have completed 2 consecutive OPCs before again being eligible to exercise 2 licence endorsements.

5.2.7.2 Training and Checking

Before commencing training for and operation of another type, flight crew members must have completed 3 months and 150 hours flying on the base aeroplane, which must include at least one proficiency check.

After completion of the line check on the new type, 50 hours flying or 20 sectors must be achieved solely on aeroplanes of the new type rating.

Annual emergency and safety equipment training and checking must cover all requirements for each type.

The following recency requirements apply to the operation of more than one type not within a single licence endorsement:

- a. Flight crew members must complete a minimum of 10 route sectors as pilot of the relevant type during the validity period of the type rating. In addition, a flight crew member must not be assigned to operate an aeroplane of the relevant type unless they are completed a minimum of one route section in each function PF and PM on the relevant type within the previous 90 days.

5.2.7.3 Aeroplanes Not With the Same Approved Grouping

NTA does NOT authorise such operations.

5.2.7.4 Aeroplanes in More than Two Approved Groupings

NTA does NOT authorise such operations.

5.3 Cabin Crew Qualification Requirements

5.3.1 Senior Cabin Crew Member

(ORO.CC.100(c))

Whenever more than one CCM is assigned to a flight, one must be designated as the Senior Cabin Crew Member, as described in [4.1.7 Designation of Senior Cabin Crew Member](#).

As described in OM-D, training for SCCM is incorporated in the initial training course and first conversion training for all NTA Cabin Crew.

The designated Senior Cabin Crew Member must have at least one year's experience as an operating CCM and must have completed a senior cabin crew training course and associated checks.

5.3.2 Cabin Crew Member

(ORO.CC.110) (CC.GEN.025)

5.3.2.1 Required Cabin Crew Member

1. A CCM shall:

- Be at least 18 years old;
- Hold a valid attestation issued in accordance with PART-CC;
- Hold a cabin crew medical report in accordance with PART-MED;
- Remain medically fit to discharge the duties specified in the OM;
- Before operating as a CCM, have successfully completed:
 - the appropriate aeroplane type specific training;
 - the NTA conversion course and familiarisation training;
 - the appropriate difference training.
- Be competent to perform duties as specified in the OM.

2. The following medical requirements are applicable to each CCM:

- Good health;
- Freedom from any physical or mental illness which might lead to incapacitation or inability to perform CCM duties;
- Normal cardio-respiratory function;
- Normal central nervous system function;
- Adequate visual acuity (6/9 or 14/20, with or without glasses);
- Normal function of ear, nose, and throat.

5.3.2.2 Additional Cabin Crew Members and Cabin Crew Members During Familiarisation Flights

For operational reasons, it may be necessary to supplement the cabin crew by additional crew members solely assigned to specialist duties, as described in [Chapter 4- Crew Composition](#).

The requirements of [5.3.2 Cabin Crew Member \(ORO.CC.110\) \(CC.GEN.025\)](#) are not applicable to these additional crew members.

After completion of conversion training and prior to operating as one of the minimum required cabin crew, a CCM must successfully undertake familiarisation training as described in the OM-D.

5.3.3 Operation on More than One Type or Variant

(ORO.CC.130) (ORO.CC.250)

A CCM shall not operate on more than 3 aeroplane types. They may operate on 4 types only if for at least two of the types:

- a. safety and emergency equipment and type-specific normal and emergency procedures are similar; and
- b. non-type-specific normal and emergency procedures are identical;
- c. after approval from ANAC.

BD-700-1A10 (Global 6000) and BD-700-1A11 (Global 5000) are the same type hence considered a variant for the purpose of CCM training and qualification.

5.4 Training, Checking, and Supervision Personnel

5.4.1 Flight Crews

5.4.1.1 Conversion Training and Checking

(ORO.FC.120)

The Operator's Conversion Course enables a flight crew member to acquire the minimum qualification for the operation of Company aeroplanes. It may be combined with training for a type rating.

Before commencing unsupervised line flying, a flight crew member must complete a conversion course:

- When joining the company;
- When changing to an aeroplane for which a new type or class rating is required.

An FCM will not be assigned flying duties on another type or class until the Operator Conversion Course is completed or terminated. Before operating their previous equipment type, they must complete refresher training approved by the Authority.

Conversion training will be conducted by suitably qualified persons in accordance with a detailed course syllabus included in OM-D. The amount of training required will be determined after due note has been taken of an FCM previous training and experience recorded in their training and flight records.

5.4.1.1.1 Composition of the Conversion Training Course

(ORO.FC.220)

Conversion Training is composed of:

- a. Ground training and checking, (with or without flight simulator / cockpit procedure trainer, computer based training, or other training device).
- b. Emergency and safety equipment training and checking, including instruction on first aid. This must be completed before flying training on an aeroplane commences.
- c. Aeroplane / Flight Simulator training and checking.
- d. Line Flying Under Supervision and Line Check.
- e. Crew Resource Management Training.

5.4.1.1.2 All Weather Operations

NTA will ensure that flight crew's knowledge and ability to perform the tasks associated with low visibility take-off (LVTO) is trained and checked. The training and checking are prescribed in OM-D. The flight crew qualification is specific to NTA and the type of aeroplane. NTA is not approved to conduct Category II and III approaches.

5.4.1.2 Route / Area and Aerodrome Competence Qualification

In regard with route, area or aerodrome knowledge, training and qualifications, NTA will comply with the national training and checking requirements published in the aeronautical information publication (AIP).

5.4.1.2.1 Training Definitions

Briefing

Information presented orally, on paper, and / or by computer screen.

Self-training

Self-training is accomplished by either:

- a. Reading the area, route, or special aerodrome briefing supplied by NTA , or such a briefing(s) as designated in the OM-C; or
- b. Completing an online briefing / CBT / WBT and, if the online briefing / CBT / WBT has a completion test or test questions, successfully pass the online test and / or test questions.

Ground Training

The Crew Member receives instruction by an authorised person (as indicated in OM-C) on the ground. This Ground Training may be conducted by telephone.

Observation Flight

The FCM shall complete an observation flight on the jump seat for the applicable area, route, or aerodrome for which the qualification is required. Observation flights may be accomplished on other types and / or other operators (unless otherwise indicated). During the observation flight, the operating Flight Crew shall actively show and explain all relevant aspects as detailed in the OM-A / OM-C of the area, route, or aerodrome.

In-flight Familiarisation

In-flight Familiarisation shall be accomplished in either the LHS or RHS, during positioning or revenue flights as indicated in the OM-C. In-flight familiarisation is approved for Flight Crew during LIFUS, unless otherwise indicated.

In-flight Familiarisation is accomplished by either:

- a. A Familiarisation Flight performed with a commander without an 'inexperienced' designation in accordance with the OM-C requirements.
- b. A Familiarisation Flight overseen by a PIC without an 'inexperienced' designation or a Line Training Captain. The overseeing PIC or Line Training Captain may occupy the jump seat, in which case they shall not be designated the Commander.

Manoeuvres, flight patterns, or actions required for the in-flight familiarisation shall be detailed in the OM-A or OM-C for that specific area, route, or aerodrome and shall not include manoeuvres, flight patterns, or actions that are not required during a normal flight or create unnecessary risk.

In case both FCM at the controls of the aeroplane are NTA PIC or one of them is under training, both FCM shall be considered meeting the familiarisation requirements on completion of the flight.

Flight Training

Flight Training is performed in accordance with training requirements defined in OM-C and after completion of a self-instruction and / or ground instruction. Flight training shall be conducted by a Type Rated Instructor (TRI), a Line Training Captain, or designated Supervisory Captain in accordance with the requirements and syllabus published in OM-D.

The FCM under training and the Line Training Captain or Supervisory Captain conducting the training shall occupy the seat specified by the applicable qualification requirements.

Where the TRI, Line Training Captain, or Supervisory Captain conducting the training occupies the jump seat, they shall not be designated the Commander.

Where applicable, special flight procedures and go-around shall be performed with strict adherence to the training syllabus.

Simulator Training

Simulator Training is performed in accordance with training requirements defined in OM-C. The simulator training is performed in a NTA approved certified simulator for the purpose after completion of a self-instruction and / or ground instruction. Training is conducted by a SFI, TRI, or a person acceptable to the NTA and the Authority requiring the qualification.

5.4.1.2.2 Route / Area Competence Qualification

Flight Regions

For the purpose of route qualification, the world has been divided into the following regions (based on App 1 to ICAO Doc 8144):

- AFI (Africa-Indian ocean region): The area embracing Africa and associated oceanic areas and land masses between 25° W and 75° E.
- ASIA (Asia region): The area south of the Asian part of the former Soviet republics, and associated oceanic areas and land masses eastward of 90° E to 175° W to South to 60° S.
- CAR (Caribbean region): The area embracing Mexico, Central America, the Bahamas, and the West Indies.
- EUR (European region): The area embracing Europe and the Asian part of the former Soviet republics, north to 78° N, and including Turkey.
- MID (Middle East region + India / Pakistan): The area embracing that part of Asia west of 90° E, but excluding Turkey and the former Soviet republics.
- NAM (North American region): The area embracing the United States, Canada, and north to 78° N.
- NAT (North Atlantic region): The North Atlantic area not covered by NAM, CAR, SAM, POL, and ASIA regions.
- NAT (North Atlantic) Light: Only those routes, route structures, and transition areas within and adjacent to the NAT HLA specifically detailed in Chapter 3 of the North Atlantic Operations and Airspace Manual (NAT Doc 007).
- PAC (Pacific region): The Pacific area not covered by the NAM, CAR, SAM, POL, and ASIA regions.
- SAM (South America area): The area embracing South America and the associated oceanic areas and land masses between 25° W and 90° W and south to 60° S.
- POL (Polar region): Area lying North of 78° N and south of 60° S.

Route / Area Knowledge

Area and route training and familiarisation shall include knowledge of:

- i. terrain and minimum safe altitudes;
- ii. seasonal meteorological conditions;
- iii. meteorological, communication, and air traffic facilities, services, and procedures;
- iv. search and rescue procedures where available; and
- v. navigational facilities associated with the area or route along which the flight is to take place.

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5.4.1.2.3 Initial Route / Area Qualification

- a. Route qualification requirements shall be applicable to PIC and SIC, unless otherwise indicated.
- b. For regions EUR, MID, and AFI, route competence qualification shall be achieved by self-training using route documentation from the OM-C and Jeppesen Route Manual.
- c. For all other regions, completion of a Route Qualification Course module for the relevant region shall be required. The course may be delivered as ground training in a classroom or as self-training online. Additionally, a route qualification flight event may be required according to region, as described below.
 - For NAT region, two route qualification Flight Training sectors. The flight training sectors shall be conducted by a NAT qualified Line Training Captain or designated Supervisory Captain in accordance with the requirements and syllabus published in OM-D. The FCM receiving the training shall always occupy an operating seat. The Line Training Captain or Supervisory Captain conducting the training may occupy the jump seat, provided a PIC already qualified for the route occupies an operating seat for that portion of the flight which is within the NAT region.
 - For ASIA region and for PIC only, one route flight familiarisation sector. The flight familiarisation sector shall be conducted by an ASIA qualified PIC. The FCM receiving the familiarisation shall occupy an operating seat or the jump seat.
- d. Qualification for regions CAR and NAM shall only be assigned to FCMs who have completed the relevant Route Qualification Course module and hold a NAT route qualification.

5.4.1.2.4 Route / Area Re-qualification

- a. For regions EUR, MID, and AFI, route / area competence re-qualification shall be achieved by self-training using route documentation from the OM-C and Jeppesen Route Manual.
- b. For all other regions, completion of a route / area Qualification Course module for the relevant region shall be required. The course may be delivered as ground training in a classroom or as self-training online. The course shall be valid for a period of 12 calendar months in addition to the remainder of the month of completion. If completed within the final three calendar months of validity of the previous course, the period of validity shall extend from the date of completion until 12 calendar months from that previous date.
- c. For NAT and ASIA re-qualification, a periodic flight within the region is additionally required, as follows:
 - For the NAT region, the FCM shall operate a minimum of one sector, occupying an operating seat for that portion of the flight which is within the NAT region.
 - For the ASIA region, the FCM shall operate a minimum of one sector, occupying an operating seat or jump seat for that portion of the flight which is within the ASIA region.
 - The validity of this re-qualifying flight shall be 36 calendar months in addition to the remainder of the month of completion.

5.4.1.2.5 Route / Area Re-qualification Following Expiry

In the event of an FCM's route qualification expiring, re-qualification training shall be conducted in accordance with OM-D.

5.4.1.2.6 Ad Hoc Route / Area Qualification

Exceptionally, flights may be dispatched within a flight region with one or more crew members who do not hold a current qualification for that region.

Such flights shall be subject to written authorisation by the DO or the NJE Chief Pilot in accordance with [OM-A 2.3 Operational Control and Supervision](#). The requester shall establish, in coordination with the DT, the route qualification requirements to ensure that the flight crew have the required competence to safely operate on the route, considering the route and area knowledge requirements laid in ORO.FC.105.. The authorisation shall only be valid for a specified flight or series of flights.

Where such authorisation applies to flights which include routes within the NAT region, the planned flight(s) shall be restricted to routes defined in the North Atlantic Operations and Airspace Manual (NAT Doc 007).

5.4.1.2.7 Upgrade from SIC to PIC Route / Area Qualification

SICs who hold a current route qualification and subsequently upgrade to PIC shall have their qualifications treated as follows:

- a. All route / area qualifications except NAT and ASIA shall be retained upon upgrade without additional requirements.
- b. For NAT and ASIA qualification, the pilot shall be required to complete the route / area qualification flight(s) specified in [5.4.1.2.3 -c](#).

5.4.1.2.8 Fleet Change Route / Area Qualification

- a. PICs and SICs who hold a current route qualification and subsequently change fleet in their current rank shall have their qualifications treated as follows:
 1. All route / area qualifications except NAT shall be retained upon fleet change without additional requirements.
 2. For NAT qualification, the pilot shall be required to complete the route qualification flight(s) specified in [5.4.1.2.3-c](#), with the exception that only one qualification flight is required.
 3. For a period of six months after the first commercial flight following the introduction of a new aeroplane type, the HOFO and DOT may grant an exemption to paragraph 2. above. The HOFO and DOT shall establish the NAT route qualification requirement for this period.
- b. PICs and SICs who do not hold a current NAT or ASIA qualification and subsequently change fleet in their current rank shall gain the qualifications in accordance with [5.4.1.2.3](#). Ground and flight training events shall be scheduled according to current needs.

Additionally, a PIC shall not be designated as the Commander for flights on the NAT or ASIA route / area until it is assessed that the FCM meets the respective route experience in accordance with OM-D. Until the respective route / area restriction is removed, they may be designated as the pilot relieving the Commander.

5.4.1.2.9 Aerodrome Competence Qualification

All aerodromes are categorised as Category A, B, or C, where Category A is the least demanding, according to the method detailed in OM-C, [Chapter 11](#). Prior to operating to a Category B aerodrome, the commander shall be briefed, or self-briefed, on the Category B aerodrome concerned in accordance with the instructions in the appropriate aerodrome briefing in the OM-C, or by self-briefing of the appropriate sections of the Jeppesen Route Manual if the aerodrome briefing in the OM-C is not published. Prior to operating to a Category C aerodrome, the commander shall complete the training detailed in the appropriate aerodrome briefing in the OM-C.

Aerodrome Knowledge

Aerodrome training and familiarisation shall include knowledge of:

- i. obstructions;
- ii. physical layout, lighting, approach aids, and arrival;
- iii. departure;
- iv. holding and instrument approach procedures;
- v. applicable operating minima; and
- vi. ground movement considerations.

5.4.1.2.10 Validity

The period of validity of the route or aerodrome competence qualification shall be in accordance with OM-C (normally 12 calendar months), in addition to the remainder of:

- The month of qualification; or
- The month of the latest operation on the route or to the aerodrome; or

Route or aerodrome competence qualification shall be revalidated by:

- Operating on the route or to the aerodrome within the period of validity; or
- Completing the appropriate re-qualification requirements; or
- In accordance with the OM-C, whichever is the most restrictive.

If revalidated within the final 3 calendar months of validity of the previous route or aerodrome competence qualification, the period of validity shall extend from the date of that previous route and aerodrome competence qualification.

5.4.1.3 Differences Training

(ORO.FC.125)

Every FCM must complete differences training before operating another variant of an aeroplane of the same type or another type of the same class currently operated.

Introducing a change of equipment and / or procedures on types or variants currently operated may require additional knowledge or additional training on the aircraft, or an appropriate training device, or both.

The OM-D includes specific details of differences training.

5.4.1.4 Familiarisation

(ORO.FC.125)

Familiarisation is required when operating another aeroplane of the same type.

Introducing a change of equipment and / or procedures on types or variants currently operated may require additional knowledge or additional training on the aircraft, or an appropriate training device, or both.

The OM-D includes the specific details of familiarisation training.

5.4.1.5 Recurrent Training and Checking

(ORO.FC.230)

1. Each FCM must undergo recurrent training and checking.
2. The training and checking shall be relevant to the type or class of aeroplane on which the crew member is certificated to operate.
3. Recurrent training and checking programmes for FCM are established in OM-D.
4. Recurrent training will comprise:
 - Ground and Refresher training;
 - Flight Simulator training;
 - Emergency and Safety Equipment training;
 - CRM training.
5. Recurrent checking will comprise:
 - A Licence Proficiency Check;
 - An Operator Proficiency Check;
 - A Line Check;
 - An Emergency and Safety Equipment Check.
6. Periods of validity.
 - **Licence Proficiency Check:** The period of validity of a Licence Proficiency Check shall be 12 calendar months in addition to the remainder of the month of issue. If issued within the final 3 calendar months of validity of a previous Licence Proficiency Check, the period of validity shall extend from the date of issue until 12 calendar months from the expiry date of that previous Licence Proficiency Check.
 - **Operator Proficiency Checks:** The period of validity of an Operator Proficiency Check shall be 6 calendar months in addition to the remainder of the month of issue. If issued within the final 3 calendar months of validity of a previous Operator Proficiency Check, the period of validity shall extend from the date of issue until 6 calendar months from the expiry date of that previous Operator Proficiency Check.
 -

Line Check: The period of validity of a Line Check shall be 12 calendar months in addition to the remainder of the month of issue. If issued within the final three calendar months of validity of a previous Line Check, the period of validity shall extend from the date of issue until 12 calendar months from the expiry date of that previous Line Check.

7. An FCM's knowledge and ability to perform the tasks associated with low visibility procedures including LVTO and where applicable take-off with RVR less than 150 m are trained and checked in conjunction with the normal recurrent training and proficiency checks.
8. For FCM who are qualified to operate in either pilot's seat, additional training and checking conducted from the RHS, as described in OM-D.

5.4.1.6 Long Term Absence from NTA Flying

FCM who have been absent from NTA flying duties for more than 120 days must undergo re-qualification training as described in OM-D. This training programme ensures that only those pilots who are trained and checked on all procedures which are necessary for a safe operation will be assigned to flight duty.

5.4.1.7 Command Course

The NTA Command Course is specified in OM-D.

5.4.2 Cabin Crew

They must also have passed the prescribed checks covering the training received in order to verify proficiency in carrying out safety and emergency procedures.

For qualification to operate as one of the minimum required cabin crew members, each CCM must have completed:

- Initial Safety Training;
- Conversion and Differences Training;
- Familiarisation training and familiarisation flights.

They must also have passed the prescribed checks covering the training received in order to verify proficiency in carrying out safety and emergency procedures.

Full details are contained in OM-D.

5.4.2.1 Initial Training Course

(APPENDIX 1 TO PART-CC)

Initial Safety Training and Checking shall include:

- General theoretical knowledge of aviation and aviation regulations covering all elements relevant to the duties and responsibilities required from cabin crew;
- Communication;
- Introductory course on Human Factors (HF) in aviation and CRM;
- Passenger handling and cabin surveillance;
- Aero-medical aspects and first-aid;
- Dangerous Goods in accordance with the applicable ICAO Technical Instructions;
- General security aspects in aviation;
- Fire and smoke training;
- Survival training.

Detailed content is described in OM-D.

5.4.2.2 Aeroplane Type Specific and Operator Conversion Training and Differences Training

(ORO.CC.125)

Before undertaking assigned duties, each CCM must have completed the appropriate aeroplane type specific and operator conversion training and checking and / or differences training, as specified in the OM-D.

5.4.2.2.1 Aeroplane Type Specific Training

(ORO.CC.125)

Cabin crews' first and subsequent aeroplane type-specific training and checking is as specified in OM-D. Subsequent conversion training is undertaken when changing to a different aeroplane type. Aeroplane type specific training shall cover:

- Aeroplane description;
- Safety and emergency equipment and aeroplane systems installed;
- Operation of doors and exits;
- Fire and smoke protection equipment;
- Operation of equipment related to pilot incapacitation.

Detailed content is described in OM-D.

5.4.2.2.2 Operator Conversion Training

(ORO.CC.125)

NTA operator conversion training includes:

- Description of the cabin configuration;
- Safety and emergency equipment;
- Company SOPs for CCM (normal and emergency procedures);
- Passenger handling and crowd control;
- Fire and smoke training;
- Evacuation procedures;
- Pilot incapacitation procedures;
- Applicable security requirements and procedures;
- CRM.

Detailed content is described in OM-D.

5.4.2.2.3 Differences Training

(ORO.CC.130)

Differences training must be completed before being assigned to operate on a variant of an aeroplane type currently operated, or with different equipment, equipment location, or safety procedures on currently operated aeroplane types or variants.

5.4.2.3 Familiarisation Training

(ORO.CC.135)

Following completion of aeroplane type specific and operator conversion training and prior to operating as one of the minimum number of cabin crew required, each new entrant CCM must undertake familiarisation training in the form of aeroplane visits and familiarisation flights on the type to be flown as specified in OM-D.

During familiarisation flights, the CCM should participate in uniform as part of the operating crew, but they shall be in addition to the minimum number of required cabin crew. CCM assigned to operate subsequent aeroplane types shall undertake a familiarisation flight on the type to be flown.

5.4.2.4 Senior Cabin Crew Member Training

(ORO.CC.200)

- i. The minimum requirements for nomination as Senior Cabin Crew Member are at least one year's experience as an operating CCM and the completion of appropriate training as specified in OM-D, which will emphasise:
 - Crew coordination and pre-flight briefing;
 - Review of company regulations and legal requirements;
 - Responsibilities, human factors, and crew resource management;
 - Chain of command;
 - Accident and incident reporting;
 - Flight and duty time limitations and rest requirements.
- ii. Where practicable, participation in a flight crew's Line Orientated Flying Training in a simulator will be included.

5.4.2.5 Recurrent Training

(ORO.CC.140)

CCM must undergo Recurrent Training every 12 months to ensure continued proficiency with all equipment relevant to the aeroplane types they operate. The detailed content is specified in OM-D.

Training will cover the actions assigned to each crew member in normal and emergency situations and the appropriate procedures and drills.

The period of validity of annual recurrent training and the associated check is 12 calendar months in addition to the remainder of the month of issue. If issued within the final 3 calendar months of validity of a previous check, the period of validity will extend from the date of issue until 12 calendar months from the expiry date of that previous check.

Annual recurrent training includes:

- Emergency procedures including pilot incapacitation;
- Evacuation procedures including crowd control techniques for applicable aeroplane types;
- Touch-drills for opening normal and emergency exits;
- Location and handling of emergency equipment, including oxygen systems, and the donning by each cabin crew member of life jackets, portable oxygen, and protective breathing equipment (PBE);
- Aero-medical aspects, first aid, and the contents of the first-aid kit(s);
- Stowage of articles in the cabin;
- Security procedures;
- Incident and accident review;
- CRM;

- Procedures related to aeroplane surface contamination.

Every two years, this training shall also include:

- Dangerous Goods procedures as prescribed in OM-D.

The period of validity of the two-yearly recurrent training and the associated check is 24 calendar months. If issued within the final 3 calendar months of validity of a previous check, the period of validity will extend from the date of issue until 24 calendar months from the expiry date of that previous check.

Every three years, this training will also include:

- The operation and actual opening of normal and emergency exits for passenger evacuation in an aeroplane or representative training device in the normal and emergency mode;
- Demonstration of all other exits including flight deck windows, where applicable;
- The donning and use of protective breathing equipment by each CCM in an enclosed, simulated smoke filled environment;
- Use of pyrotechnics (actual or representative devices);
- Demonstration of the use of the life-raft, if fitted;
- Actual operation in a representative training device or in the actual aeroplane of seat restraint system and practical demonstration of the oxygen system equipment relevant to pilot incapacitation;
- Practical fire and smoke training;
- Practical demonstration of the use of flight crew check list.

The period of validity of three-yearly recurrent training and the associated check is 36 calendar months in addition to the remainder of the month of issue.

5.4.2.6 Refresher Training

(ORO.CC.145)

A cabin crew member will complete a refresher training course after a period of absence from all NTA flying duties for more than six months as specified in OM-D.

5.4.2.7 Checking

Checks are prescribed for CCM after completion of:

- Initial Training;
- Conversion and Differences Training;
- Recurrent Training.

5.4.3 In-flight Service Representative

There are no regulatory training or qualification requirements for Inflight Service Representatives.

Without prejudice to the previous statement, NTA requires ISRs to complete the SEP layout for any aircraft type they will operate on, as well as other training considered relevant for their roles, by the decision of the Training Department.

5.5 Qualification Requirements for Other Operations Personnel

5.5.1 Flight Dispatcher

A Flight Dispatcher will not be assigned too operational duties, until a sufficient level of knowledge is demonstrated in all subjects below and has successfully passed the Flight Dispatch Initial Training (described in NTA-FD-001 Flight Dispatch Initial Training) On-Job-Training, and Competency Check.

- The use of NTA Manuals;
- Air Law and Operating Procedures, relevant to the Operational Control of Flights;
- Aeroplane and Airport Performance Analysis and the correct use of PDCS charts;
- Aeronautical Meteorology;
- Mass and Balance - The basics of Mass and Balance and preparation of load sheets;
- Jeppesen Airway Manual and Air Navigation;
- Awareness on the transportation of cargo and dangerous goods;
- Flight Planning tools and Flight Planning Considerations, such as MEA and MSA, Alternate Airports selection, Route selection, and Fuel Policy;
- Flight Monitoring;
- Air Traffic Flow and Capacity Management (ATFCM) and related procedures for tactical flow management;
- Communications via radio or SAT phone and general AFTM messaging;
- Special Airports procedures;
- Emergency and Abnormal Procedures; Activation of the Emergency Response Plan or Event Notification;
- De-icing / Anti-icing procedures;
- Security Awareness;
- Minimum Equipment List limitations and procedures to be followed;
- Aeroplane Indoctrination;
- Winter Operations and weather hazards.

A Flight Manager must have gained experience of at least 6 months as an assistant in the dispatching of air transport or must have completed a course of approved training prior to performing the tasks of dispatching aeroplanes.

5.5.2 Flight Dispatcher Monitor and Examiner Certification

5.5.2.1 Monitor Qualification

1. The competence to monitor Flight Dispatchers on-job training requires a Flight Dispatcher Monitor Qualification, issued by ANAC and being integrated in an ATO with a quality management system.
2. The Flight Dispatch Monitor is responsible for the supervision of internships with the purpose of issuing and revalidating a Flight Dispatcher Licence(s).
3. The candidate for a Monitor Qualification must hold a Flight Dispatcher Licence.
4. Additionally, the candidate shall:
 - a. Hold a Flight Dispatcher Licence for a period of no less than three years;
 - b. Hold a CCP (*Certificado de Competências Pedagógicas*), issued according to the Portuguese law; or
 - c. Have an approval to conduct on-job training, as per item 1, and / or supervise internships, as per item 2;
 - d. Have dispatched a minimum of 120 flights after obtaining a Flight Dispatcher Licence, 12 of which must have occurred in the 12 months preceding the application of Monitor Qualification;
 - e. Has supervised the internship of a Flight Dispatcher candidate, on-job training, under the supervision of an approved Flight Dispatcher Monitor.
5. The Monitor Qualification is valid for a period of three years and is renewable if the applicant is integrated into an ATO and listed as a qualified monitor in the respective manuals as approved by ANAC.
 - a. Where the applicant is not integrated in an ATO, they must qualify for the renewal through an examination in an ATO;
6. The request for a Monitor Qualification shall be submitted to ANAC by the candidate or their representative, together with the following documentation:
 - a. Copy of proof of identity of the candidate, or by providing this proof in person;
 - b. Any documents evidencing the requirements laid in point 4. above.
7. The validity of the Monitor Qualification is conditional to the validity of the Flight Dispatcher licence.

5.5.2.2 Examiner Certification

1. According to Decree Law 840/2010, Article 16, it is the responsibility of an examiner authorised by ANAC to evaluate competency checks as per Article 9 of the same Decree Law.
2. ANAC may grant Examiner certification to a licensed Flight Dispatcher, who has proven their competence and ability performing their job, and fulfils the following requirements:
 - a. Holds a valid Flight Dispatcher Licence;
 - b. Holds a Monitor certification for at least 3 years.
3. Examiner certification is valid for a 3-year period and can be renewable if the candidate fulfils the pre-requisites for an initial certification, and:
 - a. Has at least evaluated one competency check per year during the validity of their Examiner certification.
 - b. During the period of validity of their certification, was evaluated by an ANAC examiner, or by an examiner appointed by ANAC.
4. For the purpose of item b. of the previous line 3., ANAC may, in case of force major, nominate, as a replacement of an Inspector or an Examiner, a Licensed Flight Dispatcher whose competence is recognised.

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Chapter 6 - Crew Health Precautions

6.1 General

(CAT.GEN.MPA.100)

Impairment of crew performance as a result of illness, fatigue, medication, alcohol, or other drugs may have a direct impact upon flight safety. All crew members have a personal responsibility to maintain a healthy lifestyle and not to report for duty unless they are fit to perform their assigned duties.

This chapter provides information and guidance on a range of issues that may affect crew's health, as well as guidance on the company's policies and procedures on matters of crew health.

- A crew member shall not perform duties on an aeroplane if they are unfit due to injury, fatigue, medication, sickness, or other similar cases and if they are in any doubt about being able to accomplish their assigned duties.
- A crew member who believes that another crew member is unfit to operate must raise their concerns with the NJE Chief Pilot, FOSM team, or Crew Support (possibly with the Commander, if deemed appropriate).

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6.1.1 Decrease in Medical Fitness

Should a crew member be in any doubt about their fitness to carry out flight duties, they shall seek the advice of a medical doctor as soon as possible. Crews are to follow the procedures below and / or in [6.1.4.1](#) when assessing the requirement for an additional medical examination.

At the conclusion of any illness or pregnancy as stated below, or at the request of NTA, the crew member is responsible for obtaining a certificate issued by an Aviation Medical Examiner (AME), or OHMP for cabin crew, clarifying fitness for flight duty and ensuring it is received by NTA.

Crew members shall not exercise the privileges of their licence or CCATT at any time when they:

- Are aware of any decrease in their medical fitness which might render them unable to safely exercise those privileges;
- Take or use any prescribed or non-prescribed medication which is likely to interfere with the safe exercise of the privileges of the applicable licence;
- Receive any medical, surgical, or other treatment that is likely to interfere with flight safety.

In addition, licence and CCATT holders shall, without undue delay, seek aero-medical advice when they:

- Have undergone a surgical operation or invasive procedure;
- Have commenced the regular use of any medication;
- Have suffered any significant personal injury involving incapacity to function as a member of the flight crew;
- Have been suffering from any significant illness involving incapacity to function as a member of the flight crew for 21 days;
- Are pregnant;
- Have been admitted to hospital or medical clinic;
- First require correcting lenses and / or spectacles.

In the case of injury or illness, the medical certificate may be re-issued by an AME, and in consultation with the issuing authority, should the crew member be deemed by the AME to be medically fit to resume flight duties.

In the case of pregnancy, the pregnant crew member shall inform the company and the issuing authority of her condition in writing as soon as she is aware of her condition. The pregnant crew member's medical certificate shall be deemed to be suspended upon confirmation of the pregnancy. The Authority may lift the suspension of the medical certificate subject to such conditions as it thinks fit and may approve certification of pregnant crew members up to 26 weeks of gestation.

After the pregnancy has ended, the crew member may resume flying duties once the suspension of her medical certificate has been lifted.

6.1.2 Regular Medical Examinations

Crew members shall be physically and mentally fit to safely exercise the privileges of their applicable licence or CCATT. In order to exercise these privileges, they shall hold a valid medical certificate issued in accordance with PART-MED appropriate to the privileges of the licence or CCATT.

Contractually, it is the personal responsibility of the crew member to maintain a valid medical certificate. Medical certificates can be renewed at any time from 45 days before the expiry date to the expiry date itself. Should a crew member allow their medical certificate to lapse, they are ineligible for flying duties until such time as a valid medical certificate is obtained and is confirmed as received by NTA. NTA reserves the right to adjust the roster pattern of crew members who are unable to perform any duty resulting from the lack of a valid medical certificate.

6.1.3 Compulsory Additional Examinations

In circumstances involving any of the following diseases, irrespective of the duration of illness, crew members shall subject themselves to a medical assessment with an AME, or OHMP for cabin crew, that certifies their fitness for flight duties, prior to further duties:

- All organic heart diseases;
- Conditions resulting in sudden unconsciousness;
- All diseases of the brain, the central nervous system, or mental illness;
- All diseases which, after the expiry of the notification of sickness, still require treatment with further medication, or cause the crew member to question their ability to work;
- Admission to hospital or clinic for more than 12 hours;
- After any surgical procedure requiring subsequent time off work;
- After suffering from any infectious disease treated with antibiotics for a period of time in excess of 2 weeks;
- After treatment for any addiction;
- After suffering from any infectious disease which affects fitness to work or subject to health regulations involving quarantine;
- After suffering a smoke / fume incident that required medical intervention (see [6.3 Smoke / Contaminated Air / Fumes](#)). In such instances, crew members must be certified fit to fly before their next tour;
- After suffering a laser attack that required medical intervention (see [6.4 Laser Beam Exposure Procedures](#)). In such instances, crew members must be certified fit to fly before their next tour.

6.1.4 Notification of Sickness

Crew members shall inform NTA without delay in the event of illness, or after having regained fitness for duty after previous illness, in accordance with the Crew Absence Policy in force.

To enable reasonable planning, anticipated changes in the course of an illness (for example, extension of the notification of sickness beyond the last assumed day) should be notified as soon as possible, using the same method above mentioned. The Crew Absence Policy in force is also applicable for the submission of doctor's certificates or for notification of fitness to fly to NTA before returning to duty.

6.1.4.1 Sickness During a Duty Period

In case of sickness during a period of duty, the affected crew member shall immediately notify the Company in accordance with the current Crew Absence Policy.

The NJE Chief Pilot and the respective team, namely the FOSM team and Crew Support, are responsible for arranging medical treatment if necessary. They must also coordinate with HR and MedAire to ensure payment for medical services or any required evacuation and make any transportation plans back to the gateway for the affected crew member, taking into consideration the crew member's medical condition.

6.1.5 Psychoactive Substances

(CAT.GEN.MPA.170) (AMC1 CAT.GEN.MPA.100(c)(1))

Safety-critical staff, including flight and cabin crew, shall not perform any duties if under the influence of, or dependent on psychoactive substances.

A crew member shall not:

- Consume or be under the influence of psychoactive substances during any duty period;
- Consume or be under the influence of psychoactive substances when in uniform;
- Consume alcoholic beverages less than 10 hours prior to the specified reporting time for a flight duty period or the commencement of standby;
- Consume alcoholic beverages in such quantity that results in a breath alcohol concentration in excess of 0.2 grams of blood alcohol concentration (BAC) per litre of blood prior to the specified show time for flight duty or commencement of standby duty as depicted in the Psychoactive Substances Misuse Policy, notwithstanding being in respect of the previous paragraph.

The requirements above do not apply to drugs, medication or pharmaceutical preparations prescribed and approved by an AME.

Any crew member identified as breaching the above requirements will be removed from all duties and subject to the respective procedures in accordance with the Psychoactive Substances Misuse Policy in force.

NTA is committed to offering support to crew members in accordance with the Crew Support Programme. All Crew Members are encouraged to report any psychoactive substance dependency or misuse to the appropriate channels set out in the referred policy.

6.1.6 Medicine(s) and / or Pharmaceutical Preparations

For the purposes of this chapter, the term 'medicine(s)' is deemed to include medicine(s) and / or any pharmaceutical preparations.

Before using any medicine(s), crew members are to obtain advice from an AME or a MedAire Doctor on the effects of said medicine(s) on the ability of the crew member to perform their duties. Crew members shall not report for duty while under the influence of medicine(s) that might impair their ability to undertake their duties.

Many medications, including prescription or 'over the counter' medicines, anti-depressants, and 'alternative' medications, such as herbal medicines, may impair performance. In addition, in many instances, the condition for which medication is taken may itself impair performance.

A crew member shall not report for duty whilst taking any medication unless they have taken the advice of an AME or a MedAire Doctor.

Additional information on widely used medicines and a description of their compatibility with performing aviation safety related duties, is available on EASA list document 'Use of medication - Guidance for pilots, ATCOs and Cabin Crew Members'.

6.1.7 Blood and Bone Marrow Donations

(CAT.GEN.MPA.100(c)(2))

If a crew member acts as a blood donor, they shall not report for any FDP or period of stand-by for at least 24 hours following the donation.

If a crew member acts as a bone marrow donor, once they commence pre-donation treatment, they should inform their AME and will be assessed as unfit. Once fully recovered from the donation the crew member should contact their AME for advice on fitness to return to operational duties.

6.1.8 Immunisation

NTA strongly encourages all crew members to be suitably protected by way of immunisations when on company duty. Crew members have a personal responsibility to ensure they are suitably protected, as per the following guidelines, to allow NTA to employ them without restriction in accordance with their contracts of employment.

6.1.8.1 Immunisation Requirements

Crew members are to be aware that there are numerous dangerous diseases against which vaccination is not possible. General guidance for health-conscious behaviour when travelling is as follows:

- Observe strict discipline when consuming food and beverages, particularly with regard to seals on bottled beverages.
- Do not bathe in stagnant water.
- In areas of bush and / or jungle, wear long-sleeved shirts and long trousers to minimise the risk of bites by disease-carrying insects or by poisonous insects and animals.
- Use an effective insect repellent in areas affected by mosquitoes. In general, the higher the concentration of the active ingredient in a product the longer it lasts.
- Use of air conditioned environments when in mosquito-affected areas whenever possible.
- Refer to World Health Organisation (WHO) for guidance on pandemic situations and recommended preventive behaviours in high risk areas.

All crew members (except those on maternity leave) are to comply with the following:

- Crew members scheduled for duty abroad must satisfy any requirement(s) for immunisation(s), to have themselves immunised in time, and to be able to produce, on demand, the appropriate WHO-approved certificate of immunisation or re-immunisation.

NTA requires all crew members to have a valid certificate of immunisation or exemption from immunisation for Yellow Fever.

- In addition, NTA recommends crew members are immunised against the following:
 - Cholera;
 - Typhoid;
 - Poliomyelitis;
 - Tetanus;
 - Hepatitis A;
 - Hepatitis B;
 - COVID-19.
- Crew members will arrange for any immunisations and / or boosters to take place a minimum of 48 hours before the reporting time for any flight duty or stand-by duty.
- In the event of an adverse reaction to an immunisation, the crew member shall obtain medical advice in a timely fashion to ascertain any possible impairment of their fitness for flight duty.

Crew members scheduled for duties in malarial zones shall be offered the means to obtain and employ appropriate medication as prescribed by a medical doctor, in time to be covered for the period of their duty. It should be noted that risk of infection is low if advice offered in [6.1.8.1](#) is followed.

Crew members are responsible for arranging immunisation appointments in their own time. NTA will reimburse costs of immunisations and / or anti-malarial medication (but not travel). Claims should be entered via Concur using the cost definition 'Medical (NJE)'.

6.1.8.2 Company Monitoring of Immunisations

NTA will record and monitor immunisation records only if required, for business purposes and visibility on availability from Crew Members. NTA will treat and store all the data in accordance with General Data Protection Regulation (GDPR).

Each crew member is responsible to ensure that valid yellow fever or any other required immunisation currency is maintained at all times in accordance with the employment contract. Should a crew member fail to renew this currency before validity lapses, the crew member will be removed from flying duties. NTA reserves the right to adjust the roster pattern of crew members who are unable to perform any duty resulting from the lack of a valid yellow fever or any other required immunisation certificate or valid certificate of exemption.

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6.1.9 Crew Meals and Precautions

Crews are paid a Per Diem allowance in order to provide for day-to-day expenses, including meals, when on NTA duty. By virtue of the unique nature of NTA's business, the company wishes crew members to remain either on board the aeroplane or within immediate reach of the aeroplane during FDPs and / or hotspare duties, such that operational flexibility can be maintained. Accordingly, the company chooses to provide meals to crew members specifically to ensure crews' continued availability for flight duty.

The Logistics Support department is not authorised to deviate from the Crew Meal Policy. Any request for deviation shall be made directly to the FOSM team (NJE Chief Pilot or DO if the FOSM team is not available). If the FOSM team (NJE Chief Pilot or DO if the FOSM team is not available) agrees to the request, they will liaise directly with the Logistics Support department.

6.1.9.1 Meal Precautions

In the interest of safety, crew members are urged not to partake of identical dishes prepared by the same caterer, restaurant, or person, or meals containing shellfish, for a period spanning 3 hours before commencing flight duties until the end of the current FDP. In-flight, Flight Crew at the controls shall eat their crew meals at different times.

6.1.10 Sleep and Rest

Flight safety requires crew members to receive regular and sufficient sleep and that they are well rested when commencing any FDP or period of stand-by. Legal provisions laid down in [Chapter 7 - Flight Duty Time Limitations](#) prescribe maximum duty and minimum rest times accordingly.

All crew members are expected to use their rest periods to relax and regain full fitness for the next allocated duty period.

6.1.11 Surgical Procedures and / or Medical Intervention

Crew members planning or requiring medical treatment are required to obtain information as to the implications of the proposed operation / intervention with their AME and report the possible consequences and restrictions, in terms of their availability for duty, to Crew Support

Following a surgical operation or intervention, the crew member shall inform Crew Support of their fitness to perform duties and detail any restrictions. Crew Support will relay this information to Crew Resource Planning and the NJE Chief Pilot.

6.1.12 Vision

Any flight crew member required to wear corrective spectacles or lenses in accordance with their medical certificate shall carry a spare set of spectacles at all times when exercising the privileges of their licence and said spare set of spectacles shall be readily available.

6.1.13 Deep Diving

(CAT.GEN.MPA.100(c)(2))

Crew members who undertake activities including diving with the assistance of compressed air bottles (SCUBA), irrespective of the depth of water concerned, shall not report for any FDP or commence a period of stand-by less than 48 hours after completing such activity.

6.2 International Health Regulations

The WHO publishes regulations and advice to ensure maximum security against the international spread of disease with minimum interference to world traffic. These publications are regularly updated and the most recent advice (including the latest edition of the International Health Regulations) can be accessed via the WHO website at <http://www.who.int/en/>.

Health requirements for individual nation states can be found in the Jeppesen Airway Manual, in the Entry Requirement - State Rules and Procedures section.

6.3 Contaminated Air (Smoke, Fumes or other)

Symptoms

There is no common pattern of symptoms that can be readily identified as characteristic of 'cabin air quality incidents' and the symptoms are not in themselves suggestive of any specific form of chemical toxicity (source: UK CAA).

If a crew member suspects or detects abnormal smoke, contaminated air, or fumes, they are to deal with the situation in accordance with Safety Equipment and Procedures Manual.

NTA has adopted UK CAA advice on the procedures to be followed after crews experience a smoke / fumes incident. The SEP Manual directs the use of onboard oxygen and this should be maintained at high concentration for 20 minutes.

Flight and cabin crew members shall take the following post-flight actions following a smoke / fumes incident:

- The Commander shall inform AMCC, the NJE Chief Pilot or FOSM team without delay and then review the incident in consultation with the crew and passengers (if on a revenue flight);
- The Commander's review / debrief shall discuss the type of smell or visible smoke experienced and if its appearance could be linked to a specific event (i.e. turning on a piece of equipment), thereby narrowing the focus for subsequent engineering investigation;
- Determine whether any crew member or passenger experienced any symptoms, felt unwell, or whether their performance was adversely affected;
- If the crew were symptom-free following the 20 minutes' use of oxygen, the Commander shall notify the NJE Chief Pilot or FOSM team, and the crew will be subject to a minimum of 12 hours rest before the next duty period may commence.

If any of the crew members experience symptoms following the incident, the FOSM team or the NJE Chief Pilot will assist the Commander in arranging medical intervention for the crew in accordance with [6.1.4.1](#).

Additionally:

- The Commander (or, in their absence, any crew member) shall recommend to the passengers that they undergo a medical assessment.
- Crew members will be grounded for a minimum of 24 hours. The NJE Chief Pilot, with the assistance of the FOSM team and Crew Support, will implement a recovery plan in accordance with this restriction.

- Crew members shall be offered examination by a medical doctor and shall inform the doctor that they have been exposed to smoke, contaminated air, or fumes. Crew members may be requested to submit blood samples as part of the examination.

Following any such incident, the crew may rest at suitable accommodation as defined in [Chapter 7- Flight Duty Time Limitations](#).

Crew members may only be released to fly or for other duties after a minimum period of 24 hours and once the individual feels fit to operate.

A NetJets Safety report shall be submitted by the Commander in accordance with [OM-A 11.6](#). The report shall detail as much information as possible to assist NTA in diagnosing the cause of the fault, such that expeditious and thorough rectification work can take place.

6.4 Laser Beam Exposure Procedures

Laser attacks on aeroplanes started some years ago when laser pointers became readily available and have now escalated such that thousands of events occur each year. The vast majority of attacks take place when aeroplanes are over or near large centres of population. In addition, ATC towers have also been targeted by lasers.

According to a study by the FAA, the risk of damage to the eyes due to laser exposure is very limited. For example, if exposed to a normal laser pointer, damages may occur only if exposed for several seconds on one spot at a distance of less than 50 ft.

The main threat posed by a laser attack on an aeroplane is from the reaction of the pilot to the laser light. Most laser are red or green coloured. The human eye is much more sensitive to green light so, for an equivalent laser output power, the green light appears to be much brighter.

The immediate effects of a laser attack are distraction and anxiety. The following are characteristics of a laser attack:

- It is always very sudden;
- It is always very bright;
- It is distracting;
- The glare may obscure many (if not all) instruments;
- Night vision may be disrupted;
- Even if the eyes are not directly illuminated, there will be a temptation to look into the beam;
- Laser illumination can result in minor an transient visual impairment, such as a retinal after-image remaining visible and / or camera flash-type blindness. Usually, these symptoms subside after a period of time, provided the individual does not look at the beam.

6.4.1 Actions on Suspecting a Laser Attack

During and Immediately After an Attack

In the event that you believe your aeroplane is being deliberately illuminated (attacked) by a laser, the following immediate actions are recommended:

- Look away from the beam and shield eyes to the maximum extent possible;
- Inform ATC as soon as possible and, in particular, if a decision has been made to diverge from the cleared flight path;
- Consider re-engaging the autopilot (if disengaged), or handing control of the aeroplane to the other pilot (if there is another pilot in the cockpit and they are less affected by the attack);
- If the aeroplane is on the approach, consider executing a missed approach;
- Turn up cockpit lighting;
- Avoid rubbing eyes to reduce the potential for corneal abrasion (see the instructions that accompany the self-assessment tool).

After the Attack

- As soon as possible after the attack, provide ATC with as much detail as possible concerning the event so that law enforcement organisations can take appropriate action. If possible, include a description of the location of the source of the laser beam, its direction, colour, and length of exposure, but do not attempt to find the light source by staring at the laser.
- Once on the ground, use the NJE self-assessment guide ([6.4.3 Self-assessment Guide for Aviation Laser Exposure](#)) to examine eyesight and, if necessary, seek assistance from an optometrist or ophthalmologist.
- If, after using the self-assessment guide, it is necessary to obtain medical assistance, then follow [Medical Intervention Following a Laser Attack](#).
- If, after using the NJE self-assessment guide, it is not necessary to obtain medical assistance (permanent eye damage is not known or is extremely unlikely to occur in this situation), the affected flight crew is / are not restricted as the experienced exposure is unlikely to have an operational impact or be amenable to treatment. In this instance, the crew may continue their duties.
- Submit a NetJets safety report.
- The form 'Suspected Laser Beam Incident and Exposure Questionnaire' shall be completed and returned to the Safety department.

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6.4.2 Medical Intervention Following a Laser Attack

If, after using the NJE self-assessment guide, the affected crew member(s) is recommended to consult an eye specialist, the following procedure is to be followed:

- The affected crew member(s) is (are) grounded for a minimum of 24 hours.
- The Commander is to inform Scheduling without delay to allow a recovery plan to be implemented.
- The Commander shall contact the FOSM team or the NJE Chief Pilot to obtain assistance in coordinating medical intervention.

NJE suggests that you consult an eye specialist as there may be a possibility of eye damage. However, if you do not have any symptoms, there is no urgency.

6.4.3 Self-assessment Guide for Aviation Laser Exposure

The following guide is designed to help crew members who have been exposed to a laser beam in making a decision if to consult or not an optometrist or an ophthalmologist.

If you have been exposed to laser and experience any of the following health issues, then please follow the procedure described in [6.4.2 Medical Intervention Following a Laser Attack](#) to arrange an appointment with an eye specialist:

- Eye problems: swelling, pain, itching, watering, discharge, dryness, or redness of the eye.
- Visual disturbance: blurring, black spot, trouble reading, loss of peripheral vision, floaters, halos, poor night vision, sensitivity to light.

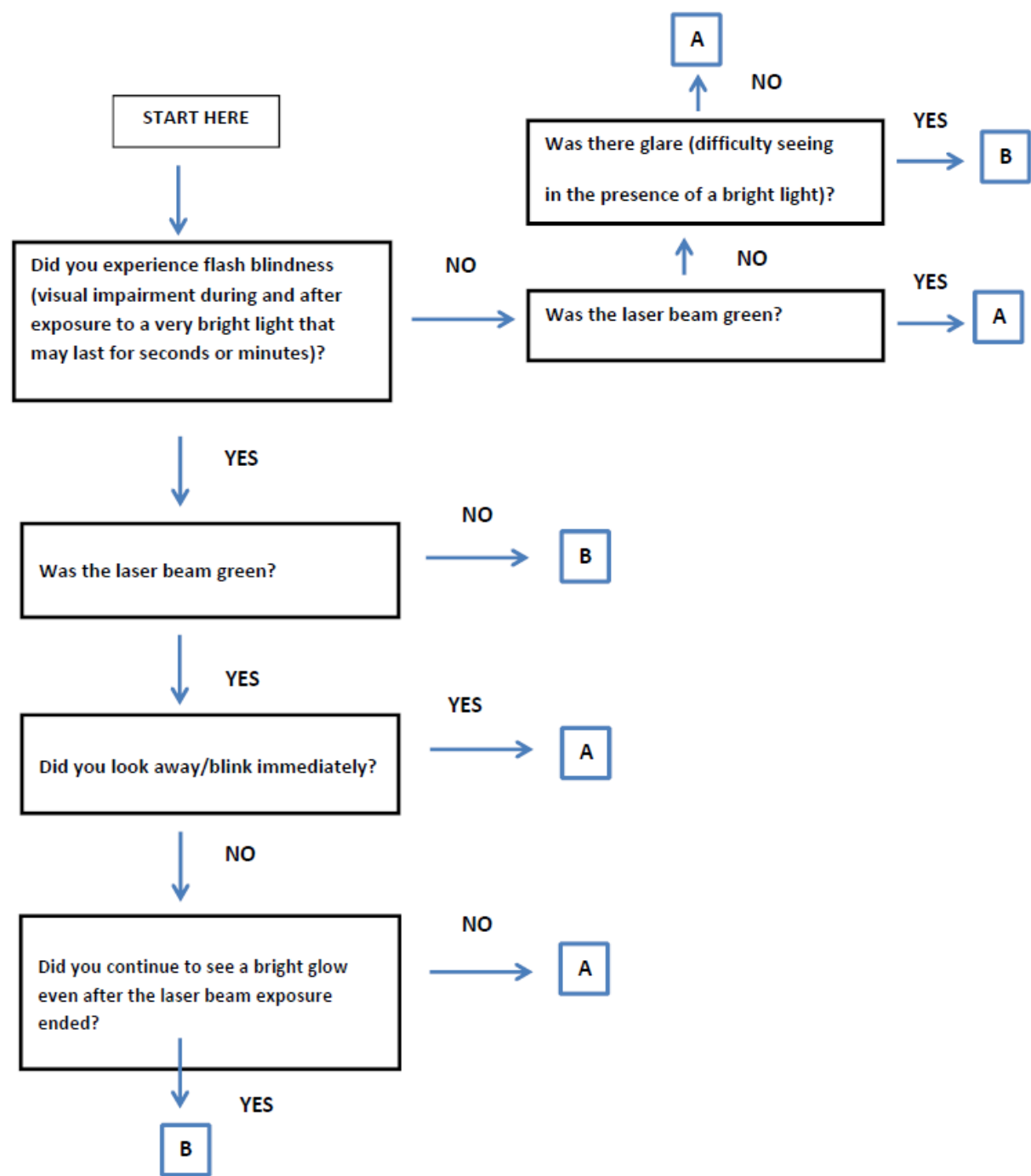
These symptoms may not appear until hours after the incident and may not be related directly to laser exposure, but could reflect other eye issues perhaps not previously noticed.

Additionally, if there is an indication that the laser was high power and capable of causing eye damage, NJE strongly suggests that you consult an eye specialist.

Advise the specialist that the evaluation should include ophthalmoscopy, visual acuity testing, and central visual field testing with the Amsler Grid.

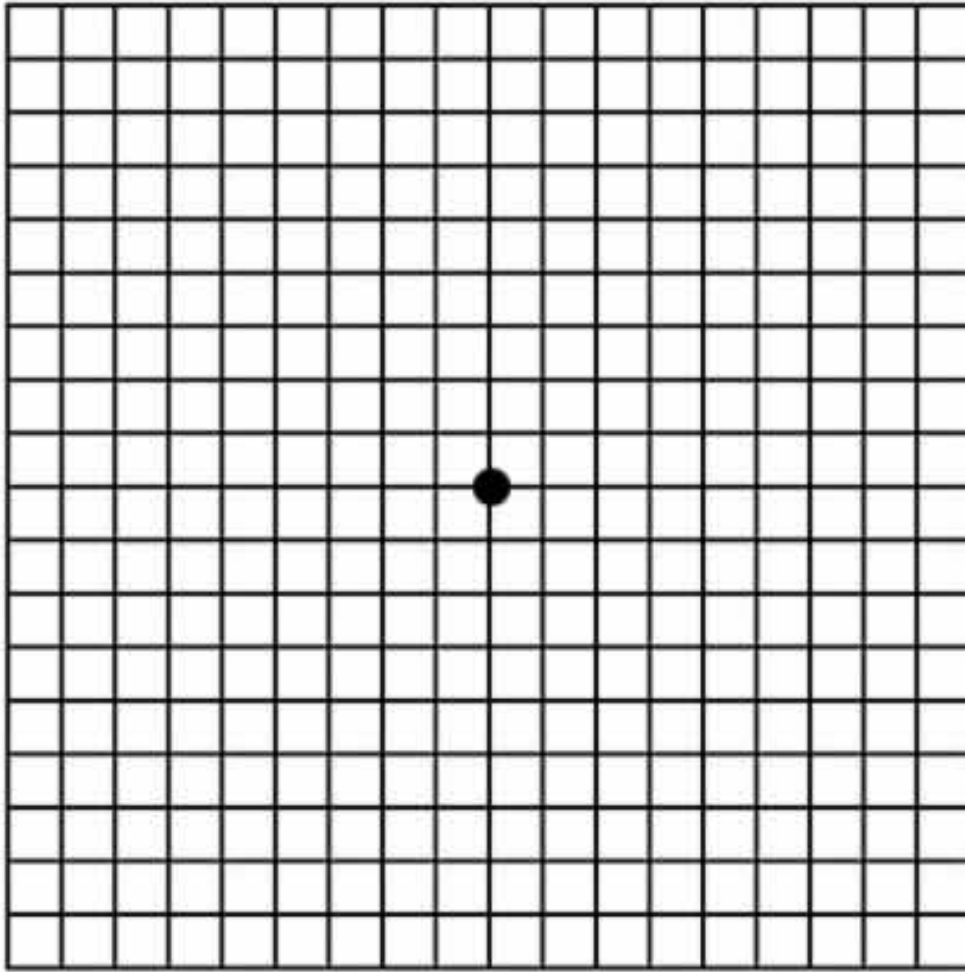
In any other case, please do the two tests that you can find on the following pages.

TEST 1



A	Permanent eye damage is not known or is extremely unlikely to occur in this situation.
B	NJE suggests that you consult an eye specialist, as there may be a possibility of eye damage. However, if you do not have any symptoms, there is no with urgency.

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TEST 2: AMSLER GRID

Use this grid in dimensions 10 cm x 10 cm. While viewing the grid from 30 cm in front of your eyes, please test one eye at a time to answer the following questions:

1. Can you see a dot in the centre of the grid?
2. While looking at the centre dot, can you see all four sides and corners of the grid?
3. While looking at the centre dot, do all lines appear straight with no distortions or blank or faded areas?

If you answered NO to any of the above questions, then you may wish to remove yourself from flying as soon as it is safe to do so, contact the FOSM team or the NJE Chief Pilot, and consult an eye specialist.

6.5 Traumatic Events

Any event that could cause fear of death or injury should be considered a potentially traumatising event. Emotionally traumatised individuals can have varying levels of reaction to the event dependant on severity and duration. Furthermore, a traumatised individual shall not be placed in a situation where clear thinking and reaction time are important for their own well-being and the well-being of others.

Following an abnormal event during flight operations, there is the likelihood of an occurrence of stress for one or more crew members. As such, the management of this risk is important to ensure minimal time away from work and safety of flight operations.

It is important to remember that not all crew members will react the same way to the same event. For example, one pilot involved in an in-flight shutdown event may have previous experience and feel much more comfortable than their fellow crew member. Additionally, the onset of stress reaction can vary significantly. Some people react immediately to stressful situations; others may have a delayed reaction; others have no reaction.

Often, the focus following these types of events is solely on the recovery of the revenue flight, with less emphasis on the crew's well-being. To ensure that the highest level of safety is maintained, crew members involved in traumatic events are not required to immediately resume flight duties and should resort to the NJE Chief Pilot or FOSM team in order to coordinate a possible change in duty that provides time to recover. In some cases, other crew movement activities, such as by airline, may need to be reconsidered.

Regardless of whether there is any removal from flight duty or not, if the crew is to continue to fly together, they shall perform a full debrief of the event. These events include, but are not limited to the following:

- An accident or incident that requires a declaration of an emergency situation to Air Traffic Control by the flight crew;
- Serious injury, illness, or death of any passenger or member of the crew;
- An engine failure, un-commanded and / or commanded shutdown in flight;
- A multiple system failure with no clear abnormal procedure in which an emergency was declared;
- Actual or perceived threat of loss of life (e.g. near mid-air collision or runway incursion);
- Severe turbulence or wake turbulence encounter in which aeroplane control is momentarily lost and / or crew member or passenger injury occurs;
- In-flight fire or perceived fire or smoke;
- Any event involving an emergency evacuation;
- Any event after which the crew member requests to be removed from duty for the remainder of the day or tour.

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6.6 Critical Incident Stress Management (CISM)

6.6.1 Passengers Exposed to High Emotional Impact Events

An individual's reactions to critical in-flight situations (see event list in [6.5 Traumatic Events](#)) are driven by emotions and may result in post-traumatic stress.

It is the Commander's responsibility to assess and manage the situation. Whenever they feel the need to offer assistance to passengers and / or colleagues following an incident, they are urged to notify the NJE Chief Pilot or the FOSM team as soon as possible (also consider the Hotline if needed). They will then activate the Event Notification process, which may subsequently lead to those affected being offered CISM assistance. The CISM team can offer immediate assistance, normally in the individual's mother language. This process allows NTA to follow up on the mental health of the individual and show compassion.

The NJE Chief Pilot or FOSM team shall contact the Owner Services manager on duty (if applicable), and together, they shall decide if CISM support should be offered. If so, the team managing the case shall then contact the CISM hotline and request a peer, preferably speaking the language of the affected passenger. After having the name and contact details from the peer, they will notify the Owner Services manager on duty who then offers the CISM measures to the affected passengers. It is then the responsibility and choice of the affected passengers whether or not to continue.

6.6.2 Operational Debriefing

The commander shall conduct an 'Operational Debriefing' with all crew members concerned after any incident, which may have led to differing perceptions about the event.

Such a non-routine event is defined by 3 criteria:

- It is not observed during daily operation;
- It leads to a significant deviation from normal work-routine;
- A differing perception between crew members is likely.

An 'Operational Debriefing' should take place as soon as possible after the incident and be as short as possible, covering only the factual side of the event. Uninvolved persons should not take part or listen in. Examples are diversion, go-around, etc. During this debriefing, the following questions shall be dealt with:

- What are the technical procedural facts of the event?
- Are there differing perceptions among the crew members?
- Is there a need for CISM debriefing measures?
- How do we proceed?

An 'Operational Debriefing' which lasts longer than 15 minutes or involves strong physical, behavioural, and / or emotional reactions among the crew members indicates that a CISM-debriefing may be necessary.

6.6.3 Critical Incident Stress Debriefing for Crew Members

CISM-debriefing measures should be conducted after an 'Operational Debriefing' whenever any crew member has been exposed to an extreme level of stress. Stress levels can be extremely high in situations which are perceived as a high threat to life or health.

Examples of situations that may generate an extreme level of stress are:

- Extreme turbulence;
- Engine failure, rejected T/O;
- Near collision;
- Fire or smoke on board, severe bird-strike;
- Emergency landing and / or evacuation (prepared or actual);
- Death on board or during layover;
- Hijacking;
- In-flight violence, sex crimes, kidnapping, or hold-ups;
- Natural disasters.

Immediate reactions may be lack of attention, decrease of information processing and performance, inability to concentrate, confusion, symptoms of shock, and / or withdrawal. Crew members and their family members may require individual assistance in dealing with the physical and emotional impacts of extremely high stress levels after involvement in an incident or accident. NJE has established a CISM-team to provide such assistance to crew members. Specially-trained CISM Coordinators define and initiate the appropriate intervention measures. NJE is responsible for the provision of CISM qualified personnel to conduct CISM interventions. CISM debriefing must be performed by trained and assigned CISM personnel only.

Contact:

CISM website: www.stiftung-mayday.de/en/.

■ CISM contact number: +49 151 2240 7703

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Chapter 7 - Flight Duty Time Limitations

7.1 Flight Duty Time Limitations and Rest Requirements

This chapter describes the duty, flight duty, and rest time limitations / requirements for crew members before commencing any flight on an NTA aeroplane.

Pursuant to paragraph 2 of Article 8 of Cover Regulation to Commission Regulation (EU) No 965/2012 (derogation for air taxi), this chapter is fully compliant with Subpart Q of Annex III to Regulation (EEC) No 3922/91 (EU-OPS) and, where required, supplemented by Portuguese Decree Law no. 25/2022, which has retained parts of Portuguese Decree Law no. 139/2004.

According to the Principle of Precedence of Community Law, in any conflicting provision, EU-OPS Subpart Q prevails over the Portuguese Law.

7.1.1 Company Responsibility

(EU-OPS)

The company shall not schedule or request a crew member to perform any duty contradictory to the procedures and limitations stipulated in this chapter. If any company employee becomes aware or suspects that any flight is operated contradictory to the procedures and limitations stipulated in this chapter, they shall immediately inform the DO.

7.1.1.1 Fatigue Risk Management

NTA has a very complex on-demand operation with crew members living in different time zones across Europe. This presents a challenging task for fatigue management. Consequently, managing crew fatigue risk has become an important component of NTA's SMS. The NTA Fatigue Risk Management System (FRMS) is an evidence-based, data-driven system with reactive, proactive, and predictive elements that are used to continuously monitor and control fatigue risk to a level that is as low as reasonably practical. The FRMS provides a mechanism by which appropriate measures, supporting procedures, and training ensure that crew members are not subjected to unacceptable levels of work-related fatigue. Employees also have a personal obligation to minimise fatigue so that they are fit for duty, and shall not perform any duty if they consider their fatigue level to be unacceptable. There are currently no EASA regulatory requirements for air taxi operators to have an FRMS. Nevertheless, the one that NTA has implemented meets or exceeds industry best practices.

With the increasing focus on fatigue issues, NTA has deployed a computerised bio-mathematical model to assist in the analysis of fatigue risks. The System for Aircrew Fatigue Evaluation (SAFE) is a computer programme that includes a suite of algorithms that allow a range of factors influencing alertness in aircrew to be evaluated. It has been validated for use by the airline industry from a huge data set that has been compiled over decades of studies with commercial aircrew. It is used by many regulators including UK Civil Aviation Authority, CASA (Australia), CAAS (Singapore), and GCAA (Dubai). Integrating this software into our FRMS processes and scheduling software allows us to proactively target both systemic and individual fatigue risks.

NetJets supports a 'Just Culture' and guarantees that no disciplinary action will be taken as a result of reporting fatigued, a fatigue related event, or a potential fatigue contributing factor, unless there is clear evidence of wilful misconduct or gross negligence.

7.1.2 Crew Member Responsibility*(EU-OPS)*

A crew member shall not operate an aeroplane if they know, or reasonably believe, that they are suffering from, or are likely to suffer from, fatigue or any condition that makes them unfit for duty to the extent that flight safety may be endangered.

Crew members shall make optimum use of the opportunities and facilities for rest provided and shall properly plan and use their rest periods.

A crew member shall not accept or perform, nor allow another crew member to accept or perform, any duty that is contrary to the conditions stipulated in this chapter.

When notified that a Crew Alertness Risk Assessment (CARA) is required for a given duty period, a crew member shall complete the CARA no later than 30 minutes after reporting time for the applicable duty period, and if not the Commander, shall notify the Commander that the CARA has been completed.

7.1.3 Definitions*(EU-OPS)*

ANAC	Autoridade Nacional de Aviação Civil, the Portuguese National Aviation Authority.
Augmented flight crew	A flight crew which comprises more than the minimum number required for the operation of the aeroplane and in which each flight crew member can leave their post and be replaced by another appropriately qualified flight crew member. For a flight crew member to be considered part of an augmented flight crew, they must be part of an augmented flight crew since the moment of show-time of the augmented flight crew.
Basic FDP	An FDP planned in accordance with the limitations shown in 7.2.10.1 , Table 1 .
Block Time	The time between an aeroplane first moving from its parking place for the purpose of taking off until it comes to rest on the designated parking position and all engines or propellers are stopped.
Break	A period free of all duties, which counts as duty, being less than a rest period. The crew member shall be contactable by phone during this period.
CARA	Crew Alertness Risk Assessment: A tactical real-time risk assessment conducted before the scheduled flight duty period if predictive alertness thresholds suggest a potential fatigue risk. This assessment integrates data from predictive biomathematical model and crew feedback to determine whether the duty plan should be maintained or adjusted to mitigate fatigue risk.
Day	A twenty four-hour period, starting at 00:00 local time.
Day Off	A twenty-four-hour period free of any kind of duty for the crew member.
Duty	Any task that a crew member is required to carry out associated with the business of an AOC holder.

Duty Period (DP)	A period which starts when a crew member is required by an operator to commence a duty and ends when the crew member is free from all duties.
Extended FDP	An FDP where the normal maximum daily FDP has been extended (subject to the provisions of 7.2.10.1.1) by one hour during the planning phase.
Extended weekly rest period	A seventy-two-hour period, including two consecutive local nights, free from any duty for the crew member.
Flight Duty Period (FDP)	A period during which a person operates in an aeroplane as a member of its crew. The FDP starts at the briefed Show-time (or briefed Reporting Time if the operating sector is preceded by positioning within the same duty period); it finishes at the end of the last flight (blocks-on) on which they are an operating crew member or at the end of an assigned Hotspare Duty, whichever is later.
Flight Series	A number of flights between which no rest periods has taken place.
Gateway	The location, permanent or temporary, selected by the crew member for the day they start or ends a tour. The company will assign home base as being the crew member chosen gateway.
Home base	The location, assigned by the operator to the crew member, from where the crew member normally starts and ends a duty period or a series of duty periods and where, under normal circumstances, the operator is not responsible for the accommodation of the crew member concerned.
Home standby	Standby at a location within one hour travelling time from gateway.
Local Night	A period of 8 hours falling between 22:00 hours and 08:00 hours local time.
Minimum Crew	Such crew (Flight or Cabin) as defined by the relevant certifying aviation authority for each aeroplane type for the purpose of ensuring safety of the flight, as stated in the Airplane Flight Manual.
Operating crew member	A crew member who carries out their duties in an aeroplane during a flight or during any part of a flight.
Positioning	The transferring of a non-operating crew member from place to place, at the behest of the operator, excluding travelling time.
Reporting time	Time when a crew member is required to present themselves for duty.
Rest Period	An uninterrupted and defined period of time during which a crew member is free from all duties and airport standby.
Sector	Route performed, from the moment the aeroplane moves from its parking area to a location where it parks, including one take-off and the respective subsequent landing.
Show-time	The time for a crew member to present themselves in order to start the preparation for a flight or series of flights, airport standby, or hotspare.

Single Day Free of Duty	A single day free of duty shall include two local nights. A rest period may be included as part of the day off.
Split flight duty	A Flight Duty Period containing a break.
Standby	A defined period of time during which a crew member is required to be available to receive an assignment for a flight, positioning, or other duty without an intervening rest period.
Suitable accommodation	An individual room, duly furnished, providing means for resting in a horizontal resting, subjected to a minimum noise level, well ventilated, and permitting control of light and temperature levels.
Suitable rest location	Location where a suitable accommodation is available for the crew member.
Time-off	Time-off is not briefed, does not have a predefined ending time and is not considered duty, hence does neither count towards any rest period nor is part of any rest period. If the crew member has not received a briefing, by the end of a duty period, the crew member is considered on time-off.
Time Zone	An extension of the globe, generally coinciding with a time zone, corresponding to 1/24 of the terrestrial sphere and having an extension of 15° longitude, that is, one hour of time. Refer to 7.3 Establishment of Time Zone .
Tour	A series of consecutive duty days that normally begins after a period of more than 3 consecutive days off and that starts when the crew member first leaves their gateway for the purpose of performing a flight duty period positioning (including any standby period). A crew member is considered to be away from the gateway for the entire duration of the tour. If, during a tour, a crew member ends a duty period at their gateway, NTA shall continue to provide suitable accommodation unless the crew member is assigned home standby duty. A crew member that wishes not to use the suitable accommodation provided by the company during a tour shall inform the Travel department as soon as possible. In these cases, a crew member is still considered to be away from the gateway.
Travelling Time	Travelling time is defined as: • time from home to a designated place to show for a duty period and vice versa; • time for local transfer from a place of rest to the commencement of duty and vice versa.
Turn time	The time between blocks-on time of a flight and blocks-off time of the next flight within the same flight duty period. Flight times used to estimate the available turn time shall be based on conditions not less than 12 hours before the flight or 18 hours if the estimated total flight time is longer than 6 hours.

Weekly rest period	A thirty-six-hour period, including two consecutive nights, free of any duty for the crew member.
Window of Circadian Low (WOCL)	The Window of Circadian Low (WOCL) is the period between 02:00 hours and 05:59 hours. Within a band of three time zones, the WOCL refers to the gateway time. Beyond these three time zones, the WOCL refers to gateway time for the first 48 hours after departure from the gateway time zone, and to local time thereafter.
Working time	A period during which the Crew Member is working, at the employer's disposal and carrying out their activity or duties, in accordance with national laws and / or practice.
Year	A period of 52 consecutive weeks.

7.1.4 Acknowledgment of Briefings

A crew member that reads the briefing in the company mobile phone is equivalent to it being acknowledged. If the briefing is not accepted, the crew member shall contact the Scheduling department within a maximum of 30 minutes after reading the briefing.

7.1.5 Pre-Tour

7.1.5.1 Tour-Start Briefing

On the day before a crew member is rostered to start a tour, they shall be contactable on the company mobile phone (or other agreed method of communication). This other method must each time be approved by the Chief Pilot or Cabin Crew Fleet Chief and Scheduling in order for a crew member to be briefed for the first day of the tour.

The tour-start briefing shall be accomplished a minimum of 12 hours prior to start of the next duty, airline check-in time, or ground transportation departure / pickup time.

A show-time at the crew's home base shall not be scheduled before 04:59 local time unless the crew member agrees to an earlier show-time.

7.1.5.2 Briefings Changes

Any changes to the times (e.g. airline check-in, ground transportation time), which were previously briefed, shall be communicated to the crew member at least 12 hours before the new planned start of duty time, airline check-in, or ground transport time. The company mobile phone briefing shall be updated with these changes and they shall be communicated by telephone, unless the crew member has read the briefing prior to the 12 hours before the new planned start of duty. Other changes not affecting the crew member's start of duty time, check-in time, or ground transportation time may be entered and subsequently sent to the crew member by company mobile phone briefing at any time.

7.1.5.3 Home Standby

(EU-OPS & PORTUGUESE DECREE LAW no. 25/2022)

Home standby duty can be assigned when the crew member is scheduled to start a tour or when they are positioned to their gateway during a tour. No accommodation is provided when a crew member is assigned home standby duty.

From the time of notification, the crew member has 1:45 hours for arrival at the gateway.

When the crew member is notified to execute a FDP, the FDP starts:

- a. at reporting time or show time (as applicable), if the notification and FDP are outside the WOCL;
- b. at the time of notification, if the notification or FDP is within the WOCL.

NOTE

1. Crew member positioning followed by a flight duty is considered an FDP.
2. The time between notification and the reporting time or show-time under a) above is considered standby.

The time spent on standby counts 25% towards the limits in [7.2.12 Accumulated Duty, Flight, and Working Time Limitations](#).

Home standby is indicated in the crew member's briefing as 'STBY@HM'.

Automatic home standby procedure

If a crew member has not received a briefing by 19:00Z (DST 18:00Z) then they will automatically be scheduled for a 12-hour home standby that commences at 04:00Z (DST 03:00Z) or at the earliest legal start time if later.

If the crew member is re-briefed before the standby began, there shall be a minimum of 12 hours between the moment of the re-brief and the time the crew member must check in for the airline or any other duties. The re-brief shall be done by telephone.

During the Day Light Savings period (summer time / DST based on switch-over times in Portugal) all times under this paragraph are one hour earlier (as indicated above).

7.2 On Tour

The following assignments can be given to a crew member while on tour:

a. Duty	See definition of Duty .
b. Flight duty	7.2.10 Flight Duty Period (FDP)
c. Rest	7.2.21 Rest
d. Time-off	7.2.4 Time-off during a Tour
e. Travelling	See definition of Positioning
f. Positioning	7.2.19 Crew Positioning
g. Hotspare	7.2.14 Hotspare Duty
h. Standby at suitable rest location	7.2.17.1 Standby at a Suitable Rest Location

7.2.1 Travelling Time to Gateway

(Portuguese Decree Law no. 139/2004)

All limits in this chapter are based on the crew member not spending more than one hour in travelling time before arriving at their gateway. It is the crew member's sole responsibility to ensure that this time is not exceeded.

7.2.2 Briefing on Tour

7.2.2.1 Briefing by Proximity

If during the tour a complete crew is in close proximity and a crew member from the first flight of that complete crew acknowledges by email that the complete crew is briefed, there is no requirement for the other crew members to acknowledge the brief or to be called.

7.2.2.2 Changes to a Briefing

Once a crew member has been briefed and is in rest, any changes to the reporting time are only permitted as follows:

- The period between the notification of the change and the revised reporting time must be greater than or equal to the minimum rest period the crew member is entitled to.^{1 2}
- The company mobile device briefing shall be updated with the change, and it shall also be communicated by telephone, unless the crew member has already read the briefing prior to the start of the minimum rest period according to the revised reporting time.^{1, 2}

¹ In cases where the crew member is entitled to a rest period of 24 hours or greater, then the period between the notification and the revised reporting time may be reduced to no less than 11 hours, provided the notification occurs between 07:00LT and 21:00LT (this refers to local time where the crew member is resting, not WOCL Time Zone). This does not modify the rest period entitlement, but a notification in accordance with this note is not considered to interrupt a crew member's rest period.

² With the agreement of the crew member, the rest period may be reduced in accordance with [7.2.21.5 Commander's Discretion to Increase / Reduce a Rest Period \(On Tour\)](#), thereby reducing the minimum time prior to the show that the crew member must be notified.

NOTE

The crew member is not considered to have been notified until they have either read the briefing or been notified by phone.

Other changes not affecting the crew member's reporting time, check-in time, or ground transportation time may be entered and subsequently sent to the crew member by company mobile phone briefing at any time.

7.2.3 Company Provided Suitable Accommodation

(Portuguese Decree Law no. 139/2004)

When the crew member does not wish to use company provided accommodation (whether at their gateway or any other location), the crew member shall ensure that they will be able to be at the aerodrome within the same time as from the company assigned accommodation, taking into account traffic, parking time, etc., but never to exceed one (1) hour travelling time. If a crew member is on standby at a suitable rest location and they do not use the company provided accommodation, the crew member shall ensure that the total time from activation to the show time is not more than one (1) hour.

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7.2.4 Time-off During Tour

There are two requirements for a crew member during a time-off period:

- The crew member shall be contactable by company mobile phone to receive a briefing;
- The crew member is responsible to be fit and available for duty after a minimum rest period at the location where the time-off period started (a different location may be agreed with the Scheduling department).

If, due to the location, no mobile phone communication is possible, the crew member and Scheduling shall agree on (a) time(s) where the crew member will contact the Scheduling department for an update. Other methods of communication can be agreed, i.e. hotel room telephone. If the crew member does not respond to a phone call for a briefing, the respective department shall send an email requesting the crew member to update the briefing or contact Scheduling. The time the email was sent shall count as the time the crew member is considered briefed and the rest period started.

Subsequent to receiving a new briefing, a minimum rest period, as described in [7.2.21.2](#), before the show-time or the commencement of standby shall be scheduled.

A suitable rest location shall be provided during the time-off period unless otherwise agreed with the crew member.

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7.2.5 Scheduled Minimum Show-times

All show-times shall, as a minimum, be according to the table below, unless exempted by this chapter.

Conditions	Minimum Show-time
New aeroplane and Revenue	1:45
New aeroplane and Non-revenue	1:15
Revenue	1:30
Non-revenue	1:00

NOTE

A new aeroplane is an aeroplane that since the last acceptance, hotspare or flight has been used by another crew or crew member or has been in unscheduled maintenance (longer than 24 hours scheduled down time). If the aeroplane is new for the cabin crew member, it is new for the whole crew. If one of the cockpit crew members accepted the aeroplane or performed a previous FDP on the aeroplane, the aeroplane is not considered new for the whole crew, unless the cabin crew member is new.

After scheduled maintenance (longer than 24 hours scheduled down time), 15 minutes shall be added to the minimum new aeroplane show-time.

The Scheduling department shall plan sufficient time between the departure of the crew member from their home base and the show-time on the first day of a tour. If, due to an actual delay in positioning, there will be less show-time available than indicated in the table above, the passenger(s) shall be briefed of a new ETD that will allow the showtime as is applicable in the table above without applying paragraph [7.2.5.1 Reduced Scheduled Show-times](#).

Accumulated minimum scheduled show-times shall not exceed 2:45 hours.

If any crew member or a complete crew is swapped on a revenue flight (same passengers inbound as outbound) the show time for the outbound crew shall be at least 30 minutes prior to the on-blocks time of the inbound flight. If crew assess that they require more show time, they shall inform the Scheduling department of the extra show time requirement as soon as they receive the briefing. The planned time between blocks-on of the inbound flight and blocks-off of the outbound flight shall be at least 1:15. The requirement of [7.2.5.2-a](#). is not applicable at the show time as the aeroplane has not yet arrived.

An unscheduled situation requiring the swap of a crew member or aeroplane does not fall under this paragraph and crew members are expected to safely prepare the next flight.

7.2.5.1 Reduced Scheduled Show-times

The minimum Show-times defined in [7.2.5 Scheduled Minimum Show-times](#) may be reduced according to the following procedures and limitations.

CAUTION

Crew members shall note that when any reduced show is planned, it is **expected** that the departure time has a very high probability of being later than planned and that service to the Owners may be impacted.

The focus shall be on achieving the preparation and departure in a safe and unhurried manner, ensuring all SOPs are complied with and tasks completed fully prior to departure; this shall be the driving force for the actual departure time, **not** the ETD referenced in the briefing.

Notification

In addition to the briefing, an email shall be sent by scheduling prior to the show time, notifying the Commander that a reduced show is planned, copying lis-opssupport@netjets.com.

Since a reduced show requires the previous duty period to be extended by the amount the show was reduced, the crew will be notified by scheduling prior to the end of the previous duty period. The crew is expected to use the additional time to prepare what they can in advance of the subsequent reduced show.

NOTE

Any reduced show notified after the end of the previous duty period will require commanders' acceptance, as this will require the increase in the previous duty period to be waived.

As the crew are under no obligation to reply to emails while on rest, Scheduling will email the crew with this request. However, if no positive response is received within 1 hour, it will be assumed the crew have not accepted the request, and the schedule will be adjusted accordingly.

Acceptance

Certain alleviations, as detailed below, require the Commander's acceptance. The request and response shall be by email, copying lis-opssupport@netjets.com. When such a request is made, the Commander shall consult with the other crew members and take all aspects of the operation, including crew experience and fatigue levels, into consideration when assessing whether or not the request can be safely accommodated and respond promptly by email to allow the Scheduling department to plan accordingly.

Reduced Show Procedures and Limitations

The scheduled show-time may be reduced by a maximum of 15 or 30 minutes (with or without refuelling required, respectively) to a minimum of one hour, only if the following conditions have been met:

- a. The aircraft is not considered to be a 'new aircraft' to the crew, according to [OM-A 7.2.5](#);
- b. A reduced show with a crewmember under LIFUS (PIOE or SIOE) will only be permitted with explicit acceptance of the Line Training Captain. The request and reply emails shall copy lis-opssupport@netjets.com;

- c. A reduced show with an SIC who is 'inexperienced' according to [OM-A 4.1.8](#) will only be permitted with explicit acceptance of the PIC. The request and reply emails shall copy lis-opssupport@netjets.com;
- d. A reduced show is prohibited where the nominated PIC is 'inexperienced' according to [OM-A 4.1.8](#);
- e. Passengers are not escorted to or from the aeroplane by any crew member, except in locations without FBO assistance;
- f. The previous duty period shall be increased by the amount the showtime was reduced. However, with the commander's acceptance and with agreement of all the concerned crew members, the increase of the previous duty period may be waived.

BD-700-1A10 and BD-700-1A11 show-times shall not be reduced to less than 1 hour and 15 minutes, considering the technical pre-flight requirements.

When the Scheduled Minimum Show-time exceeds 1 hour and 30 minutes, the NJE Chief Pilot or DO may, in exceptional circumstances, and taking into consideration all flight preparation and pre-flight requirements, waive the other requirements and limitations on this section and authorise, in accordance with [OM-A 2.3 Operational Control and Supervision](#), a reduction in the minimum show time, but never below 1 hour and 30 minutes. Such a reduction must be acceptable to the commander and all concerned crew.

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7.2.5.2 Show-time Requirements

The following minimum requirements shall be present at show-time:

- a. Immediate access to the aeroplane;
- b. All (pre)flight documentation available to the Commander, including but not limited to, weather (as applicable), NOTAMs and OFP;
- c. Handling agent present at such a time as to be able to provide the crew with the required facilities and support;
- d. Fuel available before ETD - 30 minutes (if fuelling required).

NOTE

1. If the Handling agent will not be present at show time then the commander shall be briefed on the alternative arrangements that have been made and the expected show time of the handling agent.
2. If it is expected that the (pre)flight documentation might not be present at show-time, then in lieu of (b) above, the (pre)flight documentation may be sent to the rest location of the Commander. This shall be briefed to the Commander before the rest period and the (pre)flight documentation shall be available at the rest location a minimum of one hour before departure time from the rest location.

If these minimum requirements are not available, the Commander shall delay the flight appropriately and ensure that all pre-flight requirements and preparations are fully complied with. This delay shall be immediately communicated to Scheduling.

7.2.5.3 Increased Showtime Due to Aerodrome Logistics

If the Commander, at the end of a DP, determines that at the beginning of the following FDP the aerodrome logistics will take longer than normal (e.g. there is a requirement to move the aeroplane under its own power for fueling / passenger loading, or customs and immigration issues etc), they shall notify Scheduling of the requirement for an increased showtime for the next FDP, including the reason and amount of increase in show time required. Scheduling shall increase the showtime as indicated by the Commander. The Commander shall notify Scheduling of this requirement before the end of the preceding DP.

At locations where this is a common occurrence, an IJET note may be added stating the increment, and that should be added to all show times at that location. Crews should be aware that this is not automatically added and requires manual intervention. In case this fact has been missed, and crews notice, they should notify scheduling as soon as possible.

7.2.5.4 Increased Show-time Due to Cold or Adverse Winter Operations

The show-time shall be increased by 15 minutes if one or more of the following conditions are present and the aeroplane is not parked in a hangar (15 minutes will be added only one time):

- Aeroplane contaminated or expected to be contaminated;
- Expected temperature at show-time less than -10°C;
- Any expected frozen precipitation or heavy rain during the period from show-time to STD.

The new show-time shall be reflected in the briefing before the rest period commences.

7.2.5.5 Minimum Show Time for In-flight Services Representatives (ISR)

The minimum show time for an ISR is 1 hour, which may be reduced to 30 minutes with the approval of both the ISR and the commander.

7.2.6 Scheduled Minimum Turn Times

All turn times shall be scheduled according to the table below, unless exempted by this chapter.

Conditions	Minimum Show-time
Turn time before Revenue	1:15
Turn time before Non-revenue	1:00
Turn time between two Revenue	1:30

Turn times can be reduced according to the circumstances outlined within [7.2.6.1](#) and [7.2.6.2](#).

It is understood that the Commander is fully authorised to depart if able, as soon as the turnaround services and procedures are completed, or, in the interest of safety, to extend the actual turnaround time on the day as required to safely prepare the flight.

No crew member or Commander will be held at fault for extending a turn time for safety purposes.

NOTE

In all cases, and due to limitations of IJET, a turn time which is scheduled to be up to a maximum of 5 minutes shorter than the applicable requirement is still considered to meet that requirement. This tolerance shall only be used to avoid unnecessary disruption due to IJET limitations (resolution of 0.1 hours / 6 minutes on flight time) and shall not be used for any other purposes.

7.2.6.1 Reduced Scheduled Turn Times Due to Aerodrome Slot, Operating Hours or Parking Restrictions

The minimum Turn Times defined in [7.2.6 Scheduled Minimum Turn Times](#) may be reduced according to the following procedures and limitations.

CAUTION

Crew members shall note that when any reduced turn is planned, it is **expected** that the actual turn time has a very high probability of being longer than planned and that service to the Owners may be impacted.

The focus shall be on achieving the turnaround in a safe and unhurried manner, ensuring all SOPs are complied with and tasks completed fully prior to departure; this shall be the driving force for the actual departure time, **not** the ETD referenced in the briefing.

Permitted use of a reduction:

1. **Aerodrome slot allocations, aerodrome or FBO operating hours or parking restrictions**

Where aerodrome slot allocations, aerodrome or FBO operating hours or parking restrictions do not permit the desired schedule for a turnaround in accordance with 7.2.6 plus any applicable additions, the turn time may be reduced in accordance with the

provisions contained in this section by no more than 30 minutes and the scheduled turn time can never be less than 45 minutes prior to a ferry, and 60 minutes prior to a revenue.

2. Aerodrome or FBO mandates a turn time less than permitted by 1. above

Only where an aerodrome or FBO mandates a turn time less than that defined in point 1 above can the turn time be reduced further, but only to that mandated turn time, and never below 30 minutes.

3. Back to Back Revenue Flights

A Reduced Turn may also exceptionally be applied to situations not limited to those described in points 1. and 2. above, where there is an opportunity to link two separate revenue flights back-to-back. Where this is applied, in addition to the stipulations below, the maximum reduction shall be 30 minutes and the minimum turn shall be 1 hour. In addition, the turn prior or following the flight shall not be reduced.

4. Following a Reduced Show

A Reduced Turn may follow a Reduced Show, but the starting show time for the reduced show shall be the required show according to 7.2.5 with the 15 or 20-minute increment required by a reduced turn added to it. The reduction shall be in accordance with [7.2.5.1 Reduced Scheduled Show Times](#).

The tables below show the minimum reductions (assuming no increase is required for 'remote operations' or other factors).

Show Prior to Single Reduced Turn				
Conditions	Standard Minimum Show Time i.a.w 7.2.5	Show Time with +15min for 1 Reduced Turn	Minimum reduced show prior to a reduced turn	
			Refuel Required	No Refuel Required
Revenue	1:30	1:45	1:30	1:15
Non-revenue	1:00	1:15	1:00	1:00

Show Prior to Two Consecutive Reduced Turns				
Conditions	Standard Minimum Show Time i.a.w 7.2.5	Show Time with +20min for 2 Reduced Turns	Minimum reduced show prior to a reduced turn	
			Refuel Required	No Refuel Required
Revenue	1:30	1:50	1:35	1:20
Non-revenue	1:00	1:20	1:05	1:00

Notification and Acceptance

i. Notification

Reduced turn times require prior Commander notification, which will take the form of a briefing containing the reduced turn followed by an email sent prior to show time notifying the Commander that a reduced turn is planned, copying lis-opssupport@netjets.com. In cases where a briefing changes to incorporate a reduced turn after show time, then the email will be sent concurrently with the new briefing.

ii. Acceptance

Certain alleviations as detailed below require the Commander's acceptance. The request and response shall be by email copying lis-opssupport@netjets.com. When such a request is made, the Commander shall consult with the other crew members and take all aspects of the operation, including crew experience and fatigue levels, into consideration when assessing whether the request can safely be accommodated and respond promptly by email so scheduling can plan accordingly.

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Limitations and Procedures

Reduced turns as described above may only be applied subject to the following:

- a. Fifteen min shall be added to the show or turn prior to the reduced turn, or in case two consecutive reduced turns are required, then this is increased to 20 minutes;
- b. A maximum of two consecutive reduced turns may be planned (note the additional increase to the show or turn prior). Three consecutive reduced turns are prohibited;
- c. None of the crew members shall be under LIFUS (PIOE or SIOE designation on IJET), unless the Line Training Captain explicitly accepts the reduced show in accordance with the guidance given by the Training Department, and only a single reduced turn is planned;
- d. Reduced turns are prohibited where the nominated commander is 'inexperienced' according to OM-A 4.1.8;
- e. Reduced turns with an SIC classified as inexperienced according to OM-A 4.1.8 are only permitted if the commander explicitly accepts the reduced turn, having taken into consideration the crew's experience and all other safety aspects, and only a single reduced turn is planned;
- f. The flight package for the sector following the reduced turn (and the following sector if two consecutive reduced turns are planned) shall be sent to the FBO of departure for the flight preceding the reduced turn(s) no less than 30 minutes prior to the ETD of that flight. This allows the crew to have it printed and on board prior to departure;
- g. Except in locations without FBO assistance, passengers shall not be escorted to or from the aeroplane by any crew member;
- h. Where possible, without contravening fuel uplift requirements (e.g. Refuel EU) and subject to operational constraints (e.g. performance or mass and balance), fueling during a reduced turn should be avoided irrespective of the fuel index.

DO or NJE Chief Pilot Approval for Reduced Turn

The Director of Operations or NJE Chief Pilot may also authorise a reduced turn outside the terms of this section, but only for exceptional and specific circumstances, on a case-by-case basis and where the planned operation is confirmed to be acceptable to the Commander.

Where such approvals are given, they shall be in accordance with [OM-A 2.3 Operational Control and Supervision / DO/CP Supervision and approval of non-routine operations](#).

A record shall be maintained of such turn reductions.

7.2.6.2 Minimum Turn Time for Passengers Drop-off / Pick-up / Fuel Stop / Custom Stop

For a flight that is either picking-up an extra passenger(s), dropping-off a passenger(s), making a fuel stop, or a custom stop, the scheduled turn time can be reduced to:

- a. 30 minutes if no refuelling is required;
- b. 45 minutes if refuelling is required.

The scheduled minimum turn time shall be increased by 15 minutes for BD-700-1A10, BD-700-1A11 and aeroplanes requiring overwing refuelling or flights with a trainee under initial line training or conversion training.

The following shall be adhered to:

- a. If catering upload is required, 10 minutes shall be added to the scheduled turn times in a. and b. above.
- b. Passengers are not escorted to or from the aeroplane by any crew member.

The turn time or show-time before the flight with the minimum turn time shall be increased by 15 minutes. Upon crew request, these 15 minutes may be provided after the flight with the minimum turn time.

A maximum of two consecutive minimum turn times involving passenger drop-off / pickup / fuel stop / custom stop can be scheduled within one FDP. When two consecutive minimum turn times are scheduled, 30 minutes will be added to the turn time or showtime before the first flight with the minimum turn time.

7.2.6.3 Scheduled Turn Times for Revenue-Recovery Flights

Revenue-recovery flights shall be planned with the turn-times in [7.2.6 cheduled Minimum Turn Times](#).

The Commander shall ensure that the required time is taken by all crew members to safely prepare the flight according to all legal and company requirements. The emphasis during a recovery is on safety and not time.

7.2.6.4 Increased Scheduled Show / Turn Times Due to MCF

A minimum of 45 minutes before and 15 minutes after the flight will be added to the normal show / turn-around times. Scheduled EET will vary according to the aeroplane type and maintenance action performed.

7.2.6.5 Increased Scheduled Show / Turn Time Due to Remote Operations

When an FDP includes operations to / from / within a remote location, the following applies:

- a. If the first flight of an FDP is to / from / within a remote location, the respective time below shall be added to the show time; or
- b. If the applicable flight is not the first flight of the FDP, then the respective time below shall be added to either the show time or to the turn time prior to the applicable flight; or
- c. If an FDP has multiple flights operating to / from / within remote locations, then the longest time from below shall be added to either the show time or to the turn time prior to the first of the applicable flights;
- d. If more than one of the below additions is applicable for one flight, only the longest time shall be added:

To Israel:..... 15 minutes

Crossing the Atlantic:..... 30 minutes

Crossing the Pacific:..... 30 minutes

Crossing the Indian Ocean:..... 30 minutes

To, within, and out of Africa except North Africa:..... 30 minutes

Out of or within Central and South America (excluding the Caribbean):..... 30 minutes

To, within, and out of Asia:..... 30 minutes

North Africa for this paragraph is: Morocco, Algeria, Libya, Tunisia, Egypt.

- e. If the Commander determines that the conditions for a given flight do not require the additional time given by this paragraph, then they may contact scheduling to request it to be waived or reduced. This shall only be on the request of the Commander and shall not be requested by scheduling who shall plan in accordance with this paragraph.

7.2.7 Reduced Actual Times

If, due to actual delays, the subsequent turn time is 15 minutes less than stipulated in [7.2.6 Scheduled Minimum Turn Times](#), the appropriate arrangements for the passengers to be escorted to or from the aeroplane and notification of the Commander with the details will be made (all attempts shall be made to contact the Commander inflight, if required).

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7.2.8 Aerodromes with Known Frequent Delays

For aerodromes where it is known that procedures (arrival, departure, taxi, ramp movement, customs, immigration, handling, fuelling, etc.) take longer than normal, the appropriate extra turn time or show-time shall be added. A list of aerodromes where these additions are applicable shall be maintained by International Planning and Ground Services department.

Crew members that experience delays at aerodromes must email lis-aerodromedelays@netjets.com with the following information:

- a. Airport ICAO identifier;
- b. Length of the delay;
- c. Cause of the delay;
- d. Why, from the crew perspective, it probably is a frequent delay and not an infrequent delay.

The information will then be validated and the list updated, if necessary. Ops Aerodata will add a feasibility note in IJet (QC and / or CREW) stating the requirement for the increased show-time and / or turn time for that particular aerodrome.

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7.2.9 Training

For a flight crew member or cabin crew member on their first operator's conversion course in NTA or following a ZFTT type rating course, the following minimum show and turn times apply.

For the purpose of this section, the following definitions shall apply:

- 1st tour is the first tour of an FCM or CCM at NTA. A trainee remains on 1st tour status until they have completed a tour including at least 5 duty days and 4 sectors.
- ZFTT is Line Flying Under Supervision with a TRI following a ZFTT type rating course. A trainee remains on ZFTT status until they have completed a minimum of 4 take-offs and landings as PF to proficiency.

Event		Minimum Times	
		FCM	CCM ²
1st tour or ZFTT	Show for Revenue	2:00	2:00
	Show for Non-revenue	1:30	1:30
	Turn-around before Revenue	1:30	1:30
	Turn-around before Non-revenue	1:00	1:00
	Hotspare	Not Allowed	No additional restrictions under this section
Subsequent tour	Show for Revenue	1:30	
	Show Non-revenue	1:00	
	Turn-around before Revenue	1:30 ¹	
	Turn-around before Non-revenue	1:00 ¹	

NOTE

1. The minimums marked with ¹ can be reduced as follows:

- Delays on the day: If delays occur on the day after show-time (weather, late passengers, etc.) and the departure and destination airports as previously scheduled have not been changed, Scheduling may contact the respective Line Training Captain (LTC) to request a reduction in the turn time. In this case, it is at the sole discretion of the LTC whether to accept a reduction or not based on the performance level of the trainee.
- Changes to the planned schedule on the day: If changes to the planned schedule occur on the day (different airports), Scheduling will plan on the turn times according to this paragraph. However, if a reduction to these turn times is required, then Scheduling will request a reduction to the Duty Training Manager (DTM). It is the responsibility of the DTM to discuss with the LTC concerned and to decide on whether to accept this reduced turn time or not.

2. For CCM, if there is a planned delay or schedule change on the day, after show time, Scheduling may contact the aircraft commander to request a reduction in turn time. In this case, the aircraft commander will discuss with the LT the individual situation (trainee performance, business need, etc.), but will retain overall responsibility for accepting a reduced turn time.

7.2.10 Flight Duty Period (FDP)*(EU-OPS)*

Each FDP shall be preceded by a minimum rest period as described in [7.2.21.2](#) except for the first day of a tour. The crew member shall ensure that are adequately rested before the first day of a tour.

If the FDP is preceded by a type of duty that is not described in this chapter or that type of duty's relationship towards an FDP is not described in this chapter, then the FDP, for the purpose of its limitations, shall start at the moment the duty started and that duty shall count 100% towards the limits of the FDP, including the effect of the hour the duty started towards the limits dictated by the WOCL.

7.2.10.1 Daily FDP (Non-augmented Flight Crew)*(EU-OPS)*

The maximum daily FDP for flight and cabin crew is as follows:

Table 1 - Maximum Basic Daily FDP

Maximum FDP (Non-WOCL Encroachment)						
Sectors	1	2	3	4	5	6 and more
Max. FDP (hh:mm)	13:00	13:00	12:30	12:00	11:30	11:00

NOTE

1. During landing or instrument approach training (base check), each 4 landings shall count as a sector and each 2 instrument approaches count as one sector.
2. For the purpose of counting sectors, a crew member only counts a sector if they are actively exercising the privileges of their licence or training onboard the aeroplane (i.e. passive positioning does not count as a sector).

When the FDP starts in the WOCL, the maximum FDP as stated above shall be reduced by 100% of its encroachment, up to a maximum of two hours.

When the FDP ends in or fully encompasses the WOCL, the maximum FDP as stated above will be reduced by 50% of its encroachment.

7.2.10.1.1 Extended Daily FDP (Non-augmented Flight Crew)*(EU-OPS)*

An Extended FDP may be used subject to the Limitations and Rest Requirements shown below.

When an Extended FDP is used: Table 1 - Maximum Basic Daily FDP in [7.2.10.1](#) is replaced by Table 2 - Maximum Extended Daily FDP below.

The notes and WOCL encroachment adjustments described in [7.2.10.1](#) remain applicable when an Extended FDP is used.

Table 2 - Maximum Extended Daily FDP

Maximum Extended FDP						
Sectors	1	2	3	4	5	6
Max. FDP (hh.mm) (Non-WOCL Encroachment)	14:00	14:00	13:30	13:00	12:30	
Max. FDP (hh.mm) (WOCL Encroachment up to 2 hours)	14:00	14:00	13:30	13:00		

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Maximum Extended FDP						
Sectors	1	2	3	4	5	6
Max. FDP (hh.mm) (WOCL Encroachment >2 hours)	14:00	14:00			Extended Daily FDP not permitted	
FDP starts in the period 22:00 to 04:59 hours	Maximum FDP will be as per this table according to the amount of WOCL encroachment, with the appropriate WOCL adjustments from 7.2.10.1 applied or 11:45 hours, whichever is less.					

Limitations to the use of an Extended FDP

- Use of the Extended FDP shall be planned prior to the beginning of the affected FDP - if at the beginning of the FDP the maximum basic daily FDP from [7.2.10.1](#) is not planned to be exceeded then an Extended FDP shall not be used, and any subsequent exceedance of the basic FDP limits shall be according to [7.2.18 Commander's Discretion to Increase / Reduce an FDP](#).
- The maximum number of Extended FDPs is two in any 7 consecutive days.
- An Extended FDP may be used for consecutive FDPs.
- If an Extended FDP was planned but the actual FDP does not exceed the limits for the maximum basic daily FDP as shown in [7.2.10.1](#) (i.e. the planned extension is not used) then the FDP shall not be considered as an Extended FDP.
- The use of Commanders Discretion to increase the FDP in accordance with [7.2.18 Commander's Discretion to Increase / Reduce an FDP](#) is permitted when an Extended FDP is used.

Extended FDP Rest Requirements

Rest Period prior to Extended FDP	Increase in Minimum Rest Period requirement following an Extended FDP
• Extended FDP is on first day of tour.	2 hours
• Rest period prior to Extended FDP at least 2 hours more than the Minimum Rest period applicable to the previous DP.	
• Rest period prior to Extended FDP less than 2 hours more than the Minimum Rest period applicable to the previous DP.	4 hours

Where the extensions are used for consecutive FDPs the pre and post rest between the two operations shall run consecutively.

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7.2.10.1.2 Daily FDP (Augmented flight Crew / In-flight Rest)

(PORTUGUESE DECREE LAW no.139/2004)

A flight or cabin crew member can only exceed the daily FDP limits in [7.2.10.1](#) if they are part of or have flown with an augmented flight crew. The required onboard resting facilities for one crew member shall be available during each sector and during the whole flight within the FDP for which they are exceeding the limits in [7.2.10.1](#).

The resting onboard the aeroplane and the time at the controls must be equally divided among all the flight crew members and controlled by the Commander. The resting facility shall not be used by any other person than the flight crew member assigned to rest at that time. In case the cabin is assigned and required as a resting area, the Flight crew member(s) and other crew member(s) assigned to rest shall be the only ones occupying the cabin. Due regard shall be given to the resting requirements of the other crew members concerning noise, light, and temperature.

Each crew member cannot be part of an augmented crew for more than 3 sectors within one continuous FDP. Sectors flown when not actively exercising the privileges of one's license or training onboard the aeroplane are not counted as a sector (i.e. passive positioning does not count as a sector). In case a fourth sector is planned, the maximum FDP for the augmented crew is reduced by 2 hours.

The resting areas must be free of any items that limit the full movement of the seat and beds or limit the position of the flight crew member (i.e. shall not be used as storage areas during any part of the flight for which an augmented crew has been scheduled and required).

At the planning phase, the difference between the planned FDP according to this paragraph and paragraph [7.2.10.1](#) shall, for each individual crew member, not exceed 8 hours per 7 consecutive days. The planned block times shown on IJET shall be used for establishing compliance with this limitation, and not the actual block times (i.e., no more than 8 hours difference may be planned, but in the event of unforeseen delays or flights taking longer than planned, the actual difference may exceed 8 hours). Additionally, it should be noted that the planned FDP ends at the planned on-block time for the final operating sector of the duty period, not the duty close time.

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7.2.10.1.2.1 Augmented Flight Crew on Revenue Flights*(PORTUGUESE DECREE LAW no.139/2004)*

The maximum FDP for revenue flights is as per the table below:

Aeroplane Type	Max FDP (hours)	Max FDP (hours)
	3 Flight Crew	4 Flight Crew (only applicable to aeroplanes with 2 crew rest seats)
BD-700-1A10	16:30	18:00

On flights with a planned FDP that requires an augmented crew, the following procedures shall be adhered to:

1. The augmented crew shall consist of 3 or 4 flight crew members as per table above and 1 cabin crew member.
2. The Commander shall agree on a rest plan that includes all flight crew members and the cabin crew member before departure.
3. The rest period shall be equally divided between each flight crew member and the cabin crew member.
4. For long revenue sectors on single bunk aircraft, the rest period per flight crew member and cabin crew member shall be calculated as the flight time minus a minimum of one hour and thirty minutes per sector, this period beginning thirty minutes after take-off and ending a minimum of one hour prior to landing equally divided between all the crew members (e.g. on an 11-hour and 30-minutes sector (3 FC and 1 CC), 10 hours is available for in-flight rest which equates to 2-hours 30-min per crew member).³
5. For long revenue sectors on double bunk aircraft, the rest period per flight crew member shall be calculated as the flight time minus a minimum of one hour and thirty minutes per sector, this period beginning thirty minutes after take-off and ending a minimum of one hour prior to landing equally divided between all the flight crew members (e.g. on an 11-hour and 30-minutes sector (3 FC), 10 hours is available for in-flight rest which equates to 3-hours and 20-minutes per flight crew member). The cabin crew member shall be allocated the remaining bunk for their sole use and shall take rest equivalent to that allocated to the flight crew at their discretion according to a rest plan agreed with the PIC.¹
6. If the FDP consists of more than 1 sector, the cabin crew rest period shall be taken as much as possible during any non-revenue sector.
7. The cabin crew member's rest period shall be, as much as possible, during the time their services are not required by the passengers. While the cabin crew member is at rest or otherwise unavailable, the Commander shall designate a flight crew member who is not at rest to perform and be responsible for cabin safety, surveillance, and service duties, as if operating without a cabin crew member.

³ With agreement from all affected crew members, the rest period may start earlier than 30 minutes after takeoff, consequently increasing the overall rest period available, provided all OM-A limitations from [8.3.10 Crew Members at Their Stations](#) are respected.

7.2.10.1.2.2 Augmented Flight Crew on Non-revenue Flights*(PORTUGUESE DECREE LAW no.139/2004)*

For the purpose of this paragraph, the cabin is considered as a suitable rest facility.

For non-revenue flights where no other persons (including passengers, additional crew members, or other crew members) are carried for the entire FDP, the maximum FDP is as per the following table:

Aeroplane Type	Max FDP (hours) 3 Flight Crew		Max FDP (hours) 4 Flight Crew	
		Min Required Rest Facilities		Min Required Rest Facilities
DA-2EASy	16:30	1 bed in the cabin	18:00	2 beds in the cabin
CL-600-2B16	16:30	1 bed in the cabin	18:00	2 beds in the cabin
BD-700-1A10	16:30	1 crew rest or 1 bed in the cabin	18:00	2 resting facilities: any combination of crew rest or bed(s) in the cabin
BD-700-1A11	16:30	1 bed in the cabin	18:00	2 beds in the cabin

NOTE

If cabin crew are carried (either on duty or positioning), then an additional suitable resting facility (bed in the cabin or crew rest area) shall be available for each cabin crew member. Aircraft configuration shall be such that sufficient resting facilities are available for all crew members.

When the cabin is used as a suitable resting facility, crew shall ensure that the cabin is free of any noise and / or disturbance in order to facilitate a proper rest opportunity for the resting crew member.

For the purpose of this paragraph, the bed(s) as required by the number of resting crew members shall be installed and made up for the flight crew to be used as resting facility.

The Commander shall agree on a rest plan that includes all flight crew members and the cabin crew member before departure.

For long non-revenue sectors, the rest period per flight crew member shall be calculated as the flight time minus a minimum of one hour and thirty minutes per sector, this period beginning thirty minutes after take-off and ending a minimum of one hour prior to landing equally divided between all the flight crew members (e.g. on an 11-hour and 30-minutes sector (3 FC), 10 hours is available for in-flight rest which equates to 3-hours and 20-minutes per flight crew member). The cabin crew member shall be allocated a bunk for their sole use and shall take rest equivalent to that allocated to the flight crew at their discretion, according to a rest plan agreed with the PIC.⁴

⁴

With agreement from all affected crew members, the rest period may start earlier than 30 minutes after takeoff, consequently increasing the overall rest period, provided all OM-A limitations from [8.3.10 Crew Members at Their Stations](#) are respected.

7.2.10.1.2.3 Cabin Crew Show-time Difference*(EU-OPS)*

The cabin crew shall show at the same time as the flight crew. After prior CP / ACP approval and under exceptional circumstances, the cabin crew may show later to accommodate a longer FDP. For cabin crew being assigned to a flight or series of flights, the FDP of the cabin crew may be extended by the difference in reporting time between cabin crew and flight crew, as long as the difference does not exceed one hour.

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7.2.11 Maximum Scheduled FDPs Encroaching on the WOCLs in Seven Consecutive Calendar Days

	2 non-consecutive	2 consecutive ¹	3 non-consecutive
Limitation	Nil	Followed by $\geq 36h$ rest + 2 local nights 1) 2) 3)	Requires $\geq 36h$ rest + 2 local nights after the 2nd non-consecutive FDP encroached WOCL 1) 2)

¹ A series of back-to-back WOCL FDPs is not considered to be 'consecutive' when a complete WOCL period without any duty falls between them.

1. Time-off provision is not applicable prior to the start of the rest.
2. Briefing shall be sent $\geq 11h$ prior to the following FDP and $< 21:00LT$ if $\geq 21:00LT$ crew member will be 12h on standby after the 36h rest period.
3. It does not apply when a crew member is positioning back to Home Base, with no further duty is planned for that tour and the crew member will have at least 36 hr + 2 local nights rest at Home Base prior to their next duty containing an FDP.

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7.2.12 Accumulated Duty, Flight, and Working Time Limitations

(EU-OPS)

Total duty periods to which a crew member is assigned shall not exceed:

- a. 60 duty hours in any 7 consecutive days, except as in b.;
- b. The 60 duty hour limit in a. may be extended to a maximum of 70 duty hours in 7 consecutive days provided that any duty above 60 hours shall only be for positioning to the gateway and shall be followed by an extended weekly rest period;
- c. 190 duty hours in any 28 consecutive days, spread as evenly as practicable throughout this period.

The total block times of the flights on which an individual crew member is assigned as an operating crew member shall not exceed:

- a. 900 block hours in a calendar year;
- b. 100 block hours in any 28 consecutive days.

The Scheduling department shall keep record of these limits and shall not schedule a crew member if these limits are exceeded. The crew member shall be advised if these limits are reached.

When a crew member is performing flight duty (FDP), including hot spare duty or standby duty, any previous duty must be accounted for in the cumulative limits. Previous duty includes, but is not limited to, Office work, GRT, simulator, or classroom training. When such duty is not followed by an FDP, the cumulative limits do not apply.

This protects the crew member from cumulative fatigue during an FDP but allows an exceedance during other types of duty.

NOTE

- When [7.2.18 Commander's Discretion to Increase / Reduce an FDP](#) is applied, the limits in this paragraph are allowed to be exceeded by the same time that the FDP is exceeded.
- Working time shall not exceed 2000 hours per calendar year.

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7.2.13 Split Duty*(PORTUGUESE DECREE LAW no.139/2004)*

The maximum FDP limits as described in paragraph [7.2.10.1](#) can be augmented according to the table below if adhering to the following:

- a. The crew shall be notified of a split duty before the sector preceding the break at the latest.
- b. The FDP contains a break of at least 3 hours.
- c. Post-flight activities are accounted for as 15 minutes (break less than 6 hours) and 30 minutes (break equal or more than 6 hours) after blocks-on of the sector preceding the break.
- d. Pre-flight activities are according paragraph [7.2.5 Scheduled Minimum Show-times](#) or longer (to accommodate owner / card holder flexibility). The pre-flight activities are reduced by 15 minutes if refuelling is required and by 30 minutes if no refuelling is required if the break is less than 6 hours.
- e. If the break is longer than or equal to 6 hours, the break must be in a suitable rest location. In this case, the travelling time is not considered part of the break.
- f. When the break is less than 6 hours, the break shall be in a comfortable location where temperature and light can be controlled and that is not open to the public.
- g. If the rest, as per f) above, is taken onboard an aeroplane, there shall be no passengers and an alternative power source may be used to provide control of light and temperature. The use of the APU is not authorised for this purpose, except where no other option for a comfortable location is available.
- h. An operating crew member shall not do more than 2 sectors before the break and not more than 1 sector after the break (airline positioning is not considered a sector).
- i. A split flight duty period that includes partly or wholly a WOCL may only be scheduled twice in a 7 consecutive day period and a rest period that includes a local night shall occur between the two split flight duty periods.
- j. The sum of the duty time of the sector(s) before the break or after the break shall not exceed 10 hours each (positioning is counted as duty time) and the total duty time including the break shall not exceed 20 hours.
- k. There shall not be more than 3-hour time zones between the location of the showtime and the location of the break.

Duration of the break (consecutive) hh:mm	Increase of the maximum FDP
00:00-02:59	None
03:00-06:59	Half the time of the break
07:00-09:59	2 / 3 the time of the break

For paragraph [7.2.12 Accumulated Duty, Flight, and Working Time Limitations](#) the time of the break equal to or less than 8 hours counts as 100% and the time of the break above 8 hours counts as 50%.

NOTE

1. If, due to a delay on the sector before the break, the actual break will not be sufficient or will be too long to complete the scheduled flight(s) within the remaining FDP, the schedule shall be rearranged so that the scheduled flights are scheduled to be completed within the allowable maximum FDP time(s) and [7.2.18 Commander's Discretion to Increase / Reduce an FDP](#) cannot be used to increase the planned maximum FDP. Commander's discretion may be used if subsequent unforeseen operational reasons occur after the schedule re-arrangement.
2. Once the break has started, it is possible to change the departure time to an earlier time.
 - When this is notified to the crew (via a briefing that has been read by the crew member on the waypoint application or by phone):
 - on a scheduled break of less than 6 hours; or
 - the time already spent on the break is less than 6 hours.

The FDP is considered as one continuous period with no increase in maximum FDP for the period already spent on the break as per above table. Normal FDP and limitations as per [7.2.10 Flight Duty Period \(FDP\)](#) apply and [7.2.18 Commander's Discretion to Increase / Reduce an FDP](#) can be used.

- When this is notified to the crew (via a briefing that has been read by the crew member on the waypoint application or by phone) after the time already spent on the break is 6 hours or more, the maximum FDP can be extended for the period already spent on the break as per above table. From the time of notification, the crew member shall be ready to travel within 45 minutes from the rest location to the show location.

Once the break has started, it is not possible to change the departure time to a later one and extend the break. When a later departure is required, crews will show at the original scheduled time. The additional show will count normally towards the maximum FDP and [7.2.18 Commander's Discretion to Increase / Reduce an FDP](#) cannot be used to increase the planned maximum FDP. Commander's discretion may be used if subsequent unforeseen operational reasons occur after the schedule re-arrangement.

7.2.14 Hotspare Duty

Hotspare duty is duty that serves to quickly react to contingencies in any part of the operation (aeroplanes going unserviceable, weather problems, ATC problems, etc.) or to accommodate passengers that need to depart earlier. Hotspare duty can be scheduled as a single continuous period before, in between, or after flights, and is 100% part of an FDP and all its associated limits, including the provisions of FDP Extension as per [7.2.10.1.1](#).

A maximum of one hotspare duty can be scheduled per FDP.

The DP terminates when the hotspare duty ends and no additional 30 minutes close down time will be allocated to the crew.

Off-blocks time for hotspare activation shall not be scheduled earlier than:

- 30 minutes from the call to the Commander notifying them of the activation of the hotspare;
- 30 minutes after all information and required paperwork is in possession of the flight crew;
- 60 minutes after show time unless the crew is new to the aeroplane, in which case 75 minutes will be scheduled;
- For the BD-700-1A10 and BD-700-1A11 - 75 minutes after show time unless the crew is new to the aeroplane, in which case 90 minutes will be scheduled.

The following limitations are applicable to hotspare duty:

- can only commence after a minimum of 30 minutes after actual on-block time (LSZH and LSGG: 45 minutes);
- can only be a maximum of 3 hours if any other flight(s) is (are) scheduled within the same FDP;
- If no other flights are scheduled, a maximum of 8 hours.

A list of aerodromes that are not suitable for hotspare duty longer than 3 hours shall be maintained by the Scheduling department.

If a flight is scheduled after the hotspare, the departure time will be scheduled at least 30 minutes after the hotspare has ended.

Crew members that experience an aerodrome that is not suitable for hotspare duty shall immediately notify by email lis-hotspare@netjets.com with the reasons why the aerodrome is not suitable for hotspare duty and submit a crew report.

Hotspare duty on non-suitable aerodromes is limited to a maximum of 3 hours.

7.2.15 Back-up Flight

A Back-up Flight is an option used to back-up a flight.

They shall be planned and prepared as a normal flight and the aeroplane shall be ready to depart at the scheduled time of departure.

If a Back-up Flight does not depart, it shall be treated as a cancelled flight.

All the provisions of [Chapter 7 - Flight Duty Time Limitations](#) apply to Back-up Flights.

7.2.16 Aeroplane Acceptance

A crew or part thereof, (consisting of at least one of the FCM and, if a Cabin Crew is scheduled to be carried on the following FDP, at least one of the CCM) can be scheduled for aeroplane acceptance duty to check an aeroplane before a rest period preceding an FDP to reduce the show time at the beginning of the next FDP.

The checks shall include all checks necessary to ensure that the aeroplane can be used for a revenue flight to include interior and exterior inspections and systems checks.

Time allocated for aeroplane acceptance duty is one hour.

Aeroplane acceptance counts for 100% as duty and 100% as flight duty (if followed by a hotspare duty).

7.2.17 Standby

(EU OPS and PORTUGUESE DECREE LAW no.139/2004)

The following provisions are applicable to any type of standby:

- a. A briefing for standby duty shall include a specific beginning and ending time;
- b. The maximum standby period is 12 consecutive hours;
- c. Crew activation from standby is done by a phone call to each crew member;
- d. All show-times shall be according to [7.2.5 Scheduled Minimum Show-times](#) and [7.2.5.1 Reduced Scheduled Show-times](#) shall not be applied;
- e. The time between the start of the standby period and the end of an FDP shall not be more than 20 hours.

When the crew member is notified to execute an FDP, the FDP starts:

- a. at reporting time or show time (as applicable) if the notification and FDP are outside the WOCL;
- b. at the time of notification if the notification or FDP is within the WOCL.

NOTE

1. The time between notification and the reporting time or show-time under a) directly above is considered standby.
2. The crew member may be notified of the ensuing FDP by an update of the company mobile device briefing. If the crew member acknowledges the brief, there will be no phone call as under c) above and the crew member is responsible to be on time at the designated aeroplane. If the crew member does not acknowledge the briefing, a phone call as under c) will be used for activation.

7.2.17.1 Standby at a Suitable Rest Location*(EU-OPS and PORTUGUESE DECREE LAW no.139/2004)*

From the time of notification the crew member shall be ready to travel within 45 minutes from the rest location to the show location. Show-time will be the briefed show-time or arrival at the show-time location, whichever is later.

The time spent on standby counts 25% as duty towards the limits in [7.2.12 Accumulated Duty, Flight, and Working Time Limitations](#).

If the standby is followed by an FDP, the time on standby counts as 25% towards the limits in [7.2.10 Flight Duty Period \(FDP\)](#).

Standby at a suitable rest location is indicated in the crew member's briefing as 'STBY@RL'.

7.2.18 Commander's Discretion to Increase / Reduce an FDP*(EU-OPS and PORTUGUESE DECREE LAW no.25/2022)*

In the case of unforeseen circumstances occurring after the FDP started, an aircraft Commander may, after consideration of their and their crew's fatigue levels, increase or reduce the FDP limits as detailed below.

Unforeseen reasons could be (but are not limited to):

- a. Technical failure on any aeroplane on the NTA AOC or any aeroplane hired by NTA to perform its flights which require recovery by the crew exercising discretion.
- b. Any ATC delay or imposed restriction.
- c. Denial of parking positions by the airport authorities.
- d. Weather related phenomena.
- e. Late arrival of passenger.
- f. A new booking received after the crew show time.

1. FDP Increase

The FDP may be increased in accordance with the table below:

Planned FDP Limit	Maximum Permitted FDP Increase
7.2.10.1 Daily FDP (Non-augmented Flight Crew)	2 Hours
7.2.10.1.1 Extended Daily FDP (Non-augmented Flight Crew)	
7.2.13 Split Duty	
7.2.10.2 Daily FDP (Augmented Flight Crew / In-flight Rest)	3 Hours

If on the final sector within an FDP unforeseen circumstances occur after take-off that will result in the permitted increase being exceeded, the flight may continue to the planned destination or alternate.

A. Consent Requirement

- a. If the unforeseen reason requires an additional or different flight to be planned (e.g. Denial of parking positions by the airport authorities requiring an additional ferry flight) that will result in the newly briefed plan requiring the use of Commanders Discretion to increase the FDP, then the OCC shall contact the commander either by email or phone to establish whether the required increase is acceptable.
- b. An FDP increase in accordance with [7.2.18.1](#) must be acceptable to the Commander. However, there is no requirement for the OCC to contact the Commander except as described in a) above. If the Commander notices that they are likely to go into discretion and does not consider this to be acceptable they shall notify the OCC at their earliest convenience so that they can plan accordingly.

NOTE

There is no requirement for a discretion or fatigue report in the event that the Commander does not accept to increase an FDP in accordance with this paragraph.

B. Reporting Requirement

- The Commander shall submit a PIC Discretion Report via the link provided by General Message or email reminders whenever an FDP is increased by their discretion.
- If the Commander does not receive this reminder, they shall send a PIC Discretion Report to lis-PICDiscretion@netjets.com. The report shall be submitted by no later than 48 hours after the event.
- Where the increase of an FDP due to the application of this paragraph exceeds one hour, a copy of the report sent by the Commander shall be sent to ANAC no later than 28 days after the event.
- In addition to the Commander's report, the Company shall add other comments that are applicable.

C. Rest Requirement

Type of FDP increased by use of Commanders Discretion	Rest period requirement
7.2.10.1 Daily FDP (Non-augmented Flight Crew)	As per 7.2.21 Rest which may be reduced in accordance with 7.2.21.5 Commander's Discretion to Increase / Reduce a Rest Period (on Tour)
7.2.10.2 Daily FDP (Augmented Flight Crew / In-flight Rest)	
7.2.13 Split Duty (Portuguese Decree Law DL139/2004)	
7.2.10.1.1 Extended Daily FDP (Non-augmented Flight Crew)	As per 7.2.10.1.1 Extended Daily FDP (Non-augmented Flight Crew) which may be reduced in accordance with 7.2.21.5 Commander's Discretion to Increase / Reduce a Rest Period (on Tour)

2. FDP Reduction

In case of special circumstances that could lead to severe fatigue, and after consultation with the crew members affected, the Commander shall reduce the actual flight duty period in order to eliminate any detrimental effect on flight safety. The Commander shall notify NTA immediately via the email address Lis-Fatigue@netjets.com.

Each individual crew member shall also file a Fatigue Report (accessed via the safety reporting system) within 48 hours of the FDP reduction. If, however, an individual member of the crew is not fatigued then they may notify NTA via the email address Lis-Fatigue@netjets.com and may then be assigned duty according to the FDP limits in this chapter.

The additional rest provisions of [7.2.28 Fatigue Reporting](#) will also apply as applicable.

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7.2.19 Crew Positioning*(EU-OPS)*

All the time spent on positioning is counted as duty.

Positioning within an FDP shall not count as a sector.

Required (scheduled) Show Time after positioning is as per [7.2.5 Scheduled Minimum Show-times](#). This may not be reduced.

All required crew members must be present at show time.

Method of Positioning	Minimum Reporting Time prior to Positioning	Required time from arrival of a positioning airline prior to continued positioning	Minimum time from positioning scheduled arrival time to required Show Time for operating sector
Airline - short haul	1 hr ¹	As required for the planned connection	1 hr
Airline - long haul ²	2 hr		1 hr
Train International / Hire Car / Passenger on NetJets aircraft	30 min	1 hr	As required to ensure arrival at the FBO at or prior to required show time
Train National / Bus / Boat	15 min	45 min	
Chauffeured Car / Taxi	0 min	30 min	

¹ If it is mandatory to check-in more than 60 minutes before an airline (as verified either on ticket documentation or airline website), the mandatory check-in time shall be used.

² Long Haul is defined as outside Schengen and >5 hr scheduled flight time.

When positioning by Hire Car, Chauffeured Car or Taxi, the journey time shall be calculated using <https://maps.google.com> (using the 'quickest' option) between the journey start point and journey end point (for areas not covered by Google Maps, an appropriate alternative shall be used). The calculated journey time should allow for the hour of the day, road construction works, weather conditions, method of transport, etc.

When positioning via airline during a tour and no other duty is planned:

- The duty period shall normally end 30 minutes after the scheduled airline arrival time.
- If the airline is delayed and will depart > 30 minutes after the scheduled departure time, the crew member shall notify Scheduling by email of the expected delay **prior to departure** and provide updates on the revised ETA where possible. On arrival, the crew member shall advise Scheduling by email of the actual on-blocks time of the airline and the duty period shall end at that time.
- Time spent travelling to the resting location after the duty period ends shall be considered 'travelling time' and [7.2.21.6](#) shall apply.

When positioning via airline to Home Base and no further duty will be planned for that tour:

- The duty period shall normally end at the scheduled arrival time of the airline.

- b. If the airline is delayed and the crew member advises the Scheduling department by email, then the duty period shall end at the actual on-blocks time of the airline.

7.2.20 End of Duty Period Following an FDP

A Duty Period terminates 30 minutes after blocks-on of the last sector.

If more time is required or is anticipated to be required, crew members shall use the Duty End time Extension function, in the waypoint application on the company mobile device, to notify duty closing extensions up to 30 minutes. If more time is required, the Commander or respective crew member shall then call / e-mail the Scheduling department within the standard closing time plus extensions (if used).

If the Duty End time Extension function in the waypoint application is not available, the scheduling department shall be informed by email or telephone.

If the procedure above is not followed, closing times will not be adjusted retroactively.

NOTE

When using the Duty End Time Extension function in the waypoint application, each crew member is required to extend the closing time as the application does not adjust the times for the other crew members that are part of the crew.

7.2.21 Rest

7.2.21.1 Rest Conditions

(EU-OPS and PORTUGUESE DECREE LAW no.139/2004)

Rest shall be in a suitable accommodation.

The crew member is required to program any company electronic devices so that the reception of emails, SMS, PIN, general messages, briefing messages or other messages do not interrupt their rest period when so required. However, the company mobile phone shall remain on, with phone call notifications activated, so that the crew member is contactable in case of a briefing change in accordance with 7.2.2.2 Changes to a Briefing

Reading a briefing that is sent to the company's mobile phone during the rest period is a prerogative of the crew member, but is not required by the company. A crew member who reads the briefing on the company's mobile device is equivalent to it being acknowledged.

If the briefing is not accepted, the crew member shall contact the Scheduling department within a maximum of 30 minutes after reading the briefing. A crew is only considered briefed if each individual scheduled crew member has read the briefing. A crew member-initiated telephone contact with the company is not considered an interruption of rest, and a briefing could be given during that phone contact.

7.2.21.2 Minimum Rest Period*(EU-OPS and PORTUGUESE DECREE LAW no.139/2004)*

Minimum rest shall be equal to 11 hours or the duration of the previous duty period, whichever is longer, and is reducible to 10 hours (reference [7.2.21.5](#)).

The 'previous duty period' as described above can be separated from the rest period by a time-off period (time-off is neither considered rest nor duty and therefore is not part of any rest / duty period).

Minimum rest following a standby period not followed by an FDP or DP is 11 hours reducible to 10 hours (reference [7.2.21.5](#)).

Standby at home or at a suitable rest location followed by an FDP or DP is counted 25% as duty for calculating the minimum rest period.

7.2.21.3 Weekly Rest Period*(EU-OPS)*

The minimum rest provided as outlined above is increased periodically to a weekly rest period, being a 36-hour period including two local nights, such that there shall never be more than 168 hours between the end of one weekly rest period and the start of the next. The second of those local nights may start from 20:00 hours if the weekly rest period has a duration of at least 40 hours. The weekly rest period may not be reduced.

7.2.21.4 Rest Period Increase Due to Time Zone Crossings*(Portuguese Decree Law no.139/2004)*

If the time zone of the location where the FDP started and the time zone of the location where the next rest period is taken exceeds or is equal to three time zones, minimum rest, as referred to in [7.2.21.2](#) shall be according to the table below.

	Time Zones				
	3	4	5	6	>6
Minimum rest	14	14:30	15	24	24

7.2.21.5 Commander's Discretion to Increase / Reduce a Rest Period (On Tour)

(EU-OPS)

In the case of unforeseen circumstances occurring after the FDP started, an aircraft Commander may, after consideration of their and their crew's fatigue levels, increase or reduce the rest requirements as detailed below.

Unforeseen reasons could be (but are not limited to):

- i. Technical failure on any aeroplane on the NTA AOC or any aeroplane hired by NTA to perform its flights which requires recovery by the crew exercising discretion.
- ii. Any ATC delay or imposed restriction.
- iii. Denial of parking positions by the airport authorities.
- iv. Weather related phenomena.
- v. Late arrival of passenger.
- vi. A new booking received after crew show time.

1. Rest Reduction

The rest period following the FDP may be reduced up to a maximum of 2 hours but never be less than 10 hours, as long as the previous rest period was not reduced. The total rest reduction shall be added to the next rest period which shall not be reduced.

A. Consent Requirement

- i. If the Commander and the crew (in part or as a whole) are staying together for the subsequent FDP, then Scheduling shall contact the Commander prior to the start of the rest period to establish whether a rest reduction is acceptable.
- ii. If a crew member will be joining a crew with a different Commander on the subsequent FDP, then that crew member shall have the authority to decide whether a rest reduction is acceptable to them. In this case, Scheduling shall contact the crew member to establish whether the rest reduction is acceptable.
- iii. In the case of ii. above, the crew member, at their earliest convenience shall advise the Commander of the new crew of their rest reduction, no later than the applicable show time.
- iv. In the case of ii. & iii. above, should the Commander of the new crew consider the crew member who reduced rest is not fit for the planned duty, they shall reduce the FDP in accordance with the provisions [7.2.18](#).

NOTE

There is no requirement for a discretion or fatigue report when a rest reduction is declined prior to the affected rest period.

B. Reporting Requirement

- A PIC Discretion Report shall be submitted by the Commander no later than 48 hours after the event via the link provided by General Message or email reminders. If the Commander does not receive this reminder, they shall send a PIC Discretion Report to lis-PICDiscretion@netjets.com.

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- Where a crew member who has reduced rest joins a new crew as per A ii. above, it shall be the Commander of the new crew who shall file the PIC Discretion Report.
- Where the reduction in rest period exceeds one hour, a copy of the report sent by the Commander shall be sent to ANAC no later than 28 days after the event.
- The Company shall add other comments to the Commander's report as applicable.

2. Rest Increase

In case of special circumstances that could lead to severe fatigue, and after consultation with the crew members affected, the Commander shall increase the actual rest period in order to eliminate any detrimental effect on flight safety.

The Commander shall notify NTA immediately via the email address Lis-Fatigue@netjets.com.

Each individual crew member shall also file a Fatigue Report (accessed via the safety reporting system) within 48 hours of the rest increase. If, however, an individual member of the crew is not fatigued then they may notify NTA via the email address Lis-Fatigue@netjets.com and may then be assigned duty after a normal or reduced rest period.

The additional rest provisions of [7.2.28 Fatigue Reporting](#) will also apply as applicable.

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7.2.21.6 Transportation Times to and from Rest Locations (Travelling Time)*(EU-OPS)*

Travelling time is not counted as duty. However, where travelling time is excessive, to ensure adequate time at the resting location, the applicable rest period is increased as follows:

Travelling time between duty location and rest location	
Calculated or Actual travelling time	Rest period increase (per journey)
30 min or less	None
> 30 min	Calculated or Actual travelling time minus 30 min

Calculated travelling time shall be established prior to the start of the journey as described below.

Calculated travelling time shall be equal to, or exceed, the time calculated using <https://maps.google.com> (using the 'quickest' option) between the aerodrome exit point and the hotel and vice versa (for areas not covered by Google Maps, an appropriate alternative shall be used). The calculated travelling times should allow for the hour of the day, road construction works, weather conditions, method of transport, etc.

NOTE

1. If transportation from the duty end location to the suitable accommodation is not available at the end of the duty period, the time between the end of the duty period and the transportation being available is considered to be travelling time and shall be accounted for as described in this subsection.
2. Travelling time greater than 30 minutes or additional travelling time as defined in [7.2.21.6.1](#), is not considered 'duty', but is added to the crew member's rest period so as to ensure they will have the same amount of time at the rest location as they would have had if the travelling time had been 30 minutes.

7.2.21.6.1 Availability of suitable accommodation at the rest location

If, within 15 minutes of arriving at a rest location, suitable accommodation (typically a hotel room) is not available, then the time between arrival at the rest location and the suitable accommodation being available will be additional travelling time, and the minimum rest period will be increased by this amount.

Additionally, if upon arriving at the rest location it becomes apparent that a change to a different rest location will be required (e.g. unsuitable rest location, lack of room availability, etc.), then the time between arriving at the first rest location and arriving at the rest location where the crew will take their rest period will be additional travelling time and the minimum rest period will be increased by this amount.

7.2.22 Minimum Rest / Maximum DP Before Proceeding to the Gateway

- a. If the last airline to the gateway is scheduled:
 - i. When arriving on an NTA aircraft to the airport from which that airline will depart, a minimum of 2 hours shall be scheduled between the estimated on block time of the NTA aircraft and the scheduled departure time of that airline.
 - ii. When arriving on an NTA aircraft to an airport other than where that airline will depart, a minimum of:
 - 30 minutes between the estimated on block time of the NTA aircraft and the departure time of the ground transport; and
 - 1 hour and 30 minutes between the arrival of the ground transport at the airport from which that airline will depart, and the scheduled departure time of that airline, shall be scheduled.
- b. The crew member shall be scheduled to arrive before 23:59 (gateway time) at the gateway on the last rostered day of a tour.
- c. The maximum planned duty period on the last day of a rostered tour shall be as follows, unless the crews consent has been obtained to exceed i. or ii. as applicable:
 - i. 16 hours, if it includes an FDP as per [7.2.10.1](#);
 - ii. 18 hours, if it includes an FDP as per [7.2.10.2](#);
 - iii. a DP, which does not include an FDP and has been preceded by a minimum of 11 hours rest (reducible to 8 hours by agreement of the crew member) between the previous DP and the start of the current DP, is not limited.
- d. If on the last rostered day of tour the DP exceeds the limits given in c.i. or c.ii. above as applicable either due to the crew members consent at the planning stage or due to unforeseen operational reasons during the DP which includes an FDP, the crew member may elect to:
 - i. Rest at the location where the last duty period ended for a minimum of 8 hours at a suitable resting location and proceed to the gateway following those 8 hours. Accommodation shall be provided by the Company and a per diem shall be paid;
 - ii. Proceed to the gateway following the duty period and rest at suitable accommodation for a minimum of 8 hours at Company expense before travelling;
 - iii. Position to gateway following the duty period and request ground transportation at the gateway, at the expense of the Company, up to one hour normal travelling time;
 - iv. Position to gateway following FDP and self-arrange a pickup where the crew member is not actively operating the means of transportation.
- e. If a crew member is scheduled to proceed to their gateway to end a tour prior to the last rostered day of the tour or to be assigned home standby, then paragraph d. above applies.
- f. Without prejudice to the paragraphs above, upon arrival at the gateway, if a crew member feels too fatigued to travel home, they may request that the company provides suitable accommodation in order to rest prior to proceeding home.

Late-Home schedule duty is when, due to unforeseen NTA operational circumstances or by prior agreement between scheduling and the crew member, a crew member's planned return to the gateway is beyond midnight of the final rostered duty day.

The Late-Home Schedule duty shall not include any flight duty and shall end prior to 1200 gateway local time.

The Weekly Rest Period requirement is still applicable.

7.2.23 Non-flying Duty

(EU-OPS)

Non-flying duties are to be counted 100% towards the limits of this chapter.

NOTE

Personnel assigned primarily to office duties that also perform duties as a crew member shall inform the Scheduling department of the total time they performed duties or working time after the last FDP ended, the latest before the rest period preceding the FDP. The Scheduling department shall add these times to the times limited by this chapter.

7.2.24 Crew Member Off-days

(PORTUGUESE DECREE LAW no.25/2022)

The crew member will have a minimum of:

- 7 local days off in each calendar month;
- 96 local days off in each calendar year.

7.2.25 Flight Duty for Other Entities or Personal Purposes

(EU-OPS)

The provision of any type of duty for another entity is subject to an agreement by NTA. A crew member that exercises duty period outside of NTA shall inform NTA of the flight duty or duty activity, to include details of flight duty periods, show-times, end times, and total flying times in order to comply with the provisions of this chapter.

If performing flying duties for other entities or personal purposes, they shall not limit the availability of the crew member for NTA and permission shall be requested in advance in writing to lis-crewsupport@netjets.com.

This is applicable to a crew member that performs duty:

- On any aeroplane whether it is for pleasure or not;
- All flight instruction regardless of the mass of the aeroplane;
- Simulator instruction; and
- All military flying activity.

No crew member is allowed to perform any duty or flight duty as described above within 36 hours, including 2 local nights, of the first day of a rostered tour.

A crew member that performs duty and flight duty outside of NTA as described above, with exception of performed flying duties on aeroplanes below 2750 kg for pleasure, shall send a detailed report of the duty and flight duty to Lis-FlyingoutsideNTA@netjets.com, including the following information as a minimum:

- Name;
- NTA Fleet;
- Date of duty period and flight duty period;
- Start time of the duty period;
- End time of the duty period;
- Total duty time;
- Total flight time.

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Sample Report

NAME:					
NTA FLEET:					
<i>Note: One row per duty period.</i>					
START OF DUTY		END OF DUTY		TOTAL DUTY	TOTAL FLIGHT TIME
DATE	TIME	DATE	TIME		

This report shall be sent no later than 36 hours prior to the start of first day of a rostered tour.

7.2.26 Vacation

(PORTUGUESE DECREE LAW no.25/2022)

Crew members are entitled to a minimum of 22 days of vacation per year. Full details may be found in NJCSP 1.01 - Crew Vacation Procedure.

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7.2.27 Nutrition Opportunity

(EU-OPS)

Where the FDP exceeds 6 hours, an opportunity to eat shall be scheduled. An opportunity to eat is:

- a. 30 minutes free of all duties;
- b. A flight where the IJet block time is equal or longer than 1.7 hours.

With the consent or at the request of the crew, a. and b. above may be reduced or waived provided the crew member is content that the planned schedule allows an adequate opportunity to eat.

The opportunity to eat will not be explicitly scheduled or briefed, but when the minimum required scheduled show or turn time is exceeded by 30 minutes or IJet block time is 1.7 hours or greater, this automatically adheres to a. and b. above.

Food should be available to the crewmember at the beginning of the opportunity to eat, crew should liaise with catering and the FOSM to ensure this will be the case.

This opportunity to eat shall be scheduled to begin no earlier than 3 hours after the reporting time. It may be less with the consent or at the request of the crew.

7.2.28 Fatigue Reporting

If a crew member considers they are, or will be, unfit to fly due to fatigue level, they shall notify NTA immediately via the email address Lis-Fatigue@netjets.com. The respective crew member shall also file a Fatigue Report (accessed via the safety reporting system) within 48 hours of reporting fatigued.

Rest Period after Reporting Fatigued

After reporting fatigued, the minimum rest period in order to recover from the fatigue level experienced will vary with each individual since each situation is different. In order to ensure that fatigue risk is mitigated as far as possible and to allow for the planning of the following duty period, the following applies:

- a. Unless the crew member assesses that they need additional rest time than stipulated below, the rest period after reporting fatigued shall be as follows:
 - The minimum rest period shall be in accordance with [7.2.21.2](#), except that NO rest reduction is authorised;
 - The rest period shall include a full WOCL period;
 - The show time for the duty period after a crew member reported fatigue shall not be before the end of the applicable WOCL period plus 2 hours (not applicable if the crew member reports fatigued on the last day of the tour). If the rest period includes two full WOCL periods, then the additional 2 hour requirement does not apply.
- b. If the crew member assesses that they need additional rest time, at the time of reporting fatigued (via Lis-Fatigue@netjets.com) they shall inform the company of their planned rest time requirement.
- c. If during the rest period a crew member assesses that the rest period will not be sufficient and that they require more, they shall advise the company via Lis-Fatigue@netjets.com.

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7.3 Establishment of Time Zone

The time zone(s) that are published in the Jeppesen airport directory shall be used for all provisions in this chapter that refer to or are dependent on time zones.

These are the official time zones that are published by the respective country's AIP and that are used in IJet.

Local time(s) shall not be used for any time zone calculations.

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Chapter 8 - Operating Procedures

8.1 Flight Preparation Instructions

NTA follows a process of shared responsibility between Flight Dispatch and the Commander of a flight. It is their responsibility to ensure that the flight is planned and only conducted along routes and within areas that meet all:

- Safety;
- Legal; and
- NTA requirements.

Flight preparation shall include:

- a. The preparation of an OFP considering all aspects such as aeroplane performance, minimum flight altitudes, routing, weather forecasts for en-route, destination and alternate airports, fuel planning, etc.;
- b. The preparation of an ATS flight plan;
- c. The preparation of a loadsheet (and trimsheet, if applicable);
- d. The determination of the usability of aerodromes. This includes the evaluation of possible performance limits, and the calculation of take-off and landing speeds;
- e. The relevant AIS briefing - either by appropriate AIS documentation, personal AIS briefing, or by an aeroplane dispatcher;

The primary source of NOTAMs shall be that sent by NTA. If another source is referred to and relevant discrepancies are noted then dispatch shall be contacted.

Department of Defence (DOD) NOTAMs

Any NOTAM with the prefix DOD PROCEDURAL NOTAM is applicable to military operations only and shall be disregarded for NTA operations.

- f. The relevant MET briefing - either by MET documentation, personal MET briefing, or by an aeroplane dispatcher;
- g. A check on whether 'special loads' such as dangerous / hazardous goods or heavy cargo are to be carried, and whether safety-related handling instructions are being followed;
- h. Ensuring that commercial and / or other company requirements are met;
- i. A check on whether de-icing / anti-icing procedures - if necessary - have been carried out properly;
- j. Ensuring availability of the appropriate maps and charts;
- k. Ensuring that the ground facilities and services, including meteorological services, adequate for the planned operation are provided;
- l. Ensuring that the performance of the aeroplane is adequate to comply with the minimum flight altitude requirements;
- m. Ensuring that the aeroplane equipment meets the minimum requirements for the planned operation;

- n. Checking the technical status of the aeroplane, its components, and equipment;
- o. Ensuring that operations are conducted in accordance with any restriction on the routes or the area of the operations specified by the competent authority;
- p. Checking the relevant IJet Feasibility Notices - either on the company mobile device or faxed / emailed copy.

A flight shall not be commenced unless the commander is satisfied that:

- a. The aeroplane is airworthy;
- b. The aeroplane is duly registered and that appropriate certificates with respect thereto are aboard the aeroplane;
- c. Instruments and equipment required for the execution of that flight are installed in the aeroplane and are operative, unless waived by the applicable Minimum Equipment List (MEL) or equivalent document;
- d. The mass of the aeroplane and centre of gravity location are such that the flight can be conducted within limits prescribed in the airworthiness documentation;
- e. All cabin baggage, hold luggage, and cargo is properly loaded and secured; and
- f. The aeroplane operating limitations will not be exceeded at any time during the flight;
- g. The aeroplane is not operated contrary to the provisions of the CDL;
- h. The parts of the operations manual that are required for the conduct of the flight are available;
- i. The documents, additional information, and forms required to be available are on board;
- j. Current maps, charts, and associated documentation or equivalent data are available to cover the intended operation of the aeroplane, including any diversion that may reasonably be expected;
- k. Space-based facilities, ground facilities, and services required for the planned flight are available and adequate;
- l. Any navigational database required for performance-based navigation shall meet the requirements specified in [8.3.2.6](#);
- m. The provisions specified in the operations manual in respect of fuel, oil, oxygen, minimum safe altitudes, aerodrome operating minima and availability of alternate aerodromes, where required, can be complied with for the planned flight; and
- n. Any additional operational limitation can be complied with.

By signing the OFP, the commander acknowledges that the items above are complied with.

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8.1.1 Minimum Flight Altitudes

No flight, except for the take-off / departure or the approach / landing or descending in accordance with the procedures approved by the competent authority, shall be operated below the minimum flight safe altitudes as described below. This includes any foreseeable contingencies along the planned route.

NOTE

Local regulations may require higher minimum flight altitudes (for State deviations, see Jeppesen Airway Manual, Section AIR TRAFFIC CONTROL).

When establishing the minimum flight altitudes, the following shall be considered:

- The accuracy with which the position of the aeroplane can be determined;
- The probable inaccuracies in the indications of the altimeters used;
- The characteristics of the terrain, such as sudden changes in the elevation, along the routes or in the areas where operations are to be conducted;
- The probability of encountering unfavourable meteorological conditions, such as severe turbulence and descending air currents; and
- Possible inaccuracies in aeronautical charts.

8.1.1.1 On VFR-Flights or on VFR Flights Segments of IFR-Flight

An altitude where the flight path clears all obstacles or any terrain by at least 1000 ft vertically. Whenever cities or other densely populated areas are overflown, then the minimum vertical clearance shall be 2000 ft above the highest terrain / obstacle within a radius of 600 m from the aeroplane.

8.1.1.2 On IFR-flights

An altitude which clears all obstacles by at least 1000 ft vertically over terrain / obstacles within a radius of 8 km from the estimated position of the aeroplane. The estimated position of the aeroplane will take account of the navigational accuracy which can be achieved on the relevant route segment, having regard to the navigational facilities available on the ground and in the aeroplane.

For the practical application of those basic requirements (ICAO Annex 2), the following terminologies are used:

8.1.1.2.1 Minimum En-route IFR Altitude (MEA)

The lowest published altitude between radio fixes that meets obstacle clearance requirements between those fixes and in many countries assures acceptable navigational and radio signal coverage. The MEA applies to the entire width of the airway, segment, or route between the radio fixes defining the airway, segment, or route. The published MEA's provide a terrain clearance of at least 1000 ft (over high terrain: 2000 ft) over the highest obstacle within the normal airway width (5 nm each side of the airway centreline). On the Jeppesen En-route charts, the MEA is given without any suffix (except as an m if given in metric).

As a company policy, the MEA shall be used as the minimum altitude for both flight planning and actual flight execution.

Most flight operations will be carried out by using the flight level system. The cruising levels to be observed - even or odd levels - are either published in the en-route charts / AIPs or assigned by ATC and shall meet the ICAO Annex 2 requirements.

8.1.1.2.2 Minimum Off-route Altitude

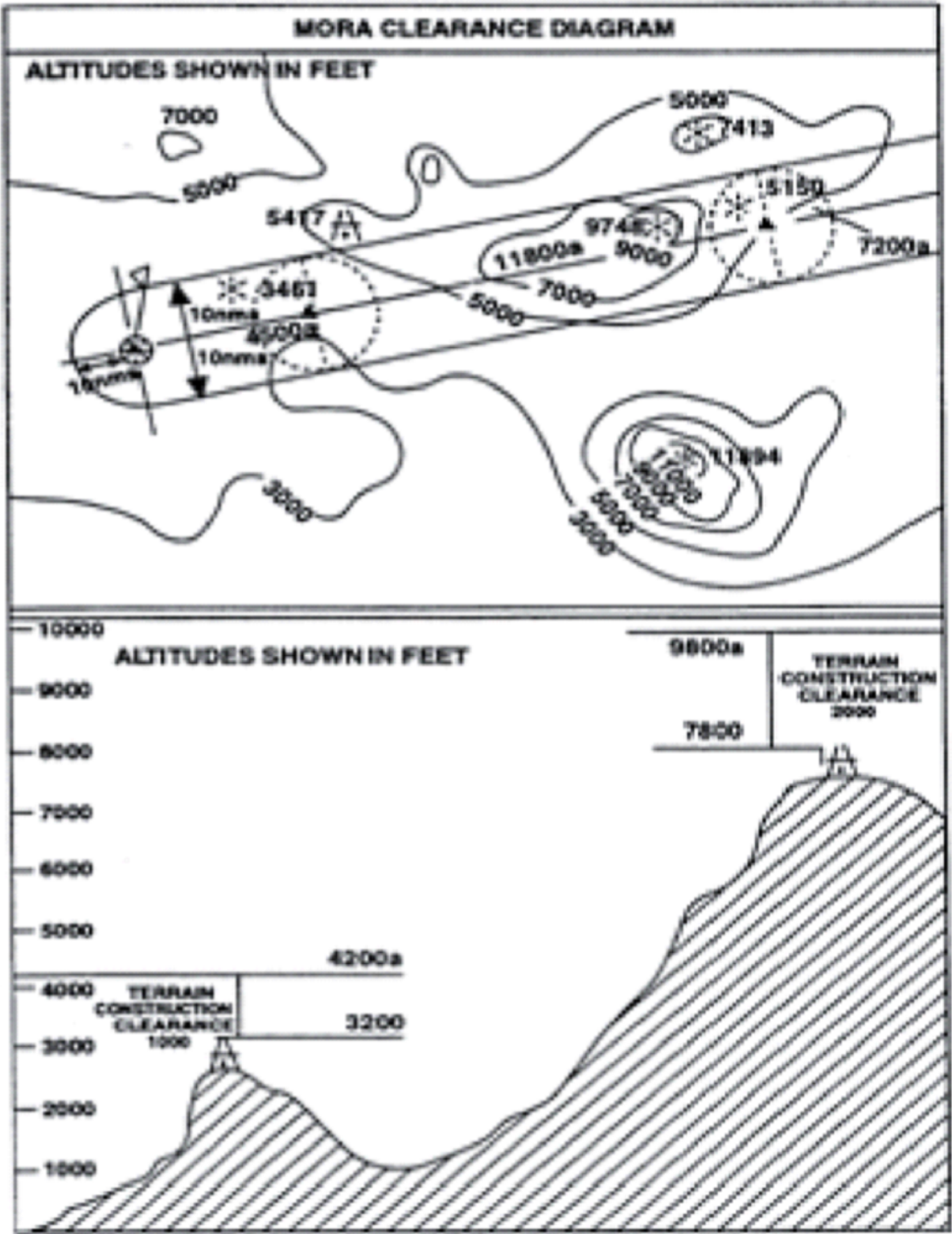
MORA is a minimum flight altitude computed by Jeppesen from current operational navigation charts (ONCs) or world aeronautical charts (WACs). Two types of MORAs are charted, which are:

- i. route MORAs, e.g. 9800a; and
- ii. grid MORAs, e.g. 98.

MORA values clear all terrain and man-made obstacles by 1000ft in areas where the highest terrain elevation or obstacles are up to 5000 ft. A clearance of 2000 ft is provided above all terrain or obstacles that are 5001 ft and above.

Route MORA values are computed on the basis of an area extending 10 nm to either side of route centreline and including a 10 nm radius beyond the radio fix / reporting point or mileage break defining the route segment.

A grid MORA is an altitude computed by Jeppesen and the values are shown within each grid formed by charted lines of latitude and longitude. Figures are shown in thousands and hundreds of feet (omitting the last two digits so as to avoid chart congestion). Values followed by $\pm$ are believed not to exceed the altitudes shown.



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8.1.1.2.3 Inoperative Engines - Performance Requirements

In addition to the minimum flight altitude requirements stated above, aeroplanes operated to performance Class A standards (i.e. NTA aeroplanes) shall meet the following performance requirements.

The SID ends at a point no lower than the MEA and usually above it. The MEA does not only provide a 1000 ft or 2000 ft vertical obstacle clearance, as applicable, but normally also assures that COM / NAV signals are received so as to ensure track guidance. From the basic performance policy, it is evident that the MEA - or MRVA (minimum radar vectoring altitude) - shall be maintainable with one engine out. A driftdown procedure shall be developed and observed in case the aeroplane gross weight is such that the single engine service ceiling is below MEA / MRVA or MORA.

A. One-engine Inoperative (OEI) En-route Flight Path

- a. The OEI en-route net flight path data shown in the AFM, appropriate to the meteorological conditions expected for the flight, shall allow demonstration of compliance with b. or c. at all points along the route. The net flight path shall have a positive gradient at 1500 ft above the aerodrome where the landing is assumed to be made after engine failure. In meteorological conditions requiring the operation of ice protection systems, the effect of their use on the flight path shall be taken into account.
- b. The gradient of the net flight path shall be positive at least 1000 ft above all terrain and obstructions along the route within 9.3 km (5 nm) on either side of the intended track.
- c. The net flight path shall permit the aeroplane to continue flight from the cruising altitude to an aerodrome where a landing can be made in accordance with the landing performance requirements.

The net flight path shall clear vertically, by at least 2000 ft, all terrain and obstructions along the route within 9.3 km (5 nm) on either side of the intended track in accordance with the following:

1. the engine is assumed to fail at the most critical point along the route;
 2. account is taken of the effects of winds on the flight path;
 3. fuel jettisoning is permitted to an extent consistent with reaching the aerodrome with the required fuel reserves, if a safe procedure is used; and
 4. the aerodrome where the aeroplane is assumed to land after engine failure shall meet the following criteria:
 - i. the performance requirements at the expected landing mass are met; and
 - ii. weather reports and / or forecasts and field condition reports indicate that a safe landing can be accomplished at the estimated time of landing.
 5. if the AFM does not contain en-route net flight path data, the gross OEI enroute flight path shall be reduced by a climb gradient of 1,1% for two-engined aeroplanes or 1,4 % for three-engined aeroplanes.
- d. The width margins of b. and c. shall be increased to 18.5 km (10 nm) if the navigational accuracy does not meet at least required navigation performance 5 (RNP5).

8.1.2 Aerodrome Selection Policy - Adequacy Of Aerodromes

8.1.2.1 General

1. Operation into any aerodrome is subject to prior approval from Flight Operations. As a general policy, an operation to or from an aerodrome will only be permitted provided that standard operating procedures can be used. Such procedures shall apply not only to the approach / landing and take-off phase, but shall also cover all forms of ground handling and operation. In approving an aerodrome for Company operations, the following aspects shall be considered:
 - a. Aerodrome dimensions with regard to performance requirements;
 - b. Obstacle situation in the approach;
 - c. Missed approach and departure sectors approach facilities (electronic and visual aids);
 - d. Local conditions such as special weather situations;
 - e. Night flying restrictions or even political aspects which might affect operations;
 - f. Ground service facilities for fuelling, loading, de- / anti-icing, fire fighting and rescue, catering, general handling, and the availability of immigration authorities;
 - g. Fuel freeze considerations.

The table below states the ICAO Aerodrome Reference Code for NTA aeroplane types:

Aeroplane Type	Aerodrome Reference Code
EMB-505	2B
CE-560XLS	2B
CE-680A	2B
BD-100-1A10	3B
DA-2000EASy	3B
CL-600-2B16	3B
BD-700-1A10	4C
BD-700-1A11	4C

The aerodrome reference code is determined by:

1. (Code number) Aeroplane reference field length; and
2. (Code letter) Aeroplane wingspan and outer main gear wheel span.

The essential information to note is the code letter, as it is often used to determine taxiway restrictions (e.g., a taxiway is restricted to aircraft code B). These restrictions may be indicated on Jeppesen aerodrome charts, taxi charts, or in the aerodrome briefing pages.

The aerodrome approval process is detailed in NTA-FO-PR-001_Aerodrome_Initial_Approval and NTA-FO-PR-002_Aerodrome_Approval_Review.

8.1.2.1.1 Isolated Aerodromes

An isolated aerodrome is one for which the alternate and final reserve fuel required to the nearest adequate destination alternate aerodrome is more than the fuel required to fly for 2 hours at normal cruise consumption above the destination aerodrome, including final reserve fuel.

The use of an isolated aerodrome as a destination aerodrome requires prior approval from the competent authority.

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8.1.2.2 Rescue and Fire Fighting Services (RFFS)

1. ICAO-Annex 14 specifies the aerodrome requirements for rescue and fire fighting services. Different aerodrome categories are laid down, depending upon the aeroplane's overall length, max fuselage width, and the number of movements of individual aeroplane types.
2. For explanation of Airport Category for Rescue and Fire Fighting, refer to the AIP and / or Jeppesen Airway Manual, 'AIRPORT DIRECTORY', Legend and Explanation.
3. It is NTA policy not to operate to aerodromes with inadequate fire and crash facilities and, when assessing suitability, the following factors are taken into account:
 - The RFFS category for each of the aerodromes on the operational flight plan for a given flight should be equal to or better than the aeroplane RFFS category.
 - Operations at aerodromes and selection of alternate aerodromes with RFFS categories below that of the aeroplane category to the minimum categories stated in the table below, may be authorised only after an NTA internal procedure including a risk assessment has determined that the available RFFS category is acceptable.

Aeroplane Type	Aeroplane RFFS Category	Minimum Aerodrome RFFS Category	Minimum Aerodrome RFFS Category (Temporary Downgrade ⁽¹⁾)	Minimum Aerodrome RFFS Category for operations without Passengers
EMB-505	3	2 ⁽²⁾	1	1
CE-560XLS	3	2 ⁽²⁾	1	1
CE-680A	4	2	2	2
BD-100-1A10	4	2	2	2
DA-2EASy	4	2	2	2
CL-600-2B16	4	2	2	2
BD-700-1A10	6	3	3	2
BD-700-1A11	6	3	3	2

NOTE

1. **Temporary Downgrade:** RFFS category as notified, including by NOTAM, and resulting from the downgrade of the level of RFFS protection available at an aerodrome, for a period of time not exceeding 72 hours.
2. The approval of an Alternate Aerodrome with a RFFS category of 1 may in exceptional cases be authorised only after an NTA internal procedure has determined that the available RFFS category is acceptable.

4. In flight, the pilot-in-command may decide to land at an aerodrome regardless of the RFFS category if, in the pilot's judgement after due consideration of all prevailing circumstances, to do so would be safer than to divert.

Aerodromes with reduced or inadequate facilities will accept an aeroplane making an emergency landing or a landing where the commander judges that a diversion or holding delay may be a greater potential hazard.

8.1.2.3 Aerodrome Categorisation for Flight Crew Competence

Aerodromes for NTA operations are categorised in ascending order of difficulty, from Category A to Category C, according to their characteristics, as listed below. In some cases, aerodromes might be categorised differently for different aeroplane types (e.g. due to aeroplane size or equipment fit). Additionally, others might change categorisation for different periods (e.g. summer vs winter). Such variations in categorisation will be detailed in the applicable aerodrome briefing.

Commanders shall ensure that their aerodrome qualifications fulfil the briefing and / or recency requirements before operating to Category B or Category C aerodromes.

Category A

An aerodrome which satisfies all of the following requirements:

- A straight-in 3D instrument approach procedure with a glide path angle of not more than 3.5 degrees to each runway expected to be used for landing;
- At least one runway with no performance limited procedure for take-off and / or landing, such as no requirement to follow a contingency procedure for obstacle clearance in the event of an engine failure on take-off from any runway expected to be used for departure; and
- Night operations capability.

Category B

An aerodrome which does not satisfy the Category A conditions or which requires extra considerations due to:

- Non-standard approach aids and / or approach patterns, such as restrictions on the availability of straight-in instrument approach procedures;
- Unusual local weather conditions, such as environmental features that can give rise to turbulence, windshear or unusual wind conditions;
- Unusual characteristics or performance limitations, such as unusual runway characteristics in length, width, slope, markings or lighting that present an atypical visual perspective on approach;
- Any other relevant considerations including obstructions, physical layout, lighting, etc., such as restrictions on circling in certain sectors due to obstacles in the circling area;
- Training or flight crew experience requirements stipulated by the competent authority responsible for the aerodrome that do not include instruction in an FSTD or visiting the aerodrome.

Prior to operating to a Category B aerodrome, the pilot-in-command shall:

1. Comply with any requirements stipulated by the competent authority responsible for the aerodrome; and
2. Be briefed, self-briefed by means of programmed instruction, about the additional considerations applicable to operations to that aerodrome or by self-briefing of the appropriate sections of the Jeppesen Manuals if the NTA aerodrome briefing is not published for the aerodrome(s) concerned. The commander certifies that these instructions have been carried out by signing the OFP prior to departure.

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Category C

An aerodrome is categorised as category C when:

- It requires considerations in addition to those for a Category B aerodrome.
- Flight crew experience or qualification requirements stipulated by the competent authority responsible for the aerodrome include instruction in an FSTD or visiting the aerodrome.

Prior to operating to a Category C aerodrome, the Commander shall ensure that all requirements below are met:

1. Meet any requisites stipulated by the competent authority responsible for the aerodrome;
2. Have received a briefing or have self-briefed by means of programmed instruction, about the additional considerations applicable to operations to that category C aerodrome;
3. Have completed the appropriate training or familiarisation and be current in accordance with the recency requirements as detailed in the OM-C;
4. Hold a current Advanced Handling Qualification (AHQ).

8.1.2.4 Performance Considerations Performance Class A Aeroplanes

(CAT.POL.A.100)

8.1.2.4.1 General

The performance regulations applicable for Performance Class 'A' under PART-CAT shall be complied with. It states, as a basic principle, that the flight shall be dispatched in such a way that in the event of an engine failure the aeroplane can clear all obstacles throughout all phases of flight by the margins prescribed by the regulations.

Take-off and landing masses (weights), stated in the PDCS for the applicable runway, shall not be exceeded.

- a. The mass of the aeroplane:
 1. at the start of the take-off, or
 2. in the event of in-flight replanning, at the point from which the revised operational flight plan applies, shall not be greater than the mass at which the requirements of [8.1.2.4](#) can be complied with for the flight to be undertaken. Allowance may be made for expected reductions in mass as the flight proceeds and for fuel jettisoning.
- b. The approved performance data and supplemental data contained in the AFM / AOM shall be used to determine compliance. Data to be used shall be selected in accordance with the Table 1 below.
- c. Due account shall be taken of aeroplane configuration, environmental conditions and the operation of systems that have an adverse effect on performance. ¹
- d. Charting accuracy shall be accounted for when assessing the take-off requirements.

¹ See [Technical Annex](#) for how to relate RSD to a RWYCC.

Table i. - Definitions and Terminology

ALT RWYCC	Alternative Runway Condition Code	Logic similar to RWYCC. The ALT RWYCCs were derived together with NJE OEMs and represent a conversion of non-GRF landing performance datasets to GRF format. The objective is to facilitate the assessment of RCR on legacy performance fleets. These are labelled as 'ALT RWYCC X' on PDCS.
GRF	Global Reporting Format	A set of rules and guidelines to facilitate and uniformise the dissemination of information regarding runway actual landing performance and contamination.
OEM	Original Equipment Manufacturer	
RCC (or RWYCC)	Runway Condition Code	Numerical code identifying the expected landing performance on a certain runway, ranging from 6 to 0. Landing performance usually degrades when moving from 6 to 0. For those OEMs that provide landing performance datasets compliant with GRF, PDCS will be labelled as 'RWYCC X'.
RCR	Runway Condition Report	A report issued by an aerodrome describing the performance status of a runway. A runway condition report issued under GRF format usually contains a section with Runway Condition Codes (RCC), Contaminant Depth and Runway Surface Descriptors (RSD). The report divides the runway in thirds.
RSD	Runway Surface Descriptor	Text descriptor of the main contaminant present on the runway surface (e.g. SLUSH, DRY SNOW, ICE, ...).
Legacy Performance Feet	A fleet that does not have or partially has complete performance datasets published under GRF format.	
Runway Condition Upgrade	A RCR where the actual landing performance is deemed by the airport to be better than the reported RSD.	
Runway Condition Downgrade	A RCR where the actual landing performance is deemed by the airport to be worse than the reported RSD.	

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Table 1 - Instructions to obtain performance data

Item to determine	Instructions
Take-off performance	Use legacy performance data based on reported RSD (Runway Surface Descriptor)
Dispatch assessment of landing performance at destination and destination alternate	Establish data to be used from Table A
In flight assessment of landing distance at time of arrival	Establish data to be used from Table B
Crosswind Limitation according to Crosswind Operation	In accordance with Tables C and D

Table A- Data for Dispatch assessment

Dispatch assessment of landing performance at destination and destination alternate			
Runway reports include a RWYCC	OEM ¹ RWYCC Data for Dispatch	Performance data matching RSD - Upgrade or downgrade	Data to be used
Y	Available		Dispatch RWYCC Data
	Not available	RSD Perf data available AND No upgrade or downgrade of RWYCC	Legacy performance data based on reported RSD
		RSD Perf Data NOT available	'ALT RWYCC X'[3] selection according to reported RWYCC.
		RWYCC upgraded / downgraded ²	
N	Available		Derive a RWYCC using the RSD + braking action [4], then use Dispatch RWYCC Data for the derived RWYCC
	Not available	RSD Perf data available	Legacy performance data based on RSD
		RSD Perf Data NOT available [2]	Derive a RWYCC using the RSD + braking action , then use 'ALT RWYCC X ' ³ selection according to derived RWYCC.

¹ Where RWYCC data is provided for normal approaches but is not valid for Steep approaches, Landing on Wet Grooved Runways or Landing with system failures, it will be necessary to revert back to legacy performance data – see OM-B or Aircraft assumptions document for more details.

² Not applicable to Steep Approaches.

³ For conversions tables behind 'ALT RWYCC X' PDCS selections see Technical Annex

⁴ Use worst case of RSD and Braking Action (if provided) to determine RWYCC – See Technical Annex for more detail.

Table B- Data for FLDTA assessment

In flight assessment of landing distance at time of arrival (FLDTA)			
Runway reports include a RWYCC	OEM ¹ RWYCC Data for Time of Arrival	Performance data matching RSD- Upgrade or downgrade	Data to be used
Y	Available		FLDTA RWYCC Data
	Not available	RSD Perf data available AND No upgrade or downgrade of RWYCC	Legacy performance data based on reported RSD
		RSD Perf Data NOT available	'ALT RWYCC X' ³ selection according to reported RWYCC.
		RWYCC upgraded / downgraded ²	
N	Available		Derive a RWYCC using the RSD + braking action [4] , then use FLDTA RWYCC Data for the derived RWYCC
	Not available	RSD Perf data available	Legacy performance data based on RSD
		RSD Perf Data NOT available [2]	Derive a RWYCC using the RSD + braking action [4], then use 'ALT RWYCC X'[3] selection according to derived RWYCC.

¹ Where RWYCC data is provided for normal approaches but is not valid for Steep approaches, Landing on Wet Grooved Runways or Landing with system failures, it will be necessary to revert back to legacy performance data – see OM-B or Aircraft assumptions document for more details.

² Not applicable to Steep Approaches.

³ For conversions tables behind 'ALT RWYCC X' PDCS selections see Technical Annex.

⁴ Use worst case of RSD and Braking Action (if provided) to determine RWYCC – See Technical Annex for more detail.

Table C- Instructions for determining crosswind limitation

Determining Crosswind Limitation for Contaminated Runways	
Runway reports include a RWYCC	Instructions (see 8.1.3.8.2 for further information)
Y	Use RWYCC on Table D
N	Use Braking action if reported, otherwise convert RSD to RWYCC using table and use this derived RWYCC on Table D

Table D- Contaminated Runway Crosswind Limitation (reproduced from 8.1.3.8.2)

Crosswind Limitation on Contaminated Runway				
RWYCC	Reported Brake Action		Reported Runway Friction Coefficient	Max Cross Wind
	Rep Brake Action (SNOWTAM Index)	EUR-RODEXOPMET		
5	Good (5)	95	0.4 and above	Aeroplane Limit
4	Medium to Good (4)	94	0.36 - 0.39	20
3	Medium (3)	93	0.30 - 0.35	15
2	Medium to Poor (2)	92	0.26 - 0.29	5
1	Poor (1)	91	0.25 and below	Prohibited

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8.1.2.4.2 Take-off

- a. The take-off mass shall not exceed the maximum take-off mass specified in the AFM for the pressure altitude and the ambient temperature at the aerodrome of departure.
- b. The following requirements shall be met when determining the maximum permitted take-off mass:
 1. the accelerate-stop distance shall not exceed the accelerate-stop distance available (ASDA);
 2. the take-off distance shall not exceed the take-off distance available, with a clearway distance not exceeding half of the take-off run available (TORA);
 3. the take-off run shall not exceed the TORA;
 4. a single value of V1 shall be used for the rejected and continued take-off; and
 5. on a wet or contaminated runway, the take-off mass shall not exceed that permitted for a take-off on a dry runway under the same conditions.
- c. When showing compliance with b., the following shall be taken into account:
 1. the pressure altitude at the aerodrome;
 2. the ambient temperature at the aerodrome;
 3. the runway surface condition and the type of runway surface;
 4. the runway slope in the direction of take-off;
 5. not more than 50% of the reported headwind component or not less than 150% of the reported tailwind component (these components are already accounted for in the performance data provided in the AFM / AOM and subsequently in the PDCS data and therefore no additional corrections required by flight crew); and
 6. the loss, if any, of runway length due to alignment of the aeroplane prior to take-off.

8.1.2.4.3 Take-off Obstacle Clearance

- a. The net take-off flight path shall be determined in such a way that the aeroplane clears all obstacles by a vertical distance of at least 35 ft or by a horizontal distance of at least 90 m plus $0.125 \times D$, where D is the horizontal distance the aeroplane has travelled from the end of the take-off distance available (TODA) or the end of the take-off distance if a turn is scheduled before the end of the TODA. For aeroplanes with a wingspan of less than 60 m, a horizontal obstacle clearance of half the aeroplane wingspan plus 60 m, plus $0.125 \times D$ may be used.
- b. When showing compliance with a.:
 1. The following items shall be taken into account:
 - i. the mass of the aeroplane at the commencement of the take-off run;
 - ii. the pressure altitude at the aerodrome;
 - iii. the ambient temperature at the aerodrome; and
 - iv. not more than 50% of the reported headwind component or not less than 150% of the reported tailwind component.

2. Track changes shall not be allowed up to the point at which the net take-off flight path has achieved a height equal to one half the wingspan but not less than 50 ft above the elevation of the end of the TORA. Thereafter, up to a height of 400 ft it is assumed that the aeroplane is banked by no more than 15°. Above 400 ft height bank angles greater than 15°, but not more than 25° may be scheduled.
3. Any part of the net take-off flight path in which the aeroplane is banked by more than 15° shall clear all obstacles within the horizontal distances specified in a., b.6. and b.7. by a vertical distance of at least 50 ft.
4. Application of increased Bank angles of not more than 20° between 200 ft and 400 ft, or not more than 30° above 400 ft, requires prior approval from the competent authority.
5. Adequate allowance shall be made for the effect of bank angle on operating speeds and flight path including the distance increments resulting from increased operating speeds.
6. For cases where the intended flight path does not require track changes of more than 15°, the obstacles that have a lateral distance greater than i) and ii) below do not need to be considered:
 - i. 300 m, if the pilot is able to maintain the required navigational accuracy through the obstacle accountability area; or
 - ii. 600 m, for flights under all other conditions.
7. For cases where the intended flight path requires track changes of more than 15°, the obstacles that have a lateral distance greater than i) and ii) below do not need to be considered:
 - i. 600 m, if the pilot is able to maintain the required navigational accuracy through the obstacle accountability area; or
 - ii. 900 m, for flights under all other conditions.
- c. Contingency procedures to satisfy the requirements in a. and b. and to provide a safe route, avoiding obstacles, to enable the aeroplane to either comply with the en-route OEI requirements, or land at either the aerodrome of departure or at a take-off alternate aerodrome is accomplished by the use of the PDCS data, the Jeppesen Flight Planning system and Jeppesen charts.

The iPreFlight application on the portable EFB can assist in calculating take-off performance provided that all obstacles are considered. This assessment is particularly relevant since obstacles can be added at short notice via NOTAM. More information on determining if obstacles must be manually added to iPreFlight, enabling its use for take-off performance, can be found in TA 32.

8.1.2.4.4 Landing - Destination and Alternate Aerodromes

A. General

The landing mass of the aeroplane determined in accordance with [8.1.2.4.1](#) shall not exceed the maximum landing mass specified for the altitude and the ambient temperature expected for the estimated time of landing at the destination aerodrome and alternate aerodrome.

B. Missed Approach

- a. For instrument approaches with a missed approach climb gradient greater than 2.5%, the expected landing mass of the aeroplane shall allow for a missed approach with a climb gradient equal to or greater than the applicable missed approach gradient in the OEI missed approach configuration and at the associated speed.
- b. For instrument approaches with DH below 200 ft, the expected landing mass of the aeroplane shall allow a missed approach gradient of climb, with the critical engine failed and with the speed and configuration used for a missed approach of at least 2.5%, or the published gradient, whichever is greater.

C. Landing - Dry Runways

- a. The landing mass of the aeroplane determined in accordance with [8.1.2.4.1](#) for the estimated time of landing at the destination aerodrome and at any alternate aerodrome shall allow a full stop landing from 50 ft above the threshold:
 1. For turbojet-powered aeroplanes, within 60% of the landing distance available (LDA);
 2. By way of derogation from point (a)(1), for aeroplanes that are approved for reduced landing distance operations, within 80% of the LDA.

NOTE

As per [8.1.2.4.6](#) no NTA aircraft is currently approved for reduced landing distance operations.

- b. For steep approach operations, the operator shall use the landing distance data factored in accordance with point (a)(1), based on a screen height of less than 60 ft, but not less than 35 ft, and shall comply with [8.3.28.3](#).
- c. For short landing operations, the landing distance data shall be factored in accordance with point (a)(1).

NOTE

As per [8.3.28.2](#) NTA does not currently hold an approval for Short Landing Operations.

- d. When determining the landing mass, the following shall be taken into account:
 1. not more than 50% of the headwind component or not less than 150% of the tailwind component (these components are already accounted for in the performance data provided in the AFM / AOM and subsequently in the PDCS data and, therefore, no additional corrections required by flight crew); and
 2. corrections as provided by the AFM.

- e. For dispatching the aeroplane it shall be assumed that:
1. the aeroplane will land on the most favourable runway, in still air; and
 2. the aeroplane will land on the runway most likely to be assigned, considering the probable wind speed and direction at the time of arrival, the ground handling characteristics of the aeroplane, and other conditions such as landing aids, ATC, noise abatement procedures, and terrain.

The landing mass for dispatch shall be based on the lowest masses of 1. or 2. above.

- f. If unable to comply with (e)(2) for a destination aerodrome, the aeroplane shall be dispatched only if an alternate aerodrome is designated that allows full compliance with one of the following:
1. Points a. to d. above if the runway at the estimated time of arrival is dry.
 2. [8.1.2.4.4](#) D points a. to d. if the runway at the estimated time of arrival is wet or contaminated.

D. Landing - Wet and Contaminated Runways

- a. When the appropriate weather reports and / or forecasts indicate that the runway at the estimated time of arrival may be wet, the LDA shall be one of the following distances:
1. A landing distance provided in the AFM for use on wet runways at time of dispatch, but not less than that required by [8.1.2.4.4 C.a.1](#);
 2. If a landing distance is not provided in the AFM for use on wet runways at time of dispatch, at least 115% of the required landing distance, determined in accordance with [8.1.2.4.4 C.a.1](#);
 3. A landing distance shorter than that required by point (a)(2), but not less than that required by point [8.1.2.4.4 C.a.1](#), if the runway has specific friction-improving characteristics (e.g. grooves) and the AFM includes specific additional information for landing distance on that runway type;
 4. By way of derogation from points (a)(1), (a)(2) and (a)(3), for aeroplanes that are approved for reduced landing distance operations under point [8.1.2.4.6](#), the landing distance determined is in accordance with this paragraph.

Additionally, it is strongly recommended to check the FLDTA in accordance with [8.3.28.1](#) to avoid the possibility of dispatching an aerodrome where the FLDTA exceeds the LDA.

- b. When the appropriate weather reports and / or forecasts indicate that the runway at the estimated time of arrival may be contaminated, the LDA shall be one of the following distances:
1. At least the landing distance determined in accordance with point (a), or at least 115% of the landing distance determined in accordance with approved contaminated landing distance data or equivalent, whichever is greater;
 2. On specially prepared winter runways, a landing distance shorter than that required by point (b)(1), but not less than that required by point (a), may be used if the AFM includes specific additional information about landing distances on contaminated runways. Such landing distance shall be at least 115% of the landing distance contained in the AFM.

- c. By way of derogation from point (b), the increment of 15% does not need to be applied if it is already included in the approved landing distance data or equivalent.
- d. For points (a) and (b), the following shall apply accordingly:
 - 1. For steep approach operations, landing distance data shall be based on a screen height of less than 60 ft, but not less than 35 ft, and shall comply with [8.3.28.3](#).
 - 2. For short landing operations, shall comply with [8.3.28.2](#).
 - 3. When determining the landing mass, the operator shall take into account the following:
 - i. not more than 50% of the headwind component or not less than 150% of the tailwind component (these components are already accounted for in the performance data provided in the AFM / AOM and subsequently in the PDCS data and, therefore, no additional corrections required by flight crew);
 - ii. corrections as provided in the AFM.
- e. For dispatching the aeroplane it shall be assumed that:
 - 1. the aeroplane will land on the most favourable runway, in still air; and
 - 2. the aeroplane will land on the runway most likely to be assigned, considering the probable wind speed and direction at the time of arrival, the ground handling characteristics of the aeroplane, and other conditions such as landing aids, ATC, noise abatement procedures, and terrain.

The landing mass for dispatch shall be based on the lowest masses of 1. or 2. above.
- f. If the operator is unable to comply with point (e)(1) for a destination aerodrome where the appropriate weather reports or forecasts indicate that the runway at the estimated time of arrival may be contaminated and where a landing depends upon a specific wind component, the aeroplane shall only be dispatched if two alternate aerodromes are designated.
- g. If the operator is unable to comply with point (e)(2) for the destination aerodrome where the appropriate weather reports or forecasts indicate that the runway at the estimated time of arrival may be wet or contaminated, the aeroplane shall only be dispatched if an alternate aerodrome is designated.
- h. For points (f) and (g), the designated alternate aerodrome or aerodromes shall allow compliance with one of the following:
 - 1. [8.1.2.4.4 -C](#) points (a) to (d), if the runway at the estimated time of arrival is dry;
 - 2. [8.1.2.4.4 -D](#) points (a) to (d), if the runway at the estimated time of arrival is wet or contaminated.

8.1.2.4.5 Contaminated Runway Limitations

Regardless of the AFM / AOM performance data, a take-off or landing shall not be made using any portion of a runway where the:

- i. Reported runway condition code (RWYCC) is 0 or 1;
- ii. The braking action (BA) is 'Poor';
- iii. The braking coefficient is equal to or less than 0.25; or
- iv. Where no RWYCC or BA is reported, and the runway surface descriptor (RSD) gives a condition which relates to an RWYCC of 0 or 1;
- v. Runway is contaminated ² and loose contaminant depth exceeds the limits show in the tables below ³

² See Contaminated Runway Guidance Material.

³ There are no depth limits applicable to solid contaminants Compact Snow and Ice.

Take-off - Loose contaminant maximum depth limitations

Contaminated	Challenger 350	XLS	Latitude	Phenom300	Falcon 2000EX	Challenger 650	Global 5000 / 6000
STANDING WATER	1/4 in (6.4 mm)	1/2 in (12.7 mm)	0.3 in (7.6 mm)	1/2 in (12.7 mm)	0.35 in (8.9 mm)	1/4 in (6.4 mm)	1/4 in (6.4 mm)
SLUSH	1/4 in (6.4 mm)	1/2 in (12.7 mm)	0.4 in (10.2 mm)	0.59 in (14.9 mm)	0.45 in (11 mm)	1/4 in (6.4 mm)	0.3 in (7.6 mm)
WET SNOW	1/2 in (12.7 mm)	2 in (50.8 mm)	1/2 in (12.7 mm)	1 in (25.3 mm)	0.7 in (19 mm)	1/2 in (12.7 mm)	0.6 in (15.2 mm)
DRY SNOW	2 in (50.8 mm)	2 in (50.8 mm)	2 in (50.8 mm)	2.5 in (63.4 mm)	2.5 in (63 mm)	1.2 in (30.5 mm)	1.2 in (30.5 mm)

Landing - Loose contaminant maximum depth limitations

Contaminated	Challenger 350	XLS	Latitude	Phenom300	Falcon 2000X	Challenger 650	Global 5000 / 6000
STANDING WATER	1/2 in (12.7 mm)	1/2 in (12.7 mm)	1/2 in (12.7 mm)	1/2 in (12.7 mm)	1/2 in (12.7 mm)	1/2 in (12.7 mm)	1/2 in (12.7 mm)
SLUSH	1/2 in (12.7 mm)	1/2 in (12.7 mm)	1/2 in (12.7 mm)	0.59 in (14.9 mm)	0.59 in (14.9 mm)	1/2 in (12.7 mm)	0.6 in (15.2 mm)
WET SNOW	1 in (25.4 mm)	1 in (25.4 mm) ¹	1/2 in (12.7 mm)	1 in (25.4 mm)	1 in (25.4 mm)	1 in (25.4 mm)	1.2 in (30.5 mm)
DRY SNOW	3 in (76.2 mm)	2 in (50.8 mm) ¹	2 in (50.8 mm)	2.5 in (63.5 mm)	2.5 in (63.5 mm)	2.4 in (61.0 mm)	2.4 in (61.0 mm)

¹ Labeled SNOW in Citation XLS performance dataset.

Normally within the RCR, aerodromes will report the RWYCC and RSD of each third of the runway (start point, mid-point, and stop end), in states where GRF has not yet been implemented fully, braking action may still be reported as well and the RWYCC omitted.

When considering the RWYCC, RSD or braking action to apply to a given runway, the report for the most limiting 1/3 shall be applied to the whole runway, unless operating in accordance with the exception detailed below.

Operation using only the first 2/3 of the runway

When operating at a large airfield and the stop end of the runway is limiting the operation, the stop end may be disregarded provided the operation (1) (including any applicable safety factors) can be accomplished within the first two thirds of the runway.

NOTE

'Operation' is defined as follows depending on the phase of flight:

- **Take-off** – the longest of TORR, ASDR or TODR.
- **Landing** – the LDR including all operational factors according to the phase of flight ([8.1.2.4.4](#) or [8.3.28.1](#)) as applicable.

For Practical Application - See [Technical Annex](#).

8.1.2.4.6 Reduced Required Landing Distance Operations

(CAT.POL.A.255)

NTA does not currently hold approval to conduct Reduced Required Landing Distance Operations.

8.1.3 Aerodrome Selection Policy - Operating Minima (CAT.OP.MPA.182)

(CAT.OP.MPA.110)

NOTE

For special ETOPS requirements, refer to [Extended Range Operations with Two-Engined Aeroplanes \(ETOPS\)](#).

8.1.3.1 General

The aerodrome operating minima detailed here are the absolute minima authorised by NTA for each departure, destination, or alternate aerodrome. If the minima presented in the Jeppesen Route Manual, any other Aeronautical Information Publication or in the RTOW / M charts of the NTA, AOM, or RM, etc., are higher, then those higher minima will apply, unless special approval is given by the competent authority for a particular aerodrome. Minima shall not be lower than any that have been established by the competent authority in which the aerodrome is located. NTA complies with the requirements for the variations to a basic fuel scheme, as defined in CAT.OP.MPA.182.

The method used to establish aerodrome operating minima takes into account:

- a. the type, performance, and handling characteristics of the aircraft;
- b. the equipment available on the aircraft for the purpose of navigation, acquisition of visual references, and / or control of the flight path during take-off, approach, landing, and the missed approach;
- c. any conditions or limitations stated in the AFM;
- d. the relevant operational experience of the operator;
- e. the dimensions and characteristics of the runways / final approach and take-off areas (FATOs) that may be selected for use;
- f. the adequacy and performance of the available visual and non-visual aids and infrastructure;
- g. the obstacle clearance altitude / height (OCA / H) for the instrument approach procedures (IAPs);
- h. the obstacles in the climb-out areas and necessary clearance margins;
- i. the composition of the flight crew, their competence and experience;
- j. the IAP;
- k. the aerodrome characteristics and the available air navigation services (ANS);
- l. any minima that may be promulgated by the State of the aerodrome;
- m. the conditions prescribed in the operations specifications including any specific approvals for low-visibility operations (LVOs) or operations with operational credits;
- n. any non-standard characteristics of the aerodrome, the IAP or the environment.

All approaches shall be flown as stabilised approaches (SAp) unless otherwise approved by the Authority for a particular approach to a particular runway.

All non-precision approaches shall be flown using the continuous descent final approaches (CDFA) technique unless otherwise approved by the Authority for a particular approach to a particular runway. When calculating the minima in accordance with [8.1.3.6](#), the applicable minimum RVR is increased by 200 metres (m) for CAT A/B aeroplanes and by 400 m for CAT C/D aeroplanes for approaches NOT flown using the CDFA technique, providing that the resulting RVR / CMV value does not exceed 5000 m.

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8.1.3.1.1 Aeroplane Category*(AIR OPS and ICAO)*

Aeroplane performance has a direct effect on the airspace and visibility needed to perform the various manoeuvres associated with the conduct of instrument approach procedures. The most significant performance factor is aeroplane speed. The aeroplane category shall be based on the VAT. The table below shall be used to determine the aeroplane category.

Table 8.1.3.1.1 A - Aeroplane categories

Aeroplane Category	VAT (VREF) KIAS	Range of Speeds for Initial Approach	Range of Final Approach Speeds	Maximum Speeds for Visual Manoeuvring (Circling)	Max Speeds for Missed Approach	
					Intmdt	Final
A	<91	90-150 (110) ¹	70-100	100	100	110
B	91-120	120-180 (140) ¹	85-130	135	130	150
C	121-140	160-240	115-160	180	160	240
D	141-165	185-250	130-185	205	185	265
E	166-210	185-250	155-230	204	230	275

¹ Maximum speed for reversal and racetrack procedures.

VAT or **VREF** - Indicated air speed at threshold (VAT) which is equal to the stalling speed (VSO) multiplied by 1,3 or VS1G multiplied by 1,23 in the landing configuration at the maximum certificated landing mass. If both VSO and VS1G are available, the higher resulting VAT shall be used.

NTA may apply a lower landing mass for determining the VAT if approved by the competent authority. Such a lower landing mass shall be a permanent value, independent of the changing conditions of day-to-day operations. Refer to the Aeroplane OM-B for Maximum Landing Mass Limitations.

All aeroplanes shall apply the aerodrome procedures and minima according to the aeroplane category in the OM-B and cannot use procedures and minima applicable to a lower aeroplane category, except at aerodromes where the AIP explicitly permits aeroplanes of higher categories to utilise procedures for lower categories. In such cases, operations are feasible but must take into consideration the performance requirements, particularly obstacle clearance limitations, operating speed limitations or any other limitations described in the AIP.⁴

⁴ This exceptional approval does not lower the aeroplane category and, therefore, is outside the scope of CAT.OP.MPA.320.

8.1.3.2 Selection of Aerodromes

- a. If it is not possible to use the departure aerodrome as a take-off alternate aerodrome due to meteorological or performance reasons, another adequate take-off alternate shall be selected and specified in the OFP. The take-off alternate aerodrome, in relation to the departure aerodrome, shall be located within:
 - i. For two engine aeroplanes, either:
 1. One hour flight time at a one-engine-inoperative cruising speed according to the AFM in still air standard conditions based on the actual take-off mass; or
 2. The approved ETOPS diversion time, subject to any MEL restriction, up to a maximum of two hours, at one-engine-inoperative cruising speed according to the AFM in still air standard conditions based on the actual take-off mass of the aeroplane and crews authorised for ETOPS;
 - ii. For three engine aeroplanes, two hours flight time at a one-engine-inoperative cruising speed according to the AFM in still air standard conditions based on the actual take-off mass;
 - iii. If the AFM does not contain a one-engine inoperative cruising speed, the speed to be used for calculation shall be that which is achieved with the remaining engine(s) set at maximum continuous power.
- b. At least one destination alternate shall be selected for each IFR flight unless:
 - i. Both:
 1. The duration of the planned flight from take-off to landing does not exceed 6 hours or, in the event of in-flight re-planning, in accordance with [In-Flight Replanning Fuel](#), the remaining flying time to destination does not exceed 4 hours; and
 2. Two separate runways are available and usable at the destination and the appropriate weather reports and / or forecasts for the destination aerodrome, indicate that for the period from one hour before until one hour after the expected time of arrival at the destination aerodrome, the ceiling will be at least 2000 ft or circling height + 500 ft, whichever is greater, and the ground visibility will be at least 5 km, provided that a single event (such as an aircraft accident) does not eliminate both options.
 - or
 - ii. The destination aerodrome is classified as an isolated aerodrome.
- c. Two destination alternates shall be selected when:
 - i. The appropriate weather reports and / or forecasts for the destination aerodrome, indicate that during a period commencing one hour before and ending one hour after the estimated time of arrival, the weather conditions will be below the applicable planning minima; or
 - ii. No meteorological information is available.

NOTE

When two alternates are required, fuel requirements to reach the most distant alternate shall be taken into account.

- d. Any required alternate aerodrome(s) shall be specified in the OFP.
- e. Any required destination alternate shall be selected based upon the following criteria:
 - i. Shall be suitable in accordance with [Aerodrome Selection Policy - Adequacy of Aerodromes](#).
 - ii. Shall, if possible, be within 50 nm from the destination aerodrome.
 - iii. If no suitable aerodrome is available within 50 nm from the destination aerodrome, the closest suitable aerodrome shall be selected.
 - iv. If suitable, the departure aerodrome can be selected as the destination alternate, when complying with the selection criteria above.
 - v. Shall meet the weather requirements detailed in [8.1.3.3](#).
- f. Ensuring sufficient means to navigate to and land at the destination on IFR flights:
 - i. when intending to use PBN operations, an instrument approach which does not rely solely on GNSS must be available either at the destination aerodrome or at the destination alternate aerodrome. This restriction does not apply to take-off alternates or ERA aerodromes or if the flight does not require a destination alternate.
 - ii. When 'operational credits' (e.g. EFVS, SA CAT I, etc.) are being used, it must be ensured that an instrument approach procedure that does not rely on the same 'operational credit' is available either at the destination aerodrome or at the destination alternate aerodrome.

8.1.3.3 Planning Minima for IFR Flights

a. Planning minima for a take-off alternate aerodrome.

An aerodrome shall only be selected as a take-off alternate aerodrome when the appropriate weather reports and / or forecasts indicate that, during a period commencing one hour before and ending one hour after the estimated time of arrival at the aerodrome, the weather conditions will be at or above the applicable landing minima specified in accordance with [8.1.3.6](#). The ceiling shall be taken into account when the only approaches available are NPAs and / or circling approaches. Any limitation related to one-engine-inoperative operations shall be taken into account.

b. Planning minima for a destination aerodrome other than an isolated destination aerodrome.

A destination aerodrome shall only be selected when:

- i. The appropriate weather reports and / or forecasts indicate that, during a period commencing one hour before and ending one hour after the estimated time of arrival at the aerodrome, the weather conditions will be at or above the applicable planning minima as follows:
 1. RVR / visibility specified in accordance with [8.1.3.6](#); and
 2. For a non-precision approach or a circling approach, the ceiling at or above MDH; or
- ii. Two destination alternate aerodromes are selected under [Selection of Aerodromes - c](#).

c. Planning minima for a: destination alternate aerodrome, isolated aerodrome, fuel ERA aerodrome, en-route alternate aerodrome required at the planning stage.

An aerodrome for one of these purposes shall only be selected when the appropriate weather reports and / or forecasts indicate that, during a period commencing one hour before and ending one hour after the estimated time of arrival at the aerodrome, the weather conditions will be at or above the planning minima in Table [8.1.3.3 A](#) or in Table [8.1.3.3 B](#) below. Any limitation related to one-engine-inoperative operations shall be taken into account.

Table 8.1.3.3 A - Planning minima: Destination alternate aerodrome, isolated destination aerodrome, fuel ERA, and en-route alternate aerodrome.

Type of approach operation	Aerodrome ceiling (cloud base or vertical visibility)	RVR / VIS
Type B instrument approach operations (DH < 250ft)	DA / H + 200 ft	RVR / VIS + 800 m
Type A instrument approach operations (DH / MDH ≥ 250ft)	DA / H or MDA / H + 400 ft	RVR / VIS + 1 500 m
Circling approach operations	MDA / H + 400 ft	VIS + 1 500 m
Crosswind planning minima: see Table 1 in 8.1.6.5.6		
Wind limitations should be applied taking into account the runway condition (dry, wet, contaminated).		
NOTE Type A instrument approach operation means an instrument approach operation with an MDH or a DH at or above 250 ft; Type B instrument approach operation means an operation with a DH below 250 ft.		

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Table 8.1.3.3 B - Planning minima: Destination alternate aerodrome, fuel ERA, when the duration of the planned flight from take-off to landing does not exceed 6 hours or, in the event of in-flight re-planning, the remaining flying time to destination does not exceed 4 hours.

Type of approach operation	Aerodrome ceiling (cloud base or vertical visibility)	RVR / VIS
Type B instrument approach operations	DA / H + 200 ft	RVR / VIS + 550 m
3D Type A instrument approach operations, based on a facility with a system minimum of 200 ft or less ⁴	DA / H + 200 ft	RVR / VIS + 800 m
Two or more usable type A instrument approach operations ³ , each based on a separate navigation aid	DA / H or MDA / H ¹ + 200 ft	RVR / VIS ² + 1 000 m
Other type A instrument approach operations	DA / H or MDA / H + 400 ft	RVR / VIS + 1 500 m
Circling approach operations	MDA / H + 400 ft	VIS + 1 500 m
Crosswind planning minima: see Table 1 in 8.1.6.5.6		
Wind limitations should be applied taking into account the runway condition (dry, wet, contaminated).		
NOTE - When an aerodrome verifies in more than one row, the most convenient planning minima can be used. For example: if an aerodrome has 2 usable type A approaches (3rd row) but one of those is an ILS CAT I (2nd row), minima from row 2 can be used. - NTA has established an operational control system that includes flight monitoring, therefore fulfilling the requirements for using this table.		

¹ The higher of the usable DA / H or MDA / H.

² The higher of the usable RVR or VIS.

³ Sufficient means to navigate to and land should be ensured as per [8.1.3.2](#) (f).

⁴ System minima depends only on approach facilities (guidance equipment) at the aerodrome. This dependency is shown in Table [8.1.3.6.1 A](#). If the landing option at the aerodrome uses a technology approved for a DH of 200 ft or less, then this row can be used.

An instrument approach operation is considered usable for planning minima when the approach facilities are available, the aircraft is equipped to perform such an approach, the flight crew is accordingly trained, and the runway is available for landing.

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8.1.3.4 Meteorological Conditions

- a. On an IFR flight a commander shall only:
 - i. Commence take-off; or
 - ii. Continue beyond the point from which a revised flight plan applies in the event of in-flight re-planning, when information is available indicating that the expected meteorological conditions, at the time of arrival, at the destination, and / or required alternate aerodrome(s) prescribed in [8.1.3.2](#) are at or above the planning minima, prescribed in [8.1.3.3](#).
- b. On an IFR flight, a commander shall only continue towards the planned destination aerodrome when the latest information available indicates that, at the expected time of arrival, the meteorological conditions at the destination, or at least one destination alternate aerodrome, are at or above the applicable aerodrome operating minima prescribed in [8.1.3.6](#).
- c. On an IFR flight a commander shall only continue beyond:
 - i. The decision point when using the reduced contingency fuel procedure; or
 - ii. The PNR when using the isolated aerodrome procedure;

When information is available indicating that the expected meteorological conditions at the time of arrival, at the destination, and / or required alternate aerodrome(s) prescribed in [8.1.3.2](#) are at or above the applicable aerodrome operating minima prescribed in [8.1.3.6](#).

- d. On a VFR flight the commander shall only commence take-off when the appropriate meteorological reports and / or forecasts indicate that the meteorological conditions along part of the route to be flown under VFR will, at the appropriate time, be at or above the VFR limits.

8.1.3.5 Take-off Minima

NOTE

See [Low Visibility Take-off \(LVTO\)](#) for low visibility take-off procedure.

8.1.3.5.1 General

- a. Take-off minima shall be expressed as visibility or runway visual range (RVR) limits, taking into account all relevant factors for each aerodrome planned to be used and aeroplane characteristics. Where there is a specific need to see and avoid obstacles on departure and / or for a forced landing, additional conditions, e.g. ceiling, shall be specified.
- b. The commander shall not commence take-off unless the meteorological conditions at the aerodrome of departure are equal to or better than applicable minima for landing at that aerodrome, unless a weather-permissible take-off alternate aerodrome is available.
- c. When the reported meteorological visibility is below that required for take-off and RVR is not reported, a take-off shall only be commenced if the commander can determine that the RVR / visibility along the take-off runway is equal to or better than the required minima.
- d. When no reported meteorological visibility or RVR is available, a take-off shall only be commenced if the commander can determine that the RVR / visibility along the take-off runway is equal to or better than the required minima.
- e. Before commencing a take-off, the commander shall be satisfied that the weather at the aerodrome and the conditions of the runway intended to be used would not prevent a safe take-off and departure and that the established aerodrome operating minima consistent with:
 - i. The operative ground equipment;
 - ii. The operative aircraft systems;
 - iii. The aircraft performance;
 - iv. Flight crew qualifications.

8.1.3.5.2 Visual Reference

The take-off minima shall be selected to ensure sufficient guidance to control the aeroplane in the event of both a discontinued take-off in adverse circumstances and a continued take-off after failure of the critical power unit.

For night operations, ground lights shall be available to illuminate the runway and any obstacles.

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8.1.3.5.3 Required RVR / Visibility

RVR / Visibility minima for take-off are as defined in Table [8.1.3.5.3 A](#) below.

When the reported visibility is below that required for take-off and RVR is not reported, or when neither reported visibility nor RVR is available, a take-off shall only be commenced if the commander can determine that the RVR / visibility along the take-off runway is equal to or better than the required minimum. This may require a runway visit to assess the RVR at the relevant RVR reporting points by a company dedicated observer, or Airport Authority observer. Take-off minima shall be increased whenever there is a special need to see and avoid obstacles on departure or for other performance requirements.

Table 8.1.3.5.3 A - Take-off minima

TAKE-OFF MINIMA	
Facilities (Note 6)	RVR / visibility (Note 1)
No facilities, (day only), controlled aerodromes with published SID's.	RVR 500 m / 1600 ft (Note 2)
Runway edge lights or centreline markings or runway centre line lights, without LVP. (NIGHT: End and edge lights or end and centreline lights)	RVR 400 m / 1200 ft (Note 2 and 3)
Runway edge lights and centre line markings, LVP in force. (NIGHT: additionally, end lights or centre line lights) The required RVR value has been achieved for all of the relevant RVR reporting points.	RVR 300 m / 1000 ft (Note 2 and 7)
All of the following: Runway edge lights, centreline lights, centreline markings, runway end lights. The required RVR value has been achieved for all of the relevant RVR reporting points. LVP in force.	RVR 150 m / 500 ft (Note 2 ,4 and 7)
All of the following: Runway edge lights (spaced 60 m or less), centreline lights (spaced 15 m or less), centreline markings, runway end lights. The required RVR value has been achieved for all of the relevant RVR reporting points. LVP are in force.	RVR 125 m / 400 ft (Note 4 ,5 and 7)

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NOTE

1. The reported RVR / visibility value representative of the initial part of the take-off run can be replaced by pilot assessment.
2. The required RVR values as given in the above table shall be available to the take-off run (except for the initial part which may be based on pilot's assessment) and a distance on a possible accelerate stop case, that is for the Accelerate Stop Distance (ASD). From this assumption, only that number of RVR values shall be at or above those required in the table of take-off minima as needed to cover the distance from start of take-off until the aeroplane would come to a stop in case of an aborted take-off. If, for instance, the runway is 3000 metres long, and the calculated accelerate stop distance is less than 2000 metres, only the first two values shall be at / above the RVR limits. The RVR value for the initial part of the take-off run may be replaced by pilot assessment by counting runway edge lights or runway centreline lights once the aeroplane is lined up.
3. For night operations at least runway edge and runway end lights or runway centreline lights and runway end lights are required. Without LVP, runway end lights may be substituted by colour-coded runway edge lights or colour-coded runway centre line lights.
4. The required RVR / visibility value shall be achieved for all relevant RVR reporting points with the exception given in Note 1 above.
5. Flight crew members have satisfactorily completed training in the flight simulator. LVTO procedures with 125 metres are approved as reflected on the AOC.
6. In the event that the runway edge / centreline lighting installation is partially unserviceable, it may be considered as operative provided that the minimum specification for the lighting group as detailed in [8.1.3.10](#) or the table above if more restrictive is maintained.
7. Relevant reporting points are those representative of the parts of the runway from the point at which the aircraft commences the take-off until the calculated accelerate-stop distance from that point. The reported RVR value representative of the initial part of the take-off run can be replaced by pilot assessment.

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8.1.3.6 Approach Minima

8.1.3.6.1 Determination of DH / MDH for Instrument Approach Operations

1. Decision height (DH). The decision height (DH) to be used for 3D approach operation or a 2D approach operation flown with the CDFA technique, shall not be lower than the highest of:
 - i. the OCH for the category of aeroplane;
 - ii. the published approach procedure DH , or minimum descent height (MDH), where applicable;
 - iii. For NPA approaches (2D, type A) flown with the CDFA technique, the published MDA(H) / DA(H) plus 50 ft;
 - iv. the system minimum in [Table 8.1.3.6.1 A](#); or
 - v. the minimum DH permitted for the runway specified in [Table 8.1.3.6.1 B](#); or
 - vi. the minimum DH specified in the AFM or equivalent document, if stated.
2. The minimum descent height (MDH)for a 2D approach operation flown not using the CDFA technique shall not be lower than the highest of:
 - i. the OCH for the category of aeroplane;
 - ii. the published approach procedure MDH where applicable;
 - iii. the system minimum in [Table 8.1.3.6.1 A](#);
 - iv. the lowest MDH permitted for the runway specified in [Table 8.1.3.6.1 B](#); or
 - v. the lowest MDH minimum descent height specified in the AFM, if stated.
3. Where a barometric DA / H or MDA / H is used, this should be adjusted where the ambient temperature is significantly below international standard atmosphere (ISA). [8.3.3.6 CHECKING OF TERRAIN CLEARANCE - ALTITUDE CORRECTIONS](#) provides temperature correction tables for adjustment of minimum heights / altitudes.
4. Approach operation utilising enhanced vision systems (EVS) for which an operational credit is applied to reduce the RVR minima by no more than one third of the published RVR.
 - i. The RVR / CMV values published for an approach procedure can be reduced according to [Table 8.1.3.6.3 E](#) for CAT I approach operations, APV approach operations, and NPA approach operations flown with the CDFA technique;
 - ii. For CAT I:
 - a. May continue an approach below DA to 100 ft above the runway threshold elevation, provided that a visual reference is displayed and identifiable on the EVS image; and
 - b. Shall only continue and approach below 100 ft above runway threshold elevation provided that a visual reference is distinctly visible and identifiable to the pilot without reliance on the EVS.

- iii. For APV operations and NPA operations flown with the CDFA technique:
 - a. May continue an approach below DH / MDH to 200 ft above the runway threshold elevation, provided that a visual reference is displayed and identifiable on the EVS image; and
 - b. Shall only continue an approach below 200 ft above the runway threshold elevation, provided that a visual reference is distinctly visible and identifiable to the pilot without reliance on the EVS;
 - c. The approach needs to be flown using the vertical flight path guidance mode;
 - d. The approach segment from final approach fix (FAF) to runway threshold has to be straight and the difference between the final approach course and the runway centreline is not greater than 2° ;
 - e. The final approach path is published and is not greater than 3.7° .
- iv. Circling operations require the pilot to maintain visual references which may not be directly ahead of the aircraft and may not be aligned with the current flight path. Therefore, EVS operations are prohibited for circling approaches.

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Table 8.1.3.6.1 A - System minima vs. facilities

System Minima	
Facility	Lowest DH / MDH (ft)
ILS / MLS / GLS	200
GNSS / SBAS (LPV)	200 (see note 1 ¹)
Precision approach radar (PAR)	200
GNSS / SBAS (LP)	250
GNSS (LNAV)	250
GNSS / BARO VNAV (LNAV / VNAV)	250
Localiser with or without DME	250
SRA (terminating at 1/2 NM)	250
SRA (terminating at 1 NM)	300
SRA (terminating at 2 NM or more)	350
VOR	300
VOR / DME	250
NDB	350
NDB / DME	300
VDF	350

¹ For localiser performance with vertical guidance (LPV), a DH of 200 ft may be used only if the published FAS datablock sets a vertical alert limit not exceeding 35 m. Otherwise, the DH should not be lower than 250 ft.

Table 8.1.3.6.1 B - Runway type minima

Runway type	Lowest DH/MDH (ft)	
Instrument runway	Precision Approach rwy, CAT I	200
	NPA runway	250
Non- instrument runway	Non-instrument runway	Circling Minima as per Circling Approach

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8.1.3.6.2 Reserved**8.1.3.6.3 Determination of RVR / VIS for Instrument Approach Operations**

1. The RVR or VIS for straight-in instrument approach operations should be not less than the greatest of:
 - a. the minimum RVR or VIS for the type of runway used according to Table [8.1.3.6.3 A](#);
 - b. the minimum RVR determined according to the MDH or DH and class of lighting facility according to T Table [8.1.3.6.3 B](#); or
 - c. the minimum RVR according to the visual and non-visual aids and on-board equipment used according to Table [8.1.3.6.3 C](#).

If the value determined in (a) is a VIS, then the result is a minimum VIS. In all other cases, the result is a minimum RVR.
2. For Category A and B aeroplanes, if the RVR or VIS determined in accordance with (1) is greater than 1 500 m, then 1 500 m should be used.
3. If the approach is flown with a level flight segment at or above the MDA / H, then 200 m should be added to the RVR calculated in accordance with (1) and (2) for Category A and B aeroplanes and 400 m for Category C and D aeroplanes.
4. The visual aids should comprise standard runway day markings, runway edge lights, threshold lights, runway end lights and approach lights as defined in Table [8.1.3.6.3 D](#).
5. For night operations or for any operation where credit for visual aids is required, the lights should be on and serviceable except as provided for in Table [8.1.3.6.3 F](#).
6. Where any visual or non-visual aid specified for the approach and assumed to be available in the determination of operating minima is unavailable, revised operating minima will need to be determined.

Table 8.1.3.6.3 A - Type of runway versus minimum RVR or VIS

Type of Runway	Minimum RVR or VIS (m)
PA runway Category I	RVR 550
NPA runway	RVR 750
Non-instrument runway	VIS according to 8.1.3.6.4

Table 8.1.3.6.3 B - RVR vs. DH/MDH

DH or MDH (Feet)			Class of lighting facility (Table 8.1.3.6.3 D)			
			FALS	IALS	BALS	NALS (Note 1 ¹)
			RVR (Metres)			
200	-	210	550	750	1000	1200
211	-	240	550	800	1000	1200
241	-	250	550	800	1000	1300
251	-	260	600	800	1100	1300
261	-	280	600	900	1100	1300
281	-	300	650	900	1200	1400
301	-	320	700	1000	1200	1400
321	-	340	800	1100	1300	1500
341	-	360	900	1200	1400	1600
361	-	380	1000	1300	1500	1700
381	-	400	1100	1400	1600	1800
401	-	420	1200	1500	1700	1900
421	-	440	1300	1600	1800	2000
441	-	460	1400	1700	1900	2100
461	-	480	1500	1800	2000	2200
481	-	500	1500	1800	2100	2300
501	-	520	1600	1900	2100	2400
521	-	540	1700	2000	2200	2400
541	-	560	1800	2100	2300	2400
561	-	580	1900	2200	2400	2400
581	-	600	2000	2300	2400	2400
601	-	620	2100	2400	2400	2400
621	-	640	2200	2400	2400	2400
641	-	660	2300	2400	2400	2400
661	and above		2400	2400	2400	2400

¹ No lights required for Daylight Operations.

Table 8.1.3.6.3 C - Visual and non-visual aids and / or on-board equipment versus minimum RVR

Type of Approach	Facilities	Lowest RVR
3D Operations WITH Final approach track offset Up to and incl. 15° for Category B aeroplanes Up to and incl. 5° for Category C aeroplanes	Runway Touchdown Zone Lights (RTZL) and Runway Center Line Lights (RCLL)	Not limited
	Without RTZL and / or RCLL but using either: • HUDLS or equivalent system; • Autopilot or flight director to the DH	Not limited
	Without RTZL and / or RCLL and NOT using: • HUDLS or equivalent system; or • Autopilot or flight director to the DH	750 m
3D Operations WITH Final approach track offset More than 15° for Category B aeroplanes More than 5° for Category C aeroplanes	Runway Touchdown Zone Lights (RTZL) and Runway Center Line Lights (RCLL)	800 m
	Without RTZL and / or RCLL but using HUDLS or equivalent system AND Autopilot or flight director to the DH	800 m
2D Operations	Final approach track offset $\leq$ 15° for Category B aeroplanes or $\leq$ 5° for Category C aeroplanes	750 m
	Final approach track offset > 15° for Category B aeroplanes	1000 m
	Final approach track offset > 15° for Category C aeroplanes	1200 m

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Table 8.1.3.6.3 D - Approach light systems

Class of lighting facility	Length, configuration and intensity of approach lights
FALS (full approach light system)	ICAO: CAT I Lighting System (HIALS ≥ 720 m) distance coded centreline, Barrette centreline.
	FAA: ALSF1, ALSF2, SSALR, MALSR, high or medium intensity and / or flashing lights, ≥ 720 m / 2400 ft.
IALS (intermediate approach light system)	ICAO: Simple approach lighting system (HIALS 420-719 m) single source, Barrette.
	ICAO: CAT I Lighting System (HIALS ≥ 720 m) distance coded centreline, Barrette centreline.
BALS (basic approach light system)	Any other approach lighting System (HIALS, MIALS or ALS 210-419 m).
	FAA: ODALS, high or medium intensity or flashing lights 210- 419 m / 700-1399 ft.
NALS (no approach light system)	Any other approach lighting system (HIALS, MIALS or ALS < 210 m / 700 ft) or no approach lights.

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Table 8.1.3.6.3 E - EFVS operations: allowable reduction in RVR/CMV in relation to RVR/CMV required without EFVS

RVR / CMV (m) required without the use of EFVS	RVR / CMV (m) with the use of EFVS
550	350 ¹
600	400
650	450
700	450
750	500
800	550
900	600
1000	650
1100	750
1200	800
1300	900
1400	900
1500	1000
1600	1100
1700	1100
1800	1200
1900	1300
2000	1300
2100	1400
2200	1500
2300	1500
2400	1600

¹ Reported RVR should be available (no CMV conversion).

Effect on Landing Minima of Temporarily Failed or Downgraded Ground Equipment

These procedures / instructions are applicable for dispatch as well as in flight situations. It is not expected that the commander consults these instructions after passing 1000 ft above the aerodrome. If failure of ground aids are announced at such a late stage in the approach, the approach may be continued at commander's discretion. If failures are announced before such a late stage in the approach, their effect on the approach shall be considered as described in Table [8.1.3.6.3 F](#), (i.e. minima shall be increased and / or relevant procedures shall be used), and the approach may have to be abandoned.

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Table 8.1.3.6.3 F - Effect on Landing Minima of Temporarily Failed or Downgraded Ground Equipment

Failed or downgraded equipment (Note)	Effect on landing minima		
	Type B	Type A	EFVS-A ²
NAVAID stand-by transmitter	No effect		
Outer marker	For CAT I: Not allowed, except if the required height versus glide path can be checked using other means, e.g. DME fix	APV - Not applicable	No effect if the required height versus glide path can be checked using other means, e.g. DME fix
		NPA with FAF : no effect unless used as FAF	
		If the FAF cannot be identified (e.g. no method available for timing of descent), NPA operations cannot be conducted	
		For CAT I: Not allowed, except if the required height versus glide path can be checked using other means, e.g. DME fix	
Middle marker (ILS only)	No effect	No effect unless used as MAPT	No effect
DME	No effect if replaced by RNAV (GNSS) information or the outer marker		
RVR assessment systems	No effect		On runways equipped with two or more RVR assessment units, one may be inoperative
Approach lights	Minima as for NALS		Not allowed
Approach lights except the last 210 m	Minima as for BALS		Not allowed
Approach lights except the last 420 m	Minima as for IALS		As per IAP or no effect ¹
Standby power for approach lights	No effect		As per IAP or no effect ¹
Standby power for runway lights with 1-second switchover time	No effect		RVR 550 m
Edge lights	Day - No effect Night - Not allowed, except in the case of partial unserviceability when the minimum lighting specifications for the type of operation detailed in 8.1.3.10 can still be met.		

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Failed or downgraded equipment (Note)	Effect on landing minima		
	Type B	Type A	EFVS-A ²
Threshold lights	Day - No effect Night - Not allowed, except in the case of partial unserviceability when the minimum lighting specifications for the type of operation detailed in 8.1.3.10 can still be met.		
Runway end lights	Day - No effect Night - Not allowed, except in the case of partial unserviceability when the minimum lighting specifications for the type of operation detailed in 8.1.3.10 can still be met.		No effect if centre line lights are serviceable
Centreline lights	No effect if flight director, HUDLS, or autoland; otherwise RVR 750 m	No effect, but RVR 750m.	As per IAP or - Day RVR 350 m - Night RVR 550 m (400 m with HUD or auto-land)
Centreline lights spacing increased to 30 m	No effect		As per IAP or no effect ¹
Touchdown zone lights	No effect if flight director, HUDLS, or autoland; otherwise RVR 750 m	No effect	As per IAP or - Day RVR 300 m - Night RVR 550 m (350 m with HUD or auto-land)
Taxiway light system	No effect		As per IAP or no effect ¹

¹ If no effect on IAP, then consider the failed or downgraded equipment has no effect on landing minima.

² Effect on landing minima for EFVS-A approaches laid in this table will be superseded if specific references to failed or downgraded equipment are made in the aerodrome briefing.

NOTE

Conditions applicable to Table 8.1.3.6.3 F:

- Multiple failures of runway lights other than those indicated in Table 8.1.3.6.3 F are not acceptable.
- Failures of approach and runway lights are acceptable at the same time, and the most demanding consequence should be applied.
- Failures other than ILS, MLS affect RVR only and not DH.
- Only those facilities mentioned in the table are acceptable to be used to determine the effect of temporarily failed or downgraded equipment.

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8.1.3.6.4 Circling Approach

Circling is the term used to describe the visual phase of an instrument approach to bring an aeroplane into position for landing on a runway which is not suitably located for a straight-in approach.

1. Minimum descent height (MDH). The MDH for circling shall be the highest of:
 - i. the published circling OCH for the aeroplane category; or
 - ii. the minimum circling height derived from Table [8.1.3.6.4 A](#) below; or
 - iii. the DH / MDH of the preceding instrument approach procedure (IAP).
2. Minimum descent altitude (MDA). The MDA for circling shall be calculated by adding the published aerodrome elevation to the MDH, as determined by 1. above.
3. Visibility. The minimum VIS for circling shall be the highest of:
 - i. the circling VIS for the aeroplane category, if published;
 - ii. the minimum VIS derived from Table [8.1.3.6.4 A](#) below; or
4. Circling with prescribed tracks is an accepted procedure within the meaning of this paragraph.
5. Circling at aerodromes located in a mountainous area is prohibited at night unless specifically authorised by either the OM-C briefing or an IJet Note (feasibility note).

Table 8.1.3.6.4 A - Minimum VIS and MDH for circling vs. aeroplane category

Aeroplane Category	MDH	Minimum VIS
A	400 ft	1500 m
B	500 ft	1600 m
C	600 ft	2400 m
D	700 ft	3600 m

8.1.3.6.5 FAA Terminal Instrument Procedures (TERPS) Vs. ICAO Procedures for Air Navigation Services - Aeroplane Operations (PANS OPS)

The majority of approaches within the European area are designed in accordance with PANS-OPS. Many countries throughout the world have adopted these standards. The United States, FAA, produces their own set of standards, TERPS. Some countries have adopted these standards. The design criteria used for a particular approach is marked on the left lower corner of the Jeppesen chart or may be determined by reference to that state's aeronautical information publication.

CAUTION

Some states appear to use a mix of both methods and this may not be readily apparent. One of the differences is the circling MDA and the radii of the obstacle clearance area.

For all circling approaches in Canada, Japan, and Mexico TERPS - Old Criteria shall be used.

For circling approaches in South Korea a check shall be made of the chart to establish which criteria has been used.

The FAA has revised the TERPS criteria for circling approaches. Until all procedures are revised in accordance with the new criteria, both the old and the new are used.

Jeppesen charts where the new criteria have been applied can be identified by the symbol in the CIRCLE-TO-LAND minima box.

CIRCLE-TO-LAND	
Circling not authorized East of Rwy 3R/21L.	
	
Max Kts	MDA(H)
90	1580' (495') - 1
120	
140	1580' (495') - 1½
165	1640' (555') - 2

Circling minima The Jeppesen minima in the 10-9S charts are compliant with the PART-CAT minima.

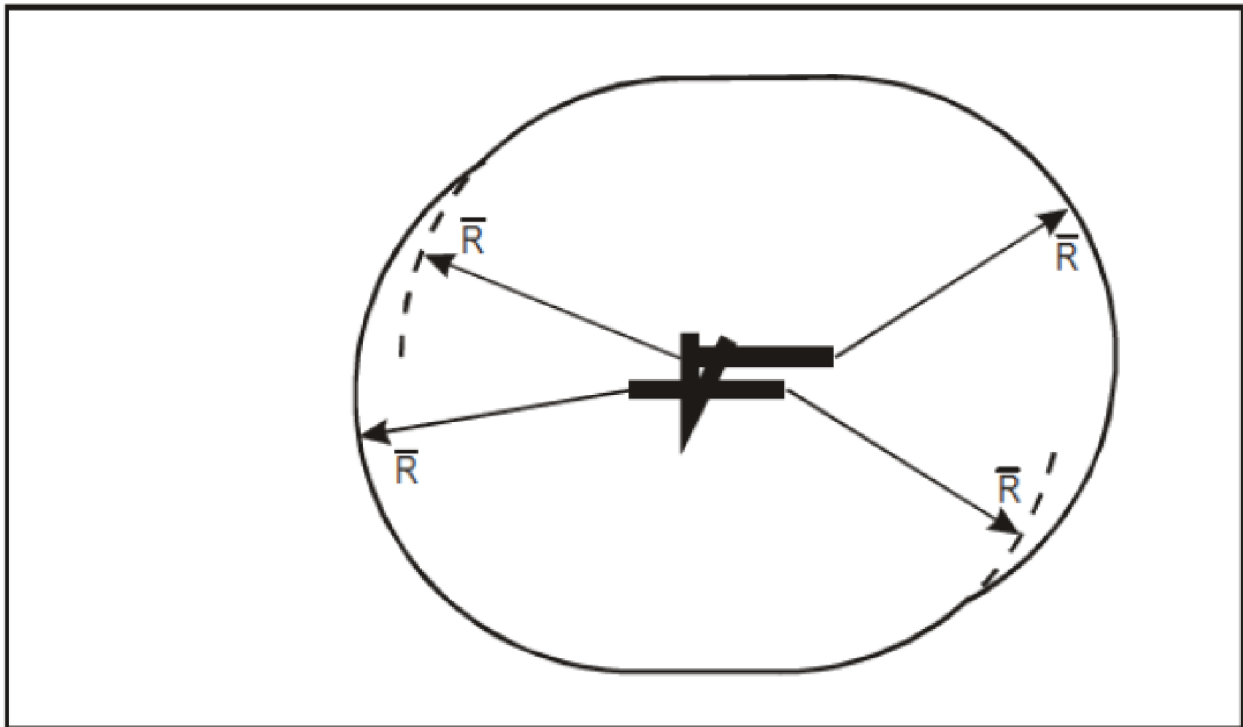
Circling Approach Obstacle Clearance

The following highlights the different radii from runway threshold used by PANS-OPS and TERPS to construct the circling obstacle clearance area.

a. Obstacle clearance:

TERPS uses a minimum obstacle clearance of 300 feet whereas PANS-OPS uses 295 feet for CAT B and 394 feet for Category C and D aeroplanes.

b. Radii from runway threshold:



ii. TERPS - New Criteria

The R varies with the circling altitude.

Circling MDA (ft-MSL)	ACFT CAT / IAS kt			
	CAT B / 1320	CAT C / 140	CAT D / 165	CAT E / 200
	R from Threshold (NM)			
1000 or less	1.7	2.7	3.6	4.5
1001 - 3000	1.8	2.8	3.7	4.6
3001 - 5000	1.8	2.9	3.8	4.8
5001 - 7000	1.9	3	4	5
7001 - 9000	2	3.2	4.2	5.3
9001 and above	2.1	3.3	4.4	5.5

A. Conducting Circling Approaches Under TERPS

The applicable Aeroplane category minima shall be used.

If the speed to be used for the circling is greater than the maximum speed stated for the specific aeroplane category on the chart, then the minima for the next higher category shall be used.

a. Flying inside TERPS circling radii:

100 ft shall be added to the published circling MDA regardless of aeroplane category.

b. Flying outside TERPS circling radii:

- If conditions require flight outside the TERPS radii flight crew shall:

Ensure they have complete awareness of obstacles within the intended circling area stated on the Jeppesen approach chart.

- Weather required for flight outside TERPS circling radii:

Day only, 5 km / 3 sm visibility, ceiling MDA + 1000 ft.

8.1.3.6.6 Visual Approach

The minimum RVR for a visual approach shall be at least 800 m.

Instructions for carrying out a visual approach are given in [8.3.28.11](#).

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8.1.3.6.7 Conversion of Reported Visibility to CMV

The following conditions apply to the use of converted meteorological visibility (CMV) instead of RVR:

1. If the reported RVR is not available, a CMV may be substituted for the RVR, except:
 - i. to satisfy the take-off minima; or
 - ii. for the purpose of continuation of an approach in LVOs
2. If the minimum RVR for an approach is more than the maximum value assessed by the aerodrome operator, then CMV should be used.
3. In order to determine CMV from visibility:
 - i. for flight planning purposes, a factor of 1.0 should be used;
 - ii. for purposes other than flight planning, the conversion factors specified in Table [8.1.3.6.7](#) should be used.

Table 8.1.3.6.7 - Conversion of reported VIS to RVR / CMV

Lighting Elements in Operation	RVR/CMV = Reported VIS multiplied by	
	Day	Night
HI approach and runway Lighting	1.5	2.0
Any type of lighting installation other than above	1.0	1.5
No lighting	1.0	Not applicable

8.1.3.6.8 Night Landings Without Vertical Guidance

Night landings to runways without vertical guidance (electronic or visual) are only authorised provided the following are complied with:

- The intended runway has a straight-in approach procedure (includes NPA CDFA and PAR) that can be conducted; and
- EGPWS is operable.

Deviations from this will only be permitted when the specific case is shown in the applicable aerodrome briefing or IJet feasibility notice.

8.1.3.7 Uncontrolled Aerodromes

When operating to uncontrolled aerodromes during

- i. Daytime with less than 8 km visibility or a ceiling below 1000' AAL; or
- ii. Nighttime with any weather conditions.

The following requirements are applicable:

- a. AFIS;
- b. Fire and rescue services have a two-way communication with the AFIS;
- c. Runway inspection has been performed within thirty minutes of the landing or take-off.

ORTS will ensure that points a., b., and c. are met and adhered to by checking with the aerodrome authorities.

Deviations from this will only be permitted when details of approval to modify these requirements, along with any applicable procedures and / or restrictions, are shown in the applicable aerodrome briefing or IJet feasibility notice.

8.1.3.8 Maximum Operating Winds

If the reported winds at an airport are above 50 kts (steady or gust) even if aligned with the RWY, the Duty Manager shall request authorisation from the DO/CP in accordance with [OM-A 2.3 Operational Control and Supervision](#) to continue operations. The DO/CP shall decide if operations are to be suspended or not, based on reports or information received from crews, from the airport itself or ATC and last but not least taking in consideration the Airport location and surrounding terrain / obstacles and shall communicate this decision in accordance with [OM-A 2.3 Operational Control and Supervision](#).

If the reported winds are above 60 Kts all operations are to be suspended.

8.1.3.8.1 Tailwind Operation

Aeroplanes will not take-off or land in tailwinds, including gusts, that exceed either the absolute or maximum demonstrated limits laid down in the AOM.

8.1.3.8.2 Crosswind Operation

Aeroplanes will not take-off or land in crosswinds, including gusts, exceeding either the absolute or maximum demonstrated limits laid down in the AOM.

In interpreting AOM limitations, pilots will also take into account associated conditions which might adversely affect the landing or take-off, such as turbulence, ice or standing water, reduced visibility, or limited runway length as follows:

- a. AOM demonstrated winds apply for all conditions except as in b. and c. below;
- b. For take-off operations when LVTO procedures are enforced and for visibility less than 400 m, 50% of demonstrated winds as per AOM or recommended AOM limitations if more restricted apply;
- c. For operations on contaminated runways and where the RWYCC, reported braking action or reported runway friction coefficients are stated, the maximum crosswind component will be the limits stated in the aeroplane specific AFM / AOM for contaminated runways or the table below if such data is not published in the AFM / AOM.

Contaminated Runway Crosswind Limitation				
RCC	Reported Brake Action		Reported Runway Friction Coefficient	Max Cross Wind
	Rep Brake Action (SNOWTAM Index)	EUR-RODEX/OPMET		
5	Good (5)	95	0.4 and above	Aeroplane Limit
4	Medium to Good (4)	94	0.36 - 0.39	20
3	Medium (3)	93	0.30 - 0.35	15
2	Medium to Poor (2)	92	0.26 - 0.29	5
1	Poor (1)	91	0.25 and below	Prohibited

NOTE

1. In case the final third of the runway is limiting, the runway may be shortened as per [Contaminated Runway Limitation](#) so as to be able to disregard the final third.
2. See [Technical Annex](#) for a description of how to establish an RWYCC from an RSD if one is not given.

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8.1.3.9 Suitable En-route Alternate Aerodromes

Planned suitable en-route aerodromes will be checked for suitability by Flight dispatch prior to any flight in accordance with the OM-A.

If, during the course of a flight, the use of an en-route alternate other than the one planned is or may be required, the flight crew shall determine the suitability of en-route alternate airports by referring to the OM-C including the Jeppesen Route Manual, PDCS data, AFM / AOM, and any other relevant document on board for information on the aerodrome. The flight crew should also contact the company to obtain assistance and further information.

OM-A sections [8.1.2](#), [8.1.3](#), [8.1.4](#), and any other relevant chapters shall be complied with when determining the suitability of the aerodrome.

NTA shall be notified by whatever means available, preferably Satellite phone call, of the intention to proceed to an en-route alternate.

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8.1.3.10 Minimum Runway Lighting Specifications for Night Operations

When one or more lights are unserviceable on a runway, this table may be used to assess whether the remaining lights will be sufficient for that lighting group to be considered operative.

Lighting group	Minimum specification to be considered operative
Runway edge lights	• Maximum runway edge light spacing for an instrument runway is 60 m. • Maximum runway edge light spacing for a non-instrument runway is 100 m. • Lights should be uniformly spaced in rows; however, at intersections to runways or due to temporary unserviceability, lights may be spaced irregularly or omitted, provided that adequate guidance remains available to the pilot.
Runway threshold lights	• A minimum of 6 threshold lights is required for a non-instrument runway. • On a precision approach runway Cat 1, at least the number of lights that would be required if the lights were uniformly spaced at intervals of 3 m between the rows of runway edge lights is required. • On a non-instrument or non-precision approach runway which has a displaced threshold, the runway threshold lights may be replaced by runway wing bar lights.
Runway wing bar lights	• Each wing bar should be formed by at least five lights extending at least 10 m outward from the runway edge lights.
Runway end lights	• A minimum of 6 runway end lights is required.
Runway centreline lights	• Maximum runway centre line light spacing is 15 m.

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8.1.4 En-route Operating Minima for VFR Flights / Portions of a Flight

NTA does not conduct VFR-only flights. For VFR flight, refer to [VFR-IFR Policy](#).

8.1.5 Presentation and Application of Aerodrome and En-route Operating Minima

The Jeppesen Airway Manual is used for presentation of en-route and aerodrome operating minima. The description of the presentation of the respective charts can be found in the introduction section of the Jeppesen Airway Manual.

1. En-route operating minima presented in the Jeppesen Airway Manual shall be used as long as they do not conflict with the minima prescribed in [8.1.1](#).
2. Aerodrome operating minima as presented in the Jeppesen Airway Manual shall be used as long as they do not conflict with the basic minima as presented under [8.1.3](#).
3. When selecting the aerodrome operating minima, the following shall be considered:
 - Crew qualifications;
 - The ground equipment required for the intended procedure are operative;
 - The aeroplane systems required for the type of approach are operative;
 - The required aeroplane performance criteria are met.
4. The higher of state or PART-CAT (described in [8.1.3](#)) minima shall be used.

NOTE

1. **Standard** - STANDARD type minima are published either on charts 10-9S, 20-9S, etc., or instrument approach charts with a Standard label on the minima box. These are in accordance with Part-CAT.
2. **TERPS** - Aerodrome operating minima in the USA that are compliant with TERPS rev 20 or later are equivalent to Part-CAT. Jeppesen depicts this compliance by placing a TERPS label in the approach minima box. For charts that do not have the TERPS label in the minima box, the minima are not Part-CAT compliant and shall not be used.
3. **JAR-OPS** - JAA-type minima may also be published either on charts 10-9X, 20-9X, etc., or instrument approach charts with a JAR OPS 1 label on the minima box. These are NOT in accordance with Part-CAT. Reference to [8.1.3.3](#) is required to establish the correct minima.
4. For Charts that do not have the label, as per 1. and 2. above, reference to [8.1.3.3](#) is required to establish the correct minima. NTA will normally provide the flight crew with the Aerodrome Operating Minima either by a feasibility note or by a NetJets Aerodrome Operating minima form.
5. Supplementary information which is either not yet included in OM-C (Route Manual), or which is of a temporary nature, is provided by NTA OM-C Temporary Revision or OM-Alert. The OM-Alert shall be studied in close connection with OM-C (Route Manual). In addition, the latest supplementary information provided by AIS NOTAM during flight preparation shall also be taken into account.

8.1.6 Interpretation of Meteorological Information

The meteorological authority designated by each State shall provide or arrange for the provision of meteorological services for international air navigation in accordance with the standards and recommended practices of ICAO Annex 3. In order to meet the requirements of aeronautical users for meteorological information on forecast en-route conditions, the World Area Forecast System (WAFS) comprising World Area Forecast Centres (WAFCs) and Regional Area Forecast Centres (RAFCs) has been established. Meteorological charts should be issued four times a day for fixed valid times of 0000, 0600, 1200, 1800 UTC and should normally be available at least nine hours before such times.

8.1.6.1 Appropriate Meteorological Information

Meteorological information used during planning or in-flight by ground personnel or flight crew members shall, under part-MET and part-CAT:

- a. cover the flight in respect of time, altitude and geographical extent;
- b. relate to appropriate fixed times or periods of time;
- c. extend to the aerodrome of intended landing, also covering the meteorological conditions expected between the aerodrome of intended landing and alternate aerodromes designated by the operator;
- d. be up to date;
- e. must originate only from authoritative sources or certified providers to avoid the risk of being tampered.

Flight crew must exercise care when using meteorological information not provided directly by NTA to ensure points a) to e) above.

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8.1.6.2 Pre-flight Planning Information

Crew members shall be supplied with the following documentation, as appropriate, before flight:

- In all cases, TAF / METARs for departure, destination, and appropriate alternate aerodromes for required periods;
- Upper wind charts in accordance with [8.1.6.3](#);
- Significant weather charts in accordance with [8.1.6.4](#);
- SIGMETs that affect the route.

This documentation shall be provided by the meteorological office or FBO at the airport of departure. If, for any reason, the documentation is not available, it shall be sent to the crew members by Flight Dispatch by fax (or email, if it can be printed).

Crew members shall check other meteorological information, as appropriate, including:

- Aerodrome warnings;
- Windshear warnings;
- Aerodrome surface and runway condition;
- RVR reports.

8.1.6.3 Charts of Upper Winds

Where upper wind and upper air temperature information is supplied in chart form to flight crew members before departure, the charts shall be fixed time prognostic charts for standard isobaric surfaces. In tropical areas, or for short flights, current charts may be provided in lieu of prognostic charts; in such cases, the level depicted shall correspond to the standard isobaric levels.

8.1.6.4 Significant Weather Charts

1. The significant weather charts should cover the following layers:

- The layer between FL 250 and FL 450;
- The layer between FL 100 and FL 250 for limited geographical areas, as determined by regional air navigation agreement.

2. Where information on significant en-route weather phenomena is supplied in chart form to flight crew members before departure, the charts shall be significant weather charts valid for a specified fixed time. Such charts shall show, as appropriate to the flight:

- Active thunderstorms;
- Tropical cyclone;
- Severe Line squalls;
- Moderate or severe turbulence (in cloud or clear air);
- Moderate or severe icing;
- Widespread sandstorm / dust storm;
- For FL 100 to FL 250, clouds associated with 1. to 6. above;
- Above FL 250, cumulonimbus cloud associated with 1. to 6. above;
- Surface position of well-defined convergence zones;
- Surface positions, speed, and direction of movement of frontal systems;
- Tropopause heights;
- Jet streams; and
- Information on the location and timing, if possible, of volcanic eruptions which are producing ash clouds of significance to aeroplane operations and a reminder to users that reference should be made to SIGMETs issued to the area concerned.

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8.1.6.5 Routine Aeronautical Meteorological Information

8.1.6.5.1 Explanation of Codes and Associate Terms Abbreviations

The METAR and TAF codes are explained in the Jeppesen Airway Manual (as part of the OM-C) (see General Airway Manual > Meteorology Service for International Air Navigation). Some codes and abbreviations are further explained as follows:

BECMG (Becoming)	Indicates an expected permanent change in the meteorological conditions occurring at a regular or irregular rate.
TEMPO	Indicates expected frequent or infrequent temporary fluctuations in the meteorological condition lasting for a period of less than one hour in each distance and, in the aggregate, cover less than one-half of the period during which the fluctuations are forecast to occur.
PROB (30 or 40)	Indicates the probability (in per cent) of the occurrence of an alternative value of a forecast element or elements.
AT	At
FM	From
TL	Until
CAVOK	Shall be used when the following conditions prevail simultaneously at the time of observation: • Visibility is 10 km or more; • No clouds below 5000 ft, or below the highest minimum sector altitude, whichever is greater, and no cumulonimbus; • No precipitation, thunderstorm, sandstorm, dust storm, shallow fog, low drifting sand, dust, or snow.
NSC	No Significant Cloud - should be used when: • No cumulonimbus; and • No cloud below 5000 ft, or below the highest minimum sector altitude, whichever is greater, are forecast and CAVOK is not appropriate.
NSW	No Significant Weather.

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8.1.6.5.2 Aviation Routine Weather Report (METAR)

At aerodromes, routine observations shall be made throughout the 24 hours each day, except as otherwise specified. Such observations shall be made at intervals of one or of one half-hour.

At other aeronautical meteorological status, such observations shall be made as determined by the meteorological authority.

Routine reports (as well as special and selected special reports) should contain the following information:

- a. Identification of the type of report;
- b. Location indicator;
- c. Time of observation;
- d. Surface wind direction and speed;
- e. Visibility (meteorological visibility);
- f. When applicable: runway visual range incl. tendency (Upward, Downward or No distinct tendency), which is an analysis of the last 10 minutes prior to the time of observation and not a forecast;
- g. Present weather;
- h. Cloud amount, type (only for cumulonimbus and towering cumulus clouds at or near the aerodrome) and height of base;
- i. Air temperature and dew point;
- j. QNH and, when applicable and for local distribution only, QFE;
- k. Supplement information comprising:
 - **RE**cent weather;
 - **W**ind **S**hear (followed by TKOF and / or LDG and RWY No.) when reported along the take-off and / or approach paths;
 - The abbreviation DENEb where fog dispersal operations are performed.

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8.1.6.5.3 Aerodrome Forecasts (TAF)

An aerodrome forecast shall be prepared by the designated meteorological office.

An aerodrome forecast shall contain the expected meteorological conditions at an aerodrome during a specified period. It shall include surface wind, visibility, weather, and cloud and expected significant changes to one or more of these elements during the period.

The period of validity of routine aerodrome forecasts is up to 30 hours, issued every three (3) hours.

The 'day of month' is included to prevent ambiguity from time periods.

Not all TAFs will have a validity of 30 hours, but any TAF of shorter duration will also follow the same format for indicating validity and change groups.

Fictitious example:

TAF LPPT 160505Z 1606/1712 13018KT 9999 BKN020 BECMG 1606/1608 SCT013CB BKN020 PROB40 TEMPO 1608/1612 21020G45 3000 TSRAGR BKN008CB TEMPO 1612/1614 20015KT FEW020 FM161700 4000 BR BKN020 TEMPO 1709/1712 BKN012 TX20/1615Z TN10/1700Z T18/1712Z=

TAF LPPT 160505Z 1606 / 1712 - TAF of Lisbon, issued at 0505 UTC on the 16th of the month, valid from 06 UTC on the 16th until 12 UTC on the 17th.

BECMG 1606 / 1608 - becoming 06 UTC on the 16th to 08 UTC on the 16th.

PROB40 TEMPO 1608 / 1612 - 40 percent chance, temporarily between 08 UTC on the 16th and 12 UTC on the 16th.

TEMPO 1612 / 1614 - temporarily between 12 UTC on the 16th until 14 UTC on the 16th.

FM161700 - From 1700 UTC on the 16th.

TX20 / 1615Z TN10 / 1700Z T18 / 1712Z - max temperature of 20oC at 1500 UTC on the 16th, and min temperature of 10oC at 0000 UTC on the 17th, temperature 18oC at 1200 UTC on the 17th.

Aerodrome forecasts and amendments thereto shall be:

- In the TAF code form prescribed by the Meteorological Organisation;
- In abbreviation plain language;

or

In teletypewriter characters and symbols.

A change described by the indicator BECMG is expected to occur at an unspecified time during the indicated time period. If a temporary fluctuation is expected to last an hour or longer, the change group BECMG should be used (instead of TEMPO).

FM on its own signifies a significant to a complete set of weather conditions from the corresponding time period onwards.

The probability group PROB (per cent) may as necessary be combined with the change indicator TEMPO; it should not be used to qualify the change indicator BECMG nor the time indicator FM.

8.1.6.5.4 TREND (Trend Type Landing Forecast)

A trend-type landing forecast is a routine, special, or selected special report for an aerodrome (METAR, SPECI) supplemented by the expected trend of the meteorological conditions at that aerodrome.

The period of validity of a trend-type landing forecast shall be 2 hours from the time of the report. The trend-type landing forecast shall indicate expected significant changes - as defined by certain prescribed values - of one or more of the elements surface wind, visibility, and cloud as well as the expected onset, cessation, or change in intensity of the following weather phenomena:

- a. Freezing precipitation;
- b. Moderate or heavy rain, snow, ice pellets, hail, small hail and / or snow pellets, rain and snow mixed;
- c. Low drifting dust, sand, or snow;
- d. Blowing dust and / or snow (including snowstorm);
- e. Dust storm;
- f. Sandstorm;
- g. Thunderstorm (with rain, ice pellets, hail, small hail and / or snow pellets, snow, or combinations thereof;
- h. Squall;
- i. Funnel cloud (tornado or waterspout).

When no change is expected to occur, this shall be indicated by the term 'NOSIG'.

When a change is expected to occur, the trend part of the trend-type forecast shall begin with one of the change indicators BECMG or TEMPO used either alone or in appropriate combination with the time indicators FM, TL, or AT. The indicator PROB shall not be used in trend-type landing forecasts.

8.1.6.5.5 Forecast for Take-off

A forecast for take-off shall be prepared by a meteorological office. It should be supplied to operators and flight crew members on request within the 3 hours before the expected time of departure.

This forecast should contain information on expected conditions over the runway complex in regard to surface wind direction and speed and any variations thereof, temperature, pressure (QNH), and any other elements as agreed locally.

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8.1.6.5.6 Application of Aerodrome Forecast (TAF & TREND) to Pre-flight Planning

APPLICATION OF AERODROME FORECASTS (TAF AND TREND) TO PRE-FLIGHT PLANNING						
(a) APPLICATION OF INITIAL PART OF TAF						
(1) Application period: from the start of the TAF validity period to the time of applicability of the first subsequent 'FM...' or 'BECMG', or if no 'FM...' or 'BECMG' is given, to the end of the validity period of the TAF.						
(2) Application of forecast: the forecast of the prevailing weather conditions in the initial part of the TAF should be fully applied, with the exception of mean wind and gusts that should be applied in accordance with the policy under column						
"BECMG AT and FM..." in the table below. However, this may be temporarily superseded by a 'TEMPO' or 'PROB**', if applicable according to the table below.						
(b) APPLICATION OF FORECAST FOLLOWING CHANGE INDICATION IN THE TAF AND TREND						
TAF or TREND for AERODROME PLANNED AS:	FM... (alone) and BECMG AT :	BECMG (alone), BECMG FM, BECMG TL, BECMG FM... TL in case of:		TEMPO (alone), TEMPO FM , TEMPO FM... TL , PROB 30/40 (alone)		PROB TEMPO
	Deterioration and improvement	Deterioration	Improvement	Deterioration	Improvement	Deterioration and improvement
				Transient/shower conditions in connection with short-lived weather phenomena, e.g. thunderstorms, showers	Persistent conditions in connection with e.g. haze, mist, fog, dust storm/sandstorm, continuous precipitations	In any case
DESTINATION at ESTIMATED TIME OF ARRIVAL (ETA) ± 1 HR	Applicable from the start of change	Applicable from the start of change	Applicable from the end of change	Not applicable	Applicable	
TAKE-OFF ALTERNATE at ETA ± 1 HR	Mean wind should be within required limits	Mean wind should be within required limits	Mean wind should be within required limits	Mean wind exceeding required limits may be disregarded	Mean wind should be within required limits	
DESTINATION ALTERNATE at ETA ± 1 HR	Gusts exceeding crosswind limits should be fully applied	Gusts exceeding crosswind limits should be fully applied	Gusts exceeding crosswind limits should be fully applied	Gusts exceeding required limits may be disregarded	Gusts exceeding crosswind limits should be fully applied	Should be disregarded
FUEL ERA at ETA ± 1 HR						Deterioration may be disregarded. Improvement should be disregarded including mean wind and gusts.
ETOPS ERA From (earliest ETA) -1h to (latest ETA) +1h as specified in 8.5.4.4	Applicable from the start of change Mean wind should be within required limits Gusts exceeding crosswind limits should be fully applied	Applicable from the start of change Mean wind should be within required limits Gusts exceeding crosswind limits should be fully applied	Applicable from the end of change Mean wind should be within required limits Gusts exceeding crosswind limits should be fully applied	Applicable if below applicable landing minima Mean wind should be within required limits Gusts exceeding limits crosswind should be fully applied	Applicable if below applicable landing minima Mean wind should be within required limits Gusts exceeding limits crosswind should be fully applied	
* The space following 'FM' should always include a time group, e.g. FM1030. Note 1: 'required limits' are those contained in the OM. Note 2: If promulgated aerodrome forecasts do not comply with the provisions of ICAO Annex 3, operators should ensure that guidance on the application of these reports is provided. Note 3: For the definitions of the meteorological terms used in this table, see ICAO Annex 3.						

8.1.6.6 Non-routine Aeronautical Meteorological Information

Non-routine aeronautical meteorological information is issued in addition to the above mentioned routine information, either when actual weather conditions change significantly, or when weather conditions are expected to change with respect to a previously issued forecast, or as warnings.

The following non-routine aeronautical meteorological information is provided:

- Special Reports (SPECI);
- Selected Special Reports (SPECI);
- Amended TAFs;
- SIGMET Information;
- Aerodrome Warnings;
- Wind Shear Warnings.

8.1.6.6.1 Special Observations and Reports

Special observations shall be made in accordance with a list of relevant criteria established by the meteorological authority in consultation with the ATS authority and the operators. Reports of special observations shall be prepared for use at the aerodrome of origin.

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8.1.6.6.2 Selected Special Reports

Reports of special observations indicating changes in accordance with the following criteria shall be prepared as selected special reports intended primarily for dissemination beyond the aerodrome of origin:

- a. When the mean surface wind direction has changed by 60 degrees or more from that given in the latest report, the mean speed before and / or after the change being 20 kts or more;
- b. When the mean surface wind speed has changed by 10 knots or more from that given in the latest report;
- c. When the variation from the mean surface wind speed (gusts) has increased by 10 kts or more from that given in the latest report, the mean speed before and / or after the change being 15 kts or more;
- d. When the wind changes through values of operational significance;
- e. When the visibility changes to or passes 800 or 1.500 m, except that such reports are not required when runway visual range is given;
- f. When the runway visual range changes to or passes 150, 350, 600 or 800 m;
- g. When the onset, cessation or change in intensity of any of the weather phenomena listed occurs;
- h. When the ceiling changes to or passes 60, 150 or 300 m (200, 500 or 1.000 ft);
- i. When the amount of a cloud layer below 450 m (1.500 ft) changes from 4 oktas or less to more than 4 oktas and v.v. (SCT to BKN to OVC).

8.1.6.6.3 Amended TAFs

Meteorological offices preparing aerodrome forecasts shall keep the forecasts under continuous review and, when necessary, shall issue amendments promptly. The criteria used for the inclusion of change groups in aerodrome forecasts or for the amendment of aerodrome forecasts should be based on the following:

- a. When the mean surface wind direction is forecast to change by 60° or more, the mean speed before and / or after the change being 5 m/s (10 kts) or more;
- b. When the mean surface wind speed is forecast to change by 5 m/s (10 kts) or more;
- c. When the variation from the mean surface wind speed (gusts) is forecast to increase by 5 m/s (10 kts) or more, the mean speed before and / or after the change being 7.5 m/s (15 kts) or more;
- d. When the surface wind is forecast to change through values of operational significance. The threshold values should be established by the meteorological authority in consultation with the appropriate ATS authority and operators concerned, taking into account changes in the wind which would:
 - i. Require a change in runway(s) in use; and
 - ii. Indicate that the runway tailwind and crosswind components will change through values representing the main operating limits for typical aircraft operating at the aerodrome;

- e. When the visibility is forecast to improve and change to or pass through one or more of the following values, or when the visibility is forecast to deteriorate and pass through one or more of the following values:
- i. 150, 350, 600, 800, 1500 or 3000 m; or
 - ii. 5000 m in cases where significant numbers of flights are operated in accordance with the visual flight rules.
- f. When any of the following weather phenomena or combinations thereof are forecast to begin or end:
- ice crystals;
 - freezing fog;
 - low drifting dust, sand, or snow;
 - blowing dust, sand, or snow;
 - thunderstorm (without precipitation);
 - squall;
 - funnel cloud (tornado or waterspout).
- g. When the height of base of the lowest layer or mass of cloud of BKN or OVC extent is forecast to lift and change to or pass through one or more of the following values, or when the height of the lowest layer or mass of cloud of BKN or OVC extent is forecast to lower and pass through one or more of the following values:
- i. 30, 60, 150 or 300 m (100, 200, 500 or 1000 ft); or
 - ii. 450 m (1500 ft) in cases where significant numbers of flights are operated in accordance with the visual flight rules;
- h. When the amount of a layer or mass of cloud below 450 m (1500 ft) is forecast to change:
- i. from NSC, FEW, or SCT to BKN or OVC; or
 - ii. from BKN or OVC to NSC, FEW, or SCT;
- i. When the vertical visibility is forecast to improve and change to or pass through one or more of the following values, or when the vertical visibility is forecast to deteriorate and pass through one or more of the following values: 30, 60, 150 or 300 m (100, 200, 500 or 1000 ft); and
- j. Any other criteria based on local aerodrome operating minima, as agreed between the meteorological authority and the operators.

8.1.6.6.4 SIGMET Information

SIGMET information shall be issued by a meteorological watch office and shall be give a concise description in abbreviated plain language concerning the occurrence and / or expected occurrence of specified en-route weather phenomena, which may affect the safety of aeroplane operations, and of the development of those phenomena in time and space. The information shall, for subsonic cruising levels, be indicated using one of the following as appropriate:

Thunderstorm	
» <i>obscured</i>	OBSC TS
» <i>embedded</i>	EMBD TS
» <i>frequent</i>	FRQ TS
» <i>Line squall</i>	LSQ TS
» <i>obscured with heavy hail</i>	OBSC TS HVYGR
» <i>embedded with heavy hail</i>	EMBD TS HVYGR
» <i>frequent, with heavy hail</i>	FRQ TS HVYGR
» <i>Line squall with heavy hail</i>	LSQ TS HVYGR
Tropical cyclone	TC (+ cyclone name)
Severe turbulence	SEV TURB
Severe icing	SEV ICE
» <i>due to freezing rain</i>	SEV ICE (FZRA)
Severe mountain wave	SEV MTW
Heavy dust storm	HVY DS
Heavy sandstorm	HVY SS
Volcanic ash	VA (volcano name)
Thunderstorm	

SIGMET messages shall be prepared in abbreviated plain language, using approved ICAO abbreviations and numerical values of self-explanatory nature.

The period of validity of a SIGMET message should not exceed 6 hours, preferably not more than 4 hours. In the special case of SIGMET messages for volcanic ash cloud and tropical cyclones, an outlook should be included giving information for up to 12 hours beyond the period of validity specified above concerning the trajectory of the volcanic ash cloud and positions of the tropical cyclone centre.

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8.1.6.6.5 Aerodrome Warnings

Aerodrome warnings shall give concise information, in plain language, of meteorological conditions which could adversely affect aeroplane on ground, including parked aeroplane, and the aerodrome facilities and services.

Aerodrome warnings should relate to the occurrence or expected occurrence of one or more of the following phenomena:

- a. Tropical cyclone;
- b. Thunderstorm;
- c. Hail;
- d. Snow;
- e. Freezing precipitation;
- f. Hoar frost or rime;
- g. Sandstorm;
- h. Dust storm;
- i. Rising sand or dust;
- j. Strong surface wind and gusts;
- k. Frost.

8.1.6.6.6 Wind Shear Warnings

Wind shear warnings should give concise information, in abbreviated plain language, of the observed or expected existence of wind shear which could adversely affect aeroplane on the approach path or take-off path between runway level and 500 m (1600 ft) above that level.

NOTE

Where local topography has been shown to produce significant wind shears at heights in excess of 500 m (1600 ft) above runway level, then 500 m (1600 ft) should not be considered restrictive.

8.1.6.7 Briefing / Consultation

Briefing and / or consultation shall be provided, on request, to flight crew members. Its purpose shall be to apply the latest available information on existing and expected meteorological conditions along the route to be flown, at the aerodrome of intended landing, alternate aerodromes, and other aerodromes as relevant.

Where local circumstances at an aerodrome make personal briefing or consultation impracticable, the meteorological office should provide those services by telephone or other suitable telecommunications facilities.

8.1.6.7.1 Pre-departure Amendments

Whenever necessary and possible, the flight documentation should be brought up to date, in writing or orally, before it is supplied to the flight crew members. In cases where a need for amendments arises after the flight documentation has been supplied, and before take-off of the aeroplane, the meteorological office should, as agreed locally, issue the necessary amendment or up-dated information to the operator or to the local air traffic services unit, for transmission to the aeroplane.

8.1.6.8 Meteorological Information for Aeroplane in Flight

Meteorological information including aerodrome reports and forecasts, SIGMET information, upper winds and upper air temperatures shall be disseminated to aeroplane in flight through HF and VHF broadcasts, in particular by one of the following means:

- a. Issuance by FIC / ACC on its own initiative (SIGMET, SPECIs, Amended TAFs);
- b. Scheduled broadcasts of aerodrome weather reports and aerodrome forecasts (VOLMET, ATIS);
- c. Issuance by the appropriate ATS unit of information in response to a request originated by an aeroplane in the assigned area.

Meteorological information, obtained from the appropriate meteorological office may also be supplied to the aeroplane by NTA either on their own initiative or in response to a request received from the aeroplane.

8.1.6.9 Aeroplane Meteorological Observations

The following aeroplane observations shall be made:

- a. Routine aeroplane observations;
- b. Special aeroplane observations;
- c. Aeroplane observations during climb-out and approach;
- d. Other aeroplane observations on request.

8.1.6.9.1 Routine Aeroplane Observations

Routine observations shall be made in relation to those air traffic services reporting points or intervals:

- At which the applicable air traffic service procedures require routine position reports.
- Which are those separated by distances corresponding most closely to intervals of one hour of flying time.

In the absence of designated reporting points, position reports should be made as soon as possible after the first half hour of the flight and at hourly intervals thereafter.

An aeroplane shall be exempted from making routine observations when:

- The flight duration is 2 hours or less; or
- The aeroplane is at a distance equivalent to less than one hour of flying time from the next intended point of landing; or
- The altitude of the flight path is below 1500 m (5000 ft).

NOTE

Under conditions specified by the competent authority, flights may be exempted from the requirement to make position reports at each designated compulsory reporting point or interval.

Exemptions may also be prescribed by regional air navigation agreement for flights over routes and areas with high density air traffic and / or with adequate synoptic networks.

When under an exemption to make position reports, account should be taken of the meteorological requirement for the making and reporting of routine aircraft observations.

In some regions, designation procedures may be used where only some aeroplanes are designated to make and transmit observations.

8.1.6.9.2 Special Aeroplane Observations

Special observations shall be made by all aeroplane operating international air routes whenever:

- Severe turbulence or severe icing (see Definitions below) is encountered; or
- Vo-eruption volcanic activity or a volcanic eruption is observed.

NOTE

Pre-eruption volcanic activity in this context means unusual and / or increasing volcanic activity which could precede a volcanic eruption.

Definitions

Turbulence: See [8.3.8.4](#).

8.1.6.9.3 Aeroplane Observation During Climb-out and Approach

Observations shall be made by all aeroplanes of meteorological conditions encountered during the climb-out or approach phases of flight, not previously reported to the Commander, which are likely to affect the safety of other aeroplane operations.

8.1.6.9.4 Other Aeroplane Observations

Observations shall also be made by aeroplane:

- If a meteorological office providing meteorological service for a flight makes a request for specific data; or
- By agreement between a Meteorological Authority and an operator.

8.1.6.10 Recording and Reporting of Aeroplane Observations

Routine and special aeroplane observations shall be reported in the AIREP format (as per the 'Model AIREP Special' contained in the Jeppesen Airway Manual / General / ICAO ATM / Appendix 1 - Instructions for air-reporting by voice communications). Special aeroplane observations of pre-eruption volcanic activity, a volcanic eruption, or volcanic ash cloud shall be reported in the Special Air Report of Volcanic Activity format (as per the 'Special Air-report of volcanic activity form (model VAR)' contained in the Jeppesen Airway Manual / General / ICAO ATM / Appendix 1 - Instructions for air-reporting by voice communications). Other aeroplane observations should only be recorded when this is specifically required.

Routine aeroplane observations shall be reported during flight as routine air reports at times of transmission of the associated position report. Special aeroplane observations shall be reported during flight as special air-reports, as soon as practicable after they have been made.

NOTE

In some areas a system of meteorological reporting points has been established. At selected ATS compulsory reporting points, aeroplanes are required to report, in addition to section 1, also section 3 of the AIREP. Such meteorological reporting points are denoted on the applicable Enroute Chart with the symbol M in a circle.

Aeroplane observations made during the climb-out and approach phase of flight shall be reported as soon as practicable.

Where wind shear conditions in the climb-out or approach phases of flight were reported or forecast but not encountered, the Commander should advise the appropriate air traffic services unit as soon as practicable.

Other aeroplane observations should be reported in flight, unless it has been agreed that such reporting is not necessary.

8.1.6.11 Post-flight Reporting of Aeroplane Observations

On arrival of a flight at an aerodrome, the completed AIREP form shall be delivered by NTA or a flight crew member, without delay, to the aerodrome meteorological office. If there is no aerodrome meteorological office, or if such office is not easily accessible to arriving flight crew members, the completed form shall be dealt with in accordance with local arrangements.

8.1.6.11.1 Surface Wind

For reports for take-off, the surface wind observations should be representative of conditions along the runway and for reports for landing the observations should be representative of the touchdown zone.

The averaging period for wind observations should be:

- i. 10 (ten) minutes for reports disseminated beyond the aerodrome, except that when the 10-minute period includes a marked discontinuity in wind direction and / or speed, only data occurring since the discontinuity should be used;
- ii. 2 (two) minutes for reports used at the aerodrome for take-off and landing and for wind indicators in air traffic services units.

In reports for take-off and landing, variations in the wind direction should be given when the total variation is 60 degrees or more with mean speeds above 3 kts; such directional variations should be expressed as the two extreme directions between which the wind has varied during the past 10 minutes.

Variations from the mean wind speed (gusts) during the past 10 minutes should be reported only when the variation from the mean speed has exceeded 10 kts; such speed variations (gusts) should be expressed as the maximum and minimum speeds attained. When the 10-minute period includes a marked discontinuity in the wind direction and / or speed, only variations in direction and speed occurring since the discontinuity should be reported.

NOTE

In METARs, TAFs, and other information received from Meteorological Authorities, the wind direction is given in degrees true north. Information on wind direction transmitted by ATC to arriving and departing aeroplane either directly or via ATIS broadcasts is expressed in degrees magnetic.

8.1.6.11.2 Visibility

For reports for take-off the visibility observations should be representative of the takeoff and climb-out area, and for reports for landing the observations should be representative of the approach and landing area. Visibility observations made for reports disseminated beyond the aerodrome should be representative of the aerodrome and its immediate vicinity; in such observations special attention should be given to significant directional variations.

8.1.6.11.3 Runway Visual Range (RVR)

Runway visual range observation should be made, and the runway visual range reported, throughout periods when either the horizontal visibility or the runway visual range is observed to be less than 1500 meters.

Runway visual range observations should be representative of the touchdown zone and, as may be selected by the authority concerned, of the middle and far sections of the runway.

In reports disseminated beyond the aerodrome, only the value representative of the touchdown zone should be given.

Runway visual range observation should be made on all runways intended for use during periods of reduced visibility and in particular on:

- Precision approach runways;
- Runways used for take-off and having high-intensity edge lights and / or centreline lights.

The reporting scale should consist of increments between 25 m and 60 m for runway visual range up to 800 m and increments of 100 m for runway visual range above 800 m.

Runway visual range should be reported to the appropriate local air traffic services units, whenever there is a change in the value to be reported in accordance with the reporting scale.

8.1.6.11.4 Clouds

Cloud observations for reports for landing should be representative of the approach area or, in the case of aerodrome with precision approach runways, of the middle marker site of the instrument landing system. Cloud observations made for reports disseminated beyond the aerodrome should be representative of the aerodrome and its immediate vicinity.

The height of the base of cloud should normally be reported above aerodrome elevation. When a precision approach runway is in use, which has a threshold elevation 15 m or more below the aerodrome elevation, local arrangements should be made in order that the height of clouds reported to arriving aeroplane should refer to the threshold elevation.

When several layers or masses of cloud are observed, their amount and height should be reported in the following order:

- a. The lowest layer or mass, regardless of amount to be reported as SCT, BKN, or OVC as appropriate;
- b. The next layer or mass, covering more than 2/8 to be reported as SCT, BKN, or OVC as appropriate;
- c. The next higher layer or mass, covering more than 4/8 to be reported as BKN or OVC as appropriate;
- d. Cumulonimbus and / or towering cumulus clouds, whenever observed and not reported in above.

8.1.7 Determination of the Quantities of Fuel and Oil Carried - Fuel Policy

8.1.7.1 General Policy

The following fuel policy is an integral part of NTA's fuel scheme and should be utilised for both pre-flight planning and in-flight re-planning. NTA complies with the requirements for the variations to a basic fuel scheme, as outlined in CAT.OP.MPA.180 and CAT.OP.MPA.181.

Preparing an operational flight plan (OFP) is the responsibility of Flight Dispatch, and it is done in compliance with the requirements outlined in this policy. It is accomplished using a computerised flight planning tool to calculate fuel quantities considering all aspects outlined in section [8.1 Flight Preparation Instructions](#) and section [8.1.11 Aeroplane Technical Log - Flight and Maintenance Report](#).

All aircraft and airspace requirements or limitations must be strictly adhered to. Certain assumptions regarding operating conditions must be made during the planning phase. These assumptions are critical since they directly influence the determination of the fuel quantities specified in this policy. Below is a non-exhaustive list of limitations and assumptions used during the planning process:

- Minimum (safe) altitudes and routing;
- The effects of deferred maintenance items;
- Air navigation services provider(s) procedures and airspace restrictions;
- Anticipated delays;
- Meteorological forecasts for departure, en-route, destination, and alternate aerodromes;
- Operating masses;
- Fuel burn data per fleet for flight phases, derived from the respective AFM;
- Standard taxi times, including APU usage;
- Fuel bias for specific aerodromes (departing and arriving).

The procedures described in Section [8.3.7 Policy and Procedures for In-flight Fuel Management](#) are integral to this fuel policy and are required to validate the assumptions made during the planning process.

A flight may not commence unless the aircraft is loaded with at least the required amount of fuel and oil to ensure it can be completed safely, considering the assumed operating conditions and this fuel policy. An example of the OFP layout is available in [8.1.10](#).

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8.1.7.2 Flight Crew Responsibility

The commander shall only commence taxi, take-off or continue a flight in the event of in-flight replanning when satisfied that the aeroplane carries at least the required amount of usable fuel and oil to complete the flight safely and in accordance with the provisions of this policy, considering the expected operating conditions.

The commander shall ensure that the OFP data matches the data in the flight release.

The PF shall check the fuel values in the OFP for reasonability and gross errors and confirm these values with the PM.

If an aeroplane has been refuelled but hasn't flown due to, for example, a flight cancellation, a non-activation during hot spare duty, or refuelling at the request of Flight Dispatch for a flight scheduled the next day, the crew is responsible for informing Flight Dispatch of the fuel quantity onboard. This ensures that the data in IJet is accurate for future planning.

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8.1.7.3 Standard Calculation

8.1.7.3.1 General

The fuel quantity to be carried onboard consists of multiple components as described in the following sections. This quantity shall never be less than any AFM minimum fuel requirement, if published.

8.1.7.3.2 Taxi Fuel (TAXI)

Taxi fuel shall not be less than the quantity expected to be used prior to take-off, considering that all engines will be running for the duration of the taxi. APU is assumed to be off during taxi.

Taxi time is variable, based on historical data from specific airports, with a minimum value of 10 minutes.

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8.1.7.3.3 Trip Fuel (DEST)

Trip fuel consists of the quantities necessary to:

1. Perform the take-off and climb from aerodrome elevation to the initial cruising altitude, considering the expected routing;
2. Fly from the top of climb to the top of descent, including any planned step climbs and descents;
3. Continue from the top of descent to the point where the approach to landing begins, considering the expected arrival procedure;
4. Conclude the approach and landing at the destination aerodrome.

The following factors are considered for the calculation of trip fuel:

1. The optimum FL;
2. Operational and airspace restrictions (e.g. when optimum FL cannot be planned due to expected flight conditions or due to airspace planning limitations on a given route);
3. Any biases required for specific airports where local procedures differ from the standard flight planning profile, such as, but not limited to:
 - a. expected low-level flying;
 - b. extensive radar vectoring after take-off or during approach. Note that arrival routings, like STARs, are included in the trip fuel calculation.

NOTE

These bias values are outlined in the Flight Dispatch Handbook and are integrated into the flight planning software database.

4. Type of flight:

- a. For revenue flights, the selected cruise schedule for the flight shall be reflected on the OFP;
- b. For non-revenue flights, the standard speed shall be selected as per the table below. However, for operational, fuel-related, performance-related, or time constraints, Flight Dispatch may use a different cruise schedule may be used by Flight Dispatch, which shall be reflected in the OFP. Flight Dispatch shall inform the crew of such changes by using the REMARKS line on the OFP.

Aeroplane Type	Speed / Thrust Setting
EMB-505S/E	M 0.70
CE-560XLS	M 0.65/NCT
CE-680A	M 0.70
BD-100-1A10	M 0.78
DA-2EASy	M 0.8
CL-600-2B16	M 0.80

Aeroplane Type	Speed / Thrust Setting
BD-700-1A10	M 0.83
BD-700-1A11	M 0.83

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8.1.7.3.4 Contingency Fuel (CONT)

During the planning stage, not all factors that may influence fuel consumption to the destination aerodrome can be anticipated. Therefore, contingency fuel is carried to mitigate the impact of such uncertainties, including, but not limited to:

1. Deviations of an individual aeroplane from the expected fuel consumption data;
2. Deviations from forecasted meteorological conditions;
3. Deviations from planned routing;
4. Deviations from planned cruising levels / altitudes.

The contingency fuel corresponds to a) or b) below, whichever is higher:

- a. Depending on the availability of en-route alternates, either i. or ii.:
 - i. provided that an en-route alternate (fuel ERA) is available according to [8.1.3.3](#) and [8.1.7.7](#) and that the aircraft is part of a Fuel Consumption Monitoring Programme, not less than 3% of the planned trip fuel or, in the event of in-flight replanning, 3% of the trip fuel for the remainder of the flight;
 - ii. if an en-route alternate is not available according to [8.1.3.3](#), 5% of the planned trip fuel or, in the event of in-flight replanning, 5% of the trip fuel for the remainder of the flight.
- b. An amount of fuel to fly for 5 minutes at holding speed at 1500 ft (450 m), above the destination aerodrome in standard conditions, calculated with the estimated mass on arrival at the destination aerodrome.

8.1.7.3.5 Destination Alternate Fuel

Destination alternate fuel is the quantity required to fly from the missed approach point or MDA/DA at the destination aerodrome to the destination alternate aerodrome.

Alternate aerodrome(s) shall be selected based upon the criteria detailed in OM-A [8.1.3.2](#):

The alternate fuel shall be the sum of the quantities necessary to:

1. Perform a missed approach from the applicable DA/H or MDA/H at the destination and climb to the missed approach altitude, considering the complete missed approach procedure;
2. Climb from missed approach altitude to cruising level / altitude, considering the expected departure routing;
3. Fly from the top of climb to top of descent, considering the expected routing;
4. Continue from the top of descent to the point where the approach to landing begins, considering the expected arrival procedure; and
5. Conclude the approach and landing at the destination alternate aerodrome.

If, per [8.1.3.2 \(c\)](#), two destination alternates are required, the alternate fuel shall be sufficient to proceed to the alternate that requires the greater amount of alternate fuel.

When a flight is operated with no destination alternate aerodrome, destination alternate fuel shall be the quantity required to hold at the destination aerodrome, while enabling the aeroplane to perform a safe landing, and to allow for deviations from the planned operation. This amount is described in point b. of [Additional Fuel](#).

NOTE

The FL indication shown in the ALTERNATE information line on the OFP is for fuel calculations ONLY and may be below MSA in some cases.

8.1.7.3.6 Final Reserve Fuel (FRSV)

Final reserve fuel is the quantity required to fly for 30 minutes at holding speed at 1500 ft (450 m), above the destination alternate aerodrome in standard conditions, calculated with the estimated mass on arrival at the destination alternate aerodrome or at the destination aerodrome when no destination alternate aerodrome is required.

8.1.7.3.7 Additional Fuel (ADDL)

Additional fuel is the quantity that shall permit:

- a. the aeroplane to descend as necessary and proceed to an adequate alternate aerodrome in the event of engine failure or loss of pressurisation, whichever requires the greater amount of fuel based on the assumption that such a failure occurs at the most critical point along the route, and
 - i. hold there for 15 minutes at 1500 ft (450 m) above aerodrome elevation in standard conditions; and
 - ii. make an approach and landing, except that additional fuel is only required, if the sum of the Trip Fuel, Contingency Fuel, Alternate Fuel, and Final Reserve is not sufficient for such an event, and
- b. when a flight is operated without a destination alternate aerodrome, holding for 15 minutes at 1500 ft (450 m) above destination aerodrome elevation in standard conditions, calculated according to the estimated aeroplane mass on arrival at the destination aerodrome.

Additional fuel is required only if the minimum quantity of fuel that is calculated for trip, contingency, destination alternate and final reserve is not sufficient to cover for point a, after such an event, or point b. as applicable.

For specific Non-ETOPS 180 fuel planning requirements, refer to [Fuel and Oil Quantities](#).

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8.1.7.3.8 Extra Fuel (XTRA)

The minimum extra fuel is an amount sufficient to accommodate anticipated delays or specific operational constraints.

Acknowledging the dynamic and ad-hoc nature of the NTA's operations, a quantity of fuel corresponding to 30 minutes of flight is carried (calculated as defined in 'Final Reserve Fuel'). This amount is deemed sufficient to limit the impact of the variations that arise from not having a fixed route network, which results in more frequent changes to flight parameters and, consequently, reduced predictability on the assumptions made during the planning stage, when compared to a typical scheduled airline operation. While this variance in the operation is common and therefore anticipated, the extent of the variability for each flight is impossible to determine. The selection of 30 minutes serves as a compromise between operational stability and environmental objectives.

The Extra fuel may be reduced to the minimum on the commander's request or when the aeroplane's performance, fuel capacity or mass and balance limits do not allow the full 30 minutes to be carried. Notwithstanding the commander's and dispatch's discretion to reduce from the standard 30 minutes, the amount carried as extra fuel shall not be less than the amount necessary to accommodate anticipated delays or other operational constraints.

When extra fuel is reduced, the commander shall closely monitor adherence to flight plan, namely by following the procedures laid in [8.3.7](#), keeping in mind the reduced tolerance for changes in the operation.

8.1.7.3.9 Discretionary Fuel

If the commander decides to take discretionary fuel (beyond the amount needed to round the TREQ fuel figure on the OFP to the nearest 200 lbs), the reason for this decision must be included in the Reply to Release message, using the features available in the crew app. After receiving the commander's request to add discretionary fuel, the OFP will be updated accordingly. Due to the limitations of the flight planning system, the discretionary quantity will then be shown in the revised OFP's XTRA fuel component.

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8.1.7.3.10 Minimum Required Fuel

The legal minimum usable fuel required to be onboard at blocks-off is:

1. The sum of:

- Taxi fuel (TAXI);
- Trip fuel (DEST);
- Contingency fuel (CONT);
- Destination alternate fuel (ALTN);
- Final Reserve fuel (FRSV);
- Additional fuel, if required (ADDL);
- The quantity of Extra Fuel (XTRA) included to accommodate anticipated delays or specific operational constraints (see [8.1.7.3.8](#)).

2. At the commander's discretion, this will normally also include the sum of (see Note):

- The quantity of Extra Fuel (XTRA) in excess of that included to accommodate anticipated delays or specific operational constraints;
- Discretionary Fuel (DSCR)

NOTE

Extra fuel in excess of that required for anticipated delays or specific operational constraints and Discretionary fuel, must be on board at release. However, at the commander's discretion, it may be used before going off-blocks (e.g., more APU burn than planned). The commander should, in such cases, closely monitor adherence to the flight plan, bearing in mind the possibly reduced tolerance for changes in the operation. Notwithstanding this discretion, all fuel quantities specified in point 1 above must be on board at blocks-off.

The minimum legal fuel required for take-off is the same as required for blocks-off minus anticipated taxi fuel (includes planned quantity plus any additional quantity for taxiing requested by the commander).

8.1.7.3.11 Release Fuel (RLS)

Release fuel is the quantity expected to be on board at the estimated time of off-blocks when released by dispatch and shall always include the following:

- Taxi fuel (TAXI);
- Trip fuel (DEST);
- Contingency fuel (CONT);
- Destination alternate fuel (ALTN);
- Final Reserve fuel (FRSV);
- Additional fuel, if required (ADDL);
- Extra Fuel (XTRA);
- Discretionary Fuel (DSCR).

8.1.7.3.12 Economic Tankering Fuel

Flight Dispatch shall examine each flight plan, including the next two subsequent planned flights, to determine the optimal tankering fuel quantity. This shall be reflected on the OFP.

Economic Tankering fuel may be used up to the lower of:

- Applicable maximum take-off mass minus the mass of two passengers;
- Applicable maximum landing mass minus the mass of two passengers;
- Applicable limitations for the subsequent takeoff, landing and payload.

Economic Tankering shall be applied when the Fuel Index at the destination aerodrome exceeds the Fuel Index at the departure aerodrome with reference to the table below:

Fuel index difference (Destination - Departure) when it is economical to tanker				
Departure Aerodrome Fuel Index	Sector length			
	< 1.5 hrs	1.5 - 3 hrs	3 - 5 hrs	> 5 hrs
≤ 150	≥ 5	≥ 10	≥ 20	≥ 35
> 150 ≤ 200	≥ 7	≥ 15	≥ 25	≥ 50
> 200	≥ 10	≥ 20	≥ 40	≥ 70

Economic Tankering fuel is always additional to the Minimum Required fuel.

Due to technical limitations of NTA's computerised flight planning system, Economic Tankering fuel is included in the XTRA field of the operational flight plan.

NOTE

Economic tankering fuel shall not be carried on flights:

- arriving at locations classified as RefuelEU union Airports or under similar minimum fuel uplift requirements from the state of the aerodrome;
- between aerodromes in Switzerland.

8.1.7.3.13 APU Fuel (APU)

APU Fuel (APU) is the fuel expected to be used by the APU prior to off-blocks. The commander shall determine the amount of APU Fuel to be loaded, considering local conditions and restrictions and referring to [8.3.35.2](#) and [8.3.35.3](#) when making their assessment.

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8.1.7.4 Reduced Contingency Fuel (RCF) Procedure

When pre-flight planning to a destination 1 aerodrome (commercial destination) with an RCF procedure using a decision point along the route and a destination 2 aerodrome (optional refuel destination), the amount of usable fuel, on board for departure, shall be the greater of a. or b. below:

- a. The sum of:
 - i. taxi fuel;
 - ii. trip fuel to the destination 1 aerodrome, via the decision point;
 - iii. contingency fuel equal to not less than 5% of the estimated fuel consumption from the decision point to the destination 1 aerodrome;
 - iv. alternate fuel or no alternate fuel if the decision point is at less than six hours from the destination 1 aerodrome and the requirements [8.1.3.2-b. i. -2](#), are fulfilled;
 - v. final reserve fuel;
 - vi. additional fuel;
 - vii. extra fuel if required by the commander; and
 - viii. discretionary fuel, if required by the commander; or
- b. The sum of:
 - i. taxi fuel;
 - ii. trip fuel to the destination 2 aerodrome, via the decision point;
 - iii. contingency fuel equal to not less than the amount calculated in accordance with [8.1.7.3.3](#) above from departure aerodrome to the destination 2 aerodrome;
 - iv. alternate fuel, if a destination 2 alternate aerodrome is required;
 - v. final reserve fuel;
 - vi. additional fuel; and
 - vii. extra fuel if required by the commander;
 - viii. discretionary fuel, if required by the commander.

8.1.7.5 Isolated Aerodrome Procedure

NTA does not operate to isolated aerodromes.

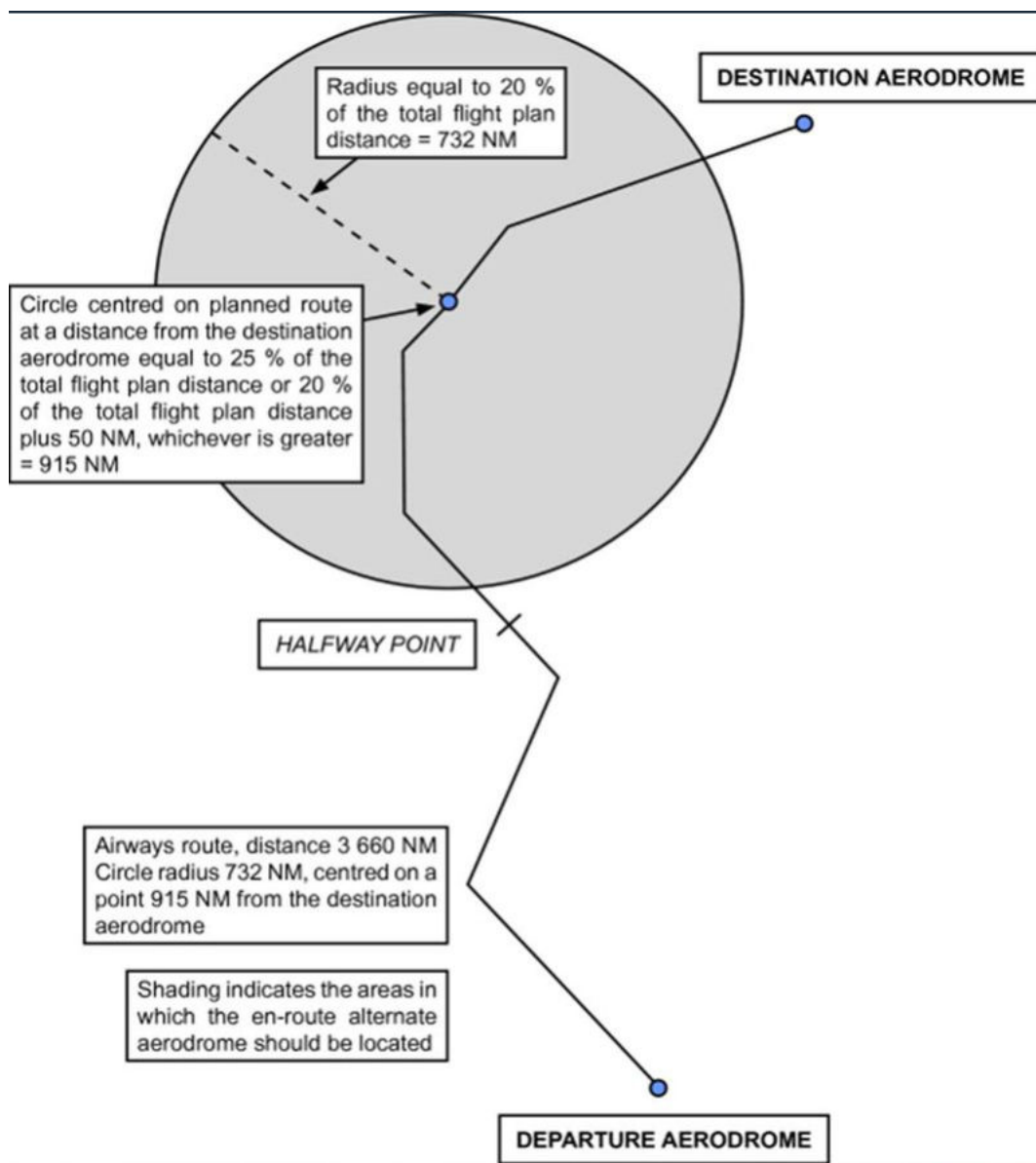
8.1.7.6 Hotspare and Back-up Fuel

Pre-emptive refuelling of Hotspare or Backup aircraft can result in excessive fuel onboard upon landing at a Union Airport and may lead to violations of the RefueLEU rules.

When assigned to Hotspare or Backup duty, do not proceed with fuel uplift until officially activated for the recovery flight unless explicitly instructed to do so by dispatch.

8.1.7.7 Location of the Fuel ERA

The fuel ERA aerodrome should be located within a circle having a radius equal to 20% of the total flight plan distance, the centre of which lies on the planned route at a distance from the destination aerodrome of 25% of the total flight plan distance, or at least 20% of the total flight plan distance plus 50 nm, whichever is greater. All distances should be calculated in still air conditions (see figure below)



8.1.7.8 In-flight Replanning Fuel

In the case of in-flight replanning, the usable fuel required when a flight proceeds along a route or to a destination aerodrome other than originally planned shall include:

1. trip fuel for the remainder of the flight; and
2. Contingency Fuel;
3. Destination Alternate Fuel;
4. Final Reserve Fuel;
5. Additional Fuel, if required;
6. Extra Fuel;
7. Discretionary fuel, if required.

8.1.7.9 Fuel Consumption Monitoring Programme

NTA has established a Fuel Consumption Monitoring Programme in accordance with the guidelines of ICAO Doc 9976 Flight Planning and Fuel Management (FPFM) Manual. This programme allows the evaluation of an individual aeroplane's performance in comparison with the data and assumptions used in planning. Analysis of the differences shall be used to increase the accuracy of future fuel planning. The functioning of this programme is a requirement for the use of 3% contingency fuel ([Contingency Fuel](#) (a)(i)) and the reduced contingency fuel procedure ([Reduced Contingency Fuel \(RCF\) Procedure](#)).

8.1.8 Mass and Centre of Gravity

8.1.8.1 Definitions

Basic Empty Mass (BEM) - The mass and CG of the aeroplane determined by the weighing of the aeroplane in accordance with [NJMP 1.14](#). This mass shall be used to calculate the DOM by adding the non-variable mass to it.

Certified Centre of Gravity Limits - These are the CG limits the aeroplane was certified with and must not be exceeded.

Dry Operating Index (DOI) - The applicable index on the aeroplane index system corresponding to the specific DOM.

Dry Operating Mass (DOM) - The total mass of the aeroplane ready for a specific type of operation - excluding all usable fuel and traffic load. The DOM includes: crew and crew baggage, catering, removable passenger service equipment, potable water and lavatory chemicals, special emergency equipment as required for the route(s), and cabin service equipment such as newspapers, pillows, blankets, etc. From the above it follows that different DOMs may be published in the AOM for different crew, catering, or emergency equipment versions.

Maximum Allowed Mass for Landing - The maximum allowed mass for landing considers: the maximum (structural) landing mass and - the maximum permitted landing mass for performance reasons.

Maximum Allowed Mass for Take-off - The maximum allowed mass for take-off for a specific flight considers:

- The maximum (structural) take-off mass;
- The maximum permitted take-off mass as limited by performance reasons and the calculated trip fuel, to assure that the maximum permitted landing mass is not exceeded.

This is the lowest of the following two masses:

- Max Mass for Take-off; and
- Max Landing Mass + Trip fuel.

Maximum (Structural) Landing Mass (MLM) - The maximum permissible total aeroplane mass upon landing under normal circumstances. This is the maximum structural limit and shall not be mistaken with the Maximum Allowed Mass for Landing, which also considers any additional performance limitations.

Maximum (Structural) Take-off Mass (MTOM) - The maximum permissible total aeroplane mass at the start of the take-off run. This is the maximum structural limit and shall not be mistaken with the Maximum Allowed Mass for Take-off, which also considers any additional performance limitations for a particular route.

Maximum Zero Fuel Mass (MZFM) - The maximum permissible mass of an aeroplane with no usable fuel. The mass of the fuel contained in particular tanks shall be included in the ZFM when it is explicitly mentioned in the Aeroplane Flight Manual limitations.

Non-variable Mass - The term used in the mass and balance form that corresponds to the mass of the operating items that are used to calculate the DOM, including: crew and crew baggage, catering and removable passenger service equipment, potable water and lavatory chemicals, special emergency equipment as required for the route(s), and cabin service equipment such as newspapers, pillows, blankets, etc.

Operational Centre of Gravity Envelope - The operational centre of gravity envelope further restricts the certified centre of gravity envelope to compensate for errors such as the differences between assumed passenger masses and actual masses, use of one common moment-arm for several seat rows, deviations in the distribution of baggage / cargo / mail in the various compartments, inaccuracies in the actual mass of baggage, deviation caused by gear and / or flap positions, in flight movements of passengers, cabin crew and pantry equipment (trolleys), deviation in fuel distribution, etc. NTA currently defines the Operational Centre of Gravity using a 5% buffer within the certified Centre of Gravity Envelope. Flight outside the Operational CofG Envelope is only permitted in accordance with [8.1.8.8](#).

Payload (PL) - The total mass of the revenue load (pax, cargo, or mail).

Traffic Load (TL) - The total mass of passengers, baggage, and cargo including any non-revenue loads.

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8.1.8.2 Methods, Procedures, and Responsibility for Preparation and Acceptance of the Mass and Centre of Gravity Calculations

During any phase of operation, mass and balance of the aeroplane shall comply with the limitations specified in the AFM or the operations manual, if more restrictive.

The preparation of mass and balance documentation is centralised through the company's operations headquarters in Portugal. The following system shall be used:

1. A mass and balance document shall be prepared for each flight conducted by NTA.
2. It is the responsibility of Flight Dispatch to prepare the mass and balance document using an appropriate fuel load based on the fuel policy, applying fuel burn, passenger numbers, and expected baggage (already available from flight reservation and flight plan records).
3. The derived Mass and Balance Form shall be retained by the Flight Dispatch and added to the flight documentation folder for each flight. The mass and balance form will state when standard masses are not used.
4. The mass and balance form shall be verified by the flight crew to ensure the Mass and Balance is within limits and ensure that the items under 6. below are correct. The commander shall check and sign the form to approve it as correct.
5. Any change in passenger manifest, masses, or baggage shall require the preparation of a revised Mass and Balance document by Flight Dispatch. The revised document shall be made available to the flight crew prior to departure, electronically via Skybook or by other approved means.
6. The Mass and Balance Form will be computer-generated. It shall contain details of:
 - Aeroplane registration and type;
 - Flight call sign and date;
 - Name of the commander;
 - Name of the person who prepared the document;
 - Dry operating mass and the corresponding CG of the aeroplane;
 - Mass of the fuel at take-off and the mass of trip fuel;
 - Mass of consumables other than fuel, if applicable;
 - Mass and disposition of all loaded items including passengers, baggage, and ballast;
 - Take-off mass, landing mass, and zero fuel mass;
 - Applicable aeroplane CG positions; and
 - The limiting mass and CG values.
7. Nothing described above prevents the Commander from carrying out a full manual Mass and Balance computation, using the Aeroplane Flight Manual or Weight and Balance Manual, where under irregular circumstances they deem this to be necessary. The mass and balance values shall be communicated to Flight Dispatch.
8. In accordance with applicable regulations, the Commander shall indicate acceptance of the Mass and Balance documentation by hand signature or equivalent, confirming the aircraft is loaded in accordance with the plan. This acceptance will be confirmed by the

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Commander by entering their name, the Flight Plan reference number, and their signature on the paper OFP Annex. Any last-minute changes shall also be recorded on the OFP Annex.

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8.1.8.3 Standard Mass Values

NTA does not operate 'holiday charter' flights and thus differentiation between these and other flights is unnecessary.

8.1.8.4 Applicable Mass Values**8.1.8.4.1 Crew Mass Values**

Standard Masses for crew are as follows:

	Crew Mass including Hand Baggage	Baggage Mass
Flight Crew	85 kg / 187 lbs	13 kg / 29 lbs
Cabin Crew	75 kg / 165 lbs	13 kg / 29 lbs

8.1.8.4.2 Passenger Mass Values**Passenger Standard Mass Values**

The standard mass values stated in the table below shall be used depending on passenger weight category. The adult masses already consider the carriage of an infant.

Passenger Classification:

- Any persons of an age of 12 years and above are considered adults;
- Children are defined as persons of an age of two years and above, but who are less than 12 years of age;
- Infants are defined as persons who are less than two years of age.

On flights where no hand baggage is carried in the cabin or where hand baggage is accounted for separately, 6 kg / 13 lbs may be deducted from the adult masses. (Articles such as an overcoat, an umbrella, a small handbag or purse, reading material, or a small camera are not considered as hand baggage for the purpose of this sub-paragraph.)

Passenger Seats	6-9	10-19
Adult female	78 kg / 172 lbs	74 kg / 163 lbs
Adult other than female	96 kg / 212 lbs	92 kg / 203 lbs
Children	35 kg / 77 lbs	35 kg / 77 lbs

On any flight identified as carrying a significant number of passengers whose masses, including hand baggage, are expected to significantly deviate from the standard passenger mass, the passenger mass shall be determined the actual weighing of such passengers.

Actual Passenger Mass Values

At the Commander's discretion or when so determined by NTA, actual mass by weighting of passengers may be used instead of the standard mass value indicated above. If actual masses are used, all passengers including personal belongings and hand baggage shall be weighed.

Where the number of passenger seats available is less than 10, the passenger mass may be established by use of a verbal statement by or on behalf of each passenger and adding to it 10 kg / 22 lbs per passenger to account for hand baggage and clothing (4 kg for clothing and 6 kg for hand baggage).

Flight Dispatch shall be informed by the appropriate department or by the commander when the passenger mass by verbal statement is to be used and shall complete the mass and balance documentation accordingly.

The passengers' verbally stated mass and the mass of passengers' clothing and hand baggage shall be checked prior to boarding and adjusted, if necessary.

NOTE

The systems of standard mass values and actual weights cannot both be used on the same flight.

8.1.8.4.3 Pets Mass

Pet Under: domestic or tamed animal kept for companionship or pleasure weighing a mass equal or less than 40 lbs. For mass and balance purposes, 40 lbs are used in the calculation.

Pet Over: domestic or tamed animal kept for companionship or pleasure weighing a mass greater than 40 lbs. For mass and balance purposes, 60 lbs are used in the calculation.

8.1.8.4.4 Baggage Mass

Baggage Standard Mass / Count Values

When a flight is confirmed, passengers are asked to provide actual baggage mass if they anticipate it will differ from the standards defined below. Where this is provided, it shall be used by dispatch in place of the standard baggage mass / count when preparing the mass and balance form.

Where the actual baggage mass is not provided, a standard mass value of 13 kg / 29 lbs shall be used for each piece of baggage, and it shall be assumed that every adult or child passenger will carry one piece of baggage each.

Actual Baggage Mass / Count Values

When loading the baggage and the mass / count differs from that shown on the mass and balance form:

- If the flight crew determine the standard baggage mass is reasonable for the bags presented, then the standard baggage mass may be used to update the mass and balance form (i.e. using baggage count multiplied by 13 kg / 29 lbs).
- If the flight crew estimates that the baggage masses will differ significantly from the standard baggage masses, then the actual mass of the baggage shall be determined by weighing or on the basis of a statement by, or on behalf of, the passenger, and the mass and balance form shall be updated with this mass.

Any changes required to the mass and balance form shall be made as described in [8.1.8.5](#).

8.1.8.5 Last Minute Changes Procedure

Any last-minute changes require the flight crew to advise dispatch in order for a new mass and balance form to be created. This shall be done by either:

- Using the Waypoint App function;
- or
- Phoning Dispatch.

If the flight crew is unable to access the revised Mass and Balance, the required changes shall be recorded in the OFP annex, and the completed document shall be retained as part of the flight documentation (uploaded to Skybook when practicable).

8.1.8.6 Specific Gravity of Fuel and Other Fluids

When entering the mass figures for the take-off fuel and trip fuel (fuel burn) the correct specific gravity shall be used to convert volumes into mass values. The same applies to any other fluids, such as water, methanol, or oil.

Whenever possible and practicable, the specific gravity of fuel - as obtained from the fuelling crew - shall be used. As this often is not practicable, the following specific gravity values may be used if no other values are published in the AOM:

(At 15 °C fuel temp)

JET A1 (Jet fuel JP-1)	0,796 kg/l
JET B (Jet fuel JP-4)	0,760 kg/l
AVGAS	0,710 kg/l
Oil	0,880 kg/l

8.1.8.7 Seating Policy

The Mass and Balance Form is prepared assuming that passengers are seated in a predetermined standard order depending on the aeroplane seating configuration.

All mass and balance computations shall be made on the above assumption, unless:

- a. the Commander notifies Flight Dispatch of a different seating plan; or
- b. the pre-determined seating plan results in the centre of gravity limits being exceeded.

In these cases, a new mass and balance form shall be computed by dispatch to ensure the correct seating allocation is applied and the applicable centre of gravities are within the limits.

The mass for each occupied seat shall be reflected in the mass and balance form.

8.1.8.8 Centre of Gravity Envelope Warnings

Warnings

Warnings are shown on the Mass and Balance Document when a flight is planned outside the Operational or Certified Centre of Gravity Envelope as follows:

- Planned C of G falls outside the operational centre of gravity envelope; 
- Planned C of G falls outside the certified centre of gravity envelope. 

Crew Actions for Operational Centre of Gravity Envelope Warning

Where practical, the passenger / baggage distribution should be adjusted to bring the centre of gravity within the operational centre of gravity envelope.

Where this is impractical, provided the flight is planned within the Certified Centre of Gravity Limit it may continue to operate provided the flight crew:

- Carefully assess the mass and the distribution of the actual baggage and passenger load against that shown on the mass and balance document and report any discrepancies to dispatch so the mass and balance document can be updated.
- Consider the possible impact of passengers / crew moving around the cabin during flight on the Centre of Gravity and if considered necessary, restrict the movement of passengers / crew during flight so as not to take the aircraft outside the Certified Centre of Gravity Limit.

8.1.9 ATS Flight Plan

An ATS flight plan shall be submitted for every flight.

NTA submits a flight plan for each flight conducted by the company. Since all operations are unscheduled to on-demand destinations, repetitive (stored) flight plans are not used.

Information submitted in the individual ATS flight plan shall be based on the Operational Flight Plan (OFP) for the respective flight.

8.1.9.1 ATS Flight Plans for Individual Flights

Unless otherwise required by circumstances, the ATS flight plan shall be completed and submitted by Flight Dispatch.

Subsequent changes which take place between the time of submission and actual departure shall be reported immediately to the appropriate ATS unit, for amending the filed flight plan.

Changes of one or more items listed below require the cancellation of a flight plan and the submission of a new flight plan:

- a. Date of flight;
- b. Flight number;
- c. Departure Airport;
- d. Destination Airport;
- e. Advanced Estimated Off-Block Time (EOBT);
- f. Any other item specified by the state concerned according to Regional Procedures / Special Instructions.

Flight crew shall retain copy of the ATS flight plan.

8.1.9.2 Repetitive Flight Plan

NTA does not use Repetitive Flight Plans.

8.1.9.3 Time Slots (CTOTS) - Calculated Take-off Time

For flights into areas or aerodromes with limited acceptance rate, a departure Calculated Take-off Time (CTOT) may be assigned by CFMU (Central Flow Management System) (i.e. a specified time or period at / during which the flight may take-off). Flights must depart within the slot tolerance window, defined as 5 minutes before to 10 minutes after the assigned CTOT.

NOTE

1. The primary source of information for ATC delay information is the Aerobrief FS app, available in the company mobile device. It must be used by FCM to self-brief any ATC Delay messages and assigned CTOT. Guidance on using the app is available in the DMS, EFB programme folder;
2. The secondary / contingency source will be Flight Dispatch or handling agent, from whom the assigned CTOT may be obtained.
3. When it becomes apparent that , for unforeseen reasons, the deadline for passenger boarding or aeroplane handling cannot be accomplished, a revised deadline plan shall be agreed upon upon between Flight Dispatch and the Commander. It shall be, who shall then determined if the assigned CTOT can still be complied with,, or if a new one has to be requested.a new TOBT / TSAT should be coordinated (in the case of A-CDM airport), or if a delay is required.

8.1.9.3.1 Ready-message

For flights having already received their slot and being in a situation to depart before their CTOT (doors closed and ready to depart), the Commander may ask the local ATC to send a Ready (REA) message. In the REA message, local ATC may also include a MINLINEUP time, to indicate the minimum time needed for that flight to get from its position to take-off.

REA may be sent between EOBT minus 15 minutes and the CTOT of the flight. When the REA is filed before the EOBT, the flight is considered as having a new EOBT at this filing time and the MINLINEUP as a revised taxi time.

NOTE

Ready-Message (REA) requests to ATC are not applicable when operating in A-CDM airports. In these cases, crews should coordinate with the ground handling agent an update of the CDM system with the revised TOBT.

Flight Dispatch and ATC will receive the Slot Revision Message (SRM) if the change is accepted. The crew may obtain the SRM via the CTOT app on the company mobile or request it from Flight Dispatch.

8.1.9.3.2 CTOT Revision Request Message

If a CTOT cannot be operationally met, Flight Dispatch needs to be notified immediately.

8.1.9.4 Aeroplane Movement Messages

The following messages are designated Aeroplane Movement Messages:

- a. Departure Message;
- b. Arrival Message;
- c. Delay Message.

They serve the purpose of controlling the punctual and regular operation of all flights and represent the basis for the coordination of aeroplane and crew rotation, and for the registration of flight duty hours by technical and commercial departments.

Flight Dispatch reports departures, arrivals, and delays by sending a departure, arrival, or delay message as appropriate. The message shall be dispatched immediately after departure or arrival of a flight or whenever a delay occurs.

8.1.9.5 Airport Collaborative Decision Making (A-CDM)

A-CDM reduces slot delays at major airports by optimising CTOTs based on real-time data. It is based on the transfer of information between operators, handling agents, and airports about the progress of a turn-around and expected taxi times. The following acronyms are in use:

TOBT: Target Off-Block Time - the time the operator or handler estimates that a flight will be ready, doors closed, and ready to start up.

TSAT : Target Start-up Approval Time- the time, provided by ATC, a flight may expect start-up clearance to achieve an optimised departure sequence at the take-off runway. TSAT takes into account the TOBT and any Eurocontrol slot (CTOT).

TTOT : Target Take-Off Time- the expected take-off time, taking into account the TOBT / TSAT, as well as taxi time, wake turbulence and SID separation for the active runway.

At CDM airports, flight crews shall liaise with the FBO / handling agent to pass on the TOBT and obtain a TSAT / TTOT. If ETD change is 30 minutes or less, the flight plan does not need to be adjusted, and a new TOBT should be coordinated with FBO. In case ETD change is more than 30 minutes, a new slot should be coordinated with Flight Dispatch team via a phone call and the flight plan should be adjusted according to the new departure airport slot. Consideration must be given to any constraints in the arrival slot. Flight Dispatch should be contacted to assure that the arrival slot is coordinated after a change in TOBT.

8.1.10 Operational Flight Plan (OFP)

8.1.10.1 General

An Operational Flight Plan shall be prepared for all flights.

The OFP shall be prepared by Flight Dispatch - using a computerised flight planning system that implements the rules described in the applicable sections of this manual. Updates of this planning system are tested before being deployed, to ensure the proper functionality.

The Operational Flight Plan (OFP) shall be made available electronically to the flight crew via the approved electronic flight folder application (Skybook) on the portable EFB.

The Skybook application is the primary platform for distribution, operational use and storage of the OFP. It provides two viewing modes:

- a static 'Briefing' OFP (identical in content and layout to traditional paper formats) and;
- an interactive 'In-Flight' Navigation Log for recording flight progress.

The OFP shall be reviewed by the Commander prior to the flight, and acknowledgement of receipt must be recorded (typically electronically, as specified in internal procedures).

The OFP shall remain accessible to the flight crew throughout flight preparation and execution. In the event of unavailability of Skybook or the portable EFB, the OFP shall be transmitted to the flight crew by email and may be printed locally at the FBO as required.

All entries on the OFP shall be made concurrently and be permanent in nature.

The official Mass and Balance (M&B) document is the Flight Release (normally a PDF file received via email), which must be loaded into the EFB PDF reader. Acceptance of the M&B and the flight release is documented by the Commander signing the declaration on the paper OFP Annex, referencing the specific Flight Plan ID shown in the Flight Release.

8.1.10.2 Common Features

The route selection shall consider:

- a. Standard routes where published by NTA;
- b. ATS-standard routes - if published;
- c. Air traffic flow management regulations;
- d. Available NAV equipment on the ground and in the aeroplane;
- e. NOTAMs;
- f. Meteorological conditions;
- g. Traffic rights;
- h. If applicable, minimum time track optimisation;
- i. If applicable, minimum en-route altitudes (for driftdown performance);
- j. The speed schedule as prescribed in the AOM for the type concerned or as required by ATC;
- k. The altitude / flight level selection considering:
 - Obstacle clearance for all phases of flight (minimum safe altitude, minimum en-route altitude);
 - ATC-regulations;
 - Economical aspects;
 - Meteorological conditions (wind and temperature);
- l. Performance aspects;
- m. The selection of alternate(s) considering:
 - Meteorological conditions;
 - Suitability of alternate(s);
 - Distance;
 - Economic / handling aspects.
- n. Fuel freezing considerations;
- o. Depressurisation over critical areas;
- p. If applicable, ETOPS and Non-ETOPS-180 requirements.
- q. If applicable, PBCS requirements.

8.1.10.3 OFP - Format and Contents

A sample of the OFP and decode is available in [8.1.10.4](#). The form shall, however, contain at least the following:

- a. Aeroplane registration;
- b. Aeroplane type and variant;
- c. Date of flight;
- d. Flight identification;
- e. Names of flight crew members (shown on the Flight Release Document);
- f. Duty assignment of flight crew members (shown on the Flight Release Document);
- g. Place of departure;
- h. Place of arrival (planned and actual);
- i. Time of departure (actual off-block time, take-off time);
- j. Time of arrival (actual landing and on-block time);
- k. Hours of flight (block time and air time);
- l. Type of operation (ETOPS, VFR, IFR, ferry flight, etc.);
- m. Route and route segments with checkpoints / waypoints, distances, time, and tracks;
- n. Planned cruising speed and flying times between checkpoints / waypoints;
- o. Estimated and actual times overhead (based on available wind data);
- p. Safe altitudes and minimum levels;
- q. Planned altitudes and flight levels;
- r. Fuel calculations (records of in flight fuel checks);
- s. Fuel on board when starting engines;
- t. Alternate(s) for take-off, en-route (including for fuel), and destination including information required in m., n., o., and p. above;
- u. Initial ATS Flight Plan clearance and subsequent re-clearance;
- v. In-flight re-planning calculations;
- w. Relevant meteorological information.

For flights in excess of 100 nm the top of climb (TOC) and top of descent (TOD) shall be calculated with the relevant distances from / to the checkpoints / waypoints and the resulting time intervals. To allow a flight planning calculation to be as accurate as possible, the take-off mass shall be calculated using the expected zero fuel mass.

8.1.10.4 OFP Decode

1
 NON-ETOP180 FLIGHT PLAN
 2 3 4 5 6 7 8
 PLAN 8040 NJE578Q EGPH TO KBGR GL6T M85/F IFR 03/20/24
 NONSTOP COMPUTED 1216Z FOR EOBT 1230Z PROGS 2006UK CSGLA LBS
 9 10 11 12 13
 14 *** KBGR TEMPO CLOSURE 1915Z-1930Z, CHECK ETA ***
 15 () WX / () NOTAMS / () F/C / () FUEL / () CAT / () LUG / () RELEASE
 16 17 18 19 20 21 22 23
 FUEL TIME DIST ARRIVE TAKEOFF LAND AV PLD OPNLWT
 24 DEST KBGR 024157 06/09 2740 1849Z 083724 059567 000464 053510
 25 CONT (3 %) 000725 00/17
 26 ALTN KBOS 001931 00/36 0181 1925Z
 27 FRSV 001244 00/30
 28 REQD 028057 07/32 38 NAM 39 AV WIND
 29 TAXI 000250 00/10 2975 256T/039
 30 ADDL 000000 00/00
 31 MREQ 028307 07/42 NAT B NA 754 42 AV ISA DEV P05
 32 XTRA 001693 00/41 40 41
 33 TREQ 030000 08/23 43 AV DESC WIND
 34 DSCR /..
 35 RLS /..
 36 APU /..
 37 TTL FUEL /..
 44 TAS 490 FL 340/BRUCE 45
 360/BALIX
 370/URTAK
 360
 46 47
 RECOMMENDED APU FUEL ALLOWANCE..450 LB FUEL INDEX 112 CONT/102
 CHEMOI
 48 EGPH GOSA1C GOSAM..GOW L602 BRUCE..SUPIT..EVBEV..BALIX..6120N..
 6230N..FIR..6240N..6050N..FIR..URTAK N754A YBC..FIR..FIR..PQI..MLT..
 KBGR

 49 CLEARANCE

 51 52 53 54 55 56

 50 T/O ALTERNATE EGPF MSA TTK DIST FL TIME ETA
 058 262 0056 100 0.10 1240
 58 59
 MSA TTK DIST FL W/C TIME FUEL
 57 ALTERNATE - 1 KBOS 049 212 0181 260 M057 0036 1931
 -N0311F260 KBGR V93 BRNNS T295 ENE V106 YUKES V3 REVER..KBOS
 60 ALTERNATE - 2 KPSM 047 219 0135 200 M053 0029 1539
 -N0293F200 KBGR..RAZZR T295 ENE..KPSM
 62
 61 ERA DIVERT PT CUMD MSA TTK DIST FL TIME ETA
 CYR N5537.7 2055 043 139 0185 100 0.33 1847
 W06354.4

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      63      64      65      66
-----
TTL(BEGIN)FUEL .... OFF-BLOCK FUEL .... ON-BLOCK FUEL .... FUEL BURN ....

      67      68      69
OFF-BLOCK TIME .... ON-BLOCK TIME .... BLOCK TIME ....

      70      71      72
TAKE-OFF TIME .... LDG TIME .... FLIGHT HOURS ....

-----
73 CMDR SIGN..... T/O: PIC/SIC 74 LDG: PIC/SIC 75
-----
76 REMARKS
ALERT FOR EGYPT: SINAI REGION PROHIBITED. WEST OF SISIK AND SOUTH OF DELNA
APPROVED AT OR ABOVE FL260. HESH AND HEGN APPROVED WITH RESTRICTIONS

-----
77 M85 FL340 026103 06/14 M85 FL360 024598 06/10
M85 FL380 023311 06/05
M85 FL360 024598 06/10
M85 FL320 027581 06/14
M85 FL300 029078 06/11

M85 FL360 024598 06/11
-----
78 WIND M038 MXSH 06/6120N 79 DISPATCHER: JP 80
-----

81 EGPW ELEV 00136FT
      82      83      84      85      86      87      88      89      90      92      93
CPT FLT T WIND S AWY MCS DIST ETE/ETR FU FR
FREQ TRO TDV COMP TAS MSA TCS GRS ETA/ATA FF/E AFR
-----
      91      94
-----

95 T/O FUEL.....
-----
UW 96 GOSA1 239 005 0001/0608 00164 27168
368.0 041 238 .... / ....
N55543 W003302

97 TOC 340 L602 297 021 0003/0555 00459 25363
058 295 .... / ....
N56120 W005408

98 ETP1 370 46 22032 01 NAT-B 271 155 0020/0259 01238 14144
31 P11 M028 500 112 252 472 .... / .... 01894 .....
N61066 W045054

100 -CZQX 360 53 24025 01 N754A 226 013 0002/0129 00102 08591
33 P03 M022 492 055 205 470 .... / .... 01847 .....
N55192 W064088

101 TOD 360 48 24055 00 .. 215 047 0006/0014 00382 03937
31 P08 M037 497 074 194 460 .... / .... 01868 .....
N46006 W068222

102 KBGR .. 216 49 0011/0000 00646 03175 103
ELEV 00192FT 074 196 .... / ....

-----
104 LANDING FUEL .....
-----

105 FIRS EGGX/1314 BIRD/1403 CZQX/1456 ADIZ/1623 CZQX/1641
FIRS CZUL/1708 CZQX/1710 CZUL/1721 CZQM/1813 KZBW/1815

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106 MSA TTK DIST FL W/C TIME FUEL
 ALTERNATE - 1 KBOS 049 212 0181 260 M057 0036 1931
 -N0311F260 KBGR V93 BRNNS T295 ENE V106 YUKES V3 REVER..KBOS

107 CPT FLT T WIND S AWY MCS DIST ETE/ETR FU FR
 FREQ TRO TDV COMP TAS MSA TCS GRS ETA/ATA FF/E AFR

T/O FUEL.....

TICBU CLB M28 23044 03 V93 242 022 0004/.... 00551 002624
 032.. M043 044 227 297 / CLB
 N4433.5W06912.1

108 (FPL-NJE578Q-IN ¹⁰⁹ YN or ZN)
 -GLEX/M-SBDE2E3FGHIJ1J3J4J5M1M3RWXYZ/LB1D1
 -EGPH1230
 -N0490F340 GOSAM DCT GOW L602 BRUCE/N0492F360 DCT SUPIT DCT EVBEV
 DCT BALIX/M085F370 NATB URTAK/N0492F360 N754A YBC DCT PQI DCT MLT
 DCT ¹¹⁰
 -KBGR0609 KBOS KPSM
 -PBN/A1B1C2C3D1L1O2S2 NAV/RNP2 DAT/1FANSP2PDC SUR/260B DOF/240320
 REG/CSGLA
 EET/EGGX0044 BIRD0133 30W0213 CZQX0226 40W0250 50W0330 ADIZ0353
 URTAK0403 CZQX0411 CZUL0438 CZQX0440 CZUL0451 CZQM0543 KZBW0545
 SEL/KPEG CODE/491D81 OPR/NJE PER/C TALT/EGPF
 RMK/TCAS EQUIPPED SATCOM 16025848193)

111 :
 : ALTIMETER CHECKS :
 :
 : CHECK : TIME (UTC) : ALT 1 : ALT 2 : STBY :
 :
 : PRIOR : : : : :
 : OCEANIC : H Z : : : :
 : : : : : :

112 ENRT ALTN
 BIKF SUITABLE 1338/1839 (/)
 CYYR SUITABLE 1640/1839 (/)

113 MOST CRITICAL FUEL SCENARIO AT : ETP 1 FUEL EXCESS OF 04463 LBS ¹¹⁴
¹¹⁵ ETP1 ¹¹⁶ BIKF/CYYR ¹¹⁷ 03/10 ¹¹⁸ 1361NM ¹¹⁹ M018/P000 ¹²⁰ BURN 11794/12099 ¹²¹ DIS 0644/0682

122 N61066 W045054 FOB: 016562 ¹²³

- ETP1 14144 + CONT 725 + XTRA 1693 = FOB 16562
- FOB 16562 - HIGHEST ETP BURN 12099 = EXCESS 4463

124 ETOPS ENTRY N5504.4W02739.1 ETOPS EXIT N5332.0W03747.0

125 EROPS DATA

- - - - -

126 DIVERSION SUMMARY - BIKF CYR
127 LAT/LONG N61066 W045054
128 TIME 02.00 @ 0341KT
129 F.L. 0100
130 FOB 016562
131 G/C DIST 0644 0682
132 CRUISE DIST 0644 0682
133 AVG W/C M018 P000
134 ENROUTE TEMP M018 M022
135 ETP TEMP @ FL100 M024 M014 M022
136 AVG GWT 066810 066967 066833 067020
BURN SUMMARY ...
137 LRC 138 1LE LRC 1LE
139 CRUISE 007133 02.40 007451 007133 007409 007031
140 DESCENT 000131 00.05 000134 000131 000134 000131
141 HOLD 000650 00.15 000650 000650 000650 000650
142 MAP 001200 001200 001200 001200 001200
143 APU 000707 000000 000707 000000 000698
144 ICE DRAG 001367 001415 001367 001409 001352
145 ANTI-ICE 000911 000944 000911 000939 000901
146 CONSERV. 000000 000000 000000 000000 000000
147 TOTAL 012099 00.00 011794 012099 011741 011963
- - - - - 148

149 DRIFTDOWN SUMMARY DATA

CRZ TO BURN FL MSA TO BURN FL MSA FOB LAT LON W
150 LRC*BIKF 007893 100 038 CYR 008054 100 053 016582 N58192 W044024 151
1LE*BIKF 005378 290 038 CYR 005423 290 053 016582 N58192 W044024

WARNING FLAGS: M-MSA, D-FUEL DUMP REQ., F-DIVERT FUEL REQ.,
S-SPIRAL DESCENT
- - - - -

DRIFTDOWN DETAIL DATA

152 DEPRESSURIZED (LRC) DIVERT SUMMARY ...

LAT/LONG N58192 W044024 BIKF CYR
ETP TIME 01.57 @ 0341KT
ETP F.L. 100
ETP FOB 016582
G/C DIST 0705 0629
ETP W/C P022 M017
DRIFTDOWN BURN DATA ... LRC LRC
DRIFT F.L. 100 100
ENROUTE TEMP M022 M022
AVG GWT 067006 066925
MSA F.L. 038 053
DRIFTDOWN DUMP FUEL 000000 000000
EMERGENCY DESCENT 000000 000000
CRUISE 007103 007265
FINAL DESCENT 000134 000134
HOLD 000656 000655
TOTAL 007893 008054
- - - - -

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153 ONE ENGINE INOP DIVERT SUMMARY ...

LAT/LONG	N58192 W044024	BIKF	CYYR
ETP TIME	01.57 @ 0341KT		
ETP F.L.	100		
ETP FOB	016582		
G/C DIST		0705	0629
ETP W/C		P022	M017
DRIFTDOWN BURN DATA ...		1LE	1LE
DRIFT F.L.		290	290
ENROUTE TEMP		M049	M052
AVG GWT		068167	068144
MSA F.L.		038	053
DRIFTDOWN DUMP FUEL		000000	000000
DRIFTDOWN DESCENT		000243	000243
CRUISE		004295	004341
FINAL DESCENT		000266	000266
HOLD		000574	000573
TOTAL		005378	005423

Legend

1. Information if OFP is 'NON-ETOPS 180'. No description if standard planning
2. OFP number
3. ATC callsign
4. ICAO departure and destination airport codes
5. ICAO Aeroplane type designator
6. Speed Profile / Optimised for Fuel, Cost or Time (Standard is Fuel)
7. Flight Rules / In case of Y or Z should be checked in ATC flight strip ([109])
8. Date OFP was computed
9. Time OFP was computed (shall be within 12 hours before ETD)
10. Estimated Off Blocks Time (EOBT)
11. Weather data collection time. Valid for 12 hours, last characters refer to source
12. Aeroplane registration
13. Weight units of measure (LBS – Pounds)
14. Dispatch comments related to the flight. If no comments, field is empty
15. Gross Error checks (to be filled by the FCM who made the check prior to flight) **WX** (Weather) / **NOTAMS** / **F/C** Feasibility & Comply365 / **FUEL** / **CAT** (Catering) / **LUG** (Luggage) / **RELEASE**
16. Trip fuel to destination
17. Estimated Elapsed Time (EET) to destination
18. Ground distance to destination in NM (Nautical Miles)

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19. Estimated Time of Arrival at destination (ETA) – Accounts for taxi time (ETD + TAXI + EET = ETA)
20. Calculated Take-Off Mass $([23]+[22]+[33]-[29])$
21. Calculated Landing Mass $([20]-[16])$
22. Expected Payload
23. Dry Operating Mass (DOM)
24. Destination Airport
25. Contingency Fuel / Endurance
26. Alternate Fuel / Endurance, distance and ETA
27. Final Reserve Fuel / Endurance
28. Minimum Fuel required for take-off / Endurance
29. Taxi fuel / Endurance
30. Additional fuel / Endurance
31. Minimum legally Required fuel to dispatch / Endurance
32. Extra Fuel
33. Total Fuel required $([31]+[32])$
34. Discretionary fuel
35. Release Fuel (TREQ plus DSCR) – this is the figure that will be reported in the reply to release
36. APU fuel – Expected APU fuel burn before off-block
37. TTL Fuel – (RLS + APU). This fuel figure should be logged post-flight as 'Begin Fuel' on the company mobile phone application
38. Air distance to destination NM. If this figure is less than DIST [18], it indicates a tailwind. If this figure is more than DIST [18], it indicates headwind.
39. Average cruise wind in direction (True) and speed
40. Indication of planned North Atlantic Track (NAT OTS). If empty, flight not planned using NAT OTS
41. Indication of North American Route (NAR), preferential airway system arriving / departing North America (i.e. N754A)
42. Average cruise temperature deviation from ISA
43. Average descent wind in direction (True) and speed
44. Estimated True Air Speed (TAS)
45. Planned cruising level and cruising level changes

46. Fleet specific Recommended APU fuel allowance in accordance with [8.3.35.2](#) and [8.3.35.3](#)
47. Fuel Index and fuel supplier details at airport of departure and destination (percentage in reference to current fuel price at LSGG – Fuel Index 100)
48. Flight plan route
49. Available to fill in pre-departure ATC clearance
50. Take-Off alternate summary
51. Minimum Safety Altitude (MSA)
52. True Track Distance
53. Distance
54. Flight Level
55. Estimated Elapsed Time
56. ETA to Alternate
57. Destination alternate data, including route. Text '*** DCT ***' in this field is not legal and an alternate route must be computed
58. Wind component in alternate route
59. Trip fuel to alternate at planned speed and altitude
60. In case of two alternates planned, a second alternate section is available. The most distant alternate should be alternate 1
61. Enroute Alternate, including diversion point coordinates. **MSA/TTK/DIST/FL/TIME/ETA** refers from diversion point to enroute alternate airport. The flight level shown in the summary is for fuel calculation purposes only and may occasionally indicate a flight level below MSA
62. Cumulative distance until diversion point
63. Total fuel on board (including APU and DSCR)
64. Off-Blocks fuel
65. On-Blocks fuel
66. Fuel used – ([63]-[65]). This will capture APU fuel burn prior to off-blocks
67. Off-Blocks time (HH:MM)
68. On-Blocks time (HH:MM)
69. Block time (in HH:MM and in decimal HH.M)
70. Take-off time (HH:MM)
71. Landing time (HH:MM)
72. Flight time (in HH:MM and in decimal HH.M)

73. Commander's signature for acceptance of OFP
74. Identify pilot flying for take-off – D (Day), N (Night). Record number of take-offs
75. Identify pilot flying for landing – D (Day), N (Night). Record number of landings
76. REMARKS – Available for either Dispatch or Crew use
77. Planned flight Level +/- 2000ft, +/- 4000ft where available. Trip fuel and trip time
78. Average en-route wind component (**P** tailwind, **M** headwind)
79. Maximum expected shear intensity and position
80. Dispatcher's name
81. ICAO identifier of airport of departure with elevation
82. Check-point name (CPT) / Frequency in case waypoint is a radio aid (FREQ)
83. Flight Level (FL) / Tropopause Level (TRO)
84. Temperature (T) / Temperature deviation from ISA (TDV)
85. Wind direction (True) and knots (WIND) / Wind component (COMP)
86. Expected shear intensity (S) / True Airspeed (TAS)
87. Airway designator (AWY) / Minimum Safety Altitude (MSA) from previous waypoint (grid minimum off-route altitude)
88. Magnetic course / True course
89. Distance / Ground speed
90. Estimated time en-route from previous waypoint (ETE) / Estimated time remaining until landing (ETR)
91. Estimate time of arrival over waypoint (ETA) / Actual time of arrival over waypoint (ATA)
92. Estimated fuel used from previous waypoint (FU) / Fuel flow per engine (FF/E)
93. Minimum fuel remaining over waypoint (FR) - includes remaining trip fuel + alternate fuel + final reserve fuel
94. Actual fuel remaining over waypoint (AFR)
95. Actual fuel at take-off
96. SID designator, if existing
97. Top of climb, estimated Level, Lat / Long and details
98. Equal time Point, calculated for NON-ETOPS 180 and Driftdown
99. NAT Track designator, in case it's planned over a NAT Track
100. North American Route designator, or airway designator
101. Top of descent, estimated level, Lat / Long and details

- 102. Destination airport, elevation and details
- 103. Minimum fuel remaining over waypoint (includes alternate fuel and final reserve fuel)
- 104. Fuel remaining upon landing
- 105. FIRs en-route with estimated time of entry
- 106. Alternate route – Same applies in case of 2 alternates
- 107. Extended route from Destination Airport to Destination Alternate – Same applies in case of 2 alternates
- 108. ATC filling strip
- 109. OFP flight rules and type of flight – Can be 'Y' (IFR to VFR) or 'Z' (VFR to IFR). In case of 'G' General aviation (Fractional), it may not reflect the type of flight
- 110. OFP Alternates – In case of 2 alternates, it also reflects in the filling strip
- 111. Time and altimeter readouts prior to entering oceanic airspace (Prior to entering RVSM Airspace – link to [8.3.2.7.2](#))
- 112. En-route alternate window of suitability
- 113. Identification of the Critical Fuel Scenario (CFS)
- 114. Fuel excess at the Critical Fuel Scenario – Calculation described accordingly
- 115. Critical point identification
- 116. ICAO identification of Diversion Airports
- 117. Flight time between Diversion Airports
- 118. Distance between diversion Airports
- 119. Average wind component from (CFS point/ETP1) to each Diversion Airport (backward / forward)
- 120. Fuel burn from (CFS to point/ETP1) to each Diversion Airport (backward / forward)
- 121. Distance from (CFS point/ETP1) to each Diversion Airport (backward / forward)
- 122. CFS point/ETP location coordinates Latitude / Longitude
- 123. Fuel on board at (CFS point/ETP1)
- 124. Location of the NON-ETOPS 180 sector, Entry and Exit point – Can be included on top of FPL ([14])
- 125. **EROPS DATA descriptor identifies that CFS is calculated based on NON-ETOPS 180**
- 126. Diversion summary with ICAO Airports for each ETP
- 127. ETP location coordinates Latitude / Longitude
- 128. Flight time from ETP to Diversion Airports, calculated at 1LE speed (One Engine Inoperative) cruise speed for the segment from ETP to the Diversion Airport

129. Flight Level used for Diversion calculation – Oxygen endurance checks should be done according to FL and diversion time
130. Fuel on board at ETP
131. Great Circle Distance, no factor applied to Cruise Distance
132. Cruise Distance from ETP to Diversion Airports 1
133. Average wind component from ETP to the Diversion Airports
134. Average temperature from ETP to the Diversion Airports
135. Average temperature at ETP FL / Average temperature at ETP FL, from each Diversion airport
136. Average Gross weight from segment from ETP to each Diversion Airports at the different speeds
137. Fuel burn summary – LRC : All Engines Operative
138. Fuel burn summary – 1LE : One Engine Inoperative
139. [139] to [147] Fuel data for the most penalizing scenario, based on the highest total ([148])
148. Most critical scenario is the one with higher total, which will reflect in [120]
- 149. DRIFTDOWN SUMMARY DATA identifies that CFS is calculated based on Driftdown (ETOPS 120)**
150. LRC : All Engine depressurization scenario / 1LE : Pressurized Engine-Out. Includes burn fuel backward and forward / FL / MSA / FOB and Lat/Long ETP position
151. Warning Flag as described in the legend. If empty, no Warning
152. Detail data form Depressurized divert summary
153. One Engine Inoperative divert summary

8.1.11 Aeroplane Technical Log - Flight and Maintenance Report

8.1.11.1 System

The NTA technical log system is a complex document / computer system using a 4 part documentation system: Aircraft Flight Log, Aircraft Technical Log, Aircraft Appearance Log, the form NTA-M 102 (CRS-Onboard version) and a computer system (Maintenance Tracking Software Application). The Aircraft Technical Log System and any revisions shall be approved by ANAC before entry into force. The system is approved through the CAME or [NJMP 1.01](#), as applicable.

The Aircraft Flight Log represents the daily flight and servicing activity of the aeroplane. Each page is designed for a maximum of 3 flights and should not be used to record data for more than one day.

The purpose of the Aircraft Technical Log is to record the aeroplane defects detected between scheduled maintenance events and allow their deferral, when allowed by the applicable MEL.

An aeroplane with defects is prohibited from flying unless the items have been deferred in accordance with the applicable MEL, applicable CDL, No Technical Objection (NTO) by a Design Organisation Approval (DOA), or with a Special Flight Permit issued by the Authority ([NJMP 1.09](#)). The Commander shall ensure that the respective completed document is on board prior to the flight with the defects.

The purpose of the Aircraft Appearance Log is to record non-safety related equipment defects, detected both in their interior and exterior of the aeroplane, but do not affect the airworthiness.

8.1.11.2 Sections

The technical log system covers the necessary details in five sections.

Section 1

Contains details of the registered name and address of the operator, the aeroplane type and the complete international registration marks of the aeroplane.

Section 2

Contains details of when the next scheduled maintenance is due including, if relevant, any out of phase component changes due before the next maintenance check. In addition this section should contain the current Certificate of Release to Service for the complete aeroplane issued normally at the end of the last maintenance check.

NOTE

The flight crew does not need to receive such details if the next scheduled maintenance is controlled by other means acceptable to the Authority.

Section 3

Contains details of all information considered necessary to ensure continued flight safety. Such details are:

- a. The aeroplane type and registration mark;
- b. The date and place of take-off and landing;
- c. The take-off and landing times of the aeroplane;
- d. The running total of flying hours, such that the hours to the next scheduled maintenance can be determined. The flight crew does not need to receive such details if the next scheduled maintenance is controlled by other means acceptable to the Authority;
- e. Details of any known or suspected defect, including emergency systems, affecting airworthiness or safe operation of the aeroplane. The commander shall sign and date the log entry for each flight including, as appropriate, a 'Nil Defect' entry. There is provision for a Certificate of Release to Service following rectification of a defect or maintenance check carried out. Such a certificate should readily identify the defect(s) to which it relates or the particular maintenance check as appropriate;
- f. The quantity of fuel and oil uplifted and the quantity of fuel available in each tank, or combination of tanks, at the beginning and end of each flight. The units for the amount of fuel planned to be uplifted and the amount of fuel actually uplifted shall be the same;
- g. The time when ground de-icing and / or ant-icing was commenced and the type of fluid applied with mixture ratio;
- h. The pre-flight inspection signature by the commander or competent and authorised person.

In addition to the above it may be necessary (on request of the technical department) to record the following supplementary information:

- a. The time spent in particular engine power ranges where use of such engine power affects the life of engine or engine module (i.e. maximum or contingency power);
- b. The number of landings where landings affect the life of an aeroplane or aeroplane component;
- c. Flight cycles or flight pressure cycles where such cycles affect the life of an aeroplane or aeroplane component.

Section 3 is designed such that one copy of each page may remain on the aeroplane and one other copy may be retained on the ground until completion of the flight to which it relates. The layout of section 3 shows clearly what is required to be completed after flight, and what is required to be completed in preparation for the next flight.

Section 4

Contains details of all deferred defects that affect or may affect the safe operation of the aeroplane and shall therefore be known to the aeroplane commander. Each page of this section will be printed with NTA name and page serial number, and makes provision for recording the following:

- a. A cross reference for each deferred defect such that the original defect can be identified in the particular Section 3 Sector Record Page;
- b. The original date of occurrence of the defect deferred;
- c. Brief details of the defect;
- d. Details of the eventual rectification carried out and its Certificate of Release to Service, or a clear cross reference back to the document that contains details of the eventual rectification.

Section 5

Contains any necessary maintenance support information that the aeroplane commander needs to know. Such information includes data on how to contact maintenance engineering if problems arise whilst operating the routes etc.

8.1.11.3 Recurrent Defect Information

The identification of recurrent events is defined in the NetJets Maintenance Procedure [\(NJMP\) 1.31](#).

The airworthiness status of the aeroplane is logged in the ATL.

In addition to checking the Aircraft Technical Log (ATL), flight crew can obtain information via NetJets portal whether recurrent defects have been identified for specific aeroplane tail numbers. This information may be found in the Maintenance section of the NetJets portal.

The information about recurrent defects is 'nice to know information'.

Whenever one of the following occurs, the event will be listed as a recurrent event:

- The rate of unscheduled component removal triggers the alert levels;
- One aeroplane reports 3 failures from the same aeroplane system within a 3 month period;
- A component or aeroplane system has been identified as a top grounding item.

8.1.12 Documents, Forms, and Additional Information to be Carried

The following documents, manuals, and information shall be carried on board for each flight, as originals or copies, unless electronically authorised as specified in the table below, and shall, within a reasonable time of being requested to do so by a person authorised by an authority, be produced by the Commander of the aeroplane to that person.

Carriage of the documents electronically is only authorised for the aeroplane types that are approved to carry electronic documents in an approved EFB.

Document	Hard copy (H) / Electronic (E)
a. Original Certificate of Registration;	H
b. Original Certificate of Airworthiness - Current;	H
c. A certified true copy of the Airworthiness Review Certificate;	H
d. Original Noise Certificate;	H
e. A certified true copy of the Air Operator Certificate (AOC) - whenever possible 3 Copies will be onboard, and a copy of the operations specifications relevant to the aeroplane type issued with the AOC;	H
f. Original Aeroplane Radio Licence - Current;	H
g. Insurance Certificate - in English, German, French & Spanish;	E or H
h. General / Basic Flight Operations Manual, parts A, B, and C, which shall be easily accessible to the crew members;	E or H
i. Technical Log / Aircraft Appearance Log / Journey Log;	H
j. Minimum Equipment List;	E or H
k. INOP Placards;	H
l. German and Austrian Mineral Tax Licence;	E or H
m. Fuel Cards;	H
n. Valid Flight Crew Licences;	H
o. Aircraft Flight Manual and Aircraft Operating Manual;	E or H
p. Quick Reference Handbook or equivalent;	H
q. Operational Flight Plan and details of the filed ATS Flight Plan;	E or H
r. Appropriate NOTAM / AIS briefing documentation;	E or H
s. Appropriate meteorological information;	E or H
t. Mass and balance documentation;	E or H
u. When applicable: 1. Notification of special categories of passenger; 2. Notification of special loads; 3. US Customs and Border Protection User Fee decal.	E or H

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Document	Hard copy (H) / Electronic (E)
v. Current suitable aeronautical maps and charts for the route of the proposed flight and all routes along which it is reasonable to expect that a flight may divert to, including: Departure, Destination, En route alternates, Emergency airfields, Documentation for en route diversions and emergency fields, and;	E or H
w. Information concerning search and rescue services for the area of the intended flight, which shall be easily accessible in the FCC;	E or H
x. Procedures and visual signals information for use by intercepting and intercepted aeroplanes;	E or H
y. Cargo and / or passenger manifest, if applicable;	E or H
z. Air Safety Report forms;	E or H
aa. On-Board Certificate of Release to Service;	H
ab. Any other documentation that may be pertinent to the flight or is required by the States concerned with the flight.	E or H
ac. OFP Annex	H

In case of loss or theft of documents a. to f. listed above, the operation may continue until the flight reaches its destination or a place where replacement documents can be provided.

In these cases, prior to the next flight, copies of the documents shall be placed onboard until the originals, if applicable, are replaced.

NOTE

Aeroplanes whose empty mass equals or exceeds 12 tons do not need to have German Mineral Tax Licence on board since they are automatically exempt according with Mineral Oil Tax Regulation in German 'Mineralölsteuerverordnung' § 21, appendix 1.

Additional documentation carried on board:

- [NJMP 0.02](#)
- [NJMP 1.01](#)
- [NJMP 1.09](#)
- [NJMP 1.11](#)
- [NJMP 1.12](#)
- [NJMP 1.13](#)
- [NJMP 1.36](#)
- [Approved Portable EFB Configuration \(iPad\)](#)
- Maintenance Check Flight Manual

8.1.12.1 FMS Databases

FMS Navigation databases shall be from an approved source that has an LOA type 2.

8.1.12.2 Electronic Flight Bags (EFB)

Refer to [Electronic Flight Bag \(EFB\)](#).

8.1.12.3 Updating of FMS and E-Chart Databases

Whenever applicable to the aeroplane type, the onboard FMS Navigation Database and E-Charts Databases shall be updated by flight crew as defined in the respective Operations Manual Part B.

Updating of FMS and E-Chart Databases by flight crew shall only be accomplished after the end of a flying duty and only when there is sufficient time before the next planned flight so as not to impact it should there be an issue.

Flight crew shall advise Scheduling if extra duty time is required to accomplish the required updates.

8.2 Ground Handling Instructions

8.2.1 Fuelling Procedures

All fuelling of aeroplanes operated by NTA will be actively supervised by a member of the flight crew. Refuelling with any engine running is prohibited.

Special refuelling or defuelling procedures and any change to them shall require prior approval by the competent authority. The following are considered special procedures:

- Refuelling with an engine running;
- Refuelling / defuelling with passengers embarking, on board, or disembarking;
- Refuelling / defuelling with wide-cut fuel.

8.2.1.1 Crew Checks

The Commander, or a crew member nominated by them, will confirm that:

- a. Particular care is taken in advising the refuelling agency of the type, grade and fuel quantity required, with special reference to the units of measurement quoted (litres, US gallons, pounds etc.);
- b. The bowser or other fuel installation is earthed to the aeroplane structure before the hose is extended, and remains so earthed until refuelling is complete;
- c. Fuel hoses are laid by the shortest route from the fuel installation to the refuelling inlets avoiding wheel brakes (by at least one metre) and APU air intakes;
- d. Smoking is not permitted within 20 metres of the aeroplane whilst refuelling is in progress;
- e. The use of mobile phones is not permitted within 25 metres of the aeroplane whilst refuelling is in progress;
- f. Except for the fuel truck, no vehicle is positioned within the venting areas;
- g. Cargo / baggage loading equipment complies with safety regulations for antisparking otherwise it shall not be operated until fuelling is complete and hoses removed;
- h. Spilled fuel is removed or dried up immediately with the fire service in attendance;
- i. The correct type, grade and quantity of fuel has been loaded;
- j. The fuel drains are operated to check for water content, and left properly closed;
- k. Where practical, a visual check of tank contents, or if specified in the check lists for smaller aeroplanes, a dipstick check reveals the correct amount of fuel on board to within reasonable tolerance;
- l. All fuel tank caps are properly secured;
- m. The aeroplane fuel gauges indicate that the tanks have been filled to the required levels;
- n. Details of the fuel uplift have been correctly entered in the Flight log and a gross error check is carried out;
- o. If an auxiliary power unit located within the fuelling zone or which has an exhaust efflux discharging into the zone is stopped for any reason during a fuelling operation, it shall not be restarted until the flow of fuel has ceased and there is no risk of igniting fuel vapours;
- p. If thunderstorms are in the vicinity, fuelling / defuelling operations are suspended;
- q. The correct quantity of anti-freeze additive is dispensed into the fuel where specified by the aeroplane manufacturer;
- r. The fuel bowser / installation readings at the start and finish of refuelling reflect accurately the fuel uplift as indicated on the aeroplane fuel quantity gauges, and a gross error check is carried out;
- s. Chocks need to be in place while refueling. If chocks are not available the Commander needs to assure that the aeroplane is on a level platform and that the parking brake is fully operational and set;

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- t. In locations where there is any doubt as to the quality of the fuel or when deemed necessary by the commander, the fuel from the fuel supplier shall be checked for contamination prior to and after the fuelling of the aeroplane.

NOTE

1. When an aeroplane is fuelled / defuelled via an overwing filler (i.e. unpressurised system) which is located closer than 10 metres from the nearest exit door, the operation shall not be carried out whilst passengers are on board.
2. During pressure refuelling electrical / electronic systems, except weather radar and HF transmitter, may be operated to the extent required for pre-flight preparation.
3. In the event of an emergency (i.e. APU fire during fuelling / defuelling), the operation shall be stopped and an immediate disembarkation initiated. The flight crew will initiate an 'emergency' evacuation.

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8.2.1.2 Passengers On Board

During normal operations it may be necessary for refuelling to take place when passengers are embarking, on board or disembarking the aeroplane. With the commander's authority, passengers may embark, disembark or remain on board during refuelling / defuelling provided that the following precautions are observed:

- a. Fuelling / defuelling with passengers on board is to be recorded in the Comments / Observations section of the Aircraft Flight Log (Journey Log) for the appropriate sector using the format 'Sector no / [ICAO location] / REFUEL CONDUCTED WITH PASSENGERS ON BOARD'
- b. Fuelling / defuelling with wide cut fuel (i.e. Jet B, JP-4, AVTAG or equivalent) or when a mixture of these types of fuel could occur is prohibited when passengers are embarking, on board or disembarking;
- c. A member of the flight crew shall remain on or near to the FCC;
- d. The Aeroplane main exit doors shall be open;
- e. Air traffic control and the aerodrome fire services are to be advised that refuelling / defuelling will be taking place with passengers on board;
- f. Crew, staff and passengers shall be advised that refuelling / defuelling is about to take place;
- g. 'Fasten Seat Belt' signs shall be off, 'No Smoking' signs shall be on together with interior lighting to enable emergency exits to be identified. For aeroplanes that are unable to individually select the 'Fasten Seat Belt' and 'No Smoking' signs, both signs shall be on;
- h. Passengers shall be briefed not to smoke at any time on the ground and to remain seated, but with seat belts / harnesses unfastened, until the refuelling has been completed;
- i. A crew member shall be stationed near the main exit door. This crew member may be the same crew member as stated in point b. When cabin crew are carried, the emergency door exits shall be guarded by a cabin crew member. These crew members shall monitor passenger safety and be prepared for an immediate emergency evacuation;
- j. The exits to be used for emergency evacuation and / or rapid deplaning shall be designated. A crew member shall ensure that the areas outside the designated exits are unobstructed and that the exit routes to the designated exits are also unobstructed;
- k. If the presence of fuel vapour is detected inside the aeroplane, or any other hazard arises, refuelling / defuelling shall be stopped immediately;
- l. No individual items of electrical equipment may be switched on or off, no strike matches or lighters may be used and no other item that may produce an ignition shall be used / operated while refuelling / defuelling is in progress;
- m. The position of the fuel bowser / installation relative to the aeroplane is to be such that it will not impede the rapid exit of passengers if an emergency evacuation becomes necessary;
- n. When refuelling / defuelling with passengers on board, ground servicing activities and work inside the aeroplane, such as catering and cleaning, shall be conducted in such a manner that they do not create a hazard and allow emergency evacuation to take place through those aisles and exits intended for emergency evacuation.

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- o. Two way communications shall be established between the flight crew and:
 - i. ground crew supervising the refuelling.
 - ii. crew member(s) positioned to monitor passenger safety.

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8.2.1.3 Refuelling / Defuelling with Wide-cut Fuel

- a. 'Wide cut fuel' (designated JET B, JP-4 or AVTAG) is an aviation turbine fuel that falls between gasoline and kerosene in the distillation range and consequently, compared to kerosene (JET A or JET A1), it has the properties of higher volatility (vapour pressure), lower flash point and lower freezing point.
- b. Wherever possible, the use of wide-cut fuel types should be avoided. If a situation arises such that only wide-cut fuels are available flight crew shall be aware that mixtures of wide-cut fuels and kerosene turbine fuels can result in the air / fuel mixture in the tank being in the combustible range at ambient temperatures. The extra precautions set out below are advisable to avoid arcing in the tank due to electrostatic discharge. The risk of this type of arcing can be minimised by the use of a static dissipation additive in the fuel. When this additive is present in the proportions stated in the fuel specification, the normal fuelling precautions set out below are considered adequate.
- c. Wide-cut fuel is considered to be 'involved' when it is being supplied or when it is already present in the aeroplane fuel tanks.
- d. Refuelling / defuelling using wide-cut fuels is prohibited with passengers onboard.
- e. When wide-cut fuel has been used, this shall be recorded in the Aircraft Technical Log in accordance with MEL Chapter 28. The next two consecutive uplifts of fuel shall be treated as though they too involved the use of wide-cut fuel.
- f. When refuelling / defuelling with turbine fuels not containing a static dissipator, and where wide-cut fuels are involved, a substantial reduction on fuelling flow rate shall be used. Reduced flow rate, as recommended by fuel suppliers and / or aeroplane manufacturers, has the following benefits:
 1. it allows more time for any static charge build-up in the fuelling equipment to dissipate before the fuel enters the tank;
 2. it reduces any charge which may build up due to splashing; and
 3. until the fuel inlet point is immersed, it reduces misting in the tank and consequently the extension of the flammable range of the fuel.
- g. The flow rate reduction necessary is dependent upon the fuelling equipment in use and the type of filtration employed on the aeroplane fuelling distribution system.

It is difficult, therefore, to quote precise flow rates. Reduction in flow rate is advisable whether pressure fuelling or over-wing fuelling is employed.
- h. With over-wing fuelling, splashing should be avoided by making sure that the delivery nozzle extends as far as practicable into the tank.
- i. When refuelling / defuelling with wide cut fuels, the aeroplane electrical supply shall be switched off before refuelling starts, and remain off until refuelling ceases and the hoses have been removed.
- j. Flight crew shall refer to the AFM and AOM for additional aeroplane type specific procedures and limitations regarding the use of wide-cut fuels.

8.2.1.4 Mixing of Fuel - Contamination

Mixing of fuel is only permitted as specified in the 'Limitations' section of the AOM / AFM. Flight and ground crews shall be aware of possible fuel contamination, particularly at smaller airports, and fuel samples shall be taken to check for contaminants. The fuel drain procedures specified in the AOM / AFM shall be followed to check for water in the fuel tanks and / or lines.

8.2.2 Aeroplane, Passenger and Cargo Handling Procedures Related to Safety

Safety shall govern the handling (i.e. servicing, of the aeroplane on the ramp, the embarkation and disembarkation of passengers and the loading and unloading of baggage, cargo and mail).

8.2.2.1 Embarkation / Disembarkation

1. During embarking and disembarkation, ground staff / crew members shall ensure that passengers are not smoking.
2. When passengers are required to walk on the ramp they shall be escorted by ground staff / flight crew to / from the aeroplane or their approved transport. Passenger routes shall whenever possible be clear of oil, ice, snow and other hazards and shall be selected in such a way as to prevent damages and accidents (i.e. No passing below wings or engines!).
3. Boarding shall not commence until clearance has been given by the commander or their representative.
4. Ground staff or a crew member should remain at the bottom of the aeroplane steps during boarding to control and direct passengers, ensure no one is smoking, to monitor the use of PED and to monitor hand baggage.
5. The crew member shall monitor the passengers as they board to check for signs of drunken / unruly passengers or anything suspicious that may jeopardise the safety of the flight and the other passengers.
6. When all the passengers have boarded, a crew member shall ensure that the total number of passengers on board is correct and shall inform the commander.
7. For emergencies during boarding, refer to Emergency Evacuation (SEP manual).
8. During transit stops, e.g. passenger drop off / pickup, fuel stop, customs / immigration stop, if passengers elect to or are required to stay onboard, a crew member shall brief them on safety and security procedure, e.g. no smoking rules, unfasten or fasten seat belts as appropriate. If passengers are to disembark and embark again, then the other points in this paragraph apply.
9. Disembarkation shall not commence until the crew has received confirmation from the ground staff that passenger steps / jetways are safely in position, if applicable to the aeroplane type, and that ground equipment will not be a hazard.
10. For embarkation / disembarkation when refuelling / defuelling is in progress, refer to [8.2.1.2](#). No embarkation / disembarkation is permissible with any engine of the aeroplane running.

8.2.2.2 Carriage of Passengers and Special Categories of Passengers (SCP)

A. Carriage of Passengers

Crew shall ensure that in flight, no person other than the operating crew is in any part of the aeroplane that is not designed for the accommodation of the persons unless temporary access has been granted by the commander:

- A. for the purpose of taking action necessary for the safety of the aeroplane or of any person, animal or goods therein; or
- B. to a part of the aeroplane in which cargo or supplies are carried, being a part that is designated to enable a person to have access thereto while the aeroplane is in flight.

B. Carriage of SCPs

The commander shall be notified in advance when any SCPs are to be carried.

When assessing the carriage of SCP, the following shall be considered and evaluated:

- The aeroplane type and cabin configuration;
- The total number of passengers onboard;
- The number and categories of SCPs, which should not exceed the number of passengers capable of assisting them in case of an emergency; and
- Any other factor(s) or circumstances possibly impacting on the application of emergency procedures by the operating crew members.

NOTE

A passenger capable of assisting in case of an emergency means a passenger who is not an SCP and has no other role or private responsibility that would prevent them from assisting the SCP. For example, an adult travelling alone has no other role or private responsibility, unlike a family travelling together with younger children.

8.2.2.2.1 Allocation of Seats, Adults, Children and SCPs

A. General

The allocation of seats to passengers by flight or cabin crew depends upon the aeroplane type, its seating version, aspects of mass and centre of gravity (refer to [8.1.8](#)). In general, the following rules apply:

- a. Each person to be carried who is 2 years of age or older shall be allocated a separate seat or berth, equipped with a proper safety belt / harness;
- b. Multiple occupancy of a seat by one adult and by one infant is permitted only if the infant is properly secured by a loop belt or other restraint device supplementary to the adults safety belt harness. In such a case, and when oxygen dispensing units are prescribed, one unit each shall be installed and available for both the adult and the infant. Refer to SEP Manual for type-specific permitted adult / infant seating configuration;
- c. For the allocation to a passenger of a seat in the FCC refer to [Admission to Flight Crew Compartment \(FCC\)](#);
- d. For the allocation to a passenger of a vacant crew seat refer to [Use of Vacant Crew Seats](#);
- e. Passengers shall be seated where - in the event of an emergency evacuation - they may best assist and not hinder evacuation. Only those persons who appear reasonably fit, strong, able, and willing to assist the rapid / emergency evacuation of the aeroplane in an emergency after an appropriate briefing by the crew shall be allocated seats that permit direct access to emergency exits. The passengers seated adjacent to emergency exits shall be made aware that in the event of an emergency and if safe to do so (no fire or hazard), they will be required to remove the exit. Should a passenger appear apprehensive after being briefed, cabin crew or appropriate crew member should consider moving them to an alternative seat;
- f. Aeroplane type-specific instructions for crew and ground staff are listed in the AOM and OM-B.
- g. The following persons shall not be allocated, nor occupy, seats that permit direct access to emergency exits, where their presence could obstruct access to emergency equipment or otherwise impede the crew in carrying out their duties, or impede the emergency evacuation of the aeroplane:
 - PRMs and passengers suffering from obvious physical or intellectual disability to the extent that they would have difficulty in moving quickly if asked to do so;
 - Passengers who are either substantially blind or substantially deaf to the extent that they might not readily assimilate printed or verbal instructions given;
 - Passengers who because of age or sickness are so frail that they have difficulty in moving quickly;
 - Passengers who are so obese that they would have difficulty in moving quickly or reaching and passing through the adjacent emergency exit;
 - Children (whether accompanied or not) and infants;
 - Deportees, inadmissible passengers or persons in custody;
 - Persons with guide dogs or animals.

For further provisions governing the allocation of seats to sick, disabled and PRMs, see [8.2.2.2.3](#) below. The prescribed seating, if any, of infants and Unaccompanied Minors (UM) and their maximum number, if any, are aeroplane type specific. They are laid down in the SEP Manual.

B. Seating of SCPs

Seating of SCPs shall be as follows:

A disability and / or restraint aid that requires to be secured around the back of the seat shall not be used if there is a person seated behind unless the seat design or installation would prevent head contact of the person seated behind.

Passenger travelling with a child of less than 12 years of age.	For take-off, landing and, if possible, for emergency situations, the child should be seated in a location close enough to the accompanying adult to ensure they can be adequately assisted.
Passenger whose physical size would possibly prevent them from passing through an emergency exit.	A passenger whose physical size would possibly prevent them from passing through an emergency exit shall be seated in the vicinity of a suitable exit, taking into account the size of the exit. Seating of more than one of such passengers in the same seat row segment should be avoided.
Passenger with physical disability of the upper limbs.	A passenger with a physical disability of the upper limbs travelling without an accompanying passenger should be allocated seats during all phases of the flight so that visual and audible communication can be established with the cabin crew or with the flight crew in the absence of cabin crew.
Mentally impaired passenger.	A mentally impaired passenger, who travels without an accompanying passenger, should be allocated seats during all phases of the flight so that visual and audible communication can be established with the cabin crew or with the flight crew in the absence of cabin crew.
Passenger travelling with a recognised assistance dog in the cabin.	Suitable arrangements shall be made between the passenger and the NTA in advance of a flight where a recognised assistance dog is to be accommodated. A suitable restraint harness should be provided by the owner to secure and restrain the dog during taxi, take-off, landing and turbulence. In cruise, it is acceptable for the dog to be subject to less restraint.

8.2.2.2.2 Unaccompanied Minors

Definition

An UM is a person under the age of 18 travelling alone, unaccompanied by parents, guardians, or family members older than 18. An Escort is any person who is not the parent, guardian or member of the family, but is 18 years of age or older and is accompanying, where required, an Unaccompanied Minor.

Unaccompanied Minor Policy

1. A minor below the age of 5 shall fly with a guardian or caretaker who is at least 16 years of age (non-NetJets staff).
2. A minor of age 5 to 11 inclusive shall be accompanied by a person aged 16 or older (family member or guardian) or an escort.
3. A minor aged 12 or older does not need to be accompanied.

With the exception of minors below the age of 5, NetJets may be able to provide an escort for small / mid-cabin aeroplanes.

Transfer of custody procedure for NetJets T.A. assigned escorts or unaccompanied minors:

1. Owner services will insert the following information under special crew instructions for the specific flight:
 - a. Name, type and number of the identification document of either the parent, guardian or family member that is delivering the minor to the escort before the flight or the Name, type of identification and identification number of the minor.
 - b. Name, type and number of the identification document of either the parent, guardian or family member that is receiving the minor from the escort after the flight.
2. The assigned escort or operating crew member (in case of unescorted minor) shall identify either the parent, guardian, family member or the minor themselves before accepting the minor into their custody.
3. The assigned escort or operating crew member (in case of unescorted minor) shall identify the parent, guardian or family member before releasing custody of the minor.

NOTE

1. Flight crew members assigned as operating flight crew members for the flight cannot be assigned escort duties.
2. Any crew member assigned as the primary escort may release this duty temporarily to another operating crew member of that flight, if this is more efficient or effective. Release of custody is not authorised to a crew member during flight that is a required flight crew member after the main entry door is closed for departure and before the main entry door is opened after arrival.

Parental consent for Unaccompanied Minors and Minors travelling with only one parent / guardian

The consent documentation requirements for unaccompanied minors or minors traveling with only one parent / guardian vary with the scenario, the responsibility for fulfilling any requirements lies with Owner Services and the Legal Department who as part of an internal procedure completed during the booking process will assess each case as required. Crews will not normally be provided with any documentation / information in this respect.

8.2.2.2.3 Sick Passengers and Persons with Reduced Mobility (PRMs)

The acceptance for transportation of sick, disabled and handicapped or passengers with a pre-existing medical condition that requires the potential need for oxygen, is restricted in the interest of their own safety and that of other passengers. A PRM is understood to mean a person whose mobility is reduced due to physical disability, sensory or locomotory, permanent or temporary intellectual deficiency, age, illness or any other cause of disability when using transport, and whose situation requires special attention and the adaptation, to their needs, of the service made available to all passengers. NTA therefore, is entitled to insist upon the production of a written report on fitness for travel, issued by a suitably qualified medical doctor.

No transportation, under any circumstances, will be provided to a person who:

- Has a contagious / infectious disease, (i.e. Open tuberculosis, infectious hepatitis; scarlet fever, diphtheria, chickenpox etc.);
- Has suffered a heart attack or stroke within the last eight weeks, unless:
 - A medical certificate from a suitably qualified specialist confirms fitness to travel.
 - The passenger is accompanied by an medical specialist.
 - All medical support equipment required is placed on board the aeroplane.
 - All the above equipment can and may be used in flight.
- Requires medical treatment by pneumatically or electrically operated apparatus which, for specific reasons, cannot or is not allowed to be operated on board.
- Has engaged in diving using self-contained underwater breathing apparatus (SCUBA) or similar equipment, within the previous 24 hours.
- Has a wet cast over a broken bone unless the cast has been split to allow expansion or has had 24 hours in place to dry.

Passengers with pre-existing medical condition that requires the potential need for supplementary oxygen can be accepted under the following pre-determined conditions:

- A medical certificate / statement from a suitably qualified specialist is presented to NTA prior to the day of the flight, confirming that the passenger is fit to fly and:
 - a. What the requirements for the supply of oxygen are, such as rate per minute;
 - b. Limitations on the cabin altitude for the duration of the flight, if applicable;
 - c. The need for a medical support specialist, e.g. nurse, to accompany throughout the flight to ensure the proper administration of such oxygen, if needed.
- A recognised medical organisation is contacted to organise and deliver at the airport of departure adequate and certified oxygen and breathing equipment for air transport.

Passengers may be allowed to carry and utilise their own oxygen and breathing equipment, under the following conditions:

- The equipment to be carried is certified to be transported in the cabin and be used during flight;
- A medical certificate / statement from a suitably qualified specialist is presented to NTA prior to the day of the flight, confirming that the passenger is fit to fly, and;
- The passenger declares that they can administer oxygen on themselves, if needed or, the passenger is accompanied by a person that can handle and administer oxygen, which can be a NTA crew member.

The appropriate crew member / cabin crew member shall inform the commander of the seating location of any PRM.

NTA operations will advise the crew of any planned PRM passengers.

For briefing of visually impaired passengers refer to [Passenger and Supernumerary Briefing Procedures](#).

8.2.2.2.4 Transport of Mothers-to-be

This is in accordance with the World Health Organisation recommendations. Pregnant passengers are allowed to be carried as follows:

- For a single pregnancy, flying is permitted up to the end of the 36th week.
- For a multiple pregnancy, flying is permitted up to the end of the 32nd week.
- A medical certificate is required over 28 weeks. NTA has to have a copy of the certificate prior to the flight.

The medical certificate shall state:

1. The passenger's term of pregnancy at the time of the flight(s).
2. The date(s) of the flight(s) concerned.

If the passenger has not already supplied information about the pregnancy to Flight Operations (and thus to the crew via briefing), it becomes a matter for the flight crew to determine if travelling is allowed. One of the crew members should politely and discreetly ask the pregnant passenger for the necessary information to assess the suitability for travel based on the restrictions above.

8.2.2.2.5 Transport of Mothers that have Just Given Birth and Newly Born Babies

A mother that has just given birth may travel 24 hours after giving birth provided the mother's medical status is stable and has no bleeding, no pain and no cramping. Any other circumstance requires a doctor's medical certificate / authorisation.

A new born baby may travel no earlier than 48 hours after being born provided the mother's pregnancy reached the full term (38 weeks or more) and no complications were encountered during birth. Any other circumstance requires a doctor's medical certificate / authorisation.

8.2.2.2.6 Infant Seating

ADDITIONAL GUIDANCE

A Child Restraint Device (CRD) is a category of restraint which includes devices such as (but not restricted to) Infant Lap belts, approved Car and Aircraft Infant seats, bassinets (skycots).

Aircraft must have an acceptable child restraint device (CRD) for each person on board younger than 24 months.

Any child younger than 24 months must be appropriately secured to an acceptable CRD, as established in this section, on take-off, landing and any time the fasten seatbelt sign is on.

A CRD is considered acceptable:

1. If it is a 'supplementary loop belt' manufactured with the same techniques and the same materials as the approved safety belts; or
2. Provided the CRD can be installed properly on the respective aircraft seat, the following CRDs are also considered acceptable:
 - a. CRDs approved for use in aircraft according to the European Technical Standard Order ETSO-C100c on Aviation Child Safety Device (ACSD);
 - b. CRDs approved by EASA through a Type Certificate or Supplemental Type Certificate;
 - c. Child seats approved for use in motor vehicles on the basis of the technical standard specified in point (i) below. The child seat must be also approved for use in aircraft on the basis of the technical standard specified in either point (ii) or point (iii):
 - i. UN Standard ECE R44-04 or -03, or ECE R129 (usually referred to as i- Size) bearing the respective 'ECE R' label; and
 - ii. German 'Qualification Procedure for Child Restraint Systems for Use in Aircraft' (TÜV / 958-01 / 2001) bearing the label 'For Use in Aircraft'; or
 - iii. Other technical standard acceptable to the competent authority. The child seat should hold a qualification sign that it can be used in aircraft
 - d. Child seats approved for use in motor vehicles and aircraft according to Canadian CMVSS 213 / 213.1 bearing the respective label;
 - e. Child seats approved for use in motor vehicles and aircraft according to US FMVSS No 213 and bearing one or two labels displaying the following two sentences:
 - i. 'THIS CHILD RESTRAINT SYSTEM CONFORMS TO ALL APPLICABLE FEDERAL MOTOR VEHICLE SAFETY STANDARDS'; and
 - ii. in red letters 'THIS RESTRAINT IS CERTIFIED FOR USE IN MOTOR VEHICLES AND AIRCRAFT';

- f. Child seats approved for use in motor vehicles and aircraft according to Australia/ New Zealand's technical standard AS / NZS 1754:2013 bearing the green part on the label displaying 'For Use in Aircraft'; and
- g. CRDs manufactured and tested according to other technical standards equivalent to those listed above. The devices should be marked with an associated qualification sign, which shows the name of the qualification organisation and a specific identification number, related to the associated qualification project.

The qualifying organisation should be a competent and independent organisation that is acceptable to the competent authority.

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CRD Location

- Forward facing CRDs may be installed on both forward and rearward facing passenger seats but only when fitted in the same direction as the passenger seat on which it is positioned.
- Rearward facing CRDs can only be installed on forward facing passenger seats.
- A CRD shall be positioned in a way it will not hinder evacuation for any passenger.
- A CRD shall not be located on a seat close to an emergency exit. A window seat is the preferred location (if applicable).
- Only one CRD per row segment is recommended.

CRD Installation

- CRDs tested and approved for use in aircraft should only be installed on a suitable passenger seat by the method shown in the manufacturer's instructions provided with each CRD and with the type of connecting device they are approved for the installation in aircraft. CRDs designed to be installed only by means of rigid bar lower anchorages (ISOFIX or equivalent) should only be used on passenger seats equipped with such connecting devices and should not be secured by passenger seat lap belt.
- If a forward-facing child seat with a rigid backrest is to be fastened by a seat lap belt, the restraint device should be fastened when the backrest of the passenger seat on which it rests is in a reclined position. Thereafter, the backrest is to be positioned upright. This procedure ensures better tightening of the child seat on the aircraft seat if the aircraft seat is reclinable.
- Placing the child in the seat could depress the foam filled seat and could cause a slackening of the aeroplane seat belt. Therefore, the aeroplane seatbelt shall be rechecked for tightness once the child has been placed into the seat.
- The buckle of the adult safety belt must be easily accessible for both opening and closing, and must be in line with the seat belt halves (not canted) after tightening.
- For aeroplanes that have seats with shoulder straps, crew shall ensure the shoulder strap does not prevent the loop belt from releasing.
- Forward facing restraint devices with an integral harness must not be installed such that the adult safety belt is secured over the infant.

CRD Operation

- Each CRD shall remain secured to a passenger seat during all phases of flight, unless it is properly stowed when not in use.
- Where a CRD is adjustable in recline it shall be in an upright position for all occasions when passenger restraint devices are required to be used.

Crew members shall request instructions on how to release the infant / child from the seat to ensure they may do so in the case of an emergency.

In the event of an evacuation, the child shall be released from the seat. No attempt to release the seat shall be made. The seat shall remain onboard where it was secured.

For taxiing, take-off, landing and emergency situations, the person responsible for the infant / child should occupy a seat that allows them to assist the infant / child with placing an oxygen mask on, donning a life vest, and to be responsible for them in the event of an evacuation.

Although the use of a CRD is not mandatory for children over the age of 24 months, the available harnesses may not correctly position on the children's bodies, due to their size, in which cases the use of a CRD or booster seat may be desirable for the child's comfort and safety.

Children over the age of 24 months may therefore be secured to the seat in the following ways:

- By the use a CRD that complies with the requirements laid in this section;
- By using a booster seat as described below;
- Using the standard seat belt.

Booster Seats

a. **Booster seats approved as a CRD**

Some high-back booster seats may be approved for use in aircraft as CRD. Look for the approvals and markings required for an acceptable CRD and follow the manufacturer's instructions. Their use is permitted as for other types of acceptable CRD.

b. **Booster seats not approved as a CRD**

Booster seats are designed to allow the use of a safety belt by children who no longer fit in or require a CRD. They modify the position of the child in the seat so that the straps (lap and shoulder) are correctly positioned against the child's body.

For children older than 24 months, the use of a booster seat is acceptable if:

- seats are equipped with lap strap combined with a shoulder harness;
- its use helps to correctly position child in the seat, in relation to the seat belts.

For aircraft seats equipped with only a lap strap, use of booster seats are not permitted.

8.2.2.2.7 Transport of Inadmissible Passengers, Deportees, or Persons in Custody

NTA does not normally carry passengers who fall within these categories. In the event, however, that circumstances at some time oblige the company to deal with such passengers, the notes in [In-Flight Supplies and Cleaning](#) will provide guidance.

8.2.2.3 Permissible Size and Weight of Hand Baggage

1. The secure stowage of hand baggage, as prescribed in [8.2.2.4.3](#), limits, by inference, the permissible size and weight of hand baggage.
2. These limits vary of course with each aeroplane type and cabin layout combination (see AOM). Crew members shall visually scan the hand baggage held by passengers, and satisfy themselves that such baggage can be securely stowed in the facilities available in that aeroplane type.

8.2.2.4 Loading and Securing of Items in the Aeroplane

8.2.2.4.1 Loading

1. The loading of the aeroplane shall be accomplished with the supervision of the operating crew.
2. All baggage and cargo shall be handled carefully to avoid damage to the aeroplane or the load. All baggage and cargo on board that might cause injury or damage, or obstruct aisles and exits if displaced shall be stowed so as to prevent movement.
3. Only the appropriate compartments shall be used for loading.
4. All aeroplane specific limitations and instructions shall be observed (see AFM / AOM) including any structural limits such as the floor strength limitations, the maximum load per running metre, the maximum mass per cargo compartment and the maximum seating limit.
5. Special handling instructions (i.e. 'This Side Up') shall be observed for sensitive shipments which shall be loaded carefully to prevent damage by other items. Heavy and solidly packed pieces shall be loaded at the bottom.
6. Cargo shall be excluded from carriage if:
 - It is not properly packed;
 - It may damage or contaminate the aeroplane or other load;
 - Special handling instructions / equipment cannot be observed / supplied.
7. Carriage of items in the cabin are only authorised if it can be securely stowed e.g. with seatbelts, that will secure the baggage so that it cannot move vertically or horizontally and it's mass does not exceed the structural loading limits of the floor or seats.
8. Carriage of baggage in the toilets is authorised provided:
 - the items can be securely stowed;
 - the doors or access to the toilets prevent the movement of the baggage outside of the toilet area;
 - the toilet is not a designated emergency exit or is part of the route to a designated emergency exit.

The commander shall be satisfied that the size and the weight of the baggage is such that in case of movement the toilet doors and area will not be damaged.

NOTE

1. Any damage to the aeroplane skin or structure, however slight, which occurs or is noticed during loading / unloading, shall be reported immediately.
2. Each AOM details special procedures to be followed during loading / unloading so as to prevent the aeroplane tipping onto its tail. The centre of gravity shall be maintained forward of the main wheels and loading should start in the forward compartments and continue aft. Unloading should start in the aft compartments and continue forwards. The same sequences apply to catering of galleys. Movement of passengers to achieve ground stability is prohibited.

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8.2.2.4.2 Unloading

For unloading the same safety-relevant principles are valid as for loading.

Personnel and equipment should be available at the parking position upon arrival of the aeroplane. Passenger baggage should normally be unloaded first. Cabin load, if any, shall be unloaded after all passengers have disembarked. After completion of unloading, all compartments used shall be carefully checked by a crew member for damages, spilled liquids, or contamination.

8.2.2.4.3 Securing of Load, Cabin

For hand baggage:

- a. Each item carried in a cabin shall be stowed in a location that is capable of restraining it. Baggage behind passenger legs is prohibited;
- b. Mass limitations placarded on or adjacent to stowages shall not be exceeded;
- c. Under seat stowage shall not be used unless the seat is equipped with a restraint bar and the baggage is of such size that it may adequately be restrained by this equipment;
- d. Items shall not be stowed in toilets or against bulkheads that are incapable of restraining articles against movement forwards, sideways or upwards unless the bulkheads carry a placard specifying the greatest mass that may be placed there;
- e. Baggage placed in lockers shall not prevent doors from being closed securely;
- f. Baggage shall not be placed where it can impede access to emergency equipment;
- g. Items carried in passenger or crew compartment (i.e. hand baggage of crew or passenger, each item of galley equipment, any medical or other apparatus providing in flight medical aid for a passenger) shall be secured in such a way as not to become a hazard by shifting under the appropriate load factors corresponding to the ultimate inertia forces specified in the emergency landing condition of type certification.

Cargo may only be carried in a passenger compartment if it is stowed and secured in an approved cargo bin or seat container certificated to withstand specific load factors; the bin shall be attached to the seat tracks / floor structure, the seat container to the passenger seat; maximum loading limits shall be observed.

Installations are not permitted in a position that obscures any passenger's view of any required 'seat belt' or 'no smoking' or 'exit' sign, nor in a location that restricts access to or use of any required emergency exit, or of the aisle.

8.2.2.4.4 Securing of Load, Holds

All individual items of load which, by their nature, shape or density, may constitute a hazard shall be restrained. Restraint can be achieved by filling the compartment, or net section volumetrically, or by tie down. Compartments and net sections which are filled up to three-quarters of their height are considered to be volumetrically full.

Particular attention shall be given to dangerous goods. All packages containing such goods shall be stowed in an upright position if indicated so, have the hazard label visible and shall be restrained to prevent their shifting or any damage by other load.

Whenever the available volume of the compartment / net section is not completely used, additional securing is necessary for:

- Small pieces with high individual weight;
- Load which is sensitive against jolts or tilting;
- Pipes, bars, planks, machinery, etc.

Heavy items with an individual weight of 150 kg or more as well as wet freight shall always be lashed.

Loads in individual net sections are secured by compartment separation nets; after completion of loading the nets shall be fastened to the corresponding attachment fittings and tightened.

Limitations and instructions on baggage compartment placards shall be adhered to.

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8.2.2.5 Special Loads and Classification of Load Compartments

PART-CAT (CAT.GEN.MPA.200) accepts the carriage of articles and substances which would be otherwise be classed as dangerous goods from the provisions of [Dangerous Goods and Weapons](#), to the extent specified in the ICAO Technical Instructions for the Safe Transport of Dangerous Goods by Air (ICAO Doc 9284) provided that:

- a. They are required to be aboard the aeroplane in accordance with the relevant AIR OPS or for operating reasons;
- b. They are carried as catering or cabin service supplies;
- c. They are carried for use in flight as veterinary aid or as a humane killer for an animal;
- d. They are carried for use in flight for medical aid for a patient, provided that:
 - Gas cylinders have been manufactured specifically for the purpose of containing and transporting that particular gas;
 - Drugs, medicines and other medical matter are under the control of trained personnel during the time when they are in use in the aeroplane;
 - Equipment containing wet cell batteries is kept and, when necessary, secured in an upright position to prevent spillage of the electrolyte;
 - Proper provision is made to stow and secure all the equipment during take-off and landing and at all other times when deemed necessary by the commander in the interest of safety.
- e. They are carried by passengers or crew members.

NOTE

Articles and substances intended as replacements for those listed under the first point above shall be transported on an aeroplane as specified in the Technical Instructions.

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8.2.2.5.1 Battery Operated Wheelchairs

Before loading a wheelchair, it shall be ascertained:

- a. That the battery is securely attached to the wheelchair;
- b. That the battery is disconnected;
- c. That the battery terminals are insulated in order to prevent short-circuits.

Wheelchairs which cannot be loaded, stowed, secured, and unloaded in an upright position shall have the battery removed, and the removed battery shall be carried in strong, rigid packaging conforming to the following specifications:

- a. Packaging shall be leak tight and impervious to battery fluid;
- b. Batteries shall be protected against short circuits, secured upright in their packaging and surrounded by compatible material sufficient to absorb their total liquid contents;
- c. Packaging shall be marked 'BATTERY, WET, WITH WHEELCHAIR' and be labelled 'corrosive';
- d. Packaging shall be stowed / secured in accordance with the provisions of section [8.2.2.5](#) and [Operating Procedures](#).

The commander shall be informed, by Special Loads Notification of the location of the battery aboard. Flight Operations shall send a message to the destination, notifying the handling agent of the passenger's name and the location of wheelchair / battery.

NOTE

Battery driven wheelchairs with dry cell batteries or non-spillable wet cell batteries may be carried in a baggage / cargo compartment provided that the battery has been disconnected, securely attached to the wheelchair, and that the poles have been insulated.

8.2.2.5.2 Wet Cargo

'Wet Cargo' designates shipments containing liquids or which, by their nature, may produce liquids and which are not subject to the Dangerous Goods regulations.

Watertight containers shall meet the specifications of the ICAO Technical Instructions. They shall be able to withstand the variations in atmospheric pressure and temperature encountered in the course of flight, without rupture or leakage.

Other containers shall be of high quality waterproof material. Containers with cargo which may produce liquids shall be leak proof or contain sufficient absorbent material. Packing shall allow for the maximum angles of roll and bank the aeroplane may encounter during flight without release of the liquid contents. Plastic tarpaulins shall be spread out on the aeroplane's floor and walls so as to catch spillages.

Containers shall be secured in an upright position. For wet cargo in containers which are not watertight or waterproof, secondary measures shall be taken to ensure that any spillage is contained, by placing the shipment in a container of sufficient volume to contain any spillage, and turning up the edges of the tarpaulin against the aeroplane's walls or against other cargo so as to create a second waterproof container around the shipment.

8.2.2.5.3 Live Animals

The policies outlined herein are based on the nature of the operations carried out by NTA, which is that of operating aeroplane owned by individuals or corporations, or supplying ad-hoc charter services to similar individuals or organisations.

In this context, it may be assumed that all the passengers on board any particular flight are known to each other, being either the aeroplane owner and their family, friends, or associates, or business colleagues within a corporation, etc. As such, the carriage of pets in the cabins of aeroplane operated by NTA is a commonplace occurrence, and normally of no inconvenience to any of the passengers.

Carriage of live animals in the cabin is thus limited only by considerations of passenger and aeroplane safety.

NTA crews shall adhere to the following:

1. Within reason, any pets accompanying the passengers may be carried in the cabin, provided that the cabin is large enough to accommodate the particular size of pet (and its cage if applicable) comfortably, and, in the judgement of the Commander, the animal can be carried without prejudice to the safety of the passengers or the aeroplane.

Should the Commander judge that it is not possible to safely carry an animal which is presented at the aeroplane for carriage, then the Commander has the authority to refuse the carriage of such an animal.

2. It is preferable for pets to be housed in a suitable leak-proof container. However, dogs and cats may be carried without such containers, provided all the requirements in this paragraph are verified (points a to g below). Note the importance of point b., requiring that the animal have a harness or collar and be secured in a way that prevents it from reaching the flight deck during all stages of flight.

NOTE

In the following paragraphs, whenever the term 'dog' is used, it should be understood to refer to any animal being carried).

- a. The number / size of the dogs to be carried does not exceed that permitted by the OM-B for a given aircraft type. For this purpose, dogs are classified into the following size groups:

Very small	11kg (25lb) or less
Small	More than 11kg (25lb) up to 25kg (55lb)
Medium	More than 25kg (55lb) up to 39kg (85lb)
Large	More than 39kg (85lb) up to 45kg (100lb)
Extra large	More than 45kg (100lb) up to 68kg (150lb)

- b. For any phase of flight a dog may be seated on the cabin floor which may include being seated at a passengers feet in a location authorised in the OM-B for that purpose, as long as:

1. The dog does not extend into the aisle that would be used to access an emergency exit, emergency equipment or the foot space of another passenger;

2. The following securing arrangements are complied with:
 - i. The leash or harness are secured to a seatbelt. If this is the seatbelt of a passenger, then it shall be attached so that it does not interfere with the passenger's ability to unfasten the belt and evacuate in an emergency;
 - ii. Secure the leash or harness to a seatbelt by passing the seatbelt through the 'closed-loop' end of the leash or harness. If there is no loop, then it may be necessary to tie one;
 - iii. The leash or harness shall restrict the movements of the dog within the authorised location in the cabin. It may be necessary to tie off the leash to shorten its length.
- c. For all phases of flight, including ground movement, a very small dog may be seated on a passenger's lap as long as that passenger is not carrying an infant on their lap and is not seated adjacent to an emergency exit. The dog shall be secured to the passenger's seat belt so that it does not interfere with the passenger's ability to unfasten the belt and evacuate in an emergency;
- d. Unless otherwise restricted in the OM-B, a dog up to 150lbs / 68kg may be carried on an empty seat provided this seat is not adjacent to an emergency exit and they are secured to the seat in such a manner that restricts them to that seat. The seat should be protected from damage using the best means available; the blankets carried on board can be used for this purpose;
- e. In flight, when the fasten seat belt sign is switched off, a dog may be permitted to leave the locations described above only if it is possible to prevent it from reaching the FCC. This means the dog must be kept attached to a seatbelt (or other suitable structure if one is defined in the OM-B) by a leash or harness, or it is held by a passenger;
- f. A maximum ratio of 3 un-caged dogs per passenger shall not be exceeded;
- g. Crew shall ensure that, before proceeding airside, the dog is secured by a leash so that it cannot run loose around the apron.
3. If there is any risk of the animal biting a passenger, crew member, or any other person associated with the flight and its handling, the animal shall be suitably muzzled.
4. The passenger, under whose care the pet travels, shall be in possession of all documents required by the authorities at destination.
5. No animal may be carried in the cabin in such a location or position that permits direct access to emergency exits, where their presence could obstruct access to emergency equipment or otherwise impede the crew in carrying out their duties, or impede the emergency evacuation of the aeroplane.
6. The carriage of caged animals in Cargo and Luggage compartments will be subject to the following considerations:
 - Crews shall take the specific needs of the animals into account;
 - The basic environmental requirements shall be ascertained from all sources available;
 - The temperature range and oxygen requirements shall be matched by the heating and ventilation capability of the cargo compartment provided;

- Feeding requirements shall be met, i.e. for carriage of birds on long haul flights the cargo compartment lights shall be left switched on to enable the birds to feed during flight;
 - Stowage and loading of animals shall follow the principles outlined in section [Loading and Securing of Items in the Aeroplane](#);
 - Containers shall be stowed in such a manner as to guarantee sufficient air circulation;
 - Containers shall be accessible, without needing to be offloaded when care of the animals is required at transit stations;
 - In the event of excessive delays, special care, according to passenger or specialist instructions, shall be taken of the animal(s);
 - Containers shall normally not be loaded directly in front of / below air ventilation outlets or internal lighting;
 - Animals which are natural enemies shall not be loaded in close proximity to each other;
 - Male and female animals should be stowed as far apart as possible;
 - In general, live animals shall not be loaded in close proximity to any other load which might have a negative effect on their well-being or health.
7. The carriage of caged animals in the cabin will be subject to the considerations listed in (6) above and in addition the following shall apply:
- Size of the container shall be appropriate to place where it will be secured as detailed in [Permissible Size and Weight of Hand Baggage](#).
 - The container shall be secured as detailed in [Loading and Securing of Items in the Aeroplane](#).
 - If the cage is to be strapped onto a seat then consideration should be given to the possibility of leakage and where possible the seat should be protected (a bin liner might provide a liquid proof barrier).
8. Certain types of dog are considered dangerous in some countries; this can mean restrictions may be placed on owning, importing, exporting or breeding them, for example the following types are considered to be 'banned' in the UK;
- Pit Bull Terrier.
 - Japanese Tosa.
 - Dogo Argentino.
 - Fila Brasileiro.

Whereas in France some of these are still classed as dangerous but are legal to own, provided they have a pedigree, are kept muzzled and on a lead at all times in public places - regulations in other European countries and even regions with these countries also differ considerably.

It is therefore not feasible to provide an exhaustive worldwide list of these regulations.

If crew members have concerns regarding possibility the dog could become dangerous on board then as detailed in (3) above they may require it to be muzzled.

If there are concerns regarding the legality of a dog in either the country of departure or destination then these should be addressed to the Owner Services team.

9. When it is seen by the crew on the briefing or at the aeroplane that animals will be carried in the cabin, consideration should be given to requesting ad-hoc cleaning at destination, stating in this request that the reason is the carriage of animals in the cabin.

All devices used in carriage shall be thoroughly cleaned and disinfected immediately after unloading.

10. The type, size and caged / uncaged status of the animal will be included in the passenger manifest contained within the crew's flight briefing.

In the event that passengers arrive intending to travel with animals which are not included on the manifest, the crew shall notify Flight Dispatch so that they can be added to the manifest, documentation checked and the suitability of the destination aerodrome checked. This shall take place prior to departure.

11. The carriage of animals in cargo compartments will be subject to the following considerations:

- Crews shall take the specific needs of the animals into account;
- The basic environmental requirements shall be ascertained from all sources available;
- The temperature range and oxygen requirements shall be matched by the heating and ventilation capability of the cargo compartment provided;
- Feeding requirements shall be met; i.e.:, for carriage of birds on long haul flights the cargo compartment lights shall be left switched on to enable the birds to feed during flight;
- Stowage and loading of animals shall follow the principles outlined in section [8.2.2.5](#) and [8.2.2.5.3](#);
- Containers shall be stowed in such a manner as to guarantee sufficient air circulation;
- Containers shall be accessible, without needing to be offloaded when care of the animals is required at transit stations;
- In the event of excessive delays, special care, according to passenger or specialist instructions, shall be taken of the animal(s);
- Containers shall normally not be loaded directly in front of / below air ventilation outlets or internal lighting;
- Animals which are natural enemies shall not be loaded in close proximity to each other;
- Male and female animals should be stowed as far apart as possible;
- In general, live animals shall not be loaded in close proximity to any other load which might have a negative effect on their well-being or health.

Flight Operations shall brief the commander before departure of all live animals intended to be carried.

All devices used in carriage shall be thoroughly cleaned and disinfected immediately after unloading.

8.2.2.5.4 Perishable Cargo

Perishable goods are those whose condition or suitability may deteriorate if exposed to undue changes in temperature or humidity, or delay in carriage.

NTA does not carry perishable cargo - the only perishable goods carried are those associated with catering on a flight, and such goods are replaced with fresh supplies on completion of the flight.

8.2.2.5.5 Human Remains

NTA does not normally carry human remains. On the extremely rare occasion that this might prove necessary, in an applicable aeroplane type, flight crew will adhere to the following guidelines:

1. Non-cremated human remains shall be contained in a hermetically sealed inner coffin of lead or zinc inside a wooden coffin.
2. The wooden coffin may be protected by outer packing and should be covered by canvas or tarpaulins in such a way that the nature of its contents is not apparent.
3. Such human remains shall not be loaded in close proximity to food for human or animal consumption or edible materials (refer to [8.2.2.5.8](#)).
4. Mourning ceremonies on the apron should be avoided.

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8.2.2.5.6 Classification of Load Compartments

For the designation of aeroplane holds, compartments, bays and cabin the following definitions shall be used:

Bay - A subdivision of a containerised / palletised compartment, (i.e. ULD position).

Cabin - The compartment of an aeroplane where passenger seats are installed.

Cabin Section - Resulting from division of the cabin into zones for the purpose of balance.

Compartment - A space designated within a hold.

Deck - A structural floor level. For aeroplanes having one structural level only, this floor level shall be referred to as the main deck. For aeroplanes having more than one structural floor level the different levels shall be referred to as lower deck, main deck and upper deck, starting from bottom to top.

Hold - The space confined by ceiling, floor, walls and bulkhead, used for carrying load.

Left / Right - To be understood as left and right in the direction of flight.

Section - A subdivision of a non-containerised / palletised compartment, (i.e. net section).

Each (type-specific) AOM shall provide information on which cargo compartment classification (A, B, C, D, E, and F) is applicable for an individual compartment.

EASA provides the following cargo compartment classification:

- **Class A.** A Class A cargo or baggage compartment is one in which:
 - The presence of a fire would be easily discovered by a crew member while at their station; and
 - Each part of the compartment is easily accessible in flight.
- **Class B.** A Class B cargo or baggage compartment is one in which:
 - There is sufficient access in flight to enable a crew member standing at any one access point and without stepping into the compartment, to extinguish a fire occurring in any part of the compartment using a hand fire extinguisher;
 - When the access provisions are being used, no hazardous quantity of smoke, flames or extinguishing agent will enter any compartment occupied by the crew or passengers; and
 - There is a separate approved smoke detector or fire detector system to give warning to the pilot or flight engineer station.
- **Class C.** A Class C cargo or baggage compartment is one not meeting the requirements for either a Class A or B compartment but in which:
 - There is a separate approved smoke detector or fire detector system to give warning to the pilot or flight engineer station;
 - There is an approved built-in fire extinguishing or suppression system controllable from the cockpit;
 - There are means to exclude hazardous quantities of smoke, flames, or extinguishing agent, from any compartment occupied by the crew or passengers; and

- There are means to control ventilation and draughts within the compartment so that the extinguishing agent used can control any fire that may start within the compartment.
- **Class D.** A Class D cargo or baggage compartment is one in which:
 - A fire occurring in it will be completely confined without endangering the safety of the aeroplane or the occupants;
 - There are means to exclude hazardous quantities of smoke, flames or other noxious gases, from any compartment occupied by the crew or passengers;
 - Ventilation and draughts are controlled within each compartment so that any fire likely to occur in the compartment will not progress beyond safe limits;
 - Reserved;
 - Consideration is given to the effect of heat within the compartment on adjacent critical parts of the aeroplane;
 - The compartment volume does not exceed 1000 cubic ft;
 - For compartments of 500 cubic ft or less, an airflow of 1500 cubic ft per hour is acceptable.
- **Class E.** A Class E cargo compartment is one on aeroplanes used only for the carriage of cargo and in which:
 - Reserved;
 - There is a separate approved smoke or fire detector system to give warning at the pilot or flight engineer station;
 - There are means to shut off the ventilating airflow to, or within, the compartment, and the controls for these means are accessible to the flight crew in the crew compartment;
 - There are means to exclude hazardous quantities of smoke, flames, or noxious gases, from the flight-crew compartment; and
 - The required crew emergency exits are accessible under any cargo loading condition.
- **Class F.** A Class F cargo or baggage compartment is one in which:
 - There is a separate approved smoke detector or fire detector system to give warning at the pilot or flight engineer station;
 - There are means to extinguish or control a fire without requiring a crew member to enter the compartment; and
 - There are means to exclude hazardous quantities of smoke, flames, or extinguishing agent from any compartment occupied by the crew or passengers.

8.2.2.5.7 Dangerous Goods

See [Dangerous Goods and Weapons](#).

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8.2.2.5.8 Incompatibility Charts

Incompatibility charts in [Dangerous Goods and Weapons](#) and the AOM provide guidance on load incompatibilities.

Some dangerous goods may either be incompatible with other loads or react dangerously with each other in case of damage.

As already outlined above, certain types of special loads may also not be loaded close to each other.

Aeroplane type specific load incompatibility charts (AOM) shall also specify permissible loading positions, or conversely, prohibited positions.

8.2.2.5.9 Inflatable Items in Unpressurised Hold

When loading items that are inflated (e.g. bicycle tyres, wheelchair tyres, footballs, etc.) into an unpressurised hold, due consideration should be given to whether it is necessary to deflate or reduce the pressure in the item in order to prevent it being damaged by the increase in differential pressure that it will experience at altitude. In these cases the following should be considered:

- In a standard atmosphere the increase in differential pressure an item would experience between Sea-Level and 45,000 ft is 13 psi (e.g. a bicycle tyre inflated to 30 psid at Sea Level would experience 43 psid at 45,000 ft pressure altitude). If a change in pressure of this amount might damage the item, then the pressure should be reduced to an acceptable level before flight.
- Availability of a means to re-inflate the item on arrival at destination should be checked, particularly in the case of a wheelchair that the passenger might need to use on arrival.

Generally most items will tolerate a 13 psid increase in pressure, but if there is any doubt, then the situation should be discussed with the passenger prior to loading the item and an appropriate course of action agreed.

8.2.2.5.10 Large PEDs

Any large PED that cannot be carried in the passenger cabin (e.g. due to its size) and, therefore, must be carried in the hold shall be:

- Completely switched off and effectively protected from accidental activation. To ensure the device is never powered on during its transport, any application, alarm, or pre-set configuration that may activate it shall be disabled or deactivated;
- Protected from the risk of accidental damage by applying suitable packaging or casing or by being placed in a rigid bag protected by adequate cushioning (e.g. clothing);
- Not carried in the same baggage together with flammable material (e.g. perfumes, aerosols, etc.).

Furthermore, where carry-on bags are put in the hold (e.g. due to the lack of space) crew are reminded to ensure that passengers are requested to remove from the bag any spare batteries or e-cigarettes.

8.2.2.6 Positioning of Ground Equipment

1. Only appropriately trained Company staff or handling agents may operate mobile ground equipment.
2. The pattern to be followed, for a given type of aeroplane, in positioning loading and servicing equipment is published in the respective AOM.
3. Ground equipment shall not approach the aeroplane until all engines have been cut or propellers stopped and the aeroplane's parking brakes have been set or the chocks are in position.
4. Steps / passenger jetways and catering trucks shall principally be positioned at the aeroplane prior to opening the respective cabin doors, see [8.2.2.8](#).
5. Sufficient distance between ground equipment and the aeroplane shall be maintained in order to avoid damage caused by vertical movement of wings / fuselage during unloading / loading / refuelling / defuelling. Utmost care shall be taken in shifting ground equipment in the aeroplane's vicinity.
6. Fuel hoses (see also [Fuelling Procedures](#)) and connections shall never be run over by ground equipment. Loading and servicing equipment shall not be positioned or manoeuvred under the wings.
7. With the exception of fuel trucks, mobile equipment shall not be positioned within the venting areas during fuelling / defuelling (see also [8.2.1.2](#)).
8. Equipment when parked away from, or positioned at, the aeroplane shall have parking brakes set.
9. Equipment approaching, manoeuvring at or leaving the aeroplane shall not be driven faster than at walking speed.
10. Aeroplanes and pedestrians have the right-of-way. Equipment should never move across the path of taxiing aeroplanes or of embarking / disembarking passengers.
11. Personnel shall not ride on elevating platforms of moving ground equipment.

8.2.2.7 Operation of Aeroplane Doors

Aeroplane type-specific normal, abnormal and emergency procedures, concerning the operation of the cabin and compartment doors, are specified in the respective AOM. The following general guidelines shall be observed:

- Normally, cabin and compartment doors, upon arrival, shall not be opened until all engines have been cut or propellers stopped and the aeroplane's parking brakes have been set or the chocks are in position;
- Conversely, all doors shall be closed and locked before start of engines;
- All cabin doors (passenger and service) shall normally be opened and closed by members of the operating crew only. In exceptional cases, trained handling staff may operate these doors;
- When opening such a door from outside, the respective staff member shall ascertain by knocking at the door, that nobody is standing in the danger area on the inside and that the escape slide, if applicable, is disarmed;

- Whenever steps / passenger jetways, servicing vehicles (see also [8.2.2.6](#)) or catering trucks are used, the design of which does not permit opening or closing the doors when the equipment is in position, the respective doors shall only be opened immediately prior to the positioning of the equipment and closed immediately after the equipment has been removed;
- Main dock cargo doors, cargo and lower compartment doors shall normally be operated by the operating crew.

Manually Operated Doors

Their operation does not require special training, but shall have been demonstrated to the staff concerned before they are permitted to operate such doors.

Electrically, Pneumatically, or Hydraulically Operated Doors

No person shall operate such doors without having received prior theoretical / practical training by properly qualified staff. For such training, NTA, (via the respective AOMs), shall provide detailed descriptions of such doors and their operation.

NOTE

For the operation of aeroplane doors in strong winds refer to the AOM which specifies maximum permissible wind speeds and other particulars to be observed. If difficulties occur when attempting to close doors in strong winds, the aeroplane shall be moved in order to position the doors concerned on the downwind side.

8.2.2.8 Safety on the Ramp, Including Fire Prevention, Blast and Suction Areas

The provisions of [8.2.1](#), [8.2.2.1](#), [8.2.2.6](#), [8.2.2.8](#), [8.2.2.9](#), and [Dangerous Goods and Weapons](#) have all been developed in order to secure the safety on the ramp of all aeroplanes, crews, passengers, staff and load. The following additional provisions shall be observed by all handling agents contracting services to NTA.

- No unauthorised persons shall enter the ramp;
- Smoking and the use of open fire is strictly prohibited;
- Ramp surfaces shall be frequently checked to prevent accidents caused by slipping / skidding on oil, ice or snow; such checks shall, as far as possible, also serve to detect foreign objects on the ramp surfaces which shall be removed in order to prevent foreign object damage to the tyres and other parts of the aeroplane, and to preclude foreign object ingestion by an engine;
- Ramp personnel shall wear ear protection permanently;
- It is strongly recommended that all personnel, while on the ramp at night, wear high visibility clothing;
- The responsible handling staff member in charge shall ensure that the security zones around the suction and blast areas of the individual aeroplane are observed and that no personnel or equipment are within such zones when the signals for engine start-up are given (for safety distances refer to AOM);
- As outlined in [8.2.1.2](#), during fuelling / defuelling no vehicles (except fuel trucks) shall be positioned within the venting areas. In addition, whenever fuels other than Jet A1 are being uplifted, no electrical appliances shall be connected or disconnected within the venting areas. Normally, the fuel vents are located at the wing tips and for aeroplanes with tail tanks at the tip of the right-hand part of the stabiliser. For details see AOM;
- Whenever fuel has been spilled, fuelling / defuelling shall be stopped immediately. It shall be removed or dried up immediately in the presence of the fire service (see [8.2.1.2](#));
- Certain systems of the aeroplane shall not be operated on the ramp, (i.e. weather radar, or while fuelling / defuelling, HF radio transmitter). Refer to respective AOM.

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8.2.2.9 Start-up, Ramp Departure and Arrival Procedures

The marshaller's signals specified by ICAO Annex 2 are depicted in the Chapter 'Rules of the Air' of the RM under tab AIR TRAFFIC CONTROL and the chapter 'Signals for Aerodrome Traffic'.

Signals for engine start-up shall only be given after the handling staff member in charge has ascertained that the security zones around the suction and blast areas have been cleared (refer to [8.2.2.13](#)) and after they have given 'clearance for start-up'.

Depending on the type of aeroplane, the ground to cockpit communication may be performed by means of a headset. If that is impossible, the hand signals of the RM shall be used. After disconnection of the headset, hand signals only apply. For type-specific procedures and provisions see AOM.

Upon arrival, hand signals only shall be used from ground to cockpit for 'Chocks are in position', and 'Shut down engines'. This does, of course, not preclude the cockpit personnel from re-checking, via the headset or visually, that chocks indeed are in position.

After arrival at ramp position parking brakes shall not be released until all engines have been shut down and until the cockpit personnel have ascertained that chocks have been inserted, and that the aeroplane is not moving.

The FCC personnel shall carefully evaluate the situation around the aeroplane, particularly the distances to other aeroplane and objects, select an adequate taxi speed and handle throttles accordingly to minimise blast effects and noise, particularly on start of taxi-roll from standstill.

A high degree of awareness is required for all low visibility taxi operations. In order to render the aeroplane visible to other traffic, display of all lights (including strobe lights) is recommended unless own vision is thereby impaired. However, care shall be taken not to blind other traffic or ground handling staff.

NOTE

1. Company aeroplanes may be taxied on the movement area of an aerodrome only by:
 - a. An appropriately qualified flight crew; or
 - b. A person authorised and found competent by the Company (see [NJMP 1.13](#)), and
 - i. is competent to taxi the aeroplane and to use the required means of communication, and
 - ii. is instructed in respect to aerodrome layout, routes, signs, marking, lighting, ATC signals and instructions, and all applicable phraseology and procedures, and
 - iii. is able to conform to the operational standards required for the safe aeroplane movement at the aerodrome.
2. For ramp operations in ice, snow or freezing precipitation refer to [De-icing and Anti-icing on the Ground](#).
3. For pushback and towing procedures refer to [8.3.23.2](#) and [NJMP 1.13](#).

8.2.2.10 Servicing of Aeroplanes

Potable Water - Potable water systems are susceptible to contamination by bacteria and other microorganisms. It is therefore essential that such water is free from chemical substances / micro-organisms which might cause illness, and that it is chlorinated. NTA will ensure that suitable bacteriological examinations of water samples taken from water supply systems, servicing vehicles and aeroplane water systems are carried out at least four times a year. Potable water services shall not be filled up from the same tap as toilet services. Potable water services and toilet services shall not be parked in the same area. Personnel engaged in toilet servicing shall not perform water service.

NTA, in cooperation with its handling agents, has drawn up a list of those stations where uplift of potable water is permitted.

Removal and Disposal of Waste - According to Article 14(3) of the WHO's International Health Regulations, 'every port and airport shall be provided with an effective system for the removal and safe disposal of excrement, refuse, waste water, condemned food, and other matters dangerous to health'. The Company's handling agents will provide warning when the removal of waste at a specific aerodrome is not assured. Such information will be relayed, by Operations Planning and Control, to the crew.

Oxygen - Special safety provisions shall be observed when oxygen bottles of the aeroplane are being filled or exchanged (i.e. connection / disconnection to / from the system):

- No passenger shall be on board;
- No ground power unit shall be connected or disconnected;
- The AOM specifies which electrical systems shall be 'off' or, alternatively, shall not be operating;
- No fuelling / defuelling is permitted;
- Filling / exchanging is not permitted during a thunderstorm.
- No activation of electrical switches;
- No radio communication, in particular HF;
- No smoking;
- No loading and on board services closer than a radius of 15 m from the filling connection, with the exception of work inside the cabin;
- Vehicle traffic must remain at a radius of 15 m from the filling connection;
- Galley power (shall be disconnected prior to refilling).

NOTE

The flight crew shall inform the cabin crew, if applicable, prior to commencement of refilling.

Cleaning of Cabin - Cleaning should have been finished before passenger embarkation. If passengers stay on board during transit, cabin cleaning should be performed in such a way as not to disturb the passengers.

The FCC may only be cleaned under supervision of an authorised company employee, or a handling agent appointed by NTA.

8.2.2.11 Document and Forms for Aeroplane Handling

See paragraph [Documents, Forms, and Additional Information to be Carried](#).

8.2.2.12 Multiple Occupancy of Aeroplane Seats

Multiple occupancy of crew seats, whether by crew members or by passengers, is strictly prohibited.

For multiple occupancy of a passenger's seat, refer to [8.2.2.2.1](#).

8.2.2.13 Smoking Regulations

Airport authorities prohibit smoking in designated areas of the terminal buildings as well as on the ramp. As far as smoking on board is concerned, refer to [Flight Procedures](#).

8.2.3 Procedures for the Refusal of Embarkation**8.2.3.1 General**

A person shall be refused carriage and / or removed from the aeroplane when, in the exercise of reasonable discretion, the Commander decides that:

- a. Such action is necessary in the interest of safety of the aeroplane or its occupants; or
- b. Such action is necessary to prevent violation of laws, regulations or decrees of any country to be flown from, into or over; or
- c. The age or mental or physical condition of the person is such as to require special assistance which cannot be provided; or
- d. The person can be considered to be abusive or disrupted; or
- e. The person recklessly or negligently acts or omits to act so as to:
 - Endanger the aeroplane or person therein; or
 - Cause or permit an aeroplane to endanger any person or property.

8.2.3.2 Alcohol and Drugs

Any person who appears to be under the influence of alcohol or drugs to the extent that the safety of the aeroplane or its occupants is likely to be endangered shall be refused embarkation.

8.2.3.3 Removal

Whenever it becomes necessary to remove a passenger from an aeroplane in accordance with [8.2.3.1](#) and [8.2.3.2](#), the flight crew shall inform Flight Dispatch prior to taking any action. Flight Dispatch will coordinate the necessary action with the handling agents.

8.2.4 De-icing and Anti-icing on the Ground

Introduction

Every year, winter confronts flight crews with the typical problems associated with low temperatures, especially conditions conducive to aircraft icing exist on the ground, freezing fog, freezing precipitation, frost, rain or high humidity (on cold soaked wings), hail, ice pellets, snow or mixed rain and snow, etc. To deal with these critical weather conditions, special cold weather procedures have been established. They supplement normal procedures and shall be observed when applicable. The intention of this section is to define both general and specific information.

General

The procedures stated herein establish the minimum requirements for de-icing and especially anti-icing of aeroplanes on the ground to provide an aerodynamically clean aeroplane for take-off.

Exterior inspection will ensure that vital parts of the aeroplane are free of frost, ice, slush and snow. The anti-icing procedure is to be used by ground crews to inform flight crews about the quality of protection the aeroplane has received against the forming or accumulation of frost, ice or snow on the surface of the aeroplane. Experience shows that the higher the precipitation rate and the nearer the temperature is to 0°C, the more problematic it is to keep the aeroplane free of frost, ice, slush or snow.

8.2.4.1 Definitions

Anti-icing - The process of protecting the aircraft to prevent contamination due to existing or expected weather, typically by applying anti-icing fluids on uncontaminated aircraft surfaces.

Check - An examination of an item against the relevant standard by a trained and qualified person.

Critical Surfaces - should be determined by the aircraft manufacturer, but may include the following: wings, control surfaces, propellers, horizontal stabilisers, vertical stabilisers or any other stabilising surface on an aircraft. These surfaces should be completely free of ice, snow, slush or frost before take-off.

De-icing - the process of eliminating frozen contamination from aircraft surfaces, typically by applying de-icing fluids.

De-icing / Anti-icing - is the combination of de-icing and anti-icing performed in either one or two steps.

One-step de-icing / anti-icing means that the de-icing and anti-icing are carried out at the same time using a mixture of anti-icing fluid and water.

Two-step de-icing / anti-icing means that de-icing and anti-icing are carried out in two separate steps. The aeroplane is first de-iced using heated water only or a heated mixture of de-icing fluid and water. After completion of the de-icing operation a layer of a mixture of anti-icing fluid and water, or of anti-icing fluid only, is to be sprayed over the aeroplane surfaces. The second step shall be applied before the first step fluid freezes, typically within three minutes and, if necessary, area by area.

Freezing Drizzle - Fairly uniform precipitation composed exclusively of fine drops (diameter less than 0.5 mm [0.02 in]) very close together which freezes upon impact with the ground or other exposed objects.

Freezing Fog - A suspension of numerous minute water droplets which freezes upon impact with ground or other exposed objects, generally reducing the horizontal visibility at the earth's surface to less than 1 km (5/8 mile).

Freezing Precipitation - Corresponds to freezing rain or freezing drizzle.

Holdover Time (HoT) -The period of time during which an anti-icing fluid provides protection against frozen contamination to the treated aircraft surfaces. It depends among other variables, on the type and intensity of the precipitation, OAT, wind, the particular fluid (or fluid Type) and aircraft design and aircraft configuration during the treatment.

Light Freezing Rain - Precipitation of liquid water particles which freezes upon impact with exposed objects, either in the form of drops of more than 0.5 mm (0.02 in) or smaller drops which, in contrast to drizzle, are widely separated. Measured intensity of liquid water particles are up to 2.5 mm (0.10 in) / hour or 25 grams / dm² / hour with a maximum of 2.5 mm (0.10 in) in 6 minutes.

Lowest operational use temperature (LOUT): the lowest temperature at which a fluid has been tested and certified as acceptable in accordance with the appropriate aerodynamic acceptance test whilst still maintaining a freezing point buffer of not less than:

1. 10°C for a Type I fluid; or
2. 7°C for Type II, III or IV fluids.

Pre Take-off Check - The flight crew should continuously monitor the weather conditions after the de-icing / anti-icing treatment to assess whether the applied holdover time is still appropriate. Within the aircraft's HOT and prior to take-off, the flight crew should check the aircraft's wings or representative aircraft surfaces for frozen contaminants.

Pre-take-off contamination check: A check of the treated surfaces for contamination, performed when the HOT has been exceeded or if any doubt exists regarding the continued effectiveness of the applied anti-icing treatment. It is normally accomplished externally, just before commencement of the take-off run.

Rain or High Humidity (On Cold Soaked Wing) - Water forming ice or frost on the wing surface, when the temperature of the aeroplane's wing surface is at or below 0°C (32°F). See [Cold Soaked Surface Frost \(CSSF\)](#).

Sleet - Precipitation in the form of a mixture of rain and snow. For operation in light sleet treat as light freezing rain.

Slush - Snow or ice that has been reduced to a soft watery mix by rain, warm temperatures and / or chemical treatment.

Snow - Precipitation of ice crystals, most of which are branched, star-shaped or mixed with unbranched crystals. At temperatures higher than minus 5°C (23°F), the crystals are generally agglomerated into snowflakes.

8.2.4.2 Types of Icing

Frost - Ice-crystal deposits formed on cold, clear nights by sublimation on surfaces which have a temperature lower than the surrounding air. Such deposits on leading edges upper and lower surfaces, even when they are very thin (hoar frost) can seriously affect an aeroplane's performance. Frost 3mm or less on the lower surface of a wing has no effect and may be discounted. The OM-B specifies limits of frost deposits for take-off.

Frost / Hoar Frost - Ice crystals that form in saturated air at temperatures below 0°C (32°F) by direct sublimation on the ground or other exposed objects.

Rime Ice - An opaque white or milky deposit formed by the rapid freezing of supercooled water droplets, trapping air between them, as they impinge upon an exposed aeroplane. In fog, mist or drizzle conditions with temperature at or below freezing point, rime ice can form on any part of the aeroplane and may become very thick on the windward side.

Clear Ice - A coating of ice, generally clear and smooth, but with some air pockets. It forms on exposed objects, the temperatures of which are at, below or slightly above the freezing temperature, by the freezing of super-cooled drizzle, droplets or raindrops.

Crews shall be especially aware of the difficulty of detecting clear ice and, in some situations, its presence may only be detected by touch. It can form on aeroplane surfaces below a layer of snow and rime ice. Significant deposits can form on upper wing surfaces in the vicinity of fuel tanks after refuelling with low temperature fuel or when sufficient supercooled fuel remains in tanks after a long flight at altitude.

This has occurred with ambient temperatures as high as 14°C. Conversely, refuelling with relatively warm fuel can cause dry falling snow to melt with the danger of refreezing on the upper surface of the wing.

Cold soaked surface frost (CSSF) - frost developed on cold soaked aircraft surfaces by sublimation of air humidity. This effect can take place at ambient temperatures above 0° C. Cold soaked aircraft surfaces are more common on aircraft that have recently landed. External surfaces of fuel tanks (e.g. wing skins) are typical areas of CSSF formation (known in this case as cold soaked fuel frost (CSFF)), due to the thermal inertia of very cold fuel that remains on the tanks after landing.

Dry Snow - Fine, powder like snow which does not stick and may be blown or brushed away.

Wet Snow - Has a much higher liquid content and tends to stick on airframe / engine components and may freeze.

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8.2.4.2.1 Responsibility

The Commander shall check the aeroplane for the need to de-ice / anti-ice. They will, based on their judgement, initiate de- and anti-icing, if required and they are responsible for the correct and complete de- / anti-icing of the aeroplane.

Completion of the correct process has to be entered in the Aircraft Flight Log by entering the start time, the type of fluid used and the mix ratio, whenever required for the current flight leg.

As the final decision rests with the Commander, their request to de- / anti-ice will supersede any ground crew member's judgement not to de- / anti-ice.

The Commander is responsible for the anti-icing condition of the aeroplane during ground manoeuvring prior to take-off.

The commander is responsible for the calculation of the Holdover Time (HOT) and shall prior to take-off ensure that the applied HOT is still appropriate.

The commander shall only commence a take-off if the aeroplane is clear of any deposit that may adversely affect the performance or controllability of the aeroplane.

CAUTION

Taking into account weather conditions, taxi-conditions, taxi-times, hold-over time and other relevant factors, the Commander shall, when in doubt about the aerodynamic cleanliness of the aeroplane, have a visual inspection performed or return to the ramp.

8.2.4.3 Minimum Runway Clearance

Whenever snow has been removed from a runway the cleared width should normally be 45 meters. The Commander may accept a width of less than 45 meters for take-off and landing, taking into account the:

1. Minimum acceptable value as published in OM-B;
2. Aeroplane status;
3. Weight;
4. Braking conditions;
5. Crosswind;
6. Snow banks.

8.2.4.3.1 Inspection of Aeroplane Surfaces and Engine Inlets

The critical surfaces have to be checked for possible frost, ice, slush or snow accumulation. Clear ice, which is very difficult to detect, may have formed on the upper and lower side of the wing as a result of:

- a. Freezing rain;
- b. Cold fuel causing cold wing surfaces, and precipitation (i.e. rain) freezing above and below the fuel tank area;
- c. Snow melting on a warm wing, but refreezing as the wing cools down, or melted snow running to a colder part of the wing.

If frost or ice has formed on the lower wing surface tank area and the aeroplane has been subjected to precipitation on the ground, (rain, drizzle, fog), or when otherwise there is doubt about clear ice having been formed on the wing upper surface, the wing upper surfaces shall be checked using a suitable means of access.

CAUTION

Heavy freezing has been reported during drizzle / rain at temperatures as high as +15° C, due to cold fuel.

It shall not be assumed that light snow will be blown off the wings during take-off acceleration. You may also anticipate that partly frozen ice or snow patches on the wings may have severe effects on aeroplane aerodynamics (i.e. pitch and / or roll upsets), whereas clear ice built up during flight may have no great influence despite the loss in performance due to the weight of the ice. It shall always be remembered that below a snow / slush layer there can be clear ice.

CAUTION

As clear ice is very difficult to detect visually, it may be necessary to touch the areas to be inspected.

During snowfall, freezing rain and drifting snow, snow and melting ice could penetrate slots, balance bays, drainage openings, hinges and operating linkages, and then refreeze. Therefore the above-mentioned areas shall be checked carefully, especially when the aeroplane has been parked outside for long periods under the conditions described above.

This is also applicable to landing gear areas, including landing gear latching mechanisms, operating mechanisms, electrical switching elements, inlet scoops, pilot probes, air inlet and outlet openings of the APU, and the air-conditioning system, as well as all adjacent areas.

Fuselage areas in front of the cockpit windows have to be free of all ice and snow, even very thin layers.

NOTE

The only method currently known to positively ascertain that an aeroplane is aerodynamically clean is by close visual inspection and / or by touching the relevant surfaces. Check also that inlet areas are free of ice or snow.

When a ground ice detection system (GIDS) is used to perform an aeroplane surfaces check prior to and / or after a treatment, the use of GIDS by suitably trained personnel should be part of the procedure.

8.2.4.3.2 De-icing and Anti-icing Procedures

To preserve holdover times, the de- / anti-icing process shall be continuous and as short as possible. Ensure that the procedures are performed at the latest possible time prior to take-off.

For aeroplane type specific procedures e.g. applicable surfaces to be de-iced / anti-iced, refer to aeroplane AFM / AOM.

8.2.4.3.3 De-icing

De-icing is a procedure by which frost, ice, snow and / or slush is removed from all surfaces, openings and hinge points of an aeroplane by means of physical removal of snow and ice deposits and / or by spraying of the aeroplane with hot water.

8.2.4.3.4 One Step De- / Anti-icing

One-step de- / anti-icing is done with a mixture of hot anti-icing fluid / water appropriate to the ambient temperature, plus anti-icing content appropriate to weather conditions (to obtain adequate holdover time). In the same process, frost, ice, slush or snow are removed from the aeroplane and, a protective film is applied to the aeroplane surfaces.

8.2.4.3.5 Two Step De- / Anti-icing

The first step, de-icing, is done with either hot water or a hot mixture of anti-icing fluid / water appropriate to the ambient temperature, followed by anti-icing, which takes the form of an overspray with a cold mixture of anti-icing fluid / water, or 100% cold antiicing fluid appropriate to ambient temperature and weather conditions. The second step provides anti-icing protection and shall be performed within 3 minutes from the beginning of the first step, if necessary area by area.

NOTE

It is permitted to perform de-icing with hot water, using Type II fluid for subsequent antiicing only.

8.2.4.3.6 Anti-icing

Anti-icing is a precautionary measure which prevents frost, ice, slush or snow to form or accumulate on unprotected surfaces of an aeroplane. Anti-icing is applied to an aeroplane already free of frost, ice, slush or snow.

CAUTION

If an additional de- / anti-icing treatment is required before take-off, anti-icing alone is not allowed. The residues of the de- / anti-icing fluid applied earlier may have frozen and have to be removed by a new complete de-icing procedure. The aeroplane has to be treated as if it were covered by frost, ice, slush or snow, employing either a one-step or two-step procedure.

Crews shall supervise all de / anti-icing activities to ensure that all surfaces are free of frost, ice, slush or snow. The anti-icing process is not properly accomplished if an insufficient amount fluid has been used and which results in incomplete or inadequate coverage of the surfaces to be treated. While thickness will vary in time over the profile of the wing surface, the anti-icing fluid should be distributed uniformly. To control the uniformity, all horizontal aircraft surfaces should be visually checked during application of the fluid (either by the crew or the de-icing service provider). As a rule of thumb, a sufficient amount of fluid has been applied when it can be visually confirmed that the fluid is just beginning to run off the leading and trailing edges of the surfaces. Where possible, a delivery note (anti-icing code) should be obtained from the ground personnel. This code provides the flight crew with the minimum details necessary to estimate a HOT and confirms that the aircraft is free of contamination.

This information is provided in the following sequence:

1. The fluid type.
2. The fluid name (except for Type I).
3. Concentration (except for Type I).
4. Local time at start, either
 - a. for a one-step de-icing / anti-icing: at the start of the treatment, or
 - b. for a two-step de-icing / anti-icing: at the start of the second step (anti-icing)
5. Date (optional).
6. The statement 'post- de-icing / anti-icing check completed' (if check completed).

Example: TYPE II / MANUFACTURER, BRAND X / 75% / 1335 / 15FEB23 / POST-DE-ICING / ANTI-ICING CHECK COMPLETED.

Before reconfiguring or moving the aircraft, the flight crew should receive a confirmation from the personnel involved in the operation that all de-icing and / or anti-icing operations are complete and that all personnel and equipment are clear of the aircraft.

When a two-step de-icing / anti-icing operation has been carried out, the anti-icing code should be determined by the second step fluid.

The type of fluid, mix ratio and the de-ice / anti-ice start time shall be recorded in the Aircraft Flight Log.

8.2.4.3.7 Handling

De- / anti-icing shall be accomplished by authorised and trained personnel familiar with critical aeroplane sections in general, and with the equipment available. Depending on their judgement, the Commander shall inform the ground crew about specific ground and spray procedures for the particular aeroplane type.

The type of fluid used will vary between the aerodromes, depending on availability and de-icing equipment in use. Fluids used may vary also at the same aerodrome within a winter season. However, all of the fluids applied to NTA aeroplanes at regularly used aerodromes have been approved after thorough testing. The de-icing supervisor will normally determine the concentration of the applied fluids, with regard to weather conditions, traffic conditions, expected take-off time, etc. However, the Commander may request a different concentration of Type II or IV fluids, if they desire a higher holdover time. Determination of the correct concentration of Type I fluids is normally the sole responsibility of the ground crew.

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8.2.4.3.8 De-icing and Anti-icing Fluids

The application of de-icing / anti-icing fluid is the most common means of effecting ground de-icing and anti-ice protection. These fluids are water / glycol solutions, broadly classified into four groups: Type I, Type II, Type III and Type IV.

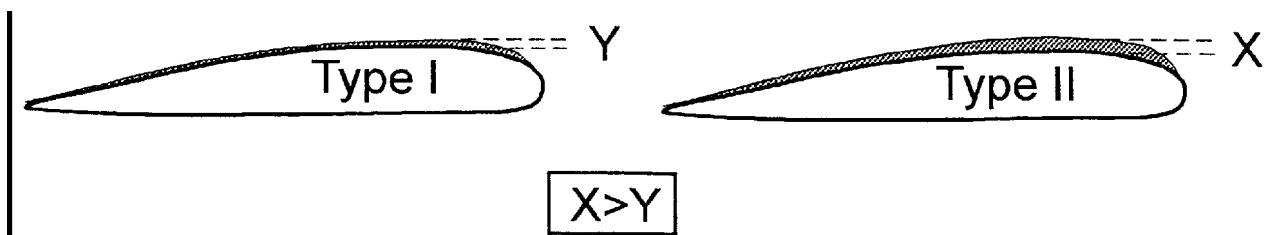
NOTE

Both the Society of Automotive Engineers (SAE), and the International Standards Organisation (ISO), issue specifications defining the composition and characteristics of Type I, II, III and IV fluids. For the practical purposes of flight operations, these respective specifications are identical. In North America, fluids other than Type I, II and IV have been used extensively and are still available. The tables in [Holdover Time Guidelines](#) also contain brands and types approved in addition to generic Type I, II and IV fluids.

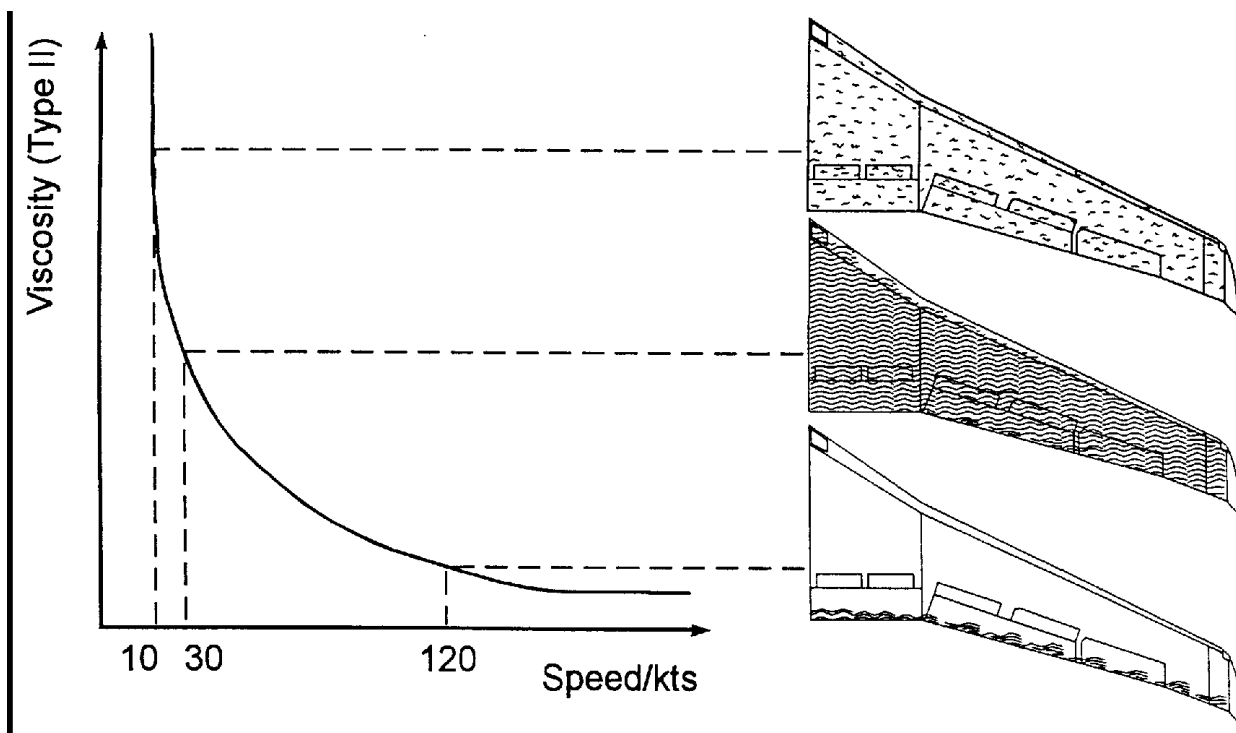
Type I Fluids

In concentrated form, these fluids contain glycols to a minimum concentration of 80%, but with no thickening agents. Their resulting low viscosity and very short holdover time limits them to use as de-icing agents providing a short period of anti-ice protection.

Comparison of Type I and II thickness



Viscosity Breakdown Illustration



Type II Fluids

These fluids contain glycols to a minimum concentration of 50% as well as thickening agents. Their relatively high viscosity permits the application of a layer of fluid that is effective in de-icing and persists for a significant holdover time to provide anti-icing. During take-off, the airstream imparts a shear stress to the fluid layer causing it to flow off the surface to which it was applied.

Type III Fluids

These fluids provide less protection than Type II but are formulated to be blown off during a short, low speed, take-off (low 'shear speed' for turbo-prop aeroplane).

Type IV Fluids

These fluids provide greater protection than Type II in most circumstances. They are the most used ones in western / central Europe. These fluids provide greater protection than Type II in most circumstances. They are the most used ones in western / central Europe.

Comparison between Type I and Type II / IV Fluids.

Type II / IV fluids provide protection against refreezing, during precipitation (see guideline to holdover times).

In addition to water and a minimum of 50% glycol, Type II / IV fluids also contain a thickener system. Due to their pseudo plastic characteristics they form a uniform film on the aeroplane surfaces. This layer is thicker than that of a Type I fluid and therefore more resistant to precipitation.

The viscosity of Type II / IV fluids mainly depends on the shearing forces. When, during take-off run, Type II / IV fluid is exposed to the shearing forces of the airstream, the fluid's viscosity collapses at a speed of about 30 knots. It liquefies abruptly and flows off the wing, thus providing an aerodynamically clean wing for take-off.

NOTE

The anti-icing performance of all fluids not qualified as Type II or Type IV shall be judged as Type I quality.

CAUTION

Due the viscosity characteristics of Type II / IV fluids it may become necessary to perform a complete de- / anti-icing treatment after an aborted take-off run.

Fluid behaviour on aeroplane surfaces

Fluids have been tested and manufactured to perform in a certain way. This has been aerodynamically tested and it is evident that the fluid is sheared from the wing at takeoff leaving only a marginal film of fluid that does not constitute a notable aerodynamic effect. Some fluids are thicker than others (Type-I vs. Type-II / III / IV) and less viscous fluid drains off easier than thicker fluids.

Failed fluid no longer performs aerodynamically as expected. At this point the fluid is comparable to any other contamination. Also old fluid (e.g. after a flight) sticking to the surface should be removed.

8.2.4.4 Holdover Time

For the latest holdover time tables, refer to [Holdover Time Guidelines](#). This document is a joint publication by The Federal Aviation Administration (FAA) and Transport Canada (TC), with the assistance of the members of the SAE Holdover Time Subcommittee. It contains the evaluation of the results from the tests to available anti-icing fluids.

Holdover time is the estimated time that an application of anti-icing fluid is effective in preventing frost, ice, slush or snow from adhering to treated surfaces. Holdover time is calculated as the beginning with the final application of the anti-icing fluid, and as expiring when the fluid is no longer effective, as measured in endurance time tests and published in 'Holdover Time Guidelines'.

CAUTION

The time of protection will be shortened in heavy weather conditions, or in the presence of high wind velocities and jet blast, which cause a degradation of the protective film. Under such circumstances, the time of protection can be shortened considerably. This is also the case when the fuel temperature is lower than OAT by 10°C or more.

Holdover time is thus only one of the relevant factors in determining the eventual deicing status and consequent aerodynamical cleanliness of an aeroplane.

The following factors will have an influence on holdover times:

- Ambient Temperature;
- Aeroplane Surface Temperature;
- Fuel Temperature significantly lower than OAT (>10°C);
- Presence of De- / Anti-icing Fluid;
- De- / Anti-icing Fluid Type;
- De- / Anti-icing Fluid Aqueous Solution (Concentration);
- Precipitation Type and Rate;
- Proper De / Anti-icing application procedure;
- Relative Humidity;
- Solar Radiation;
- Operation in close proximity to other Aeroplanes, Equipment and Structures;
- Operation on Snow, Slush or Wet Surfaces;
- Wind Velocity and Direction;
- Aeroplane Inclination Angle, Contour and Surface Roughness.

Exceeding the holdover time means that precipitation is no longer absorbed by the de- / anti-icing fluid, making the protection ineffective. This results in a white appearance (build-up of ice crystals) in or on top of the fluid.

For Type I fluid, holdover times are applicable for all fluid / water mixtures, irrespective of their fluid to water ratio, as long as certain freezing point limitations are observed.

Type I fluids may be applied in many different concentrations, to ensure a certain margin between the freezing point of the mixture and the actual temperatures. No useful increase in holdover time is available if the percentage of de-/anti-icing fluid is increased above the value required.

The recommendations of AEA (Association of European Airlines) have been substituted by guideline from Transport Canada as the latter are more exhaustive.

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8.2.4.4.1 Procedures with SAE / ISO Type Fluids

Table 1 in [Holdover Time Guidelines](#) serves as guidelines to determine the minimum concentration of Type I, II, III and IV fluids. Responsibility for the application of these data remains with the user.

CAUTION

These tables are for use in departure planning only. They should be used in conjunction with the relevant prior take-off check procedures;

Holdover time starts at the beginning of the anti-icing application; De / anti-icing fluids have not been tested for ice pellet precipitation, nor do holdover tables address ice pellet precipitation;

When ice pellet precipitation occurs after the application of de / anti-icing fluid, the de / anti-icing fluid dilutes, which results in rapid wing contamination.

WARNING

If the holdover time since the last anti-icing has elapsed, a complete new de- and anti-icing shall be performed. The fluid applied with the first de-icing / anti-icing shall be removed completely.

LIMITATIONS

Refer to OM-B.

8.2.4.4.2 Guidelines for Holdover Times

[Holdover Time Guidelines](#) contain the latest guidance from Transport Canada for generic Type I, II and IV type fluids and specific combinations of brands and types in addition to generic Type I, II and IV fluids. This information shall be used in conjunction with [Deicing and Anti-icing on the Ground](#).

Under certain meteorological conditions, de-icing and / or anti-icing procedures may be ineffective in providing sufficient protection for continued operations. Examples of these conditions are freezing rain, ice pellets and hail snow exceeding certain intensities, high wind velocity, and fast-dropping OAT. No HOT guidelines exist for these conditions.

NOTE

LOWEST OPERATIONAL USE TEMPERATURES (LOUT)

When determining the LOUT as per the tables in [Holdover Time Guidelines](#), the High Speed Aerodynamic Test temperatures may be used for all NTA aeroplanes unless it is specified in the aeroplane type specific AOM. If LOUT for the fluid to be used is not shown, then crew should request this from the de-icing personnel.

8.2.4.4.3 Final Checks after Aeroplane De-icing

Check of Aeroplane Surfaces

All surfaces have to be free of frost, ice, slush and snow as well as water accumulations.

To provide passenger comfort, cleaning of the cabin windows with hot water should be considered as soon as practicable. In most cases this could be accomplished at the end of the first leg after a preceding de- / anti-icing treatment, when all residual fluid has run back off the fuselage.

Functional Test of Flight Controls

Whenever an aeroplane is substantially covered by ice or snow, flight control movement could be restricted. A flight control check, including trim tabs shall be considered prior leaving the parking position.

NOTE

Never perform a flight control check prior to de-icing.

After treatment. Before reconfiguring or moving the aeroplane, the flight crew should receive a confirmation from the personnel involved in the operation that all de-icing and / or anti-icing operations are complete and that all personnel and equipment are clear of the aeroplane.

8.2.4.4.4 Taxi-through De-icing

De-icing with engines running may be performed in taxi-through facilities at particular airports. The AOM / AFM prescribes procedures and checklists for this procedure. Prebriefing shall be conducted between the flight crew and the ground agent personnel conducting the de-icing to ensure that the ground personnel is aware of any specific requirements called for by the flight crew or dictated by the AOM / AFM. VHF contact shall be maintained with the de-icing supervisor throughout the operation. The de-icing supervisor is responsible for confirming that the aeroplane is completely free of ice or snow prior to releasing it.

The lowest possible power setting should be used, flaps / slats as per the AFM / AOM and the air-conditioning switched OFF to prevent de-icing fluid from entering the cabin.

NOTE

Avoid use of rain repellent before, during and after de-icing. Operate windscreen wipers to prevent fluid from drying on the windscreen.

Before leaving the facility, perform a flight control and flaps check whilst ground personnel observe control surface movement.

As soon as the aeroplane is clear of the facility, the air-conditioning may be switched ON again. Verify that the taxi and take-off checklists are duly completed. When lining up for take-off and prior to releasing the brakes accelerate the engines to verify their operation.

8.2.4.4.5 Flight Log

Whenever de-icing has taken place the commander (or the person designated by the commander) shall make an appropriate entry in the flight log showing the time of start of the de-icing process. If there is a subsequent departure delay, or further deterioration in weather conditions, they shall use this information together with that from the AEA tables above, to assess whether further de-icing is required.

NOTE

Protection period (HOLDOVER TIME) is measured from time of treatment start.

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8.2.4.5 Precautions During Usage

8.2.4.5.1 Ramp and Taxi Precautions

Icy conditions on the ramp / gate area mean that a push-back vehicle, if used, may not be able to develop sufficient traction to push the aeroplane with engines running. In this case, after completing checks down to engine start, have the aeroplane pushed back, to a position from which it can taxi forward, before starting engines. Be aware that idle thrust may be sufficient to move the aeroplane forward even though the parking brake is set.

Abnormal engine indications may occur during engine start, with oil pressure exceeding normal limits until oil temperature rises. This is acceptable provided the engine is operated at no more than idle thrust until indications are normal.

Be alert to the possibility of engine inlet ice build-up during taxi and ground holding operations. Ice can form in engine inlets at temperatures above 10°C with high humidity present during extensive ground holds with the engines at idle. If visible moisture is present with a temperature below that specified in the AFM / AOM, nacelle anti-ice systems shall be selected ON after each engine is started.

Taxi with great care since rutted areas cause steering problems. Neither aeroplane nor ground vehicles are capable of stopping quickly. When guide lamp installations are available, make use of them to align the aeroplane in the gate area. Snow may cover the normal taxi markings. Be alert to the possibility of foreign object damage due to high snow in ramp areas or along narrow taxiways. Anti-skid and body-gear steering systems (if fitted) should be disarmed and use of differential brake for turns avoided. Test braking and steering capabilities frequently. Maintain a greater distance than normal from other traffic to avoid jet-blast which could adversely affect anti-ice treatment and / or blow contaminants onto the aeroplane.

Engine thrust may need to be higher than normal to overcome the drag caused by slush or snow. Changes should be made slowly and carefully to avoid blowing equipment and / or contaminants into other aeroplane. At intervals during prolonged taxi and immediately before take-off, engines shall be run up to higher settings as specified in the AFM / AOM.

Flaps shall be kept UP when taxiing through slush or standing water, with pre-take-off checks delayed until they are able to be lowered and anti-skid selected ON.

Hold well clear of an active runway as a landing aeroplane is more likely to deviate from the centreline if braking conditions are marginal. Slush thrown up by landing aeroplane can be ingested by engines, cause airframe damage and degrade anti-ice treatment.

In the event of a longer than usual taxi time or ATC hold, and before committing themselves to take-off, the commander shall re-assess the ice-free state of the aeroplane. If any doubt exists they shall return to the ramp for a complete two-step procedure (i.e. de-icing to clear any new deposits and degraded residues of previous anti-ice treatment, before re-applying a fresh anti-ice coating). A double anti-ice coating shall be avoided as this adversely affects the flow-off properties of Type II fluids.

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8.2.4.5.2 Take-off

Take-off is prohibited if any of the following conditions exist:

- Snow, ice or frost deposits are adhering to the wings, control surfaces, engines or propellers of the aeroplane;
- Heavy falls of wet snow with ambient temperature around freezing point;
- Freezing rain or freezing drizzle;
- Prevailing weather conditions classified as 'Other' which include Heavy snow, snow pellets, ice-pellets or hail;
- The runway braking action is reported as 'poor' (braking coefficient is equal or less than 0.25) unless multiple braking actions are available and operations are in accordance with [8.1.2.4.5](#);
- Any tailwind is present if the runway is contaminated.

A check of the treated surfaces for contamination must be performed when the HoT has been exceeded or if any doubt exists regarding the continued effectiveness of the applied anti-icing treatment.

Flight crew shall observe appropriate performance limitations for take-off. Acceleration will be adversely affected by slush or standing water on the runway, which cause significantly greater drag. Flight crew shall be alert for conditions which could affect stopping and directional control when an abort is required.

Line up shall be accomplished carefully and flight crew shall ensure that the nose wheel is straight, as skidding of an offset nose wheel upsets directional control. Thrust shall be applied slowly to prevent asymmetry which would also affect directional control.

During the take-off roll, positive forward pressure on the nose wheel shall be maintained and the rudder shall be used for directional control as soon as it becomes effective.

8.2.4.5.3 Runway Condition

Information on runway condition will generally be disseminated in one of the following formats:

- i. Global Reporting Format (GRF) for Runway Surface Conditions; or
- ii. EUR-RODEX (European Regional OPMET Data Exchange); or
- iii. SNOWTAM.

Details describing these formats as well as the decode can be found in the Jeppesen Airway Manual as follows:

i and ii - **Publications / Any Region (e.g. Europe) / General Airway Manual / Meteorology** and then selecting the required format.

iii - **Publications / Any Region (e.g. Europe) / General Airway Manual / Tables and Codes / SNOWTAM Reference Information / Snowtam**

The Global Reporting Format (GRF) is currently in the process of being implemented to replace all previous systems, however the implementation date will vary by state and it is not compulsory so some states might choose not to implement it.

Known dates for implementation are as follows (the rest of the world is unknown)

- **Already implemented** - USA
- **12th August 2021** - EASA and Canada
- **1st October 2021** - Iceland
- **4th November 2021** - UK and ICAO

8.2.4.5.4 Rejected / Aborted Take-off

Directional control problems may be aggravated during an aborted take-off due to excessive anti-skid cycling and / or individual wheels skidding. The rudder shall be used as the primary steering aid and, to regain directional control, be prepared to release wheel brakes, cancel reverse thrust and reselect forward idle. Brakes should be re-applied when directional control is regained. Do not 'pump' the wheel brakes when the anti-skid system is ON.

8.2.4.5.5 Rotation Technique

Despite apparently effective anti-ice treatment, the aeroplane could become contaminated during the take-off roll. With contamination on the wings, the aeroplane will behave as if it were incorrectly trimmed in the nose-up attitude as the angle of attack is increased, requiring an abnormal push force to prevent entry into the stall region. It is critically important that the rotation be performed smoothly at a rate of 2-3 degrees per second and target pitch attitude should not be exceeded. This will ensure that any abnormal pitch-up tendency is detected early and the wing angle of attack does not become higher than normal.

If pitch-up or lateral instability is experienced after lift-off, use elevator, rudder and aileron to maintain the desired attitude. Apply maximum available thrust and use smooth continuous control inputs to avoid overcontrolling.

Do not allow further increase in pitch attitude until full lateral control has been regained.

Select wing and stabiliser anti-ice systems to ON as soon as practicable after lift-off.

8.2.5 Aeroplane Handling with Strong Wind on the Ground

With strong winds (including gusts) on the ground, the aeroplane shall be secured as follows.

8.2.5.1 Windspeeds Above 25 Knots

Chock forward and aft of all wheels.

8.2.5.2 Windspeeds Above 40 Knots

In addition to previous measures:

- Parking brake set;
- Catering and similar vehicles shall not be near the aeroplane.

8.2.5.3 Windspeeds Above 60 Knots

In addition to previous measures, fill fuel tanks to the maximum levels possible in accordance with the AFM, taking into consideration flights planned, number of passengers, and landing weight at next destination.

8.2.6 Reserved

8.2.7 Personal Safety Measures

To prevent accidents with vehicles or other movable equipment, high visibility vests shall be worn day and night when on the ramp.

Crew members are prohibited from using mobile phones on the aerodrome ramp.

8.3 Flight Procedures

Language

Standard phraseology as defined in ICAO DOC 4444 shall be used for all RTF communications, unless specific regional procedures apply.

All crew members shall communicate in English while performing their duties in company aeroplane during pre-flight and post-flight, all training and evaluation activities. This policy shall include all communications, whether related to the conduct of the flight or not, during the entire period that the aeroplane is crewed for the purpose of flight. The use of a language other than English while in the aeroplane is strictly prohibited, except where the Commander deems it necessary to exercise their authority.

Instrument Departure; Arrival and Approach Procedures

The instrument departure, arrival and approach procedures established by the State of the aerodrome shall be used.

Procedures other than the above may be used provided they have been approved by the State in which the aerodrome is located and they are specified in the OM.

The commander may accept an ATC clearance to deviate from the published departure and arrival route provided that obstacle clearance is observed and full account is taken of the operating conditions. The final approach shall be flown visually as per [Visual Approaches Under IFR](#) or in accordance with the established instrument approach procedures.

8.3.1 VFR-IFR Policy

VFR flights are not authorised as a means to depart or arrive at aerodromes, unless one or more of the following cases apply:

- there is no IFR departure, or IFR approach at ETD or ETA, respectively;
- execution of training flights for the purpose of training landings;
- authorisation for the VFR departure or arrival is specifically given in the NTA Aerodrome Briefing or OM-C Briefing.

The maximum speed during VFR flight is 200 KIAS, or 160 KIAS inside an aerodrome traffic circuit, unless local airspace regulations dictate a lower speed. If radar service is available, the best available radar service shall be used to the maximum extent possible.

When departing from a VFR airport, an IFR clearance should be requested before departure. If this is not possible, then departure under VFR is authorised if VMC can be maintained until an IFR clearance has been received. However, at no time will a flight depart from a VFR airport without an IFR flight plan filed.

A VFR chart shall be used to monitor the flight's progress and locate flight hazards, terrain, and ground features.

8.3.1.1 VFR Flight Day

For the purpose of VFR flight, day is considered to be between:

- Sunrise -30 and sunset +30 minutes in non-mountainous terrain;
- Sunrise -15 and sunset +15 minutes in mountainous terrain.

8.3.1.2 VFR Flight Night

The following requirements shall be met for VFR flight during night-time:

- For night arrival, approach, and landing, a minimum altitude / height equal to or higher than MSA, or MVA if under positive radar control, shall be maintained until intercepting a visual approach system glidepath.
- VFR night take-off and departure shall be in accordance with a published departure procedure or obstacle / terrain clearance procedure. The procedures shall be compliant with the take-off obstacle clearance as per PART-CAT.
- The Aerodrome MSA shall be 3,000' AAL or less, unless specific procedure for night VFR are documented in the NTA Aerodrome Briefing or OM-C Briefing.
- Fully functional TCAS and EGPWS.

8.3.1.3 Special VFR

No flight will be conducted under Special VFR.

8.3.1.4 VFR Weather Minima

VFR flights shall be conducted so that the aeroplane is flown in conditions of visibility and distance from clouds equal to or greater than:

Flight Visibility: 8 km.

Cloud Clearance: Horizontally - 1,500 m; Vertically - 1,000'.

Except when operating at aerodromes:

- Covered in the OM-C or NTA aerodrome briefing and for which a flight visibility and / or ceiling are defined; or
- Where the local airspace requires higher minima.

8.3.1.5 Visual Lookout and Task Allocation

When weather conditions permit, flight crew shall maintain a lookout to see and avoid other aircraft during all phases of flight regardless of flight rules.

During VFR flight, the PF should maintain a continuous visual scan for other traffic, obstacles, and terrain with momentary scans of aeroplane parameters, as required. The PM is primarily responsible, besides their normal duties, for navigation and advising the PF on courses, headings, altitudes, and airspeeds to fly, if required.

When operating in class E, F, or G airspace, ATC may have no knowledge of conflicting VFR traffic. Collision avoidance is the responsibility of the crew, even when under IFR.

8.3.1.6 Operation Within Class E Airspace

Operations in Class E airspace generate a unique threat due to the specific nature of the ATC environment. When operating in Class E airspace, collision avoidance with VFR traffic is the responsibility of the crew, even when under IFR. ATC may have no knowledge of conflicting VFR traffic. Separation is based on the 'see and avoid' principle.

During flight in Class E airspace, the following procedural items are recommended to increase situation awareness of the traffic environment:

- Locate Class E airspace during flight briefing and attribute airprox threat to VFR traffic;
- Brief the approach early to aid lookout;
- Make optimal use of automation;
- Request VFR traffic information and if necessary, delay the arrival or departure to deconflict;
- Reduce speed below 200 kts;
- Minimise visual approaches;
- Minimise the time in Class E airspace.

Refer to [Operations in Class E Airspace](#) for further details.

8.3.1.7 Use of Automation

During planned VFR flight the use of autopilot is highly recommended to allow the flight crew more spare capacity to scan for other traffic. For last minute traffic avoidance, the autopilot shall be disconnected.

To aid navigation, as a minimum, the following should be programmed into the FMS / FMC:

- VFR reporting points;
- Extended runway centreline;
- Runway threshold;
- A glide path appropriate to the runway to be used.

NOTE

FMS / FMC waypoints for this paragraph are for reference only and shall not be used to fly approaches based on pilot programmed waypoints.

8.3.1.8 VFR Transponder Settings

The transponder codes listed in OM-C for the appropriate region shall be set during VFR flight.

8.3.1.9 Application of IFR

The following provisions shall govern all operations, except for training flights under [VFR-IFR Policy](#), whether conducted in controlled or uncontrolled airspace and irrespective of weather conditions:

- An IFR flight plan shall be filed for every flight.
- An IFR flight plan shall only be cancelled when required by published procedures.
- Instructions to keep 'VMC on top' or '1000 feet on top' shall not be accepted.

Clearances subject to maintaining own separation and remaining in VMC shall only be requested or accepted in the following circumstances:

- Day only.
- Only for limited portions of climb, descent, or approach.
- Not accepting such a clearance would result in an undue fuel penalty and / or delay.
- Meteorological conditions will permit operation in uninterrupted VMC.
- Traffic information is provided by an appropriate ATS unit.
- The flightpath can be managed to ensure safe separation from other traffic.

NOTE

Safety of flight shall always take precedence over economic penalties suffered through delays or additional mileage.

In addition to the provisions above, crews shall:

- Within Class F airspace, elect to use the advisory service provided, complying with the same procedures as those applying to controlled flights, except:
 - The flight plan and changes thereto are not subject to a clearance (air traffic advisory service will only provide advice on the presence of essential traffic or suggest a possible course of action). However, it is assumed that a pilot will not effect a change in the current flight plan until they have notified the intended change to the appropriate air traffic services unit and, if practicable, have received acknowledgement or relevant advice;
 - It is upon the crew to decide whether or not to comply with the advice or suggestion received, and to inform the unit providing air traffic advisory service, without delay, of their decision;
 - Air-ground contact should be made with the air traffic services unit designated to provide air traffic advisory service within the advisory airspace or respective portion.
 - When a flight is operating or about to operate within a control area—continuing eventually into an advisory area or along an advisory route—a clearance may be issued for the entire route. However, this clearance, or any revisions to it, only applies to the portions of the flight conducted within control areas and control zones. Advice or suggestions will be provided as needed for the remaining sections of the route.
- Within Class G airspace, apply IFR procedures as follows:
 - Maintain communication with the appropriate ATS unit.
 - Make position reports at specified reporting points or time intervals.
 - Request traffic information prior to any level change.
 - Report changes to the filed flight plan and request traffic information in relation to such changes.

8.3.1.10 ATC-Clearances

CAUTION

Any doubt by either crew member about the meaning of a clearance shall be clarified with ATC.

No flight requiring an ATC departure clearance shall commence take-off without such clearance.

When the initial ATC departure clearance or significant re-clearance is received, it shall be written down and read back by the pilot handling ATC, and then acknowledged by the other pilot.

When a clearance to take-off or land is received, the PM shall select the landing light to ON. The switch position acts as visual confirmation of receipt of the clearance.

Both Pilots should monitor the radio frequency when the initial ATC clearance is received. For operational reasons this may not always be possible. In such cases, the pilot receiving the clearance will fully brief the other pilot on the clearance received. If there are any doubts regarding the clearance, it shall be clarified with ATC.

When a pilot needs to leave the cockpit, the remaining pilot shall avoid any distractions. Communications shall be limited to those with ATC. Upon returning to the cockpit, the pilot shall be informed of any changes to the ATC clearance and frequency.

When a similar call sign is in use on the same ATC frequency, it shall be pointed out to ATC. Extra vigilance is required during radio transmissions.

ATC shall also be advised of blocked or simultaneous transmission.

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8.3.2 Navigation Procedures

8.3.2.1 FMS / FMC Data Entry Verification

After a crew member has entered the departure, flight plan routing and performance data into the FMS / FMC the other crew member shall verify the inputs before taxi. Refer to [Autopilot, Autothrottle, and Automation Policy / Level-Off Awareness](#).

8.3.2.2 Use of Navigational Aids

ADDITIONAL GUIDANCE

To ensure accurate navigation and monitoring of flight progress during all phases of flight, including flights being radar vectored, or operating with visual reference to the ground, full use shall be made of radio navigation and landing aids during all weather conditions.

8.3.2.2.1 Failure of Navigational Aids

In case of failure of any navigation aid or airborne radio navigational equipment essential for safe and accurate navigation, or for complying with ATS procedures, the Commander shall notify the appropriate ATS unit, making full use of any radar assistance available.

8.3.2.2.2 Unreliable / On Test

Radio navigation aids promulgated as being 'unreliable' or 'on test' shall not be used.

8.3.2.3 Final Navigation Setup

As a general rule, the 'Final Navigation Setup' is defined as both pilots having the setup of the aid to be used for the final approach completed (in the case of conventional aids this means they are tuned, identified, and track set). The latest point for such 'Final Navigation Setup' is shortly prior to the final approach segment.

Prior to intercepting the final approach, the PM shall use available resources to monitor the aircraft's position relative to the final approach; this may include but is not limited to ADF, VOR, or depiction of the final approach track by the FMS / avionics.

8.3.2.4 Display of Lights

Navigation lights - shall be displayed when in flight, whenever an APU is running, or operating on the manoeuvring area of an aerodrome at any time between sunset and sunrise, and during any other period when deemed necessary by the Commander or prescribed by the appropriate authority.

Anti-collision / Recognition / Beacon lights - shall be operated, irrespective of the period of the day and meteorological conditions, at any time any aeroplane engine is in operation or while being towed.

Strobe lights - shall be switched ON before entering the runway for take-off, switched OFF when leaving the runway after landing, switched ON prior to crossing a runway during taxi operations, and switched OFF once clear of that runway.

Taxi and / or landing lights - Taxi and / or Landing Lights shall be selected ON:

- At night to enable the crew to distinguish objects and obstructions while taxiing, taking off, and landing.
- When cleared for take-off and landing.

Tail / Logo Lights - Shall be used during night operation only and shall be turned on below 10.000 ft AAL.

The Commander may choose to switch off or reduce the intensity of any lights if they do or are likely to:

- adversely affect the satisfactory performance of duties; or
- subject an outside observer to harmful dazzle.

Towing at night - If due to aircraft design it is not possible to comply with the display of lights depicted above during towing at night (e.g. due to battery capacity with APU shut down), refer to the relevant OM-B for type specific procedures.

8.3.2.5 MNPS

SPA.MNPS.100 MNPS operations

Aircraft shall only be operated in designated minimum navigation performance specifications (MNPS) airspace in accordance with regional supplementary procedures, where minimum navigation performance specifications are established, if the operator has been granted an approval by the competent authority to conduct such operations.

NTA currently holds approval for:

- North Atlantic High-Level Airspace (NAT HLA)

8.3.2.5.1 NAT HLA

Detailed requirements, procedures, and information are provided in the 'North Atlantic Operations and Airspace Manual (NAT Doc 007)', which can be found in the fleet Onboard Docs folders in the DMS.

i. Equipment, operating limitations, and MEL entries

The equipment necessary to satisfy the navigation accuracy required for unrestricted operations in NAT HLA is described in NAT Doc 007, section 1.3; this includes but is not limited to the following:

- 2 Long Range Navigation Systems (LRNS), as per NAT Doc 007, section 1.3, routes permitted to be used by aircraft not equipped with 2 LRNS are described in NAT Doc 007, section 1.4.
- Functioning datalink equipment, as per NAT Doc 007, section 1.8. This section also includes the exceptions to this requirement;
- Altimetry and height-keeping systems that meet RVSM Minimum Aircraft System Performance Specifications. Exceptions to this requirement are detailed in NAT Doc 007, section 1.6.

The MEL will either directly reflect the NAT HLA equipment requirements, defining limits on NAT HLA operations when certain aircraft equipment is inoperative, or state, 'May be inoperative provided operating regulations do not require its use'. In the latter case, NAT Doc 007, chapter 1 must be consulted to establish whether the planned operation is permitted with the given equipment failure (equipment requirement can vary by route and level).

Flight crew should be aware that some equipment may indirectly affect NAT HLA operations, for example, equipment required for RVSM, including but not limited to autopilot, flight director and transponder systems.

Airspace and tracks designated for PBCS operations, as described in NAT Doc 007 section 1.10, require separate authorisation and contain specific operational limitations and requirements. The PBCS requirements, including equipment and MEL entries, are detailed in OM-A 8.3.34.1. Only PBCS-authorised aircraft and qualified crew shall operate on PBCS routes.

ii. Flight Crew Composition and Experience

There are no specific requirements associated with NAT HLA operations regarding crew composition.

Flight crew shall be appropriately trained and qualified to perform such operations as per OM-D; applicable qualifications are:

- **NAT HLA** - permitting unrestricted NAT HLA operation;
- **NAT HLA Light** - permitting operation only on those routes, route structures, and transition areas within and adjacent to the NAT HLA specifically detailed in NAT Doc 007, Chapter 3.

Additional details on route and area competence qualification requirements are laid in OM-A, chapters 5.4.1.2.2 to 5.4.1.2.6.

iv. Contingency procedures

A description of the contingency procedures to be used in NAT HLA operations is included in NAT Doc 007, specifically:

- a. HF communications failure: sections 6.6;
- b. Communication of equipment failures: section 9.2;
- c. Navigation system degradation or failure: sections 12.1 and 12.2;
- d. In-flight contingencies: sections 13.2 to 13.6.

Correct application of these procedures will be covered during crew members' initial and recurrent NAT HLA training.

v. Monitoring and incident reporting

Routine performance monitoring must be performed during NTA HLA operations in accordance with NTA Doc 007 section 1.9 and chapter 11.

Lateral deviations of 10 NM or more must be reported (OM-A 11.8 - 4.(g)) and further investigated. Analysis of incidents may use the data collected in routine performance monitoring and data from flight recorders if used under the conditions in OM-A 11.10.

Complementary information related to ATC procedures in NAT HLA operations can be found in the Air Traffic Data - Atlantic section of the Jeppesen Europe Airway Manual, available in the Portable EFB.

State-specific rules and procedures can be found in the State rules and procedures - Atlantic section of the Jeppesen Europe Airway Manual, available in the Portable EFB.

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8.3.2.6 Performance-Based Navigation (PBN)

CAT.OP.MPA.126

Operations in designated airspace, on routes or in accordance with procedures where PBN specifications are established, have specific requirements that must be met. The PBN Navigation specifications for which NTA meets the applicable requirements are listed in [Approvals by Aeroplane Type](#). Instructions for meeting these requirements are described in this section.

All FCM shall each hold a current PBN endorsement to the instrument rating, trained, and qualified in accordance with the OM-D.

8.3.2.6.1 Required Navigation Performance

RNP is the navigation specification required under PBN which permits the operation of an aircraft along a precise flight path with a high level of accuracy and the ability to determine aircraft position with both accuracy and integrity. RNP refers to the level of performance required for a specific procedure or a specific airspace block.

[ADDITIONAL GUIDANCE](#) on RNP specifications.

A. Flight Preparation for PBN

In addition to the requirements shown in [Flight Preparation Instructions](#), preparation for a flight involving PBN operations shall also include:

- a. Checking the procedures, routes and airspace planned to be used during the flight to establish the required PBN navigation specification(s) for the flight.
- b. Checking [Appendix 2 to Chapter 8](#) - to confirm the aircraft is authorised by NTA to operate to the required navigation specification(s) and checking:
 - i. Aircraft Technical Log to ensure no defects have been deferred which might affect the aircraft's ability to carry out the required PBN operations; applicable restrictions will be detailed in the MEL.
 - ii. Feasibility notes to ensure that applicable procedures, runways, or waypoints have not been excluded from the navigation database.
 - iii. NOTAMs to establish whether navigation aids critical to the operations of the intended PBN procedure are available (e.g. conventional NAVAIDS required for a missed approach or sensor requirements, RAIM prediction, GPS jamming, etc.).
 - iv. Any PBN procedures critical to the flight are available and selectable in the navigation database.
- c. Checking that RAIM will be available for any portion of the flight for which RAIM is required.
 - [Additional Guidance](#) for the RAIM function;
 - RAIM availability prediction (RAIM Check) is normally provided by JetPlan when the OFP is run. If there is no indication on the OFP that the RAIM check has failed, this indicates a successful check for the entire route;
 - In case the RAIM check is shown as fail on the OFP, refer to [PBN Operation with RAIM Fail on OFP](#).

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- d. Checking that the navigation database is current and will remain current for the duration of the flight - if the database validity has expired or will expire during the flight then refer to [PBN Operations with Database out of Currency](#).
- e. Following navigation system initialisation, ensuring that the aircraft position is correct in accordance with OM-B / AFM procedures.
- f. Checking that the active flight plan in the navigation system accurately represents the OFP routing (e.g. waypoint sequence, reasonableness of track angles and distances, etc.).

NOTE

Pilots may notice a slight difference between the navigation information portrayed on the chart and their primary navigation display. For all phases of flight, differences of 3 degrees or less may result from the equipment manufacturer's application of magnetic variation and are operationally acceptable.

B. PBN Operating Procedures

- When operating in PBN airspace and on PBN routes, the aircraft shall be operated in accordance with relevant navigation specifications and limitations in the AFM. Use of lateral / vertical deviation indicators, RNP selections and flight guidance modes for the applicable Navigation Specification shall be as per the OM-B / AFM. The flight crew should maintain route / procedure centrelines unless: Performing offset / contingency procedures permitted by that airspace;
- Authorised to deviate by air traffic control (ATC); or
- Under emergency conditions.

8.3.2.6.2 PBN SID/STAR

ADDITIONAL GUIDANCE

A. PBN SID/STAR Preparation

When accepting a clearance for a PBN SID or STAR:

- a. The navigation specification, sensor, and NAVAID requirements shown on the SID/STAR chart must be authorised and available according to the checks made under [Required Navigation Performance](#).
- b. The SID/STAR must be retrievable by route name from the onboard navigation database; it is not permitted to manually insert the waypoint sequence for a PBN SID/STAR.

NOTE

An exception is that RNAV-5 (B-RNAV) PBN SID/STARs may be entered manually.

- c. After selecting the cleared SID/STAR in the navigation system, pilots shall crosscheck the SID/STAR for waypoint sequence, reasonableness of track angles and distances, any altitude or speed constraints, and, where possible, which waypoints are flyby and which are fly-over by comparing charts or other applicable resources with the navigation system textual display and the aircraft map display, if applicable.
- d. User-defined data may be entered and used for waypoint altitude / speed constraints on a procedure where said constraints are not included in the navigation database coding.

B. PBN SID/STAR Operating Procedures

- a. Where GNSS is used, the signal shall be acquired before the take-off roll commences.
- b. At the start of the take-off roll, prior to commencing a take-off, the indicated aircraft position shall be confirmed to be consistent with the actual aircraft position.
- c. If ATC issues a heading assignment taking the aircraft off a route, the pilot should not modify the flight plan in the navigation system until a clearance is received to rejoin the route or the controller confirms a new route clearance.
- d. The route may be modified through the insertion or deletion of specific waypoints in response to ATC clearances. The creation and insertion of new waypoints by manual entry (latitude and longitude or radial and bearing) is not permitted. Additionally, pilots must not change any RNAV SID/STAR database waypoint type from a fly-by to a fly-over or vice versa.
- e. Any published altitude and speed constraints shall be complied with unless exempted by ATC.
- f. Any minimum obstacle / terrain clearance altitudes shall be corrected when required by and in accordance with [Checking of Terrain Clearance - Altitude Corrections](#).

C. PBN SID/STAR Contingency Procedures

See [PBN Contingency Procedures](#).

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8.3.2.6.3 PBN En-Route

ADDITIONAL GUIDANCE

A. PBN En-route preparation

When accepting a clearance into airspace or onto a route requiring PBN operations:

- a. The navigation specification, sensor, and NAVAID requirements for that airspace / route must be authorised and available according to the checks made under [Flight Preparation for PBN](#).
- b. The route must be entered into the navigation system respecting the following limitations according to the applicable navigation specification:
 - i. RNAV-10 (RNP-10), RNAV-5, RNP-4: Pilots may make manual entries using waypoints from the database and create new waypoints by manual entry of latitude and longitude or radial / distance values.

CAUTION

RNP-4: Pilots should not modify waypoints that have been retrieved from the database.

- ii. RNAV-1, RNAV-2, and RNP-2: Whenever possible, routes in the en-route domain should be extracted from the database in their entirety, rather than loading individual waypoints from the database into the flight plan. However, it is permitted to select and insert individual, named fixes / waypoints from the navigation database, provided all fixes along the published route to be flown are inserted. Moreover, the route may subsequently be modified through the insertion or deletion of specific waypoints in response to ATC clearances. Manually created waypoints shall only be used as follows:

CAUTION

RNAV-1 and RNAV-2: The creation of new waypoints by manual entry of latitude and longitude or radial / distance values is not permitted.

- RNP-2: On flexible route structures, creation of new waypoints by manual entry of latitude and longitude is permitted.
- c. After entering the route in the navigation system, pilots shall cross-check the route for waypoint sequence, reasonableness of track angles and distances, any altitude or speed constraints, and, where possible, which waypoints are fly-by and which are fly-over by comparing charts or other applicable resources with the navigation system textual display and the aircraft map display, if applicable.

B. PBN En-route Contingency Procedures

See [PBN Contingency Procedures](#).

8.3.2.6.4 PBN Approach (RNP APCH)

ADDITIONAL GUIDANCE

A. PBN Approach Preparation

When accepting a clearance for a PBN approach:

- a. The navigation specification, sensor, and NAVAID requirements for that approach must be authorised and available according to the checks made under [Flight Preparation for PBN](#).

Navigation specifications / functionalities applicable to PBN Approaches:

- RNP-APCH:
 - LNAV Minima;
 - LNAV/VNAV Minima;
 - LPV Minima.

CAUTION

The following RNP approaches are not authorised for any NTA aircraft:

- LP Minima
- RNP AR APCH
- RF (Radius to Fix)

Sensors normally applicable to RNP APCH:

- GNSS (GPS);
- DME/DME;
- SBAS (e.g. EGNOS or WAAS).

Conventional navigation aids may also be required or require excluding for certain segments of the approach.

The chart will state, if applicable, which sensors (or combination) and / or conventional navigation aids are required, and which are not permitted to be used. OM-B / AFM procedures shall be complied with to meet these requirements.

- b. The approach shall be retrievable by route name from the onboard navigation database;
 - It is not permitted to manually insert the waypoint sequence for a PBN approach that is not available in the navigation database.
 - It is not permitted to insert nor modify waypoints by manual entry into a procedure that has been retrieved from the database.
 - The lateral and vertical definition of the flight path between the FAF and the missed approach point (MAPt) retrieved from the database shall not be revised by the flight crew.

- User-defined data may be entered and used for waypoint altitude / speed constraints on a procedure where said constraints are not included in the navigation database coding.
- c. Prior to commencing the approach operation (before the IAF), the flight crew should verify the correctness of the loaded procedure by comparison with the appropriate approach charts. This check should include:
- the waypoint sequence;
 - reasonableness of the tracks and distances of the approach legs and the accuracy of the inbound course; and
 - the vertical path angle, if applicable.

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B. PBN Approach Operating Procedures

1. Vectoring and Positioning

- a. ATC tactical interventions in the terminal area may include radar headings, 'direct to' clearances which bypass the initial legs of an approach procedure, interceptions of an initial or intermediate segment of an approach procedure, or the insertion of additional waypoints loaded from the database.
- b. 'Direct to' clearances may be accepted to the IF provided that it is clear to the flight crew that the aircraft will be established on the final approach track at least 2 NM before the FAF.
- c. 'Direct to' clearance to the FAF shall not be accepted. Modifying the procedure to intercept the final approach track prior to the FAF should be acceptable for radarvectored arrivals or otherwise only with ATC approval and subject to AFM limitation.
- d. The final approach trajectory should be intercepted no later than the FAF in order for the aircraft to be correctly established on the final approach track before starting the descent (to ensure terrain and obstacle clearance).

2. Altimeter Settings

BARO VNAV (LNAV/VNAV minima)

As per [Altimeter Setting Procedures](#) except:

- a. BARO VNAV approaches using QFE are prohibited.
- b. A current local altimeter setting source shall be used - a remote or regional altimeter setting source shall not be used.
- c. A cross-check of the primary altimeters shall be made prior to the FAF and they shall not differ more than 100 ft.

Non-precision PBN approach without vertical guidance (LNAV minima)

As per [Altimeter Setting Procedures](#).

NOTE

QFE approaches are permitted.

SBAS PBN Approach (LPV minima)

As per [Altimeter Setting Procedures](#).

NOTE

AFM limitations for conducting LPV approaches when using QFE shall be observed.

3. Cold Temperature Corrections

[ADDITIONAL GUIDANCE](#)

For all PBN approach types, when outside of the final approach segment cold temperature corrections shall be applied, in accordance with [Checking of Terrain Clearance - Altitude Corrections](#).

The final approach segment is defined as the segment including all points between the FAP/FAP and the DA (not including either).

When inside the final approach segment, the following shall apply:

BARO VNAV (Charted LNAV/VNAV minima)	WITHIN PROMULGATED TEMPERATURE LIMITS i. Do not apply temperature correction to the FAF altitude. ii. Temperature corrections to altitudes in the final approach segment are not required; however, they may be applied if the aircraft uses an approved and available temperature compensation system. iii. Corrections shall be applied to the DA if the aerodrome is 0°C or colder.
	OUTSIDE PROMULGATED TEMPERATURE LIMITS i. The approach can be flown ONLY if the aircraft uses an approved and available temperature compensation system. Operating outside promulgated limits without a temperature compensation system is prohibited. ii. Corrections shall be applied to the DA if the aerodrome is 0°C or colder.
	NOTE The flight crew should consider the effect of temperature on terrain and obstacle clearance when outside the final approach segment. Only the final approach segment is protected by the promulgated aerodrome temperature limits. This protection does not encompass the DA.
Non-precision PBN Approach without vertical guidance (Charted LNAV minima)	Flight crew shall make cold temperature corrections to all altitudes in accordance with 8.3.3.6 .
SBAS PBN Approach (Charted LPV minima)	Flight crew shall make cold temperature corrections to all altitudes in accordance with 8.3.3.6 .

C. PBN Approach Contingency

Unless the flight crew has sufficient visual reference to continue the approach operation to a safe landing, an RNP APCH operation shall be discontinued if:

- a. A navigation system failure is annunciated (e.g. warning flag). See note below.
- b. Sustained lateral deviations exceed $\frac{1}{2}$ the cross track error associated with a procedure.
- c. Vertical deviations exceed the tolerances for an approach. During 3D approaches, when at or below 1000 ft above aerodrome level, deviations above or below the vertical path should not exceed 75 ft at any time, or half scale deflection where angular deviation is indicated.
- d. There is a loss of the onboard monitoring and alerting system.

NOTE

Where vertical guidance is lost while the aeroplane is still above 1000 ft AAL, the flight crew may decide to continue the approach to LNAV minima when supported by the system.

The missed approach shall be flown in accordance with the published procedure. Use of the RNAV system during the missed approach is acceptable provided:

- a. The RNAV system is operational (e.g. no loss of function, no RAIM alert, no failure indication, etc.).
- b. The whole procedure (including the missed approach) is loaded from the navigation database.

8.3.2.6.5 PBN Contingency Procedures

A. In-flight Contingency Procedures

- a. If the aircraft suffers any loss of RNAV capability, or is unable to comply with the requirements of a RNAV route, ATC shall be notified as soon as possible, along with a proposed course of action. The loss of RNAV capability includes any failure or event causing the aircraft to no longer satisfy the RNAV requirements of the route (e.g. loss of RAIM when relying on GPS without SBAS, notification by the navigation system of degraded performance, etc.).
- b. In the event of communications failure, the pilot should continue with the RNAV route in accordance with the established lost communications procedures for the given aerodrome or state.

B. PBN Operations with Database out of Currency

1. Operations with an expired database

An expired database may only be used if the following conditions are satisfied:

- a. The crew shall confirm that the parts of the database which are intended to be used during the flight and any contingencies that are reasonable to expect are not changed in the current version;
- b. Any NOTAMs associated with the navigational data are taken into account;
- c. Maps and charts corresponding to those parts of the flight are current and have not been amended since the last cycle;
- d. Any MEL limitations are observed; and
- e. The database has expired by no more than 28 days.

2. AIRAC cycle change during flight

If the AIRAC cycle is due to change during the flight and part of the flight will be conducted with an expired database, then the procedures detailed above shall apply.

C. PBN Operation with RAIM Fail on OFP

For aeroplanes that have a functioning SBAS receiver operating within an SBAS coverage area, a RAIM check / RAIM is not required. If SBAS (EGNOS, WAAS, etc.) is lost or unavailable, a RAIM check must be performed and RAIM shall be available for the required PBN operation.

If predictions indicate that the maximum allowable Fault Detection and Exclusion (FDE) outage as per the table below will be exceeded, the operation shall be rescheduled to a time when the required FDE is available.

Alternatively, the flight may be operated on the planned schedule using an alternative navigation specification for which the required FDE outage is not exceeded.

RNAV 10	34 Minutes
RNAV 5	5 Minutes
RNAV 2	5 Minutes
RNAV 1	5 Minutes
RNP 4	25 Minutes
RNP 2	5 Minutes
RNP 1	5 Minutes
RNP APCH	5 Minutes for any part of the RNP Approach Procedure

8.3.2.7 RVSM

(SPA.RVSM.100)

RVSM operations are conducted in designated airspace where a reduced vertical separation minimum of 300 m (1,000 ft) applies between FL 290 and FL 410, inclusive. Approval to perform RVSM operations is detailed in the NTA AOC.

Details of the regions in which RVSM has been implemented and associated procedures are contained in the Jeppesen airways manual and Jepp FD Pro en-route chart notes.

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8.3.2.7.1 Pre-flight**ALTIMETER SETTING PROCEDURES**

RVSM pre-flight procedures require that the readings of both primary altimeters shall agree with the altitude of the apron within a 75 ft (23 m) tolerance, or AFM/AOM limitations if more restrictive.

Flight into RVSM airspace imposes additional restrictions in accordance with the MEL.

Before take-off, equipment required for flight in RVSM airspace shall be operative, and any indications of malfunction shall be resolved.

The following equipment shall be operating normally at entry into RVSM airspace:

- Two primary altitude measurement systems.
- One automatic altitude-control system.
- One altitude-alerting device.
- One ATC Transponder with altitude reporting that can be connected to the altitude measurement system in use for altitude keeping.

8.3.2.7.2 Prior to Entering RVSM Airspace

Should any of the required equipment fail prior to the aircraft entering RVSM airspace, the pilot should request a new clearance to avoid entering this airspace.

Flight into RVSM airspace requires the two primary altimeter systems to be crosschecked and agree within ± 200 ft. Failure to meet this condition will require that the altimetry system be reported as defective and ATC notified.

While in level flight and just prior to entering oceanic (NAT HLA or other) or remote continental airspace, the altimeter readings, including standby instruments, shall be recorded on the OFP, for use in any possible contingency situations.

8.3.2.7.3 In-flight Procedures

All the required equipment shall be monitored to ensure satisfactory operation before and within RVSM airspace.

The automatic altitude control system shall be operative and engaged during level cruise, except when circumstances such as the need to re-trim the aeroplane or turbulence require disengagement. Adherence to cruise altitude shall be done by reference to one of the two primary altimeters. In normal operations, the altimetry system being used to control the aeroplane shall be selected for the input to the altitude reporting transponder transmitting information to ATC.

Crosschecks of the primary altimeters shall be made at intervals of approximately one hour and they shall agree within 200 ft of each other. The usual scan of FCC instruments should suffice for altimeter cross-checking on most flights.

In level cruise it is essential that the aeroplane is flown at the cleared flight level. The aeroplane shall not intentionally depart from cleared flight level without a positive clearance from ATC unless the crew are conducting contingency or emergency manoeuvres.

When changing levels, the aeroplane shall not overshoot or undershoot the cleared flight level by more than 150 ft. The altitude capture feature shall be used for the level off.

If notified by ATC of a deviation from an assigned altitude exceeding ± 300 ft then action shall be taken to return to the cleared flight level as quickly as possible.

Any equipment failure which might adversely affect the aircraft's height-keeping performance should be advised to ATC immediately, particularly if the crew believes the aircraft may be non-compliant with RVSM requirements. ATC may require non-compliant aircraft to descend out of designated RVSM airspace.

Crews shall notify ATC when encountering greater than moderate turbulence.

If unable to notify ATC and obtain an ATC clearance prior to deviating from the cleared flight level, flight crew shall follow any established contingency procedures for the region of operation and obtain ATC clearance as soon as possible.

8.3.2.7.4 Post-flight Procedures

Any malfunctions in height-keeping systems shall be recorded in the aeroplane technical log and reported via an NTA safety report.

[NTA Safety Reporting Requirement for RVSM](#)

8.3.2.7.5 Phraseology

Circumstance	Phraseology
ATC wish to know RVSM status of flight.	Confirm RVSM Approved.
Pilot indication that flight is RVSM approved.	Affirm RVSM.
Pilot indication that flight is non-RVSM approved.	Negative RVSM.
Pilot of State aeroplane indicating that flight is non-RVSM approved.	Negative RVSM. State Aeroplane.
ATC denial of clearance into RVSM Airspace.	Unable Clearance into RVSM Airspace. Maintain (or Descend to, or Climb to) FL...
Pilot reporting severe turbulence / weather affecting ability to maintain RVSM height-keeping requirements.	Unable RVSM due Turbulence.
Pilot reporting equipment degraded below RVSM requirements.	Unable RVSM due Equipment.
ATC requesting pilot to report when able to resume RVSM.	Report able to resume RVSM.
Pilot ready to resume RVSM after equipment / weather contingency.	Ready to resume RVSM.

8.3.2.8 Areas of Magnetic Unreliability (AMU)

Operations in Areas of Magnetic Unreliability are authorised in accordance with the Aeroplane type specific AFM, AOM and respective OM-B limitations and procedures.

8.3.2.9 Polar Navigation

Operations in the Polar Region are prohibited except for operations to aerodromes specifically approved for 'High Latitude' operations. Routing to such aerodromes shall be arranged to minimise the enroute time spent in the Polar Region (i.e. entry / exit of the Polar region shall be as close as possible to the intended landing aerodrome).

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8.3.2.10 In-flight Re-planning

Also refer to [Diversion of Flight En-route](#).

For in-flight re-planning, flight crew shall:

- Ensure that the fuel requirements will be met - refer to [Policy and Procedures for In-flight Fuel Management](#);
- Ensure that the available navigational aids for the re-planned route and / or the replanned destination as well as the airborne equipment will be sufficient and satisfactory for a safe conclusion of the flight;
- Ensure that the applicable meteorological and aerodrome operating minima will be met;
- Ensure that the applicable performance requirements will be met;
- Contact the OCC in-flight, if time permits, or immediately after landing, once any urgent matters have been addressed.

8.3.2.11 Procedures in the Event of System Degradation

For aeroplane type specific navigation system degradation, refer to OM-B / AFM / AOM.

For regional contingency procedures, refer to specific contingency procedures in OM-A, OM-C and / or Jeppesen Airway Manual.

8.3.3 Altimeter Setting Procedures

The following procedures describe the methods to be used to ensure adequate vertical separation between aeroplanes and adequate terrain clearance during all phases of a flight.

If the AAD does not allow selection of the exact altitude / height depicted on a chart, it shall be set to the next settable altitude / height above the one depicted.

Altimeter settings to be used are:

Altimeter Setting	Reference Data	Altimeter Indication
Standard	1013.2 hPa / 29.92 in	FLIGHT LEVEL
QNH	QNH Pressure Level (approx. Mean Sea Level)	ALTITUDE
QFE	QFE Reference Level	HEIGHT above QFE Reference Level

8.3.3.1 QNH - QFE Policy

QNH shall be used for operations at locations where QNH is used as the primary altimeter setting.

QFE shall be used for operations at locations where QFE is used as the primary altimeter setting, except for those fleets listed below.

For the CE-680A and EMB-505 types only, QNH shall be used for operations at locations where the primary altimeter setting is QFE in accordance with [QFE Operations](#) and under the following conditions:

- Jeppesen charts are available and have the QFE-QNH conversion tables / data published or a NetJets issued Alt / Height conversion chart (AHCC) is available.
- If TCAS or EGPWS is inoperative, QNH operations in QFE environment are not permitted.

8.3.3.2 Normal Procedures

Ground

Before leaving the ramp all altimeters shall be set to the reported QNH of the aerodrome, except for QFE operations as per [QFE Operations](#).

To ensure that the tolerances specified in the respective OM-B are not exceeded, the altitude indicated on the altimeters shall be compared with each other and checked against the nearest known elevation on the aerodrome.

Before take-off, each time a new altimeter pressure setting is reported by the aerodrome, the above procedure shall be followed.

Airborne

Climb: When above the final altitude / height restriction on a SID, or contained in an ATC clearance, and cleared to a flight level, both primary altimeters shall be set to standard and compared.

Unless prescribed by AFM / OM-B procedures, the standby altimeter shall remain set to QNH of the departure aerodrome until passing 10,000 ft, transition altitude, or MSA, whichever is highest, when standard shall be set.

Cruise: At level-off and regularly during the cruise (at least every hour), all altimeters shall be checked to verify compliance with the RVSM requirements and to assess the accuracy of the standby altimeter as a backup system.

The appropriate transition level / altitude (aeroplane specific) shall be inserted in the FMS during arrival preparation, to provide warning against an altimeter setting error below transition level.

Descent: Destination QNH shall be obtained by ATIS or ATC, and both pilots shall compare this figure with an appropriate secondary source (e.g. electronic information, release message METAR, etc.). Unless prescribed by AFM/OM-B procedures, the QNH shall then be set on the standby altimeter by the PM.

When cleared to descend to an altitude, the QNH given by ATC issuing the descent clearance shall be compared with the value set on the standby altimeter for gross error detection and set on the primary altimeters using the procedure prescribed below.

For setting of standby altimeter for QFE operations, refer to [8.3.3.4](#).

When approaching MEA or MSA, each time a new altitude is selected, the altimeters' settings and minimum safe altitude shall be checked to provide early detection of altimeter setting errors and protection against an unsafe descent clearance.

SITUATION	PF	PM
ALTIMETER SETTING TO STD PF: Verifies conditions for setting altimeters to STD are met and sets STD. PM: Verifies conditions for setting altimeters to STD are met, sets STD and crosschecks the altimeter pressure settings. PF: Verifies both altimeters are set STD.	'SET STANDARD'	'STANDARD SET AND CROSSCHECKED'
ALTIMETER SETTING TO QNH/QFE PF: Verifies conditions for setting altimeters to QNH/QFE are met. PM: Verifies conditions for setting altimeters to QNH/QFE are met. Both FCM set the QNH/QFE value. PM: Crosschecks the altimeter settings (with the standby altimeter, if applicable). PF: Silently verifies both altimeters are set to correct QNH/QFE.	'SET QNH/QFE'	'QNH/QFE XXXX SET AND CROSSCHECKED'

8.3.3.3 QNH Operations in QFE Environment

Restrictions

The following restrictions apply to QNH operations in QFE environment:

- Jeppesen charts with the QFE to QNH conversions shall be available.
- Manual calculations of QFE heights to QNH Altitudes are prohibited, only converted QNH altitudes that are displayed on the Jeppesen charts or AHCC shall be flown. Flight crew shall not accept clearances to QFE heights that are not on the Jeppesen charts or AHCC.
- AHCC Charts shall only be used converting ATC given heights; they shall not be used for establishing approach minima.
- Altitudes shall be flown in feet (depicted on Jeppesen charts or AHCC).
- EGPWS and TCAS shall be operative for dispatch, except as detailed in the '[QNH - QFE Policy](#)' at the beginning of this section.

Normal Procedures

Normal Crew briefings shall include a discussion of these procedures.

QNH altimeter setting and cross-checking procedure

If the QFE/QNH altimeter setting is given in Millimetres Mercury, flight crew shall request hPa. Calculation of the hPa is also available in the conversion section ([Conversions](#)).

Departure

Prior to engine start, flight crew shall obtain the QNH from the ATIS, from ATC, or by converting the provided QFE to QNH (see conversions section [Conversions](#)).

The QNH shall be set on the standby Altimeter.

If there are any doubts as to the QNH altimeter setting, flight crew shall seek correct QNH setting from ATC and cross-check again.

All altimeters shall be cross-checked for correct setting and reading.

Arrival

Prior to descent or as soon as possible after the beginning of the descent, flight crew shall obtain the QNH from the ATIS, from ATC, or by converting the provided QFE to QNH (see conversions section [Conversions](#)).

The QNH shall be set on the standby Altimeter.

When cleared to descend below the transition level, flight crew shall request QNH from ATC and cross-check it with the previously obtained QNH.

If there are any doubts as to the QNH altimeter setting, flight crew shall seek correct QNH setting from ATC and cross-check again.

All altimeters shall be cross-checked for correct setting and reading.

ATC will normally provide clearances to QFE height in Metres. Jeppesen charts will depict the converted altitude in feet adjacent to the QFE height.

NOTE

There are no conversions to altitudes in metres and thus operations in metres are prohibited.

Example of [Jeppesen Chart With Conversions](#).

8.3.3.4 QFE Operations

QFE altimetry is normally used in conjunction with the metric system (heights are given in metres instead of feet). QFE uses the elevation of the airport as the zero reference line instead of MSL. As a result, the altimeters on the ground read ZERO and not the field elevation.

Limitations

1. QFE operations are only authorised for aeroplane types that are permitted to do so. Refer to OM-B.
2. EGPWS (TAWS)

Flight crew shall refer to the OM-B / AFM for procedures and limitations on the use of EGPWS.

Generally, if the aeroplane uses geometric altitude then automatic correction for altimeter settings is used.

NOTE

The Terrain display will show elevations in feet referred to the MSL regardless of altimeter setting.

Below Transition Height and Transition Level, adding the airfield elevation to the current QFE height will provide the altitude that can be used to reference the TAWS altitudes (if displayed).

Setting QNH on the Standby Altimeter enhances awareness of the terrain depicted on the TAWS.

3. FMS / VNAV

When using QFE, the use of VNAV is prohibited since the FMS depicts QNH altitudes and not QFE Heights.

General QFE Procedures

QFE OM-B / AFM procedures shall be complied with if published.

Normal Crew briefings shall include a discussion of these procedures.

Verify with ATC the QFE altimeter setting.

Verify with ATC that vertical clearances are referenced to QFE.

If the QFE altimeter setting is given in Millimetres Mercury, flight crew shall request hPa from ATC. Calculation of the hPa is also available in the conversion section ([Conversions](#)).

Metres shall be used if the aeroplane is capable of selecting metres in the AAD and ATC are giving instructions in metres.

If the aeroplane is not capable of selecting metres, then conversion to feet is required as follows:

- The Jeppesen Charts (STAR, SID, and Approach) normally have a conversion table showing the normal ATC height assignments used and show QFE feet.
- FMS Conversion page (for aeroplanes that have this function), or
- Tables provided in [Conversions](#).

AIP charts and charts made available by navigators will depict QFE heights only.

Ground

Before leaving the ramp the pressure setting of all primary altimeters shall be set to the reported QFE and the standby altimeter shall be set to the QNH to improve awareness of the MSA.

To ensure that the tolerances specified in the respective OM-B are not exceeded, the height indicated on the primary altimeters shall be compared with each other, and zero; the altitude indicated on standby altimeter shall be checked against the nearest known elevation on the aerodrome.

RVSM pre-flight procedures require that the readings of both primary altimeters shall agree with the zero-height indication within a 75 ft (23 m) tolerance, or AFM / AOM limitations if more restrictive.

Before take-off, each time a new altimeter pressure setting is reported by the aerodrome, the above procedure shall be followed.

Airborne

Climb: As per general procedures.

Cruise: As per general procedures.

Descent: Prior to descent or as soon as possible after the beginning of the descent, flight crew shall obtain both QFE and QNH from the ATIS or from ATC.

QNH shall be set on the standby altimeter and the QFE shall be noted in order to crosscheck the QFE provided by ATC on transition to heights.

If either is not available in the ATIS, flight crew should use the latest available QNH/QFE to determine the expected QNH/QFE. This is accomplished by adding / subtracting the RWY hPa, depicted in the AHCC or from the Hectopascal / Millibar Equivalent for Airfield Elevation table in [Conversions](#), to / from the respective pressure.

When cleared to a height, both primary altimeters shall be set to QFE and compared.

If there are any doubts as to the QFE altimeter setting, flight crew shall seek correct QFE setting from ATC and cross-check again.

When approaching MEA or MSA, each time a new a new descent clearance is received, the standby altimeter and minimum safe altitude shall be checked to provide early detection of an unsafe descent clearance.

8.3.3.5 Conversions

[ADDITIONAL GUIDANCE](#)

Aeroplanes equipped with FMSs that have conversions pages / function, can obtain the majority of the conversions on the respective page(s).

8.3.3.5.1 Metres to Feet (Nearest Hundred)

[METERS TO FEET CONVERSION TABLE](#)

CAUTION

Aircraft types unable to use QFE shall not use this table for operation in the Metric / QFE environment and should instead refer to the aerodrome specific AHCC chart.

8.3.3.5.2 Millimetres Mercury to Hectopascals / Millibars Conversion

Conversion from Millimetres Mercury to hPa / Mb may be accomplished as follows:

- FMS Conversion page (for aeroplanes that have this function), or
- Conversion Table provided as [ADDITIONAL GUIDANCE](#).

8.3.3.5.3 QFE to QNH Altimeter Setting Conversion

If QNH altimeter setting is required but not available, convert the QFE altimeter setting to a QNH altimeter setting as follows:

- FMS Conversion page (for aeroplanes that have this function), or
- Convert QFE to QNH altimeter setting by referring to the Hectopascal / Millibar Equivalent for Airfield Elevation tables provided as [ADDITIONAL GUIDANCE](#).
- Formula: QFE altimeter hPa + Airport elevation hPa = QNH altimeter hPa.

8.3.3.6 Checking of Terrain Clearance - Altitude Corrections

This chapter deals with safe altitude corrections to take account of altimeter error due to pressure, temperature and, where appropriate, wind / terrain effects.

The pilot is responsible for making these corrections except when under radar vectoring. In that case, the radar controller shall issue clearances such that the prescribed obstacle clearance will exist at all times, taking the cold temperature correction into account. Once released from radar vectors (e.g. to intercept an approach) the pilot is again responsible for cold temperature corrections.

The PIC is responsible for ensuring safe obstacle clearance by making any required corrections during all phases of flight.

PRESSURE CORRECTIONS

When flying at a Flight Level, and the local QNH is lower than 1013.2 hPa / 29.92 inches, a correction shall be added to the minimum safe altitude to provide a minimum safe Flight Level.

The table below may be used to determine the applicable correction. Alternatively, it may be calculated manually using 30 ft (10 m) per hPa.

QNH of nearest station (hPa / Inches)	Correction to MSA	QNH of nearest station (hPa / Inches)	Correction to MSA
1013 / 29.92	0 ft	975 / 28.79	1070 ft
1010 / 29.83	90 ft	970 / 28.64	1210 ft
1005 / 29.68	230 ft	965 / 28.50	1350 ft
1000 / 29.53	370 ft	960 / 28.35	1490 ft
995 / 29.38	510 ft	955 / 28.20	1630 ft
990 / 29.23	650 ft	950 / 28.05	1780 ft
985 / 29.09	780 ft	945 / 27.91	1920 ft
980 / 28.94	920 ft	940 / 27.76	2070 ft

Temperature Corrections

[ADDITIONAL GUIDANCE](#)

When to Apply Temperature Correction

- When airport temperature is 0° C or colder.

Where to Apply Temperature Correction

- MSA (corrections of 200 ft or less may be disregarded).
- If below the corrected MSA:
 - All altitudes from the IF to the FAF, or where an IF is not published all altitudes from the IAF to the FAF.
 - All altitude restrictions on the missed approach up to the final missed approach altitude.
 - Final Missed Approach altitude, all SID, STAR, Transition and en-route altitudes (corrections of 200 ft or less may be disregarded).
- Step-down fixes after the FAF.
- Approach Minima.
- EFFRA (Engine failure flap retraction altitude / acceleration altitude).
- Exceptions:
 - do not apply correction to the FAF/FAP altitude when on a BARO VNAV approach and the temperature is within promulgated limits (for PBN-specific temperature corrections, see [8.3.2.6.4, B.3, Cold temperature corrections](#));

- when flying to US Cold Temperature Restricted Airports (CTRA), corrections are only necessary if the temperature is below the published temperature limit, as shown in the notes section of the procedure chart. Example: '*Cold temperature altitude correction required at or below -22°C*'. EFFRA shall always be corrected when temperature aerodrome temperature is 0°C or colder.

ATC Communications

When pilots intend to apply corrections to the FAF crossing altitude, procedural altitudes prior to the FAF or missed approach altitudes, they must advise ATC of their intention and the correction to be applied.

Pilots may refuse IFR assigned altitudes if altimeter temperature error will reduce obstacle clearance below acceptable minima. However, once an assigned altitude has been accepted, it must not subsequently be adjusted to compensate for temperature error.

How to apply temperature correction

FMS Temperature Compensation: Use the FMS Cold Temperature Compensation Function, if approved and available. Refer to OM-B for limitations and procedures.

Manual Temperature Correction: Use the appropriate correction charts as per the links below, when selecting a chart, round down to the next lowest airport elevation as this is more conservative (e.g. for Elev 1,417 ft use the chart for Elev 1,000 ft).

When using the charts below, the temperature refers to the reported temperature at the aerodrome and the corrections are given in ft.

[Temperature Correction Chart - Elevation Sea Level](#)

[Temperature Correction Chart - Elevation 1,000 ft](#)

[Temperature Correction Chart - Elevation 2,000 ft](#)

[Temperature Correction Chart - Elevation 3,000 ft](#)

[Temperature Correction Chart - Elevation 4,000 ft](#)

[Temperature Correction Chart - Elevation 5,000 ft](#)

[Temperature Correction Chart - Elevation 6,000 ft](#)

[Temperature Correction Chart - Elevation 7,000 ft](#)

[Temperature Correction Chart - Elevation 8,000 ft](#)

[Temperature Correction Chart - Elevation 9,000 ft](#)

[Temperature Correction Chart - Elevation 10,000 ft](#)

[Temperature Correction Chart - Elevation 11,000 ft](#)

[Temperature Correction Chart - Elevation 12,000 ft](#)

[Temperature Correction Chart - Elevation 13,000 ft](#)

[Temperature Correction Chart - Elevation 14,000 ft](#)

For further guidance on the application of temperature corrections, see [ADDITIONAL GUIDANCE](#).

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WIND CORRECTION MOUNTAINOUS TERRAIN - TERMINAL AREAS

The combination of strong winds and mountainous terrain can cause local changes in atmospheric pressure due to the Bernoulli effect which can result in an altimeter error. This occurs particularly where the wind direction is across mountain crests or ridges.

It is up to the pilot-in command to evaluate whether the combination of terrain, wind strength and direction are such as to make a correction necessary, where corrections are likely to be required this will normally be indicated in the OM-C.

Any corrections for wind speed should be applied in addition to the standard corrections for pressure and temperature, and ATC should be advised, the table below gives some guidance for appropriate corrections.

Wind Speed	Altimeter error
20 kt	53 ft
40 kt	201 ft
60 kt	455 ft
80 kt	812 ft

8.3.4 Altitude Alerting System Procedures

Setting of Altitude Alert

Until an ATC clearance is received, the Altitude Alerting Device shall be set in accordance with the OM-B.

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8.3.5 (Enhanced) Ground Proximity Warning System (EGPWS) Procedures

(CAT.OP.MPA.290)

8.3.5.1 General

The EGPWS should always be operational, except for the circumstances outlined in the MEL. When undue proximity to the ground is detected by any flight crew member or by the EGPWS, the commander or PF shall make immediate positive corrective action to establish safe flight conditions.

It is mandatory to respond to all EGPWS alerts. If the PF does not respond to an EGPWS alert, the PM shall immediately take control and initiate the manoeuvre.

8.3.5.2 EGPWS Caution Alerts

The PF shall take immediate positive corrective action to adjust aircraft flight path as necessary to eliminate the caution alert.

Refer to the Ground Proximity Warning section on Operations Manual Part B, Chapter 3.

8.3.5.3 EGPWS Warning 'Pull Up' Alerts

Refer to the Ground Proximity Warning section on Operations Manual Part B, Chapter 3 for the type specific EGPWS 'PULL UP' escape manoeuvre.

During night or IMC operations, the EGPWS 'PULL-UP' escape manoeuvre shall be initiated immediately. Do not delay for diagnosis.

During day VMC conditions, the response to an EGPWS Warning Alert may be as for a Caution Alert provided that:

- Terrain and obstacles are clearly in sight; and
- It is immediately obvious to the flight crew that the aeroplane is in no danger in respect of its configuration, proximity to terrain or current flight manoeuvre;
- If either FCM has any doubt about the most appropriate course of action or the EGPWS warning alert persists, either FCM shall call 'TERRAIN GO'.

NOTE

Crews shall ensure that any likely alert scenarios are identified and fully briefed prior to commencing the approach or departure, particularly those described in the respective OM-C or NTA Aerodrome Briefing. When a steeper-than-normal descent is necessary during the approach, the higher rates of descent near terrain increase the likelihood of EGPWS alerts. These circumstances do not diminish the necessity for an immediate response from the PF or the commander upon the activation of an alert. Regardless of any procedures outlined in the operations manual or AFM for conducting steep approaches, there is no alleviation to the policies established in this subsection.

8.3.5.4 Reporting

For every EGPWS Warning 'PULL UP' alert encountered, the crew shall submit a NetJets Safety report.

Following the recovery from an EGPWS alert or caution requiring deviation from an ATC clearance, the appropriate ATSU shall be informed.

8.3.6 ACAS / TCAS

(CAT.OP.MPA.295)

ADDITIONAL GUIDANCE

8.3.6.1 ACAS / TCAS Procedures

Resolution Advisories (RA) shall always be complied with using fleet-specific procedures. Pilots are authorised to deviate from their current ATC clearance to comply with a TCAS RA.

If conflicting instructions are received from an RA and ATC, the instruction given by the RA shall be followed.

Stall warnings, ground proximity warning system / enhanced ground proximity warning system (EGPWS) warnings, and wind shear warnings shall take precedence over TCAS advisories.

TCAS shall only be inhibited when airborne when the use of 'TA only' is called for by non-normal QRH drills.

Whenever an RA requires 'CLIMB' while the aeroplane is in landing configuration, a go-around shall be initiated.

ACAS / TCAS does not relieve the crew of the need to maintain a visual lookout when meteorological conditions permit.

The ACAS / TCAS display should not be relied upon when assessing whether other traffic is a factor; some traffic might not show on the display until an RA is unavoidable.

If ACAS / TCAS equipment does not aurally announce system failures, a periodic check shall include the checking the serviceability of the system and its associated equipment. Any failures shall be communicated to each ATC unit that controls the aeroplane.

8.3.6.2 TA / RA Actions to be Taken

Refer to OM-B, section 3.10 TCAS.

8.3.6.3 Reporting

For every Resolution Advisory encountered, the crew shall submit a *NetJets Safety Report*.

8.3.6.4 RTF Phraseology for Notifying RAs

RAs shall be reported to the ATC unit using the phraseology below.

Circumstances	Phraseology	
	PILOT	CONTROLLER
After a flight crew starts to deviate from any ATC clearance or instruction to comply with an ACAS resolution advisory (RA).	'FRACTION XXXX TCAS RA'	'ROGER'
After the response to an ACAS RA is completed and a return to the ATC clearance or instruction is initiated.	'FRACTION XXXX CLEAR OF CONFLICT, RETURNING TO, (assigned clearance)'	'ROGER' or Alternative Instructions
After the response to an ACAS RA is completed and the assigned ATC clearance or instruction has been resumed.	'FRACTION XXXX CLEAR OF CONFLICT, (assigned clearance) RESUMED'	'ROGER' or Alternative Instructions
After an ATC clearance or instruction contradictory to the ACAS RA is received, the flight crew will follow the RA and inform ATC directly.	'FRACTION XXXX UNABLE TCAS RA'	'ROGER'

8.3.6.5 Rate of Climb and Descent

Refer to [Rate of Climb and Descent Approaching Level-off](#).

8.3.7 Policy and Procedures for In-flight Fuel Management

This policy is part of NTA's fuel scheme. It shall be used to ensure that the usable fuel expected to remain upon landing at the destination aerodrome is at least the amount required by NTA's fuel policy.

NTA complies with the requirements for the variations to a basic fuel scheme, as defined in CAT.OP.MPA.180 and CAT.OP.MPA.181.

The pilot-in-command shall continually ensure that the amount of usable fuel remaining on board is not less than the fuel required to proceed to an aerodrome where a safe landing can be made with the planned final reserve fuel remaining upon landing.

The fuel quantities available on board at the start of every flight shall be in accordance with [Determination of the Quantities of Fuel and Oil Carried - Fuel Policy](#).

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8.3.7.1 In-flight Fuel Checks

The commander shall ensure that fuel checks are carried out in-flight at regular intervals as appropriate to the length of the flight, but not less than every 60 minutes, or at specified points indicated in the operational flight plan and these checks shall be documented on the OFP. Unless unable to do so due to the nature and length of the flight, flight crew shall as a minimum carry out fuel checks as follows:

- At or abeam an appropriate waypoint as soon as reasonably practical after reaching initial cruise altitude;
- At or abeam an appropriate waypoint as close as practical to the TOD (slightly before or after the TOD); and
- If diverting, at the time of initiating the diversion and regularly during the diversion.

The usable remaining fuel and ATA over or abeam the waypoint shall be recorded and evaluated to:

- a. compare actual consumption with planned consumption;
- b. check that the usable remaining fuel is sufficient to complete the flight, in accordance with [In-flight Fuel Management](#); and
- c. determine the expected usable fuel remaining on arrival at the destination aerodrome.

8.3.7.2 In-flight Fuel Management

The flight crew shall continuously monitor the fuel situation, taking into consideration changes in the operational status of the aerodrome of intended landing and alternate aerodromes, as well as deviations from the flight plan affecting the range capability of the aeroplane.

The flight shall be conducted so that the expected usable fuel remaining on arrival at the destination aerodrome is not less than:

- a. the required alternate fuel plus final reserve fuel; or
- b. the final reserve fuel if no alternate aerodrome is required.

When unforeseen circumstances may result in not being able to comply with the conditions a. or b. above, the commander must obtain delay information described in [Protecting the Final Reserve](#).

If a fuel check shows that the aircraft will arrive at destination with less than a. or b. above, action should be taken to reduce fuel consumption for the rest of the flight, and / or select a closer destination alternate.

If further checks continue to show that the aircraft has insufficient fuel to meet the minimum fuel requirements, flight plan changes shall be initiated as soon as practicable for a refuelling stop or diversion.

On a flight using the RCF procedure, to proceed to the destination 1 aerodrome, the commander shall ensure that the usable fuel remaining at the decision point is at least the total of:

- a. Trip fuel from the decision point to the destination 1 aerodrome;
- b. Contingency fuel equal to 5% of the trip fuel from the decision point to the destination 1 aerodrome;
- c. Destination 1 alternate fuel, if a destination 1 alternate aerodrome is required;
- d. Additional fuel, if required; and
- e. Final reserve fuel.

On a flight using the isolated aerodrome procedure (includes isolated aerodromes) to proceed to the destination aerodrome, the commander shall ensure that the usable fuel remaining is at least the total of:

- a. Trip fuel from the PNR to the destination aerodrome;
- b. Contingency fuel from the PNR to the destination aerodrome; and
- c. Additional fuel, as described in [Isolated Aerodrome Procedure](#).

8.3.7.3 Procedures in the Event of a Foreseen Fuel Shortage

Whenever the flight crew foresees a shortage of fuel in-flight that might endanger the safe conduct of the remaining portion of the flight, flight plan changes shall be initiated as soon as practicable to take the necessary precautions for a refuelling stop or diversion.

If a fuel check shows that the aircraft will arrive at destination with less than alternate plus final reserve fuel in tanks, action should be taken to reduce fuel consumption for the remainder of the flight, and / or to select a closer destination alternate.

8.3.7.4 Fuel Management Approaching Destination Aerodrome

ATC shall be advised of a MINIMUM FUEL state when, having committed to land at a specific aerodrome, any change to the existing clearance to that aerodrome may result in landing with less than planned final reserve fuel.

Minimum fuel: The term used to describe a situation in which an aircraft's fuel supply has reached a state where the flight is committed to land at a specific aerodrome and no additional delay can be accepted.

NOTE

The declaration of MINIMUM FUEL is not declaring an emergency but informs ATC that:

1. all planned aerodrome options have been reduced to a specific aerodrome of intended landing; and
2. as long as the current clearance is not modified, the flight should be able to proceed as cleared without compromising the PIC's responsibility to protect final reserve fuel.

ATC are not required to provide priority to pilots of aircraft that have declared MINIMUM FUEL.

The commander shall declare a fuel emergency when the calculated usable fuel on landing, at the nearest adequate aerodrome where a safe landing can be performed, is less than final reserve fuel.

Single Runway Available at Destination

When the destination aerodrome is operating a single runway, crew should request delay information from ATC and determine the latest approach time to ensure landing with not less than alternate and final reserve fuel. If ATC is not able to guarantee an approach time to meet this requirement, an immediate diversion to the planned alternate is mandatory.

Two or More Runways Available at Destination

If, due to unexpected delays encountered while approaching a destination aerodrome, the remaining fuel reaches the required alternate fuel plus final reserve fuel (when a destination alternate is required), the flight may be continued to the destination aerodrome provided that at least 2 separate runways are available. This should be understood as committed to land and ATC shall be advised of a MINIMUM FUEL state. This decision shall only be made if it provides the same, or a higher degree of safety, than the initiation of a diversion to the alternate aerodrome considering especially:

- The meteorological conditions at destination and alternate at decision time and for the expected time of landing;
- The number and state of suitable runways at destination and alternate and approach aids available;
- The expected approach time issued or anticipated.

The term 'separate runways' in this context means:

- Separate landing surfaces which may overlay or cross such that if one of the runways is blocked, it will not prevent the planned type of operations on the other runway, and;
- Each of the landing surfaces has a separate approach procedure based on a separate aid.

Regardless of Available Runways at Destination

Irrespective of the number of runways available; if the approach has commenced and additional delays are encountered whereby the remaining usable fuel at landing will be less than the required alternate fuel plus the final reserve fuel, this should be understood as 'committed to land'.

8.3.7.4.1 Protecting the Final Reserve

The purpose of Final Reserve Fuel (FRF) is to ensure that a safe landing is always possible when unforeseen circumstances occur.

During all phases of flight, the Commander shall continuously assess whether a safe-landing option exists that can be achieved with more than the FRF remaining. When conditions change, priority shall be given to identifying a safe landing option rather than continuing as originally planned.

If, based on the Commander's assessment, it becomes apparent that landing with more than the FRF cannot be guaranteed, a fuel emergency shall be declared without delay as per the escalation process shown below.

Once FRF protection is no longer guaranteed, the primary objective is to land the aircraft safely as soon as practicable. All reasonable landing options shall be considered, including aerodromes not normally planned or assessed, and deviations from normal rules, procedures, or methods are permitted when required in the interest of safety.

If FRF cannot be protected, decisive action to achieve a safe landing takes precedence over all other considerations.

1	REQUEST DELAY INFORMATION FROM ATC when unanticipated circumstances may result in landing at the destination aerodrome with: a. less than final reserve fuel, plus fuel required to proceed to an alternate aerodrome; or b. the fuel required to operate to an isolated aerodrome. Example ATC responses: 'NO TRAFFIC DELAY EXPECTED / DELAY LESS THAN [NUMBER] MINUTES, EXPECT [NUMBER] HOLDING PATTERNS / EXPECTED APPROACH TIME [TIME] / DELAY NOT DETERMINED [REASON FOR THE DELAY]' NOTE Pilots should use this delay information to determine if a diversion is required.
2	DECLARE MINIMUM FUEL: 'FRACTION XXX, MINIMUM FUEL' a. when committed to land at a specific aerodrome; and b. any change to the existing clearance to that aerodrome may result in landing with less than the planned final reserve fuel. NOTE Do not expect priority. This is not an emergency situation but an indication that an emergency situation is possible should any additional delay occur. Example ATC responses: 'ROGER, NO TRAFFIC DELAY EXPECTED / EXPECT [NUMBER] MINUTES DELAY / [NUMBER] MILES FROM TOUCHDOWN' NOTE Pilots should use this delay information to determine if a fuel emergency should be declared.

3	DECLARE A FUEL EMERGENCY: 'MAYDAY MAYDAY MAYDAY FUEL...' when the calculated usable fuel predicted to be available upon landing at the nearest aerodrome where a safe landing can be made is less than the planned final reserve fuel. ATC Response: <i>Subsequent ATC action shall be based on the intentions of the pilot and the overall traffic situation.</i>
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8.3.8 Adverse and Potentially Hazardous Atmospheric Conditions

8.3.8.1 Significant Weather Avoidance Policy

Flights through areas with known or forecast thunderstorms, severe icing, severe turbulence or wind shear should be avoided whenever possible. Flight crew shall avoid significant weather en-route by deviating from the flight path (horizontally and / or vertically) if operationally feasible and possible.

[Reporting of Hazardous Flight Conditions](#)

[General Guidelines For Operation In Turbulence, Windshear, And Near Thunderstorms](#)

8.3.8.2 Thunderstorms

ADDITIONAL GUIDANCE

A take-off shall not be performed during active thunderstorm activity over the departure aerodrome.

An approach shall be delayed or a diversion to an alternate aerodrome shall be initiated rather than penetrate or fly under an active thunderstorm in the approach and landing area.

In the case of detection of a significant cell or storm, the flight crew should observe the following recommendations:

- To avoid a large storm, the flight crew should make decisions while still 40 nm from it;
- Deviate upwind instead of downwind of a cell (less probability of turbulence or hail);
- For storm-avoidance planning, consider the height of the storm:
 - Avoid all yellow, red, or magenta weather radar returns by at least 10 nm;
 - Avoid all green, yellow, red, and magenta weather radar returns from cells taller than 25,000 ft by at least 20 nm;
 - Cells exceeding 35,000 ft should be considered extremely hazardous and further separation (in addition to the 20 nm) should be used.
- If the top of cell is at or above 25,000 ft, do not overfly it to minimise the possibility of encountering turbulence;
- Do not attempt to penetrate a cell or clear its top by less than 5,000 ft vertically, otherwise the aircraft may encounter severe turbulence;
- Avoid flying under a thunderstorm or a cumulonimbus overhang because of possible windshear, microbursts, severe turbulence, or hail.

If the aeroplane is not equipped with radar or it is inoperative, avoid by 10 miles any storm that by visual inspection is tall, growing rapidly, or has an anvil top.

Lightning

ADDITIONAL GUIDANCE

Approaching an aeroplane on the ground during a thunderstorm shall be avoided. The human body may provide the grounding that is otherwise lacking. If you are in the FCC, do not attempt to communicate with ground personnel through a wire handset, and stay inside the aeroplane until the storm has passed.

In flight, if a lightning strike is likely to occur, all cockpit lights should be set to full brightness. FCMs should consider wearing sunglasses to protect their eyes from the flash, or have one pilot keep eyes downward.

If the aeroplane suffers a lightning strike, refer to OM-B for type specific in-flight procedures.

If a lightning strike has occurred, or is suspected to have occurred, it shall be recorded in the Aircraft Technical Log and maintenance shall be informed. A proper maintenance release shall be obtained before further flight.

8.3.8.3 Icing Conditions

(CAT.OP.MPA.255)

ADDITIONAL GUIDANCE

The Commander shall only commence a flight or intentionally fly into expected or actual icing conditions if the aircraft is certified and equipped to cope with such conditions.

All NTA aeroplanes shall be operated in accordance with the in-flight icing procedures, limitations, and restrictions according to the AFM, OM-B, and MEL. Furthermore, any recommendations by the manufacturer in any official Aeroplane Operating Manual (or equivalent) shall be followed.

If severe icing is encountered or icing exceeds the intensity of icing for which the aircraft is certified, the Commander shall exit the icing conditions without delay, by a change of level and / or route, if necessary by declaring an emergency to ATC.

NOTE

Consult the AFM/AOM for specific information regarding handling techniques if inadvertently encountering severe icing conditions.

Flight into any intensity of freezing drizzle or freezing rain is prohibited.

Flight in known severe icing conditions is prohibited.

Ice accumulation rates

Trace: Ice becomes perceptible but is of no consequence and does not affect the performance of the aeroplane.

Light: The rate of accumulation may create a problem if extended flight in this condition occurs. It can be safely handled by the aeroplane's anti/de-icing equipment. No restriction to operations, provided the systems are used and does not necessitate a change of heading or altitude.

Moderate: The rate of accumulation is such that even short encounters become potentially hazardous. The aeroplane's anti/de-icing equipment will safely handle it. However, for practical purposes, it should be a signal to the pilot to alter their flight plan in order to avoid further exposure.

Severe: Adverse icing condition in which the rate of accumulation is such that the anti / de-icing equipment fails to reduce or control the hazard. Pilots shall change the flight path immediately, to establish more favourable conditions, or land as soon as possible.

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8.3.8.4 Turbulence

ADDITIONAL GUIDANCE

Flight crew shall comply with the turbulent air penetration procedures and limitations stated in the respective AFM / AOM.

The intensity specifications for turbulence reporting are defined as follows:

- Light: Slight discomfort;
- Moderate: There may be moderate changes in aeroplane attitude and / or altitude, but the aeroplane remains in positive control at all times, usually with small variations in airspeed. Changes in accelerometer readings of 0,5 g to 1,0 g at the aeroplane's centre of gravity. Walking is difficult, loose objects move around. Occupants feel strain against seat belts.
- Severe: Abrupt changes in aeroplane attitude and / or altitude, aeroplane may be out of control for short periods accompanied by large variations in airspeed. Changes in accelerometer readings greater than 1.0 g at the aeroplane's centre of gravity. Occupants are forced violently against seat belts. Loose objects are tossed around.
- Excessive: Excessive turbulence means severe turbulence possibly exceeding the g-limits for the specific aeroplane.

8.3.8.5 Windshear

ADDITIONAL GUIDANCE

Windshear encountered on take-off or approach must be reported to ATC immediately by R/T if considered a hazard, giving details of height and severity.

Windshear standard response

If any Red Windshear warning or negative windshear is experienced, any pilot shall call 'WINDSHEAR GO' and the windshear recovery procedure in the OM-B/AFM shall be initiated immediately.

If any Amber Windshear caution or positive windshear is experienced and depending on the prevailing weather conditions, the current flight path may be maintained unless the procedure in the OM-B/AFM requires that the recovery shall be initiated immediately.

CAUTION

Following an Amber Windshear caution or positive windshear, flight crew shall be prepared for a possible RED Windshear Warning or negative windshear.

During the windshear recovery procedure, the flight crew shall:

1. Control the flight path through pitch attitude - resist a downward change in pitch attitude as normal response to low airspeed.
2. Expect that unusual stick forces may be required to maintain pitch attitude during airspeed variations away from normal target airspeed.
3. Accept low airspeed. Flying at airspeeds below normal reference speeds may be required in order to utilise the full performance capability of the aeroplane.

Precautions

Avoidance is the best precaution. If windshear is anticipated during take-off or on approach, special precautionary techniques can be applied by the flight crew to mitigate the effects of windshear. The following precautions shall be taken into consideration:

- Thrust setting:

Maximum take-off thrust should be used for take-off. It shortens the take-off roll and provides the best climb performance which leads to increasing altitude available for recovery if required. During approach thrust reductions should be minimised.

- Runway selection:

The longest suitable runway should be used taking into consideration crosswind and tailwind limitations, and obstacles in take-off or landing path.

- Flap selection:

Take-off: unless dictated by field length, optimum flap should be used, which may offer the best performance and higher margin to flap limiting speed should windshear be encountered on the runway or just after take-off.

Landing: use the AOM recommended flap for landing.

For both situations, consider flap selection against airspeed recommendations below.

- Airspeed

Take-off: increased airspeed at lift-off increases the safety margin above the stall should a windshear be experienced at or shortly after take-off, but due consideration needs to be given to available field length by using the V_r at MTOM as an upper limit.

Landing: increased airspeed improves climb performance capability, but due allowance needs to be made for the increase in LDR.

Use of Auto-throttle, Autopilot, and Flight Director

For take-off only, speed-referenced flight directors with windshear recovery guidance should be used.

During approach, flight director, autopilot and auto-throttle should be used to the maximum extent practical. This will relieve the workload of the flight crew and give them more time to monitor instruments and weather conditions.

8.3.8.6 Jetstream

[ADDITIONAL GUIDANCE](#)

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8.3.8.7 Volcanic Ash[ADDITIONAL GUIDANCE](#)

Flight in visible / discernible ash must be avoided.

When operating in areas where volcanic activity may occur pilots should be vigilant and aware of their responsibilities to report any activity immediately. Flights planned near or through any areas of forecast contamination will comply with the NTA Safety Risk assessment accepted by ANAC and the operating crew will be provided with the latest information. The responsibility to avoid flight in visible / discernible ash remains with the operating Commander.

8.3.8.8 Heavy Precipitation[ADDITIONAL GUIDANCE](#)

Limitations and procedures in type-specific AFM and OM-B must be used for operation in heavy precipitation.

8.3.8.9 Sand Storms[ADDITIONAL GUIDANCE](#)

Limitations and procedures in type-specific AFM and OM-B must be used for operation in Sand Storms.

8.3.8.10 Mountain Waves[ADDITIONAL GUIDANCE](#)**8.3.8.11 Significant Temperature Inversion**[ADDITIONAL GUIDANCE](#)

In case of known or forecast temperature inversions, corrections to the performance limits shall be applied as per the AOM.

8.3.8.12 Hot and High Operation[ADDITIONAL GUIDANCE](#)**Introduction**

This section contains guideline information that should be considered when operating at hot and high airfields, as a clear understanding of the performance limitations of your aeroplane when compared to ISA conditions is highly desirable.

8.3.9 Wake Turbulence

ADDITIONAL GUIDANCE

8.3.9.1 Probability of a Wake Turbulence Encounter

Most low altitude wake turbulence encounters occur when the winds are light. Even a momentary wake turbulence encounter may be hazardous in the area up to 1,000 ft below and behind a Super or Heavy aeroplane.

Application of separation minima cannot entirely remove the possibility of a wake turbulence encounter. The objectives of the minima are to reduce the probability of encountering wake turbulence to an acceptably low level, and to minimise the magnitude of the upset should an encounter occur.

NTA aircraft types will generally have significantly better take-off and climb performance than a typical airliner style aircraft which might, for example, be departing with reduced thrust. This gives rise to additional risk factors when following such an aircraft:

- When making an intersection departure with minimum separation, NTA aircraft types may cross the flight path / wake turbulence of the leading aircraft from below;
- When SIDs contain an intermediate level off, NTA aircraft types are likely to reach this level off earlier and accelerate sooner, thereby eating into the separation and potentially intersecting the leading aircraft flight path / wake turbulence.

During the approach phase, particular care shall be exercised where the leading aeroplane has followed the glide path on final approach from an extended range (e.g. continuous descent approach). Significant wake turbulence encounters have been reported where, although ATC have applied the appropriate separation minima, the following aeroplane is vectored onto final approach behind and below a substantially heavier aeroplane.

NOTE

ATC may impose only the minimum separation criteria. They may be increased at the discretion of the controller or at the request of the pilot. However, any pilot request shall be made before entering a runway or commencing final approach, to avoid a departure delay or an avoidable missed approach. Where an aeroplane has lined up on a runway and take-off clearance is issued, the aeroplane is expected to commence take-off without delay.

8.3.9.2 Operational Aspects

As a general rule, avoid the area below and behind the generating aircraft, especially at low altitude, where even a momentary wake encounter could be hazardous.

FCM should be particularly alert in calm wind conditions and situations where the vortices could:

- Remain in the touchdown area.
- Drift from aircraft operating on a nearby runway.
- Sink into take-off or landing path from a crossing runway.

8.3.9.3 Wake Vortex Categories

Two systems of categorising aircraft wake characteristics are now in use: ICAO standard and RECAT-EU. European airports are free to choose which system to use. Crews should be aware that airports adopting RECAT-EU procedures may provide reduced separation compared to the ICAO wake turbulence scheme.

Table 1: Standard ICAO Wake Turbulence Categories

Category	Maximum Take-off Mass (MTOM) in kg
Super (J)	Airbus 380-800 ¹
Heavy (H)	≥ 136,000 ²
Medium (M)	> 7,000 & < 136,000
Light (L)	≤ 7,000

¹ Super category: All those aircraft listed as such in ICAO Doc 8643. Currently, only the A380-800 meets that criterion.

² Except if categorised as Super.

Some States establish differences from the ICAO standards, described in the Air Traffic Services section of the Jeppesen manual.

Table 2: RECAT-EU Wake Turbulence Categories

Category	Max take-off mass (Tons)	Wing span
Super heavy (A)	Greater than 100 T	> 72 m
Upper heavy (B)		60 - 72 m ¹
Lower heavy (C)		< 52 m ¹
Upper medium (D)	15 - 100 T	> 32 m
Lower medium (E)	15 - 100 T	Up to 32 m
Light (F)	Up to 15 T	

¹ Aircraft with MTOM above 100T with wing spans between 52 m and 60 m are subject to a specific analysis to determine if their category should be B or C.

[Example List of Aircraft Types Assigned to RECAT-EU Categories](#)

NTA aircraft types are categorised for wake turbulence as follows:

Table 3: NTA WT Categories

Aircraft type	ICAO WT category	RECAT-EU WT category
EMB- 505	Medium (M)	Light (F)
CE-560 XLS	Medium (M)	Light (F)
CE-680A	Medium (M)	Light (F)
BD-100-1A10	Medium (M)	Lower Medium (E)
Falcon 2000EX	Medium (M)	Lower Medium (E)
CL-600-2B16	Medium (M)	Lower Medium (E)

Aircraft type	ICAO WT category	RECAT-EU WT category
BD-700-1A11	Medium (M)	Lower Medium (E)
BD-700-1A10	Medium (M)	Lower Medium (E)

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8.3.9.4 Departure

Wake Turbulence Separation - Departures

Details of fleet specific separation standards are published in the relevant OM-B, section 2.7.1 Take-off.

NOTE

1. *Some airports may use distance to separate aircraft on departure instead of time;*
2. *When distance is used to separate aircraft on departure, this can result in a time separation that is less than that stated in the fleet specific separation tables. Crew may either accept this distance based separation or request additional separation (e.g. 2 min instead of 4 miles) for wake turbulence avoidance. This request should be made as soon as practical and at least before taxiing onto the runway. Full details of these differences can be found in the Jeppesen Airway Manual under the Air Traffic Control section.*

Wake turbulence minima on departure are applied by ATC by measuring airborne times between successive aeroplanes. Take-off clearance may be issued with an allowance for the anticipated take-off run on the runway. This may result in a take-off clearance being issued at less than prescribed time interval. However, the airborne time interval will reflect a difference of at least the required time separation.

8.3.9.5 Approach

Wake Turbulence Separation - Distance Based

Details of fleet specific separation standards are published in the relevant OM-B, section 2.12 Approach.

Wake Turbulence Separation - Time Based

As an alternative to Distance Based separation, some airports (EGLL for example) have introduced Time Based Separation (TBS) in order to increase capacity during strong headwind conditions.

Details of fleet specific separation standards are published in the relevant OM-B, section 2.12 Approach.

8.3.10 Crew Members at Their Stations

(CAT.OP.MPA.210)

During taxi, take-off, and landing, each flight crew member required to be on duty in the FCC shall be seated at their assigned station.

During the en-route phase of the flight, each flight crew member required to be on duty in the FCC shall remain at their station unless their absence is necessary for the performance of duties in connection with the aeroplane's operation, for physiological needs, or passenger service-related reasons, provided that at least one suitably qualified pilot remains at the controls of the aeroplane at all times. A required flight crew member, absent for one of the reasons stated above, shall return to their station as expeditiously as possible.

When a pilot transfers control of the aeroplane or leaves the FCC, the nominated PF shall remain alert, maintain unobstructed access to the flight controls and monitor, respond to and act upon any ATC instructions. Any secondary communications (ATIS, SATCOM, etc.) shall only be conducted when both pilots are seated at their assigned stations in the FCC.

No FCC seat changes shall be made during critical phases of flight, during flight below 10,000 feet above the aerodrome of departure after take-off and the aerodrome of destination before landing, or MSA, whichever is higher, unless required in the interest of safety.

In the interest of safety, the commander may request supernumerary flight crew members to occupy an assigned station in the FCC.

During all phases of flight, each flight crew member required to be on duty in the FCC shall remain alert unless taking controlled rest in accordance with [Fatigue Management \(Controlled Rest on FCC\)](#) (cabin crew aircraft only).

8.3.10.1 Cabin Crew Seating

The cabin crew member shall preferentially occupy a cabin seat, with the lead passenger's permission. If a cabin seat is not available, they shall occupy an FCC jump seat or crew rest seat.

If occupying a cabin seat, the order of preference is as follows:

1. Emergency exit, forward-facing seat;
2. Emergency exit, rear-facing seat;
3. A forward-facing seat;
4. A rear-facing seat;
5. A sideways-facing seat.

In case two cabin crew members are onboard, the same principle applies and the option of one cabin crew member seated on the FCC jump seat or crew rest seat and one cabin crew member on a cabin seat shall be offered to the lead-passenger.

When reporting cabin secure to the FCC, a cabin crew member shall indicate which seat(s) will be occupied by the cabin crew member(s).

During critical phases of flight, each cabin crew member shall be seated at the defined seat and shall not perform any activities other than those required for the safe operation of the aeroplane.

During the taxi, take-off, and landing Cabin Crew shall:

- Review their emergency procedures.
- Remain alert to any hazards inside or outside the aeroplane.
- Should an incident arise the Cabin Crew shall remain at their stations, be alert and await instructions from the commander. In the absence of commands from the flight crew, if the situation clearly presents an imminent threat to life, the Cabin Crew shall initiate an evacuation.

During planned emergency, a cabin crew member shall be seated in the cabin (if seats are available) in a seat adjacent to an emergency exit (aft facing, if available) regardless of owner / passenger consent.

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8.3.11 Use of Safety Belts for Crew and Passengers

(CAT.OP.MPA.225)

This policy for the use of safety belts for crew and passengers applies to all NTA aircraft types. If due to type specific technical reasons, the use of the fasten seatbelt sign as described in [Cabin Safety Requirements](#) and [Passenger and Supernumerary Briefing Procedures](#) is not possible, an alternative means of ensuring compliance with [Use of Safety Belts for Crew and Passengers](#) will be described in the OM-B for that aircraft type.

8.3.11.1 Crew Members and Occupants of the FCC

During the following phases / conditions of flight, all crew members and other occupants of the FCC will be secured with all the provided safety belts and harnesses for their station:

- a. Taxi: Cabin crew may walk about the cabin for cabin preparation for take-off.
- b. Take-off and departure until a minimum of 10000' AAL (or cruise below FL100/10,000'): After take-off and when deemed safe by the commander, cabin crew may be released.
- c. Approach and landing and below 10,000' AAL: Cabin crew may walk about the cabin for cabin preparation below 10,000' AAL, but shall be seated and secured with the safety belts and harnesses for landing.
- d. Turbulence: For cabin crew, this is dependent on the level of turbulence. See [Turbulence](#).
- e. Abnormal / emergency situation, unless securing or being secured distracts from handling the abnormal / emergency situation.
- f. When one required flight crew member leaves their station in the FCC.
- g. When requested by the PIC.

During all other phases of the flight, each flight crew member and other occupants of the FCC shall keep their safety belt fastened.

8.3.11.2 Passengers and Supernumeraries

The commander shall be satisfied that each passenger or supernumerary on board occupies a seat or berth with the safety belt or restraint system properly secured, for the following situations:

- During taxiing;
- Before and during take-off and climb to the first planned cruising level (TOC);
- During descent from cruising level to landing;
- Prior to and / or during turbulence;
- Whenever deemed necessary in the Interest of safety.

It shall be recommended to passengers and supernumeraries to keep their safety belt / restraining belts / harnesses secured at all times during the flight when they are seated.

Multiple occupancy of an aeroplane seat by one adult and one infant is only allowed on specified seats, and with the infant properly secured by a supplementary loop belt or other restraint device.

8.3.11.3 Securing of Belts, Harnesses

At each unoccupied seat or berth, the safety belt / restraining belt and shoulder harness, if installed, shall be secured so as not to interfere with crew members in the performance of their duties or with the rapid egress of occupants in an emergency.

8.3.12 Admission to Flight Crew Compartment (FCC)

None of the aircraft types operated by NetJets are fitted with a flight crew compartment door, so the flight crew compartment shall be defined as the area where the flight crew and flight deck are within easy reach of a person in that area.

Subject to the acceptance of the commander, the following persons may be granted admission to the FCC:

- a. Operating crew members (e.g. cabin crew, ISR, specially appointed operations expert);
- b. A representative of the competent or inspecting authority, if required to be there for the performance of their official duties;
- c. Other persons performing official and specific duties:
 - i. Employees of national or foreign aeronautical authorities or organisations;
 - ii. Employees of the NetJets business group, if required for observation, inspection or training and qualification (e.g. flight dispatchers, safety pilots, compliance inspections);
- d. Passengers not meeting the criteria contained in (a) to (c) above, but only when permitted by [8.3.12.1 Additional Procedures for Passenger Access to the FCC](#) and in accordance with the additional instructions contained therein.

In all cases where a person other than the operating flight crew assigned to a flight are granted access to the FCC, the commander shall ensure that:

1. admission to the FCC does not cause distraction or interference with the operation of the flight; and
2. all persons carried in the FCC are made familiar with the relevant safety procedures.

In all cases the commander shall make the final decision regarding admission to the FCC.

8.3.12.1 Additional Procedures for Passenger Access to the FCC

Given the unique nature of the NetJets business (sole use by known customers) and the type of aircraft operated (no cockpit doors and no regulatory requirement for cabin crew) it may be necessary and / or desirable for the passenger to enter the FCC area as defined in [8.3.12 Admission to Flight Crew Compartment \(FCC\)](#). This is permitted only in accordance with the following instructions and with the agreement of the commander.

During Cruise above FL100

Once level at cruising altitude (provided this is above FL100), at the commander's discretion, passengers may be granted short-term access to the FCC without using the FCC jump seat. Provided this does not cause distraction or interference with the operation of the flight.

Flights without cabin crew

On flights where no cabin crew member is present, the flight crew is responsible for the passengers but may need to remain seated at their stations. Exceptionally, passengers may be granted temporary admission to the FCC below FL100 if necessary to ensure their safety. This admission must be limited in time to only that which is sufficient to provide the required information to the passenger(s) (e.g., emergency communications, use of safety equipment, clarification of procedures, use of cabin equipment, providing instructions).

During Taxi, Take-off and Landing (Seated in the FCC jump seat)

Passengers may be granted admission to the FCC for the purpose of remaining in the FCC jump seat for Taxi, Take-off and Landing, only when complying with the following policy and procedures:

Passenger Selection

- A passenger may only occupy the FCC jump seat if they meet all the requirements to be allocated an emergency exit seat as detailed in [8.2.2.2.1 Allocation of Seats, Adults, Children and SCPs / A. General / \(g\)](#).
- No passenger shall be permitted to occupy the FCC jump seat if they are under the influence of any psychoactive substance (including alcohol).
- The lead passenger must explicitly agree to the passenger being carried on the FCC jump seat.
- Where a passenger was considered suitable to occupy the FCC jump seat for departure, this does not guarantee they will be suitable for the landing (e.g. if they have consumed alcohol during the flight or displayed unsuitable behaviour during the departure)

Limitations

- The Commander shall always have the discretion to not permit a passenger to be carried on the FCC jump seat or for them to be removed from the FCC jump seat.
- In addition to the Commander, all other operating crewmembers must be comfortable with the passenger being on the FCC jump seat.
- A passenger may only be carried on the FCC jump seat where a flight has an operating cabin crew member.
- A passenger may not be carried on the FCC jump seat during any planned response to a non-normal situation (e.g., landing after an abnormal situation such as a flap failure or a flight with an MEL which impacts operational procedures).
- In all cases where a passenger is carried on the FCC jump seat, they must have an operable / belted passenger seat allocated to them in the cabin.

Procedures

- Where a passenger is to be in the FCC jump seat for departure, they shall not be permitted to occupy that seat until such time as all crew flight preparation, walk-arounds and briefings have been completed. Once completed the flight crew will advise the cabin crew that they may bring the passenger to the FCC.
- In the event of an abnormal or emergency situation, when safe to do so, the passenger shall leave the FCC jump seat and return to the cabin. The flight crew will advise the cabin crew when this applies and the cabin crew will facilitate the passenger's return to the cabin.

Briefings

A passenger carried on the FCC jump seat shall receive:

- All safety briefings required for passengers.
- The briefing outlined in [8.3.13.2 FCC jump seat briefing](#).
- An additional briefing highlighting:
 - i. risks associated with critical controls in the FCC and hand / foot placement.
 - ii. the need to avoid distracting the crew during normal operations focusing on sterile cockpit in critical phases of flight.

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8.3.13 Use of Vacant Crew Seats

Crew seats may only be used by authorised persons, as described in the table below.

Seat type	Authorised persons	Exceptions
Pilots seats	Operating flight crew	As per 4.3 Flight Crew Incapacitation
Jump seat	As per 8.3.13.1 Authorisation for the use of the FCC Jump Seat	As per 8.3.12.1 Additional Procedures for Passenger Access to the FCC
Crew rest seat	Operating crew	None

8.3.13.1 Authorisation for the use of the FCC Jump Seat

At the commander's discretion, the persons that may be granted admission to the FCC as defined in [8.3.12 Admission to the Flight Crew Compartment \(FCC\)](#) may also be authorised to occupy the FCC jump seat.

Persons carried according to the criteria listed in [8.3.12 Admission to the Flight Crew Compartment \(FCC\)](#)(a) to (c), may be carried in the FCC without having a seat in the cabin, provided the FCC jump seat is equipped with a seat belt. Passengers carried according to [8.3.12 Admission to the Flight Crew Compartment \(FCC\)](#)(d) shall always also have a seat in the cabin.

Any person authorised to use the FCC jump seat shall:

- be briefed as per [8.3.13.2 Jump Seat Briefing](#);
- not be classified as an SCP (their ability to operate the jump seat and evacuate in case of an emergency must be assured);
- be able to speak and understand English;
- wear the seat belt at all times; however, in non-critical phases of flight, only a lap belt is required;
- not be under the influence of psychoactive substances;
- not consume beverages or food of any kind in the FCC.

Persons should be instructed not to approach the FCC while the FSB sign is on, unless their assigned seat is the jump seat; in this case, they must immediately assume their station when the FSB sign is turned on.

In addition to above, when passenger who does not meet any other criteria listed in [8.3.12 Admission to the Flight Crew Compartment \(FCC\)](#)(a) to (c) is permitted to occupy the FCC jump seat, then the provisions contained in [8.3.12.1 Additional Procedures for Passenger Access to the FCC](#) shall also apply.

8.3.13.2 FCC Jump Seat Briefing

If the occupant of the jump seat is a person who is not an active crew member in NTA, and therefore not familiar with the operation of the seat and relevant company procedures, the commander shall ensure that they are briefed on the following items:

- a. Seat operation and stowage;
- b. Handling of seat belt and harness;
- c. Position and handling of fire extinguisher;
- d. Position, handling, and time of use of the oxygen masks;
- e. Position and use of the life vests (if required);
- f. Sterile FCC Procedures;
- g. Inter comm and radio operations, if available from the seat;
- h. Actions in the event of an emergency;
- i. Task allocation in Non-normal Procedures:
 - i. Crew incapacitation;
 - ii. Emergency descent;
 - iii. Emergency evacuation;
 - iv. Ditching.
- j. Evacuation routes and exits;
- k. Operation of normal and emergency exits.

8.3.14 Incapacitation of Crew Members

Refer to SEP Manual. Adherence to the 2 communication rule as described in the SEP Manual shall serve as an incapacitation check during all stages of flight.

8.3.15 Cabin Safety Requirements

It shall be noted that a number of aeroplane types operated by NTA do not have a cabin crew as part of the crew complement, unless specifically requested by an owner or customer. The details that follow below involve many actions associated with cabin crew, rather than FCC crew.

On an aeroplane type that does not carry a cabin crew member, one of the flight crew, (described as 'an appropriate crew member') shall be responsible for ensuring compliance with the various provisions listed in this and any other relevant section. Ultimate responsibility rests with the aeroplane Commander.

Those aeroplanes on which a cabin crew member forms part of the crew normally carry a single cabin crew member. Therefore any reference to a 'Senior Cabin Crew' may be taken to mean the sole cabin crew member in such cases.

The provisions of sections 8.2 ([Allocation of Seats](#)), ([Sick, Disabled and Handicapped Passengers, PRMs](#)), ([Transportation of Inadmissible Passengers, Deportees](#)), ([Permissible Size and Weight of Hand Baggage](#)), ([Securing of Load, Cabin](#)), ([Live Animals](#)), ([Procedures for the Refusal of Embarkation and Removal of Passengers](#)), and 8.3 ([Seat Belts / Passengers](#)), ([Use of Vacant Crew Seats](#)), ([Passengers Briefing](#)) relate, more or less directly, to cabin safety.

The procedures to be followed during passenger embarkation and disembarkation are outlined in [Embarkation / Disembarkation](#), the operation of aeroplane doors in [Safety on the Ramp, Including Fire Prevention, Blast and Suction Areas](#).

The procedures in the event of fuelling with passengers embarking or disembarking or on board have been described in [Passengers On Board](#).

Of cabin safety relevance, in the wider sense, are [Operating Procedures](#), section [Oxygen Requirements](#), [Dangerous Goods and Weapons](#), [Security](#), OM-B Chapter 3 - Abnormal and Emergency Procedures, provisions relating to fire and smoke drill, unpressurised / partly pressurised flight), OM-B Chapter 11 - Emergency Evacuation Procedures and the SEP Manual.

The policy for the use of safety belts for crew and passengers described in [Use of Safety Belts for Crew and Passengers](#) applies to all NTA aeroplane types. If due to type specific technical reasons the use of the fasten seatbelt sign as described in [Cabin Safety Requirements](#) and [Passenger and Supernumerary Briefing Procedures](#) is not possible, an alternative means of ensuring compliance with [Use of Safety Belts for Crew and Passengers](#) will be described in the OM-B for that aircraft type.

For aeroplane that normally carry cabin crew or ISRs, when no Cabin Crew or ISR are carried, the use of the oven and microwave in-flight is prohibited. The exception to this is the use of the microwave by the passengers on the BD-100-1A10 when the procedures in the Challenger 350 In-flight Service Handbook are complied with.

The use of the oven and microwave is permitted on the ground provided the procedures in the OM-B are complied with.

8.3.15.1 General

Legal provisions require all crew members not to perform any activities during critical phases of the flight other than those required for the safe operation of the aeroplane.

This means that all cabin crew members shall, in the critical phases of flight, concentrate on their designated task of securing the cabin and shall, if necessary, firmly and politely refuse service requests by the passengers.

Additional restrictions apply in respect of contacting the flight crew. See [Sterile Flight Crew Compartment](#).

The commander shall ensure that before take-off and landing, and whenever deemed necessary in the interest of safety, all equipment and baggage are properly secured.

8.3.15.2 Prior to Boarding of Passengers

Crew shall ensure that inoperative emergency exits are marked in accordance with the provisions laid down in the MEL (see AOM).

On taking over their aeroplane at the beginning of a day's flight or series of flights the cabin crew shall check the type-specific (see SEP Manual) cabin emergency equipment using a check list. Equipment found at a wrong place shall be re-stowed at the appropriate location. Switching and thereby checking of emergency exit lights shall be performed as early as possible to ensure sufficient time for repair in case of malfunction.

On completion of the check list the senior cabin crew member will advise the commander of any discrepancies so that these may be rectified or, if this is not practicable, entered in the Aeroplane Technical Log.

NOTE

MEL No-Go deficiencies shall be rectified before flight.

In the interest of hygiene and passenger comfort / well-being, the cabin crew or an appropriate crew member shall check that the cabin (inclusive of lavatories and galleys) has been cleaned and tidied up appropriately. They shall keep in mind that the appearance of the cabin's interior at boarding profoundly affects the company's reputation with its customers.

Safety and courtesy render it necessary to ensure that no unstowed catering, galley or other equipment / load inconveniences boarding passengers nor impedes the orderly flow of boarding.

8.3.15.3 Prior to Taxiing

Cabin crew or an appropriate crew member shall check if the seating of handicapped passengers, children, or infants complies with the provisions of [Carriage of Passengers and Special Categories of Passengers](#); if not, they shall arrange for their reseating.

Cabin crew or an appropriate crew member shall ensure that all passengers are seated with their safety belt / harness properly secured (see [Passengers and Supernumeraries](#)), that no food, beverage or tableware is located at any passenger seat and that food and beverage trays as well as passengers' seat backrests are in the upright position.

Seats that are not occupied, that are not on an emergency exit row and do not obstruct the paths to and the use of the emergency exits, are allowed to be transformed into beds for take-off and landing.

In addition, it shall be ascertained that all passenger and crew baggage, catering equipment and other cabin load is properly secured and stowed where it cannot impede evacuation or cause injury by falling or other movement. Overhead bins, if installed, shall be closed. All baggage shall be removed from exit rows. This includes loose and bulky items on passengers' laps, e.g. handbags, blankets and large hardback books.

By an announcement and by visual checks, cabin crew or an appropriate crew member shall ensure that passenger use PEDs is in accordance with the PED Policy.

Cabin or flight crew shall strictly enforce the 'NO SMOKING' instruction, given in accordance with [Passenger and Supernumerary Briefing Procedures](#).

All exits and escape paths shall be unobstructed. If it is necessary to pass through a doorway or curtain to reach any required emergency exit from any passenger seat, the door or curtain shall be secured in the open position.

All relevant emergency equipment shall remain easily accessible for immediate use. Furthermore, any activity which could distract any flight crew member from their duties is not authorised. Thus cabin crew are not to enter or call the FCC except in the event of an emergency or for a safety related problem.

8.3.15.4 Prior to Take-off

Prior to take-off, the cabin crew shall occupy their assigned stations and fasten their safety belts and, if installed, their harnesses.

At night, the cabin lights shall be dimmed in order to improve the night vision of cabin staff and passengers. The senior cabin crew member, having received the appropriate cabin report from each cabin crew member, where applicable, shall report CABIN SECURE to the flight crew. Refer to [Cabin Secure Checks](#).

The CABIN SECURE report shall be acknowledged by the commander. Lining up without the CABIN SECURE report is prohibited.

Prior to lining up, the PM shall cycle the No Smoking sign to alert the cabin crew that take-off is imminent and that they shall take their seats for take-off.

8.3.15.5 In-flight

After take-off, all cabin crew shall remain seated with seat belt / harnesses fastened until the NO SMOKING signs have been switched off, unless permission of the commander to start the service earlier has been obtained.

Having heard the signal from the flight crew the decision to leave their seats to commence the cabin service remains with the cabin crew as conditions in the cabin, e.g. the steep angle of climb, may indicate that waiting a few moments would be preferable.

For fire avoidance / detection and cabin safety, cabin crew shall conduct regular checks and observe the following provisions:

They shall:

- a. Enforce strict observation of the 'NO SMOKING' sign.
- b. Look out for passengers who have fallen asleep while smoking.
- c. Search trays for cigarettes and cigars before stowing.
- d. Never dispose of burning cigarettes / cigars in waste boxes.
- e. Restrict spreading of papers by passengers on the cabin floor, (where discretion and courtesy can be maintained).
- f. Check lavatories for signs of smoke, and open flaps of waste boxes (automatic fire extinguishers).
- g. Check the temperature of the hot water and the proper working of electric toilet appliances. (These checks shall be made when infrequent use of lavatories may prevent detection by passengers.)
- h. Conduct regular checks in order to ensure that passenger use of PEDs is in accordance with the PED policy.
- i. Ensure aisles are free from obstruction.
- j. Emergency and Safety Equipment stowages are accessible and free from obstruction.
- k. All service equipment is secured and stowed when not in use.
- l. Conduct regular galley checks in order to prevent a fire resulting from the oven or microwave from occurring.

If it becomes necessary to administer first aid oxygen, the cabin crew shall inform the flight crew who will switch on the NO SMOKING sign.

If discovering a container with combustible or quickly vaporising fluid, cabin crew shall place the container into crushed ice (not dry ice!) and inform the commander who will switch on the NO SMOKING sign and request passengers verbally to stop smoking.

NOTE

Also refer to [Marking and Labelling of Packages](#).

All breakdowns or malfunctions of electrical equipment in the cabin shall be immediately reported to the flight crew. Cabin crew shall be aware of the fact that resetting circuit breakers, without knowing the reason for the failure, increases the fire risk. Cabin crew shall not reset any tripped circuit breakers unless explicit permission for each CB is given by the commander.

Cabin Crew shall remain vigilant and all occurrences and abnormalities which may affect the safety of the operation and the well-being of the passengers shall be immediately reported to the flight crew (i.e. illness on board, unusual noises, odours and other observations).

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8.3.15.5.1 Turbulence

The flight crew will, wherever possible, give advice on the actions that should be taken by cabin crew. The advice will depend on the severity of the turbulence.

If turbulence is anticipated before departure:

- The commander will advise the cabin crew member of its expected severity and duration and whether or not the cabin crew should remain seated.

If turbulence develops during the flight:

- The commander will switch on the FSB sign.
- The CCM or appropriate crew member will inform the passengers asking them to return to their seats and fasten their seatbelts and ensure that they have fastened their seat belts.
- CCM will report to the FCC to ascertain if the service can continue.

If moderate or severe turbulence is encountered during the flight:

- The commander will switch on the FSB and NS sign.
- The CCM or appropriate crew member will inform the passengers asking them to return to their seats and fasten their seat belts and ensure that they have fastened their seat belts.
- The Cabin Crew shall cease the cabin service, restrain any equipment, sit immediately with SBF and communicate with the flight crew if possible.

Unless the degree of suddenly encountered turbulence renders it too dangerous, cabin crew shall, in case of turbulence, ensure that baggage and other load is re-stowed and secured so as not to cause injury by falling or other movement.

8.3.15.6 Prior to Landing

The FSB sign shall normally be illuminated when the aeroplane commences its descent from the cruise level. In the descent, passing 10,000 ft AAL, the commander shall cycle the NS sign on then off, to inform the CCM to adopt limited service and to prepare the cabin for landing. If no cabin crew is on board, the FSB sign will normally be illuminated when the aeroplane commences its descent from the cruise level and the NS sign illuminated when passing 10,000 ft AAL.

The cabin crew or an appropriate crew member, after securing and locking galleys and lavatories, shall ascertain that all passengers are safely seated and that all baggage / load has been properly re-stowed. All baggage shall be removed from exit rows. This includes loose and bulky items on passengers' laps, e.g. handbags, blankets and large hardback books.

At this stage of the flight, all provisions outlined in [Incapacitation of Crew Members](#) and [Cabin Safety Requirements](#) shall again be observed by the cabin crew. They shall, on receipt of instruction from the FCC, occupy their designated stations and fasten seat belts / harnesses (see [Use of Safety Belts for Crew and Passengers](#) and [Admission to Flight Crew Compartment \(FCC\)](#)).

The senior cabin crew, having received the appropriate cabin report from each cabin crew, shall report CABIN SECURE to the flight crew. The CABIN SECURE report shall be acknowledged by the commander.

Shortly before landing: The 'No smoking sign' will be turned ON by the FCC crew as a sign for the cabin crew to sit down immediately if that is not the case yet.

8.3.15.7 After Landing

After the aeroplane has touched down, cabin crew shall ensure that the passengers remain seated with their seat belts fastened until the final parking position has been reached and the engines have been switched off and the FASTEN SEAT BELT sign has been switched off.

After landing and when the Commander determines that it is safe for the cabin crew to move in the cabin, the PM will cycle the NO SMOKING sign to indicate that cabin crew can get out of their seats.

Cabin crew should remain in their seats as long as is practicable, ideally until the aeroplane has arrived on stand.

Immediately before reaching that parking position, and upon instruction from the FCC, each cabin attendant shall ensure that the automatically deploying means for emergency evacuation at the door(s) / emergency exit(s) they have been designated responsible for, if applicable, are armed or disarmed as prescribed in the type-specific instructions of the AOM. For opening of doors refer to [Multiple Occupancy of Aeroplane Seats](#), for disembarkation to [Embarkation / Disembarkation](#).

8.3.15.8 Cabin Secure Checks

CABIN SECURE checks require the following to be checked:

- a. Passengers and supernumeraries are seated with seat belts fastened, backrests upright and tables / trays in stowed position;
- b. Rear facing seats that contain headrests, that the headrests are extended;
- c. Equipment and baggage properly secured;
- d. Galleys and lavatories are checked, locked, and secured;
- e. Galley electrical equipment is off (large cabin);
- f. Stowage spaces and bins closed;
- g. All exits and escape paths unobstructed;
- h. Window blinds raised;
- i. Curtains and doors secured in the 'open' position;
- j. All doors in 'FLIGHT' condition, where applicable;
- k. Cabin lights dimmed (night);
- l. Personal electronic devices are in the condition as required by the PED Policy;
- m. In-flight entertainment system is off and personal screens stowed;
- n. Passenger briefing completed.

Exits

All baggage shall be removed from exit rows. This includes loose and bulky items on passengers' laps, e.g. handbags, blankets and large hardback books.

Seatbacks / Arm Rests

Passenger seats shall be in a vertical position with the armrests down and the table stowed for both take-off and landing. (A vertical seat facilitates the preferred bracing position; it also provides the maximum structural support).

Window Blinds

For take-off and landing, window blinds of the exits or those adjacent to the exits shall be in the open position, to allow cabin crew and passengers to observe exterior conditions. In case of inoperative emergency exit window blinds, always observe restrictions imposed by the MEL.

Catering Supplies

All catering supplies and in-flight service equipment shall be securely stowed in approved areas.

Galley Electrics

Galley power for take-off / landing shall be in accordance with the OM-B / AFM.

Cabin Lights

During the hours of darkness cabin lights should be dimmed for both take-off and landing. Should an incident occur which results in a failure of the aeroplane main lighting system, passengers shall be able to adjust to the lower level of emergency lighting.

Before dimming the cabin lighting, the passengers should be informed by the cabin crew member.

8.3.15.9 Smoking On Board

In order to minimise fire hazards smoking is prohibited in an aeroplane:

- Whilst it is parked on the ground;
- During taxiing, take-off and landing;
- During periods of moderate or severe turbulence;
- Outside designated smoking areas⁵ ;
- In the aisle(s) and the toilet(s);
- In cargo / baggage compartments and / or other areas where cargo / baggage is carried;
- In those areas of the cabin where oxygen is being supplied;
- When there is any smell of fuel or other flammable liquid;
- When there is no crew member on board the aeroplane;
- During any emergency;
- Whenever deemed necessary by the commander in the interest of safety.

All crew members, supernumeraries and NetJets personnel are prohibited from smoking on board a NTA aeroplane.

The use of e-cigarettes shall follow the normal smoking on board procedures.

E-cigarettes shall not be carried in checked baggage. Passengers shall be briefed to ensure that any e-cigarettes are removed from the checked baggage and are carried in the cabin. Charging of e-cigarettes on board is prohibited.

⁵ A designated smoking area is defined as being seated in any passenger seat certified to be occupied for take-off and landing that has an ashtray within easy reach and is not be within a toilet, the FCC or any other area prohibited in the list above.

8.3.15.10 Portable Electronic Devices (PED) Policy

8.3.15.10.1 PED Types

Portable Electronic Devices (PEDs) are any electronic devices brought on board by crew, passengers, or loaded as cargo. They are not part of the approved aircraft configuration.

PEDs are classified as NIT-PEDs, T-PEDs, C-PEDs, or Large PEDs, as defined below.

PED Category	Definition	Examples	Operational Notes
NIT-PED	Device not capable of intentional radio transmission	Calculators; cameras; audio / video players; toys without transmitters.	May be used as permitted by 8.3.15.10.3 Use of PEDs During Flights .
T-PED	Device capable of intentional radio transmission	Mobile phones; satellite phones; two-way radios; laptops / tablets with Wi-Fi / Bluetooth / cellular.	May be used as permitted by 8.3.15.10.3 Use of PEDs During Flights . If transmitting functions are deactivated (flight mode), they can be used as a NIT-PED.
C-PED	PED under NetJets administrative control	EFB	Governed by EFB Policy in OM-A. It may be NIT-PED or T-PED.
Large PED ¹	Device too large to be operated with one hand (not a handheld device)	Laptops; large tablets.	Must be stowed as prescribed in 8.3.15.10.3 Use of PEDs During Flights , as it may become a projectile; see 8.2.2.4.3 Securing of Load, Cabin ; May be NIT-PED or T-PED.

¹ Large PEDs are not a transmission category — it refers to size and weight (e.g., laptops) and relates to cabin safety and stowage requirements rather than RF emission type.

8.3.15.10.2 Procedures for PEDs

8.3.15.10.2.1 General

Crew shall take all reasonable measures to ensure that passengers comply with the policy for the use of PEDs on board, to prevent any adverse effect on the performance of the aeroplane's systems and equipment.

Crew shall comply with any procedures that are stated in the AFM, AOM and QRH / FCOM.

Any cables associated with the use of PEDs, e.g. earphones or charging cables, must not be allowed to obstruct access to aisles and exits during taxi, take-off and landing.

All PEDs shall be switched off in any emergency situation unless their use is required in the interest of safety.

For any fire related to PEDs, including Lithium batteries, refer to the SEP Manual.

8.3.15.10.2.2 Charging of PEDs

PEDs must be disconnected from power outlets or other electrical power supplies during taxi, take-off, in-flight below 10,000 ft, landing roll-out, and in any abnormal or emergency situation. Passengers shall receive instructions on this matter, and crew members must know how to switch off the in-seat power system when necessary. Aircraft power outlets shall be used only to charge PEDs and only when such PEDs are being monitored at all times by the passenger.

Power banks

Power banks can only be carried in accordance with [9.2 Provisions for Dangerous Goods Carried by Passengers or Crew](#). Additionally, the following provisions apply:

- a. Power banks must not be recharged while on board the aircraft;
- b. Charging of PEDs using power banks must be kept to a minimum, and aircraft power outlets should be used whenever available. However, it is authorised under the conditions i. to iv. set out in the restricted use policy below, designed specifically to control for the risks associated with the use of lithium batteries on board:
 - i. Charging is attended by a responsible person. Unattended charging is not permitted.
 - ii. When in use, the power bank is fully visible at all times (not in a bag, pocket, or covered).
 - iii. Not in use during taxiing, take-off, in-flight below 10,000 ft, landing roll-out or during abnormal or emergency conditions. (Note that this also applies to using the power outlets).
 - iv. Passengers are briefed on the conditions of this policy.

8.3.15.10.2.3 In-flight monitoring of PEDs

8.3.15.10.2.3.1 General Monitoring Requirements

Crew shall continuously monitor the use of all PEDs in the cabin to ensure that their operation complies with this policy.

Particular attention shall be given to T-PEDs, ensuring that all transmitting functions (e.g., mobile networks, Wi-Fi, Bluetooth) remain disabled when required. If a failure to comply with the policy is identified, the crew shall:

1. Instruct the passenger(s) to immediately disable all transmitting functions or switch off the identified device(s).
2. If the passenger(s) refuse(s) to comply, the commander shall be notified without delay, if applicable.

8.3.15.10.2.3.2 Monitoring for Thermal Runaway

Crew shall actively monitor all PEDs for signs of overheating or thermal runaway. Extra vigilance is required for devices lost or trapped in seat mechanisms. Thermal events can occur with any lithium-battery-powered PED, regardless of whether it is charging. Early indicators of lithium battery failure include:

- Swelling, bulging or deformation;
- Smoke or fumes;

- Hissing, popping, or crackling sounds;
- Unusual heat on touch.

If a PED shows signs of overheating or thermal runaway, the crew shall follow the instructions laid out in the SEP manual. If the flight is operated with cabin crew, the commander shall be notified without delay.

Although not formally considered PEDs, items such as e-cigarettes and spare lithium batteries must be kept in the passenger's immediate possession. If this is not possible, they can also be stored in a location within the cabin where quick assistance can be provided in case of thermal runaway, and preferably away from flammable items or pressurised cans. If more than one item containing lithium batteries is to be carried, they must not be stored in the same location. See [9.2 Provisions for Dangerous Goods Carried by Passengers or Crew](#) for a full list of restrictions for carrying batteries or battery-operated devices.

8.3.15.10.2.4 Use of PEDs during Passenger Briefings

If passengers' attention to the safety briefing and / or crew instructions is being affected by their continued use of a PED then they shall be asked to refrain from using it or to switch it off.

8.3.15.10.3 Use of PEDs During Flights

Whilst parked on stand prior to engine start all types of PED may be used with no restrictions. Once engine start is commenced the tables below shall be used to determine the permitted use of PEDs:

Table 1 - Use of type of PEDs by phase of flight

PED Type	Phase of flight	
	On the ground and Airborne, below 10,000 ft	Airborne, above 10,000 ft
NIT-PEDs	Unrestricted	Unrestricted
T-PEDs ¹	Restricted as per Table 2 ² (Unrestricted on taxi after landing)	Restricted per Table 2 ²
C-PEDs	In-flight use only in accordance with portable EFB policy	
Large PEDs ³	Stowed (Unrestricted on taxi after landing)	Unrestricted

¹ T-PEDs with the transmitting functions off are considered NIT-PEDs.

² Table 2 lists allowed technologies (type of connection). Use is not permitted for technologies other than those listed per aircraft type.

³ Large PEDs are also either NIT-PEDs or T-PEDs, and the applicable restrictions shall also be applied.

Table 2 - Fleet status of T-PED tolerance demonstration in compliance with CAT.GEN.MPA.140

Fleet	Allowed Technologies
BD-7001A10	Wi-Fi
CL-600-2B16	Wi-Fi and Bluetooth
DA-2EASy	Bluetooth and / or Wi-Fi
BD-100-1A10	Wi-Fi / GSM / Cell phone technologies in 800 to 1800 MHz bands
CE-680A	Wi-Fi
CE-560XLS	Wi-Fi
EMB-505	Wi-Fi and Bluetooth

Use of PEDs during LVO approach operations

For LVO approaches, all PEDs must be switched off.

For aeroplanes operating with cabin crew, the commander shall inform the cabin crew when an LVO approach will be conducted to ensure compliance with the restriction above.

8.3.15.10.4 Exceptions to PEDs Rules

The following exceptions apply to the rules detailed in the previous paragraphs:

- Medical equipment necessary to support physiological functions does not need to be switched off at any stage of flight.
- PEDs not completely disconnected from the internal power source when switched off may remain in this stated at any stage of flight.
- Devices having no transmit capability and operated by coin cells without further deactivation capability, e.g. wrist watches do not need to be switched off at any stage of flight.
- If the aeroplane is stationary during long departure delays, the commander may permit the use of any type of PED provided that there will be sufficient time to check compliance with the requirements [Use of PEDs During Flights](#) before departure.
- The Commander may, for any reason and during any phase of flight, require deactivation and stowage of PEDs.

8.3.15.10.5 PEDs Not Accessible During Flight

Any PED not carried on the passenger's person or contained within their carry-on baggage must be switched off. Crew members shall brief passengers and instruct them to switch off all PEDs not in their immediate possession or stowed in their carry-on baggage.

8.3.15.10.6 Use of PEDs in the Flight Crew Compartment

The use of PEDs in the flight crew compartment is restricted due to the increased risk of electromagnetic interference and potential distraction for the flight crew. PED use in the flight crew compartment is permitted only when required for operational purposes and when it does not compromise safety.

- a. Any PED used in the flight crew compartment shall be securely stowed, mounted, or held so that it does not create a hazard or impede access to controls.
- b. During critical phases of flight (taxi, take-off, approach, landing), only C-PEDs approved by NTA based on demonstrated acceptable RF interference levels may be operated.
- c. T-PEDs shall not be operated in the flight deck during pre-flight checks, navigation database loading, monitoring fuel loading, or any other activity where RF transmission could adversely affect aircraft systems. In all other situations, the flight crew shall comply with the same T-PED restrictions applicable to passengers, including the restrictions by type of connection.

These limitations do not prohibit the use of a T-PED (e.g., mobile phone) to manage an emergency, although reliance on T-PEDs shall not be assumed for contingency planning.

- d. Flight crew must make sure that using a PED doesn't distract them from safely performing their operational duties. This includes diverting their attention or causing mental distraction through content. Crew should stop using PEDs immediately if their workload increases, their situational awareness might decrease, or if any unusual or urgent situation occurs.
- e. Applications such as email clients, messaging apps or any communication platform that may prompt extended reading and typing, as well as those requiring multi-step interaction, pose a high risk of prolonged distraction. These applications often require sustained attention, diverting attention away from the crew's operational tasks. Additionally, they generate notifications or prompts that may diminish situational awareness. For these reasons, the use of email or messaging applications in the flight crew compartment is prohibited unless required for operational purposes. Exceptionally, during the cruise phase, the use of such applications is allowed under the restrictions set out in paragraph (d). In the context of the current paragraph, cruise is understood as any phase of levelled flight above 10,000 ft during which no crew actions are expected for at least 10 minutes.
- f. Crew members must actively monitor their workload and task engagement, stop PED interaction immediately when operational conditions change, and avoid 'task spillover', where an operationally initiated PED interaction (e.g., checking navigation charts) turns into unrelated or extended use.
- g. NIT-PEDs may be used during non-critical phases of flight for verifying information relevant to the flight (i.e. information from the briefing).

8.3.15.10.7 PED Reportable Events

Events of suspected or confirmed PED interference, passengers refusing to comply with the PED policy and smoke or fire caused by a PED shall be reported by means of a NetJets Safety report.

8.3.16 Passenger and Supernumerary Briefing Procedures

(CAT.OP.MPA.170)

The rationale underlying cabin briefing procedures is to enhance owner safety.

Turbulence, whether wake or weather related, may be experienced during any phase of flight. Consequently, cabin safety briefings shall highlight the requirement to observe the FSB sign. Additionally, the safety benefits of keeping seat belts (loosely) fastened when the FSB sign has been turned off shall be emphasised.

The policy for the use of safety belts for crew and passengers described in [Use of Safety Belts for Crew and Passengers](#) applies to all NTA aeroplane types. If due to type specific technical reasons the use of the fasten seatbelt sign as described in [Cabin Safety Requirements](#) and [Passenger and Supernumerary Briefing Procedures](#) is not possible, an alternative means of ensuring compliance with [Use of Safety Belts for Crew and Passengers](#) will be described in the OM-B for that aeroplane type.

8.3.16.1 Passenger Briefing Procedures

Passengers shall be verbally briefed and given demonstrations relating to safety, including observation of the FASTEN SEAT BELT and NO SMOKING signs, and also be provided with a safety briefing card. The safety briefing card shall provide, by means of pictorial presentations, information and detailed instructions on the use of emergency equipment and exits intended to be used by passengers in an emergency situation.

Pictorial placards (pictograms) and picture type instructions shall indicate the location and the use of emergency installations.

Prior to engine start, the following non-safety related items shall be discussed with the passengers:

- Destination
- Expected flight time
- En-route and destination weather
- Loaded luggage
- Services requirements

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8.3.16.1.1 Prior to Take-off

When the aeroplane's doors have been closed, and preferably before taxiing has commenced, a designated crew member shall brief the passengers on the following provisions / regulations / procedures, if applicable:

1. Requirement to comply with crew member instructions.
2. Observation of the FSB sign when switched on; to remain seated with the seat belt secured.
3. The use of safety belts or restraint systems, including how to fasten and unfasten the safety belts or restraint systems.
4. Observation of NO SMOKING signs when switched on, prohibition of smoking on the ground, prohibition of smoking during flight in the lavatories and baggage compartments, including the use of e-cigarettes.

Passengers shall be briefed to ensure that any e-cigarettes are removed from the checked baggage and are carried in the cabin and that charging of e-cigarettes on board is prohibited.

5. Cabin secure items to include, as applicable: Backs of the seats to be in the upright position, tables stowed, footrests stowed, window blinds open.
6. Location, operation, and use of the emergency exits. When cabin crew are carried and they occupy an emergency exit seat, then the briefing of operation and use of the emergency exit is not required.
7. Location of floor proximity escape path markings and exit signs.
8. Correct stowage of hand baggage and the importance of leaving hand baggage behind in case of an evacuation.
9. The use of PEDs in accordance with the PED policy, including a reminder to immediately inform the flight crew / cabin crew when a device is damaged, hot, produces smoke, is lost, or falls into the seat structure. Passengers shall be advised that power banks must not be carried in checked baggage and must be transported in the cabin. If a power bank is used during the flight, it must remain under constant supervision and kept visible at all times. The charging of power banks onboard is strictly prohibited.
10. Location and presentation of the safety briefing card, the importance of its content, and the need for passengers to review it prior to take-off.
11. The location and use of oxygen equipment if such is required (see [Oxygen Requirements](#)). Passengers shall also be briefed to extinguish all smoking material when oxygen is being used or supplied.
12. The location and use of life jackets for flights over water and at a distance of more than 50 NM from the shore, or where the take-off or approach path at an aerodrome is so disposed over water that in the event of an emergency there is a likelihood of ditching.

When infants are carried in the circumstances above:

- a. The commander shall ensure that an infant life jacket is distributed to and placed within reach of the passenger responsible for the infant. The responsible passenger(s) shall be briefed on the use of the infant life jacket, and
- b. On flights where cabin crew are carried, the Commander shall advise the cabin crew member when this procedure is required, and

- c. At the end of the flight or at a time when the requirement is no longer necessary, crew are to ensure that the infant life jackets are returned to their normal stowing location.

8.3.16.1.2 Climb, Cruise, and Descent

After take-off, the passengers shall be briefed, either verbally or by switching on or off the illuminated signs, on the following procedures if applicable:

1. The smoking ban or permission to smoke.
2. The observation of the FSB sign and use of safety belts and / or safety harnesses during:
 - CLIMB – FSB will remain illuminated until reaching first planned cruising level;
 - CRUISE – the FSB sign shall be turned off at the discretion of the Commander. However, it shall be recommended that passengers keep their seat belts fastened even if the FSB sign has been switched off, in case unexpected turbulence (weather or wake) is encountered;
 - DESCENT – the FSB sign shall be illuminated when the aeroplane commences its descent from the cruise level. At approximately 5 mins to TOD, the owners shall be informed that the FSB sign will be turned ON, for owner safety.

NOTE

In case of turbulence, the NO SMOKING signs may additionally be switched on in order to reduce the risk of fire, or for the well-being of the passengers since nicotine increases susceptibility to air-sickness.

8.3.16.1.3 Prior to Landing

The FASTEN SEAT BELT sign shall be switched on in due time prior to landing in order to ensure the cabin is properly secured for landing.

The designated crew member shall verbally brief the passengers on the following procedures if applicable:

- The imminence of approach and landing;
- The requirement to observe the NO SMOKING and FASTEN SEAT BELT signs as soon as they are switched on again;
- Cabin secure items to include, as applicable:
 - backs of the seats to be in the upright position, tables stowed, footrests stowed, window blinds open;
 - re-stow hand baggage in its appropriate location and the importance of leaving hand baggage behind in case of evacuation.
- Restrictions on the use of portable electronic devices in accordance with the PED Policy.
- The location of the safety briefing card and the importance of its content and review.

NOTE

On aeroplanes where cabin crew is not carried, the above briefing may be carried out by the playing of a pre-recorded audio or video message, or by a flight crew member via the PA system.

Shortly before landing, the NO SMOKING sign shall be switched on.

8.3.16.1.4 After Landing

After landing, the passengers shall be reminded, either verbally or by means of the illuminated signs, about:

- Observing the smoking ban;
- Keeping safety belts and / or safety harnesses fastened until the aeroplane has come to a full stop, and the engines have been shut down and the FSB sign has been turned off;
- Restrictions on the use of portable electronic devices in accordance with the PED Policy.

In addition, they shall be informed of all other provisions relevant to their safety during disembarkation.

8.3.16.1.5 Emergency Situations

In an emergency, passengers shall be instructed in such emergency action as may be appropriate to the circumstances (see AOM and *SEP Manual*).

8.3.16.2 Supernumerary Briefing Procedures

All supernumeraries that are carried in the passenger cabin shall be briefed by the cabin crew or appropriate crew member on the onboard smoking policy, compliance with the NO SMOKING sign, and compliance with the FASTEN SEAT BELT sign.

Supernumeraries shall also be briefed and instructed to ensure that they do not interfere with or impede any flight crew or cabin crew in the performance of their duties.

8.3.16.3 Briefing of SCPs

The briefing of SCPs shall be accomplished by a crew member once the SCP(s) is / are onboard. The information to be provided is as follows:

The SCP shall be briefed that crew can only assist the SCP once the cabin has been evacuated. The following table contains additional information by SCP category.

SCP category	Type of information
Adult travelling with an infant	Information on brace position for adult with lap-held infant. Information on the use of the loop belt, in case of a lap-held infant. Information to fit own oxygen mask before fitting the infant's oxygen mask. Information on how to evacuate when carrying an infant. In case of ditching, how to fit and when to inflate infant flotation aid (e.g. life vest, flotation device).
Physically disabled passenger (aided walking)	Inform the SCP to leave mobility aid behind in an emergency evacuation.
Passenger with disability of upper limbs	Inform the accompanying passenger to: a. fit the life jacket on the SCP, in case of a ditching evacuation; b. first put on their own oxygen mask before fitting the SCP's oxygen mask, in case of decompression; and c. secure / release the SCP's seat belt, if necessary.
Passenger with disability of lower limbs	Inform the SCP: a. on the location of the nearest suitable exit; and b. that mobility aids might not be accessible in an emergency evacuation.

SCP category	Type of information
Passenger with disability of both upper and lower limbs	Inform accompanying passenger to secure / release the SCP's seat belt. Inform the SCP: a. in case of an evacuation, on the location of the nearest suitable exit; b. in case of a ditching evacuation, that the accompanying passenger should fit the life jacket on the SCP; and c. in case of a decompression, that the accompanying passenger should first put on their own oxygen mask before fitting the SCP's oxygen mask.
Visually impaired passenger	Depending on the level of impairment, inform the visually impaired passenger on the following: a. seat and seat belt operation; b. location of the nearest exit (e.g. number of seats rows to the nearest exit); c. oxygen mask deployment; d. location of life jacket; e. brace position; and f. location of cabin crew call button. If available, take the aeroplane demonstration equipment to the passenger for tactile assistance.
Passenger travelling with a recognised assistance dog in the cabin	Advise how to evacuate the dog.

8.3.17 Procedures for Cosmic or Solar Radiation Detection Equipment

NTA provides crew members with workplace protection against the dangers arising from exposure to ionising radiation by complying with Council Directive 2013/59/EURATOM.

When crew members' work-related exposure to cosmic radiation (including positioning flights) is liable to be above 1 mSv per year, NTA shall, for the crew concerned:

- a. Assess their exposure;
- b. Take into account assessed exposure when organising working schedules, with a view to reduce the doses of highly exposed crew members;
- c. Inform the crew members concerned of the health risks their work involves and their individual dose;
- d. Ensure that crew members, once they have notified NTA that they are pregnant, keep the equivalent dose to the fetus as low as reasonably achievable and unlikely to exceed 1 mSv during at least the remainder of the pregnancy.

NTA shall, in accordance with the applicable regulation, keep a record of the individual monitoring results, which will be made available to the crew member, upon request.

The active cosmic radiation exposure monitoring is performed by Global Aviation Data A/S (GloboLog Service). The report provided by the GloboLog service consists of the following:

- Exposure of a flight;
- Total accumulated exposure (lifetime or since start of using GloboLog);
- Total exposure accumulated over the calendar year;
- Accumulated exposure during the last 12 months;
- Accumulated exposure since declaration of pregnancy.

8.3.18 Autopilot, Autothrottle, and Automation Policy / Level-off Awareness

The autopilot and autothrottle shall be used according to manufacturer recommendations and, if provided, procedures in the OM-B.

Automation Levels & Situational Awareness

Flight related tasks shall be accomplished in accordance with the following priorities:

Fly (Aviate): Control flight parameters to capture and maintain the desired targets, vertical and lateral flight paths.

Navigate: Select the desired modes for vertical and lateral navigation, being aware of surrounding terrain and MSA.

Communicate: Involves communications between flight crew and ATC and between flight crew members.

Manage: Includes managing aeroplane systems and performing the applicable checklists to manage the continuation of the flight.

The purpose of automation is to assist pilots in flight path control and navigation, freeing their mental resources for other tasks and enhancing their situational awareness. The degree of automation appropriate to the prevailing circumstances will vary and pilots should use automation accordingly.

In high workload situations, it is strongly recommended that flight crew use the highest level of automation appropriate to the situation.

Nevertheless, like any other resource, automation shall be managed and pilots shall be proficient in:

- Operating the aeroplane at all levels of automation;
- Selecting the appropriate level of automation.

Automated Cockpit

Regardless of the automation level selected the following shall apply:

- For departure the PF shall maintain one hand on the control column and one hand on the thrust levers, as systems operation permit, until the aeroplane is clean and has reached 3,000 ft AAL on departure.
- On arrival the PF shall maintain one hand on the control column and one hand on the thrust levers, as systems operation permit, at all times below 3,000 ft AAL.
- This is to achieve the fastest reaction in the event of an unexpected aeroplane upset.

Interfacing with Automation - Data Check

All commands or data shall be checked before being allowed to affect the operation of the aeroplane.

Flight Crew shall ensure:

- Before making any flight guidance selection, that they check that the knob or push button or menu selection is the correct one for the desired function;
- That after each action they verify the result of this action on:
 - PFD/FMA (i.e. for arming or engagement of modes); and / or,
 - PFD/MFD data (i.e. for selected targets, like altitude pre-selection and NAV data); and,
 - By reference to the aeroplane flight path and airspeed response.
- That they announce all changes in accordance with standard call-outs;
- When changing the selected altitude that they cross-check the altitude indication on PFD;
- During descent, that the selected altitude is not below the MEA or MSA unless under published approach procedures or under radar vectoring or performing a VFR approach in VMC.
- Whenever possible, the FMS is prepared for arrival before starting the descent;
- In case of a routing change that they always cross-check the new route selection;
 - Caution is essential during descent in mountainous areas. Ensure that the new track and assigned altitude are not below the sector safe altitude;
 - If ATC routing changes are near a restricted area, assess their impact in the portable EFB enroute charts by introducing the new flight route, before activating any change to the FMS;
 - If under radar vectors, be aware of the sector minimum vectoring altitude;
 - If necessary, the selected heading mode can be used with reference to Conventional NAVAIDS data, while verifying the new route and / or requesting confirmation from ATC;
- Before using lateral NAV mode, that the correct active way point (i.e. TO waypoint) is displayed;
 - Monitor the correct interception of the FMS lateral flight plan;
- That in the case of a late routing or runway change, a reversion to basic guidance modes and raw data may be considered; and FMS reprogramming should be avoided.

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Supervising Automation

Supervising automation is simply monitoring FCC displays and indications to ensure that the aeroplane response matches mode selections and guidance target entries, and that the aeroplane attitude, speed and trajectory match the crew's expectations, e.g.:

- The autopilot shall only be engaged if the FD is centred (i.e. the current flight path matches the selected / desired flight path);
- During capture phases, observe the progressive centring of FD bars and the progressive centring of deviation symbols (i.e. during localiser and glideslope capture);
- Enhancing the supervision of automation during capture phases and crosscheck with raw data, as applicable enables the early detection of a false capture or of the capture of an incorrect beam (e.g. ILS in maintenance mode emitting a permanent on-glideslope signal);
- Do not attempt to analyse or rectify an anomaly by reprogramming the flight guidance modes or FMS, until the desired flight path and / or airspeed are restored;
- At any time, if the aeroplane does not follow the desired flight path and / or airspeed, revert to a more direct level of automation, e.g.:
 - revert from FMS-managed modes to basic modes;
 - disconnect AP and follow FD guidance (if correct); or
 - disengage FD and hand fly the aeroplane, using raw data.
- When operating with the autopilot not engaged and the FD displayed, the FD shall be followed unless the FD is indicating a flight condition not desired by the PF; in this case the FD shall be switched off. The cause shall only be investigated by the PM once the aeroplane has been returned to the desired flight condition

Use of Automation / Flight Director when intercepting an ILS

It is recommended to always intercept an ILS glideslope from below, when doing so the glideslope capture mode shall be activated in accordance with the relevant OM-B.

If it becomes necessary to intercept an ILS glideslope from above with the autopilot engaged then the glideslope capture mode shall only be activated once the aircraft is established on the glideslope and the aircraft has been confirmed (e.g. By DME fix) to be on the correct glideslope.

NOTE

Activating the glideslope capture mode with the autopilot engaged when above the glideslope can lead to the capture of a false glideslope with a reversed signal, which may cause a severe pitch up or down.

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Task Allocation

Management of the AP/FD control panel is based on whether or not the AP is engaged.

PF	PM
AP ENGAGED	
• Selects FD modes (Note 2); • Sets headings (and courses when using raw data); • Sets target altitude / FLs; • Makes long and short term navigation inputs to the FMS when workload permits, otherwise calls for long or short term navigation inputs to the FMS as dictated by workload; • Long term navigation inputs by PF are prohibited during critical phases of flight (Note 1).	• Crosschecks PF inputs (Note 2). • Makes long or short term navigation inputs to the FMS when called for by the PF (Note 1).
AP NOT ENGAGED	
• Calls for AP modes if needed; • Calls for FD modes (Note 2); • Calls for mode value changes e.g. heading, course, and target altitudes / FL's settings; • Calls for ALL FMS inputs (long and short term); • Monitors and checks settings and inputs made by PM, then adjusts flight path accordingly.	• Selects called-for AP modes; • Selects called-for requested FD modes (Note 2); • Sets mode value changes e.g. headings and courses, target altitudes / FL's when called for by the PF or to comply with ATC instruction (depending on context); • Inserts ALL navigation inputs in the FMS when called for by PF.

Short term navigation inputs are those with an immediate effect on the flight path, such as inserting a 'direct to'. These are inputs that do not require browsing between different pages of the FMS via the menu system.

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Long term navigation inputs are those which are required for major reprogramming of the FMS requiring access to different pages and lengthy verification of the route. Examples are inserting waypoints not contained in the flight plan, programming an arrival or holding pattern or programming a runway centreline.

NOTE

1. The PF shall ensure that FMS programming does not interfere with the primary task of flying the aeroplane. During times of high workload, particularly during critical phases of flight, it may be more appropriate to request all FMS inputs from the PM.
2. Modes shall be selected on both mode control panels for aeroplanes that have dual mode control panels.

CAUTION

The commander shall ensure that during all phases of flight, one pilot is supervising and monitoring flight progress and aeroplane systems.

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FMA Callouts

Each pilot shall confirm the selected modes by reading the mode annunciation on their PFD, not by announcing the desired selection.

The Standard Procedures and Callouts that are listed below shall be used and are applicable to all fleets.

For other type specific callouts refer to the respective aeroplane OM-B.

FD Mode Selection

SITUATION	PF	PM
AUTOMATIC CHANGE OR SELECTION OF FD MODE (AP ON) PF selects mode or observes automatic change and calls out PFD annunciation. PM verifies selected mode.	'[XX]GREEN / WHITE'	
SELECTION OF FD MODE (AP OFF) PF requests mode selection. PM selects FD mode and calls out PFD annunciation. PF verifies selected mode.	'SELECT [XX]'	'[XX] GREEN / WHITE'

FD values setting

SITUATION	PF	PM
SETTING FD MODE VALUE ON ATC INSTRUCTION (AP ON) PM acknowledges ATC instruction. PF makes selection of new value. PM verifies correct value selected.		(ATC READBACK)
SETTING FD MODE VALUE ON ATC INSTRUCTION (AP OFF) PM acknowledges ATC instruction and then makes selection of new value. PF verifies correct value selected.		(ATC READBACK)
SETTING FD MODE VALUE ON REQUEST OF PF (AP OFF) PF requests FD value change. PM selects new FD value and calls out new FD value (e.g. 800 fpm).	'SELECT [XXX]'	'[XXX]'

Altitude setting in ALT selector (Altitude Alert Device)

SITUATION	PF	PM
SETTING OF NEW ALTITUDE OR FLIGHT LEVEL (AP ON) PM acknowledges ATC instruction. PF sets new value. PM verifies correct value is set.	'FL [XXX] SET'	'CHECKED'
SETTING OF NEW ALTITUDE OR FLIGHT LEVEL (AP OFF) PM acknowledges ATC instruction and then sets new value. PF verifies correct value is set.	'CHECKED'	'FL [XXX] SET'

Level-off Awareness

To promote situational awareness of the vertical rate approaching level-off, the following callouts shall be used:

SITUATION	PF	PM
2000 FT TO GO (QFE/QNH SET)		'TWO THOUSAND TO GO'
2000 FT TO GO (STD SET)		'TWO TO GO'

The callout is not required:

1. During a departure with a level-off at or below 3,000 ft AAL, as this procedure shall be briefed in accordance with OM-B;
2. Whenever the change in altitude is 2,000 ft or less;
3. Whenever communicating with ATC.

The vertical rate shall be adjusted to meet a maximum 1,500 fpm by 1,000 ft to go to cleared altitude or level.

During the last 1,000 ft to-go before a level-off, non-essential communications, conversations, and activities should be avoided and attention should be focussed on monitoring the level-off.

Autothrottle

Refer to the aeroplane specific OM-B.

8.3.19 Crew Coordination

Crew members shall use all resources available to them to ensure that communication paths are established and kept open.

Although the Commander is the final authority, all crew members should be encouraged to contribute to the decision-making process.

8.3.19.1 Closed Loop Policy

The 'closed loop' method shall be used for configuration, altimeter and critical system changes.

The method is as follows:

- PF checks the conditions are appropriate for an action and calls for the action.
- PM confirms the conditions are appropriate and verbalises the condition or deviation.
If conditions are appropriate, they perform the action.
- PM announces the end result of the action and thus closes the loop.
- PF verifies the end result meaning both crew members are now aware of the status of a particular system.

8.3.19.2 'By Exception' Policy

The 'by exception' rule shall be used for less critical items whereby the pilot makes a call-out only if the system or procedure has deviated from the assigned parameters.

Example: When monitoring the engine start, communication shall be kept to a minimum but in the case of a deviation, either pilot can call-out the deviation such as 'No oil pressure'.

The 'by exception' rule can also be used in times of high workload in order to allow the PF to concentrate on the primary task of flying the aeroplane and the PM to monitor.

Example: During a PAR approach, heading and VS changes made by the PF are monitored silently by the PM. Only deviations are called by the PM such as 'Check heading' to alert the PF if the commanded input differs from the ATC instruction.

Non-Standard

In cases where crews decide to use non-standard procedures, the following requirements must be observed:

- Is agreed by both FCM; and
- It has an operational benefit; and
- The non-standard procedure is as safe as following the SOP; and
- It is a one-off event, and should not routinely replace an SOP action; and
- It may be specific to a location or event and briefed beforehand. For example, delaying the retraction of flaps departing EGKB RWY 21.

8.3.19.3 Monitoring and Crosschecking

The PF is responsible for ensuring the aircraft is correctly configured for each phase of flight, and for utilising the most appropriate handling methods to ensure the desired parameters are achieved. Both PF & PM must monitor the flight path, background radio communications, each other's actions and input selections throughout all flight phases, using their own instruments as the primary source of information.

A crew member noting a deviation from the intended ground or flight path at any time shall immediately call it to the attention of the other flight crew member. PF shall verbally state their intention and initiate appropriate actions.

Flight Path Deviation	Callout	
	PF	PM
When bank angle exceeds 30°.		BANK ANGLE
When IAS is more than +10/-5 knots from the PF nominated ¹ value.		SPEED
More than 100 ft from planned height / altitude / FL.		ALTITUDE
1 dot or more from the ILS LOC.		LOCALISER
1 dot or more from the ILS GS.		GLIDEPATH
Excessive rate of descent (according to phase of flight & altitude).		SINK RATE
Acknowledges and initiates corrective action.	CORRECTING	

¹ Unless the PF nominated value is less than 5 kt greater than the Safe speed during take-off, approach, or landing in which case a tolerance of +10/-0 KT shall apply.

Should a crew member not respond to two verbal communications, or not respond to any verbal communication associated with a significant deviation from intended ground or flight path or standard operating procedures, the challenging crew member will take control of the aeroplane.

When addressing a deviation, the crew should use the following escalation path:

1. **Prompt** (Low assertiveness)

Goal: Raise awareness of the problem and allow timely correction.

Examples: 'check speed' / 'the stop bar is still lit' / 'crosswind is xx, out of limits'.

2. **Direct** (Clear and assertive)

Goal: Clearly state the required action. (Repeatedly stating the problem will not help if the PF has degraded situational awareness and / or excessive workload).

Examples: 'unstable – go-around' / 'stop stop' / 'crosswind out of limits, go-around'.

3. **Intervene** (Take action)

Goal: Prevent or stop an unsafe situation.

Action: Take control, stating - 'I have control'.

NOTE

In some safety-critical circumstances, it may be appropriate to jump directly to step 2. or 3. as applicable.

The following critical crew actions shall be crosschecked by both flight crew members.

- Performance calculations and entries into the FMS;
- Mass and Balance Calculations.

A pilot changing the status of a system outside the sequence of a standard procedure shall bring the change to the attention of the other crew member before proceeding.

Movement of switches or levers that are critical to the safe operation of the flight during abnormal / non-normal and emergency procedures refer to [Critical Items](#).

No movement of any switch normally associated with an abnormal or emergency procedure shall be made until called for by the checklist, unless included in the 'memory' items.

FMS inputs, including route and performance modifications during flight shall only be executed after confirmation by both pilots.

Whenever FMS is being used as the primary means of en route navigation, the PM should reference appropriate NAVAIDS and a suitable navigation chart to monitor the flight's progress.

8.3.19.4 Transferring Control of the Aeroplane on Request of the Commander / Take Over for Safety Reasons

The following standard procedure and callouts shall be used:

SITUATION	PF	PM
COMMANDER OR PM NEEDS TO TAKE CONTROL PM or Commander announces. PF acknowledges handover.	'YOU HAVE CONTROL'	'I HAVE CONTROL'
TRANSFER OF CONTROL ON REQUEST OF PF PF gives a description of the flight path, configuration, and modes where relevant. (Example): PM acknowledges handover.	'HEADING [XXX], FL [XXX], YOU HAVE CONTROL'	'I HAVE CONTROL'

The PF shall transfer control each time they need to engage in an activity that may compromise their ability to perform their primary tasks or when the failure of primary flight instruments on their side requires them to do so.

Whatever the automation level, all transfers of control will result in a swap in task allocation, ensuring no doubt exists over which crew member is responsible for control of the aeroplane.

Depending on the situation, consideration shall be given to transferring the AP input to the side of the pilot assuming control.

CAUTION

If flying manually, the PF shall make every effort to ensure that the aeroplane is correctly trimmed before handing over control, to avoid any potential for an aeroplane upset.

8.3.19.5 Transfer of Radio Communications

For transfer / handover of radio communications, the following standard procedure and callouts shall be used.

SITUATION	PF	PM
COMMUNICATIONS TRANSFER Pilot specifies any significant points or changes.	'YOU HAVE COMM ONE'	'I HAVE COMM ONE'

8.3.19.6 Sterile Flight Crew Compartment

AMC1 ORO.GEN 110(f)

General

Sterile FCC procedures are applied to ensure that flight crew are not disturbed or distracted, except for matters critical to the safe operations of the aeroplane or the safety of the occupants, and are focused on the essential operations and safety critical activities.

Cabin crew shall be aware of the sterile FCC procedures and, during these flight phases, refrain from distracting the flight crew members from concentration on their duties (unless safety requires such distraction), and shall assist the flight crew by providing, in due time, the 'CABIN SECURE' or the 'CABIN NOT SECURE' report.

Sterile Flight Crew Compartment procedures shall be applied:

- a. During critical phases of flight;
- b. During taxiing;
- c. Below 10,000 feet above the aerodrome of departure after take-off and the aerodrome of destination before landing, or MSA whichever is higher, except for cruise flight above MSA; and
- d. When determined by the pilot in command.

Flight crew activities

When sterile FCC procedures are applied, communications and actions in the FCC shall be limited to those directly related to the safe operation of the aeroplane, ensuring the flight crew remain focused on their essential operational activities without being disturbed by non-flight related matters.

Examples of activities not directly related to the safe operation of the aeroplane include:

- a. Non-critical communications (FBO, passengers, CCM, etc.);
- b. Non-critical administrative duties (Flight logs, W&B, etc.);
- c. Operating equipment not related to the conduct of the flight (phones, cameras, etc.);
- d. Eating, drinking, sightseeing.

Cabin crew communication to the FCC crew

Cabin crew shall use their own discretion to determine whether the situation is related to safety and security and whether to call or interrupt the flight crew.

Critical situations requiring the flight crew to be contacted may include:

- a. Any outbreak of fire inside the cabin or in an engine;
- b. A burning smell in the cabin or presence of smoke inside or outside;
- c. Fuel or fluid leakage;
- d. Localised extreme cabin temperature changes;
- e. Evidence of airframe icing;
- f. Cabin / galley equipment or furniture malfunction / breakage posing a hazard to the occupants;
- g. Suspicious object;
- h. Disruptive passenger;
- i. Security threat;
- j. Abnormal vibration or noise;
- k. Medical emergency;
- l. General drop-down of the oxygen masks in the cabin;
- m. Any other condition deemed relevant by a cabin crew member; and
- n. Reporting of CABIN SECURE to the flight crew.

Examples of situations that do not warrant sterile cockpit Interruption:

- Non-safety-related cabin defect reports;
- Meal preferences;
- Requests to FBO;
- Catering issues;
- Passenger accommodations;
- Rude passengers.

8.3.19.7 Fasten Seat Belt and No Smoking Sign Procedure (Aeroplane Without Cabin Crew)

FSB=Fasten Seat Belt

NS=No Smoking

SBF=Seat Belt
Fastened

The table below shows the standard general procedures for the use of the FSB and NS signs. Refer to the respective OM-B for type specific differences and procedures.

COMMUNICATION	SIGNAL	ACTION / CONDITION
Cockpit Preparation	NS Sign ON	
Engine Start	FSB sign ON NS sign ON NS sign ON (already ON)	1. PAX seated with SBF 2. Ensure Cabin is Prepared for taxiing and take-off
Imminent take-off	FSB sign cycled (chime)	Ensure Cabin is secured for take-off
In climb passing 10000 ft AAL or MSA, whichever is higher End of Sterile FCC	NS sign OFF	PAX may smoke
Top of climb	FSB sign OFF	PAX may move around the cabin
Turbulence	FSB sign ON	Ensure PAX are seated with SBF
Moderate or severe turbulence	FSB sign ON NS Sign ON	1. PAX seated with SBF 2. PAX stop smoking
Top of descent	FSB sign ON	PAX seated with SBF
In descent passing 10000 ft AAL or MSA, whichever is higher Start of Sterile FCC	NS sign ON FSB sign ON (already ON)	1. Ensure Cabin is Prepared for landing 2. PAX stop smoking
After Engine shutdown	FSB OFF	PAX may move around the cabin

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8.3.19.8 Fasten Seat Belt and No Smoking Sign Procedure (Aeroplane With Cabin Crew)

FSB=Fasten Seat Belt

NS=No Smoking

SBF=Seat Belt
Fastened

Refer to the respective OM-B for type specific differences and procedures.

Effective communication shall be used between the FCC and CCM when the tables below do not cover the situation as described. This includes but is not limited to:

1. CCM sitting down with SBF and stopping any service, if conditions so dictate (this shall be communicated to the FCC).
2. The suspension of certain services (hot meal or drinks) if turbulence or manoeuvring could endanger passengers or crew.
3. FCC advising CCM of possible turbulence or flight manoeuvres.

Normal Procedures

COMMUNICATION	SIGNAL	CABIN CREW ACTION / CONDITION
Cockpit preparation	NS sign ON	
Engine Start	FSB sign ON NS sign ON (Already ON)	1. PAX Seated with SBF 2. Limited Service 3. Prepare Cabin for Take-off
Cabin secure		Cabin secure reported to FCC and inform CCM Seating location
Imminent take-off (Prior to lineup)	NS sign cycled (chime)	CCM seated with SBF
After take-off CCM release	NS sign OFF	CCM can move in the cabin and galley and begin service
In climb passing 10,000 ft AAL or MSA, whichever is higher End of Sterile FCC	NS sign cycled ON and then OFF (chime)	
Top of climb	FSB sign OFF	PAX may move around the cabin
Turbulence	FSB sign ON	Ensure PAX are seated with SBF
Moderate or severe turbulence	FSB sign ON NS sign ON	1. Interrupt service immediately 2. PAX seated with SBF 3. PAX stop smoking 4. CCM seated with SBF

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COMMUNICATION	SIGNAL	CABIN CREW ACTION / CONDITION
Top of descent	FSB sign ON	PAX seated with SBF
In descent passing 10,000 ft AAL or MSA, whichever is higher (see note) Start of Sterile FCC	NS sign cycled ON and then OFF (chime)	1. Limited Service 2. Prepare Cabin for landing
Cabin secure		Cabin secure reported to FCC and inform CCM Seating location
Cabin Crew take their seats for landing	NS sign ON	1. PAX stop smoking 2. CCM seated with SBF
After landing and safe for CCM to move in the cabin	NS cycled (chime)	CCM can move about the cabin
After Engine shutdown	FSB OFF	1. PAX may move around the cabin 2. Commander advises when the main door can be opened
CCM to FCC	NS sign cycled twice (2x chime) or another method agreed between the FC and CC	CCM goes to the FCC to receive brief

Abnormal / Emergency Procedures

COMMUNICATION	SIGNAL	CABIN CREW ACTION / CONDITION
Rapid Decompression	FSB sign ON NS sign ON O2 masks drop down	1. Interrupt service immediately 2. PAX and CC don nearest O ₂ mask and strap into the
Unlawful interference	No Signal	In the case of unlawful interference crew should use all possible means to communicate the situation
Other abnormal and emergency procedures including Crew Incapacitation	PA 'Cabin Crew to the FCC'	1. Interrupt service immediately 2. CC goes to the FCC to receive brief

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8.3.19.9 Cabin and Hold Loading / Unloading and Securing

Before every revenue, one of the flight crew members, or the OSR with agreement from the crew, must meet the passengers at the agreed meeting point if applicable, count all hold luggage and confirm with passengers.

The following general guidelines should always apply:

- The crew member loading the luggage should be the one unloading the luggage.
- The Cabin must be searched immediately after each flight by at least one crew member.
- The flight crew member (or OSR) escorting the passengers must wait for confirmation (Thumbs-up) that the Cabin and the luggage hold are clear of all passenger luggage.
- Special attention must be paid when passengers have access to the luggage hold that nothing is left behind in this area.

Receiving and Loading Luggage:

The escorting flight crew member (or OSR) should inform the remaining flight crew member on the aeroplane of the total number of hold luggage items being sent to the aeroplane. This may be by means of a message or phone call, or the FBO may forward a luggage slip (example as below) in accordance with the *Ground Handling Manual*, 6.3 Aircraft Load / Unloading Specifications.

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Netjets Luggage Log

(Crew Copy)

Airport

C/S

N/E

A/C

CS-

Nbr of Items

Date

Time

Agent Name

Signature

Netjets Luggage Log

(FBO Copy)

Airport

c/s

N/E

A/C

CS-

Nbr of Items

Date

Time

Agent Name

Signature

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The flight crew member on the aeroplane will compare the amount of luggage notified with the luggage received and load and secure the luggage in the hold. If a cabin crew member is present, they may assist.

Upon passengers boarding, both flight crew members should cross-check verbally the number of hold luggage onboard, inform the cabin crew member (if present) and confirm with the passengers before closing the aeroplane's door.

If a cabin crew member is present, it is their responsibility to secure all passenger hand luggage in the cabin. The cabin crew member must inform the flight crew about any luggage loaded into the cabin.

If a cabin crew member is not present, the flight crew member assisting the passengers to board is also responsible for securing the passenger hand luggage in the cabin for taxi, take-off, and landing.

Luggage Unloading

The flight crew member unloading the luggage should be the one who loaded the luggage.

On aeroplanes without cabin crew, one flight crew member must then proceed with a cabin check ensuring the passengers have not left any of their belongings behind. On aeroplanes with cabin crew, the cabin crew member should normally do this check. This check shall be performed as soon as possible and should consist of the following as a minimum:

Toilet Area

- Check that there are no passengers' garments remaining on any hanging rails.
- Check that no passengers' bags are remaining in the toilet area or on the toilet seat.
- Check the floor area for any fallen items (e.g. jewellery).
- Check inside vanity cupboards for passengers' personal items.

Cabin - Rear to Front

- Check down the side of each seat by sliding it to the inboard position and checking between the seat and the side-wall.
- Check on top of each seat.
- Recline seats to confirm nothing has fallen within the seat.
- Check the entire cabin floor area.
- Check the ashtrays, cup-holders, and side compartments.
- Check magazine holders.

Other Luggage Hold

- Confirm there are no passengers' bags remaining in the rear luggage hold. Given our passengers have access to the baggage hold (on some aeroplane types), it may occur that luggage is added during the flight which is not accounted for during the initial load.
- The crew member in charge of unloading the baggage compartment should check with the cabin crew if any luggage has been moved from the cabin to the luggage compartment, and, if so, where this luggage has been placed.
- Check that there are no passengers' garments remaining on any hanging rails.

Upon completing the cabin check, confirmation that no items were left behind must be given to the flight crew member escorting the passengers or OSR by suitable means, such as a thumbs-up signal.

If any items are found, these must be returned to the passengers immediately.

If for any reason the passengers have departed, the quickest and most reasonable point of contact must be established with either:

- Flight crew member (or OSR) escorting the passengers;
- FBO handling agent;
- Owner Services Team;
- Flight Operations - Chief Pilot or Cabin Crew Assistant Fleet Chief.

8.3.19.9.1 Manual Handling of Luggage

Guidelines to prevent back injuries

Loading and unloading of luggage can pose a risk of back injuries for crew members, leading to immediate pain and discomfort, as well as long-term issues if not properly managed. Accessing luggage compartments can be challenging, and assistance may not always be readily available. To mitigate these risks, NetJets offers operational guidelines and training during GRT.

The main risk areas during the baggage handling operation are:

- Handling items inside the luggage compartment, in particular stacking bags;
- Frequency and weight of items being handled;
- Handling items with arms extended away from the low back;
- Handling items below knee height and above shoulder height;
- Pushing and pulling equipment.

Risk reduction measures

As for the operational guidance, the following applies:

- Crew members must avoid loading items that can foreseeably be challenging or pose a high risk of injury, either from their weight or volume. Before any luggage is loaded on board it must be weighted. This can be done via personal judgement or by using the onboard scale while using the proper lifting technique as trained.

If the luggage appears too heavy for an individual's personal ability or exceeds 20 kg, it must be loaded with the assistance of at least one supporter.

- Assistance can be provided by the other crewmember or a service provider.
- When handling luggage, it is crucial to maintain a correct body posture at all times. In particular, ensure that you are using safe lifting techniques while moving items into the luggage hold. If you find that you need assistance, don't hesitate to ask for help. If, despite receiving support from others, it is not possible to move the luggage safely with the correct posture, the items shall not be loaded. Instead, notify the NJE Chief Pilot and request further guidance and assistance.
- Allow sufficient breaks. The table below gives you an idea on the mitigation of risk when waiting in-between lifting luggage (MAC).

MAC lifting score for one operator outside the aircraft				
Bag Weight	One lift every			
	5 seconds	9 seconds	14 seconds	30 seconds
Less than 10 kg	Green	Green	Green	Green
10 kg	Amber	Amber	Green	Green
15 kg	Amber	Amber	Amber	Amber
20 kg	Red	Red	Amber	Amber
25 kg	Red	Red	Red	Red
32 kg (IATA max)	Purple	Purple	Red	Red

Colour coding:

Green	Low level of risk.
Amber	Medium level of risk, task should be examined closely.
Red	High level of risk, prompt action needed.
Purple	Very high level of risk, such operation may represent a serious risk of injury and should come under close scrutiny, particularly when the entire weight of the load is supported by one person.

- Use heavy bags as the base of the stack to reduce the need to lift bags into place.
- Avoid moving luggage while kneeling.
- If no support is available, the crew member should notify the NJE Chief Pilot to ask for further guidance and support.

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8.3.20 Briefing Procedures

8.3.20.1 Policy

The crew member leading the briefing should use an open style that encourages participation of all other crew members. The briefing should confirm the agreement and understanding of the PM and provide an opportunity for the PF and PM to correct each other.

Briefings should be structured and follow a logical sequence of events and actions.

The boxed structure below offers a basic framework of mandatory items to be used as a template for departure and arrival briefings:

1. WNPFA
2. PLAN
3. HOW TO
4. CONTINGENCIES
5. HAZARDS AND MITIGATION

The brief shall be tailored so as to give special attention to hazard avoidance, and mitigation of the threat posed by such hazards, by highlighting special aspects of the procedure being briefed.

1. WNPFA

- WX: review for the relevant airports of the flight.
- NOTAMS: review and discuss all applicable NOTAMS for possible operational impact.
- Performance: runway conditions, performance data, MATOM, MLM, Speeds.
- Fuel: fuel on board, minimum diversion fuel, and applicable holding fuel and time.
- Aeroplane status: review the aeroplane status and discuss the possible consequences in terms of operation and performance.

2. PLAN

Taxi, SID, STAR, and approach type to be expected. This should be cross-checked by the PM in the FMS and any deviations pointed out and acted on.

3. HOW TO

Take-off and approach configuration, profiles, and operating techniques. Flight path management. Special airports brief.

4. CONTINGENCIES

Specific configuration and profiles for one engine inoperative procedure, missed approach procedure, and balked landing.

5. HAZARDS AND THREAT MITIGATION

Hazards and threat mitigation should be pointed out by the PF during 1-4 and then summarised by the PM to ensure a common understanding of the threats. The PF should then assess if every relevant hazard was covered.

To avoid repetitive briefing of standard operating actions and procedures, these may be substituted by the phrase 'standard procedures', e.g. '**Standard procedures to deal with an abort before V1**'.

The FMS, NAVAIDS, and displayed data, including the EFB, should be referred to whenever possible to ensure correct data entry and to verify crew member understanding. As such, cockpit equipment should be used as a visual aid to briefing.

In addition to the flight crew briefing outlined above, an owner experience briefing shall be conducted, at the discretion of the PIC, that follows NSSM, 3.3 Complete the Owner Experience Briefing guidelines. This briefing shall include potential hazards contributing to crew distraction, including but not limited to commercial pressure.

Should a CCM be operating as part of the crew they shall be included in the first flight of tours, daily briefing, end of day briefing, turn-around briefing, and any briefing incorporating owner experience aspects.

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8.3.20.2 Instructions for Crew Briefings

Flight briefings shall be performed in accordance with the following matrix:

Type of briefing	Led by	Attended by	Frequency	Reason for the briefing
First flight of Tour ¹	Commander	All crew ²	Before first flight of tour or following a change in crew composition	To establish an overview of the day and provide an opportunity to identify particular hazards to the safe and efficient operation of the flight(s). The briefing should include non-technical and owner experience aspects and may be performed as part of the daily briefing.
Daily Briefing	Commander	All crew ²	Before first sector of each day	The daily briefing should establish an overview of the day, with a focus on task allocation, hazards, operational, and owner experience aspects.
Turn-around briefing	Commander	All crew ²	At the beginning of each turnaround once tasks related to the previous flight are completed.	The briefing should establish an overview of the turnaround, with a focus on task allocation, hazards, operational, and owner experience aspects.
Departure briefing	PF	PF, PM	Before every departure	To inform the PM of the planned course of actions for both normal and abnormal conditions during the take-off and departure phase.
Arrival briefing	PF	PF, PM	Before every arrival	To inform the PM of the planned course of action for the arrival and should it be required, the go-around.
End of Day De-brief	Commander	All crew ²	After last flight of the day	To openly discuss the operational aspects of the day in order to highlight any positives as well as lessons learned, as well as owner experience aspects in order to ensure all feedback is gathered and passed on in accordance with NSSM, paragraph 7.4 Ensure Future Success guidelines.

¹ The first flight of tour briefing shall include the following mandatory items: rejected take-off, evacuation and engine failure after V1. Other items such as go-around actions may be discussed at the discretion of the commander.

² All crew defined as PF, PM, and where applicable CCM, and augmented FCM.

8.3.20.3 Scope of Crew Briefings

The departure and arrival briefings should be adapted to the specific conditions of the flight and focus on the items that are relevant for the particular phase of the departure or approach and landing. All flight briefings shall be structured in accordance with the guidance in [Policy](#).

8.3.20.4 Briefing Style

The importance of briefing techniques is often underestimated; effective briefings contribute to enhancing crew standardisation and communication.

The style and tone of the briefing play an important role; interactive briefings - i.e. confirming the agreement and understanding of the other crew members after each phase of the briefing - provide more effective and productive communication than an uninterrupted recitation terminated by the final query 'Any questions?'

An interactive style better fulfils an important purpose of the briefings; that is to provide the crew members with an opportunity to correct each other (e.g. confirming the use of the correct and effective departure / approach chart, confirming the correct setup of NAVAIDs for the assigned take-off / landing runway, etc).

Briefings should be concise and structured (i.e. follow the logical sequence of the takeoff / departure or approach / landing).

The routine and formal repetition of the same points on each sector may become counterproductive; adapting and expanding a routine briefing by highlighting the special aspects of the airport and departure / approach procedure, or the prevailing weather conditions and circumstances, results in more effective communication.

In short, the briefing should encourage effective listening by both crew members.

The briefing should be conducted when the workload and availability of the crew members permits effective communication.

Any aspect that may affect normal operation (e.g. system failures, weather conditions or other particular conditions) should be carefully evaluated and discussed.

An effective briefing should create a common mental model of the event being briefed, enhancing the ability of the crew members to visualise the sequence of events and actions, as well as recognise any associated hazards or special circumstances.

Whether anticipated or not, changes in an ATC clearance, weather conditions or landing runway require reviewing part of the initial briefing.

8.3.20.5 Hazard Identification and Avoidance and Threat Mitigation

A hazard is anything that increases operational complexity of the flight and requires crew attention and action to maintain safety margins.

Flight briefings should be seen as a tool that seeks to identify hazards (such as adverse weather, terrain), and errors (such as the selection of the wrong automation mode during a complex departure), and mitigate the threat they pose, to prevent the occurrence of an undesired aeroplane state (such as an altitude deviation or runway incursion). By tailoring flight briefings to give special attention to hazard avoidance and threat mitigation a crew increases their potential for maintaining adequate safety margins. Crews that plan ahead during low-workload phases of flight reduce task loading during high workload phases, thus increasing safety margins.

A crew should use all available resources to identify possible hazards before flight or during low workload phases (e.g. cruise flight), and manage these hazards BEFORE they negatively impact safety. Available resources include:

- Flight release
- Weather briefing
- NOTAMs
- Dispatch
- Hotline
- Airport briefing

Should any crew member have any concerns relating to any aspect of the flight(s), they should raise it to the attention of the commander. Where the commander is unable to mitigate a threat or safety concern, they should contact the NJE Chief Pilot or FOSM team for guidance.

A hazard may be expected or unexpected.

Expected Hazards - these are hazards for which the pilot has knowledge and can plan in advance how to mitigate them. For example:

- Severe weather
- Complex airspace and procedures
- Crew factors (time of day, fatigue)
- Unfamiliar airports
- Contaminated runways
- International procedures

Unexpected Hazards - these are unanticipated threats for which the pilot must act upon their occurrence. For example:

- Passenger distractions
- Commercial pressure considerations
- Similar sounding callsigns
- Medical emergencies
- Non-normal situations
- Aeroplane malfunctions

Once identified, the crew should discuss the hazard and put in place a plan to mitigate its effect.

For example, a hazard may be identified as '**thunderstorms during departure**'. Mitigation could be to ask **ATC for a track deviation** or to use the **weather radar early to identify a safe route around the weather**.

Threat mitigation shall be reinforced, at the request of the PF, by the PM summarising the hazards and the planned mitigation at the end of each departure and / or arrival brief.

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8.3.20.6 Briefing Requirements for a Change of Operating Crew in the FCC

When there is a change of operating crew in the FCC in accordance with [Flight Duty Time Limitations](#) augmented crew requirements, the Commander, or in their absence, the designated pilot relieving the Commander, shall brief the incoming pilot on the following:

Mandatory items

1. Crew task allocation (Delegation of pilot relieving the commander, PF / PM).
2. Current ATC frequency.
3. Current ATC clearance.
4. Next FMS waypoint and next reporting point, if applicable.
5. Aircraft technical status.
6. Any actual and forecast significant en-route weather.
7. Destination, en-route, and potential diversion airports - availability and weather.
8. Any other items that the Commander considers necessary for the safe operation of the flight during their absence.
9. Procedures to be used to summon them to the flight deck.

Items to be briefed if relevant

1. Any required or desired flight level change anticipated during the Commander's absence.
2. Secondary frequency.
3. Current fuel state and the fuel plan.
4. Weather radar serviceability and setting.
5. MSA (both off-route and en-route).
6. Transponder setting requirements.
7. Any special weather deviations, traffic separation, or contingency procedures applicable to the airspace through which the flight is being conducted.
8. Cabin status.

The Commander shall also brief the CCM on any relevant items to include as a minimum the delegation of the pilot relieving the Commander.

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8.3.21 Checklists

A checklist is an aid to memory to ensure that critical items necessary for the safe operation of an aircraft are not forgotten.

8.3.21.1 Checklist Methodology

Checklist management shall follow a Trigger, Flow, Checklist (TFC) methodology. A defined trigger will initiate a flow, followed by a checklist covering any critical items within the flow. On occasion, a trigger may directly initiate a checklist without the necessity for a preceding flow. Similarly, some flows might not need to be followed by a checklist.

The following definitions shall apply:

- **Trigger:** A trigger is a clear cue that cannot be easily removed, obstructed, or forgotten, such as the occurrence of a particular event or action.
- **Flow:** An ergonomically efficient sequence of actions performed from memory.
- **Challenge & Response Checklist:** Pilot A reads the challenge from the checklist, both Pilot A and B verify the item is set properly, and Pilot B responds with the verified status of the item, using the listed response. When the listed response is 'A/R' the actual value or switch / control position shall be stated.
- **Read & Do Checklist:** One pilot reads the challenge from the checklist, accomplishes or verifies its status, and calls out the verified status of the item.

The TFC checklist management philosophy shall be applied to all checklists from engine start to shutdown inclusive. This includes non-normal checklists, where the trigger will be the initiation of the procedure, the flow will encompass the memory items, and the checklist will be used to ensure full and correct completion of the non-normal procedure.

Regarding checklist methodology, the following principles apply:

- A specified crew member will be responsible for initiating each checklist;
- Not all items encompassed by a flow need to be included in the checklist;
- To initiate a flow or checklist, a trigger is defined that takes into account FCC workload;
- Flows may be accomplished simultaneously by PF and PM;
- A specified crew member is responsible for accomplishing each item on the checklist;
- The challenging crew member shall wait for a positive response (and shall crosscheck the validity of the response, as required) before moving to the next item.
- From the initiation of the engine start checklist to the completion of the engine shutdown checklist, any interrupted checklist shall be restarted, except where electronic checklists are employed that display the status of open items;
- Upon completion of any checklist, the crew member conducting it will report its completion;
- All crew members are responsible for highlighting any deviations from the checklist.

8.3.21.2 Checklist Procedures

Checklist procedures shall be conducted in a calm and deliberate manner, adhering to the methodology explained above, and ideally should be carried out during low-workload periods.

The following standard callouts shall be used:

SITUATION	PF	PM
PF calls for a checklist ¹	AFTER TAKE-OFF CHECKLIST	
PM starts checklist		AFTER TAKE-OFF CHECKLIST
Checklist accomplished down to the line		AFTER TAKE-OFF CHECKLIST TO THE LINE
Checklist completed		AFTER TAKE-OFF CHECKLIST COMPLETE
Reminder if not called for at associated trigger ²		STANDING BY FOR (...) CHECKLIST

¹ PF should allow sufficient time for relevant flows to be completed before calling for the checklist. This terminology should also be used if PF notices that PM has not initiated a flow following a defined trigger.

² Should a PF flow not be automatically conducted following a defined trigger, this terminology should be used to prompt the completion of the flow and the subsequent checklist.

8.3.22 Aeroplane Pre-flight

An external pre-flight inspection must be conducted by qualified crew members in accordance with the respective OM-B and AFM.

One FCM (FCM 1) shall independently complete the inspection, ensuring all gear pins and protective covers are removed and stowed. If interrupted, the inspection shall be commenced from the beginning. FCM 1 shall sign for the pre-flight inspection in the AFL.

Once all pre-flight servicing is complete (fuel, toilet service, catering loaded, etc.), the other FCM (FCM 2) shall conduct an independent external pre-flight inspection, ensuring that all servicing equipment has been removed and all panels are secure; no panels should be opened during this inspection.

The FCM responsible for loading the baggage (normally FCM 2 – see OM-A 8.3.19.9) shall, upon securing the baggage access door, ensure that no hazards are present that may affect the safety of the aircraft during taxi. If there is no baggage to load, this verification shall be carried out during the independent external pre-flight inspection.

Immediately before closing the main passenger door, FCM 1 shall conduct a final visual check of the aeroplane, ensuring all panels (including baggage access door) are secure, the nose wheel torque-link pin is correctly configured for taxi/pushback as applicable, and the intended taxi route is clear of any obstructions.

Flight crew members who have not completed the initial line check on the aeroplane type are not permitted to sign for the pre-flight inspection in the AFL but may perform it under supervision.

A flashlight must be used when conducting pre-flight inspections in poor lighting conditions. Aeroplane emergency flashlights are part of the emergency equipment and must only be used in exceptional circumstances.

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8.3.22.1 Emergency Equipment

For aeroplane type specific equipment and checks, refer to the SEP Manual.

All emergency equipment carried in the aeroplane shall be checked. It shall be correctly stowed and serviceable.

To ensure the availability, accessibility, and serviceability of aeroplane FCC and cabin emergency systems and equipment, a pre-flight inspection of the respective systems and equipment shall be performed prior to the first flight:

- a. Of the flight crew on an aeroplane during a duty period;
- b. After a new cabin crew has assumed control of the aeroplane cabin;
- c. After an aeroplane has been left unattended by flight crew or cabin crew for any period of time.

Swapping of emergency equipment between aeroplanes, even if the aeroplane is of the same type, is prohibited unless M.A.305, M.A.403, and M.A.501 of Regulation (EU) No 1321/2014 are complied with and it has been authorised by the maintenance engineering department.

8.3.22.2 PA and Emergency Lights

PA and Emergency lights shall be checked to be serviceable.

When cabin crew are carried, these checks are to be carried out in conjunction with the Flight Crew and are usually done simultaneously. Each check will be initiated by the Flight Crew.

8.3.22.3 Crew Seats

Each Crew member shall check the serviceability of their own crew seat to ensure that the seat is functional and the seatbelt / harness is adjusted to fit.

8.3.22.4 Waste Bins

For aeroplanes that have automatic fire extinguishers in the waste bin, the respective waste bin flaps shall be checked to ensure that they close properly, to enable correct operation of the automatic fire extinguisher.

8.3.22.5 Security Checks

Security checks are required for every flight (refer to [Aeroplane Security](#)).

8.3.23 Start, Pushback, and Towing

CAT.OP.MPA.205

8.3.23.1 Start

Engine start shall be conducted with the aid of ground personnel whenever marshalling is available. Communications with ground personnel and engine start procedures are outlined in the OM-B.

Engine starting during pushback and towing is prohibited, except when authorised in the OM-B and in accordance with any local restrictions. OM-B procedures relating to aeroplane doors and internal stairs shall be observed.

8.3.23.2 Pushback and Towing Operations

Pushback and towing procedures, including bar-less towing, are detailed in OM-B / AFM / AOM and [NJMP 1.13](#).

Whenever a pushback or tow is required with crew onboard:

- The crew shall ensure that adequate marshalling personnel are available to move the aeroplane safely.
- At least one flight crew member shall remain in the FCC to monitor the procedure.

Prior to towing the crew shall ensure that ground personnel are familiar with the marshalling signals. Standard hand signals shall be used, as published in ICAO Rules of the Air - Annex 2 (Jeppesen Airway Manual -> General Airway Manual -> Air Traffic Control -> ICAO Rules of the Air - Annex 2) -> 5 Marshalling Signals. If there is any doubt the crew member will pre-brief the signals with the person leading the pushback or towing operation.

When a clearance is required for pushback or towing, it shall be obtained prior to movement.

8.3.24 Taxiing

CAT.GEN.MPA.124, AMC1 CAT.GEN.MPA.124, GM2 CAT.GEN.MPA.125

Taxiing is a high workload phase of flight that requires the full attention of the flight crew. It shall be treated as a safety-critical activity and sterile cockpit procedures shall be applied.

The following procedures apply for taxiing:

- a. Standard phraseology as defined in ICAO DOC 4444 shall be used for all RTF communications;
- b. The necessary aerodrome layout charts shall be available to both flight crew members;
- c. All taxi clearances shall be heard and understood by both flight crew members, and recorded. If any doubt exists as to the clearance, the aeroplane shall be stopped and clarification shall be requested from ATC;
- d. All taxi clearances shall be cross-checked against the aerodrome chart and aerodrome surface markings, signs, and lights, with special attention to runway incursion hot spots;
- e. Conditional clearances to enter or cross a runway such as 'behind landing traffic' present a significant threat as they rely on correct traffic identification. Misidentification, poor visibility, high workload, unclear ATC phraseology and crews acting on expectation rather than positive visual confirmation can lead to runway incursions. To mitigate these risks,

the runway must not be entered unless the correct traffic is clearly identified independently by each FCM; any doubt must be clarified with ATC before entering the runway.

- f. If either flight crew member is unsure of the aeroplane position the aeroplane shall be stopped and ATC shall be contacted;
- g. The parking brake shall be set whenever the aeroplane is brought to a complete stop;
- h. The PF shall announce their intentions to the PM in advance and the PM shall monitor the taxi progress and adherence to the clearance and assist the PF;
- i. Unless the aeroplane is stationary checklists shall not be initiated or ATC clearance obtained until clear of congested areas;
- j. The PF shall be looking outside at all times, and any action which may distract from the taxi activity shall be avoided or, if it is essential, the aeroplane shall be stopped and the parking brake set;
- k. Except for approved EFBs, the use of mobile phones and PEDs (including the company mobile device) is prohibited during taxiing unless it is essential for safety or is operationally vital, and the aeroplane is stopped with the parking brake set;
- l. Any runway (active or inactive) shall only be crossed or entered if explicitly cleared to do so by ATC. Any checklist activity shall be halted when crossing a runway;
- m. Prior to entering or crossing any runway (active or inactive) the PF shall repeat back the 'enter or cross runway' clearance to the PM and the flight crew shall make a visual check of the runway and final approach for conflicting traffic and be aware that entering from rapid exit taxiways or oblique or angled taxiways may limit the effectiveness of this visual check;
- n. If cleared to enter, cross or take-off from a runway prior to a point that would be considered appropriate to switch on the Strobe / Anti-collision or Landing lights as applicable, then the clearance shall be re-confirmed with ATC prior to entering, crossing or initiating take-off.
- o. The aeroplane shall stop and hold at all lighted stop bars and will only proceed further when an explicit clearance to enter or cross a runway has been issued by ATC and the stop bar lights have been switched off. If a clearance to cross or enter the runway has been issued by ATC and the stop lights remain on, flight crew shall question ATC and shall not taxi across the lit stop bar unless complying with all of i to iii below:
 - i. ATC have confirmed the stop bar is inoperative (they are unable to switch it off), and
 - ii. ATC have confirmed the instruction to cross the lit stop bar, and,
 - iii. A 'follow me' car or marshaller is provided and the crossing is conducted under their instruction.

Alternatively, if contingency procedures have been established and notified by the aerodrome, examples of which can be found in [OM-A 3.2.2 Right-of-Way \(3\)](#), crew may follow these;

- p. Use of aeroplane lights will be as follows:
 - i. The TAXI lights shall be switched ON whilst the aeroplane is moving and OFF when the aeroplane is stopped;

- ii. Strobe / Anti-collision lights shall be turned ON prior to crossing or entering a runway (active or inactive) and turned OFF when the runway is vacated;
 - iii. Landing lights shall be turned ON when cleared to take-off and turned OFF in case of a cancelled take-off clearance or in accordance with OM-B
- q. If there is any doubt with regards to the safety of the aeroplane when taxiing in and out of the parking area, the aeroplane shall be stopped until the situation is resolved. Before taxiing in or out of the parking area, crews shall identify any potential hazards on or adjacent to the planned taxi route, discussing mitigation as necessary. If at any stage a hazard is identified which encroaches within, or is likely to encroach within 3m (10ft) of any part of the aeroplane, the aeroplane shall be stopped and one or more of the following measures shall be taken:
- i. Request removal of the hazard
 - ii. Request towing
 - iii. Request ground personnel guidance
 - iv. Use an alternative taxi route
- The Commander remains ultimately responsible for the safety of the aeroplane.
- r. If workload increases during taxi to the extent that either crewmember is rushed or situational awareness is reduced, particularly following receipt or amendment of ATC clearance, the crew should use the following tool to regain situational awareness:
1. Stop
 - Stop the aircraft and apply the parking brake so both FCM can give their full attention.
 2. Set
 - Ensure all settings match the received clearance: FMS, altitude selector, FMA.
 - Ensure 'How to' has been briefed.
 3. Verify
 - Both FCM cross-check the clearance and settings.
- s. Taxi speed shall not exceed 30 knots (based on a Ground Speed readout) and shall be reduced as necessary for, passenger comfort, weather, surface conditions, or traffic, to ensure the aeroplane can be stopped safely;
- t. Taxi speed on slippery surfaces and during LVP should not exceed 10 kts.

For maximum taxi speeds during MCFs, refer to [Maintenance Check Flights \(MCF\)](#).

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8.3.24.1 Taxi and Runway Entry - Crew Coordination

SITUATION	PF	PM
Taxi clearance with 'hold short instruction (runway / intersection / other limit)'	Reads back the hold short instruction to the PM 'HOLD SHORT [stated taxi limit]'	Reads back taxi clearance to ATC in full Verifies PF readback is aligned with ATC clearance Uses airport diagram to identify the hold short point
Clearance to enter runway	Reads back the clearance to enter the runway and if applicable cleared entry point to the PM 'CLEARED TO ENTER¹ RUNWAY [XX] VIA [XX]'	Reads back clearance to ATC in full Verifies PF readback is aligned with ATC clearance. Uses airport diagram to identify the entry point
Conditional Clearance to enter runway	Reads back the clearance to enter the runway including the conditions and if applicable cleared entry point to the PM 'AFTER THE [condition] CLEARED TO ENTER¹ RUNWAY [XX] VIA [XX] BEHIND' Visually identifies the conditional traffic and states '[condition] in sight ENTER RUNWAY [XX] BEHIND'	Reads back clearance to ATC in full Verifies PF readback is aligned with ATC clearance Uses airport diagram to identify the entry point Independently visually identifies the conditional traffic and if aligned with the PFs statement 'CONFIRMED'

SITUATION	PF	PM
Crossing the final holding point to enter a runway	States that the aircraft is entering a runway and if applicable the entry point with reference to airport signage 'ENTERING RUNWAY [XX] AT [XX]'	Verifies runway entry is aligned with ATC clearance 'CONFIRMED'
Approaching the runway after crossing the holding point	Checks the runway / area to their side of the cockpit is clear '[LEFT / RIGHT] CLEAR'	Checks the runway / area to their side of the cockpit is clear '[LEFT / RIGHT] CLEAR'

¹ In case the phraseology used by ATC is different (e.g. 'Line up and wait' instead of 'Cleared to enter' then align the callouts with the phraseology used by ATC.

8.3.24.2 Taxi During LVO

Refer to [Low Visibility Take-off Procedure](#).

8.3.24.3 Jet Blast

Congestion and jet blast are a risk during taxiing. The Commander shall carefully assess the situation around the aeroplane and shall ensure appropriate separation from all obstructions and other taxiing aeroplanes, select an appropriate taxi speed and handle throttles / thrust levers accordingly considering blast effect and noise.

8.3.24.4 Thrust Reverser Use During Taxi

Use of reverse thrust is authorised in accordance with the AFM and OM-B. The potential FOD hazard of using reverse thrust during taxi shall be considered.

8.3.24.5 Clearance from Runway

The aeroplane shall not hold closer to the runway than the designated holding position. In the absence of such markings or visual aids the aeroplane shall be held at least 70 metres clear of the active runway.

8.3.24.6 Parking Brake

After reaching the final parking position, the parking brake shall not be released until all engines have been shut down and the flight crew have confirmed chocks are in position.

When the parking brake is released the pilot at the controls shall ensure the aeroplane is not moving.

8.3.24.7 Ramp Signals and Guidelines

Marshalling signals are in the Jeppesen Airway Manual, ATC section, ICAO Rules of the Air. Taxi guidelines vary between aerodromes and do not always ensure clearance from obstructions, especially in congested areas. Even when guided by apron guide lines or marshalling signals the Commander must exercise caution.

8.3.24.8 Change of Departure Runway or Clearance

Pilots shall not accept a runway or departure change unless there is sufficient time to re-brief, set up / crosscheck the avionics, and recalculate performance prior to departure.

The aeroplane shall be stopped and parking brake set so these tasks can be performed with attention from both pilots and without hazard to the taxi activity. ATC should be advised of any delay expected.

If the Commander requests a change of take-off runway for safety reasons, the safety reason shall be declared to Air Traffic Control.

8.3.24.9 Departure Review

Prior to lining up the flight crew shall review the following items:

1. Initial heading or track;
2. First turning point;
3. Initial assigned altitude, height, or flight level;
4. Emergency special procedure.

8.3.24.10 Runway Status Lights (RWSL)

For details and a description of RWSLs, please see [Technical Annex TA34](#).

The following policy shall be adhered to with respect of RWSLs:

Runway Entrance Lights (RELs)

- When RELs are illuminated red, the runway is unsafe to enter or cross. Stop immediately.

Take-off Hold Lights (THLs)

- When lined up and waiting on the runway with THLs illuminated red, the runway is not safe for take-off - hold position.
- If the take-off roll has commenced and THLs illuminate red below V1, the take-off must be aborted.

Runway Intersection Lights (RILs)

- When RILs are illuminated red, this indicates the presence of high-speed traffic on the intersecting runway and that it is unsafe to enter or cross the runway.

When RWSL indications contradict an ATC clearance, flight crew shall respond to the RWSL indication first, as RWSL are intended to provide an independent safety barrier against runway incursions and human error. The absence of RWSL indications shall never be interpreted as an ATC clearance. RWSL have only two states :

- ON – red lights illuminated (follow the guidance above);
- OFF – lights extinguished (operate in accordance with ATC clearance and normal procedures).

8.3.25 Line-up

Flight crew should not enter a runway for departure until both:

- Are ready to take-off; and
- Clearance to enter the runway has been issued.

ATC shall be advised without delay if additional time on the runway is required.

Prior to entering a runway, the flight crew shall make a visual check of the runway threshold and final approach area, and be aware that accepting rapid exit taxiways or oblique or angled taxiways for line-up may limit the ability to check these areas.

Before lining up, the flight crew shall ensure the cabin is secure.

For aeroplanes carrying cabin crew, line-up is not authorised until the 'CABIN SECURE' call from the cabin crew has been received.

During the line-up and prior to initiating take-off, the crew shall positively identify the runway, either visually or in accordance with [8.4.2.1 Low Visibility Take-Off Procedure](#), to verify line-up has been performed on the correct runway.

If cleared to 'line up and wait', unless ATC advises a longer wait time, or there is a clear and obvious reason for one, notify the tower if holding on the runway for more than one minute.

8.3.26 Take-off

CAT.OP.MPA.265

All mobile electronic devices shall be switched off or turned to 'flight safe mode', except for those specifically authorised in the OM to be switched on during flight.

Shortly before take-off, the Commander shall verify that the actual weather conditions are similar to or more conservative than the weather data used for the take-off performance calculations, and the actual take-off mass does not exceed the maximum allowable take-off mass. If this is not the case, then the take-off data shall be recalculated.

Manufacturer's recommendations for standard take-off flap settings shall be followed.

QRH / AFM derived computations do not take into account obstacle clearance and shall not be used solely to determine the flap configuration for take-off.

Performance Data Computation System (PDCS) charts for the chosen flap configuration shall be available.

If, during take-off, the runway condition differs from that reported, ATC shall be advised as soon as practicable.

8.3.26.1 Commencement of Take-off Roll

The take-off should normally be commenced at the beginning of the runway (maximum TORA / TODA for the applicable conditions).

Intersection take-offs are permitted, provided due consideration is given to take-off performance limitations, wake turbulence separation requirements, and local noise abatement requirements. There is no requirement to have specific PDCS intersection data available, provided Actual Take-off Mass is equal to or below the PDCS limiting mass for a more restrictive intersection on the specific runway.

8.3.26.2 Callouts During Take-off Roll

Refer to the respective Standard Operating Procedures in OM-B.

8.3.26.3 Rejected Take-off

Refer to the respective Standard Operating Procedures in OM-B.

8.3.26.4 Engine Failure / Loss of Thrust on Take-off

In the event of an engine failure on take-off, the Commander shall, after consideration of all relevant factors, take the course of action which offers the highest possible degree of safety.

At those runways where obstacles are a factor, follow the specified Engine Failure procedure on the applicable PDCS chart. This chart provides the lowest obstacle route after engine failure, specified for each relevant runway.

The specified engine failure procedure must be flown accurately to minimise the risk of EGPWS Cautions or Warnings. If an EGPWS Caution or Warning does occur, the crew must continue to fly the specified engine failure procedure and utilise maximum climb performance whilst observing a minimum speed of V_2 .

For an engine failure at V_1 , the SID or ATC departure clearance may be followed only if it coincides with the applicable PDCS engine failure procedure. If not, ATC must be notified of the deviation as soon as safely possible.

The safest course of action will normally be to immediately return to the airport of departure.

8.3.26.5 Initial Climb-out for Noise Abatement

CAT.OP.MPA.130

For noise abatement purposes the initial climb out shall be performed as specified in OM-B.

The altitude for acceleration to en-route climb speed as specified in OM-B shall be used, unless higher altitudes are listed in the aerodrome noise abatement procedures.

8.3.26.6 Turns After Take-off

Turns up to a bank angle of 15° may be executed between a height of 100 and 400 feet.

Turns up to a bank angle of 25° may be executed above a height of 400 feet.

8.3.26.7 Communication with ATC

Communication with departure ATC should be established as specified by the SID or ATC, preferably after climb power has been set, but not before the landing gear has been retracted.

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8.3.27 Climb, Cruise, and Descent

8.3.27.1 Rate of Climb and Descent Approaching a Level-off

When approaching 1,000 ft to go to an assigned flight level or altitude, in the climb or descent, the vertical speed shall be reduced to a maximum of 1,500 ft per minute.

If performing a departure with a level-off at or below 3000 ft AAL, adhering to the procedures and guidance in the OM-B shall take precedence.

Whilst operating in busy airspace where the possibility of conflict with other traffic is high, unless otherwise instructed by ATC or required by the flight profile, it is strongly recommended that the rate of climb / descent is managed accordingly to avoid TCAS RAs.

To promote situational awareness of the vertical rate approaching a level-off, and to ensure that a maximum rate of climb or descent of 1,500 ft per minute is achieved with 1,000 ft to go, the callouts detailed in [Level-off Awareness](#) shall be used.

8.3.27.2 Maximum Bank Angle

The recommended maximum bank angle for normal operations is 25° and should not exceed 30°.

NOTE

During operation with the Autopilot ON, the bank angle could temporarily exceed 30°. It is recommended not to disconnect the autopilot if a timely correction is observed.

8.3.27.3 Maximum IAS Below 10.000' / FL100

The maximum IAS shall not be greater than 250 knots below 10.000' / FL100 unless requested by ATC or emergency procedures. The ATC statements 'No ATC Speed' or 'high speed approved' are not authorisation to exceed 250 KIAS.

8.3.27.4 Adherence to Level Assignments

As a safeguard against inadvertent reduction of vertical separation standards, the cruising level(s) assigned by ATC or, in uncontrolled airspace as selected in accordance with the applicable cruising level system, shall be maintained as accurately as possible.

8.3.27.5 Maximum Flight Altitude / Level

If the maximum certified altitude / flight level of the aeroplane is above 49.000'/FL490, the maximum altitude / Flight level shall not exceed 49.000'/FL490 (or equivalent in meters) unless required for during an emergency.

8.3.27.6 Flight Below Minimum Altitudes During Climb and Cruise

Aeroplanes proceeding in accordance with published departure routes are safe in respect of terrain and other restrictions on any level included in the departure route or otherwise assigned by ATC.

Minimum Safe En-route Altitudes published for subsequent segments of a departure route shall be observed by commencing the climb sufficiently in advance to ensure that the point from which the higher minimum safe en-route altitude applies is crossed not below such altitude.

During an off-route departure, level flight may only be accepted at or above the minimum sector altitude or minimum safe grid altitude. An aeroplane being radar vectored may be flown below these altitudes if the pilot is able to monitor its position with reference to the Radar Minimum Altitude chart.

8.3.27.7 Cruise Procedures

The FSB sign shall be turned-off at the discretion of the Commander. However, it shall be recommended that passengers keep their seat belts fastened even if the FSB sign has been switched off, in case unexpected turbulence (weather or wake) is encountered.

Cross-check that altimeters are within RVSM limits.

The OFP shall be used to record fuel checks as specified in [In-flight Fuel Checks](#).

121.5 MHz shall be monitored on the secondary com box when not required for other communications.

Flight crew members shall continuously monitor aeroplane position / navigation, altimetry, aeroplane systems and perform periodic system checks during the cruise phase of the flight in order to provide an early detection of abnormal situations.

8.3.27.8 Planning of Descent

Passenger comfort, efficiency and economy in flight operations very much depend on a well planned and conducted descent. The descent shall be performed as efficiently and smoothly as possible taking into account flight safety, local conditions, ATS procedures, meteorological factors (turbulence, icing), and passenger comfort.

In order to obtain descent clearance in time, adequate advance notice should be given to ATC. Where available, full use shall be made of DME and INS / IRS / FMS during descent.

Descent from the cruising altitude shall be planned and conducted using the procedures, profiles and rates of descent prescribed in the OM-B so as to:

- Ensure arrival at the destination or radio fix serving the destination in a configuration and at an altitude permitting an instrument or VMC / visual approach procedure.
- Allow the greatest fuel economy.
- Allow sufficient time in the descent for satisfactory performance of the cabin pressurisation system.

8.3.27.9 Arrival Briefing

Also refer to [Instructions for Crew Briefings](#).

Arrival Briefing should be done as soon as practical, which may be prior to a flight on very short sectors. Flight Crew shall follow the idea of 'brief what you can brief' at every step while allowing the required crosschecks, preserving the PF / PM task allocation and avoiding distractions:

1. PF may remain PF while performing the briefing, which they will interrupt as necessary.
2. PF will brief the approach at their own rhythm, using the appropriate charts.
3. PM should make all necessary FMS insertions on request of the PF.
4. PM will check the routing and approach that is briefed by the PF by use of the FMS.
5. Entries in the FMS are crosschecked by the PF/PM as appropriate.

8.3.27.10 Maximum Permissible Rate of Descent

During descent down to 10,000 ft. above the minimum safe en-route altitude or minimum safe grid altitude, there are no limitations with regard to the rate of descent.

During descent conducted below the altitudes specified above the rate of descent shall, for safety reasons, not exceed the following values:

Maximum Vertical Speed	Down to Altitude
5,000 ft/min	5,000 ft above the terrain
4,000 ft/min	4,000 ft above the terrain
3,000 ft/min	3,000 ft above the terrain
2,000 ft/min	2,000 ft above the terrain
In accordance with stabilised approach criteria but not >1,500 ft/min	Below 2,000 ft above terrain

NOTE

Adherence to published approach profiles may, in exceptional cases, (such as a steeper than standard glideslope), require higher rates of descent than permitted above. Every effort shall be made to reduce the Ground Speed in such a manner as to avoid higher sink rates.

8.3.27.11 Descent Below Minimum Safe En-route Altitude / Minimum Safe Grid Altitude

Descent below the minimum safe en-route altitude / minimum safe grid altitude to the minimum sector altitude may be made when approaching the navigation aid / RNAV fix from which an approach-to-land will be conducted, provided the aeroplane's position can be accurately established as being within 25 NM or respective distance from the reference point aid upon which the minimum sector altitude is based on.

8.3.27.12 Descent Below Minimum Sector Altitude

When being radar vectored, clearance to descent below the minimum sector altitude may be accepted, provided the flight crew is able to monitor the aeroplane's position in relation to terrain and obstacles and with reference to the Radar Minimum Altitude chart if available, or with sufficient visual reference to terrain and obstacles.

In certain instances the minimum sector altitude for a given sector may be higher than the minimum safe en-route altitude established for a particular route segment between fixes, or for a holding area within that sector. In such cases descent below the minimum sector altitude down to the minimum safe en-route altitude is permitted, provided the flight is conducted along the respective route or within the holding area.

Definition:

Minimum sector altitude is the lowest altitude which will provide a minimum clearance of 300 m (1,000 ft.) above all objects located in an area contained within a sector of a circle of 46 km (25 NM) radius centred on a radio aid to navigation.

8.3.27.13 Descent Below Minimum Vectoring Altitude

Descent below the Minimum Vectoring Altitude is only permitted when:

- Established on a published IFR procedure, or
- Established on a Visual approach in DAY VMC when not being radar vectored.

8.3.27.14 Holding

Official IFR holding patterns and manoeuvring procedures are in the OM-C Route Manual (Jeppesen Airway Manual).

The entry procedure and the holding pattern shall be briefed and acknowledged by both flight crew members. Special attention is required when the procedure is programmed in the FMS to ensure that the side of the turn is correct.

Fuel status shall be monitored closely.

When instructions to hold have been received, the aeroplane is considered to be in the holding pattern upon arrival over the respective fix or facility. Therefore, the airspeed at that time shall not be higher than the maximum holding speed.

After departing the holding fix, normal speed should be resumed, limited only by airspace speed restrictions, if any, or ATC constraints.

8.3.27.15 Fatigue Management (Controlled Rest on FCC)

Proper management of rest / fatigue shall be observed by all crew.

During all phases of flight, each flight crew member required to be on duty in the FCC shall remain alert unless taking controlled rest in accordance with this chapter (cabin crew aircraft only).

If the pilot fatigue level is impeditive for the flight, no flight will occur.

If fatigue level increases in-flight to cause one of the pilots to approach a drowsiness state, follow the procedures below:

a. Aeroplanes with no cabin crew:

Fatigue-induced diversion should be initiated immediately in the interest of safety.

b. Aeroplanes with cabin crew:

Follow controlled rest on FCC procedures stated below.

8.3.27.15.1 Controlled rest on FCC

Controlled rest may be used at the discretion of the Commander to manage both sudden unexpected fatigue and fatigue which is expected to become more severe during higher workload periods later in the flight.

Controlled rest shall be used in conjunction with other on-board fatigue management countermeasures such as physical exercise, bright cockpit illumination at appropriate times, balanced eating and drinking, and intellectual activity.

This procedure is concerned with controlled rest taken by the minimum certificated flight crew. It is not related to planned in-flight rest of an augmented crew.

The maximum rest time has been chosen to limit deep sleep with consequent long recovery time.

1. It is the responsibility of the crew members to be properly rested before flight.
2. Controlled rest means a period of time 'off task' some of which may include actual sleep.
3. Controlled rest cannot be planned before flight.
4. Controlled rest periods should be agreed according to individual needs and the accepted principles of CRM; when the involvement of the cabin crew is required, consideration should be given to their workload.
5. Only one crew member at a time shall take rest, at their station; the harness shall be used and the seat positioned so as to cause minimum potential interference with the controls.
6. The Commander shall ensure that the other(s) crew member(s) is (are) adequately briefed to carry out the duties of the resting crew member.
7. One pilot shall be fully able to exercise control of the aeroplane at all times.
8. Any system intervention which would normally require a cross check according to multi crew principles should be avoided until the resting crew member resumes their duties.

9. The rest period shall be no longer than 45 minutes (in order to limit any actual sleep to approximately 30 minutes) to limit deep sleep and associated long recovery time (sleep inertia).
10. After this 45 minute period there shall be a recovery period of 20 minutes to overcome sleep inertia during which control of the aeroplane shall not be entrusted to the pilot who has completed their rest. At the end of this recovery period an appropriate briefing shall be given.
11. The Commander shall inform the cabin crew member of the intention of the flight crew member of taking controlled rest, and of the ending time of that rest.
12. The cabin crew shall establish contact with the non-resting FCC every 30 minutes, and at the end of the resting period.
13. A minimum of 20 minutes should be allowed between rest periods to overcome the effects of sleep inertia and allow for adequate briefing.
14. If necessary, a flight crew member may take more than one rest period if time permits on longer sectors, subject to the restrictions above.
15. Controlled rest periods shall terminate at least 30 minutes before TOD.

8.3.27.15.2 Crew Monitoring

Mandatory mutual monitoring shall be implemented during all phases of duty to support crew well-being and maintain operational safety. This process is intended to enable early detection of fatigue, distress, reduced alertness, or incapacitation.

Crew monitoring shall be conducted through scheduled well-being checks between the flight crew and cabin crew, combining direct observation, verbal interaction, and task confirmation as appropriate.

CCM to FCM	The CCM shall perform well-being checks of the FCM at intervals not exceeding 30 minutes.
FCM to CCM	The FCM shall perform well-being checks of the CCM at intervals not exceeding 60 minutes.

Any indication of fatigue, distress, or potential incapacitation shall be promptly communicated, and appropriate mitigation actions taken.

8.3.28 Approach and Landing

- Before commencing an approach operation, the commander shall be satisfied that:
- the meteorological conditions at the aerodrome and the condition of the runway intended to be used will not prevent a safe approach, landing or go-around, considering the performance information contained in the operations manual, including carrying out a landing distance assessment in accordance with [In-flight Determination of Landing Distance at Time of Arrival](#); and
 - the selected aerodrome operating minima are consistent with all of the following:
 - the operative ground equipment;
 - the operative aircraft systems;
 - the aircraft performance;
 - flight crew qualifications.

Every approach shall be planned and conducted in accordance with the procedures prescribed in the OM-B.

- During the approach briefing crew shall brief the planned runway exit and initial taxi route.

An approach chart shall be available during every approach (refer to [Autopilot, Autothrottle, and Automation Policy / Level-off Awareness](#)).

- Flight crew shall not accept a runway or approach change unless there is sufficient time to set up, brief the approach and check landing performance. A clearance that requires a change of landing runway shall not be accepted once the aeroplane has descended below 1000 ft above runway threshold elevation.

Altimeter setting during the approach to land shall be in accordance with [Altimeter Setting Procedures](#).

- Landing lights shall be turned ON when cleared to land and turned OFF in case of a go-around or cancelled landing clearance or in accordance with OM-B.

8.3.28.1 In-flight Determination of Landing Distance at Time of Arrival

CAT.POL.A.105; CAT.OP.MPA.303

No approach to land shall be continued unless both:

- The landing distance available (LDA) on the intended landing runway is at least 115% of the landing distance required (LDTA) at the estimated time of landing, determined in accordance with the performance information for the assessment of the landing distance at time of arrival.
- The aerodrome and intended runway have previously been assessed and found to meet the dispatch criteria described in [Landing - Destination and Alternate Aerodromes](#), either at the time of dispatch or in the event of an in-flight change of destination and / or alternate, prior to that change being confirmed.

NOTE

For the requirements applicable at the time an **in-flight** change of destination or alternate is initiated see **table 1** below.

Table 1

Aircraft Status	Performance requirement at time an in-flight change of destination and / or alternate is initiated ²
Normal	As per Normal dispatch assessment described in Landing - Destination and Alternate Aerodromes
In-Flight ¹ Technical failure affecting performance	More limiting of: a. Dispatch assessment as per Landing - Destination and Alternate Aerodromes - without considering the performance penalty resulting from the technical failure, and b. FLDTA - with full consideration of the performance penalty resulting from the technical failure.

¹ Only applicable to an **in-flight** case, if dispatching with a technical failure affecting performance under MEL then the dispatch requirements according to [Landing - Destination and Alternate Aerodromes](#) shall be applied with full consideration of the performance impact of the technical failure.

² In all cases a subsequent 'In-flight check of landing distance at time of arrival' will also be required for the landing runway, in case of a short diversion this may have to be carried out concurrently with the dispatch assessment described in this table.

The in-flight landing distance assessment should be based on the latest available weather report and runway condition report (RCR) or equivalent information based on RCR. It will usually be carried out around top of descent, however if the planned duration of the flight does not allow the assessment to be carried out during non-critical phases of flight, the assessment should be carried out before departure.

Normally the In-flight check of the landing distance at time of arrival will be carried out using the PDCS application on the portable EFB which will use the same data as specified below to calculate the LDTA as well as applying the required 115% factor to give a FLDTA.

When meteorological conditions may lead to a degradation of the runway surface condition, the assessment should include consideration of how much deterioration in runway surface friction characteristics may be tolerated, so that a quick decision can be made prior to landing, if the flight crew of the preceding aircraft landing at the same runway provides a special AIREP of worse than expected braking action.

The flight crew should monitor the evolution of the actual conditions during the approach, to ensure that they do not degrade below the condition that was previously determined to be the minimum acceptable.

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In-flight check of the landing distance at time of arrival based on dispatch criteria

The required landing distance for dry runways determined in accordance with [Landing - Destination and Alternate Aerodromes - C.-a.](#) contains adequate margin to fulfil the intent of [In-flight check of the landing distance at time of arrival](#) on a **dry runway**, as it includes allowance for the additional parameters considered in that calculation.

When at time of arrival the runway is dry and the overall conditions, including weather at the aerodrome and runway condition, have been confirmed as not changed significantly compared to the time of dispatch, the in-flight check of the landing distance at time of arrival may be carried out by confirming that the assumptions made at time of dispatch are still valid.

In all other cases the LDTA shall be calculated and the requirements of [In-flight check of the landing distance at time of arrival](#) shall be met.

Calculation of the LDTA

The specific AFM or supplemental data to be used for calculating the LDTA for each aircraft type will be shown in the type specific Aircraft Calculations and Assumptions document and / or OM-B.

When calculating the LDTA the crew shall take account of the following:

- Runway surface condition
- Aeroplane Mass
- Aeroplane configuration
- Pressure Altitude
- Wind⁶
- Outside Air Temperature
- Runway Slope
- Approach Speed Increments
- Reverse Thrust
- Use of autobrake⁷
- Any technical failure / abnormal condition resulting in a performance penalty⁸

⁶

The information on wind contained in METAR/SPECI/ATIS reports (average of a 10 minute period) should be the basis for the landing performance calculations, while instant wind information reported by the Tower should be monitored during the approach to ensure that the wind speed does not exceed the assumptions made for landing performance calculations.

⁷ Autobrake usage: While autobrakes are a part of the aeroplane's landing configuration, the landing distance assessment at the time of arrival is not intended to force a selection of a higher than reasonable autobrake level. For operations when the runway is dry or wet grooved or with a porous friction course (PFC), if the manual braking distance provides at least 15% safety margin, then the braking technique may include a combination of autobrakes and manual braking even if the selected autobrake landing data does not provide a 15% safety margin.

⁸ In case the destination and / or alternate are changed, see Table 1 above for the additional performance checks required at the time the diversion / change is initiated.

Factoring of the LDТА

To comply with the requirement for the LDA to be at least 115% of the LDТА, the LDТА must be multiplied by 1.15 to give Factored LDТА (FLDТА) which can then be compared directly with the LDA.

Emergency Landings

For an emergency landing, the Landing Distance Available shall be equal to or greater than the LDТА (i.e. the 115% factor is not required).

Pilot Reporting of Runway Surface Condition

Whenever flight crew observe worse runway braking action than previously reported, they shall communicate this to ATC via an AIREPs. It is up to the pilot to assess the manner in which an aircraft responds to the application of wheel brakes.

These reports provide feedback to the aerodrome operator regarding the accuracy of the assigned RWYCCs relative to the runway surface conditions actually experienced and may be used to trigger a downgrade of the RWYCC, they may also be passed to other crews in order to help them correctly assess the applicable runway condition.

Table 2 below shows the correlation of pilot reports of runway braking action with RWYCCs (this table forms part of the overall RCAM).

Table 2

Pilot report of runway braking action	Description	RWYCC
N/A		6
GOOD	Braking deceleration is normal for the wheel braking effort applied and directional control is normal.	5
GOOD TO MEDIUM	Braking deceleration OR directional control is between good and medium.	4
MEDIUM	Braking deceleration is noticeably reduced for the wheel braking effort applied OR directional is noticeably reduced.	3
MEDIUM TO POOR	Braking deceleration OR directional control is between medium and poor.	2
POOR	Braking deceleration is significantly reduced for the wheel braking effort applied OR directional control is significantly reduced.	1
LESS THAN POOR	Braking deceleration is minimal to non-existent for the wheel braking effort applied OR directional control is uncertain.	0

8.3.28.2 Short Landing Operations (SLO)

CAT.POL.A.250

Short landing operations require approval from the competent authority in accordance with Part-CAT and NTA does not currently hold such an approval.

8.3.28.3 Steep Approach Operations

CAT.POL.A.245

For NTA aeroplane types that are approved for Steep Approach operations, see [Approvals by Aeroplane Type](#).

Steep approach operations using glideslope angles of 4.5° or more require prior approval by the competent authority.

Each AFM states the maximum approved glideslope angle, any other limitations, normal, abnormal or emergency procedures for the steep approach as well as amendments to the field length data when using steep approach criteria.

For each aerodrome at which steep approach operations are to be conducted:

- a. a suitable glide path reference system comprising at least a visual glide path indicating system shall be available;
- b. AFM procedures and limitations are to be adhered to;
- c. Specific aerodrome briefings shall be consulted before conducting steep approach operations.

When conducting a steep approach:

1. The Commander shall hold a current PF steep approach qualification.
2. The SIC shall hold as a minimum a current PM steep approach qualification.
3. For the SIC to act as PF during a steep approach, that they shall also hold a current PF steep approach qualification (unless under LIFUS to obtain the PF steep approach qualification).

NOTE

Some AFMs / OM-Bs restrict the PF role during a steep approach to the LHS. Such additional restrictions shall always be respected.

8.3.28.4 Setting and Checking of Navigation Aids

Refer to OM-B.

8.3.28.5 Positive Identification of Aerodromes

In order to preclude a safety related incident resulting from misidentification of aerodromes and / or runways, with inherent potentially serious consequences, Commanders shall take all measures to ensure positive identification of aerodromes / runways such as:

- a. Strict adherence to SOPs for every approach;
- b. When being radar vectored onto a final approach, a cross-check using available radio aids shall be made to determine that the runway ahead is the correct one;
- c. When conducting an approach subject to maintaining VMC or a visual approach:
 - ceiling and visibility shall be such that positive identification of the aerodrome / runway is possible without difficulty;
 - available navigation aids shall be used to the fullest extent (see [Use of Automations](#));
 - before accepting a visual approach, as a minimum all relevant terrain / obstacles, local restrictions such as restricted airspace and arrival noise abatement procedures shall have been briefed (see [Briefing Procedures](#)).

8.3.28.6 Descent to Prescribed Altitudes During Approach

The Altitudes and Tracks prescribed in an instrument approach procedure shall be strictly adhered to, unless given a descent clearance during radar vectoring or cleared to conduct a visual approach.

An aeroplane is considered established on final approach track if:

- a. Within $\pm 5^\circ$ of the final approach track of a non-precision approach;
- b. Within ± 1 dot of the localiser on-course signal as shown on the raw data ILS indicator.

When intercepting the final approach track of a non-precision approach procedure from an off-set position not forming part of the published procedure, descent to the applicable altitude prescribed for the next fix on final or, if no fix is provided on final, to the MDA, shall only be made provided:

- a. The position of the aeroplane is established within 10 NM from the landing threshold; and
- b. The position of the aeroplane is within $\pm 5^\circ$ of the final approach track; and
- c. The aeroplane heading is within 30° of the procedure inbound course; and
- d. The aeroplane is within the published distance for the approach.

8.3.28.7 Stabilised Approach Criteria

CAT.OP.MPA.115

All approaches shall be flown as stabilised approaches (SAp).

An approach shall be considered to be fully stabilised when the following criteria are met:

NOTE

Modified stable approach criteria may be approved for specific aerodromes and detailed within the aerodrome briefing.

- a. The aeroplane is established on the required approach path with the correct track set, approach aids tuned and identified as appropriate to the approach type flown and on the required vertical profile.
- b. The aeroplane is in an appropriate configuration for the type of approach in accordance with the procedures in the OM-B.
- c. The aeroplane speed is at a maximum of $V_{ref} + 20$ kts. In gusty conditions, minor speed fluctuations of short duration are acceptable.
- d. Wings are level, defined as ≤ 15 degrees angle of bank. For approaches that require manoeuvring below 500 ft AAL, wings shall be level by 300 ft above landing runway threshold elevation.
- e. The target rate of descent (ROD) is $\leq 1,000$ fpm except for:
 - steep approaches where the required ROD may be higher; or
 - exceptional circumstances which have been anticipated and briefed prior to commencing the approach; e.g. a strong tailwind;

Deviations from the target ROD shall not exceed ± 300 fpm. The ROD may be reduced to zero when the descent path needs to be regained from below the profile.

- f. A suitable stabilised thrust setting, usually above idle, is maintaining the desired approach path at the target speed.

These criteria represent the limits of aeroplane performance beyond which a safe landing cannot be assured. In normal operations these limits should not be approached.

Straight-in IFR Approaches

For all straight-in IFR approaches the aeroplane shall be stabilised for landing, with the exception of speed, at the latest when passing 1,000 ft above the landing runway threshold elevation. Speed stabilisation shall be achieved no later than 500 ft above the landing runway threshold elevation.

VFR, Visual, and Circling Approaches

For VFR, Visual, and Circling approaches, unless otherwise defined in the aerodrome briefing, the aeroplane shall be stabilised when passing 500 ft above the landing runway threshold elevation. In the case of circling approaches, the aeroplane shall be stabilised in the circling configuration no later than passing 1,000 ft above the landing runway elevation.

OM-B Configuration Gates

Compliance with configuration gates stipulated in the respective OM-B will assist in achieving the stabilised approach criteria.

Stabilised approach criteria not achieved or approach becomes destabilised

A go-around shall be initiated if:

- it is apparent that the stabilised approach criteria cannot safely be achieved by the required altitude / height;
- the stabilised approach criteria are not met at the required altitude / height;
- the approach becomes destabilised after the required altitude / height.

Stable approach call outs

Should a go-around be required, the PM shall call 'UNSTABLE-GO-AROUND'.

Stabilisation Gates		
Approach Type	Height above Landing RWY THR (ft)	Stable criteria
IFR Straight-in-Approach	1,000	Stable except Speed
	500	Fully stable
VFR, Visual	500	Fully stable ⁽¹⁾
Circle to land	1,000	Circling configuration
	500	Fully stable ⁽¹⁾
⁽¹⁾ Where visual manoeuvring is required below 500 ft AAL, wings-level by 300 ft above landing runway threshold elevation.		

SITUATION	PF	PM
If STABLE criteria are NOT met or maintained.	'GO-AROUND FLAPS [XX]'	'UNSTABLE - GO-AROUND'

8.3.28.8 Determination of Decision Altitude / Height or Minimum Descent Altitude / Height

The barometric altimeter, set to the current QNH/QFE of the aerodrome concerned, shall be used to indicate the applicable DA/DH or MDA/MDH.

8.3.28.9 Descent Below DA/DH or MDA/MDH During a Straight-in Approach

Descent below the relevant minima shall not be made unless:

1. The required visual references as per [Policy](#) have been acquired; and
2. The aeroplane has arrived at a position from which a normal approach profile can be followed to the runway in use.

If, at any time after descent below the minimum altitude / height, the PF cannot maintain the required visual references, an immediate go-around shall be initiated and the appropriate missed approach procedure shall be followed.

8.3.28.10 Change-over from Instrument Flying to Flying with Visual Reference

Good coordination between PF and PM is necessary during change-over from instrument flight to flying with visual reference to the ground. When, during the progress of the final approach, the required visual reference is expected to be obtained, the PM shall divide their attention between monitoring the flight instruments and lookout.

When the approach lights, runway lights, or runway markings are clearly in sight, the PM should indicate to the pilot flying where to look for visual reference (e.g. 'Runway 11 o'clock').

During transition to visual flight the pilot flying shall pay particular attention to maintaining the stabilised attitude to ensure the aircraft continues on the correct approach path.

8.3.28.11 Visual Approaches Under IFR

Visual approaches under IFR shall adhere to the basic principles of instrument approaches and are only authorised subject to ATC approval, under the conditions laid in [Operation of Air Traffic Control Service - Visual Approach](#), provided that as a minimum all relevant terrain / obstacles, local restrictions such as restricted airspace, arrival noise abatement procedures, and go-around procedures have been briefed. Additionally, either the aerodrome shall be in sight or both crew members shall be able to positively identify terrain features that will enable the visual approach to be safely carried out.

The commander is responsible for ensuring that the spacing from a preceding aircraft of a heavier wake turbulence category is acceptable. If it is determined that additional spacing is required, the flight crew should inform ATC accordingly, stating their requirements.

During straight-in visual approaches, the published instrument approach procedure shall be used to assist in achieving a visual glidepath angle of approximately 3 degrees, or such angle as is prescribed for the runway in use.

Whenever a straight in approach is not possible, the visual approach should be flown according to the profile outlined in the OM-B.

The requirements of [Positive Identification of Aerodromes](#) shall be adhered to.

8.3.28.12 Circling Approach

Use of Automation and Avionics

The use of the AP is recommended to reduce pilot workload, at least until leaving the MDA to land. The FGS mode used to fly the approach shall guarantee a level-off at the selected MDA.

Circling Configuration

If conducting a Noise Abatement Circling Pattern, the procedures in the OM-B should be followed. Similarly, if conducting a visual pattern following prescribed tracks, the published speed and configuration requirements shall be observed.

However, if weather conditions require descent to the circling MDA, the aeroplane must be fully configured for landing by the time it reaches the MDA, and maximum use of automation should be made to ensure optimum situational awareness.

Circling Patterns

Circling approaches involve manoeuvring at low altitude, often in the vicinity of noise sensitive areas. Unless the weather conditions require descent to the circling MDA to achieve adequate visual references, circling should be conducted in accordance with the Noise Abatement Circling Pattern.

Noise Abatement Circling Pattern

Unless special instructions are received or a specific noise abatement circling altitude is published, the circling should be performed at 1,500 ft AAL.

In order to minimise the impact on the environment, the noise abatement circling pattern should be used when the following conditions exist:

1. VIS > 5 Km (3 SM), procedure minima or OM-C minima, whichever is highest.
2. Ceiling > 1,500 ft, procedure minima or OM-C minima, whichever is highest.
3. No special technical or operational condition requiring early aeroplane configuration.
4. The circling pattern has been briefed accordingly.

If it is considered that a noise abatement circling is not compatible with a given procedure, then the normal circling procedure shall be flown.

Published Noise Abatement Circling Pattern

If a noise abatement circling pattern is published, it shall be followed as accurately as possible and the configuration changes schedule might have to be adapted. The procedure must be thoroughly reviewed during the arrival briefing.

8.3.28.12.1 Circling Procedures**1. Instrument approach followed by circling WITHOUT prescribed tracks.**

- a. When the aeroplane is on the initial instrument approach, before visual reference is established, but not below MDA/H, the aeroplane shall follow the corresponding instrument approach procedure until the appropriate instrument MAPt is reached.
- b. At the beginning of the level flight phase at or above the MDA/H, the instrument approach track determined by the radio navigation aids shall be maintained until the flight crew:
 - i. estimate that, in all probability, visual contact with the runway of intended landing or the runway environment will be maintained during the entire circling procedure;
 - ii. estimate that the aeroplane is within the circling area before commencing circling; and
 - iii. are able to determine the aeroplane's position in relation to the runway of intended landing with the aid of appropriate external references.
- c. When reaching the published instrument MAPt and the conditions stipulated in paragraph b. above are unable to be established by the flight crew, a missed approach shall be carried out in accordance with that instrument approach procedure.
- d. After the aeroplane has left the track of the initial instrument approach, OM-B circling procedures shall be followed to enable the aeroplane:
 - i. to attain a controlled and stable descent path to the intended landing runway; and
 - ii. to remain within the circling area and maintain visual contact with the runway of intended landing or runway environment.
- e. Flight manoeuvres shall be carried out at an altitude / height that is not less than the circling MDA/H.
- f. Descent below circling MDA/H shall not be initiated until the threshold of the runway to be used has been positively identified, and the aeroplane is in a position to continue with a normal rate of descent to land within the touchdown zone.

2. Instrument approach followed by a circling WITH prescribed tracks.

- a. The aeroplane shall remain on the initial instrument approach procedure until one of the following is reached:
 - i. the prescribed divergence point to commence circling on the prescribed track; or
 - ii. the MAPt.
- b. The aeroplane shall be established on the instrument approach track determined by the radio navigation aids in level flight at or above the MDA/H at or by the circling manoeuvre divergence point.
- c. If the divergence point is reached before the required visual reference is acquired, a missed approach shall be initiated not later than the MAPt and completed in accordance with the instrument approach procedure.
- d. When commencing the prescribed circling manoeuvre at the published divergence point, the published routing and heights / altitudes shall be adhered to.

- e. Unless otherwise specified, once the aeroplane is established on the prescribed track(s), the published visual reference does not need to be maintained unless:
 - i. required by the State of the aerodrome; or
 - ii. the circling MAPt (if published) is reached.
- f. If the prescribed circling manoeuvre has a published MAPt and the required visual reference has not been obtained by that point, a missed approach shall be executed in accordance with paragraphs 3.b and 3.c below.
- g. Subsequent further descent below MDA/H shall only commence when the required visual reference has been obtained.
- h. Unless otherwise specified in the procedure, final descent shall not be commenced from MDA/H until the threshold of the intended landing runway has been identified and the aeroplane is in a position to continue with a normal rate of descent to land within the touchdown zone.

3. Missed approach

- a. Missed approach during the instrument procedure prior to circling:
 - i. if the missed approach procedure is required to be flown when the aeroplane is positioned on the instrument approach track defined by radionavigation aids, RNAV, RNP, or ILS, MLS, and before commencing the circling manoeuvre, the published missed approach for the instrument approach shall be followed; or
 - ii. if the instrument approach procedure is carried out with the aid of an ILS, MLS or a stabilised approach (SAP), the MAPt associated with an ILS, MLS procedure without glide path (GP-out procedure) or the SAP, where applicable, shall be used.
- b. If a prescribed missed approach is published for the circling manoeuvre, this overrides the manoeuvres prescribed below.
- c. If visual reference is lost while circling to land after the aeroplane has departed from the initial instrument approach track, the missed approach specified for that particular instrument approach shall be followed.

An initial climbing turn toward the intended landing runway shall be flown to a position overhead the aerodrome in order to establish the aeroplane in a climb on the instrument missed approach segment.

- d. The aeroplane shall not leave the visual manoeuvring (circling) area, which is obstacle protected, unless:
 - i. established on the appropriate missed approach procedure; or
 - ii. at minimum sector altitude (MSA).
- e. All turns shall be made in the same direction, remaining within the circling protected area while climbing to:
 - i. the altitude assigned to any published circling missed approach manoeuvre if applicable; or
 - ii. the missed approach altitude of the initial instrument approach; or
 - iii. the MSA; or

- iv. the minimum holding altitude (MHA) applicable for transition to a holding facility or fix, or continue to climb to an MSA, or as directed by ATS.

However, when the missed approach procedure is commenced on the 'downwind' leg of the circling manoeuvre, an 'S' turn may be undertaken to align the aeroplane on the initial instrument approach missed approach path, provided the aeroplane remains within the protected circling area.

The commander shall be responsible for ensuring adequate terrain clearance during the manoeuvres described above.

- f. Because the circling manoeuvre may be accomplished in more than one direction, different patterns will be required to establish the aeroplane on the prescribed missed approach course depending on its position at the time visual reference is lost. In particular, all turns are to be in the prescribed direction if this is restricted, e.g. to the west / east (left or right hand) to remain within the protected circling area.
- g. If a missed approach procedure is published for a particular runway onto which the aeroplane is conducting a circling approach and a manoeuvre to align with that runway has been initiated, the missed approach for this direction may be accomplished. The ATS unit shall be informed of the intention to fly the published missed approach procedure for that particular runway.
- h. The commander shall advise ATS when any missed approach procedure has been commenced, the height / altitude the aeroplane is climbing to and the position the aeroplane is proceeding towards and / or heading the aeroplane is established on.

8.3.28.12.2 Circling Radii from Threshold

Refer to [FAA Terminal Instrument Procedures \(TERPS\) Vs. ICAO Procedures for Air Navigation Services - Aeroplane Operations \(PANS OPS\)](#) for information and limitations.

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8.3.28.13 Standard Call-out Procedure During Approach

Refer to the Standard Call-Out pages in OM-B and [Monitoring and Crosschecking](#).

8.3.28.14 Energy Management During Approach

In order to provide adequate handling qualities and energy margins above the stall during the approach, increments shall be added to the landing reference speed V_{REF} to compensate for wind, gusts, and shear effects. Corrections to the approach speed shall be as specified in the respective AFM, AOM, or OM-B.

Where no type-specific guidance is provided an increment of $\frac{1}{2}$ the steady head wind component plus the total gust value, up to a maximum total increment of 20 kts, shall be added to V_{REF} (e.g. for a wind of 10G20KT the increment would be $10/2 = 5$ plus $20-10 = 10$ giving an increment of 15 KT).

Landing Distance Required is normally based on the aeroplane crossing the runway threshold at V_{REF} in still air; any excess speed above V_{REF} will exponentially increase the Landing Distance Required.

Only if the flight crew determines that stall margins and / or handling qualities will not be sufficient at V_{REF} for the expected conditions, may the threshold be crossed at a speed higher than V_{REF} . The flight crew shall ensure that under no circumstances the LDR is greater than the LDA (see note below). The intention of crossing the threshold at an airspeed greater than V_{REF} shall be pre-briefed.

NOTE

Some aeroplane types publish landing performance data that provides for higher V_{REF} speeds than those normally to be used for a specific Landing Mass. If the LDR has been calculated with this performance data, the threshold may be crossed with the airspeed applicable to that landing performance data.

8.3.28.15 Use of VASIS, T- VASIS, or PAPI

On runways equipped with VASIS, T-VASIS, or PAPI and during the execution of an approach other than a precision approach, the visual glide slope shall be adhered to as closely as possible. Due to glide slope tolerances and differences in MEHT, VASIS, T-VASIS, or PAPI indications shall not be used close to the threshold (approximately below 200' above aerodrome elevation). The use of nonstandard visual glide slope indicator systems (e. g. VGPI) and of abbreviated VASIS (AVASIS) or AT-VASIS is not permitted.

8.3.28.16 Precision Approach Radar / Surveillance Radar Approach

PAR / SRA approaches are authorised only to aerodromes located in the UK and Gibraltar. Prior to conducting a PAR or SRA approach the UK phraseology in the OM-C shall be read and briefed by both flight crew members (OM-C, Chapter 13 EUR-Europe, UK, section 6).

8.3.28.17 FMS / FMC Approach Based on Ground Based NAVAID

The following approaches should be flown in FMS/FMC mode (if available in the Navigation Database):

- VOR
- VOR/DME
- NDB
- NDB/DME

The following shall be adhered to:

- The approach has to be selectable from the FMS/FMC Navigation Database.
- The FMS/FMC Navigation Data Base shall be in date.
- The navigation sensors used by the FMS/FMC are appropriate for the type of approach.
- If using GPS as a navigation sensor, RAIM shall be available during the entire approach.
- No part of the approach may be altered in the FMS/FMC except correcting altitudes for cold temperatures (See [Checking of Terrain Clearance - Altitude Corrections](#)).
- The FMS approach waypoints and altitudes shall be compared with the approach chart.

NOTE

Localiser based approaches (ILS Localiser and back course localiser, SDF, LDA, MLS, and IGS) shall not be flown from the FMS/FMC mode beyond the FAF.

8.3.28.17.1 FMS / FMC PFD Approach Display

FMS / FMC vertical and horizontal guidance shall be displayed in accordance with the OM-Bs including, as a minimum, the VOR or NDB bearing-pointer displayed as means of FMS / FMC data accuracy verification and backup in case the FMS / FMC data becomes unreliable. If DME is required for the approach, an integrated or separate display of the DME shall be used.

Where DME distances are not published on the approach plate, the FMS range may be used to assist in achieving a constant descent profile.

If either PFD is incapable of displaying the VOR or NDB bearing-pointer, the PM PFD shall display conventional VOR or NDB information.

8.3.28.17.2 FMS / FMC Approach Procedures

Refer to AFM/OM-B.

8.3.28.17.3 FMS / FMC Approach Contingency

Any time the tolerances for deviations of the VOR or NDB approach are exceeded a missed approach shall be initiated unless the landing runway visual references as per [Visual References for Instrument Approach Operations](#) are in sight.

The approach briefing shall include the course of actions in case the FMS/FMC data becomes unreliable.

8.3.28.18 Continuous Descent Final Approach (CDFA)

CAT.OP-MPA.115

Continuous Descent Final Approach (CDFA) is the term for a specific technique for flying the final approach segment of a non-precision instrument approach procedure as a continuous descent, without level-off, from an altitude at or above the Final Approach Fix altitude to a point approximately 50 ft above the landing runway threshold.

All non-precision approaches shall be flown using the CDFA technique unless a different technique is approved by ANAC and stated in the respective OM-C briefing or NTA aerodrome briefing.

NOTE

For NPAs that do not meet the criteria in [Limitations](#) to be flown as a CDFA but can be flown as circle to land or visual approaches, refer to [NPAs that do not meet CDFA Criteria - Non-CDFA](#).

The CDFA descent gradient is shown on the approach chart together with a table giving the rate of descent required to maintain this for a range of ground speeds. The descent path shall be monitored with reference to the check altitudes / heights associated with the corresponding range (DME or FMS), observing any stepdown crossing altitudes if applicable.

In all cases where the FMS is used to provide guidance for NON-RNAV (GNSS) approach procedures, the primary source of navigational information is the conventional navigation aid data. In the event of a discrepancy between the conventional navigation data and the FMS guidance revert to conventional navigation data.

There is no level flight at MDA(H) when using the CDFA technique. To ensure sufficient obstacle clearance and avoid descent below MDA(H) during a missed approach an increment of 50 ft shall be added to the published MDA(H). The minima obtained by this calculation are to be treated as a DA(H).

The manoeuvre associated with the vertical profile of the missed approach shall be initiated not later than reaching the MAPt or the DA(H) specified for the approach, whichever occurs first. In the majority of cases the DA(H) associated with a CDFA will be reached some distance prior to MAPt. Since the missed approach routing may well have terrain or obstacle considerations, it is important that no turns are initiated until reaching the MAPt.

An approach shall not be continued below DA(H) unless the visual references specified in [Visual References for Instrument Approach Operations](#) are acquired. Visual contact with the ground is not sufficient criteria for continuation of the approach below DA(H).

8.3.28.18.1 Limitations

A CDFA approach shall only be conducted if:

- a. The approach procedure is based on NDB, NDB/DME, VOR, VOR/DME, LOC, LOC/DME, VFD, SRA, GNSS/LNAV; and
- b. The final approach track offset is:
 - $\leq 5^\circ$ for Category C and D aeroplanes; or
 - $\leq 15^\circ$ for Category A and B aeroplanes;unless a larger offset is permitted for approaches where this is specifically authorised in a NetJets aerodrome briefing.
- c. The angle of the approach does not exceed 4.5° (the optimum angle is 3°); and
- d. A FAF, or another appropriate fix, e.g. final approach point, where descent initiated is available; and
- e. The distance to the THR is available by FMS/GNSS or DME; or
 - the distance from the FAF or another appropriate fix to the threshold (THR) is less than or equal to 8NM in the case of timing; or
 - the minimum final segment of the designated constant angle approach path shall not be less than 3 NM from the THR unless approved by the authority.

NOTE

For NPAs that do not meet these criteria above, refer to [NPAs that do not meet CDFA Criteria - Non-CDFA](#).

8.3.28.18.2 CDFA Operational Procedures

- a. The decision altitude will be obtained by adding 50 ft to the published straight in landing DA/MDA.
- b. Cold temperature correction shall be applied to all minimum altitudes.
- c. The descent should be initiated approximately 0.3 NM before the published descent point in order to become established on the desired glide path.

NOTE

If flying an approach subject to cold temperature corrections and the platform altitude is not corrected due to being above MSA, the descent point will need to be delayed by an appropriate distance in order to intercept the correct vertical profile.

- d. During the descent the PM shall make target altitude calls to maintain the vertical profile. This will normally be accomplished by using the appropriate DME distances or distance to threshold as depicted on Jeppesen approach plates.

NOTE

If flying an approach subject to cold temperature corrections, then each of these altitudes will need to be corrected in order to maintain the correct profile. It is advisable to calculate and note the corrected altitudes prior to initiating the approach.

- e. 100 ft above decision altitude, the PF will start their transition to acquire visual references.
- f. On reaching DA, if adequate visual reference has been acquired the PF may continue the approach to land.
- g. If adequate visual reference has not been acquired the PF shall execute a go-around and follow the published missed approach procedure.
- h. Missed approach procedure:
 - The missed approach shall be initiated not later than reaching the MAPt or the DA(H), whichever occurs first.
 - The lateral profile shall be flown via the MAPt unless otherwise stated on the approach chart.

For aeroplane type specific procedures refer to OM-B.

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8.3.28.18.3 NPAs that Do Not Meet CDFA Criteria - Non-CDFA

NPAs that do not comply with the CDFA criteria as per [Limitations](#) are designated as Non-CDFAs (NTA abbreviation for ease of use) and shall not be flown to the published straight-in minima.

The minima and procedures below allow for either circle to land or visual approach procedures to be used. These procedures are, therefore, not considered NPAs that require the application of the CDFA technique.

NOTE

These are NOT NPAs that require approval from ANAC to be flown to the straight-in minima using a technique other than CDFA.

A. Non-CDFA NPAs WITH published circle to land minima

If circle to land minima are published for the specific procedure, then the approach may be conducted to the circle to land minima and the respective circle to land procedures shall be applied.

B. Non-CDFA NPAs WITHOUT published circle to land minima

1. If circle to land minima are not published for the procedure then flight crew should consider performing, if available, a different approach to the runway or fly an approach to a different runway and circle to land to the runway in use.
2. If no other approach procedure is available, then the approach may be flown to the beginning of the final approach segment (FAS)*. Beyond that point, the approach shall only be flown as a visual approach and the respective visual approach procedures shall be applied. At beginning of the FAS, the weather shall be such that a visual approach can be conducted.
 - Minimum weather for dispatch shall be: 5000 m visibility and ceiling above the beginning of the FAS height.
 - The FAS normally begins the FAF or, if no FAF is defined, when the aeroplane is established on the final approach track in accordance with [8.3.28.5 c](#).

Flight crew will normally be advised that an NPA is Non-CDFA via an IJet feasibility note as follows:

- that the particular approach is Non-CDFA and that the approach may be flown to the circling minima; or
- that the particular approach is Non-CDFA and that a visual approach is required for the FAS and includes the dispatch weather minima.

8.3.28.19 Instrument Guidance System (IGS) Approaches

The IGS is a system that uses ILS components, but an IGS approach is not an ILS procedure. An IGS approach may be flown as an ILS procedure, but the published altitudes at the published DME/LOC steps shall strictly be observed and complied with.

The IGS may be offset from the landing direction and when reaching the DA/H the runway can still be several NM away and may not yet be in sight. From DA/H the flight will need to be continued visually maintaining visual ground contact and lining up visually with the runway when in sight.

If visual flight conditions are not achieved upon reaching the DA/H, an immediate missed approach procedure as published on the approach charts shall be initiated.

For aerodrome specific procedures refer to the OM-C.

8.3.28.20 Commencement and Continuation of Approach (Approach Ban)

CAT.OP.MPA.305

The commander or the pilot to whom conduct of the flight has been delegated may commence an instrument approach regardless of the reported RVR/VIS.

However, if the reported RVR/VIS* is less than the applicable minimum the approach shall not be continued:

- a. below 1,000 ft above the aerodrome; or
- b. into the final approach segment in the case where the DA/H or MDA/H is more than 1000 ft above the aerodrome.

**Where the RVR is not available, RVR values may be derived by converting the reported visibility (CMV). If CMV is equal or greater than the applicable minimum, then the instrument approach can be continued to the DA/H or MDA/H, even when the reported VIS is less than the applicable minimum.*

If, after passing 1000 ft above the aerodrome, the reported RVR/VIS falls below the applicable minimum, the approach may be continued to DA/H or MDA/H.

The approach may be continued below DA/H or MDA/H and the landing may be completed provided that the visual reference adequate for the type of approach operation and for the intended runway is established at the DA/H or MDA/H and is maintained.

The touchdown zone RVR shall be controlling. If the touchdown zone RVR is not reported, then the midpoint RVR should be the controlling RVR. In either case, if reported and relevant⁹, the midpoint and stopend RVR shall also be controlling. The minimum RVR value for the midpoint shall be 125 m or the RVR required for the touchdown zone if less, and 75 m for the stopend.

Where the RVR is not available, CMV should be used except for the purpose of continuation of an approach in LVO.

Where a State Approach Ban is more restrictive, the published State Approach Ban applies (Ref. Route Manual, section AIR TRAFFIC CONTROL).

⁹

⁹ 'Relevant', in this context, means that part of the runway used during the high speed phase of the landing down to a speed of approximately 60 knots.

8.3.28.20.1 Visual References for Instrument Approach Operations

For instrument approach operations Type A and CAT I instrument approach operations Type B, at least one of the visual references specified below should be distinctly visible and identifiable to the pilot at the MDA/H or the DA/H:

- i. elements of the approach lighting system;
- ii. the threshold;
- iii. the threshold markings;
- iv. the threshold lights;
- v. the threshold identification lights;
- vi. the visual glide path indicator;
- vii. the touchdown zone or touchdown zone markings;
- viii. the touchdown zone lights;
- ix. runway edge lights.

For EVS operations to a runway, the approach may be continued below the DA/H provided that the pilot can identify on the EVS image either:

- i. the approach light system; or
- ii. both of the following:
 - a. the runway threshold identified by the beginning of the runway landing surface, the threshold lights or the runway end identifier lights; and
 - b. the TDZ identified by the TDZ lights, the TDZ runway markings or the runway edge lights.

A missed approach should be executed promptly if the required visual reference is not distinctly visible and identifiable to the pilot without reliance on the EFVS by the height determined in [Determination of DH/MDH for Instrument Approach Operations, point 4](#) and the OM-B.

The required visual reference referred to in the paragraph above should be the same as the one required for the same approach flown without the use of EFVS.

8.3.28.21 Land and Hold Short Operations (LAHSO)

Land and Hold Short Operations (LAHSO):

- involve an aircraft landing and holding short of an intersecting runway, taxiway, or a designated point on a runway.
- runways are identified by yellow, parallel hold short markings painted across the runway. In pavement lighting may also be provided.
- are conducted in North America.

NTA aircraft are not authorised to participate in LAHSO operations. Refer to OM-C 13.12.1.27 – Land and Hold Short Operations (LAHSO) for further details.

8.3.29 Threshold Crossing Height and Touchdown

8.3.29.1 Threshold Crossing Height

The landing threshold (runway threshold or displaced threshold) should normally be crossed either on the electronic glide path, visual approach slope indicator or between 50 ft (60 ft for steep approaches) and 35 ft.

8.3.29.2 Touchdown

Normal Approach Aim-Point

A normal approach aim-point should be such that main wheel touchdown occurs at 1,000' after the runway threshold.

Due to environmental factors touchdown could occur slightly before or after 1,000' after the threshold. Nevertheless, touchdown should not occur outside 500' to 2,000' from the runway threshold. A go-around should be initiated if 2,000', after the runway threshold, has been passed and the main wheels have not touched-down.

WARNING

During Fractional Operations landing-distance-factoring or on short runways a touchdown point of more than 1000' after the runway threshold could lead to insufficient runway distance available and result in a runway overrun.

NOTE

Notwithstanding the above, the commander is responsible for initiating or commanding a go-around if the touchdown point is such that the landing cannot be accomplished within the remaining landing distance.

8.3.29.3 Noise Abatement After Landing

After each landing the thrust levers shall be brought into reverse thrust as soon as practicable.

When landing at night, reverse thrust should only be applied to the extent required by safety, paying due regard to landing weight, length of runway, wind, and runway conditions.

During daytime, reverse thrust should be applied as deemed adequate by the PF for the prevailing conditions (i.e. runway length, runway status), with maximum reverse thrust only being used in the interest of safety.

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8.3.30 Balked Landing

Definition: A balked landing is a landing manoeuvre that is unexpectedly discontinued at any point below the OCA/DA/MDA.

Before touchdown:

Procedures as detailed in the OM-B / AFM shall be followed. In the absence of procedures for a go-around or balked landing, the procedures for missed approach shall be followed.

NOTE

Some aerodromes might have different ATC procedures for a missed approach or a balked landing. The flight crew shall be familiar with both before commencing the approach. Flight crew may coordinate with ATC for different procedures if published procedures are less safe or less efficient. This could include, but is not limited to, a visual circuit.

After touchdown:

After touchdown the decision to discontinue a landing depends on many factors and is therefore not proceduralised. The decision to discontinue a landing rest solely with the flight crew.

Factors to be taken into consideration for the decision to discontinue a landing are, but not limited to:

- a. Flight crew proficiency (a balked landing after touchdown is not a trained manoeuvre);
- b. Runway length remaining (runway length remaining on most runways is not indicated);
- c. Runway length required (there is no published runway length required data after touchdown);
- d. Runway contamination / condition (unexpected and adverse runway conditions might increase the stopping distance required. Although normally the aeroplane should be able to stop in the runway length remaining if the landing was performed according to standard and the runway condition was as expected);
- e. Non-standard landings (when a touchdown occurred too fast and / or too long down the runway then the landing data published by the manufacturer are not valid anymore and determining whether or not the aeroplane can come to a full stop at the end of the runway is impossible);
- f. Obstacles / terrain in the departure area (aeroplane performance might not be sufficient to clear terrain or obstacles in the departure area);
- g. Aeroplane malfunctions (Aeroplane malfunctions can either increase runway required to decelerate to a stop or accelerate to V_{rotate});
- h. Aeroplane mass (the higher the landing mass of the aeroplane the more runway is required to accelerate to V_{rotate});
- i. Aeroplane performance;
- j. Aeroplane lift device settings (flap / slats), depending on manufacturer guidance and the landing configuration the flap / slats maybe need to be repositioned. This repositioning requires time during which the flaps / slats could still keep creating disproportional drag decreasing aeroplane acceleration);

- k. Thrust-reverser position (stowing thrust-reversers and spooling up the engines from idle takes significant time during which the aeroplane will continue to decelerate and reduce runway length available);
- l. Pressure altitude (as pressure altitude increases the performance of the engines is decreased, unless the engines are flat rated). The TAS required to rotate is higher and therefore requires more runway length to attain);
- m. Temperature (as temperature increases the performance of the engines decreases, unless they are flat rated. The TAS required to rotate is higher and therefore requires more runway length to attain);
- n. Current airspeed (slower airspeeds require exponentially more runway to accelerate);
- o. Wind condition (less headwind or more tailwind exponentially increases the runway required to accelerate to Vrotate);
- p. Weather (Windshear, thunderstorms, icing might adversely impact the aeroplane on the departure track);
- q. Manufacturer guidance on Vrotate after touchdown is not always available.

8.3.31 Go-around and Missed Approach

1. A go-around is a normal procedure which should be applied without hesitation if required.
2. A go-around has to be executed as soon as any active flight crew member calls out GO-AROUND, regardless of who is PF or PM. The PF shall immediately initiate the missed approach procedure in accordance with the requirements of the approach being flown or ATC instructions if different.
3. Adequate protection from obstacles is assured if the missed approach procedure is initiated not lower than the decision height / decision altitude / minimum descent altitude (DH/DA/ MDA), and not later than the missed approach point (MAPt) where one is specified. For a CAT I and APV approach procedure the MAPt is the point where the glide path intersects the decision height.
4. In the event a go-around is initiated prior to arriving at the MAPt, the pilot should, while climbing (or descending if required by the missed approach procedure), proceed towards the MAP and then follow the missed approach procedure in order to remain within the protected airspace. In a CAT I approach the Commander should proceed to the middle marker or equivalent fix before following the missed approach procedure.
5. Once the missed approach has been initiated, it shall be completed unless otherwise instructed by ATC.
6. When performing a visual approach, whenever possible flight crew shall ask ATC for missed approach instructions.

8.3.31.1 Reason for Go-around

A go-around shall be initiated for the reasons below:

1. If the required visual reference as per [Commencement and Continuation of Approach \(Approach Ban\)](#) has not been established upon reaching the decision height / decision altitude / missed approach point, the prescribed missed approach procedure shall be initiated:
 - When executing a CAT I or APV approach immediately upon reaching the decision height / decision altitude;
 - When executing a non-precision approach procedure using CDFA technique immediately upon reaching the derived decision height / decision altitude;
 - Where no missed approach point is specified the missed approach shall be initiated at a suitable fix (middle marker, DME fix, etc.);
 - If no suitable fix is available, or if the MAP facility is unserviceable, the missed approach shall be initiated when time to runway threshold has expired. The foregoing method (timing to threshold) shall not be used if the missed approach procedure requires a turn prior to the threshold.

NOTE

During a non-precision straight-in approach, a landing shall only be made if visual reference to the ground is established early enough to permit a safe landing without exceeding a maximum rate of descent of 1,000 ft/min., and paying due regard to the runway conditions and the runway length available for landing.

When using a rate of descent of close to 1,000 ft/min, special attention shall be given to the flare-out characteristics of the aeroplane.

2. If at any time after descent below the decision height / decision altitude or minimum descent altitude, the PF cannot maintain the required visual reference, they shall immediately execute or command a go-around. The appropriate missed approach procedure shall be followed, and ATC shall be contacted.
3. When instructed by the appropriate ATC;
4. If the specified approach aid used is suspected to be malfunctioning or due to navigational signal unreliability, unless the visual references required by [Commencement and Continuation of Approach \(Approach Ban\)](#) have been attained, and the approach can be continued visually and both flight crew members agree to continue.

NOTE

For specific details relating to malfunctions and degradations during PBN approaches see [PBN Approach Contingency](#).

5. If required in accordance with the Approach Ban ([Commencement and Continuation of Approach \(Approach Ban\)](#));
6. If at any time during the approach, there is any confusion, crew coordination breakdown or disagreement between crew members on course, heading, altitude, position, procedures, or configuration, a missed approach shall be initiated. Any discrepancies between crew members or instruments will then be resolved at a safe altitude.

A go-around / missed approach is not required for the items in this point if the visual references have been attained and the approach can be continued visually and both flight crew members agree to continue.

7. There is uncertainty about situational awareness;
8. Checklists are being conducted late or the crew is task saturated;
9. Any malfunction threatens the successful completion of the approach;
10. The approach becomes destabilised or ATC changes will result in an destabilised approach;
11. During an instrument approach when:
 - More than half-scale deflection of the localiser or glideslope;
 - More than half-scale deflection of the CDI on a VOR approach;
 - More than half-scale deflection of CDI on RNP APP approaches (also refer to other situations in [PBN Approach \(RNP APCH\)](#));
 - More than 5 degrees bearing deviation on an NDB approach;
 - Loss of visual contact with the surface during a circling manoeuvre.

A go-around / missed approach is not required for the items above if the visual references required by [Commencement and Continuation of Approach \(Approach Ban\)](#) have been attained, the approach can be continued visually and both flight crew members agree to continue.

12. A RED Windshear Warning is triggered or negative windshear is encountered - proceed in accordance with the aeroplane specific OM-B;
13. EGPWS warning - proceed in accordance with [\(Enhanced\) Ground Proximity Warning System \(EGPWS\) Procedures](#).

8.3.31.2 After Go-around

At a safe altitude clear of obstacles, the PF shall call for the appropriate checklist.

A safe altitude will be: The published missed approach altitude when flying the published missed approach, the minimum vectoring altitude when being vectored by ATC, or the traffic pattern altitude when remaining in the pattern in VMC.

8.3.31.3 Approach Following a Go-around

More than two approaches may only be made if attempting a third approach would be no less safe than diverting to an alternate aerodrome.

If a go-around has been executed for meteorological reasons, another approach shall only be commenced if the Commander has reasons to believe that a second approach will lead to a successful landing. More than two approaches shall only be made if there is an indication that meteorological conditions have considerably improved, giving greater probability of a successful landing, and making a third approach would be no less safe than diverting to an alternate aerodrome.

8.3.32 Noise Abatement

At many locations throughout Europe the close proximity of aerodromes to densely populated areas has created considerable problems resulting from noise by aeroplanes landing and taking off. In many instances this has led Air and State authorities to enforce noise abatement rules. These are published in OM-C (Route Manual) as appropriate.

Pilots shall take all possible measures during take-off and landing to ensure that noise created be kept to a minimum, and shall comply with relevant noise abatement procedures and ATC instructions. Safety of flight operations shall always take precedence over compliance with noise abatement rules, and the final decision shall always rest with the Commander.

At aerodromes where the preferential use of specific runways is prescribed, pilots should normally use the preferential runway for landing or take-off provided:

- All requirements in respect of runway length and obstacle clearance are met; and
- Compliance with the above is not in conflict with any other provision relating to tail wind / crosswind limitations (e. g. associated with the application of reduced meteorological minima).

However, the Commander has the authority to request a runway other than the aerodromes preferential runway if they consider this is preferable from a safety perspective.

The commander has the authority to decide not to execute a noise abatement departure procedure (NADP), if conditions prevent the safe completion of the procedure:

1. Flight into icing conditions, when they dictate thrust settings and / or speed ranges different to the NADP recommendation, rendering the latter inapplicable.
2. Early level-offs below 3000 ft AGL which preclude the use of a steep climb.
3. Conduct of noise abatement climb procedures is secondary to the satisfaction of obstacle requirements. The respective departure climb gradients shall always be observed. Particularly when a reduction below maximum climb power might be prescribed by a manufacturer procedure or when flying into icing conditions, climb gradient compliance shall be carefully assessed.
4. The power settings to be used after the failure or shutdown of an engine or any other apparent loss of performance, at any stage in the take-off or NADP are at the discretion of the commander, and noise abatement considerations no longer apply.
5. NADP is not to be used in conditions where a wind shear warning exists, or the presence of wind shear or downburst activity is suspected.
6. Refer to the aeroplane respective OM-B for the NADP.

8.3.33 Communications

Air traffic services (ATS) appropriate to the airspace and the applicable rules of the air, shall be used on all flights whenever available.

Phraseology

Standard ICAO phraseology shall always be used.

Crews shall become familiar with the procedures and phraseology used by ATC within all the areas of operation approved for use in the AOC.

Monitoring

Flight Crew shall report to ATS during the period 20 to 40 minutes following the time of the last contact, whatever the purpose of such contact, merely to indicate that the flight is progressing according to plan, such report to comprise identification of the aircraft and the words 'operations normal'.

Headset Requirements

Flight crew shall utilise headsets that fulfil the following requirements:

- Earphones and microphone shall match the communication system's characteristics and the flight crew compartment environment.
- The headset shall be sufficiently adjustable to fit the pilot's head.
- Headset boom microphones shall be of the noise cancelling type.
- If noise cancelling (ANR) earphones are used, these shall not attenuate any aural warnings or sounds necessary for alerting the flight crew on matters related to the safe operation of the aeroplane.

In the event that flight crew experience any difficulties using a specific type of headset (e.g. lack of compatibility between headsets or with the aircraft communications systems, unreliability etc.) then this shall be reported in a NetJets Safety Report.

Headset Use

Headsets with boom microphones shall be worn by all flight crew members required to be on duty in the FCC when:

- on the ground when receiving ATC departure clearance and when engines are running;
- in flight below transition altitude, 10,000 ft AAL or aerodrome MSA, whichever is higher; and
- Whenever deemed necessary by the commander.

Headset use at other times is at the discretion of the commander.

The boom microphone shall be in a position that permits its use for two-way radio communications.

8.3.34 Data-link Communications and Surveillance

All terminology regarding Data-Link Communications is derived from the ICAO Global Operational Data Link Document (GOLD) Manual.

8.3.34.1 Data-link Communications and Surveillance - Operational Approval

Data-link operational approval For NTA aeroplane types that are approved for Data-Link Communications by Autoridade Nacional de Aviação Civil (ANAC), see [Approvals by Aeroplane Type](#)

Performance-based Communication and Surveillance (PBCS) operational approval

PBCS has a separate and specific operational approval. For NTA aeroplane types that are approved for PBCS by the Autoridade Nacional de Aviação Civil (ANAC), see Approvals by Aeroplane Type.

8.3.34.1.1 Performance-Based Communication and Surveillance (PBCS)

Performance-Based Communication and Surveillance (PBCS) comprises Performance-Based Communication (PBC) and Performance-Based Surveillance (PBS), which refer to communication and surveillance based on performance specifications applied to the provision of air traffic services. The standards and procedures for an air traffic management (ATM) operation predicated on communication and surveillance capabilities, such as the application of reduced separation minima, must refer to the appropriate Required Communication Performance (RCP) and Required Surveillance Performance (RSP) specifications. The RCP and RSP specifications are a set of requirements for air traffic service provision and associated ground equipment, aircraft capability, and operations needed to support performance-based communication and surveillance. The PBCS concept enables reduced separation minima and occurs in the NAT HLA Organised Track System and elsewhere around the world as identified by the Jeppesen Airway Manual. PBCS is fully described in ICAO PBCS Manual doc 9869, available in the Jeppesen Airway manual.

8.3.34.1.2 PBCS Operational Approval

NTA has PBCS operational authorisation which permits operations that take advantage of reduced separation standards, in airspace where they are available, to RCP and RSP specifications.

Operations into PBCS airspace shall be with PBCS qualified crew and PBCS authorised aircraft.

The aircraft types authorised for PBCS are shown in OM-A Appendix 2 to Chapter 8 – Approvals by Aeroplane Type.

Crew, dispatch, and engineering personnel shall be PBCS qualified as detailed in OM-D.

PBCS flight plans shall be filed to show required RCP/RSP capabilities.

NTA shall participate in the PBCS monitoring programme (PBCS Charter) via the FANS 1/A Central Reporting Website (www.fans-cra.com).

8.3.34.1.3 RCP and RSP Specifications Supporting ATM Operations

RCP specification is identified by a designator (e.g. RCP 240). The RCP designator represents the maximum communication transaction time in seconds, after which the initiator should revert to an alternative procedure.

RSP specification is identified by a designator (e.g. RSP 180). The RSP designator represents the value for the surveillance data delivery time in seconds when the surveillance data delivery is considered overdue. To perform ATM operations within a performance-based

airspace, functional, safety, and performance criteria need to be specified for the applicable communication, navigation, and/or surveillance (CNS) elements. RCP and RSP specifications, in conjunction with RNP/RNAV specifications, provide these criteria and are intended to facilitate the development of standards and procedures for ATM operations.

For example, to utilise reduced separation minima within the NAT HLA Organised Track System (OTS), the requirements include: CPDLC with RCP 240, ADS-C with RSP 180 and RNP 4

8.3.34.1.4 PBCS Required Equipment

When a flight is planned to operate within PBCS airspace, a check of the MEL in respect of any unserviceable equipment shall be made prior to departure to ensure that the RCP, RSP, and RNP specification requirements for that airspace are met.

8.3.34.1.5 Operating within PBCS airspace.

Crew are expected to respond to CPDLC messages as soon as practicable after they are received, and when within an RCP 240 environment it is expected that the response will be within one minute. If more time is required to action higher priority tasks for example, then crew should send an RSPD-3 STANDBY response.

PBCS requires no change to published emergency/contingency procedures; however, crews should be aware of the reduced separation.

8.3.34.1.6 Loss of data link capability whilst in, and prior to entering the airspace requiring specific RCP/RSP specifications.

In the event of an aircraft data link system failure, the flight crew should notify the ATS unit of the situation using the following voice phraseology:

Flight crew CPDLC FAILURE. CONTINUING ON VOICE

Controller ROGER. CONTINUE ON VOICE

NOTE

The flight crew continues to use voice until the functionality of the aircraft system can be re-established.

Since RCP/RSP affords reduced separation, the loss of either of these capabilities may lead to ATC having to alter the ATC clearance to maintain required separation.

8.3.34.2 Data-link Communications and Surveillance - Aeroplane Configurations

For NTA aeroplane types that are approved for Data-Link Communications operations, see [Approvals by Aeroplane Type](#).

Aeroplane type	Data Link System	Designator of subnetwork	RCP	RSP
BD-700-1A10 & BD-700-1A11	FANS 1/A+	SATCOM (Inmarsat) VDL M2 VDL M0/A	RCP 240	RSP 180
	PM-CPDLC ATN B1	VDL M2	N/A	N/A
CL-600-2B16	FANS 1/A+	SATCOM (Iridium) VDL M2 VDL M0/A	RCP 240	RSP 180
	PM-CPDLC ATN B1	VDL M2	N/A	N/A
BD-100-1A10	FANS 1/A+	SATCOM (Iridium) VDL M2 VDL M0/A	RCP 240	RSP 180
	PM-CPDLC ATN B1	VDL M2	N/A	N/A
CE-680A	PM-CPDLC ATN B1	VDL M2	N/A	N/A
EMB-550	PM-CPDLC ATN B1	VDL M2	N/A	N/A

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8.3.34.3 Areas of Operation

In accordance with the § 03. Areas of Operation of the NetJets Air Operator Certificate the area of operation for Data-link Communications will be WORLDWIDE anywhere Data-link communications have been implemented.

8.3.34.4 Pre-flight Procedures

8.3.34.4.1 ICAO Flight Plan Annotation

ATS systems use Item 10 (Equipment) of the standard ICAO flight plan to identify an aeroplane's data link capabilities. Flight Dispatch is responsible for inserting the following items in the ICAO flight plan:

- Item 10 - One or more of the descriptors, as appropriate, listed in the table below to identify aircraft's data link equipment and capabilities:

Item 10a – CPDLC equipment and capabilities	Descriptor
CPDLC ATN VDL Mode 2 (ATN B1)	J1
CPDLC FANS 1/A HFDL	J2
CPDLC FANS 1/A VDL Mode 0/A	J3
CPDLC FANS 1/A VDL Mode 2	J4
CPDLC FANS 1/A SATCOM (INMARSAT)	J5
CPDLC FANS 1/A SATCOM (MTSAT)	J6
CPDLC FANS 1/A SATCOM (Iridium)	J7

Item 10a – RCP capability	Descriptor
CPDLC RCP 240	P2

Item 10b – ADS-C equipment and capabilities	Descriptor
ADS-C with FANS 1/A capabilities	D1
ADS-C with ATN capabilities	G1

- Item 18 – SUR/RSP180 shall be inserted to designate RSP 180 capability.

RCP/RSP capabilities required for PBCS authorisation are demonstrated as follows:

- Item 10a - CPDLC descriptors (J1-J7); RCP capability 'P2'; and
- Item 10b - ADS-C descriptors (D1 or G1); and
- Item 18 - 'SUR/RSP180' to show RSP capability

8.3.34.4.2 Notifications of Outages

Operations Planning and Control (Dispatch) continuously monitors all NOTAMs, thus any Data Link failure will be transmitted to all applicable crews via the established communication system. To notify any aeroplane while airborne, Operations Planning and Control (Dispatch) will use a non-Data-Link network SATCOM, AFIS messages via GDC or Stockholm radio.

8.3.34.4.3 Reporting of issues with datalink

The Avionics Manager will be the primary point of contact for any issues with datalink. All the reports related to datalink will be processed by the Avionics Manager and then forwarded to the NJE Chief Pilot.

Any issues with datalink communications shall be reported using the Safety Report form available in the Safety Reporting System.

Apart from the normal data on the Safety Report form, the following items shall be stated in the Safety Report:

- Active ACC (CDA) - ICAO identifier of the ACC controlling the aeroplane at the time of the event.
- Next ACC (NDA) - If the problem involves a handover between ACCs, or occurs close to the time of a handover, then this should contain the ICAO identifier of the ACC to which control was being handed over.

The description of the issue should include:

- a. Type of event, e.g.:
 - Aeroplane DLIC (LOG-ON) failure;
 - Aeroplane CPDLC failure;
 - Unexpected CPDLC Provider abort;
 - Unexpected or Missing CPDLC message displayed;
 - Erroneous CPDLC message received;
 - Delayed CPDLC message receipt.
- b. Details of the event:
 - A complete description of the problem that is being reported;
 - Any FCC indications;
 - Any indications provided to the controller when the problem occurred;
 - Any additional information that the reporter considers might be helpful and is not included on the list above.

8.3.34.5 Flight Procedures

Data-link communications is an international program mainly guided by ICAO recommendations. As such there are significant regional differences. These differences shall be confirmed in the OM-C - Jeppesen Airway Manual in the en-route section.

The following procedures should apply for all the areas of data-link communications.

8.3.34.5.1 General Rules

The same principle as in Voice communication is applied, i.e. the PM is responsible for communications, nevertheless it is necessary to confirm the understanding of PF of all received and sent messages.

- Reception of an ATC MSG/ALERT should be announced automatically; if the announcement is not made, or not immediately recognised by both FCMs, the first FCM to recognise an ATC MSG/ALERT shall announce 'ATC MESSAGE'.
- UPLINK ATC MESSAGES shall be read silently by each pilot.
- PF will verbally transmit their understanding to the PM regarding the ATC MESSAGE.
- For DOWNLINK messages the PM will type the message.
- PM will type the messages using as much as possible pre-formatted messages; free text messages should be used as a last resource.
- Once the message is typed and before it is sent the PF will confirm the message.

8.3.34.5.2 Use of Voice vs CPDLC

When operating within airspace where CPDLC is available and local ATC procedures do not state otherwise, CPDLC should be the primary means of communication. Voice should be used as the backup communication medium. During an emergency, the flight crew would normally revert to voice communications. However, the flight crew may use CPDLC for emergency communications if it is either more expedient or if they are unable to establish voice contact.

8.3.34.5.3 Notification of Frequencies to the Preceding ATSU

ATSUs shall advise frequencies to the preceding ATSU, in accordance with the appropriate letters of agreement. The uplink CPDLC frequency transfer message elements can accommodate only one frequency variable. Due to this limitation, the controller will insert the primary HF frequency in these messages.

8.3.34.5.4 Approval of Request or Clearance / Instruction

The **WILCO** downlink message indicates that the pilot will comply fully with the clearance / instruction contained in the associated uplink message. The readback of a clearance or instruction issued by CPDLC is not required. The **ROGER** or **AFFIRM** uplinks are not appropriate responses to a clearance request and shall not be used for this purpose. The controller shall only approve a clearance request by uplinking a message containing an actual clearance.

8.3.34.5.5 Negative Response to a Clearance Request

When a clearance request is denied, the controller shall use the element **UNABLE** (not **NEGATIVE**) in the uplink response. The aeroplane's current clearance shall not be restated. Pre-formatted elements such as **DUE TO TRAFFIC** (or a free text element) should be added to the response message if clarification is considered necessary. Additional elements (including free text elements) in the form of an explanation shall be included when responding to a multiple clearance request where some, but not all clearance requests can be granted.

8.3.34.5.6 HF Communications for Oceanic and Remote Airspace

High Frequency radio (HF) is required in all controlled oceanic FIRs, whether or not CPDLC or ADS-C is in use. This means that HF shall be operative for departure to remote and oceanic areas. It is imperative that crews comply with the transfer-of-control uplink by establishing HF contact or HF monitoring with each general purpose (GP) radio facility providing communications services with ATS in each of the FIRs through which the flight operates. In the data link environment, this requirement is met by calling the GP radio facility serving the FIR that the flight is entering, determining the primary and secondary HF frequency assignments, and obtaining a SELCAL check. If this is not accomplished, the GP radio facility has no way of knowing of the flight's existence and required communications may be compromised.

8.3.34.5.7 Source for CPDLC Operating Procedures

For standardisation purposes, operating procedures have been adopted in all oceanic airspace. These procedures are detailed in the ICAO Global Operational Data Link Document (GOLD).

8.3.34.5.8 AFN Logon

Successful logon is required for CPDLC. Logon should be initiated between 15 and 45 minutes before entering an area in which CPDLC is being used from an area where CPDLC is not in use. On the ground, 15 to 45 minutes prior to departure if the ATSU is within the CPDLC FIR of the departure airport.

The flight identification and the registration used for the logon shall be exactly the same as that filed in the ATS flight plan.

If a logon attempt is not successful, wait at least 5 minutes before making a second attempt. ATS ground systems use the CPDLC logon as a source of aeroplane address for initiation of ADS-C services in the oceanic environment.

8.3.34.5.9 CPDLC Use for Weather Deviation Messages

As expected, given the more direct and reliable nature of CPDLC compared with HF third person voice messaging, improvements in weather deviation request responses have been noted in airspace where CPDLC is used. Priority will be given to aeroplanes appending 'DUE TO WEATHER' to requests or which use the message REQUEST WEATHER DEVIATION.

Basic procedures are published in individual state-published Aeronautical Information Publications (AIPs) and in the Pacific Chart Supplement. As noted in the Supplement, aeroplanes with Satellite Communications (SATCOM) voice capability may utilise published General Purpose Radio telephone numbers to expedite weather deviation clearances if CPDLC response is not timely.

8.3.34.5.10 Multiple Weather Deviations

The distance off track contained in a weather deviation request or clearance is measured reference the nominally cleared track of the aeroplane. Subsequent weather deviations or route clearances supersede any previous weather deviation clearance.

8.3.34.5.11 PF and PM Coordination and Awareness

Both pilots shall review uplinked messages separately and silently. PF shall state to the PM the understanding of the uplink message. Once agreed the PM responds or executes the message. In the case of downlink messages the PM shall input the message and the PF shall confirm it before the PM sends it.

8.3.34.5.12 Transfer of Aeroplane Control to and Augmented Crew

Any augmented crew that takes the position of PM or PF shall be brief about the data link status specifically, ADS-C, CPDLC, CDA, NDA, and any pertinent message that requires attention.

8.3.34.5.13 SELCAL Watch and Guard Frequency

Regardless of whether CPDLC is being used, SELCAL watch shall be maintained on the specified backup or secondary frequency/ies. The VHF guard / emergency frequency shall be monitored and if possible it is recommended that the HF guard frequencies also be monitored. Details of the HF guard frequencies can be found in the Jeppesen Airway Manual: Emergency / Distress and Urgency Radiotelephony Communications Procedures.

8.3.34.5.14 Position Reporting

The flight crew shall ensure that waypoints are sequenced correctly. When an aeroplane passes abeam a waypoint in excess of the defined sequencing parameter, the FMS may not sequence the active waypoint (aeroplane type dependent). If the flight crew does not sequence the waypoint, incorrect information will be contained in ADS-C reports and CPDLC position reports - the next waypoint in these reports will actually be the waypoint that the aeroplane has already passed. In ADS-C environment flight crew should be aware that any point inserted in the active leg will be reported automatically by the ADS-C. In non-ADS-C environment the flight crew shall ensure that position reports are done by either voice or CPDLC.

8.3.34.5.15 Emergency and Non-routine Procedures

The ATSU within whose airspace the aeroplane is operating remains in control of the flight. If the flight crew takes action contrary to a clearance that the controller has already coordinated with another sector or ATSU and further coordination is not possible in the time available, then the flight crew performs this action under their emergency command authority. The flight crew will use whatever means are appropriate, i.e. CPDLC and / or voice, to communicate during an emergency. During an emergency, a controller would normally expect the flight crew to revert to voice communications. However, the flight crew may use CPDLC for emergency communications if it is either more expedient to do so or if they are unable to establish voice contact. The flight crew may use CPDLC to indicate an emergency situation or degraded operations to an ATSU by sending either a MAYDAY MAYDAY MAYDAY or PAN PAN PAN message. The flight crew should use the CPDLC emergency downlink message MAYDAY MAYDAY MAYDAY to automatically select the ADS-C function to emergency mode. The flight crew should only use ADS-C emergency mode when situations prohibit sending a CPDLC emergency message (e.g. in an ADS-C only environment).

8.3.34.5.16 Automatic Transfer of CPDLC and ADS-C Between FIRs

When transferring CPDLC and ADS-C between FIRs, the flight crew should not need to reinitiate a logon. Under normal circumstances, the current and next ATSUs automatically transfer CPDLC and ADS-C services. The transfer can be confirmed by the crew approximately 10 minutes prior to the FIR border, by verifying the Next Data Authority (NDA) becoming in the Current Data Authority (CDA).

8.3.34.5.17 Monitor Uplink Message Instruction

The uplink message will contain a voice frequency to be monitored at a specified time or position. The Flight crew **should not establish** contact with the frequency.

8.3.34.5.18 Contact Uplink Message Instruction

The uplink message will contain a voice frequency to be contacted at a specified time or position. The Flight crew **should establish** contact with the frequency. This instruction might be referred as VCI.

8.3.34.5.19 HF Backup Frequency

If the ATSU assigns a single HF frequency for backup, the flight crew should select another frequency from the same 'family' as a secondary frequency.

8.3.34.5.20 Exiting CPDLC and ADS-C Service Areas

Approximately 15 minutes after exiting CPDLC and ADS-C service areas, the flight crew should ensure there are no active CPDLC or ADS-C connections.

8.3.34.6 Flight Crew and Dispatcher Training

8.3.34.6.1 Flight Crew Training Programs

Before acting as flight crew on flights using data link communications, crews shall receive initial and recurrent training in accordance with OM-D.

8.3.34.6.2 Dispatcher Training Program

Before releasing flights to operate in data link environment, dispatchers should receive training in the following areas:

- Description of the data link network including ACARS, AFTN and SATCOM;
- Flight planning requirements for data link flights;
- Implications of flights departing under MEL relief; and
- Implications of planned and unplanned network outages on data link operations.

8.3.35 APU and GPU Policy

The aim of this policy is to provide recommended guidelines for the safe and efficient utilisation of the APU and GPU during routine NTA flight operations.

8.3.35.1 General

The use of the APU or GPU shall be guided by four basic principles:

1. Safety;
2. AFM procedures and checklists, operational requirements and limitations;
3. Owner comfort and satisfaction;
4. Economy

An operating flight crew member shall be present at the aeroplane when an APU or GPU is supplying power and be in position to respond to an APU malfunction.

Crews shall make every effort to keep APU use within the time limits shown in Table 1 below. Crews may extend or reduce the use of the APU at PIC's discretion. When the APU has been used for an extended period beyond that in the table, the OFP shall be annotated stating the reasons for the extended use of the APU.

8.3.35.2 APU Use Guidelines

Under no circumstances should safety or customer satisfaction (cabin environment) be compromised by attempting to remain within maximum APU use guidelines outlined in Table 1 below.

For specific guidance, refer to fleet APU best practice documentation.

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Table 1 - APU Recommended Maximum Use Guidelines

Type	Recommended use before flight	Recommended use after flight	Notes
CE-560XLS	20 mins		Recommended APU start is 20 mins before engine start.
CE-680A	20 mins		Recommended APU start is 20 mins before engine start.
BD-100-1A10	30 mins*		Recommended APU start is 20 mins before engine start. * 45 minutes if required for ISR.
DA-2EASy	45 mins	15 mins	Consider use of GPU to carry out post-flight cabin preparation.
CL-600-2B16	45 MINS	15 MINS	Consider use of GPU to carry out pre/post-flight cabin preparation.
BD-700-1A10 & BD-700-1A11	75 mins	30 mins	Consider use of GPU to carry out post-flight cabin preparation.

NOTE

1. PICs may exercise discretion to extend APU use when required. Examples include, but are not limited to:
 - a. extreme environmental conditions;
 - b. line training requirements, especially first tour on fleet change;
 - c. aeroplane acceptance from maintenance;
 - d. airport requirements such as mandatory security checks.
2. GPU should only be used when available within handling contract.
3. In some instances, an airport may impose a maximum APU running time which may be more restrictive than the advice in the table above.
4. First flight of the day may require extended APU use.

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8.3.35.3 Recommended APU Fuel Allowance

Calculations for APU fuel to be loaded in accordance with the APU use guidelines above can be based on the table below:

Aeroplane Type	1 hr APU Fuel (lbs)	10 mins APU Fuel (lbs)	Recommended APU Fuel Allowance	
			Time (Min)	Fuel (lbs)
CE-560XLS	105	18	20	35
CE-680A	101	17	20	34
BD-100-1A10	130	22	30	65
DA-2EASy	180	30	45	135
CL-600-2B16	180	30	45	135
BD-700-1A10	360	60	75	450
BD-700-1A11	360	60	75	450

For ease of reference, the recommended APU fuel allowance for a given aircraft type is shown on the front page of the OFP.

8.3.35.4 Ground Operations

The APU shall only be used as necessary to prepare for flight and shall be in accordance with specific airport restrictions for the use of an APU.

The APU shall only be used for pre and post-flight activities that are related to flight duties.

The use of bleed air shall be minimised and only used for owner comfort or cabin conditioning.

When on Hotspare duty, the use of the APU shall be limited to preparing the aeroplane for flight.

When on a Static Demo, the use of the APU shall be coordinated with the on-site sales representative.

8.3.35.5 In-flight Operations

The APU should only be used in-flight in accordance with operating procedures.

The APU shall be shut down prior to departure unless needed for reasons of safety of flight or for operational reasons such as but not limited to:

1. Operational restrictions imposed by the MEL, AFM, or other manuals.
2. Passenger medical condition.
3. Check flights.
4. Training flights.

During descent for landing, the APU shall not be routinely started except for flight safety, operational or technical reasons.

After landing, if the APU is required in accordance with the checklist, it shall only be started as late as is considered practical before reaching the parking position.

8.3.35.6 Use of GPU

Under normal circumstances, a GPU out of contract is more expensive than using the APU.

Certain FBOs provide a GPU within the contracted turn round service. Where this is the case, the GPU should be used in preference to the APU after flight. Issues with electrical power interruptions and voltage stability should be considered if the GPU is used preflight.

The availability of GPUs at each location and FBO is documented in an internal database. The feasibility notes in the company mobile phone crew brief will also inform crews that a contract GPU is available.

If a crew member elects not to use the contracted GPU or the GPU is unavailable, the OFP shall be annotated stating why the GPU was not used.

Consider the following recommendations when using a contracted GPU:

- a. Pre-flight preparation such as FMS programming can be done utilising GPU power.
- b. Post-flight actions such as cabin cleaning, especially on the last flight of the day, should be completed using GPU power (large cabin) or without power (mid cabin).
- c. When arriving on stand and after passenger has disembarked, consider shutting down the APU and transferring to GPU.
- d. Careful consideration should be given to re-fuelling operation whilst the GPU is connected and providing power. Some fuelling contractors may require the GPU to be shut down during re-fuel operations.

8.3.36 Emergency and Abnormal Situation Management

8.3.36.1 Policy

Workload has the biggest effect on the ability to manage any situation effectively. In any situation, but most critically during emergency / abnormal situations, the severity of the problem and the perceived time available to manage it may increase the workload to an unmanageable level, leading to an unfavourable outcome.

In order for a trained crew member to effectively manage a situation and minimise its effect on their available capacity, there should be a method the crew can follow that ensures the correct order of priority of action. This prescribed framework must be simple, easy to understand and applicable to any perceived situation regardless of the time available or the severity of the situation and must allow them to effectively manage workload to an acceptable level in order to achieve the goal.

This system is complemented by crew training, CRM training, standard operating procedures, standards call-outs, standardised communications phraseology and standard checklists. They all aim to reduce the 'thought' effort required to manage any situation, reducing workload and therefore increasing the crew's capacity to maintain S.A, enhance decision making, promoting the successful achievement of the goals.

It is further complemented by briefings. These pre-define the standard reactions, task allocation and forward planning of anticipated time-critical 'immediate action events' or threats. These apply in the situation that immediate action is required to ensure the safety of the crew / pax and / or aeroplane, and considering the available options and deciding on a plan during the event, or when encountering a threat, may increase workload to an unmanageable level and risk implementing an irrational decision, and lead to an unfavourable outcome.

The goal following any emergency / abnormal event must be to ensure the safety of the passengers and crew and then if possible, the aeroplane and the environment. Therefore the system must also give the crew direction in managing the aftermath following an incident or accident.

8.3.36.2 Method of Managing an Unexpected Event

The system has three phases:

1. Initial Actions Phase;
2. Situation Management Phase;
3. Post Incident Management Phase.

Initial Actions Phase

The system shall respect the following order of initial actions (RSIC):

- **Recognise** an event has occurred;
- **Safety** of flight is assured;
- **Identify**, share and confirm the cause of the event;
- **Checklists** conducted.

Recognise

The first crew member (FCM or CCM) to recognise an unexpected event shall state it clearly. The detail of the announcement used shall depend on the phase of flight and the event itself, e.g. During take-off, below V1 'STOP, STOP'; during take-off above V1 'Master Warning / Caution'.

The other crew member(s) shall confirm they have understood the event by repeating the message (e.g. 'Master Caution, GEN 1 FAIL'), or where applicable, initiating an immediate action (such as RTO).

Safety

The first priority is always to maintain positive control of the aeroplane's desired flight path. The system of managing an unexpected situation must promote the correct priority of always having one pilot dedicated to flying the aeroplane on the desired flight path (AVIATE). There may be situations which require a handover of control. This includes immediate actions as specified in OM-B, or other scenarios where a change of control is needed to keep the aeroplane on a safe flight path.

Maintaining a desired flight path must include placing the aeroplane in a safe position relative to obstacles, terrain and traffic (NAVIGATE).

To COMMUNICATE with ATC is the last priority; however exactly when a distress or urgency call is made will depend on the situation.

Identify

The cause of the event is identified, shared and confirmed.

When time permits PM / CCM shall identify the event: e.g.

PM:	Identifies and states the indication - 'Red CAS message - right engine fire,'
PF:	Visually confirms and states the indication - 'Red CAS message - right engine fire confirmed.'
PF:	Commands - 'Cancel the Master Warning / Caution'
PM:	Cancels the warning / caution.

NOTE

Warning and caution systems vary across aeroplane types. Respecting the sequence listed above, OM-B shall specify the most effective method for managing the system ensuring that all crew members arrive at a common understanding of the nature of the problem.

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Checklists

Appropriate checklists are conducted. The checklist title shall be stated by PM. PF shall confirm it is the correct checklist for the situation.

Management of an Unexpected Event		
Initial Actions Phase (RSIC)		
Recognise	An unexpected event occurs.	First to recognise event states clearly.
Safety	- Aviate; - Navigate; - Communicate.	- PF maintains positive aeroplane control including Immediate Actions & Memory Items; - Desired safe flight path achieved; - Distress call made.
Identify	- The fault, failure or event shall be Identified; - Shared; - Confirmed; - Master Warning / Caution managed.	- PM / CCM identifies the event; - PM / CCM states nature of event; - PF confirms; - PF requests 'Cancel Master Warning / Caution'; PM cancels.
Checklists	- Appropriate checklist(s) conducted; - Priority of checklist used.	- PM selects checklist; - PF confirms correct checklist; - PM conducts checklist; - Emergency; - Normal; - Abnormal.

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Situation Management Phase

Once the immediate actions and checklists have been completed the situation is managed following a standard framework (DORDAR).

Diagnose

An analysis of the severity of the event and the time criticality must be done first.

The impact of the failure, fault or event shall be considered in the context of safe flight, i.e. Can the flight be continued to destination, or does it require diversion or a return to departure?

Options & Risk / Benefits

The subsequent amount of time spent considering the available options and the associated risks / benefits will depend on the severity of the event and its time criticality. Risks / Benefits should be considered with respect to: Weather, NOTAMs, Performance, Fuel (WNPF).

The task allocation options shall also be considered.

Decide

The final decision on the course of action and the task allocation is the responsibility of the designated commander, however it should be arrived at using input from the rest of the crew and, time permitting other 'external' resources, e.g. ATC, MedAire, NTA, even pax.

At this point the crew must determine if the situation is time-critical or not.

Time-Critical: The nearest suitable airfield should be identified and its suitability determined. This needs to include Weather and NOTAM information. The decision should be based on the situation and the crew RSIC and DORDAR process. The company should only be contacted when time permitting. The passengers' preference should not be considered.

Non-Time-Critical: The Company should be contacted and their view and guidance considered in the decision making. This input may be based upon passenger convenience (recovery option available) or the best Maintenance facility. The office may be able to assist in the trouble shooting and in the overall handling of the situation. The passengers should be advised of the situation and course of action.

Apply

The plan of action shall be implemented and tasks allocated and conducted. This shall include a NITS briefing. The NITS shall include briefing the procedures to be applied after landing. In all cases, proper means of disembarking the passengers and the crew from the aeroplane shall be briefed.

Review

Once a plan has been implemented and applicable checklists and communications accomplished a review of the situation should be conducted to evaluate the effectiveness of the actions of the plan. It is an on-going review and should be conducted considering the future phases of flight. The review is best accomplished by following the DORDAR process again as many times as necessary until the safe completion of the flight.

Situation Management Phase (DORDAR)		
Diagnose	Time available Criticality of event: Land as soon as possible (LASAPOSS), or Land as soon as practical (LASAPRAC).	Can the flight be: Continued to destination; Diverted; Returned to departure.
Options & Risks / Benefits	- Available options considered. - Task allocations considered.	Weather; Notams; Performance; Fuel. - PF / PM.
Decide	- Decide on Time-critical vs Non-time-critical timeline - Decide on course of action	- Land as soon as possible (next suitable field, don't consider Hotline or Pax convenience), or - Land as soon as practical (involve the Hotline and consider Pax convenience and or MX facilities).
Apply	- Implement plan. - Task allocate. - Communicate plan.	- Communication priority: - Other crew members (NITS); - ATC; - Pax (NITS); - Company.
Review	- Management of present situation. - Management of future situation.	- Is situation currently stable? How is the plan working now? - Is event manageable for remainder of the flight? How will the plan work for the rest of the flight?

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Situation Closure Phase		
Determine	• Rapid Disembarkation, or • Emergency Evacuation.	• Consider situation (Taxi clear off runway / Fire, Smoke, Fumes).
Apply	• Gather and count pax and crew at Safe Area; • On Site Commander approaches authorities; • Request spare room or seek medical assistance for all.	• Anybody missing? • Who is the Onsite Commander? • Never mention details or allocate blame and guilt. • Apply Operational debriefing and remind of CISM.
State only the minimum. Never discuss details or allocate blame or guilt. Contact the company for further guidance. Never give away your passports.		

Situation Closure Phase

Leaving the Aeroplane

Brief and apply the procedures in the SEP Manual. The nature of the abnormal / emergency situation shall determine whether a Rapid Disembarkation or an Emergency Evacuation is required.

Regardless of crew composition (FCM, CCM, augmented crew) the plan for disembarkation and evacuation of the aeroplane should be part of the normal briefing routine and of any DORDAR assessment. The plan should make use of a 3rd FCM and / or additional CCM when carried.

Rapid Disembarkation

Should the nature of the abnormal / emergency situation allow to taxi the aeroplane clear of the runway, or if the commander judges that the passengers should disembark without requiring an emergency evacuation, then a 'Rapid Disembarkation' should be considered.

For example, after landing with a hydraulic leak it may be prudent to disembark the passengers rapidly due to the hazard of hydraulic fluid leaking onto hot brakes.

If after landing the aeroplane is able to taxi, it should be stopped clear of the runway. On taxi out the aeroplane should be stopped on the taxiway. In both cases the aeroplane should not continue or return to the stand. The aeroplane should be shut down using normal procedures and the passengers disembarked in a prompt but orderly manner. Passengers may take hand luggage only. Rapid Disembarkation shall not be initiated with the 'Evacuate' command.

Note that in the presence of Fire, Fumes and Smoke when the aeroplane on stand, the commander may consider a Rapid Disembarkation or an Emergency Evacuation, according to circumstances.

Emergency Evacuation

Should the nature of the situation require the immediate evacuation of the aeroplane then an Emergency Evacuation should be made and announced with the 'Evacuate' command. The presence of Fire, Fumes or Smoke on the ground shall be considered a factor. The SEP Manual gives further guidance.

During the emergency evacuation the selection of emergency exits shall depend upon structural damage and / or the presence of fire. After evacuation all personnel should be guided to the safe area and accounted for. The commander (then other PICs, then SICs, then CCM) should assume the role of Onsite Commander until the fire brigade is present and a handover is done with the Fire Chief. All first aid measures should be coordinated by the Onsite Commander until the fire brigade is available.

If there is any doubt whether to use a Rapid Disembarkation or an Emergency Evacuation, the commander should initiate an Emergency Evacuation.

Post Emergency Handling

The commander should keep the authority over the crew and be the sole point of contact for all queries from the authorities. In their absence the authority shall fall (in this order) to additional PIC(s), then SIC(s) then CCM. It may be difficult to identify the official authorities amongst all the various authorities present. The commander shall try to keep the crew together at all times. All other crew members should refer to the commander for any queries. Only name, rank and position should be stated.

The commander should identify themselves to the authorities. They then should request a quiet room for the crew or transfer the entire crew to the medical team. Crew members should not make any statement about the cause of the incident or accident at this time. Special care has to be taken not to discuss any blame or guilt ('It was my mistake'. 'I knew this was happening', 'It's NetJets fault...').

Shall the authorities require any kind of identification then the passport shall never be given away. Rather consider ID cards or Crew Cards to hand out.

After making contact with the authorities the commander should contact NetJets via the Hotline. The NJE Chief Pilot and respective team, as delegated, will take over the situation and provide further guidance. Once the crew is back together the procedure as per [Crew Health Precautions](#) (Operational debriefing, CISM) should be applied.

NOTE

1. If in doubt, the commander should consider themselves and the crew to be suffering from shock and request medical assistance before making any statements to the authorities.
2. Additional guidance is given in [Post-Serious Incident / Accident Guidelines for Crew Members](#).

8.3.36.3 Immediate Action Events

There are identifiable emergency situations that require the crew to take immediate action from memory and / or fly a manoeuvre to ensure the immediate safety of the aeroplane during the initial action phase. These are learned and trained and / or pre-briefed as an automatic response.

The situations that fit this category are:

- RTO;
- EFATO;
- Emergency Descent;
- Smoke / Fumes;
- Control system issues;
- Engine failure in cruise;
- EGPWS recovery;
- Windshear recovery;
- Any memory items as defined by OM-B.

8.3.36.4 Checklist Philosophy

All emergency and abnormal checklists require positive confirmation by another crew member that the correct checklist is selected. The checklist items shall be read aloud as Read & Verify / Do by PM and an awareness of the progress maintained by the PF.

The priority of the order checklists are to be conducted:

1. Memory items;
2. Emergency checklist;
3. Normal checklist;
4. Abnormal checklist.

The order listed above should be modified where necessary to conform to the manufacturer's recommendations and / or to optimise aeroplane system features such as electronic checklists.

8.3.36.5 Critical Items

Critical items refer to certain switches / levers², which when operated incorrectly could have consequences which are unrecoverable or otherwise safety-critical. Critical items are, but not limited to: Engine Thrust levers, Fuel master or control switches, Engine Fire Handles or switches, Engine fire extinguisher switches, Electric Master Switches including GEN switches. Type specific critical items are listed in the relevant aeroplane OM-B.

In mitigation, prior to moving a critical item, the PM¹ shall place their hand or finger on the critical item, stating which item is dictated by the checklist and the action that they intend to accomplish. PF¹ shall confirm it is the correct item and the intended action is correct, responding to the PM¹ accordingly. In the case of Engine Thrust Levers and Fuel Control switch / levers the PF¹ shall guard the operative switch / lever. Only then shall the PM¹ conduct the checklist action.

NOTE

1. In some cases on specific aircraft types it will not be possible to comply with the prescriptions contained of this paragraph due to aircraft design (e.g. when a given critical switch is out of reach of the RHS pilot and they are the PM), these cases and the alternate procedures will be contained in the relevant OM-B.
2. The term items / switches / levers is generic and is used in the subchapter to refer to any items that can change the status of a system as defined in the aeroplane type specific OM-B / AFM.

8.3.36.6 Landing Gear Malfunctions or Structural Damage

A. Landing Gear Malfunction

If, following a landing gear indication problem or a landing gear malfunction, execution of the abnormal or emergency checklist has not resulted in an all-gear-down indication, and use of a secondary system or alternate means has not resulted in a confirmation that the landing gear is down, the flight crew may decide to execute a low pass at an aerodrome to allow tower personnel or apron personnel to visually verify the status of both main gears and the nose gear provided the fuel status of the aeroplane allows the flight crew to do so.

WARNING

It can only be visually verified if either gear is in an 'UP' or 'NOT UP' condition. A 'DOWN AND LOCKED' confirmation cannot be obtained by means of a low pass.

B. Structural Damage

If the aircraft has, or is suspected to have structural damage, and it is in the interest of safety to do so, the flight crew may determine that a low pass at an aerodrome is required to allow tower personnel or apron personnel to visually verify the status and extent of the structural damage.

A low pass should only be executed after the flight crew have contacted the hotline and agreed with the responsible manager that it is required or advisable to do so.

C. Conduct of a Low Pass

A low pass and the flight profile thereafter shall meet the following conditions:

1. Coordination with and approval from ATC;
2. ATC shall be briefed on the nature of the problem requiring a low pass;
3. Precise instructions on what needs to be checked and verified shall be given to ATC;
4. Weather limitations are visibility 8 km or more with ceilings at 1000 ft AAL or above;
5. The minimum altitude during the low pass is 500 ft AAL;
6. The aeroplane shall fly over the runway in use and on runway heading unless otherwise instructed by ATC;
7. Latest at DER the aeroplane will start a climb for the flight profile as instructed by ATC;
8. Only one low pass will be executed, unless after the first one the condition of the aeroplane changes, justifying any further low passes.

If uncertainty regarding the status of the aircraft still exists after the low pass, and provided the fuel status of the aeroplane allows them to do so, the crew shall join a holding pattern or an overhead pattern at a safe altitude and call the hotline.

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D. Precautionary Measures for Landing with Unsafe Landing Gear Indications

In addition to guidance provided by AFM and other operating manuals the following precautions shall be considered when landing with an unsafe landing gear indication or unknown landing gear position:

1. Divert to an airport with the best emergency services available at the estimated time of landing, taking into consideration increased fuel burn, weather and performance criteria.
2. Choose or divert to a runway which is the longest and / or widest (depending on the malfunction), taking into consideration increased fuel burn, performance criteria and weather (especially crosswind and headwind considerations).
3. Burn off the extra fuel to increase survivability in a possible post-crash fire, taking into account other malfunctions and possible structural damage.

WARNING

Fuel consumption is significantly higher when flying with the landing gear down or partially down during troubleshooting or positioning the aeroplane for landing. Fuel starvation may occur if fuel reserves are not carefully monitored and flying distances / times until touchdown are not carefully planned.

4. Declare an emergency to have all available emergency services at the highest state of readiness and receive traffic priority.

NOTE

Consider the following additional topics in the DORDAR process:

1. Use of Thrust Reversers after landing.
2. Offset from the Centreline in case of Main Landing Gear malfunction.
3. Use of the Emergency Exits.

8.3.37 Simulated Abnormal and Emergency Situations

Simulated abnormal and emergency situations, deactivation of systems operating normally, or pulling circuit breakers is prohibited on all flights with the exception of training flights under the supervision of an instructor or line trainer approved by NTA and only as documented in the relevant company approved training syllabus.

Simulated flight in IMC by artificial means is prohibited for all flights.

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8.3.38 Communication Protocol for Technical and Cabin Appearance Issues

8.3.38.1 In-flight: Technical / Cabin Appearance Issues

- Crew member detects technical issue / cabin appearance issue that potentially leads to AOG or could have a negative impact on customer service.
- Designated crew member calls HOTLINE +351 21 446 9090.
- Flight Dispatcher initiates a conference call with the CP team and the Maintenance department.
- CP informs the OCC / DM about a possible AOG or diversion and remains available to advise and assist as necessary.

8.3.38.2 On Ground: Technical / Cabin Appearance

- Crew member detects technical / cabin appearance issue that potentially leads to AOG or or could have a negative impact on customer service.
- Designated crew member calls (A) MCC +351 214 468 449.
- (A) MCC analyses information.
- (A) MCC controller may initiate a conference call which will include the following participants - PIC (SIC), Maintenance (A) FM and the Flight Operations (A) CP.
- PIC (SIC) writes the technical item / appearance item into the ATL or AAL after coordinating with (A) MCC.

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8.3.39 Automatic Dependent Surveillance-Broadcast (ADS-B)

Automatic Dependent Surveillance-Broadcast is a surveillance technique that relies on aircraft broadcasting their identity, position and other aircraft information. This signal can be captured for Surveillance purposes on the ground (ADS-B out) or on board other aircraft / vehicles (ADS-B in). The latter will enable airborne traffic situational awareness (ATSAW), spacing, separation and self-separation applications. ADS-B is an enabling technology that allows the periodic transmission of parameters, such as identification, position and position integrity, via a broadcast-mode data link. Any user, either airborne or ground-based within range of this broadcast may choose to receive, process and display this information. ADS-B information is broadcast without any knowledge of which users may be receiving it and without the expectation of an acknowledgment or reply.

ADS-B is automatic in the sense that no flight crew or controller action is required for the information to be transmitted (apart from pilot procedures for turning on the transmitter). It is dependent surveillance in the sense that the surveillance-type information so obtained depends on the suitable position source and broadcast capability.

ADS-B enhances safety by making an aircraft visible, real time, to ATC and to other appropriately equipped ADS-B aircraft with position and velocity data transmitted every second in areas of weak or poor radar coverage.

ADS-B uses GPS as its navigation source and a data-link operating at 1090 MHz to relay information to ADS towers on the ground.

ADS-B is currently operational in several areas around the world and being implemented on a worldwide bases.

8.3.39.1 Applicability

For NTA aeroplane types that are approved for ADS-B operations, see [Approvals by Aeroplane Type](#).

8.3.39.2 Emergency Squawk Codes (7500, 7600, 7700)

Setting emergency squawk codes (7500, 7600, 7700) sends a discrete emergency code to the controlling agency. Table below outlines discrete codes sent to the controlling agency.

Emergency Squawk Codes

Squawk Code	ADS-B Code Sent	Surveillance Status Parameter
7500	UNLAWFUL INTERFERENCE	PERM ALERT
7600	NO COMMUNICATIONS	
7700	GENERAL EMERGENCY	

8.3.39.3 ADS-B Phraseology

Flight through ADS-B airspace is similar to that of radar airspace. Pilots will not be advised they have entered ADS-B surveillance airspace from a radar coverage area, or that they have exited ADS-B surveillance airspace and entered a radar coverage area.

The phraseology is very similar to the current radar phraseology. The term 'RADAR' is replaced by 'SURVEILLANCE' in ADS-B areas.

Below is the standard ICAO Doc 4444 ADS-B Phraseology:

Voice

Circumstances	Phraseologies
TO REQUEST THE CAPABILITY OF THE ADS-B EQUIPMENT.	a. ADVISE ADS-B CAPABILITY; b. *ADS-B TRANSMITTER (<i>data link2</i>); c. *ADS-B RECEIVER (<i>data link2</i>); d. *NEGATIVE ADS-B. * <i>Denotes pilot transmission.</i>
TO REQUEST RESELECTION OF AIRCRAFT IDENTIFICATION.	RE-ENTER [ADS-B] AIRCRAFT IDENTIFICATION.
TO REQUEST THE OPERATION OF THE IDENT FEATURE	TRANSMIT ADS-B IDENT
TO REQUEST TERMINATION OF TRANSPONDER AND / OR ADS-B TRANSMITTER OPERATION (<i>Note 1</i>)	a. STOP SQUAWK [TRANSMIT ADS-B ONLY]; b. STOP ADS-B TRANSMISSION [SQUAWK (<i>code</i>) ONLY].
TO REQUEST TRANSMISSION OF PRESSURE ALTITUDE	TRANSMIT ADS-B ALTITUDE
TO REQUEST TERMINATION OF PRESSURE ALTITUDE TRANSMISSION BECAUSE OF FAULTY OPERATION	STOP ADS-B ALTITUDE TRANSMISSION [(WRONG INDICATION, <i>or reason</i>)].
TERMINATION OF ADS-B SERVICE	a. IDENTIFICATION TERMINATED [DUE (<i>reason</i>)] (<i>instructions</i>); b. WILL SHORTLY LOSE IDENTIFICATION (<i>appropriate instructions or information</i>); c. IDENTIFICATION LOST [<i>reasons</i>] (<i>instructions</i>).
ADS-B EQUIPMENT DEGRADATION	ADS-B OUT OF SERVICE (<i>appropriate information as necessary</i>).

NOTE

1. Independent operations of Mode S transponder and ADS-B may not be possible in all aircraft (e.g. where ADS-B is solely provided by 1 090 MHz extended squitter emitted from the transponder). In such cases, aircraft may not be able to comply with ATC instructions related to ADS-B operation, advise ATC and request other instruction.
2. Typically for large transport category aircraft this would be 'Ten Ninety datalink' which refers to 1090 MHz extended squitter.

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CPDLC

	Message intent / use	Message Element	URG	ALRT	RESP
239	Instruction that the ADS-B transmissions are to be terminated.	STOP ADS-B TRANSMISSION	N	M	W/U
240	Instruction that the ADS-B transmissions should include level information.	TRANSMIT ADS-B ALTITUDE	N	M	W/U
241	Instruction that the ADS-B transmissions should no longer include level information.	STOP ADS-B ALTITUDE TRANSMISSION	N	M	W/U
242	Instruction that the 'ident' function of the ADS-B emitter is to be activated.	TRANSMIT ADS-B IDENT	N	M	W/U
244	ATS advisory that the radar and / or ADS-B service is terminated.	IDENTIFICATION TERMINATED	N	L	R
210	ATS advisory that the aircraft has been identified on radar and / or ADS-B at the specified position	IDENTIFIED (<i>position</i>)	N	M	R
193	Notification that radar and / or ADS-B identification has been lost.	IDENTIFICATION LOST	N	M	R

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8.3.40 Runway Situational Awareness Systems

8.3.40.1 General

CAUTION

Runway Situational Awareness Systems do not replace normal procedures for avoiding take-off on the incorrect runway (e.g. visual identification). They are a backup in case these procedures fail.

Runway Situational Awareness Systems (RSAS) is the generic term used by NTA to describe systems installed on some aeroplane types to prevent:

- Take-off or landing not on a runway;
- Take-off or landing not on the intended runway;
- Runway incursions;
- Runway excursions.

These systems provide information and alerts by means of aural and visual advisories based on the position of the aeroplane during ground and flight operations.

RSAS shall not be used as a primary means of navigation and crew shall not depend on such system for aeroplane manoeuvring or positioning relative to a taxiway or a runway.

Refer to the table below for a list of NTA aeroplane types equipped with RSAS and the respective system designations as per their AFM.

TCDS / Model	System Designation
BD-100-1A10	Runway Awareness Advisory System (RAAS)
BD-700-1A10	Surface Management System / Take-off and Landing Alert Function (SMS/ TLAF)
BD-700-1A11	Surface Management System / Take-off and Landing Alert Function (SMS/ TLAF)
CE-680A	SurfaceWatch
CL-600-2B16	Runway Awareness Advisory System (RAAS)
Falcon 2000EX	Runway Awareness Advisory System (RAAS)

Types of advisories

Check AFM / AOM / OM-B for more details on the applicable system advisories and callouts for each specific aircraft type.

A. Non-Routine Advisories

Runway Situational Awareness Systems will provide non-routine advisories including but not limited to the following scenarios which will not normally be heard during day-to-day operations:

- Take-off not on a runway (e.g. from a taxiway);
- Take-off runway too short;
- Take-off runway disagrees with FMS departure runway;
- Landing not on a runway;
- Landing runway too short or long landing.

B. Routine Advisories

Some Runway Situational Awareness Systems will additionally provide routine advisories as part of normal operations. The crew shall use the information provided by Routine Advisories to enhance their situational awareness and support decision-making. Advisories associated with the following scenarios are considered routine:

- Approaching a runway;
- When entering or on a runway;
- Distance remaining on landing roll-out;
- Distance remaining following a rejected take-off.

8.3.40.2 System Limitations, Restrictions, and Procedures

Where a Runway Situational Awareness System is available, it shall be switched on at all times. However, when the crew is aware in advance that it will cause spurious Non-Routine advisories (e.g. some runways with a displaced threshold), then the system may be inhibited in accordance with type-specific procedures.

Where operation requires the Runway Situational Awareness System to be inhibited for take-off, this requirement shall be briefed during the departure briefing.

For type-specific system limitations, restrictions, and procedures, refer to the respective OM-B and applicable AFM / AOM.

8.3.40.3 Crew Response to a Non-routine Advisory

DURING TAKE-OFF (LOW SPEED REGIME)¹

If any Non-Routine advisory is received during the take-off while in low-speed regime, the crew shall immediately abort the take-off and promptly bring the aircraft to a stop.

DURING TAKE-OFF (HIGH SPEED REGIME)²

CAUTION

Aborting a take-off during the High-Speed Regime is a high-risk manoeuvre.

- All Non-Routine advisories are triggered by conditions occurring in the **Low Speed Regime** (see OM-B for type-specific triggers); therefore, the crew shall not abort a take-off solely based on a **Non-Routine** advisory which occurs during the High-Speed Regime.
- Any aborted take-off in the High-Speed Regime shall only be performed in accordance with the standard abort criteria, for example, 'Any other situation that it is deemed safer to remain on the ground'.
- **DURING Approach to LAND**

If any Non-Routine advisory is received prior to touchdown, the crew shall initiate a go-around if they cannot immediately confirm aircraft is in position to safely land on the planned runway and that the landing performance assessment shows it has sufficient length.

Only once an aborted take-off or a go-around has been completed, then the crew can investigate the cause of the advisory and inhibit the system for the next take-off or approach if they determine it was a spurious activation.

NOTE

1. Take-off Low Speed Regime is the part of the take-off from setting take-off thrust until the speed defined in the OM-B where the take-off abort criteria change from 'any abnormality' to only major items.
2. Take-off High Speed Regime is the part of the take-off beyond the Low-Speed Regime up until V_1 .

8.3.40.4 Reporting

Whenever a non-routine advisory is received, the crew shall submit a NetJets Safety report detailing the occurrence.

ATC shall be informed of any response to an advisory which required deviation from an ATC clearance.

8.4 Low Visibility Operations (LVO)

8.4.1 General

NTA shall conduct the following operations only if they are approved by the competent authority:

- a. take-off operations with visibility conditions of less than 400 m RVR;
- b. instrument approach operations in low-visibility conditions; and
- c. operations with operational credits, except for EFVS 200 operations, which shall not be subject to a specific approval.

NTA LVO approvals are as per the AOC.

Prior to commencing an LVO, the commander shall be satisfied that:

- a. the status of the visual and non-visual facilities is sufficient;
- b. appropriate LVPs are in force according to the information received from ATS;
- c. flight crew members are properly qualified;
- d. flight crew members seat are adjusted to give the correct eye position.

8.4.1.1 Training and Qualifications

The initial and recurrent training programme and qualification requirements for all aspects of Low Visibility Operations, are prescribed in the OM-D and / or [Qualifications Requirements](#) of this manual.

8.4.1.2 Minimum Equipment

- The AFM / AOM MEL prescribes the minimum equipment that shall be serviceable for the applicable LVO;
- The commander shall satisfy themselves that the status of the aeroplane and of the relevant airborne systems is appropriate is functioning for the specific operation to be conducted.

8.4.1.3 Aerodrome Requirements

An aerodrome shall not be used for LVOs below a visibility 800 m unless:

- a. the aerodrome has been approved for such operations by the state of the aerodrome; and
- b. LVP have been established;
- c. If an aerodrome where the term LVP is not used is selected, NTA shall ensure that equivalent procedures that adhere to the requirements of LVP at the aerodrome are established. Other terms used that are not listed below shall be noted in the OM-C, NTA aerodrome briefing, or an iJet Note (feasibility note).

The following terms used are equivalent to LVP:

- Low Visibility Operations (LVO/LVOP);
- Reduced Visibility Operations (RVO/RVOP);
- Surface Movement Guidance Control System (SMGCS);
- Special Safeguards and Procedures (SSP).

LVO at CAT II/CAT III Aerodromes

For aerodromes that have CAT II / CAT III Operations, LVP or its equivalent are required and, therefore, are available.

8.4.1.3.1 Suitability of Aerodromes

NTA will ensure that only suitable aerodromes are used for LVO. The process includes a verification of the adequacy of the runway to the type of operation, including:

- the navigation facilities and associated instrument flight approach procedures;
- the aerodrome operating procedures, including LVP and respective compatibility with the approved operations;
- the physical runway and runway environment characteristics and facilities.

This verification is done via a desktop assessment using aerodrome data, published instrument flight procedures and aircraft data and capabilities. The factors to be considered in the review vary with the complexity of the intended operations, considering the respective hazards and associated risks. The competent authority, the AFM or the AIP may require that additional elements are included in the assessment. The primary factors to be considered are:

- Runway and aerodrome facilities, namely lightning installed on the runway and obstacles;
- Obstacle protection, in particular:
 - in the visual segment below DA/H for NPA procedures, APV and CAT I PA on runways where an OFZ is not provided;
 - when approach procedures not designed in accordance with PANS-OPS (or equivalent standards) are used;
 - in the event of a balked landing.

Any restriction regarding approach procedures contained in the AFM must be followed.

If the assessment determines that obstacle protection in the approach is not assured, EVS operations shall not be allowed. Alternatively, the operation may be restricted to the type of approaches or approach procedures that provide protection.

If the assessment determines that obstacle protection in the event of a go-around is not assured, LVO operations shall not be allowed unless procedures to mitigate the risk of inadequate obstacle protection are developed and implemented.

For the approvals held by NTA, the following rules apply:

Low Visibility Take-off Operations

Given the complexity, the suitability assessment may be conducted by the crew members using the information available during flight preparation, following OM-A [8.4.2](#);

If during the aerodrome's initial assessment, it is determined that LVTO on specific runways are not allowed or recommended, that information will be included in the crew briefing; in this case, FCM shall not conduct LVTO.

EVS (EFVS-A)

Operations using operational credits with EFVS shall only be conducted on approved runways. The following rules apply:

- For runways other than those with CAT II/III approaches, suitability is identified on a case-by-case basis and will be included in the crew briefing.
- The certification rules on runways with published CAT II/III approach procedures ensure suitability for EFVS regarding obstacle protection and required runway facilities, so runways having this type of approach can be considered suitable unless listed as unsuitable in the crew briefing. The PIC must still verify that LVP are in force and that runway lighting is operational. EFVS operations can be performed even if the navigation facilities (ILS) are not operational/in use.

If a runway with CAT II/III approach procedures is found unsuitable for EFVS operations during the aerodrome's initial assessment, that information will be included in the crew briefing; in this case, FCM shall not conduct EFVS operations to that runway.

8.4.1.4 Occurrence Reporting

Flight crew shall report any unsatisfactory and discontinued approaches utilising EVS including any missed approaches via the NTA safety reporting system.

8.4.1.5 LVO Monitoring and Data Gathering

As part of its Safety Management System, NTA monitors its LVO operations via the collection of operational data with the objective of assessing the adequacy of LVO procedures and the performance of the equipment. Depending on the type of operations, this data collection is required for all flights and monitor specific risks associated with the approved operations. Notwithstanding the principles of confidentiality, data collected from the FDM programme will be used to complement the information provided by crew members.

8.4.2 Low Visibility Take-off (LVTO)

NTA is approved for LVTO down to a RVR of 125 m. Refer to AOC for the approved aeroplane types and restrictions.

The Commander shall satisfy themselves that the status of the visual and non-visual facilities is sufficient prior to commencing a Low Visibility Take-off.

For the applicable take-off minima and ground facility requirements refer to [Take-off Minima](#).

Low visibility Procedures (LVP) must be in force at the aerodrome for LVTO.

8.4.2.1 Low Visibility Take-off Procedure

- The following procedures apply when the RVR for take-off is less than 550 m.

General

The pilot flying shall be the PIC and shall occupy the LHS. The only exemption is during checking or training flights where the PIC may occupy the RHS however, the LHS pilot shall still act as the pilot flying.

The PIC shall confirm that the required equipment (lighting and markings) is in accordance with [Required RVR / Visibility](#).

Taxi during LVP

The primary role of the PF is to look outside in order to avoid any obstacles and obstructions and avoid taxi and runway incursions.

The primary role of the PM is to monitor the taxi and assist with the navigation. The PM shall use the aeroplane instruments and the airport layout charts to progressively monitor and check taxi routing. Utmost attention shall be paid to avoid possible taxi and runway incursions.

Surface markings shall be strictly followed. Lighted stop bars shall never be crossed. For contingency measures for unserviceable stop bars, refer to Chapter 12 [3.2.2 Right-of Way](#) (SERA.3210).

Special attention shall be paid to the taxi speed, which should not be more than 10 kts.

If there are any doubts or any uncertainty as to the position of the aeroplane, the aeroplane shall be immediately stopped and ATC shall be notified.

To be visible to other traffic, display of all lights (including strobe lights) is recommended unless vision is impaired, and taking into consideration those bright lights may disturb other traffic.

CLEARANCE FROM RUNWAY - During low visibility operations CAT II or III holding position markings and signs shall not be crossed without clearance.

Take-off

A static take-off shall be used unless a static-take-off would result in unacceptable delays and runway length permits a non-static take-off.

When lined up on the runway:

- Verify that the runway magnetic heading matches the aeroplane heading, the heading bugs are centred; and
- Use appropriate aids to verify line-up on the correct runway (e.g. ILS, Surface-Watch, etc.).

Take-off and rejected take-off procedures and call outs are as published in the respective OM-B.

During the take-off roll the PM should not look outside and shall closely monitor all aeroplane performance parameters.

Crosswind

For LVTO 50% of demonstrated winds as per AOM or recommended AOM limitations if more restricted apply.

8.4.3 EVS Approach Operations with Credit to Reduce RVR

For aeroplane types approved for EVS approach operations for which an operational credit is applied to reduce the RVR minima, refer to the AOC.

When conducting EVS operations to a runway:

- a. the DA/H used should be the same as for operations without EVS (required visual references need to be visible at DA/H using EVS); b
- b. the lowest RVR minima to be used should be determined:
 1. in accordance with criteria specified in the AFM for the expected weather conditions; or
 2. if no such criteria are specified, by reducing the RVR determined for operation without the use of EFVS/ CVS in accordance with [8.1.3.6](#);
- c. where the lowest RVR to be used, determined in accordance with (b), is less than 550 m, then this should be increased to 550 m unless LVPs are established at the aerodrome of intended landing;
- d. where the EFVS is part of a , it is only the EFVS element that should provide the operational credits. The other part of the CVS, the synthetic vision system (SVS), should not provide operational credits.

For aeroplane type specific EVS procedures refer to the respective OM-B.

For the conditions required to descend below the published DA/MDA refer to [Visual References for Instrument Approach Operations](#) - (Commencement and continuation of approach).

The approach operations using EVS, the following shall be complied with:

- a. for CAT I operations, a visual reference distinctly visible and identifiable to the pilot without reliance on the EVS shall be attained at least at 100 ft above the aerodrome threshold elevation;
- b. for approach procedure with vertical guidance (APV) and non-precision approach (NPA) operations flown with CDFA technique, a visual reference distinctly visible and identifiable to the pilot without reliance on the EVS shall be attained at least at 200 ft above the aerodrome threshold elevation and the following requirements are complied with:
 - i. the approach is flown using an approved vertical flight path guidance mode;
 - ii. the approach segment from final approach fix (FAF) to runway threshold is straight and the difference between the final approach course and the runway centreline is not greater than 2°;
 - iii. the final approach path is published and not greater than 3.7°;
 - iv. the maximum cross-wind components established during certification of the EVS are not exceeded.

Localiser sensitive area

If during an ILS approach under CAT I weather conditions, information indicates that the Localiser Sensitive Area is not clear, the approach may only be continued to published CAT I minima without the use of operational credit to reduce the RVR.

8.5 Extended Range Operations with Two-engined Aeroplanes (ETOPS)

Unless approved by the competent authority in accordance with PART-SPA Subpart F (ETOPS), NTA shall not operate a two-engined aeroplane over a route that contains a point further from an adequate aerodrome, under standard conditions in still air, than the distance flown in 120 minutes or, subject to approval by the competent authority, up to 180 minutes for turbo-jet aeroplanes, at the true airspeed the aeroplane can maintain with OEI, not to exceed VMO.

The table below stipulates the specific speed and distance to be used for each aeroplane type / variant for the purposes of this paragraph:

Aeroplane Type	OEI Cruise Speed (KTAS)	Maximum Distance from adequate aerodrome (120 min)
EMB-505	315	630 NM
CE-560XLS	305	610 NM
CE-680A	285	570 NM
BD-100-1A10	355	710 NM
DA-2EASy	378	756 NM
*CL-600-2B16	355	710 NM
*BD-700-1A10	337	674 NM
*BD-700-1A-11	344	688 NM

NOTE

* Aeroplane approved for NON-ETOPS-180 - see [Non-ETOPS Operations up to 180 min Operational Approval](#).

8.5.1 Non-ETOPS Operations up to 180 min Operational Approval

NTA non-ETOPS-180 operation is approved by the Autoridade Nacional de Aviação Civil (ANAC).

Non-ETOPS-180 operation is reflected in the AOC under Specific Approvals within [Approvals by Aeroplane Type](#).

8.5.2 Applicability

The following NTA aeroplane types are approved for a threshold time of 180 minutes - Non-ETOPS-180 operations:

- CL-600-2B16
- BD-700-1A11;
- BD-700-1A10.

8.5.3 Aeroplane Non-ETOPS-180 Data

The data specified below are only intended to be used for establishing the maximum distance from an adequate aerodrome for the applicable aeroplane.

Aeroplane Type	CL-600-2B16	BD-700-1A11	BD-700-1A10
Time	180 min	180 min	180 min
Reference Weight	35,000 lbs	81,483 lbs	87,954 lbs
One Engine Inoperative Cruise	365 KTAS	347 KTAS	341 KTAS
Maximum distance from an adequate aerodrome	1,095 NM	1,041 NM	1,023 NM

8.5.4 Definitions

8.5.4.1 Non-ETOPS-180

Non-ETOPS Operations up to 180 min from an adequate aerodrome for twin turbojet aeroplanes IAW PART-CAT - CAT.OP.MPA.140.

8.5.4.2 Adequate Aerodrome

To qualify as adequate for the purpose of Non-ETOPS-180, an airport shall satisfy the performance requirements applicable at the expected landing mass.

The following conditions shall also be considered at the expected time of use:

- availability of the airport;
- availability of ATC services, communications, NAVAIDs, weather reporting, lighting equipment and emergency services;
- availability of at least one IFR approach;
- overflying and landing authorisations.

8.5.4.3 Suitable Aerodrome

The aerodromes are selected and identified in the pre-flight planning phase and shall be listed in the operational flight plan. They shall meet the following requirements:

- Qualify as adequate aerodromes, as defined above;
- The latest available forecast weather conditions for a period commencing one hour before the established earliest time of landing and ending one hour after the established latest time of landing equals or exceeds the Non-ETOPS-180 alternate weather minima.

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8.5.4.4 Window of Suitability (Period of Validity)

To declare an adequate en-route alternate airport as suitable to support a given flight, the ceiling and visibility forecast shall be checked to meet the Non-ETOPS-180 dispatch weather minima during a required period of validity, also referred to as the window of suitability.

The required period of validity starts one hour before the earliest estimated time of arrival at the considered en route alternate airport and ends one hour after the latest estimated time of arrival at this airport. The resulting period of validity needs to include the interval beginning at the earliest ETA and ending at ETA + 1h.

The earliest estimated time of arrival is computed considering a 2-engine diversion at the normally planned cruise altitude and speed (i.e. assuming a diversion for any reason other than an engine or pressurisation failure).

The latest estimated time of arrival is computed considering a 2-engine diversion at FL 100 or at the MORA and at the LRC speed (i.e. assuming a pressurisation failure only).

The time window between the earliest and latest possible use of the alternates shall be indicated in the operational flight plan.

For a delay in excess of one hour, a new time window is defined.

8.5.4.5 Area of Operations

The Non-ETOPS-180 Area of Operation is the area in which it is permitted to conduct a flight under the Non-ETOPS-180 regulations. It is defined by drawing circles centred on each declared suitable en-route alternate airport with a radius equal to the Maximum Diversion Distance.

a. Maximum diversion time

The Maximum Diversion time from an en route alternate airport is 180 minutes for all the applicable aeroplanes under [Aeroplane Non-ETOPS-180 Data](#).

b. One-Engine-Inoperative Cruise Speed

The selected one-engine-inoperative speed is used for the following purposes:

- establishing the area of operation (maximum diversion distance);
- establishing the diversion fuel requirements for the single engine diversion;
- establishing the net level-off altitude to safely clear any en-route obstacle by the appropriate margin as specified in applicable operational rules;
- conducting the diversion following an engine failure.

The pilot in command has the authority to deviate from this planned speed after completion of the assessment of the actual situation.

c. Maximum diversion distance

The maximum diversion distance is the distance covered in still air and ISA conditions within the maximum diversion time at the selected one-engine-out diversion speed schedule and at the associated cruise altitude (including the descent from the initial cruise altitude to the diversion cruise altitude). It is used for dimensioning the area of operations.

8.5.4.6 Non-ETOPS-180 Sector

A Non-ETOPS-180 sector is a portion of a route located further than 120 minutes flying time (at the selected one-engine-out diversion speed, in still air) from an adequate aerodrome. The Non-ETOPS-180 sector extends between the Non-ETOPS-180 Entry Point (EET) and the Non-ETOPS-180 Exit Point (EXP). A Non-ETOPS-180 route may contain more than one Non-ETOPS-180 sector.

a. Non-ETOPS-180 Entry Point (EEP)

The Non-ETOPS-180 Entry Point (EEP) is the first point on the route, located at 120 minutes flying time (at the declared One Engine Inoperative Speed, in still air) from an adequate aerodrome.

b. Non-ETOPS-180 Exit Point (EXP)

The Non-ETOPS-180 Exit Point (EXP) is the last point on the route, located at 120 minutes flying time (at the declared one-engine-out diversion speed, in still air) from an adequate aerodrome (i.e the first suitable en-route alternate airport located beyond the Non-ETOPS-180 sector).

8.5.4.7 Equal Time Point (ETP)

An Equal Time Point (ETP) is a point on the route which is located at the same flying time (considering the day wind and temperature conditions) from the two associated suitable en-route alternate airports.

The location of the ETPs shall be defined on the OFP.

8.5.4.8 Critical Point (CP)

The Non-ETOPS-180 Critical Point (CP) is the point, along the route, for which the difference between the standard fuel expected to be on board (fuel to continue) and the required Non-ETOPS-180 diversion fuel (fuel to divert) is minimum or negative, thus requiring the carriage of additional Non-ETOPS-180 fuel reserves. The Critical Point is usually, but not necessarily, the last ETP within the Non-ETOPS-180 sector.

8.5.5 Maintenance

An aeroplane can be dispatched for a Non-ETOPS-180 flight only if a formal Release for Non-ETOPS-180 has been entered in the Aircraft Flight Log. This Release for Non-ETOPS-180 provides the assurance that:

- the aeroplane condition has been checked and confirmed to comply with the Non-ETOPS-180 dispatch requirements set forth in the applicable aeroplane's MEL.
- the Non-ETOPS-180 Pre Departure Service Check (PDSC) has been accomplished.
- Applicable Approved Aircraft Inspection Program defines the content of the Non-ETOPS-180 PDSC check and the Applicable NetJets Maintenance Procedures (NJMP) and Maintenance Contracts associated with the Non-ETOPS-180 Maintenance Release to Service.

Defects which affect the 'Non-ETOPS-180 Serviceability' of the aeroplane shall be rectified before the next Non-ETOPS-180 flight.

The malfunctions observed during Non-ETOPS-180 flights shall be reported by the flight crew using the aeroplane Technical Log to enable verification and rectification by a relevant ground or function test.

Long Range Operations shall notify the flight crew of any Non-ETOPS-180 MEL item and confirm that the flight preparation has taken this item into account.

8.5.5.1 MEL Terminology

For MEL terminology the ER (Extended Range) acronym will be use for operations beyond 120 minutes and up to 180 minutes.

8.5.5.2 PDSC for Non-ETOPS-180

A Non-ETOPS-180 PDSC shall be carried out before any Non-ETOPS-180 sector and shall be certified by a Non-ETOPS-180 approved engineer or by both flight crew members.

The PDSC for Non-ETOPS-180 is a set of maintenance tasks to be performed prior to dispatching an aeroplane for an Non-ETOPS-180 flight.

The scope of the PDSC is defined in the Applicable Approved Aircraft Inspection Program (AAIP) and also in the respective OM-B, procedures are defined regarding the conditions and terms of performance of this check.

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8.5.6 Flight Crew Training

A Non-ETOPS-180 qualified flight crew member (PIC or SIC) is a flight crew member having performed an initial Non-ETOPS-180 training as well as line training in accordance with the OM-D.

The Non-ETOPS-180 training programme is included in the following courses:

- NetJets Transportes Aéreos Non-ETOPS-180 Course.

8.5.7 Minimum Equipment List (MEL) Consideration

The MEL takes into account all relevant Items to support Non-ETOPS-180.

The MEL requirements specific to Non-ETOPS-180 flights are clearly identified. Deviations from the MEL items which relate to Non-ETOPS-180 require re-routing of the flight to follow a route that is not Non-ETOPS-180.

8.5.8 Dispatch / Flight Planning Requirements

Before each Non-ETOPS-180 flight, in addition to the normal preparation of any flight the dispatcher shall:

- ensure that for the Non-ETOPS-180 segment, the selected en-route alternate airports (adequate airports) are suitable;
- determine the position of each Equal Time Point;
- determine the position of the critical point and its associated fuel requirement.

The fuel planning should be in line with the critical fuel scenario requirement. A computerised flight plan should be used to calculate the fuel requirements.

In case of necessity the fuel calculation may be manually performed by the dispatcher on duty or the flight crew using the applicable AFM data and the Critical Fuel Tables on the respective OM-B.

8.5.8.1 Fuel and Oil Quantities

An aeroplane shall not be dispatched on an extended range flight unless it carries sufficient fuel and oil to comply with the applicable operational requirements. Before departure the flight crew has to make sure that engine and APU oil tanks are full.

a. Critical Fuel Scenario (CFS)

The critical point is the furthest point from an alternate aerodrome assuming a simultaneous failure of an engine and the pressurisation system.

For Dispatch / Release purposes the CFS is calculated considering engine failure only although the Operational Flight Plan will have the following scenarios:

- Engine Failure only
- Pressurisation failure only

b. Non-ETOPS-180 Fuel Reserves

In complement to the fuel rule defined in [Determination of the Quantities of Fuel and Oil Carried - Fuel Policy](#), Non-ETOPS-180 regulations require the addition of specific fuel reserves to the Non-ETOPS-180 diversion fuel.

For the computation of the Non-ETOPS-180 critical fuel reserves and of the complete Non-ETOPS-180 critical fuel planning, the diversion fuel shall include the following fuel provisions:

- i. Fly from the critical point to an alternate aerodrome:
 - At 10,000 ft; or
 - At 25,000 ft or the single-engine ceiling, whichever is lower, provided that all occupants can be supplied with and use supplemental oxygen for the time required to fly from the critical point to an alternate aerodrome.
- ii. Descend and hold at 1,500 feet for 15 minutes in ISA conditions;
- iii. Descend to the applicable MDA/DH followed by a missed approach (taking into account the complete missed approach procedure); followed by
- iv. A normal approach and landing.

c. Ice Protection / Ice Accretion

Icing is presumed to occur when the TAT is below 10°C at the approved one engine cruise speed.

If icing conditions are forecasted:

- Increase fuel in accordance with the aeroplane manufacturer to compensate ice accumulation on unheated surfaces.
- Increase fuel in accordance with the aeroplane manufacturer to compensate for the use of engine and wing anti-ice.

d. APU Operation

If the APU is considered as an alternate power source, the additional fuel consumption shall be added.

8.5.8.2 Communication and Navigation Facilities

They shall be available to provide, at the appropriate one engine inoperative cruise altitudes, reliable two-way voice communications between the aeroplane and ATC over the planned route of flight and over the routes to all Non-ETOPS-180 alternates in the event of a diversion.

Non-visual ground navigation aids, if required, shall be available and located so as to provide the navigation accuracy necessary for the planned route and altitude, and the routes and altitudes to be used in case of a diversion.

8.5.8.3 Aircraft Technical Log

Before departing to an extended range flight the flight crew shall review the Aircraft Technical Log for open items, to in regard of proper MEL procedures and deferred items.

All relevant pages in use (and previous book, if applicable) of the Aircraft Technical Logbook shall be review for maintenance performed which may affect the status of Non-ETOPS-180 significant systems.

Ensure that any inoperative equipment is not required for Non-ETOPS-180 in the MEL.

Check that the recorded Engine Oil quantities are full.

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8.5.8.4 Non-ETOPS-180 Dispatch Weather Minima for En-route Alternate Aerodromes

To declare an adequate en route alternate airport as suitable to support a given flight, the weather conditions forecast (TAF, SPECI and SIGMET messages), in terms of ceiling and visibility, shall be checked to comply with [Planning Minima for IFR Flights](#), item c. during the entire period of validity (see [Window of Suitability \(period of validity\)](#)).

8.5.8.5 Non-ETOPS-180 Flight Documentation

Flight dispatcher on duty has to confirm that the flight crew have the following documents before dispatching / releasing of any Non-ETOPS-180 flight:

- computerised flight plan - OFP;
- applicable plotting charts, with the area of Non-ETOPS-180 operation limits corresponding to the suitable alternates, depending on the route;
- airport charts for all adequate alternates of the route;
- weather dossier (forecasts and reports) for the route and for all selected suitable alternates, or if possible for all adequate alternate aerodromes;
- NOTAMs for all the selected suitable alternates of the route;
- any other documents provided for a normal flight.

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8.6 Use of the Minimum Equipment List(s)

8.6.1 Minimum Equipment List (MEL)

The MEL is approved by the Authority and permits operation with specific inoperative items of equipment for a period of time, or a number of flights, until repairs can be accomplished. It is important that repairs are accomplished at the earliest opportunity, but in any case within the time frame specified in the MEL for the particular equipment.

The basis for establishing and approving the MEL is the MMEL - the manufacturer's Master Minimum Equipment List. This document is approved by the airworthiness authority. Other than the MMEL the MEL also considers operational regulations and increased safety standards applicable to commercial operations.

The MEL for the type concerned is contained in the respective AOM and provides for release of the aeroplane for flight(s) with inoperative equipment. Whenever a flight, or series of flights, is released for dispatch with inoperative item(s), entries shall be made in the Technical Log containing a detailed description of the inoperative item(s) using the appropriate system and sequence number found in the MEL, any special advice to the crew, and information regarding corrective action(s) taken.

The MEL is not intended to provide for continued operation of the aeroplane for an indefinite period with inoperative items! The basic purpose of the MEL is to permit the operation of an aeroplane within the framework of a controlled and sound maintenance and repair programme.

With the exception of cabin service items listed separately, no aeroplane shall be operated with defective equipment not entered under the MEL system in the Technical Log, and no such MEL entry shall be made in the Technical Log by any crew member without the knowledge, cooperation, consent, and guidance of NTA Maintenance Department.

The decision of the commander of the flight to have allowable inoperative items corrected prior to flight will take precedence over the provision contained in the MEL. The commander may request requirements above the minimum listed whenever, in their judgement, such added equipment is essential to the safety of a particular flight under the special conditions prevailing at the time.

They shall, however, never accept requirements below the MEL minimum.

Maintenance, ferry, and training flights may be dispatched with less than the equipment specified in the MEL, provided all equipment expected to be utilised in that flight is operable. A decision for or against such a flight shall be requested from the Maintenance Department, Flight Operations and the Authority. Such permission will in no circumstance permit operation outside the constraints of the MMEL and CDL. Equipment obviously basic to the aeroplane's airworthiness (such as wings, rudders, etc.). is not listed in the MEL, and shall of course be operative for all flights.

On the other hand, equipment obviously not required for safe operation of the aeroplane, such as ashtrays, passenger convenience and service items, etc. are not listed and may be unserviceable. For all further details check the MEL(s) for the type(s) concerned (AOM).

8.7 Non-revenue Flights and Non-commercial Operations - Procedures and Limitations

All non-revenue flights shall be performed in accordance with NTA requirements. All non-revenue flights may be performed and filed as non-commercial operations. Any differences in operating procedures used for non-commercial operations are stated in this subchapter.

8.7.1 Ferry Flights

A ferry flight is a flight or series of flights to position an aeroplane for the purpose of conducting maintenance. Ferry flights may be conducted with minimum crew as specified in this manual. Ferry flights may dispatch with less than the equipment required in the aeroplane MEL provided that all the equipment expected to be utilised during the flight is operable and that any sections of the AFM, pertaining to the operation with inoperable equipment, are complied with.

Ferry flights with inoperable equipment not covered by the MEL / CDL shall only be carried out according to the provisions, procedures and limitations specified in NJMP 1.09.

Maintenance personnel may be carried on ferry flights provided they are authorised by the maintenance department and Duty Manager. They must also be listed on the manifest.

8.7.2 Demonstration Flights

A demonstration flight may be for a sales marketing purpose, or to demonstrate flight characteristics to a potential buyer, or flights with journalists and customers to introduce a new type of aeroplane.

In any case, all flights with passengers aboard require full and normal crew complement.

VFR demonstration flights are not permitted.

8.7.3 Positioning Flights

A positioning flight is a flight to position an aeroplane to an aerodrome for commercial operation.

Positioning flights may be performed with the minimum flight crew.

Positioning flights may be performed under Non-commercial operations.

Crew members of the company may be transported on the way to or from flight duty (Dead Head Crews).

The carriage of any other persons on positioning flights, including company staff members, is only permitted when in accordance with a NetJets policy permitting their carriage or where they are carried under an assignment from NTA.

In all cases, all persons on board the aircraft shall be listed on the manifest.

8.7.4 Fractional Operations

All aircraft listed in the NTA AOC are eligible for Fractional Operations.

A Fractional Operations flight may be filed as non-commercial for the following situations:

- a. Flights with passenger(s) who are owner(s) or guest(s) of owners.
- b. Positioning flights with only NetJets crew members on board.
- c. Positioning flights with NetJets staff members and crew on board.

Fractional operations may be performed either:

- a. In full compliance with the OMs without applying the differences in procedures and limitations in the Fractional Operations Manual e.g. a flight is filed as non-commercial for permit purposes only; or
- b. In compliance with the procedures and limitations in the Fractional Operations Manual.

If the flight is performed in compliance with the Fractional Operations Manual, then all stipulated procedures and limitations in the OMs shall be adhered to, except those procedures and limitations in the Fractional Operations Manual.

NOTE

A procedure or limitation in the Fractional Operations Manual only replaces exactly the same procedure or limitation in the OMs. If a procedure or limitation only replaces part of a procedure or limitation in the OMs, the remaining part(s) of that procedure or limitation in the OM shall still be applied and / or adhered to.

Also refer to *Fractional Operations Manual*.

8.7.5 Training Flights

8.7.5.1 Qualification

The commander of a training flight shall be qualified and licensed as a flight instructor on the respective aircraft type or a NTA qualified training captain.

8.7.5.2 Weather

The company weather minima (ceiling and visibility) specified for the respective type of aeroplane, approach and aerodrome are applicable.

8.7.5.3 Passengers on Training Flights

During training and proficiency check flights only flight crew members involved in the training and / or representatives of the Authority may be carried on board.

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8.7.5.4 Training Flight under the NTA ATO

Training flights performed under the NTA ATO are flown under Part-NCC. Any differences to standard operating procedures are described in the respective ATO manuals.

The NTA ATO will request that the NTA departments responsible for operational control and supervision make NTA aircraft available as needed for their activities. The NTA ATO may request and use any aircraft listed under the NTA AOC to conduct training flights. Since any training flight under the NTA ATO is flown under the operational control of NTA, no operator handover procedures are necessary. In addition to the normal NTA operational control and supervision procedures, the NTA ATO also applies its own procedures for Approval and Authorisation of Flights conducted under the ATO.

Except for those differences in procedures explicitly depicted in the ATO manuals, training flights shall comply with the requirements for CAT operations as prescribed in NTA OM's and the AFM. In particular, all procedures regarding airworthiness and notification and reporting of accidents, incidents and occurrences should follow the provisions in OM-A.

Since training flights are operated within the operational control of NTA, specific approvals in the operations specifications still apply, provided that all requirements regarding aircraft equipment and crew qualifications are met.

NOTE

The flight log and technical log system used by NTA shall be used.

8.7.6 Maintenance Check Flight (MCF)

MCFs shall be carried out in compliance with the requirements for CAT operations, as prescribed in NTA OM's and the AFM, except where differences are specified in the MCF Manual and/or the MCF Flight Program.

Refer to the MCF Manual for detailed MCF procedures and instructions.

8.7.7 Test Flights

A Test Flight is a pre-certification flight, usually carried out by an organisation holding a Design Organisation Approval (DOA) or Production Organisation Approval (POA) in accordance with that organisation's Flight Test Operations Manual (FTOM).

8.7.8 Delivery Flights

Delivery flights are flights where, after a purchasing or lease agreement, an aeroplane is flown from the manufacturer's, seller's or lessor's facility to the airline or vice versa.

Provided all normal requirements, such as crew complement and equipment requirements are met, non-revenue passengers may be carried if this is not excluded on the certificate of airworthiness and certificate of registration. Full insurance coverage shall be assured.

For some delivery flights the Authority might only issue a ferry permit in lieu of the certificate of airworthiness and the certificate of registration. This ferry permit may exclude the carriage of persons other than flight crew and engineers.

Delivery flights may be combined with training flights as long as the minimum crew requirements specified in the Aeroplane Flight Manual are met.

8.8 Oxygen Requirements

8.8.1 Definitions

First Aid oxygen - a supply of undiluted oxygen for passengers who, for physiological reasons, might require oxygen.

Supplemental oxygen - pressurised aeroplane, a supply of oxygen to the required number of occupants for the required flight time at the appropriate altitude(s), following a cabin depressurisation.

Supplemental oxygen - non-pressurised aeroplane, a supply of oxygen to be provided in unpressurised aeroplanes to the occupants whenever flight altitudes above 10,000 ft are used.

Crew protective breathing equipment (PBE) - equipment to protect the eyes, nose and mouth of each flight crew member while on FCC duty and for each cabin crew member. The PBE shall allow the crew members to continue to perform their duties even under smoke or toxic air conditions in the cabin or on the FCC. Portable PBE equipment shall allow active fire-fighting.

8.8.1.1 First Aid Oxygen

Pressurised aeroplanes operated at pressure altitudes above 25,000 ft, when a cabin crew member is required, shall be equipped with a supply of undiluted oxygen for passengers who, for physiological reasons, might require oxygen following a cabin depressurisation.

The amount shall be calculated using an average flow rate of at least 3 litres Standard Temperature Pressure Dry (STPD) per minute / person for the entire flight after depressurisation at cabin altitudes of more than 8,000 ft, for at least 2% of the passengers carried (but in no case for less than one person). The amount of first aid oxygen required for a particular operation shall be determined on the basis of cabin pressure altitudes and flight duration, consistent with the operating procedures established for each operation and route.

The amount of first-aid oxygen shall be calculated for the part of the flight after cabin depressurisation during which the cabin altitude is between 8,000 ft and 15,000 ft, when supplemental oxygen may no longer be available.

There shall be a sufficient number of dispensing units, but in no case less than two, with a means for cabin crew to use the supply.

The oxygen equipment shall be capable of generating a mass flow to each user of at least four litres STPD per minute. Means may be provided to decrease the flow to not less than two litres per minute, STPD, at any altitude.

8.8.1.2 Supplemental Oxygen - Pressurised Aeroplane

The amount of supplemental oxygen required shall be determined on the basis of cabin pressure altitude, flight duration and the assumption that a cabin pressurisation failure will occur at the altitude or point of flight that is most critical from the standpoint of oxygen need, and that, after the failure, the aeroplane will descend in accordance with emergency procedures specified in the AFM / AOM to a safe altitude for the route to be flown, that will allow continued safe flight and landing.

Following a cabin pressurisation failure, the cabin pressure altitude shall be considered the same as the aeroplane altitude, unless it is demonstrated to the competent authority that no probable failure of the cabin or pressurisation system will result in a cabin pressure altitude equal to the flight altitude. Under these circumstances, the maximum cabin pressure altitude established for the type certification of the aeroplane shall be used as a basis for determination of oxygen supply.

8.8.2 Oxygen Equipment and Supply Requirements

8.8.2.1 Flight Crew Members

- All flight crew shall use oxygen whenever the cabin altitude exceeds 10,000 ft;
- Each member of the flight crew on FCC duty shall be supplied with supplemental oxygen as specified in [Table 1](#). If all occupants of FCC seats are supplied from the flight crew source of oxygen supply, then they shall be considered as flight crew members on FCC duty for the purpose of oxygen supply. FCC seat occupants, not supplied by the flight crew source, are considered to be passengers for the purpose of oxygen supply.

Flight crew members, not covered by the sub-paragraph above, are for the purpose of oxygen supply, to be considered as:

- Cabin crew members if they are on call or are definitely going to have FCC duty before completing the flight.
- Passengers if they are not on call and will not be on FCC duty during the remainder of the flight.
- Oxygen masks shall be located so as to be within the immediate reach of flight crew members whilst at their assigned duty stations.
- Oxygen masks for use by flight crew members in pressurised aeroplanes operating above 25,000 ft shall be a quick donning type of mask.

8.8.2.2 Cabin Crew Members

- Cabin crew members and passengers shall be supplied with supplemental oxygen in accordance with [Table 1](#). Cabin crew members carried above the minimum number of cabin crew members required shall be considered as passengers for the purpose of oxygen supply.
- When operating above 25,000 ft there shall be provided sufficient spare outlets and masks and / or sufficient portable oxygen units with masks for use by all required cabin crew members. The spare outlets and / or units are to be distributed evenly throughout the cabin, to ensure immediate availability of oxygen to each required cabin crew member regardless of their location at the time of cabin pressurisation failure.
- When operating above 25,000 ft there shall be provided an oxygen dispensing unit connected to oxygen supply terminals immediately available to each occupant, wherever seated. The total number of dispensing units and outlets shall exceed the number of seats by at least 10%. The extra units are to be evenly distributed throughout the cabin.
- The oxygen supply requirements, as specified in [Table 1](#), for aeroplanes not certificated to fly at altitudes above 25,000 ft may be reduced to the entire flight time between 10,000 ft and 13,000 ft cabin pressure altitude for all required cabin crew members and for at least 10% of the passengers if, at all points along the route to be flown, the aeroplane is able to descend safely within 4 minutes to a cabin pressure altitude of 13,000 ft.
- When operating above 25,000 ft or, if operating below, and unable to descend safely within 4 minutes to 13,000 ft, the aeroplane shall be provided with automatically deployable oxygen equipment immediately available to each occupant. The total number of dispensing units shall exceed the number of seats by at least 10%. The extra units are to be evenly distributed throughout the cabin.

NOTE

When flying over high terrain where a descent to 13,000 ft may not be possible (i.e. MEA / MORA above 13,000 ft), oxygen shall be provided to all passengers and cabin attendants for the entire flight above 13,000 ft.

8.8.2.3 Passengers

The oxygen requirements for passengers are the same as per in [Cabin Crew Members](#).

8.8.2.4 Supplemental Oxygen - Pressurised Aeroplanes

Table 1: Minimum Requirements for Supplemental Oxygen for Pressurised Aeroplane

Supply For	Duration and Cabin Pressure Altitude
All occupants of FCC seats on FCC duty	Entire flying time when the cabin pressure altitude exceeds 10,000 ft and in no case less than: a. 30 minutes for aeroplanes certificated to fly at altitudes not exceeding 25,000 ft¹; b. 2 hours for aeroplanes certificated to fly at altitudes more than 25,000 ft².
All required cabin crew members	Entire flying time when the cabin pressure altitude exceeds 13,000 ft, but not less than 30 minutes ¹ . The remainder of the flying time when the cabin pressure altitude exceeds 10,000 ft, but does not exceed 13,000 ft after the initial 30 minutes at such altitudes ¹ .
100% of passengers ⁴	The entire flying time when the cabin pressure altitude exceeds 15,000 ft, but in no case less than 10 minutes ³ .
30% of passengers ⁴	Entire flying time when the cabin pressure altitude exceeds 14,000 ft, but does not exceed 15,000 ft.
10% of passengers ⁴	Entire flying time when the cabin pressure altitude exceeds 10,000 ft but does not exceed 14,000 ft after the initial 30 minutes at these altitudes.

¹ The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aeroplanes maximum certificated operating altitude to 10,000 ft in 10 minutes, and followed by 20 minutes at 10,000 ft.

² The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aeroplane's maximum certificated operating altitude to 10,000 ft in 10 minutes, and followed by 110 minutes at 10,000 ft. Oxygen provided by PBEs available may be included.

³ The required minimum supply is that quantity of oxygen necessary for a constant rate of descent from the aeroplane's maximum certificated operating altitude to 15,000 ft in 10 minutes.

⁴ For the purpose of this table 'passengers' means passengers actually carried and includes infants.

Table 2: Aeroplane Specific Supplemental Oxygen Reference Tables & Charts

Aeroplane Types	Supplemental Oxygen Reference Tables & Charts
EMB-505	EMB-505 POH Section 6-13-10 Flight crew and passenger oxygen system
CE-560XLS	AFM Section III - Normal Procedures - Supplemental Oxygen System, page 3-83 (XL) page 3-105 (XLS)
CE-680A	AFM Section III 3-810-20 - Oxygen System
BD-100-1A10	FCOM Vol 2 page 08-01-4 Chapter 8, Oxygen Duration Tables
DA-2EASy	Performance Manual 2-50 - Oxygen
CL-600-2B16	FCOM Vol.2 Chapter 8, Oxygen System
BD-700-1A10	FCOM Vol.1 Supplement - Oxygen System (G-MS35060001)
BD-700-1A11	FCOM Vol.1 Supplement - Oxygen System (G-MS35060001)

8.8.2.5 Supplemental Oxygen - Non-pressurised Aeroplanes

Aeroplanes without a pressurised cabin, or aeroplanes with inoperative pressurisation systems shall not operate at altitudes above 10,000 ft unless supplemental oxygen and oxygen equipment with dispensing units are provided. The amount of supplemental oxygen for sustenance required for a particular operation shall be determined on the basis of flight altitudes and flight duration, consistent with the operating procedures established for each operation in the Aeroplane Operations Manual and with the routes to be flown, and with the emergency procedures specified in the Aeroplane Operation Manual.

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8.8.2.5.1 Oxygen Supply Requirements

1. Flight crew members:

Each member of the flight crew on FCC duty shall be supplied with supplemental oxygen in accordance with Table 3. If all occupants of FCC seats are supplied from the flight crew source of oxygen supply, then they shall be considered as flight crew members on FCC duty for the purpose of oxygen supply.

2. Cabin crew members and passengers:

Cabin crew members and passengers shall be supplied with oxygen in accordance with Table 3. Cabin crew members carried above the minimum number of cabin crew members required shall be considered as passengers for the purpose of oxygen supply.

Table 3: Requirements for Supplemental Oxygen for Non-pressurised Aeroplane

Supply For	Duration and Pressure Altitude
All occupants of FCC seats on FCC duty and crew members assisting the flight crew in their duties	Entire flying time at pressure altitudes above 10,000 ft.
All required cabin crew members	Entire flying time at pressure altitudes above 13,000 ft and for any period exceeding 30 minutes at pressure altitudes above 10,000 ft but not exceeding 13,000 ft.
100% of passengers	Entire flying time at pressure altitudes above 13,000 ft.
10% of passengers	Entire flying time after 30 minutes at pressure altitudes greater than 10,000 ft but not exceeding 13,000 ft.

NOTE

For the purpose of this table 'passengers' means passengers actually carried, and includes infants under the age of 2.

8.8.2.6 Crew Protective Breathing Equipment (PBE)

All pressurised aeroplanes and those unpressurised aeroplanes with an MCTOM of more than 5,700 kg or having an MOPSC of more than 19 seats shall be equipped with protective breathing equipment (PBE) to protect the eyes, nose and mouth and to provide for a period of at least 15 minutes:

- a. oxygen for each flight crew member on duty in the flight crew compartment;
- b. breathing gas for each required cabin crew member, adjacent to their assigned station; and
- c. breathing gas from a portable PBE for one member of the flight crew, adjacent to their assigned station, in the case of aeroplanes operated with a flight crew of more than one and no cabin crew member.

The PBE intended for flight crew use shall be installed in the flight crew compartment and be accessible for immediate use by each required flight crew member at their assigned station.

The PBE intended for cabin crew use shall be installed adjacent to each required cabin crew member station.

Aeroplanes shall be equipped with an additional portable PBE installed adjacent to the hand fire extinguisher, or adjacent to the entrance of the cargo compartment, in case the hand fire extinguisher is installed in a cargo compartment.

When computing the total oxygen required for the non-portable equipment, then the supply for the supplemental oxygen system as required under [Oxygen Equipment and Supply Requirements](#) may be considered. The portable PBE shall have a self contained supply for 15 minutes.

Detailed descriptions of the oxygen system, PBE, first aid and therapeutic equipment are contained in the SEP Manual and the respective AOM.

8.8.2.7 Use of Flight Crew PBE (EROS Mask)

The use of flight crew PBE (EROS masks) shall be in accordance with the AFM or other official manufacturer's instructions.

In case there are no AFM or official manufacturer's instructions as to when to stop using the flight crew PBE, the following shall be adhered to as a minimum:

- a. No visible smoke or fire;
- b. Cabin pressure altitude shall be below 10,000' / FL100;
- c. The PM will remove the PBE first;
- d. The PM will breath for at least 10 seconds ambient air;
- e. The PF will remove the PBE.

NOTE

1. A more conservative course of action such as keeping one crew member on oxygen longer or even until landing should be considered.
2. Transfer of control or ATC COMMS might be required to be able to remove the PBE.

8.8.2.8 Time of Useful Consciousness (TUC)

The 'Time of Useful Consciousness' is the time in which, from the occurrence of the oxygen deficiency, the tasks vital for life can still be performed, irrespective of the flight altitude as well of the physical or psychical condition of man. The average TUC is:

Time of Useful Consciousness	
Altitude (ft)	Elapsed Time
22,000	approximately 5 min.
25,000	2 min.
28,000	1 min.
30,000	35 sec.
35,000	20 sec.
40,000	12 sec.

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8.9 Emergency Descent

An immediate reaction is required by PF manoeuvring the aeroplane and PM complementing their actions as below.

If there is no immediate reaction from PF the PM shall take control of the aeroplane and follow the procedures. When emergency descent is complete, coordinate intentions with ATC.

PF	PM
Execute Memory items Fly the A/C into the required descent profile Verify Altitude Alert set Set local QNH After levelling off Power Performance (configuration) Automation reset ASAP	Monitor the flight path Passenger Safety ON Transponder code 7700. If applicable, select the appropriate emergency mode on ADS-B/ADS-C Declare MAYDAY and advise ATC of intentions, if able Determine and call out MSA Set Altitude Alert to highest of MSA, 10,00ft or NON ETOPS 180 Restriction Obtain and set local QNH Turn on aircraft exterior lights Visual lookout for terrain, traffic and weather Read and execute required checklists Address Passengers via PA Call-out altitudes

8.10 Aeroplane Over-speed, Over-G, and Severe Turbulence Encounter

If any of the aeroplane airspeed, G limits are exceeded as stated in the AFM or OM-B or the aeroplane encountered severe turbulence and no procedure or guidance is given by the AFM or OM-B, the following procedure shall be followed:

1. Aeroplane airspeed reduce to below the airspeed limit that was exceeded (not applicable for severe turbulence).
2. Visual inspection (if possible) of affected control surfaces, or other aeroplane parts affected by the over-speed, for structural damage.
3. Aeroplane configuration left unchanged unless performance or other safety factors dictate otherwise.
4. Aeroplane controllability check performed at a safe altitude and at Vref or lowest practical speed that has no undesired handling characteristics, whichever is higher.
5. Land as soon as practical unless structural damage or other emergency or abnormal situations dictate otherwise, using a minimum speed as determined in point 4.

8.10.1 Exceedance of Aeroplane Limits

When the following structural aeroplane limits have been exceeded:

- airspeed limits;
- aeroplane configuration limits (flaps, gear);
- mass limits;
- G-limits;
- bird strike impact prevention airspeed limits.

(A)MCC shall be notified. In the absence of other written official procedures, from either the manufacturer or NTA, exceeding such an aeroplane limit shall be recorded in the Aircraft Technical Log (ATL).

Bird-strikes shall be notified to (A)MCC and written in the ATL (bird strike inspection are arranged by (A)MCC), therefore, can be performed by the crew.

[Mandatory Occurrence Report - Reporting Criteria](#) shall be followed.

8.10.2 Exceedance Avoidance

It is highly recommended to allow for a sufficient airspeed buffer, especially during the climb and descent and when operating the flaps and landing gear, to handle expected or unexpected turbulence, windshear or automation lag. (a difference of 10 knots arrival speeds changes the arrival time by approximately one minute).

8.11 Circuit Breaker Resetting

8.11.1 Tripped Circuit Breakers

A. General

There is a latent danger in resetting CB tripped by an unknown cause because the tripped condition is a signal that something may be wrong in the related circuit. Until it is determined what has caused a trip to occur, flight crew, maintenance, or aeroplane ground servicing personnel usually have no way of knowing the consequences of resetting a tripped CB. Resetting a tripped CB by an unknown cause should normally be a maintenance function conducted on the ground.

B. In-flight

A tripped CB shall not be reset in flight unless doing so is consistent with explicit procedures specified in the approved aeroplane operating manual or aeroplane flight manual used by the flight crew or unless, in the judgement of the commander, resetting the tripped CB is necessary for the safe completion of the flight.

If resetting of the tripped CB in flight is necessary for the safe completion of the flight, a crew member shall, time permitting, try to contact the company and get advice from maintenance.

If the reset clears the problem, then the circuit breaker and associated system shall be monitored closely by a crew member for the remainder of the flight.

If the circuit breaker trips again, it shall not be reset.

If a tripped CB is reset, an entry into the Technical Log is required and shall include the following:

- i. A description of the exact conditions when the trip occurred,
- ii. when the CB was reset, and
- iii. the result of resetting the CB.

C. On the Ground

A CB tripped by an unknown cause may be reset on-the-ground after maintenance has determined the cause of the trip and has determined that the CB may be safely reset.

A CB may be cycled (tripped or reset) as part of an approved trouble-shooting procedure, unless doing so is specifically prohibited for the conditions existing. If the Minimum Equipment List (MEL) contains procedures that allow a tripped CB to be reset, then the same cautions with reference to resetting tripped CBs also apply.

8.11.2 System Resets via Resetting of CBs

A. General

Aeroplane systems may be reset via the resetting (cycling) of the associated CB. The resetting of a CB that is not tripped to reset the associated system may be used as described below.

B. In-flight

A system reset by resetting the associated CB is prohibited in flight unless doing so is consistent with explicit procedures specified in the approved aeroplane operating manual or aeroplane flight manual used by flight crew or unless, in judgment of the commander, resetting the CB is necessary for the safe completion of the flight.

If resetting of the tripped CB in flight is necessary for the safe completion of the flight, a crew member shall, time permitting, try to contact the company and get advice from maintenance.

If the reset clears the problem, then the circuit breaker and associated system shall be monitored closely by a crew member for a minimum of ten minutes (depending of the flight phase) following the reset. If the system reset is not successful, the CB shall not be cycled again.

C. On-the-Ground

A system reset by resetting the associated CB may be accomplished on-the-ground by a crew member before consultation with maintenance or the Chief Pilot, one time only, unless doing so is specifically prohibited for the conditions existing.

If the reset clears the problem, then the circuit breaker and associated system shall be monitored closely by a crew member for a minimum of ten minutes (depending on flight phase) following the reset. If the system reset is not successful then the maintenance and / or the Chief Pilot shall be consulted for further action.

A CB may be cycled (tripped or reset) as part of an approved trouble-shooting procedure, unless doing so is specifically prohibited for the conditions existing.

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8.12 Life Rafts and Signalling Equipment

No aeroplane shall operate over water flights, where the distance from land which is suitable for making an emergency landing is more than 400 NM or 120 minutes at cruising speed, whichever is lesser, unless it carries life-rafts in sufficient numbers to carry all persons on board, stowed so as to facilitate their ready use in an emergency, and being of sufficient size to accommodate all the survivors in the event of a loss of one raft of the largest rated capacity.

The life-rafts shall each be equipped with:

- A survivor locator light; and
- Signalling equipment for making pyrotechnical distress signals.

The following equipment must also be carried, either within the life-raft provision or separately:

- Life-saving equipment to provide means of sustaining life as appropriate to the flight to be undertaken; and
- A total of at least two survival Emergency Locator Transmitters (ELT(S)) capable of operating simultaneously on 121.5 MHz and 406 MHz, transmitting on the distress frequencies prescribed in ICAO Annex 10, Volume V, Chapter 2.

8.13 Procedures for Dealing with Unruly (Abusive) Passengers

Due to the nature of our business any such occurrence shall be handled with the utmost sensitivity.

Disruptive passenger is defined as a passenger who fails to respect the rules of conduct at an airport or on board an aircraft or to follow the instructions of the airport staff or crew members and thereby disturbs the good order and discipline at an airport or on board the aircraft.

Unruly passengers are defined as persons who commit, on board a civil aircraft, from the moment when the aircraft door is closed prior to take-off to the moment when it is reopened after landing, an act of:

- a. assault, intimidation, menace, or wilful recklessness which endangers good order or the safety of property or persons;
- b. assault, intimidation, menace, or interference with a crew member in the performance of duties or which lessens the ability to perform duties;
- c. Willful recklessness or damage to an aircraft, its equipment, or attendant structures and equipment such as to endanger good order and the safety of the aircraft or its occupants;
- d. communication of information which is known to be false, thereby endangering the safety of an aircraft in flight; and
- e. disobedience of lawful commands or instructions for safe, orderly, or efficient operations.

In many instances, passengers behave irrationally and do not calculate the consequences of their actions.

8.13.1 Commander's Authority

If the Commander of an aeroplane in flight, wherever it may be, has reasonable grounds for believing that a passenger has done or is about to do anything that may jeopardise:

- a. The safety of the aeroplane, persons or property on board, or
- b. the good order and discipline onboard,

then the commander is entitled to take all reasonable measures to:

- a. Protect the safety of the aeroplane;
- b. Protect persons or property on board;
- c. Maintain good order and discipline on board;
- d. Disembark unruly or violent passengers.

Those reasonable measures may include restraining the unruly or violent passengers. However, the Commander should ensure that minimum force required is used against the passenger.

The same principle applies if the Commander has reasonable grounds for believing that a passenger has done any act on the aeroplane in flight that, in the opinion of the Commander, is a serious offence under the law in force in the country of registration of the aeroplane.

The Commander may take all reasonable measures, including the restraint of the passenger, to enable the passenger to be delivered into the custody of the police, or to any corresponding official, if the aeroplane lands in a country that is party to the Tokyo Convention.

8.13.2 Procedures

8.13.2.1 Prior to or After Flight on the Ground

When the aeroplane is still on the ground, if a passenger is unruly or shows that about to become unruly, the Commander should disembark the passenger at the airport.

If the passenger refuses to leave the aeroplane, the Commander should not use force but should call for police assistance to have the passenger removed.

8.13.2.2 In-flight

If a situation should arise where a passenger or passengers may become unruly, it is the responsibility of all crew members to ensure the safety of the aeroplane, its passengers and crew.

- a. Cabin crew (if carried) shall notify the commander of the potential situation as soon as possible with details.
- b. All efforts shall be made to try to prevent passengers from becoming unruly.
- c. The commander or available crew member will contact NTA by the quickest means of communication available, giving the fullest details and to seek advice.
- d. Commander shall use their discretion on the appropriate actions to be taken in order to maintain the safety of the aeroplane, passengers or crew.
- e. The Duty Dispatch Officer will immediately inform Senior Management.
- f. NTA will advise the relevant authorities of the incident.

8.13.3 Restraint of Passengers

The Commander should consider restraining a violent or unruly passenger if:

- a. All other feasible means of placating such a passenger have been exhausted, or
- b. if the urgency of the situation demands immediate restraint and
- c. if the Commander considers that the act of restraining the passenger is not likely to cause an increase in the safety hazard.

The restraint can be continued after landing and until the police arrives, providing the Commander notifies the relevant authorities, at the place of the landing of the restraint of the passenger and the reasons for doing so. The authorities shall be notified either before or as soon as is reasonably practicable after landing.

NOTE

1. With regard to threats of violence, it is not intended that the above procedure should apply in case of bomb threats which are not specific or where there are strong suspicions that the threat is a hoax.
2. Relevant authorities shall in the above context be taken to mean police, Department of Trade representatives, Embassy or Consular representatives, ANAC representatives.

8.13.4 Communication Procedures

During the flight, which for the purpose of these guidelines refers to after doors closed and prior to doors open, the Commander is 'in command' and should request police assistance solely via Air Traffic Control.

The commander should do so with the statement: 'A passenger has refused to obey the lawful command of the captain and has committed an act endangering the safety of the aeroplane'.

8.13.5 Hand-over to Police

When police assistance arrives at the aeroplane, the Commander should communicate with them using a form of words to the following:

'It is alleged that an incident has occurred on this flight threatening the safety or good order and discipline of the flight and I wish to investigate it. The following were present at the incident...'

8.13.6 Details to be Noted

The Commander should normally note the following details at the time of the incident:

- a. Aeroplane registration;
- b. Commander's name;
- c. Aeroplane position and the time of the incident;
- d. Name, nationality and details of the journey and of the person under restraint;
- e. Brief details of the incident.

NOTE

The Commander should record these details in a form that can be readily handed over to the police.

The crew member(s) on the aeroplane who actually witness the incident shall take a written note at the time of the incident, or as soon as possible thereafter, stating what happened, what was said and by whom. Such crew members should identify themselves and provide their contact details for future use.

Those involved should be clearly identified and the passengers' names and addresses should be noted.

This note shall be preserved for possible use in future proceedings, in which the prosecution may be under a duty to disclose it to the defence. At least two witnesses are required, which may either be crew members or passengers.

A NetJets Safety report is also required to be completed.

The Company will ensure that a representative of the relevant authorities is available at the crew de-briefing or at the de-briefing of any other person concerned, and at any other subsequent investigatory meetings of importance.

As soon as practicable, the Company will make available to the relevant authorities a copy of the form NTA-018A (Report for Insurance Claim) , together with the copies of other reports, aeroplane documents or records that are relevant.

8.13.7 Legal Considerations

The Commander needs to be concerned with the legal aspects of framing the charges; the police will investigate the Commander's referral in the usual way

NOTE

If the offence were to take place on an aeroplane at the time when it is stationary on foreign ground shortly before departure, the Commander would similarly have the power to disembark the passenger.

A violent or unruly passenger may be disembarked in any where the aeroplane lands; irrespective of the passenger's nationality. However, the Commander shall report any disembarkation and the reasons for it, to the appropriate authority in the country of disembarkation and to the national diplomatic or consular office of the passenger concerned.

If after having taken all reasonable steps the Commander is unable to make the report, a senior member of the airport or company staff should be asked to ensure that the report is made.

If the Commander believes the passenger has committed some serious offence, the Commander may deliver the passenger to a person having similar functions to those of a police constable or immigration officer or corresponding official.

If the Commander fails, without reasonable cause, to comply with these requirements on the notification of disembarkation or delivery, they may be guilty of an offence.

The Commander should also make available to the police the written notes and details from the witnesses.

8.14 Ramp Inspection

Any state worldwide may conduct ramp inspections under a national or international programme. Ramp inspections, regardless of location and authority, will be similar in nature and content. Ramp inspections shall not be refused. Doing so may result in the aircraft being put under detention. However, inspections should not cause delays to the flight, except when issues are found that may significantly affect flight safety or require further investigation to be clarified.

During a ramp inspection:

- The commander retains authority over all aspects of the operation and use of the aeroplane and must ensure all findings the inspecting authority raises are adequately addressed.
- If the commander is uncertain about any technical or operational issue, they should promptly contact the CAMO, Director of Operations, Chief Pilot, or Fleet Program Manager, as appropriate.

At the end of the inspection, the authority will provide a Proof of Inspection form, which the commander must sign.

Before signing the Proof of Inspection, the commander must read the description of the findings raised (if any). If necessary, comments or objections can be added to the 'Crew Comments' field. Signing the form does not imply acceptance of the findings by the operator but merely acknowledges that the inspection has taken place.

After a ramp inspection, the crew should notify the Flight Operations departments via email at lis-safa@netjets.com. A notification is not mandatory, but is essential when relevant items are raised, namely, findings with safety impact or category 3 (most serious).

The Proof of Inspection form differs between authorities, although the same type of information is recorded.

EU Ramp Inspection Programmes (SAFA/SACA/SANA)

Ramp inspections performed in states participating in the EU Ramp Inspection Programme are known by the acronyms SACA (Community), SAFA (Foreign) or SANA (National), depending on the state of registry of the aeroplane and inspecting authority.

Under this programme, tests for any prohibited psychoactive substances (alcohol and drug tests) may be conducted during ramp inspections. Substance testing is conducted in accordance with applicable regulations, which depend on the location where the test is being performed. EASA provides a reporting mechanism for alcohol testing, while drug testing procedures are subject to national requirements and oversight. This information may be included as a remark on the Proof of Inspection, but is outside the scope of the inspection programme.

The competent aviation authorities may delegate the execution of the alcohol or drug tests to local police or airport authorities. In such cases, the Proof of Inspection does not have to be issued by the entity running the test, and that responsibility is still with the competent aviation authorities.

If an alcohol or drug test yields a positive result, the crew member shall be immediately removed from duty, and the inspecting authority shall notify both the crew and the operator's competent authority. The crew member must also inform the company immediately through the following channels: lis-opssupport@netjets.com and lis-crewsupport@netjets.com.

8.15 Hotspare Procedures

8.15.1 General

A hotspare aeroplane is one that is crewed at the airport and ready to proceed at short notice.

The crew will show at the airport and perform duties as per [Operational Procedures](#).

The airport should have a suitable facility for the crew to be while on hotspare duty.

8.15.2 Scheduling

Refer to OM-A, [Flight Duty Time Limitations](#).

8.15.3 Operational Procedures

1. Flight Planning

Flight plans will be filed as non-commercial operations with code 'G' for all non-revenue flights. This is for landing performance purposes due to the pre-determined fuel loads that the aeroplane will have as per paragraph 3 below.

2. Pre-flight

A complete pre-flight will be accomplished and all covers will be removed. The aeroplane will be ready for an immediate departure.

A pre-flight will be accomplished every 3 hours to ascertain that the aeroplane is ready.

NOTE

The removal of covers shall be accomplished according to the specific weather conditions e.g. during icing conditions.

3. Fuel

Refer to [Determination of the Quantities of Fuel and Oil Carried - Fuel Policy](#).

4. APU

The APU shall only be used for pre-flight purposes if required. It shall not be on for the time the crew is on hotspare duty.

8.16 SAT Comm and AFIS / Email Messaging

No communication to an aeroplane shall occur by SATCOM, AFIS messaging, or AFIS-Email if an aeroplane is within 20 minutes after ATD or 20 minutes before ETA.

NOTE

ETA shall be the adjusted ETA for the ATD.

Satcom Communications

The SATCOM system provides in-flight access to the international telephone network. The international '+' prefix followed by the country code shall be used prior to any phone number. It is recommended to test the Satphone on the first day of a tour. It is used for two main purposes:

1. To allow passengers to make in-flight private calls.
2. For in-flight communication with NTA or service providers, such as MedLink.

Detailed description of functions of the Sat comm. system can be found in the Sat comm. user manual.

The crew can use the Sat Comm system for the following reasons:

- Emergencies / abnormal situations;
- Report aeroplane malfunctions that (could) prevent dispatch of the next flight;
- Reporting delays more than 15 minutes;
- Request services at the arrival aerodrome;
- Request weather and NOTAMs for new destinations;
- Loss of communication with ATC;
- Weather updates on long range flights.

When using the Sat comm system in-flight to call Operations Planning and Control, NJE Hotline number shall be used.

Contact	Number
NJE HOTLINE	+351 21 446 9090
MedLink (MEDLINK)	+ 1 602 239 3627

8.17 Aeroplane Disinsection

Some destinations require by law that the aeroplane is disinsected at departure or prior to arrival.

Information whether disinsection is required or not can be found in the Jeppesen Airway Manual section: Entry Requirements.

E.g. Australia, Seychelles require disinsection prior to arrival.

Operating from the countries listed below

Local authorities require disinsection when flying from certain countries. Authorities may conduct inspections and impose fines for non-compliance. Therefore disinsection shall be carried out in accordance with the guidelines provided by the World Health Organisation below, when operating from the following countries (see table).

Africa	Algeria	Eastern Mediterranean	Afghanistan
	Angola		Djibouti
	Benin		Iran
	Botswana		Pakistan
	Burkina Faso		Saudi Arabia
	Burundi		Somalia
	Cabo Verde		Sudan
	Cameroon		Yemen
	Central African Republic	European Area	Azerbaijan
	Chad		Kyrgyzstan
	Comoros		Tajikistan
	Congo		Turkey
	Côte d'Ivoire		Uzbekistan
	Democratic Republic of the Congo	The Americas	Argentina
	Equatorial Guinea		Belize
	Eritrea		Bolivia
	Ethiopia		Brazil
	Gabon		Colombia
	Gambia		Costa Rica
	Ghana		Dominican Republic
	Guinea		Ecuador
	Guinea-Bissau		El Salvador
	Kenya		French Guiana, France
	Liberia		Guatemala
	Madagascar		Guyana
	Malawi		Haiti
	Mali		Honduras
	Mauritania		Mexico
	Mayotte, France		Nicaragua
	Mozambique		Panama
	Namibia		Paraguay
	Niger		Peru
	Nigeria		Suriname
	Rwanda		Venezuela
	Sao Tome and Principe	South-East Asia	Bangladesh
	Senegal		Bhutan
	Sierra Leone		Democratic People's Republic of Korea
	South Africa		India
	South Sudan		Indonesia
	Swaziland		Myanmar
	Togo		Nepal
	Uganda		Sri Lanka
	United Republic of Tanzania		Thailand
	Mainland		Timor-Leste
	Zanzibar		
	Zambia		
	Zimbabwe		

8.17.1 Disinsection Procedure

NTA uses the World Health Organisation (WHO) 'Blocks Away' recommended disinsection procedure. This procedure is accomplished prior to taxi out at the departure aerodrome.

The disinsection aerosol spray for the procedure shall contain 2% D-penophrin. This is stated on the spray cans that are carried on-board.

Although not regarded as a preferred method, the on-arrival-method may be requested by authorities on arrival. On arrival, all exterior doors and windows must remain closed and may only be opened in accordance with directions from an agent of the Health Authority, who will board the aeroplane and perform disinsection of the cabin and FCC similar to the 'blocks-away' method. This is the system used by the Australian and New Zealand authorities.

8.17.1.1 Aeroplane with NO Cabin Access to Baggage Hold

- 1 Plastic bag required to store used cans and caps.
- 2 aerosol spray cans are required. 1 can for FCC and cabin, 1 can for the baggage compartment.
- Disinsect baggage compartment prior to the final closing of the baggage door.
- Disinsect FCC and cabin before engine start after all passengers are on-board and all doors are closed.

Disinsection of Baggage Compartment

- 1 aerosol spray can.
- Detach one number section of aerosol sticker and attach the sticker to the General Declaration form or if there is no sticker write the serial number on General Declaration form.
- Dispense the mist in the baggage compartment area.
- For inspection purposes leave the empty can in the baggage compartment.
- Take the cap with sticker on it and place it in bag.

Disinsection of FCC and Cabin

- Only 1 aerosol spray can is required for the both the FCC and cabin disinsection.
- Detach one number section of aerosol sticker - attach the sticker to the General Declaration or if there is no sticker write the serial number on the General Declaration.
- Once all passengers have boarded and the doors have been closed, turn all air-conditioning off and open all galley doors, cupboards, storage shelves etc. Pay attention not to directly spray on non-covered food items, cutlery, plates, glasses, etc.
- Brief the passengers that the disinsection will take place and advise passengers as follows:
'If you feel it may cause you inconvenience, close your eyes and cover your faces for the duration of the disinsection.'
- Beginning in the FCC, hold the can at arm's length and dispense the mist by walking at an even pace until the end of the cabin. Include Galley, toilets, and wardrobes.
- Place the used aerosol cans and caps in the same bag containing the caps of the cans used for the baggage compartment. Keep it to present to the authorities on arrival if required.

8.17.1.2 Aeroplanes with Cabin Access to Baggage Compartment

- 1 Plastic bag required to store used cans and caps.
- 1 aerosol spray can is required.
- Disinsect the FCC, cabin and baggage compartment before engine start after all passengers are on-board and all doors are closed.

Disinsection of FCC, Cabin and Baggage Compartment

- Only 1 aerosol spray can is required for disinsection of the FCC, cabin and baggage compartment.
- Detach one number section of aerosol sticker - attach the sticker to the General Declaration or if there is no sticker write the serial number on the General Declaration.
- Once all passengers have boarded and the doors have been closed, turn all air-conditioning off and open the baggage compartment access door, crew rest area door, all galley doors, cupboards, storage shelves etc. Pay attention not to directly spray on non-covered food items, cutlery, plates, glasses etc.
- Brief the passengers that the disinsection it take place and advise passengers as follows:
'If you feel it may cause you inconvenience, close your eyes and cover your faces for the duration of the disinsection.'
- Beginning in the FCC, hold the can at arm's length and dispense the mist by walking at an even pace until the end of the cabin including the baggage compartment. Include crew rest area, galley, toilets, and wardrobes.
- Place the used aerosol cans and caps in the bag and keep it to present to the authorities on arrival if required.

8.18 Electronic Flight Bag (EFB)

Crew shall ensure that required flight documents and navigation charts are current and available in either paper or electronic format (as approved).

Required paper-based information or documentation shall not be replaced by its electronic format, unless the aeroplane type has an operational approval to do so.

Any contingency procedure, non-normal use of a Portable EFB device or applications and any other type of unusual or unexplained event shall be reported as soon as possible to Lis-EFBsupport@netjets.com.

Each crew member assigned to duty shall have completed the NTA approved EFB training program prior to using an EFB.

During the operational assessment of new EFB hardware, a backup device from a previously approved model may be carried onboard if required. Instructions on where to store, update, and charge the backup device will be provided along with the communication initiating the operational assessment.

8.18.1 Introduction and System Description

Electronic Flight Bags are categorised into two classes of EFBs, according to the type of hardware:

- Installed EFB;
- Portable EFB.

Electronic Charts include:

- Terminal charts;
- En-route charts, and;
- Airway manuals.

In addition to the Jeppesen data, NTA publishes customised terminal and enroute data to the FD Pro application. Some data is fleet-specific, such as EDTO (ETOPS) rings, aerodrome category, and wingspan limitations. Therefore, it is necessary to select the correct fleet and import the flight route into the application. For further instructions and the type of published tailored data, refer to the Portable EFB User Guide.

Electronic Documents are documents required for flight that are approved to be stored electronically on the portable EFB (excludes Electronic charts).

Performance Calculator - take-off and landing performance data provided by iPreFlight Genesis application installed in the portable EFB. Any mention of iPreFlight in this manual refers specifically to iPreFlight Genesis.

Digital or paper copy AFM may serve as primary or backup to iPreFlight Genesis. Crew procedures for the use of iPreFlight Genesis:

1. Calculations are conducted independently by each crew member before data outputs are accepted for use.
2. Formal cross-check is made before data outputs are accepted for use. Such crosschecks should utilise the independent calculations described above, together with the output of the same data from other sources on the aeroplane.

3. Gross-error check is conducted before data outputs are accepted for use. Such a gross-error check may use either a 'rule of thumb' or the output of the same data from other sources on the aeroplane.
4. iPreFlight Genesis is an offline application, and for this reason:
 - i. It may not contain all the relevant obstacles in the database and output different results than flight dispatch;
 - ii. You may introduce obstacles manually on iPreFlight Genesis;
 - iii. The airfield database is valid for 28 days after a new update. For this reason, even though two iPreFlight Genesis applications may be updated and compliant, they can output different results or display different options (for example, a new RWY interception).

For further guidance, refer to OM-A TA 32 and the Portable EFB User Guide.

In-flight weather (IFW) - EFB function or application enabling the flight crew to access meteorological information. It is designed to increase situational awareness and support the flight crew when making strategic decisions. However, it should not be used as a substitute for certified aircraft systems (e.g. weather radar).

Approved IFW resources:

Weather function on Jeppesen FD Pro application	This function is not necessary for the execution of the flight, and the weather layers are overlaid on the FD-Pro enroute map. The weather function is for situational awareness only and must not be used for dispatch purposes.
SkyPath	SkyPath is an application to access meteorological information, specifically for observed and forecasted turbulence. It is designed to increase situational awareness and support the crew when making strategic decisions. Any current information from the meteorological documentation required to be carried on board or from aircraft primary systems should always prevail over the information from the SkyPath application. SkyPath displays observed and forecasted turbulence and may be updated on the ground. 1. Own-ship position shall be disabled at all times. 2. SkyPath shall not be used during critical phases of flight.
HoTs	Holdover Times Calculator

Electronic Flight Folder (EFF) - Controlled digital system used to assemble, distribute, display, and retain flight-specific operational documentation required to support a flight (e.g., e.g. Operational Flight Plan / Flight Release, NOTAM and meteorological packages, mass and balance references, fuel planning documentation, permits, and associated post-flight records).

The EFF provides the electronic flight package, which is made available to authorised users via an approved system, supports revision control when updates are issued, and ensures archiving with full traceability, including user access, document versioning, and time-stamped records of changes.

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NOTE

All conflicting information found on the EFB or other aeroplane systems during flight preparation shall be resolved before the flight is initiated. Flight crew members can contact Flight Dispatch or the FOSM team to obtain additional information to help identify which system has incorrect information.

All conflicting information found on the EFB or other aeroplane systems during flight shall be resolved using all available resources as necessary. This may include, but is not limited to, contacting the Hotline or ATC.

The following table states the approved EFB systems by aeroplane type and which required documents can be used electronically:

Table 8.18.1.A - Approved EFB Systems by Aeroplane Type

Aeroplane Type	Electronic Charts		Electronic Documents		Performance Calculator	
	Portable EFB	Installed EFB	Portable EFB	Installed EFB	Portable EFB	Installed EFB
CE-560XLS	Approved	Not Capable	Approved	Not Capable	Approved	Not Approved
EMB-505	Approved	Not Approved	Approved	Not Capable	Approved	Not Approved
CE-680A	Approved	Not Approved	Approved	Not Capable	Approved	Not Approved
DA-2EASy	Approved	Not Approved	Approved	Not Capable	Approved	Not Approved
BD-100-1A10	Approved	Approved with restrictions (see table 8.18.1.1 for restrictions)	Approved	Not Capable	Approved	Not Approved
CL-600-2B16	Approved	Not Approved	Approved	Not Capable	Approved	Not Approved
BD-700-1A10	Approved	Approved with restrictions (see table 8.18.1.1 for restrictions)	Approved	Not Capable	Approved	Not Approved
BD-700-1A11	Approved	Approved with restrictions (see table 8.18.1.1 for restrictions)	Approved	Not Capable	Approved	Not Approved

There are several possible combinations of two approved and operable EFB units (Portable or Installed) with all the required for flight electronic documentation for flight dispatch.

Table 8.18.1.B - Authorised EFBs Combinations EFB Function, Solution, and Flight Dispatch Minimum Requirements

EFB Function	Application	EFB Solution	Flight Dispatch Minimum Requirements
ELECTRONIC CHARTS	FD Pro X	Installed EFB and Portable EFB	Any combination of two approved EFBs (e.g. Installed EFB with redundancy; two Portable EFB units; one non-redundant installed EFB with one Portable EFB unit).
ELECTRONIC DOCUMENTS	Comply365	Portable EFB	One approved Portable EFB unit.
PERFORMANCE CALCULATOR	iPreFlight Genesis	Portable EFB	- Two approved Portable EFB units; OR - OEM performance data in PDF format on at least one approved portable EFB.
IN-FLIGHT WEATHER	SkyPath	Portable EFB	Not required for dispatch.
	FD Pro X	Portable EFB	Not required for dispatch.
HOLDOVER TIME GUIDELINES	HOTs	Portable EFB	Not required for dispatch.
ELECTRONIC FLIGHT FOLDER	Skybook	Portable EFB	Two approved Portable EFB units.

In the absence of any of the above EFB combinations, flight crew shall revert to the Contingency or Non-normal Operating Procedures, per Phase of Flight, [table 8.18.4.A](#).

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8.18.1.1 Installed EFB

For Installed EFBs approved hardware and software, refer to the respective aeroplane AFM, OM-B, MEL and NJMP 1.36.

The following table details installed EFBs approved per fleet and respective configuration. The Airway Manual is not available in any of the approved EFB applications.

Table 8.18.1.1 - Authorised Installed EFBs, Software, and Restrictions

Aeroplane (M / M / S)	Hardware (Manufacturer and Model)	Configuration	Software	EFB Applications	Restrictions and Limitations	Remarks
CE-560XLS	N/A	N/A	N/A	N/A	N/A	-
EMB-505	Garmin 3000	Dual File Server Unit (for backup purposes)	Refer to AFM	N/A	Not approved EFB system	-
CE-680A	Garmin 5000	Dual File Server Unit (for backup purposes)	Refer to AFM	N/A	Not approved EFB system	-
DA-2EASy	Honeywell EPIC	Single File Server Unit	Refer to AFM	N/A	Not approved EFB system	-
BD-100-1A10	Rockwell Collins Pro Line 21 Advanced IFIS-5000	Dual File Server Unit (for backup purposes)	Refer to AFM	Electronic Terminal Charts	Refer to AFM	En-route charts not available.
BD-700-1A10	Rockwell Collins Fusion / IFIS 6000	Triple File Server Unit (for backup purposes)	Refer to AFM	Electronic Charts	Refer to AFM	-
CL-600-2B16	Rockwell Collins Pro Line 21 Advanced IFIS-5000	Single File Server Unit	Refer to AFM	N/A	Not approved EFB system	
BD-700-1A10	Rockwell Collins Fusion / IFIS 6000	Triple File Server Unit (for backup purposes)	Refer to AFM	Electronic Charts	Refer to AFM	-
BD-700-1A11	Rockwell Collins Fusion / IFIS 6000	Triple File Server Unit (for backup purposes)	Refer to AFM	Electronic Charts	Refer to AFM	-

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8.18.1.2 Portable EFB

Portable EFBs are controlled portable electronic devices (C-PED) and are part of the crew kit.

A portable EFB hosts EFB software applications and may additionally host miscellaneous (non-EFB) software applications.

The Portable EFB must strictly match the list of hardware, software and other restrictions approved by the EFB Administrator.

Each Portable EFB configuration is tracked remotely by NTA for auditing and configuration control purposes. Use of a personal electronic device as an EFB is not authorised.

If the Portable EFB device is lost, stolen, or broken, the crew member shall inform NTA IT Helpdesk as soon as possible (+351 214 469 099 / NJE-Wireless@netjets.com) and complete the electronic form NTA-FO 049. If this occurs while on tour, the crew member shall activate contingency procedures (refer to [Contingency or Non-normal Procedures](#)) and inform the company by sending an e-mail to Lis-EFB-INOP@netjets.com.

The Portable EFB User Guide provides user-oriented instructions on operating the portable EFB. The EFB administrator is responsible for maintaining the guide, which is published in NTA DMS.

8.18.2 Onboard Utilisation Rules

8.18.2.1 Installed EFB

Onboard utilisation of Installed EFBs shall be done in accordance with the approved AFM and other OEM approved reference manuals.

Prior to each flight, flight crew shall check the database revision of Electronics Charts.

If the database is not current, the crew shall follow MEL dispatch procedures.

For the approved functionalities, an approved Installed EFB, if available and operable, shall always be used as a primary display.

The use of EFB on board does not replace any of the aeroplane information display.

Whenever there are concerns about the operability of the Installed EFB, the crew member shall perform an EFB Operational Check as per [EFB Operational Check](#).

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8.18.2.2 Portable EFB

Onboard utilisation of Portable EFBs shall be done in accordance with NTA Operations Manual Part A and Part B.

Charging the Portable EFB

The charging of Portable EFBs from the aeroplane's power outlets is only authorised outside critical phases of flight and when it is:

- Regularly and consistently monitored for adequate operating conditions, where a crew member shall ensure that:
 - a. no system overheating warning is displayed by the EFB's operating system.
 - b. there are no signs of increased temperature, such as decreased responsiveness, unanticipated reduction in screen brightness or high temperature to the touch.
 - c. the device is not left charging in direct sunlight.

If the device displays an overheating warning or the crew identifies signs of increased temperature, the EFB shall be disconnected from the power outlet immediately, removed from direct sunlight or heat sources, and allowed to cool before being returned to service.

NOTE

Temperature protection is provided by the EFB's integrated thermal management system. The EFB's operating system continuously checks the device's temperature and, after a threshold is reached, automatically lowers system performance, restricts charging, dims the screen, or, at more severe levels, displays a warning message (*'iPad needs to cool down before you can use it'*). This alert indicates that the device has exceeded a safe thermal limit and charging must be halted.

- Done only until the battery reaches 100% (the unit shall not remain connected to the aeroplane's power outlet after the battery reaches 100%).
- Done using only original manufacturer's accessories or certified Apple MFI accessories.
- Cables should not hang loosely in a way that compromises task performance and safety; flight crew members should be able to easily secure the cables out of the way during operations (e.g. by using cable tether straps); and
- Cables should be of sufficient length so that they do not obstruct the use of any movable device (e.g. flight controls, switches, seats, windows) in the flight crew compartment.

Portable EFB Connectivity to Aircraft Systems

Connecting to aircraft systems, whether wired or wireless, is not permitted, except for any electrical supply systems used for charging C-PEDs.

Prevention of Device Thermal Stress and Overheating

The EFB is designed to operate within a temperature range of 0°C to 35°C. When internal safety temperature limits are reached, it will automatically shut down to protect its components. Should a shutdown occur, allow the EFB to naturally return to the acceptable operating conditions.

The most common cause of temperature-related shutdowns is external heat gain, typically from direct solar radiation absorbed through the display. To reduce the likelihood of overheating when in bright sunlight, adjust the tilt of the Portable EFB to minimise the angle at which sunlight hits the display (the shallower the angle, the better).

Storage Requirements: When not in use, the Portable EFB must be kept within its non-operating temperature range of -20°C to +45°C . If necessary, the unit should be removed from the aircraft to meet these requirements.

Additionally, after every flight, the Portable EFB must be removed from the approved viewable stowage device, safety stored, and must not be left in any location where it may be exposed to direct sunlight or elevated temperatures, in order to prevent heat buildup and reduce the risk of automatic shutdown.

If an unexpected automatic shutdown occurs, the crew member must report the event to the EFB Support Team (Lis-EFBsupport@netjets.com), including details such as the phase of flight, solar exposure conditions, and ambient temperature.

In the event the portable EFB exhibits physical damage, and the crew suspects an over-temperature condition caused by battery thermal runaway rather than environmental factors, the crew shall immediately apply the procedures for Portable Electronic Device Fires as specified in the SEP Manual.

Device Dimming

The FD-Pro X application provides a brightness control for the application only, without the need to use the device settings.

CAUTION

During night flights, switching from the FD-Pro X may suddenly increase the screen brightness, which could be disturbing.

Management of Battery Life

Crew members shall ensure all non-EFB applications, including background applications, are closed when not needed.

Wireless connections shall be turned-off when not needed or permitted. Crew members shall ensure that the Portable EFB is in Sleep Mode when not in use (non-critical phases of flight).

Humidity

iPads shall not be operated outside its non-condensing relative humidity limits (5% to 95%).

Whenever there are concerns about the operability of the Portable EFB, the crew member shall perform an EFB Operational Check as per [EFB Operational Check](#).

8.18.3 Normal Procedures

8.18.3.1 Installed EFB

Installed EFBs shall be operated in accordance with the approved AFM, other OEM approved reference manuals, and NTA Operations Manual Part B.

8.18.3.2 Portable EFB

Each crew member is responsible for updating the EFB contents and ensuring that it is charged in accordance with the normal procedures (see [table 8.18.3.2.1.B](#)).

8.18.3.2.1 Flight Crew Portable EFB

A portable EFB can be operated inside and outside the aeroplane. Portable EFBs may be used in all phases of the flight and shall be immediately available for viewing during critical phases of flight. They shall not:

- interfere with flight control movement;
- obstruct instruments and outside view.

Flight crew members are responsible for proper use of the Portable EFB. The use of Portable EFB on board does not replace any of the aeroplane information displays.

When a Portable EFB has been designated as the primary EFB, it shall be used by the PM.

Use as a Primary or Backup EFB

An approved Portable EFB, if available and operable, may be used as a primary display or may be used with another Portable or Installed EFB as a backup display.

Table 8.18.3.2.1.A - Portable EFB - Use as Primary or Backup EFB

	Primary EFB	Backup EFB
CRITICAL PHASES OF FLIGHT	- Used as the Primary Display of navigational information. PM maintains custody and control of the Portable EFB. Continuously displays appropriate chart. - The EFB shall be secured on a viewable stowage device.	- Portable EFB may serve as a backup display to any EFB system (other Portable or Installed). - Backup Portable EFB shall be fully configured / programmed. - Backup Portable EFB may be used simultaneously with the primary portable EFB provided it is secured in accordance with B below.
NON-CRITICAL PHASES OF FLIGHT	Portable EFB may be passed to the PF for reference.	- Backup Portable EFB can be used simultaneous with primary Portable EFB. - Constant monitoring should be observed so that at least one iPad battery is kept above 50% capacity.

Securing the Portable EFB

A. Non-critical Phases of Flight

During Non-Critical Phases of Flight the Portable EFB shall be:

- In the designated securing area; or
- Secured to an approved viewable stowage device; or
- Handheld.

B. Critical Phases of Flight

During Critical Phases of Flight, the Portable EFB shall be:

- In the designated securing area; or
- Secured to an approved viewable stowage device.

If there is any concern that the Portable EFB interferes with the flight controls, it should be secured in the designated securing area next to the PM. Note that charging is not allowed during critical phases of flight, namely, to avoid interference of the cables with flight controls.

If secured in the designated securing area and momentarily required during the critical phase of flight, it shall be used briefly and not be handheld for the entire duration of the procedure.

Specific procedures and how to use the approved viewable stowage devices are documented in the Portable EFB User Guide, published in the company document management system.

Designated Securing Areas

CAUTION

To avoid solar exposition, the EFB shall not be placed on the glare shield at any time.

Figure 1: CE-560XLS Securing Area



Figure 2: DA-2EASy Securing Area



Figure 3: BD-700-1A10 and BD-700-1A11 Securing Area



Figure 4: EMB-505 Securing Area



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Figure 5: BD-100-1A10 Securing Area



Figure 6: CE-680A Securing Area

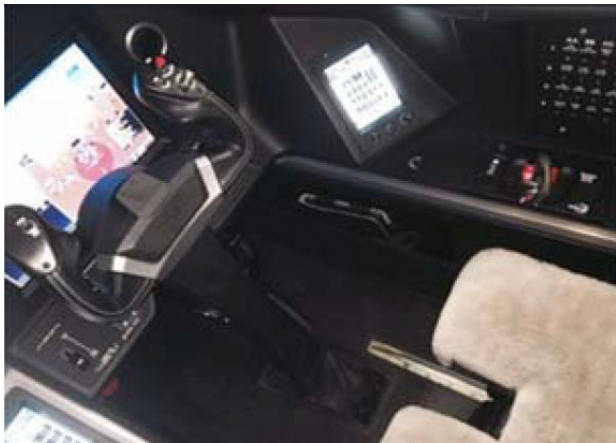


Figure 7: CL-600-2B16 Securing Area



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Normal Operating Procedures**Table 8.18.3.2.1.B: Portable EFB - Normal Operating Procedures, per Phase of Flight**

PHASE OF FLIGHT	NORMAL PROCEDURES
Critical Phases of Flight	• Portable EFBs shall be secured using one of the approved methods.

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PHASE OF FLIGHT	NORMAL PROCEDURES
Pre-Flight	• Battery Status: Ensure both Portable EFBs have a battery status of at least 50% prior to aeroplane movement. If not, charging of the Portable EFBs to 50% is required. • EFB Databases / Contents: Ensure all EFB databases are current and Electronic Charts areas of intended operation are covered. If the data base is not current, the crew shall update it before the flight. • Check Comply365 for any documents relevant to the flight e.g. aerodrome briefings. • Configuration: verify Portable EFB configurations and setting: • Battery Status.....Check 50% or more • Date / Time.....CORRECT • Airplane Mode.....ON • Wi-Fi..... ON (if onboard Wi-Fi is available) • Bluetooth..... ON (if required by other procedures) • Cellular Data.....Ensure OFF • Non-EFB apps.....CLOSED • EFB databases / contents.....CURRENT (in-flight software or database updates are prohibited) • Alarms and Timers..... ALL OFF • FD-Pro Sleep Mode..... OFF (in app settings) • Electronic charts: Ensure that the electronic charts are updated and have the appropriate flight plan route, departure, destination, and alternate airports charts programmed as appropriate. • Electronic Flight Folder: Before accepting the flight documentation as valid for use, the flight crew shall verify that Skybook is synchronised and that the flight package displayed is the latest released version by cross-checking the OFP/Release number displayed in Skybook against the OFP number shown in the Waypoint application for the same sector. Verification steps: 1. Open the sector in Skybook and identify the OFP / Release number. 2. Open the same sector in Waypoint and identify the OFP number. 3. Confirm the identifiers match.

PHASE OF FLIGHT	NORMAL PROCEDURES
	Mismatch / uncertainty (mandatory action): If the OFP / Release numbers do not match, or if the latest package cannot be retrieved, the crew shall not use the Skybook package for operational decision-making and shall contact Flight Dispatch to confirm the latest release and reissue / re-sync as required before departure.
Prior Taxi	• Secure devices: All Portable EFBs and accessories shall be secured. • Power: Ensure FCC / cabin power outlet is available for Portable EFB devices when required to power or charge Portable EFB devices during the flight. • Display: The departure airport diagram shall be continuously displayed on the Primary EFB screen.
Take-off	• The appropriate departure procedure shall be continuously displayed on the Primary EFB screen.
Terminal Area / Departure Procedure	• The appropriate departure procedure shall be continuously displayed on the Primary EFB screen until no longer required. • When the procedure is complete, the chart is no longer required; the Portable EFB may be placed in Sleep Mode and secured until required for use. • Review en-route charts as required
En-route	• Review destination charts as required. • When not in use, ensure that Portable EFBs are in Sleep Mode and secured. • Portable EFBs may be charged, if monitored. • Flight progress (time, fuel quantities remaining, and flight levels) must be recorded in the Skybook Nav Log. ATIS and ATC clearances may be recorded in either Skybook or on the OFP Annex, provided one source contains all the entries.
Approach and Landing	• The Primary EFB screen shall continuously display: • The appropriate Instrument Approach Procedure (IAP); • The appropriate Visual Approach Procedure (VAP) if no IAP is available; • The Airport Diagram (if no IAP or VAP available).
Shutdown	• Place the Portable EFBs in Sleep Mode and secure the devices. • Remove the Portable EFBs from the aeroplane if the devices will be exposed to conditions outside of the 0°C to 35°C range.

PHASE OF FLIGHT	NORMAL PROCEDURES
Through Flight	• The Portable EFB shall have the appropriate flight plan route, departure, destination, and alternate airports charts programmed as appropriate. • Ensure that the Portable EFBs batteries are at least 50%. • Charge and monitor the Portable EFBs batteries as required. • Place the Portable EFBs in Sleep Mode and secure until required for use. • The Backup Portable EFB shall be operational and immediately available for viewing. • If ATC routing changes are near a restricted area, assess their impact in the portable EFB enroute charts by introducing the new flight route, before activating any change to the FMS.
Post Flight / End of Duty	• Report any discrepancies with the Portable EFB hardware or software to NTA IT Helpdesk. • At the end of the duty day, remove the device and accessories and maintain custody • Ensure the Portable EFBs batteries are fully charged prior to the next duty period. After completion of each sector, the crew shall ensure operational flight documentation is submitted and retained electronically via Skybook. This is achieved by: 1. Completing all required post-flight entries and performing Sector Sign-Off; and 2. Ensuring the sector is successfully transmitted to the Skybook server (if a sign-off is performed when Skybook is offline, the crew shall transmit once connectivity is available)

8.18.3.2.2 Cabin Crew Portable EFB

Securing the Portable EFB

A. Non-critical Phases of Flight

During Non-critical Phases of Flight the Portable EFB shall be:

- In the designated securing area; or
- Hand-held.

B. Critical Phases of Flight

During Critical Phases of Flight the Portable EFB shall be:

- In the designated securing area.

Designated Securing Areas**Figure 1: DA-2EASy Securing Area****Figure 2: BD-100-1A10 Securing Area****Figure 3: BD-700-1A10 and
BD-700-1A11 Securing Area****Figure 4: CL-600-2B16 Securing Area**

Table 8.18.3.2.2.A: Portable EFB - Normal Operating Procedures, per Phase of Flight

PHASE OF FLIGHT	NORMAL PROCEDURES
Critical Phases of Flight	Portable EFBs shall be secured using one of the approved methods.
Pre-Flight	Battery Status: Ensure that the Portable EFB has a battery status of at least 50% prior to aeroplane movement. If not, charging of the EFB to 50% is required. EFB Databases / Contents: Ensure all that the EFB databases are current. If the database is not current, the crew shall update it before the flight. Configuration: verify Portable EFB configurations and setting: - Battery Status.....Check 50% or more - Date / Time.....CORRECT - Airplane Mode.....ON - Wi-Fi..... ON (if onboard Wi-Fi is available) - Bluetooth..... ON (if required by other procedures) - Cellular Data.....Ensure OFF - Non-EFB apps.....CLOSED - EFB databases / contents.....CURRENT (in-flight software or database updates are prohibited) - Alarms and Timers..... ALL OFF
Prior Taxi	Secure devices: The Portable EFB and accessories shall be secured.
En-route	- When not in use, ensure that Portable EFBs are in Sleep Mode and secured. - Ensure that the Portable EFBs batteries are at least 50%. - Ensure cabin power outlet is available for Portable EFB devices. Portable EFBs may be charged, if monitored.
Shutdown	- Place the Portable EFB in Sleep Mode and secure the device. - Remove the Portable EFB from the aeroplane if the devices will be exposed to conditions outside of the 0°C to 35°C range.

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PHASE OF FLIGHT	NORMAL PROCEDURES
Post Flight / End of Duty	• Report any discrepancies with the Portable EFB hardware or software to NTA IT Helpdesk. • At the end of the duty day, remove the device and accessories and maintain custody • Ensure the Portable EFBs batteries are fully charged prior to the next duty period.

8.18.4 Contingency or Non-normal Procedure

If the installed EFB is inoperable, the crew shall follow MEL dispatch procedures.

If a Portable EFB is inoperable (e.g. database is out of date), the fleet approved MEL is not applicable. Crew shall follow the required contingency procedures and report this incident as soon as possible to Lis-EFB-INOP@netjets.com.

In order to identify if an EFB is operable (Installed or Portable), an EFB Operational Check shall be completed by the crew, refer to [EFB Operational Check](#).

Any combination two approved and operable EFB units (Portable or Installed) with all the required for flight and current electronic charts and documents is approved for flight dispatch. If required, crew may (in order of priority):

- revert to utilising other combinations of two EFBs, as appropriate;
- activate the spare Portable EFB (if available);
- convert the Cabin Crew assigned Portable EFB (if available) to Flight Crew Portable EFB.

An additional Portable EFB (spare or Cabin Crew Portable EFB) may be assigned to a specific flight crew member to be used only under contingency procedures. Prior to the use of the spare Portable EFB or prior to converting the Cabin Crew Portable EFB to a Flight Crew Portable EFB, the flight crew shall activate the unit by:

1. notifying the EFB Manager;
2. updating all database content, and;
3. perform an EFB Operational Check (refer to [EFB Operational Check](#)).

While not activated, the spare Portable EFB shall be stored safely in the aeroplane.

In the absence of any combination of two approved and operable EFB units (Portable or Installed) with all the required for flight and current electronic charts and documents, flight crew shall revert to the EFB - Contingency or Non-normal Procedures.

Table 8.18.4.A: Contingency or Non-Normal Operating Procedures, per Phase of Flight

PHASE OF FLIGHT	ISSUE		DESCRIPTION	PROCEDURES
Pre-flight	OUT OF DATE DATABASE	ELECTRONIC CHARTS	EFB chart contents out of date (and cannot be updated prior to departure).	• Verify currency of required charts (departure, arrival, alternate airports, and en-route) against a current database or with Flight Dispatch. • If required charts are out of date, revert to any other approved EFB system or refer to 'Single EFB' procedures.

PHASE OF FLIGHT	ISSUE		DESCRIPTION	PROCEDURES
Pre-flight	OUT OF DATE DATABASE	ELECTRONIC DOCUMENTS	Electronic documents out of date (and cannot be updated prior to departure).	- Verify currency of required documents (OMs, AFM, Jepp Airway Manual, etc.) against a current database, with 'Aircraft Docs Revision Status' (available in the waypoint application) or with CP. - If required documents are out of date, revert to any other EFB system or refer to 'All EFB devices fail'.
		PERFORMANCE CALCULATOR	Airfield and / or Aircraft database and / or W&B Tail database(s) out of date (and cannot be updated prior to departure)	- AFM, or other OEM document with performance data, in PDF format on at least one portable EFB (refer to electronic documents requirements). - Paper copy with required performance data (printer required): - Download from nje.comply365.net. PC required. Login with your NetJets credentials; - Obtained by e-mail from Flight Dispatch.
		EFF	Skybook is not synchronised (flight package displayed is not the latest released version)	- Re-sync Skybook and performed a cross-check of the OFP/ Release number displayed in Skybook against the OFP number shown in the Waypoint application for the same sector. - If the problem persists, print the flight package in the FBO.

PHASE OF FLIGHT	ISSUE		DESCRIPTION	PROCEDURES
Pre-flight	DEVICE LOCKING UP		Portable EFB locks up and becomes nonresponsive.	- Perform the device reset procedure (Hold the HOME and SLEEP buttons simultaneously until the screen goes black and Apple logo re-appears (approx. 10 sec.). - If unit remains non-responsive, revert to any other EFB system or refer to 'Single EFB' procedures.
	Single EFB Operable (Portable or Installed)	ELECTRONIC CHARTS	Only one EFB system is operating and available with electronic charts.	- Use Jepp Trip Kit to download the required charts and use the approved digital reader to view on another approved portable EFB; OR - Carry on board paper copies of all required charts (as a backup to a single EFB system); these may be obtained from Flight Dispatch; OR - Continue operations using the other operable EFB and the Normal Procedures applicable to that EFB type.
		ELECTRONIC DOCUMENTS	Only one EFB system is operating and available with electronic documents.	- Not applicable. - Normal procedure applies (Refer Table 8.18.1.B).
		EFF	Only one EFB system is operating and available with EFF functionalities.	Print the flight package at the FBO.

PHASE OF FLIGHT	ISSUE		DESCRIPTION	PROCEDURES
Pre-flight	All EFB devices fail	ELECTRONIC CHARTS	All EFBs are inoperative - no charts available.	- Carry on board paper copies of all required charts to be used as primary reference to navigation; these may be obtained from Dispatch. - In the event of a diversion to an airport other than the departure, destination, or alternate, refer to the 'EFB Backup Approach Procedures - Voice Communication checklist' (on the back of the Portable EFB).
		ELECTRONIC DOCUMENTS	All EFBs are inoperative - Electronic documents not available.	Activate the spare portable EFB if available. If not available, then flight dispatch is not permitted.
		EFF	All EFBs are inoperative - EFF functionalities not available.	Print the flight package at the FBO.

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PHASE OF FLIGHT	ISSUE		DESCRIPTION	PROCEDURES
Pre-flight	Single viewable stowage device	ELECTRONIC CHARTS	Only one portable EFB mount is operable or available	- Crew shall inform the EFB Support Team whenever a viewable stowage device is found to be unserviceable. Upon receiving the report, EFB Support will initiate the appropriate replacement plan. - CE-560XLS: Use the backup kneeboard mount, to ensure PF has visibility of chart. - Fleets with non-approved installed EFB: Cross-check the required chart availability and effective date in the installed EFB against the electronic charts from the authorised portable EFB. Use installed EFB to ensure PF visibility of chart. - Fleets with approved installed EFB: Use installed EFB as per normal operations.
In-flight	SINGLE EFB OPERABLE (Portable or Installed)	GENERAL	An EFB system fails for any reason in flight.	- Revert to the use of any remaining EFB system (Portable or Installed). - Continue operations using the other operable EFB and the Normal Procedures applicable to that EFB type. - When the flight is completed, notify EFB failure (a maintenance action may be required).
		EFF		Continue operations using the other operable portable EFB

PHASE OF FLIGHT	ISSUE		DESCRIPTION	PROCEDURES
In-flight	ALL EFB DEVICES FAIL	GENERAL	All EFBs are inoperative.	- Refer to the 'EFB Backup Approach Procedures - Voice Communication Checklist' and copy the appropriate information. - If the aeroplane is capable of receiving a fax and time permits, call Flight Dispatch to fax the required charts (crew can use AFIS or Flight Phone to contact the HOTLINE for any required electronic documentation information). - Take-off may not be initiated when all EFB devices fail unless paper copies of all required charts are carried on board.
		EFF		Take all notes in the OFP Annex. Contact Flight Dispatch post-flight to record non-submitted information.
	BATTERY LOW INDICATION		A Portable EFB indicates a low battery status.	Charge the battery from an on-board power outlet.
	BATTERY FAILURE		A Portable EFB battery fails.	Revert to any other EFB system or refer to 'Single EFB' procedures.

PHASE OF FLIGHT	ISSUE		DESCRIPTION	PROCEDURES
In-flight	Single viewable stowage device	ELECTRONIC CHARTS	Only one portable EFB mount is operable or available	- CE-560XLS: Use the backup kneeboard mount, to ensure PF has visibility of chart. - Fleets with non-approved installed EFB: Cross-check the required charts availability and effective date in the installed EFB against the electronic charts from the authorised portable EFB. Use the installed EFB to ensure PF visibility of the chart. - Fleets with approved installed EFB: Use installed EFB as per normal operations.
	SUSPECTED WI-FI or BLUETOOTH (TPED) INTERFERENCE		In case any interference is observed.	Disable Wi-Fi, Bluetooth and cellular data in all TPEDs.

Contingency Checklist (All EFB Devices Fail)

Crew shall refer to the 'EFB Backup Approach Procedures - Voice Communication Checklist' and contact the HOTLINE in the event that all EFBs are inoperative in accordance with the above non-normal or contingency procedures. Where needed, the appropriate information shall be requested from the HOTLINE or ATC.

Table 8.18.4.B: EFB Backup Approach Procedures - Voice Communication Checklist

DESTINATION AIRPORT
STAR Name
STAR Speed and Crossing Restrictions
Active Runway
Runway Length and Width
Approach Procedure
ATIS Frequency
Weather Required for Approach Category
MSA
Approach Control Frequency
Tower Frequency

DESTINATION AIRPORT
Ground Frequency
Navigation Frequencies for Approach
Final Approach Course
FAF / GSI Altitude
Step Down Altitude / Distance
DA / MDA
Missed Approach Point
Field Elevation
TDZ Elevation
Missed Approach Procedure

Contingency or Non-normal Use Reporting

Any contingent or non-normal use of any Portable EFB device or applications, or any other type of unusual or unexplainable event, shall be reported as soon as possible to Lis-EFBSupport@netjets.com.

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8.18.5 Unauthorised Use

The following procedures are prohibited:

- Change the NetJets' Portable EFB configuration.
- Install non-approved applications.
- Update approved applications / operating system to non-approved versions unless directed by the EFB Manager or IT department (this is not applicable to database, contents, or document updates). Updating Portable EFB database / contents (charts or documents) during flight.
- Attempting to disable the restrictions, jailbreak.
- Using unauthorised peripheral devices, including but not limited to GPS receivers.
- Use, or have open in the background, non-EFB applications during flight operations.

8.18.6 Updating of EFB Database and Contents

Each crew member is responsible for keeping the assigned Portable EFB databases and content up-to-date in accordance with published company procedures.

Each flight crew member shall ensure that the installed EFBs contain current databases prior to use.

If the databases are expired and cannot be updated before flight, reference the Contingency or Non-normal Procedures.

- Data Revision Mechanisms - The navigation data supplier is responsible to timely distribute new data revisions.
- Approval Workflow - The data update does not require previous approval. Each crew member should update the data as soon as possible after receiving the notification that a new cycle is available for update.
- Data Loading - The responsibility of loading data into the EFBs are:
 - Portable EFBs - crew members;
 - Installed EFBs - Flight crew members and / or Service Centres depending on specific aeroplane type and procedures).

NOTE

Updating of EFB databases by crew shall only be accomplished after the end of a flying duty and only when there is sufficient time before the next planned flight so as not to impact it should there be an issue.

Flight crew shall advise Scheduling if extra duty time is required to accomplish the required updates.

After performing a database update, each EFB shall be checked by the crew member to ensure proper device operation (refer to [EFB Operational Check](#)).

8.18.6.1 Installed EFB

This chapter is only applicable to fleet with approved Installed EFBs.

Installed EFB electronic chart databases are updated by Maintenance or by a crew member. Each crew member shall ensure the electronic database is current and follows the appropriate Maintenance procedures if the database is out of date and unable to update ([NJMP 1.36](#)).

Table 8.18.6.1: Installed EFB Contents / Database Update

Database	Contents	Current From Issue Date / Until...	Update DB / Content When...	Required Redundancy
Jeppesen	Terminal Charts	14 days after Terminal Chart Effective Date	Directed by Company	Current chart information for the flight shall be available in Installed EFB or (two operating Portable EFBs) or Paper copy.
	Airway Manuals	Not applicable		
	En-route charts	28 days after Effective Date (@23:59Z)		
Electronic Documents	N/A	N/A	N/A	N/A

A. Electronic Charts**Content / Database Currency**

Check Prior to each flight, the crew shall check the database revision of Electronics Charts in accordance with the OM-B.

If the database is not current, the crew shall update the Electronic Charts before the next flight in accordance with the OM-B.

Content / Database Revisions Notification

Electronic Charts are available from the navigation data provider.

NetJets Engineering department performs the electronic charts database download (using JSUM), checks the electronic charts database for integrity and publishes them on NetJets secure FTP server (sftp.netjets.com) and company portal. Crew can also download the electronic charts directly with JSUM.

The fleet Chief Pilot is notified by the Engineering department of the electronic charts publishing and will coordinate with the crews the electronic charts update.

Content / Database Update (Data Loading)

The electronic charts update may be performed by a service centre or NetJets Field Service Representative if the aeroplane is in maintenance or, for applicable fleet, by a crew member upon notification from the Chief Pilot. Data load instructions are defined in the respective OM-B.

After the update of the contents, each EFB shall be checked by the crew member using the EFB operational check.

Content / Database Update Reporting (after Update)

After the update is performed, the flight crew, service centre, or NetJets FSR shall report the update to NetJets for tracking purposes, using the respective MCC email address (e.g. Lis-MxCitation@netjets.com), containing the following information:

- A/C Tail Number / Registration;
- Action Performed; (Echarts Update);
- Date Performed;
- Updated by;
- Database Revision number;
- BEGIN Date (from DATABASE EFFECTIVITY MFD PAGE);
- END Date (from DATABASE EFFECTIVITY MFD PAGE).

B. Electronic Documents

Not applicable.

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8.18.6.2 Portable EFB

Portable EFB electronic contents / databases are updated by a trained crew member. Each crew member shall ensure all contents on the Portable EFB are current and follow the appropriate operating procedures if the database is out of date and unable to update.

Table 8.18.6.2: Portable EFB Contents / Database Update

EFB Application	Database	Currency		Update database How and When...
		From...	Until...	
FD-Pro X	Terminal Charts	Effective Date	14 days after Effective Date (@23:59Z)	How... - Databases are updated with app 'Updates' function. When... - Notified by application (not to exceed currency due date).
	Airway Manuals	Effective Date	14 days after Effective Date (@23:59Z)	
	En-route charts	Effective Date	28 days after Effective Date (@23:59Z)	
	Airport Moving Map (*)	Effective Date	28 days after Effective Date (@23:59Z)	
	IFW	On-demand	The date / time is displayed in the weather layer menu.	On-demand
Comply365	Electronic Documents	N/A		How... - Automatically downloaded once connected to a network and the application is opened. When... - Mandatory synchronisation before every duty day. - Notification after 3 days without synchronising the database. - E-mail alert sent once a mandatory document is published.

EFB Application	Database	Currency		Update database How and When...
		From...	Until...	
iPreFlight Genesis	Airfields	Effective Date (Effective DD MMM YYYY - DD MMM YYYY)	28 days after Effective Date (@23:59Z)	How... - Databases are updated with the app 'Data Updates' function. When... - Notified by application (not to exceed currency due date); - When directed by NetJets; - When assigned to new aeroplane. Green: Updated (Status: On Device) Yellow: Update Available (Download button displayed) Red: Not downloaded (Download button displayed)
	Historical Weather	Effective Date (Effective DD MMM YYYY)	N/A	
	Historical Winds Aloft	Effective Date (Effective DD MMM YYYY)	N/A	
	Nav Data	Effective Date (Effective DD MMM YYYY - DD MMM YYYY)	28 days after Effective Date (@23:59Z)	
	Other Flight Planning Data	Effective Date (Effective DD MMM YYYY - DD MMM YYYY)	N/A	
	Terrain Data	Effective Date (Effective DD MMM YYYY - DD MMM YYYY)	Effective End Date (Effective DD MMM YYYY - DD MMM YYYY)	
	Aircraft (e.g. Challenger 350, Citation XLS, etc.) and fleet tails	Effective version	N/A	

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EFB Application	Database	Currency		Update database How and When...
		From...	Until...	
SkyPath	Observed and forecasted turbulence	On-demand	After two hours, each SkyPath report of turbulence has diminished accuracy.	On-demand
HoTs	Holdover Times Calculator	Date of release	Date of supersession	Each year, before the start of winter, the application is updated with the latest aircraft ground de/anti-icing fluid holdover time tables and related guidance published by Transport Canada. The application's full name will include the applicable winter season for which it is approved. For example, the full name for the 2025-26 winter season will be 'HOTs 2025-26'.
Skybook	Flight Package	N/A	N/A	Verify that Skybook is synchronised and that the flight package displayed is the latest released version by cross-checking the OFP/Release number displayed in Skybook against the OFP number shown in the Waypoint application for the same sector. Verification steps: 1. Open the sector in Skybook and identify the OFP/Release number. 2. Open the same sector in Waypoint and identify the OFP number. 3. Confirm the identifiers match.

(*) Airport Moving Map (AMM) approved use and limitations:

- AMM is a situational awareness tool and is not designed for use as a sole means of navigation on an airport surface.
- Depicted data do not include knowledge of current Notice to Airmen (NOTAM) and, therefore, factors such as closure of runways or taxiways may not be reflected. Flight crew must be cognizant of prevailing NOTAM and Automatic Terminal Information Service (ATIS) data. Similarly, data on newly constructed or altered airfield geography may not necessarily be included in the database.

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- Painted hold lines may not agree with depicted data. Crews must remain behind hold lines when required and must not depend on depicted data for positioning relative to hold lines.
- Jeppesen airport diagram charts (example: 10-9, 20-9, etc.) shall be used as the primary source of information over AMM.
- The AMM database update is not required for dispatch.
- Ownership display in the AMM is not allowed.

Content / Database Currency Check

Each crew member shall ensure that the assigned Portable EFB databases are current prior to use.

If the databases are expired and are unable to be updated, refer to the contingency or non-normal procedures.

Content / Database Revisions Notification

- FD-Pro X: Automatically notifies the crew member (on the application icon) that a new database (new contents) is available for update.
- Comply365: When new content has been published the crew will be notified automatically by a daily e-mail. Once opened, Comply365 application will show an icon alerting for new publication(s).

The app will notify each user if the last database synchronisation is over 3 days and will force a new database synchronisation after 14 days.

- iPreFlight Genesis: When new database content becomes available, the application displays a notification with the title 'Critical Updates Available' when starting.

Content / Database Update (Data Loading)

- FD-Pro X: The Jeppesen electronic charts database on the Portable EFB is updated directly from the application, which uses the device's Wi-Fi capability. Data load instructions are defined in the respective Jeppesen User Guide.
- Comply365: The electronic documents on the Portable EFB are updated directly in the Comply365 application, which uses the device's Wi-Fi capability.
- iPreFlight Genesis: The performance database on the portable EFB is updated directly in the iPreFlight Genesis application, which uses the device's Wi-Fi capability.

Crew members are authorised to use any free public Wi-Fi locations (such as hotels, FBOs, homes, etc.) and mobile data connection available on the iPad EFB, following the guidelines outlined in the Portable EFB User Guide.

After the contents update, crew members shall check each Portable EFB using the 'EFB Operational Check' procedure ([8.18.9.1](#)).

Content / Database Update Reporting (after Update)

No crew action is required (content updates are automatically tracked by the system).

8.18.7 Operating System (OS) and Application Updating**8.18.7.1 Installed EFB**

Software updates on an installed EFB is a Maintenance function only.

8.18.7.2 Portable EFB

The Operating System (OS) and any other software applications shall ONLY be updated after instructions from the EFB Manager or NetJets IT Helpdesk. This is to ensure EFB system integrity and proper functioning of the applications.

Table 8.18.7.2: EFB Software Update

EFB Software	Description	Notification of New Version...	Update Application Version when...
iOS	Operating System	Notified by the system (settings icon)	Directed by Company
FD-Pro X	- Terminal Charts - Airway Manuals - En-route charts - IFW	Receipt of an update notification (EFB Manager or IT)	Directed by Company
Comply365	Document Management System	Receipt of an update notification (EFB Manager or IT)	Directed by Company
iPreFlight Genesis	Performance Calculator	Receipt of an update notification (EFB Manager or IT)	Directed by Company
SkyPath	Observed and forecasted turbulence	Receipt of an update notification (EFB Manager or IT)	Directed by Company
HoTs	Holdover Times Calculator	Every year, a new version will include the applicable winter season for which it is approved.	Directed by Company
Skybook	Electronic Flight Folder	Receipt of an update notification (EFB Manager or IT)	Directed by Company

CAUTION

Even though software update notifications will be displayed on the device, approval for the update SHALL be sent to crews via:

- An email from IT Helpdesk, or;
- An email or an Operational Alert from the EFB Manager.

After an OS or any other application version update, the 'EFB Operational Check' ([EFB Operational Check](#)) shall be accomplished.

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8.18.8 EFB Maintenance Procedures

8.18.8.1 Installed EFB

In case of EFB failure, the crew, service centre, or FSR shall report the failure to Maintenance Control (using the respective address) in accordance with the NetJets approved procedures and NTA OM-B.

Installed EFB will be maintained in accordance with the ANAC approved aircraft maintenance program and Approved Aircraft Maintenance Manual.

8.18.8.2 Portable EFB

8.18.8.2.1 EFB Support

Any Portable EFB issue shall be reported by the crew member to the IT Helpdesk. Portable EFB issues that cannot be resolved at the NetJets IT Helpdesk level shall be forwarded to the EFB Manager for further troubleshooting.

Primary point of contact for Portable EFB support:

NetJets IT - Helpdesk (NJE-Wireless@netjets.com)

In the case of a Portable EFB failure, crew shall perform the following recovery procedures (in this order):

- a. Fully charge the device.
- b. Perform a full device reset (press the 'Home' and 'Sleep' button simultaneously for a few seconds).

If the recovery procedure does not solve the failure / issue, then crew shall contact the NetJets IT Helpdesk (NJE-Wireless@netjets.com).

Portable EFBs replacements shall be shipped by the IT department directly to the crew member, after coordination with the on duty CP. The crew member shall return the failed Portable EFB as directed by NetJets IT.

8.18.8.2.2 Battery Capacity Check for Portable EFBs

A battery capacity check shall be accomplished any time the crew reports the Portable EFB battery is not able to supply power to the device for at least 5 hours when using the device under normal operating procedures. Software (applications) approved by IT will be used to test the device battery health and if below 50% of its original capacity, the device shall be replaced in accordance with the maintenance procedures.

8.18.9 EFB Operational Check

8.18.9.1 Portable EFB

Table 8.18.9.1: Portable EFB Operational Check

Situation	Check(s) required
iPadOS updated;	- Electronic Charts / FD-Pro X operational check; and - Electronic Documents / Comply365 operational check. - Performance calculator / iPreFlight Genesis Operational Check.
- FD-Pro X application updated; - Jeppesen database updated.	Electronic Charts / FD-Pro X operational check.
- Comply365 application updated; - Electronic documents database updated.	Electronic Documents / Comply365 operational check.
- iPreFlight Genesis application updated; - Performance database updated.	Performance calculator / iPreFlight Genesis operational check.

Electronic Charts / FD-Pro X Operational Check

- Ensure the application opens normally.
- Ensure that all databases (en-route, Terminal charts, and Airway Manuals) are current.
- Perform a random lookup of airports in the airport database (consistent with the data coverage).
- Ensure that terminal charts display properly, that en-route map is visible, that it is possible to zoom in / out and select information, and that Jeppesen Manuals can be accessed through FD-Pro X.
- Simultaneous functional check of multiple applications (FD-Pro X, Comply365 and iPreFlight Genesis).

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Electronic documents / Comply365 Operational Check

- Ensure the application opens normally.
- Ensure that database is current.
- Perform a random lookup of a manual, contained within Comply365, ensuring the document displays correctly.
- Simultaneous functional check of multiple applications (FD-Pro X, Comply365 and iPreFlight Genesis).

Performance / iPreFlight Genesis Operational Check

- Ensure the application opens normally.
- Ensure that database is current.
- Perform a random performance calculation.
- Simultaneous functional check of multiple applications (FD-Pro X, Comply365, and iPreFlight Genesis).

If the EFB operational check fails, contact NTA IT Helpdesk as soon as possible and the default unit shall be returned to the office. A new unit will be sent to the crews in accordance with NTA IT procedures.

Crew shall fill the Portable EFB Operational Check form every time it fails one of the above tests.

8.18.9.2 Installed EFB**Electronic Charts Operational Check**

- Select Electronic Charts Function using Avionics Flight Deck control panel;
- Ensure the application opens normally;
- Ensure that all databases (en-route - if available, Terminal charts and Airway Manuals) are current;
- Perform a random lookup of airports in the airport database (consistent with the data coverage);
- Ensure that terminal charts display properly and that it is possible to zoom in / out and select information.

If the EFB operational check fails, contact Maintenance Control for technical assistance.

If required, defer the Installed EFB discrepancy by applicable of respective MEL, follow company established deferral procedures.

Appx1 APPENDIX 1 TO CHAPTER 8 - Reserved**Appx2 APPENDIX 2 TO CHAPTER 8 -Approvals by Aeroplane Type**

Specification	Aeroplane Type							
	EMB-505	CE-560XLS	CE-680A	BD-100-1A10	DA-2EASy	CL-600-2B16	BD-700-1A11	BD-700-1A10
COMMUNICATION								
FANS 1/A CPDLC	NO	NO	NO	YES	NO	YES	YES	YES
PM-CPDLC ATN-B1	YES	NO	YES	YES	NO	YES	YES	YES
NAVIGATION								
A-RNP	NO	NO	NO	NO	NO	NO	NO	NO
Sensors	---	---	---	---	---	---	---	---
RF legs	NO	NO	NO	NO	NO	NO	NO	NO
RNAV1	YES	YES	YES	YES	YES	YES	YES	YES
Sensors	GPS	GPS, DME	GPS, DME	GPS, DME, IRU	GPS, DME, IRU	GPS, DME	GPS, DME, IRU	GPS, DME, IRU
RNP1	NO	NO	NO	YES	YES	YES	YES	YES
Sensors	---	---	---	GPS	GPS	GPS	GPS	GPS
RNAV2	NO	NO	NO	NO	NO	YES	YES	YES
Sensors	---	---	---	---	---	GPS, DME	GPS, DME	GPS, DME
RNP2 Continental	NO	NO	NO	NO	NO	YES	YES	YES
Sensors	---	---	---	---	---	GPS	GPS	GPS
RNP2 Oceanic / Remote	NO	NO	NO	NO	NO	YES	YES	YES
Sensors	---	---	---	---	---	GPS	GPS	GPS
RNP4	NO	NO	NO	YES	NO	YES	YES	YES
Sensors	---	---	---	GPS	---	GPS	GPS	GPS
RNAV5	YES	YES	YES	YES	YES	YES	YES	YES
Sensors	GPS	GPS, DME, VOR	GPS, DME	GPS, DME, IRU, VOR	GPS, DME, IRU, VOR	GPS, DME, IRU, VOR	GPS, DME, IRU, VOR	GPS, DME, IRU, VOR
RNAV10	YES	¹	YES	YES	YES	YES	YES	YES

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Specification	Aeroplane Type							
	EMB-505	CE-560XLS	CE-680A	BD-100-1A10	DA-2EASy	CL-600-2B16	BD-700-1A11	BD-700-1A10
Sensors	GPS	GPS	GPS	GPS, IRU	GPS, IRU	GPS, IRU	GPS, IRU	GPS, IRU
RNP APCH (LNAV)	YES	YES	YES	YES	YES	YES	YES	YES
RNP APCH (LNAV/VNAV)	YES	YES	YES	YES	YES	YES	YES	YES
RNP APCH (LP)	NO	NO	NO	NO	NO	NO	NO	NO
RNP APCH (LPV)	YES	NO	YES	YES	NO	YES	YES	YES
SURVEILLANCE								
ADS-B Out	YES	YES	YES	YES	YES	YES	YES	YES
OTHER								
NCC Operations	YES	YES	YES	YES	YES	YES	YES	YES
Non-ETOPS 180	NO	NO	NO	NO	NO	YES	YES	YES
PBCS	NO	NO	NO	NO	NO	YES	NO	NO
Steep Approach	YES	YES	YES	YES	NO	YES	YES	YES

¹ CS-DXN, CS-DXO, CS-DXP, CS-DXQ, CS-DXR, CS-DXS, CS-DXT, CS-DXU, CS-DXV, CS-DXZ, CS-DQA, and CS-DQB (aircraft with dual FMS/GPS).

NOTE

Where an aeroplane registration is indicated, only that aeroplane is approved.

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Chapter 9 - Dangerous Goods and Weapons

9.1 Policy on the Transport of Dangerous Goods

(CAT.GEN.MPA.200, SPA.DG.105)

9.1.1 Approval for the Transport of Dangerous Goods

Dangerous goods can only be carried according to the International Civil Aviation Organization's Technical Instructions for the Safe Transport of Dangerous Goods by Air (Technical Instructions), irrespective of whether the flight is wholly or partly within or wholly outside the territory of a State. An approval must be granted by the Competent Authority before dangerous goods can be carried on an aeroplane, except as identified in [Items That May Be Carried by Passengers and Crew](#).

NTA does not hold an EASA approval for the transport of dangerous goods by air.

9.1.2 Approval of the Operator

The acceptance and verification of DGR, when required, shall be addressed to the designated team responsible for the acceptance of dangerous goods using the following email address Lis-DangerousGoods@ntasa.pt.

For urgent queries requiring immediate attention prior to the flight, Flight Crew Members shall contact Dispatch HOTLINE +351 21 446 9090.

9.1.3 General Exceptions

9.1.3.1 Airworthiness and Operational Items

(CAT.GEN.MPA.200 (b)(1))

An approval is not required for dangerous goods which are required to be aboard the aeroplane such as:

- a. items for airworthiness or operating reasons or for the health of passengers or crew, such as batteries, fire extinguishers, first-aid kits, insecticides, air fresheners, life rafts, escape slides, life-saving appliances, portable oxygen supplies, tritium signs, smoke hoods, passenger service units;
- b. catering or cabin service supplies such as alcoholic beverages, perfumes, colognes, liquefied gas lighters, and portable electronic devices containing lithium metal or lithium ion cells, or batteries provided they meet the provisions in NOTE 1, carried aboard an aeroplane by the operator for use on the aeroplane during the flight or series of flights, but excluding non-refillable gas lighters and those lighters liable to leak when exposed to reduced pressure; and
- c. dry ice intended for use in food and beverage service aboard the aeroplane; and
- d. alcohol-based hand sanitisers and alcohol-based cleaning products carried aboard an aircraft by the operator for use on the aircraft during the flight or series of flights for the purposes of passenger and crew hygiene; and
- e. electronic devices such as electronic flight bags, personal entertainment devices, containing lithium metal or lithium ion cells or batteries and spare lithium batteries for such devices or power banks carried aboard an aeroplane by the operator for use on the aeroplane during the flight or series of flights, provided they meet the provisions in NOTE 1. Conditions for the carriage and use of these electronic devices and for the carriage of

spare batteries (including power banks) must be provided in the operations manual and / or other appropriate manuals as will enable flight crew, cabin crew and other employees to carry out the functions for which they are responsible.

NOTE

1. The following conditions must be met for lithium cells or batteries (including power banks) and the devices they power referred to in para. B) and e):
 - a. spare lithium batteries and power banks must be individually protected so as to prevent short circuits when not in use;
 - b. measures must be taken to prevent unintentional activation of the portable electronic devices; and
 - c. the batteries must:
 - i. be of a type which meets the requirements of each test in the UN Manual of Tests and Criteria, Part III, subsection 38.3; and
 - ii. for lithium metal batteries, not exceed a lithium content of 2 g and for lithium ion batteries, not exceed a Watt-hour rating of 100 Wh.
2. Dangerous goods intended as replacements for those referred to in a, b, and d above may not be carried without the approval referred to in [Approval for the Transport of Dangerous Goods](#) and unless consigned and accepted for transport in accordance with the ICAO Technical Instructions.

9.1.3.2 Veterinary Aid

(CAT.GEN.MPA.200 (b)(1))

An approval is not required for dangerous goods which are carried for use in flight as veterinary aid or as a humane killer for an animal. Such dangerous goods must be stowed and secured during take-off and landing and at all other times when deemed necessary by the pilot-in-command. The dangerous goods must be under the control of trained personnel during the time when they are in use on the aeroplane.

Dangerous goods may be carried on a flight made by the same aeroplane before or after a flight for which they are required as veterinary aid or as a humane killer for an animal, when it is impracticable to load or unload the dangerous goods immediately before or after the flight, subject to the following conditions:

- a. the dangerous goods must be capable of withstanding the normal conditions of air transport;
- b. the dangerous goods must be appropriately identified (e.g. by marking or labelling);
- c. the dangerous goods may only be carried with the approval of the operator;
- d. the dangerous goods must be inspected for damage or leakage prior to loading;
- e. loading must be supervised by the operator;
- f. the dangerous goods must be stowed and secured in the aeroplane in a manner that will prevent any movement in flight which would change their orientation;
- g. the pilot-in-command must be notified of the dangerous goods loaded on board the aeroplane and their loading location. In the event of a crew change, this information must be passed to the next crew;

- h. all personnel must be trained commensurate with their responsibilities; and
- i. the provisions of [Dangerous Goods Accident and Incident Reports](#) apply.

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9.1.3.3 Medical Aid for a Patient

(CAT.GEN.MPA.200 (b)(1))

An approval is not required for dangerous goods which are placed on board an aeroplane with the approval of the operator. These dangerous goods may also be carried on a flight made by the same aeroplane to collect a patient or after that patient has been delivered, when it is impracticable to load or unload the goods at the time of the flight on which the patient is carried.

9.1.3.4 Excess Baggage Being Sent as Cargo

An approval is not required for dangerous goods contained within items of excess baggage being sent as cargo provided that:

- i. the excess baggage has been consigned as cargo by or on behalf of a passenger;
- ii. the dangerous goods may only be those that are permitted by and in accordance with [Items That May Be Carried by Passengers and Crew](#) to be carried in checked baggage; and
- iii. the excess baggage is marked with the words 'Excess baggage consigned as cargo'.

With the aim of preventing dangerous goods, which a passenger is not permitted to have, from being taken aboard an aeroplane in excess baggage consigned as cargo, any organisation or enterprise accepting excess baggage consigned as cargo should seek confirmation from the passenger, or a person acting on behalf of the passenger, that the excess baggage does not contain dangerous goods that are not permitted and seek further confirmation about the contents of any item where there are suspicions that it may contain dangerous goods that are not permitted.

9.1.4 Reserved

9.1.5 Items That May Be Carried by Passengers and Crew

(CAT.GEN.MPA.200 (b)(2))

Passengers and crew are forbidden to carry dangerous goods either as or in carry-on baggage, checked baggage, or on their person unless the dangerous goods are:

- a. permitted in accordance with the table on chapter [9.2](#) , and
- b. carried by passengers and crew for personal use only.

The entry in the table that most appropriately describes the item or article must be selected.

NOTE

For instance, electronic cigarettes must meet the requirements of the entry for 'Battery-powered portable electronic smoking devices' not the entry for lithium batteries or non-spillable batteries.

An item or article that contains multiple dangerous goods must meet all applicable entries.

NOTE

For instance, the restrictions and conditions for entries 1) and 14) of the table apply to an avalanche backpack that contains lithium batteries and gas cartridges.

Active devices must meet defined standards for electromagnetic radiation to ensure that the operation of the devices does not interfere with aeroplane systems.

Baggage intended to be carried in the cabin that is placed in the cargo compartment must only contain dangerous goods permitted in checked baggage. When baggage intended as carry-on is taken by the operator and placed into the cargo compartment for carriage, the operator must confirm with the passenger that dangerous goods which are only permitted in carry-on baggage have been removed.

Information on the types of dangerous goods which a passenger is forbidden to carry aboard an aircraft is provided in text in the 'Customer Copy' provided by Customer Services upon booking of a flight operated by NetJets flight.

Active devices must meet defined standards for electromagnetic radiation to ensure that the operation of the devices does not interfere with aircraft systems.

NOTE

1. The following dangerous goods may be commonly carried by passengers on other modes of transport, however, they are prohibited either as or in carry-on baggage or checked baggage:

- personal medical oxygen devices that utilise liquid oxygen;
- Christmas / holiday crackers, containing explosives;
- electroshock weapons (e.g. tasers) containing dangerous goods such as explosives, compressed gases, lithium batteries, etc.;
- 'strike anywhere' matches;
- lighter fuel and lighter refills;
- premixing burner lighter (Gas lighter in which fuel and air are mixed before being supplied for combustion, such as lighters producing a blue flame) without a means of protection against unintentional activation; and
- battery-powered lighters powered by a lithium ion or lithium metal battery (e.g. laser plasma lighters, tesla coil lighters, flux lighters, arc lighters and double arc lighters) without a safety cap or means of protection against unintentional activation.

2. Exceptions found in these Instructions are not reproduced in [9.2](#). The following dangerous goods are not subject to these Instructions:

- Radio-pharmaceuticals contained within the body of a person as the result of medical treatment; and
- Energy efficient lamps when in retail packaging and intended for personal or home use.

3. States may implement additional restrictions in the interests of aviation security.

9.1.5.1 Loading of Battery Powered Mobility Aids

General requirements:

Battery-powered mobility aids are vehicles that match the description of item 4(a) in [Provisions for dangerous goods carried by passengers or crew](#). Other battery-powered vehicles aimed for leisure, like bicycles, scooters (non-mobility), or hoverboards, are not considered mobility aids and their batteries shall be treated in accordance with items 1 or 2 of the table in [Provisions for dangerous goods carried by passengers or crew](#).

To prevent accidental activation during flight, the operator is required to verify the power is disconnected and electrical circuits are isolated. To verify that electrical circuits are isolated (inactive), place the device into drive mode (i.e. not freewheel mode), see if the mobility aid will power up and if so, whether use of the joystick results in the mobility aid moving. It must also be verified that the circuits of supplemental motorised systems such as seating systems have been inhibited to prevent inadvertent operation, e.g. by the separation of cable connectors. If an electric mobility aid has not been made safe for carriage, it must not be loaded.

9.1.5.1.1 Loading of Mobility Aids Powered by Non-spillable Wet Batteries or Batteries which Comply with Special Provision A123 or A199

The operator must secure, by use of straps, tie-downs or other restraint devices, a battery-powered mobility aid with installed battery(ies). The mobility aid, the battery(ies), electrical cabling, and controls must be protected from damage including by the movement of baggage, mail, or cargo.

The operator must verify that:

- a. the passenger has confirmed that the battery(ies) is:
 1. a non-spillable wet battery that complies with Special Provision 67.
 2. a dry battery that complies with Special Provision A123; or
 3. a nickel-metal hydride battery that complies with Special Provision A199.
- b. the battery terminals are protected from short circuits (e.g., by being enclosed within a battery container).
- c. the battery is either:
 1. adequately protected against damage by the design of the mobility aid and securely attached to the mobility aid. The electrical circuits must be isolated following the manufacturer's instructions; or
 2. removed from mobility aid, following the manufacturer's instructions;
- d. a maximum of **one** non-spillable wet spare battery is carried per passenger.

The operator must ensure that any battery(ies) removed from the mobility aids and any spare battery are carried in strong, rigid packagings, protected from short circuits, and stowed in the cargo compartment.

The operator must inform the pilot-in-command of the location of any mobility aids with installed battery(ies), removed battery(ies) and spare battery(ies).

9.1.5.1.2 Loading of Mobility Aids Powered by Spillable Batteries

The operator must secure, by use of straps, tie-downs or other restraint devices, a battery-powered mobility aid with installed battery(ies). The mobility aid, the battery(ies), electrical cabling, and controls must be protected from damage including by the movement of baggage, mail, or cargo.

The operator must verify that:

- a. the battery terminals are protected from short circuits (e.g., by being enclosed within a battery container).
- b. the battery is fitted, where feasible, with spill resistant-vent caps; and
- c. the battery is either:
 1. adequately protected against damage by design of the mobility aid and securely attached to the mobility aid. The electrical circuits are isolated following the manufacturer's instructions; or
 2. removed from the mobility aid following the manufacturer's instructions.

If required by the manufacture's instructions, the operator must load, stow, secure, and unload a spillable battery-powered mobility aid in an upright position. If the mobility aid cannot be loaded, stowed, secured, and unloaded always in an upright position or if the mobility aid does not protect the battery, the operator must remove the batteries and carry them in strong, rigid packaging's, as follows:

- a. packaging must be leak-tight, impervious to battery fluid and be protected against being overturned by securing them to pallets or by securing them in cargo compartments using appropriate means of securement.
- b. batteries must be protected against short circuits, secured upright in these packaging and surrounded by compatible absorbent material sufficient to absorb its total liquid contents; and
- c. these packaging must be marked 'Battery, wet, with wheelchair' or 'Battery, wet, with mobility aid' and be labelled with a 'Corrosive' label and with package orientation labels as required by Part 5,3 of the Technical Instructions.

The operator must inform the pilot-in-command of the location of any mobility aids with installed spillable batteries and removed batteries.

9.1.5.1.3 Loading of Mobility Aids Powered by Lithium Ion Batteries

The operator must secure, by use of straps, tie-downs or other restraint devices, a battery-powered mobility aid with installed battery(ies). The mobility aid, the battery(ies), electrical cabling, and controls must be protected from damage including by the movement of baggage, mail, or cargo.

The operator must verify that:

- a. the battery terminals are protected from short circuits (e.g., by being enclosed within a battery container).
- b. the battery is either:
 1. adequately protected against damage by the design of the mobility aid and securely attached to the mobility aid. The electrical circuits must be isolated following the manufacturer's instructions; or

2. removed from the mobility aid following the manufacturer's instructions; and
- c. each removed battery does not exceed 300 Wh. A maximum of one spare battery not exceeding 300 Wh or two spare batteries each not exceeding 160 Wh may be carried.

The operator must ensure that any battery(ies) removed from the mobility aid and any spare battery(ies) are carried in the cabin and protected from damage (e.g. by placing each battery in a protective pouch) and the battery terminals protected from short circuit (by insulating the terminals, e.g. by taping over exposed terminals).

The operator must inform the pilot-in-command of the location of any mobility aids with installed lithium ion battery(ies), removed battery(ies) and spare battery(ies).

NOTE

To determine watt hours, use:

Volts x Ampere hour (Ah) = watt hours (Wh)

NOTE

When the lithium battery(ies) remain installed in the mobility aid, there is no Watt-hour limit.

9.2 Provisions for Dangerous Goods Carried by Passengers or Crew

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
Batteries				

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Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
1. Lithium batteries (including power banks) and portable electronic devices	Yes (except for h. i. And j.)	Yes	(see c. and d.)	a. each battery must be of a type which meets the requirements of each test in the UN Manual of Tests and Criteria, Part III, subsection 38.3; b. each battery must not exceed the following: • for lithium metal batteries, a lithium content of 2 grams; or • for lithium ion batteries, a Watt-hour rating of 100 Wh; c. each battery may exceed 100 Wh but not exceed 160 Wh Watt-hour rating for lithium ion with the approval of the operator; d. each battery may exceed 2 grams but not exceed 8 grams lithium content for lithium metal for portable medical electronic devices with the approval of the operator. e. no more than two spare batteries meeting the requirements of c. or d. may be carried per person. f. for portable electronic devices containing batteries: • measures must be taken to prevent unintentional activation and to protect the devices from damage; • the devices should be carried as carry-on baggage; however • if carried as checked baggage the devices must be completely switched off (not in sleep or hibernation mode) if the batteries exceed: - for lithium metal batteries, a lithium content of 0.3 grams per device; or - for lithium ion batteries, a Watt-hour rating of 2.7 Wh per device. g. batteries and heating elements must be isolated in portable electronic devices capable of generating extreme heat,

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
				which could cause a fire if activated, by removal of the heating element, battery or other components; h. spare batteries, including power banks: • must be carried as carry-on baggage; and • must be individually protected so as to prevent short circuits (by placement in original retail packaging or by otherwise insulating terminals, e.g. by taping over exposed terminals or placing each battery in a separate plastic bag or protective pouch); i. power banks: • must be carried as carry-on baggage; • must not be recharged while on board the aircraft; • Charging of PEDs must be kept to a minimum, and aircraft power outlets should be used whenever available. However, it is authorised under the conditions i. to iv. set out in the restricted use policy below, designed specifically to control for the risks associated with the use of lithium batteries on board: i. Charging is attended by a responsible person. Unattended charging is not permitted. ii. When in use, the power bank is fully visible at all times (not in a bag, pocket, or covered). iii. Not in use during taxiing, any flight operations below 10.000ft or during abnormal or emergency conditions. (Note that this also applies to using the power outlets) iv. Passengers are briefed on the conditions of this policy. • no more than two power banks may be carried per person; and • must be individually protected so as to prevent short circuits when not in use (by placement in original retail

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
				packaging or by otherwise insulating terminals, such as by taping over exposed terminals or placing each power bank in a separate plastic bag or protective pouch); j. baggage equipped with a lithium battery(ies) exceeding: • for lithium metal batteries, a lithium content of 0.3 grams; or • for lithium ion batteries, a Watt-hour rating of 2.7 Wh. must be carried as carry-on baggage unless the battery(ies) is removed from the baggage, in which case the battery(ies) must be carried in accordance with g);
				NOTE The restrictions in a) and the applicable limits in b), c), d) or e) apply to all batteries under this item, that is those contained in portable electronic devices, spare batteries, power banks and baggage equipped with lithium batteries.

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
2. Non-spillable batteries	Yes	Yes	No	a. for a non-spillable battery: 1. must meet the requirements of Special Provision A67; 2. each battery must not exceed a voltage of 12 volts and a Watt-hour rating of 100 Wh; 3. each battery must be protected from short circuit by the effective insulation of exposed terminals; 4. no more than two spare batteries per person may be carried; and 5. if contained in equipment, the equipment must be either protected from unintentional activation, or each battery must be disconnected and its exposed terminals insulated. b. for a dry battery or nickel-metal hydride battery, each battery must comply with Special Provision A123 or A199, respectively; and c. batteries and heating elements must be isolated in battery powered equipment capable of generating extreme heat, by removal of the heating element, battery, or other components.
3. Battery-powered portable electronic smoking devices (e.g. ecigarettes, ecigs, ecigars, epipes, personal vaporizers, electronic nicotine delivery systems)	No	Yes	No	a. if powered by lithium batteries, each battery must comply with restrictions of 1) a), b) and g); b. the devices and / or batteries must not be recharged on board the aeroplane; and c. measures must be taken to prevent unintentional activation of the heating element while on board the aeroplane.

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
4. Mobility aids (e.g. wheelchairs) powered by: • spillable batteries. • non- spillable wet batteries. • dry batteries. • nickel-metal hydride batteries; or • lithium ion batteries.	Yes	(See e))	Yes	a. for use by passengers whose mobility is restricted by either a disability, their health or age, or a temporary mobility problem (e.g. broken leg); b. the passenger should make advance arrangements with each operator and provide information on the type of battery installed and on the handling of the mobility aid (including instructions on how to isolate the battery); c. in the case of a dry battery or nickel-metal hydride battery, each battery must comply with Special Provision A123 or A199, respectively. d. in the case of a non-spillable wet battery: i. each battery must comply with Special Provision A67; and ii. a maximum of one spare battery may be carried per passenger; e. in the case of a lithium ion battery: i. each battery must be of a type which meets the requirements of each test in the UN Manual of Tests and Criteria, Part III, subsection 38.3; ii. when the mobility aid does not provide adequate protection to the battery: - the battery must be removed in accordance with the manufacturer's instructions; - the battery must not exceed 300 Wh; - the battery terminals must be protected from short circuit (by insulating the terminals, e.g. by taping over exposed terminals); - the battery must be protected from damage (e.g. by placing each battery in a protective pouch); and - the battery must be carried in the cabin;

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
				iii. a maximum of one spare battery not exceeding 300 Wh or two spare batteries not exceeding 160 Wh each may be carried. Spare batteries must be carried in the cabin. NOTE When the lithium battery(ies) remain installed in the mobility aid, there is no Watt-hour limit.
Flames and fuel sources				
5. Cigarette lighter Small packet of safety matches	No	(see b))	No	a. no more than one per person; b. must be carried on the person; c. must not contain unabsorbed liquid fuel (other than liquefied gas); and d. if a cigarette lighter is powered by lithium batteries, each battery must comply with restrictions of 1) a), b), and g) and 3) b), and c).
6. Alcoholic beverages containing more than 24 per cent but not more than 70 per cent alcohol by volume	Yes	Yes	No	a. must be in retail packagings; and b. no more than 5 L total net quantity per person. NOTE Alcoholic beverages containing not more than 24 per cent alcohol by volume are not subject to any restrictions.
7. Internal combustion engines or fuel cell engines	Yes	No	No	Measures must be taken to nullify the hazard. Refer to Special Provision A70 for more information.
8. Fuel cells containing fuel Spare fuel cell cartridges	No Yes	Yes Yes	No No	a. fuel cell cartridges may only contain flammable liquids, corrosive substances, liquefied flammable gas, water reactive substances or hydrogen in metal hydride; b. refuelling of fuel cells on board an aeroplane is not permitted except that the installation of a spare cartridge is allowed;

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
				c. the maximum quantity of fuel in any fuel cell or fuel cell cartridge must not exceed: - for liquids 200 ml; - for solids 200 grams; - for liquefied gases, 120 mL for non-metallic fuel cell cartridges or 200 mL for metal fuel cell or fuel cell cartridges; and - for hydrogen in metal hydride, the fuel cell or fuel cell cartridges must have a water capacity of 120 mL or less; d. each fuel cell and each fuel cell cartridge must conform to IEC 62282-6-100 Ed. 1, including Amendment 1, and must be marked with a manufacturer's certification that it conforms to the specification. In addition, each fuel cell cartridge must be marked with the maximum quantity and type of fuel in the cartridge; e. fuel cell cartridges containing hydrogen in metal hydride must comply with the requirements in Special Provision A162; f. no more than two spare fuel cell cartridges may be carried by a passenger; g. fuel cells containing fuel are permitted in carry-on baggage only; h. interaction between fuel cells and integrated batteries in a device must conform to IEC 62282-6-100 Ed. 1, including Amendment 1. Fuel cells whose sole function is to charge a battery in the device are not permitted; i. fuel cells must be of a type that will not charge batteries when the portable electronic device is not in use and must be durably marked by the manufacturer: 'APPROVED FOR CARRIAGE IN AEROPLANE CABIN ONLY' to so indicate; and

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
				j. in addition to the languages which may be required by the State of Origin for the markings specified above, English should be used.
Gases in cylinders and cartridges				
9. Cylinders of oxygen or air required for medical use	Yes	Yes	Yes	a. no more than 5 kg gross mass per cylinder; b. cylinders, valves and regulators, where fitted, must be protected from damage which could cause inadvertent release of the contents; c. advance arrangements recommended; and d. the pilot-in-command must be informed of the number of oxygen or air cylinders loaded on board the aeroplane and their loading location(s).
10. Cartridges of Division 2.2 worn for the operation of mechanical limbs	Yes	Yes	No	Spare cartridges of a similar size are also allowed, if required, to ensure an adequate supply for the duration of the journey.
11. Cartridge of hydrocarbon gas contained in hair styling equipment	Yes	Yes	Yes	a. no more than one per person; b. the safety cover must be securely fitted over the heating element; and c. spare cartridges must not be carried.
12. Cartridges of Division 2.2 with no subsidiary hazard fitted into a selfinflating personal safety device intended to be worn by a person, such as a life-jacket or vest	Yes	Yes	Yes	a. no more than two personal safety device per person; b. the personal safety device must be packed in such a manner that it cannot be accidentally activated; c. must be for inflation purposes; d. no more than two cartridges are fitted into the device; and e. no more than two spare cartridges.

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
13. Cartridges of Division 2.2 with no subsidiary hazard for other than a self-inflating personal safety device	Yes	Yes	Yes	a. no more than four cartridges per person; and b. the water capacity of each cartridge must not exceed 50 ml. NOTE <i>For carbon dioxide, a gas cartridge with a water capacity of 50 ml is equivalent to a 28 g cartridge.</i>
14. Cartridges and cylinders of Division 2.2 with no subsidiary hazard contained in an avalanche rescue backpack	Yes	Yes	Yes	a. no more than one avalanche rescue backpack per person. b. the backpack must be packed in such a manner that it cannot be accidentally activated. c. may contain a pyrotechnic trigger mechanism which must not contain more than 200 mg net of Division 1.4S; and d. the airbags within the backpack must be fitted with pressure relief valves.
Radioactive material				
15. Radioisotopic cardiac pacemakers or medical devices	n/a (see restrictions)	n/a (see restrictions)	No	Must be implanted into a person or fitted externally as the result of medical treatment.
Mercury				
16. Small medical or clinical thermometer which contains mercury	Yes	No	No	a. no more than one per person; and b. must be in its protective case.
Other dangerous goods				

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
17. Non radioactive medicinal articles (including aerosols), toiletry articles (including aerosols) and aerosols in Division 2.2 with no subsidiary hazard	Yes	Yes	No	a. no more than 0.5 kg or 0.5 L total net quantity per single article; b. no more than 2 kg or 2 L total net quantity of all articles (e.g. four aerosol cans of 0.5 L each) per person; c. release valves on aerosols must be protected by a cap or other suitable means to prevent inadvertent release of the contents; and d. the release of gas must not cause extreme annoyance or discomfort to crew members so as to prevent the correct performance of assigned duties.
18. Dry ice	Yes	Yes	Yes	a. no more than 2.5 kg per person; b. used to pack perishables that are not subject to the Technical Instructions; c. the package must permit the release of carbon dioxide gas; and d. when carried as checked baggage, each package must be marked: 'DRY ICE' or 'CARBON DIOXIDE, SOLID'; and the net weight of dry ice or an indication that the net weight is 2.5 kg or less.
19. Cartridges in Division 1.4S (UN 0012 or UN 0014 only)	Yes	No	Yes	a. no more than 5 kg gross mass per person; b. must be securely packaged; c. must not include ammunition with explosive or incendiary projectiles; and d. allowances for more than one person must not be combined into one or more packages.
20. Permeation devices	Yes	No	Yes	Instructions on how to package permeation devices for calibrating air quality monitoring equipment are found in Special Provision A41.
21. Non-infectious specimens in flammable solutions	Yes	Yes	No	Instructions on how to package and mark specimens are found in Special Provision A180.

Dangerous Goods	Location		Approval of the operator(s) is required	Restrictions
	Checked Baggage	Carryon Baggage		
22. Refrigerated liquid nitrogen	Yes	Yes	No	Must be contained in insulated packagings (e.g. dry shippers) that would not allow the build-up of pressure and be fully absorbed in a porous material so that there is no free liquid that could be released from the packaging. Refer to Special Provision A152 for more information.
23. Dangerous goods incorporated in security-type equipment, such as attaché cases, cash boxes, cash bags, etc.	Yes	No	Yes	The security-type equipment must be equipped with an effective means of preventing accidental activation and the dangerous goods incorporated in the equipment must meet the conditions of Special Provision A178.

NOTE

Table above in accordance with ICAO Doc 9284, 2025-2026 Edition.

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Special Provisions

A41	Permeation devices that contain dangerous goods and that are used for calibrating air quality monitoring devices are not subject to these Instructions when carried as cargo provided the following requirements are met: a. Each device must be constructed of a material compatible with the dangerous goods it contains; b. The total contents of dangerous goods in each device is limited to 2 millilitres and the device must not be liquid full at 55°C; c. Each permeation device must be placed in a sealed, high impact-resistant, tubular inner packaging of plastic or equivalent material. Sufficient absorbent material must be contained in the inner packaging to completely absorb the contents of the device. The closure of the inner packaging must be securely held in place with wire, tape or other positive means; d. Each inner packaging must be contained in a secondary packaging constructed of metal, or plastic having a minimum thickness of 1.5 mm. The secondary packaging must be hermetically sealed; e. The secondary packaging must be securely packed in strong outer packaging. The completed package must be capable of withstanding, without breakage or leakage of any inner packaging and without significant reduction in effectiveness: i. the following free drops onto a rigid, non-resilient, flat and horizontal surface from a height of 1.8 m: - one drop flat on the bottom; - one drop flat on the top; - one drop flat on the long side; - one drop flat on the short side; - one drop on a corner at the junction of three intersecting edges; and ii. a force applied to the top surface for a duration of 24 hours, equivalent to the total weight of identical packages if stacked to a height of 3 m (including the test sample). NOTE Each of the above tests may be performed on different but identical packages. f. The gross mass of the completed package must not exceed 30 kg.
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A67	Batteries can be considered as non-spillable provided that they are capable of withstanding the vibration and pressure differential tests given below, without leakage of battery fluid. <i>Vibration test:</i> The battery is rigidly clamped to the platform of a vibration machine and a simple harmonic motion having an amplitude of 0.8 mm (1.6 mm maximum total excursion) is applied. The frequency is varied at the rate of 1 Hz/min between the limits of 10 Hz to 55 Hz. The entire range of frequencies and return is traversed in 95 ± 5 minutes for each mounting position (direction of vibration) of the battery. The battery must be tested in three mutually perpendicular positions (to include testing with fill openings and vents, if any, in an inverted position) for equal time periods. <i>Pressure differential test:</i> Following the vibration test, the battery is stored for six hours at $24^{\circ}\text{C} \pm 4^{\circ}\text{C}$ while subjected to a pressure differential of at least 88 kPa. The battery must be tested in three mutually perpendicular positions (to include testing with fill openings and vents, if any, in an inverted position) for at least six hours in each position. NOTE Non-spillable type batteries which are an integral part of, and necessary for the operation of, mechanical or electronic equipment must be securely fastened in the battery holder on the equipment and protected in such a manner so as to prevent damage and short circuits. Non-spillable batteries are not subject to these Instructions when carried as cargo if, at a temperature of 55°C, the electrolyte will not flow from a ruptured or cracked case. The battery must not contain any free or unabsorbed liquid. Any electrical battery or battery powered device, equipment or vehicle having the potential of dangerous evolution of heat must be prepared for transport so as to prevent: - a short circuit (e.g. in the case of batteries, by the effective insulation of exposed terminals; or, in the case of equipment, by disconnection of the battery and protection of exposed terminals); and - unintentional activation. The words 'not restricted' and the special provision number A67 must be provided on the air waybill when an air waybill is issued.
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A70	Internal combustion or fuel cell engines or machinery being shipped either separately or incorporated into a vehicle, machine or other apparatus, without batteries or other dangerous goods, are not subject to these Instructions when carried as cargo provided that: a. for flammable liquid powered engines: 1. the engine is powered by a fuel that does not meet the classification criteria for any class or division; or 2. the fuel tank of the vehicle, machine or other apparatus has never contained any fuel or the fuel tank has been flushed and purged of vapours and adequate measures taken to nullify the hazard; and 3. the entire fuel system of the engine has no free liquid and all fuel lines are sealed or capped or securely connected to the engine and vehicle, machinery or apparatus. b. for flammable gas-powered internal combustion or fuel cell engines: 1. the entire fuel system must have been flushed, purged and filled with a non-flammable gas or fluid to nullify the hazard; 2. the final pressure of the non-flammable gas used to fill the system does not exceed 200 kPa at 20°C; 3. the shipper has made prior arrangements with the operator; and 4. the shipper has provided the operator with written or electronic documentation stating that the flushing, purging and filling procedure has been followed and that the final contents of the engine(s) have been tested and verified to be non-flammable. Multiple engines may be shipped in a unit load device provided that the shipper has made prior arrangements with the operator(s) for each shipment.
A123	This entry applies to Batteries, electric storage, not otherwise listed in Table 3-1. Examples of such batteries are: alkali-manganese, zinc-carbon and nickel-cadmium batteries. Any electrical battery or battery-powered device, equipment or vehicle having the potential of a dangerous evolution of heat must be prepared for transport so as to prevent: a. a short circuit (e.g. in the case of batteries, by the effective insulation of exposed terminals; or, in the case of equipment, by disconnection of the battery and protection of exposed terminals); and b. unintentional activation. The words 'not restricted' and the special provision number A123 must be provided on the air waybill when an air waybill is issued.

A152	Insulated packagings conforming to the requirements of Packing Instruction 202 containing refrigerated liquid nitrogen fully absorbed in a porous material are not subject to these Instructions provided the design of the insulated packaging would not allow the build-up of pressure within the container and would not permit the release of any refrigerated liquid nitrogen irrespective of the orientation of the insulated packaging and any outer packaging or overpack used is closed in a way that will not allow the build-up of pressure within that packaging or overpack. When used to contain substances not subject to these Instructions, the words 'not restricted' and the special provision number A152 must be provided on the air waybill when an air waybill is issued.
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A162	Fuel cell cartridges containing hydrogen in a metal hydride transported under this entry must have a water capacity less than or equal to 120 mL. The pressure in the fuel cell cartridge must not exceed 5 MPa at 55°C. The design type must withstand, without leaking or bursting, a pressure of two (2) times the design pressure of the cartridge at 55°C or 200 kPa more than the design pressure of the cartridge at 55°C, whichever is greater. The pressure at which this test is conducted is referred to in the drop test and the hydrogen cycling test as the 'minimum shell burst pressure'. Fuel cell cartridges must be filled in accordance with procedures provided by the manufacturer. The manufacturer must provide the following information with each fuel cell cartridge: a. inspection procedures to be carried out before initial filling and before refilling of the fuel cell cartridge; b. safety precautions and potential hazards to be aware of; c. method for determining when the rated capacity has been achieved; d. minimum and maximum pressure range; e. minimum and maximum temperature range; and f. any other requirements to be met for initial filling and refilling including the type of equipment to be used for initial filling and refilling. The fuel cell cartridges must be designed and constructed to prevent fuel leakage under normal conditions of transport. Each cartridge design type, including cartridges integral to a fuel cell, must be subjected to and must pass the following tests: Drop test A 1.8 metre drop test onto an unyielding surface in four different orientations: a. vertically, on the end containing the shut-off valve assembly; b. vertically, on the end opposite to the shut-off valve assembly; c. horizontally, onto a steel apex with a diameter of 38 mm, with the steel apex in the upward position; and d. at a 45° angle on the end containing the shut-off valve assembly. There must be no leakage, determined by using a soap bubble solution or other equivalent means on all possible leak locations, when the cartridge is charged to its rated charging pressure. The fuel cell cartridge must then be hydrostatically pressurised to destruction. The recorded burst pressure must exceed 85 per cent of the minimum shell burst pressure. Fire test A fuel cell cartridge filled to rated capacity with hydrogen must be subjected to a fire engulfment test. The cartridge design, which may include a vent feature integral to it, is deemed to have passed the fire test if: a. the internal pressure vents to zero gauge pressure without rupture of the cartridge; or
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- b. the cartridge withstands the fire for a minimum of 20 minutes without rupture.

Hydrogen cycling test

This test is intended to ensure that a fuel cell cartridge design stress limits are not exceeded during use.

The fuel cell cartridge must be cycled from not more than 5 per cent rated hydrogen capacity to not less than 95 per cent rated hydrogen capacity and back to not more than 5 per cent rated hydrogen capacity. The rated charging pressure must be used for charging and temperatures must be held within the operating temperature range. The cycling must be continued for at least 100 cycles.

Following the cycling test, the fuel cell cartridge must be charged, and the water volume displaced by the cartridge must be measured. The cartridge design is deemed to have passed the hydrogen cycling test if the water volume displaced by the cycled cartridge does not exceed the water volume displaced by an uncycled cartridge charged to 95 per cent rated capacity and pressurised to 75 per cent of its minimum shell burst pressure.

Production leak test

Each fuel cell cartridge must be tested for leaks at $15^{\circ}\text{C} \pm 5^{\circ}\text{C}$, while pressurised to its rated charging pressure. There must be no leakage, determined by using a soap bubble solution or other equivalent means on all possible leak locations.

Each fuel cell cartridge must be permanently marked with the following information:

- a. the rated charging pressure in megapascals (MPa);
- b. the manufacturer's serial number of the fuel cell cartridges or unique identification number; and
- c. the date of expiry based on the maximum service life (year in four digits; month in two digits).

A178	Security type equipment such as attaché cases, cash boxes, cash bags, etc., incorporating dangerous goods, for example lithium batteries, gas cartridges and/or pyrotechnic material, are not subject to these Instructions if the equipment complies with the following: - The equipment must be equipped with an effective means of preventing accidental activation; - If the equipment contains an explosive or pyrotechnic substance or an explosive article, this article or substance must be excluded from Class 1 by the appropriate national authority of the State of Manufacture in compliance with Part 2;1.5.2.1; - If the equipment contains lithium cells or batteries, these cells or batteries must comply with the following restrictions: - for a lithium metal cell, the lithium content is not more than 1 g; - for a lithium metal or lithium alloy battery, the aggregate lithium content is not more than 2 g; - for lithium ion cells, the Watt-hour rating is not more than 20 Wh; [the watt-hour rating is calculated by multiplying a cell's or battery's rated capacity, in ampere-hours, by its nominal voltage] - for lithium ion batteries, the Watt-hour rating is not more than 100 Wh; - each cell or battery is of the type proven to meet the requirements of each test in the UN Manual of Tests and Criteria, Part III, section 38.3; - If the equipment contains gases to expel dye or ink, only gas cartridges and receptacles, small, containing gas with a capacity not exceeding 50 mL, containing no constituents subject to these Instructions other than a Division 2.2 gas, are allowed. The release of gas must not cause extreme annoyance or discomfort to crew members so as to prevent the correct performance of assigned duties. In case of accidental activation, all hazardous effects must be confined within the equipment and must not produce extreme noise. - Security type equipment that is defective or that has been damaged is forbidden for transport. The words 'not restricted' and the special provision number A178 must be provided on the air waybill when an air waybill is issued.
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A180	Non-infectious specimens, such as specimens of mammals, birds, amphibians, reptiles, fish, insects and other invertebrates containing small quantities of UN 1170, UN 1198, UN 1987 or UN 1219 are not subject to these Instructions provided the following packing and marking requirements are met: a. specimens are: 1. wrapped in paper towel and/or cheesecloth moistened with alcohol, or an alcohol solution or a formaldehyde solution and then placed in a plastic bag that is heat-sealed. Any free liquid in the bag must not exceed 30 mL; or 2. placed in vials or other rigid containers with no more than 30 mL of alcohol, or an alcohol solution or a formaldehyde solution; b. the prepared specimens are then placed in a plastic bag that is then heat-sealed; c. the bagged specimens are then placed inside another plastic bag with absorbent material then heat-sealed; d. the finished bag is then placed in a strong outer packaging with suitable cushioning material; e. the total quantity of flammable liquid per outer packaging must not exceed 1 L; and f. the completed package is marked 'scientific research specimens, not restricted Special Provision A180 applies'. The words 'not restricted' and the special provision number A180 must be provided on the air waybill when an air waybill is issued.
A199	Nickel-metal hydride batteries or nickel-metal hydride battery-powered devices, equipment or vehicles having the potential of a dangerous evolution of heat are not subject to these Instructions provided they are prepared for transport so as to prevent: a. a short circuit (e.g., in the case of batteries, by the effective insulation of exposed terminals, or, in the case of equipment, by disconnection of the battery and protection of exposed terminals); and b. unintentional activation. The words 'not restricted' and the special provision number A199 must be provided on the air waybill when an air waybill is issued.

9.3 Specified Instruments Containing Dangerous Goods

The Organisation for the Prohibition of Chemical Weapons (OPCW) and government agencies listed in the table below may carry specified instruments containing dangerous goods , however NetJets prohibits the transport of the following items:

- a. Instruments containing radioactive material (i.e. chemical agent monitor (CAM) and/or rapid alarm and identification device monitor (RAID-M));
- b. A mercurial barometer or mercurial thermometer.

9.4 Provision of Information to Passengers

(CAT.GEN.MPA.200 (f))

NTA or its FBOs / Handling agents and the airport operator must ensure that notices warning passengers of the types of dangerous goods which are forbidden to transport aboard an aeroplane are prominently displayed, in sufficient number, at each of the places at an airport where passengers are welcomed, check-in areas and aeroplane boarding areas. These notices must include visual examples of dangerous goods forbidden from transport aboard an aeroplane.

NTA ensures that its passengers are informed of the dangerous goods requirements and items which cannot be carried on board a NTA aeroplane through the NetJets Ops Bulletin distributed to all customers.

FBOs are contracted to follow the NTA Ground Handling Manual, which details the passenger notification requirements with respect to dangerous goods.

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9.5 Marking and Labelling of Packages

Articles and substances meeting the dangerous goods classification criteria are assigned a 'UN Number' under the United Nations classification system. This consists of a four-digit number preceded by the capital letters 'UN'. Packages of dangerous goods must be marked with the UN Number(s) applicable to their contents.

Packages containing dangerous goods can also be identified by labels indicating the hazard of the goods by their class or division or by the presence of certain handling labels / markings.

NOTE

1. As no approval for the transport of dangerous goods is held, dangerous goods bearing any UN Number, hazard label; the radioactive material, excepted package handling label; the lithium battery handling label; the environmentally hazardous substances marking; or the excepted or limited quantities marking must not be loaded on an aeroplane (except as identified in [Items That May Be Carried by Passengers and Crew](#)).
2. When dangerous goods markings or labels are seen on items not declared as dangerous goods, it is often an indication that they do contain such goods.

CLASS 1 - EXPLOSIVE



* Division and compatibility group



** Compatibility group



CLASS 2 - GASES

Flammable gas (Division 2.1)



Non-flammable, non-toxic gas (Division 2.2)



Toxic gas (Division 2.3)



CLASS 3 - FLAMMABLE LIQUID**CLASS 4 - FLAMMABLE SOLIDS, SUBSTANCES LIABLE TO SPONTANEOUS COMBUSTION; SUBSTANCES WHICH, IN CONTACT WITH WATER, EMIT FLAMMABLE GASES**

Flammable solid (Division 4.1)



Substance liable to spontaneous combustion (Division 4.2)



Substance which, in contact with water, emits flammable gas (Division 4.3)

**CLASS 5- OXIDISING SUBSTANCES AND ORGANIC PEROXIDES**

Oxidising substance (Division 5.1)



Organic peroxide (Division 5.2) (flame may be black or white)

**CLASS 6 – TOXIC AND INFECTIOUS SUBSTANCES**

Toxic substance (Division 6.1)



Infectious substance (Division 6.2)



The bottom part of the label should bear the inscription:

'INFECTIOUS SUBSTANCE —
In case of damage or leakage
immediately notify public
health authority'

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CLASS 7 – RADIOACTIVE MATERIAL

Category I



Category II



Category III



Criticality safety index label

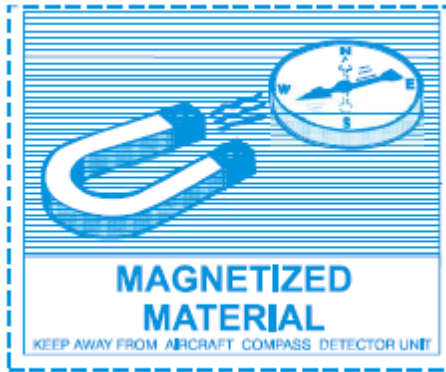
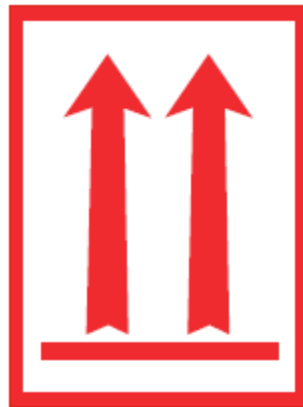
**CLASS 8 – CORROSIVE****CLASS 9 – MISCELLANEOUS**

Class 9 label for lithium cells and battery(ies) (Section I, IA and IB) or sodium ion cells and battery(ies) (Section I) shipments



HANDLING LABELS

Packages of dangerous goods may also bear labels providing handling information; these are:

Magnetized material**Cargo aircraft only****Cryogenic liquid label****Package orientation**

(red or black)

Keep away from heat

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LITHIUM BATTERIES MARK**EXCEPTED QUANTITIES MARKING**

Packages containing excepted quantities of dangerous goods can be identified from the following:

Hatching and symbol of the same colour, black or red, on white or suitable contrasting background.

* Place for class or, when assigned, the division number(s).

** Place for name of shipper or consignee, if not shown elsewhere on the package.

LIMITED QUANTITIES MARKING

Packages containing limited quantities of dangerous goods can be identified from the following:



ENVIRONMENTALLY HAZARDOUS SUBSTANCES MARKING

Packages containing environmentally hazardous substances (UN Nos. 3077 and 3082) must be durably marked with the environmentally hazardous substance mark with the exception of single packagings and combination packagings containing inner packagings with contents of 5 L or less for liquids or having a net mass per single or inner packaging of 5 kg or less for solids - such packages are not subject to the ICAO Technical Instructions other than specified packaging requirements.

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9.6 Duties of all Personnel Involved

9.6.1 Detailed Assignment of Responsibilities

(CAT.GEN.MPA.200 (d))

Duties associated with the carriage of dangerous goods for an operator not holding approval for their carriage as cargo include:

Owner Services Agents	• Ensuring procedures are implemented to ensure dangerous goods are not carried. • Recognition of undeclared dangerous goods. • Ensuring that notices, giving information about the transport of dangerous goods, are clearly displayed in sufficient number and prominence at booking process.
Persons handling passengers	• Ensuring that the provisions concerning passengers and dangerous goods are complied with. • Ensuring that notices are displayed in sufficient number and prominence at each of the places at an airport where passengers check in and aeroplane boarding areas maintained, and at any other location where passengers check in. • With the aim of preventing dangerous goods which passengers are not permitted to have from being taken on board an aeroplane in their baggage, seeking confirmation from a passenger about the contents of any item where there are suspicions that it may contain dangerous goods. • When baggage intended as carry-on is taken by the operator and placed into the cargo compartment for carriage, seeking confirmation from the passenger that dangerous goods which are only permitted in carry-on baggage (e.g. lithium batteries, including power banks) have been removed. • Ensuring that the discovery of prohibited dangerous goods (after a passenger has checked in) is reported to the appropriate Authority (see Dangerous Goods Accident and Incident Reports).

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Flight & Cabin Crew	• Recognition of undeclared dangerous goods. • Ensuring that the provisions concerning passengers and dangerous goods are complied with. • When baggage intended as carry-on is taken by the operator and placed into the cargo compartment for carriage, seeking confirmation from the passenger that dangerous goods which are only permitted in carryon baggage (e.g. lithium batteries, including power banks) have been removed. • Responding to a dangerous goods incident or accident on the aeroplane. • Ensuring that a dangerous goods incident or accident on the aeroplane, or the discovery of prohibited dangerous goods (after a passenger has boarded), is reported to the appropriate Authority (see Dangerous Goods Accident and Incident Reports).
Operations Personnel	• If there is a dangerous goods incident or accident, or if undeclared dangerous goods are detected, a report is made to the appropriate Authority (see Dangerous Goods Accident and Incident Reports).
Trainers	• Provision of initial and recurrent dangerous goods training commensurate with the responsibilities of the personnel concerned.
Compliance Monitoring Manager and Auditors	• Ensuring that activities are monitored for compliance with dangerous goods requirements and that these activities are carried out properly under the supervision of the relevant head of functional area. • Ensuring the initiation and follow-up of internal occurrence / accident investigations.
Safety Manager	• Ensuring the initiation and follow-up of internal occurrence / accident investigations.

9.7 Reserved

9.8 Recognition of Undeclared / Hidden Dangerous Goods

CAT.GEN.MPA.200(e)

9.8.1 'Hidden' Dangerous Goods

Personnel must be alert to indications that undeclared dangerous goods are present. Personnel interfacing with passengers must be alert to indications that prohibited dangerous goods are carried by passengers or within their baggage.

NOTE

The discovery of undeclared or misdeclared dangerous goods or the discovery of dangerous goods forbidden for carriage by passengers (discovered after the check-in process) must be reported to ANAC - see [Dangerous Goods Accident and Incident Reports](#).

The following is a list of general descriptions that are often used for items in cargo or in passengers' baggage and the types of dangerous goods that may be included in any item bearing that description.

Aircraft On Ground (AOG) Spares - may contain explosives (flares or other pyrotechnics), chemical oxygen generators, unserviceable tyre assemblies, cylinders of compressed gas (oxygen, carbon dioxide, or fire extinguishers), fuel in equipment, wet or lithium batteries, matches.

Automobiles, Automobile Parts / Supplies - (car, motor, motorcycle) may include engines (including fuel cell engines), carburettors or fuel tanks that contain or have contained fuel, wet or lithium batteries, compressed gases in tyre inflation devices and fire extinguishers, air bags, flammable adhesives, paints, sealants and solvents, etc..

Battery-powered Devices / Equipment - may contain wet or lithium batteries.

Breathing Apparatus - may indicate cylinders of compressed air or oxygen, chemical oxygen generators or refrigerated liquefied oxygen.

Camping Equipment - may contain flammable gases (butane, propane, etc.), flammable liquids (kerosene, gasoline, etc.), flammable solids (hexamine, matches, etc.).

Cars, Car Parts - see Automobile Parts, etc.

Chemicals - may contain items meeting any of the criteria for dangerous goods, particularly flammable liquids, flammable solids, oxidizers, organic peroxides, toxic or corrosive substances.

Consolidated Consignments (Groupages) - may contain any of the defined classes of dangerous goods.

Cryogenic (Liquid) - indicates refrigerated liquefied gases such as argon, helium, neon and nitrogen, etc.

Cylinders - may contain compressed or liquefied gas.

Dental Apparatus - may contain flammable resins or solvents, compressed or liquefied gas, mercury and radioactive material.

Diagnostic Specimens - may contain infectious substances.

Diving Equipment - may contain cylinders (such as scuba tanks, vest bottles, etc.) of compressed gas (air, oxygen, etc), high intensity diving lamps which can generate extremely high heat when operated in air. In order to be carried safely, the bulb or battery must be disconnected.

Drilling and Mining Equipment - may contain explosive(s) and / or other dangerous goods.

Dry Shipper (Vapour Shipper) - may contain free liquid nitrogen. Dry shippers are subject to these Regulations when they permit the release of any free liquid nitrogen irrespective of the orientation of the packaging.

Electrical Equipment / Electronic Equipment - may contain magnetized materials or mercury in switch gear and electron tubes, wet batteries, lithium batteries or fuel cells or fuel cell cartridges that contain or have contained fuel.

Electrically Powered Apparatus - (wheel chairs, lawn mowers, golf carts, etc.) may contain wet batteries, lithium batteries or fuel cells or fuel cell cartridges that contain or have contained fuel.

Expeditionary Equipment - may contain explosives (flares), flammable liquids (gasoline), flammable gas (propane, camping gas) or other dangerous goods.

Film Crew or Media Equipment - may contain explosive pyrotechnic devices, generators incorporating internal combustion engines, wet batteries, lithium batteries, fuel, heat producing items, etc.

Frozen Embryos - may contain refrigerated liquefied gas or Carbon dioxide, solid (dry ice).

Frozen Fruit, Vegetables, etc. - may be packed in Carbon dioxide, solid (dry ice).

Fuel Control Units - may contain flammable liquids.

Hot Air Balloon - may contain cylinders with flammable gas, fire extinguishers, engines internal combustion, batteries, etc.

Household Goods - may contain items meeting any of the criteria for dangerous goods. Examples include flammable liquids such as solvent based paint, adhesives, polishes, aerosols (for passengers, those not permitted under [Items That May Be Carried by Passengers and Crew](#)), bleach, corrosive oven or drain cleaners, ammunition, matches, etc.

Instruments - may conceal barometers, manometers, mercury switches, rectifier tubes, thermometers, etc. containing mercury.

Laboratory / Testing Equipment - may contain items meeting any of the criteria for dangerous goods, particularly flammable liquids, flammable solids, oxidizers, organic peroxides, toxic or corrosive substances, lithium batteries, cylinders of compressed gas, etc.

Machinery Parts - may contain adhesives, paints, sealants, solvents, wet and lithium batteries, mercury, cylinders of compressed or liquefied gas, etc.

Magnets and Other Items of Similar Material - may individually or cumulatively meet the definition of magnetized material.

Medical Supplies / Equipment - may contain items meeting any of the criteria for dangerous goods, particularly flammable liquids, flammable solids, oxidizers, organic peroxides, toxic, corrosive substances or lithium batteries.

Metal Construction Material - may contain ferro-magnetic material, which may be subject to special stowage requirements due to the possibility of affecting aeroplane instruments.

Metal fencing - may contain ferro-magnetic material which may be subject to special stowage requirements due to the possibility of affecting aeroplane instruments.

Metal piping - may contain ferro-magnetic material which may be subject to special stowage requirements due to the possibility of affecting aeroplane instruments.

Pharmaceuticals - may contain items meeting any of the criteria for dangerous goods, particularly radioactive material, flammable liquids, flammable solids, oxidisers, organic peroxides, toxic or corrosive substances.

Photographic Supplies / Equipment - may contain items meeting any of the criteria for dangerous goods, particularly heat producing devices, flammable liquids, flammable solids, oxidisers, organic peroxides, toxic or corrosive substances or lithium batteries.

Racing Car or Motorcycle Team Equipment - may contain engines, including fuel cell engines, carburettors or fuel tanks which contain fuel or residual fuel, flammable aerosols, cylinders of compressed gases, nitromethane, other fuel additives, wet batteries, lithium batteries, etc.

Refrigerators - may contain liquefied gases or an ammonia solution.

Repair Kits - may contain organic peroxides and flammable adhesives, solvent based paints, resins, etc.

Samples For Testing - may contain items meeting any of the criteria for dangerous goods, particularly infectious substances, flammable liquids, flammable solids, oxidizers, organic peroxides, toxic or corrosive substances.

Semen - may be packed with Carbon dioxide, solid (dry ice) or refrigerated liquefied gas. See also Dry Shipper.

Sporting Goods / Sports Team Equipment - may contain cylinders of compressed or liquefied gas (air, carbon dioxide, etc.), lithium batteries, propane torches, first aid kits, flammable adhesives, aerosols, etc.

Swimming Pool Chemicals - may contain oxidizing or corrosive substances.

Switches in Electrical Equipment or Instruments - may contain mercury.

Tool Boxes - may contain explosives (power rivets), compressed gases or aerosols, flammable gases (butane cylinders or torches), flammable adhesives or paints, corrosive liquids, lithium batteries, etc.

Torches - micro torches and utility lighters may contain flammable gas and be equipped with an electronic starter. Larger torches may consist of a torch head (often with a selfigniting switch) attached to a container or cylinder of flammable gas.

Unaccompanied Passengers Baggage / Personal Effects - may contain items meeting any of the criteria for dangerous goods, such as fireworks, flammable household liquids, corrosive oven or drain cleaners, flammable gas or liquid lighter refills or camping stove cylinders, matches, bleach, aerosols, etc. (those not permitted under [Items That May Be Carried by Passengers and Crew](#)).

Vaccines - may be packed in dry ice.

9.8.2 Safety Data Sheets

REACH (Registration, Evaluation, Authorisation & restriction of Chemicals) is an European Union regulation controlling chemicals in Europe. REACH requires for many substances and mixtures, a Safety Data Sheet (SDS) to be provided either before or at the time of first delivery. Section 14 of the EU format SDS provides basic classification information, i.e. UN number, proper shipping name, Class / Division and Packing Group.

9.8.3 Consumer Labelling (Overview)

Some everyday household items bear consumer warning labels which may or may not indicate they are classified as dangerous goods in air transport. All over the world there are different laws on how to identify the hazardous properties of chemicals (called 'classification') and how information about these hazards is then passed to users (through consumer supply labels and safety data sheets for workers). This can be confusing because the same chemical can have different hazard descriptions in different countries. For example, a chemical could be labelled for supply as 'toxic' in one country, but not in another. For this reason, the UN brought together experts from different countries to create the Globally Harmonized System of Classification and Labelling of Chemicals (GHS).

The GHS has been implemented within Europe by the Regulation on Classification, Labelling and Packaging of Substances and Mixtures (known as the CLP Regulation).

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9.8.4 GHS Labels

Products bearing the following GHS labels ARE classified as dangerous goods:



NOTE

A product bearing the GHS corrosive label (depicted far right above) is NOT classified as dangerous goods if the signal word 'Danger' and hazard statement 'causes serious eye damage' applies.

Products only bearing the following GHS labels are NOT classified as dangerous goods:



9.9 Special Notification Requirements in the Event of an Accident or Occurrence When Dangerous Goods are Being Carried or Have Been Offered for Air Transport Without Having Been Prepared and Declared in Accordance with the ICAO Technical Instructions

CAT.GEN.MPA.200(e)

9.9.1 Reserved

9.9.2 Reserved

9.9.3 Reserved

9.9.4 Reserved

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9.9.5 Removal of Contamination

(SPA.DG.105)

In the event of a spillage or leakage of undeclared dangerous goods within an aeroplane, the position where the dangerous goods or unit load device was stowed on the aeroplane must be inspected for damage or contamination and any hazardous contamination removed. Persons responding in the event of damage to or leakage of dangerous goods from packages must:

- identify the hazards and wear appropriate protective clothing;
- avoid handling the package or keep handling to a minimum;
- inspect adjacent packages for contamination and put aside any that may have been contaminated;
- arrange for decontamination of the aeroplane and equipment; and
- in the case of infectious material, inform the appropriate public health authority or veterinary authority, and provide information to any other countries of transit where persons may have been exposed to danger; and notify the shipper, and/or the consignee.

If it is evident that a package containing radioactive material is damaged or leaking, or if it is suspected that the package may have leaked or been damaged, access to the package must be restricted and a qualified person must, as soon as possible, assess the extent of contamination and the resultant dose rate of the package. The scope of the assessment must include the package, the aeroplane, the adjacent loading and unloading areas and, if necessary, all other material which has been carried in the aeroplane. When necessary, additional steps for the protection of persons, property, and the environment must be taken in accordance with provisions established by the relevant competent authority, to overcome and minimise the consequences of such leakage or damage.

An aircraft which has been contaminated by radioactive materials must be immediately taken out of service and not returned until the dose rate at any accessible surface and the non-fixed contamination are not more than the values specified in the Technical Instructions. In the event of non-compliance with any limit in the Technical Instructions applicable to dose rate or contamination, the operator must ensure the shipper is informed if the non-compliance is identified during transport; take immediate steps to mitigate the consequences of the non-compliance; and communicate the non-compliance to the shipper and relevant competent Authority(ies), respectively, as soon as practicable and immediately whenever an emergency has developed or is developing.

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Chapter 10 - Security

10.1.0 Introduction

10.1.0.1 Scope and Applicability of NTA Security Programme

Standard 3.3.1 of Annex 17 of the Chicago Convention, as well as Commission Regulation 300/2008 and references to Commission Implementing Regulation (EU) 2015/ 1998, state that each State shall require air carriers providing service from that State to establish, implement, and maintain a written air carrier security programme appropriate to meet the requirements of the national aviation security programme of that State.

10.1.1.1 Content and Purposes

The purpose of civil aviation security is to guarantee protection and safeguard of passengers, crews, ground staff and public in general, as well as aeroplanes, airport facilities, and all means supporting the air navigation against actions of unlawful interference perpetrated on the ground or during the flight.

All instructions referred in this chapter have been extracted from the NTA PT CAA approved Security Programme on the need to know basis. The Security Programme is classified as confidential, therefore, restricted for consultation.

The NTA Security Programme contains instructions and information about subjects related to aviation security activities in general, and NetJets Transportes Aéreos, S.A. (NTA) in particular. The procedures contained therein are NTA procedures designed to provide a level of safety and security to the passengers, crew, baggage, and any other items that may be transported.

1. Enclosed procedures include NTA rules as well as the national and international established regulations, namely the relevant security EU Regulations and parts of Annex 17 of ICAO and document 30 of ECAC.
2. The dissemination of the Programme are based on a 'need-to-know' principle and, therefore, the distribution of the procedures are strictly controlled. The Security Programme is NTA proprietary information and must not be copied or distributed to the general public.
3. A thorough knowledge of the procedures in this manual as they relate to the activities of each department is compulsory for all relevant staff and must be kept up to date.
4. Strict adherence to the instructions contained in this manual is compulsory.
5. Any detected discrepancies or doubt relating to the procedures contained in this chapter must be immediately communicated to the Security Manager.

10.1.1.2 Other Stations

1. At each airport the Security Manager delegates the operational security to either:
 - NTA Station Manager (if applicable);
 - NTA representative delegated to represent NTA at the specific airport, if applicable;
 - FBO Station Manager.

The NTA airport security representative will be responsible for ensuring NTA security instructions are put into practice, and are monitored, and will liaise with the airport authorities and security authorities as required.

2. At airports, the respective Station Manager or the NTA representative, as applicable, is responsible for representing NTA at all local Airport Security Committees.

10.1.1.3 General Considerations

1. All staff members whose activities are related with security must always be updated and attend programmes of in-house courses or external recognised security courses.
2. If applicable, 'on-the-job-training' specific to local needs will be used to provide recurrent training in security matters.
3. Security is both a priority for all staff members as well as a priority to passengers, as final users. Therefore, customer's information and / or data is an important issue at NTA.

10.1.1.4 Protection of Aviation Security Information

Our customer confidence in NTA approach to aviation security may be undermined by unauthorised publication of this security programme, in whole or in part. Therefore, NTA adopted protective measures for this programme in keeping a record of distributed copies to ensure that the information contained herein is not misused or disclosed to unauthorised persons. Based on the 'need to know' principle, security restricted information is restricted to those persons who require such information in the performance of their duties. If confidential / sensitive information is disseminated by the Security department, the person receiving it will be identified and made aware of security confidential information.

All documentation handled by the Security department, related to NTA flights, airports network, passenger and crew manifest and / or NetJets employees, are considered confidential and must be secured by the Security department for a period of 24 months. The release of any documents retained by the Security department needs to obtain prior approval of the Security Manager. Internal security records and documents no longer required should be destroyed, preferably by shredding. However, if deemed necessary by the Security Manager, they may be kept in the Security department but are to be classified as 'Outdated - For Reference Only'.

10.1.1.5 NetJets Cyber Security Posture

As the world-wide leader in private aviation, NetJets manages and operates the largest and most diverse private jet fleet in the world. As owners of a high-profile brand with a presence in various markets across the globe, NetJets understands the need to have a strong cyber security posture.

Accordingly, we take a very serious approach to safety and security from planning and implementation, procedures, controls, technical and non-technical policy to internal and external threats. The safety and security of our data and yours is paramount and the foundation for everything we do.

Therefore, NetJets has the following cyber security functions in place:

- Dedicated cyber security staff of certified professionals;
- Global security staff of security professionals with security expertise gained from federal, state, and local law enforcement, as well as the U.S. military;
- A comprehensive cyber security program, including:
 - Governance, risk, and compliance;
 - Data classification and retention;
 - Security architecture and engineering;
 - Identity and access management;
 - Asset and vulnerability management;
 - Cyber incident response;
 - Business continuity planning and disaster recover;
 - Supplier risk management;
 - Secure software development life cycle (S-SDLC);
 - Cyber security awareness and training;
 - Internal audits and penetration testing.
- Dedicated Aircraft cyber security, including:
 - An aviation system cyber security awareness program for crew members;
 - Documented processes and procedures for cyber security controls required to ensure the continued safe, and reliable operation of connected airplanes within NetJets' fleet;
 - Documented security controls pertaining to airplane ground support networks based on security guidance from airplane and avionics OEMs, and regulatory agencies;
 - Security requirements to assist in new aircraft purchases;
 - Active participation in industry associations and working groups to influence NetJets safety and security standards:
 - Aviation Information Sharing Analysis Center (A-ISAC) Membership;
 - RTCA Special Committee 216 - Aeronautical Systems Security;
 - OSAC - Overseas Security Advisory Council.
 - Regular cyber security training and annual attestation of cyber security policy for all employees.
 - Aviation authorities' regulations (FAA, ANAC, etc.) compliance.
 - Auditing of cyber security posture by independent assessors.
 - External penetration testing performed by third-party vendors.

- Annual entitlement reviews conducted on all financial systems.
- 24/7 system monitoring.

Further information on Cyber Security, can be found on the NTA Security Programme Appx2 - Cyber Security, available on the need to know basis.

10.1.2 Internal Organisations and Obligations

10.1.2.1 ICAO Annex 17, ECAC Doc.30, Regulation (EC) 300/2008, Commission Implementing Regulation (EU) 2015/1998, Commission Implementing Decision (C) 2015/8005

Pursuant the provisions of Article 37 of the Chicago Convention, International Standards, and Recommended Practices on Security were adopted by the ICAO Council on the 22nd March 1974, and designated as Annex 17 to the Convention with the title 'Standards and Recommended Practices - Security - Safeguarding International Civil Aviation against Acts of Unlawful Interference'.

ECAC adopted a policy statement in the field of civil aviation security called ECAC Doc.30. It contains recommendations and guidance material in the field of civil aviation security.

Pursuant to Article 80(2) of the Treaty establishing the European Community, the European Parliament, and the Council of the European Union adopted Regulation (EC) Commission Regulation 300/2008 on the common fields of civil aviation security and the Commission Implementing Regulation (EU) 2015/1998, and the Commission Decision C (2015) 8005 and all relevant EU security regulations, laying down detailed measures for the Implementation of the common basic standards on aviation security. The objective of the Regulation is to ensure the protection of the citizens within the European Community from acts of unlawful interference.

10.1.3 The Relevant Appropriate Authority in Portugal

The Chairman of the Board of the Portuguese Civil Aviation Authority (ANAC) is the Appropriate Authority for the Portuguese Civil Aviation Security, defined as the National Civil Aviation Security Authority.

Meeting their responsibilities through the Directorate of Civil Aviation Security as the executive body for Civil Aviation Security matters.

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10.1.4 NTA Security Organisation

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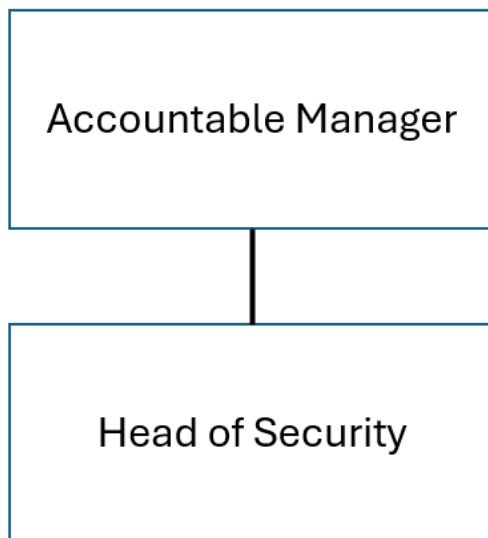
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24 hours Security Contact: NJE Security Department
Office telephone: + 351 21 446 8458
E-mail: Lis-Security@netjets.com

Security Organisational Table - NTA



10.1.4.1 Managers

All Managers within NTA are responsible for the implementation of the security measures applicable to their area of responsibility, and must be familiar with the security requirements applicable to their area of responsibility.

10.1.4.2 Staff

Staff at all levels within NTA, and / or under contract to NTA, have a duty to be familiar with and apply the security measures relevant to their specific duties. They also have an obligation to develop an awareness and sense of responsibility towards aviation security issues in general.

Staff has also the obligation to report to NTA Security department any circumstances that they believe constitute a breach of the security measures applied within the company.

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10.2 Airport Security

10.2.1 Airport Planning Requirements

The design or layout of airports / FBOs, passenger and cargo terminals and other buildings having direct airside access shall take into account the essential requirements relating to:

- a. Security controls applied to passengers, baggage, cargo, courier and express parcels, mail and air carrier catering stores and supplies;
- b. The protection and control of access to airside, security restricted areas and other sensitive airport areas and facilities;
- c. The efficient use of security equipment.

10.2.2 Access Control

Authorisations must be checked before a person is granted access to airside to ensure that it is valid and corresponds to the holder.

10.2.2.1 Access to Security Restricted Areas and Other Areas

Access to security restricted areas may only be granted if persons and vehicles have a legitimate reason to be there. Guided tours of the airport escorted by authorised persons shall be considered to have a legitimate reason.

The following conditions are applicable:

- In order to be granted access to airside a person shall carry an authorisation.
- In order to be granted access to airside a vehicle shall display a vehicle pass.
- Persons who are airside shall, upon request, present their authorisation for control.
- All staff requiring access to security restricted areas shall also receive regular training in aviation security including the risks to aviation security and be instructed to report to the Head of Security and relevant authority any incident which may pose a threat to aviation security.

In order to be granted access to security restricted areas a person shall present one of the following authorisations:

- a. a valid boarding card or equivalent; or
- b. a valid crew identification card; or
- c. a valid airport identification card; or
- d. a valid appropriate national authority identification card; or
- e. a valid compliance authority identification card recognised by the appropriate national authority.

Alternatively, access may also be granted after positive identification via biometric data verification.

10.2.2.2 Requirements for Crew Identification Cards and Airport Identification Cards

A crew identification card of a crew member employed by a Union air carrier and an airport identification card may only be issued to a person who has an operational need and has successfully completed an enhanced background check.

The enhanced background check required before issuing an NTA Crew Identification card, which allows unescorted access to the security restricted area, shall cover at least the last five years. It shall be repeated at regular intervals not exceeding 12 months for enhanced background checks, or three years for standard background checks.

Crew and airport identification cards shall be issued for a period not exceeding five years.

The identification card of a person who fails an enhanced background check shall be immediately disabled or withdrawn, as applicable, and returned to Security Department.

Airport identification cards shall be issued to all personnel working in the airport or visiting it frequently, (including airport and air carrier employees and employees of other organisations).

The identification card shall be worn in a visible place at least whenever the holder is in security restricted areas.

The identification cards shall be returned immediately to the issuing entity / air carrier:

- a. upon request of the issuing entity; or
- b. upon termination of employment; or
- c. upon change of employer; or
- d. upon change of the need to have access to areas for which an authorisation has been given; or
- e. upon expiry of the card; or
- f. upon withdrawal of the card.

The issuing entity shall be notified immediately of the loss, theft, or failure to return an identification card.

A crew identification card shall display:

- a. the name and photograph of the holder; and
- b. the name of the air carrier; and
- c. the word 'crew' in English; and
- d. the expiry date.

In order to prevent the misuse of airport identification cards, a system shall be in place to reasonably ensure that attempted use of cards that have been lost, stolen or not returned is detected. Upon detection, appropriate action shall be taken.

Access to a security restricted area by a crew member holding a valid Crew Identification Card that is not a Union Crew Identification Card and does not include a photograph of the holder, must only be granted upon presentation of a valid passport.

10.2.2.3 Requirements for Vehicle Passes

A vehicle pass must be checked before a vehicle is granted access to airside to ensure that it is valid and corresponds to the vehicle.

A vehicle pass may only be issued where an operational need has been established.

A vehicle pass shall be specific to the vehicle and display:

- a. Vehicle registration number or, if unavailable, another identifying element of the vehicle;
- b. Name of the entity in whose name the pass is issued;
- c. The areas the vehicle is authorised to access;
- d. Expiry date.

Vehicles which need to be used airside shall remain airside to the extent practicable.

Vehicles required moving between landside and airside areas shall be issued with a specific pass, issued by the appropriate authorities, and fixed in an easily visible place of the vehicle. Other vehicles requiring airside access shall be admitted only after having been inspected and issued with a temporary pass.

A vehicle may be exempted from the requirements as described above under the condition that it is escorted at all times when airside.

10.2.2.4 Escorted Access

Crew members not holding a valid airport identification card shall be escorted at all times when in security restricted areas except for:

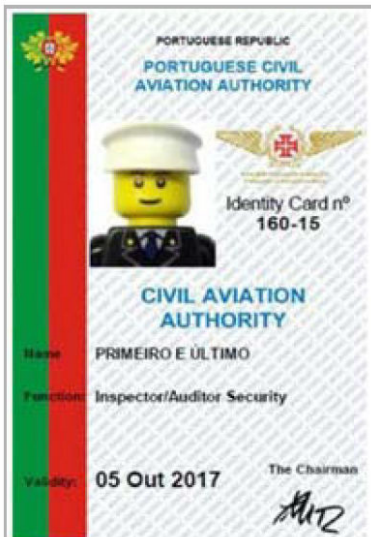
- a. areas where passengers may be present; and
- b. areas in the immediate proximity of the aircraft on which they have arrived or will depart; and
- c. areas designated for crews; and
- d. routes between the terminal or access point and the aeroplane on which crew members have arrived or will depart.

10.2.2.5 Recognised Identification Cards

Airport Cards are produced and distributed by ANAC. NetJets CREW badges are produced and distributed by the NTA Security Department.

Below are all recognisable ID cards and NetJets CREW badge.

Layout Portuguese Civil Aviation Authority (Regulation 534/2015)

 PORTUGUESE REPUBLIC PORTUGUESE CIVIL AVIATION AUTHORITY Identity Card nº 160-15 CIVIL AVIATION AUTHORITY Name: PRIMEIRO E ÚLTIMO Function: Inspector/Auditor Security Validity: 05 Out 2017 The Chairman	According to nr. 3 of Article nr. 36 of the Statutes of ANAC, approved by Decree-Law nr. 40/2015 of 16 March and for the purposes of nr. 1 and nr. 2 of the same legal provision, the holder of this identification card is legally compared to a law enforcement officer, duly accredited by ANAC, being also legally empowered to put in force the following authority powers and the following prerogatives, within the framework of inspections and audits, namely: a) To access and inspect at any time, and without previous announcement, the facilities, terrains, means of transportation, equipment, software and services of entities under ANAC's jurisdiction, as well as other related entities and of those who collaborate with them; b) To request, for further analysis, equipment, materials, documents, and any other written or digital data; c) To determine, as a preventive measure and with immediate effects, the suspension or cease of activities, as well as the closing of facilities and immediate grounding of aircrafts; d) To identify entities and persons; e) To request explaining regarding facts and documents to any legal representative, workers and employees from the entities referred to in point a), and to register the respective answers. Blood Type: _____ Doc. ID.: _____ Date of issue: 05 Out 15 SIGNATURE OF HOLDER <i>Assinatura do Titular da ANAC</i>
---	---

Layout NetJets CREW Badge (EU Regulation 2015/1998)

Exp. Date DD/MM/YYYY NETJETS Photo CREW Name ID Number	PROPERTY OF : NetJets Transportes Aéreos S.A <u>If you misplace this card Inform the Security Department immediately</u> <u>If found please return to:</u> NetJets Transportes Aéreos Security Department Rua Calvet de Magalhães, 245-B 2774-550 Paço de Arcos Portugal NetJets 24 Hour Business number + 351 21 446 8400
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10.3 Aeroplane Security

10.3.1 Searching and Checking the Aeroplane

When to perform an aircraft security search

All NTA aeroplanes shall be searched as applicable:

All NTA flights (Revenues & Ferries) require an Aircraft Security Search to be conducted.

This will be achieved by the completion of the NTA Aircraft Security Search checklist and stored electronically.

The crew is responsible for conducting the security search of the aeroplane (both internal and external areas of the aeroplane).

An aeroplane shall at all times be subjected to an aeroplane security search whenever there is reason to believe that unauthorised persons may have had access to it.

An aircraft arriving into a critical part from a third country not listed in Attachment 3-B shall be subjected to an aircraft security search any time after passenger disembarkation from the area to be searched and/or the unloading of the hold. The search may not start until the aircraft has reached its final parking position.

An aircraft arriving from a Member State where it was in transit after having arrived from a third country not listed in Attachment 3-B shall be considered as an aircraft arriving from a third country.

If the search is carried out before moving the aeroplane into a critical part, the areas of the aeroplane searched shall be either sealed, locked, or under constant monitoring by persons responsible and trained for protecting aeroplanes until the aeroplane arrives in the critical part.

Prior to commencing the search, the interior of the aeroplane must be inspected to ensure that only those persons conducting the search, or carrying out a statutory duty, are present on board the aeroplane; and

From the commencement of an aeroplane security search until push-back of the aeroplane from the airport gate, only persons with an operational need may, upon verification of their credentials, be granted access to an aeroplane.

A search of an aeroplane must be carried out only after the area of the aeroplane about to be searched has been vacated by all passengers.

How to perform an aeroplane security search

An aeroplane security search shall consist of an examination of all of the following areas, when they are accessible without the use of tools, keys, stairs or other aids, and without breaking seals, and where it could be assumed that prohibited articles may be hidden:

- overhead baggage lockers;
- cupboards and storage compartments, including crew storage areas;
- areas where passengers may have private access, including toilet compartments, showers, and wash rooms;
- cupboards and storage compartments, bars, fridges, or galley areas;
- seat and seat pockets;
- areas under seats, between seats, and between seats and the aeroplane wall;

- flight deck, if left unattended;
- between 5% and 10% of life-jacket pouches;
- aeroplane service panels and service hatches;
- aeroplane hold, unless it is sealed;
- items transported in the hold;
- wheel wells;
- all crew rest areas;
- catering supplies.

The examination of the areas shall be done by a hand search.

A visual check may be used as an alternative method for the examination of those areas that are empty.

Explosive detection dogs may be used as a supplementary method of examination.

Explosive Trace Detection (ETD) equipment may be used as a supplementary method of examination.

Exemptions and special provisions

During the examination of the areas in the cabin of the aeroplane, no passengers shall be present in the area to be examined, unless the aeroplane is in transit under special provisions as below.

Where an aeroplane in transit passengers and their cabin baggage may remain on board during the performance of a search, only if the passenger boarded the flight on a stretcher, or is the accompanying carer of such a passenger, and prior clearance for such travel was obtained from the air carrier.

From the commencement of an aeroplane security search until push-back of the aeroplane from the airport gate, only persons with an operational need may, upon verification of their credentials, be granted access to an aeroplane.

Where there is any reason to believe that unauthorised persons may have had access to the aeroplane cabin, all life jacket pouches must be physically examined as part of the aeroplane security search.

Information on the aeroplane security search

The following information on the aeroplane security search of a departing flight shall be recorded and kept at a point not on the aeroplane for the duration of the flight or for 24 hours, whichever is longer:

- a. flight number, and
- b. origin of the previous flight.

Where an aeroplane security search was performed, the information shall also include:

- a. date and time that the aeroplane security search was completed, and
- b. the name and signature of the person responsible for the performance of the aeroplane security search.

The above requirement will be achieved by the completion of the NTA Aircraft Security Search checklist (below) completed and stored electronically.

All aeroplane security searches and aeroplane security checks shall be conducted once all service providers (caterers, cleaners and others), other than those involved in the security function, have left the aeroplane and its sterility shall be maintained until and throughout the boarding process and pre-departure.

While during the search a suspect explosive device is discovered please refer to 10.13.3 Discovery of a Suspect or Prohibited Article.

AIRCRAFT SECURITY SEARCH		
TICK WHERE APPROPRIATE		
Overhead baggage lockers	☐ Checked	☐ Not Applicable
Cupboards and storage compartments, including crew storage areas	☐ Checked	☐ Not Applicable
Areas where passengers may have private access, including toilet compartments, showers, and wash rooms	☐ Checked	☐ Not Applicable
Cupboards and storage compartments, bars, fridges or galley areas	☐ Checked	☐ Not Applicable
Seat and seat pockets	☐ Checked	☐ Not Applicable
Areas under seats, between seats, and between seats and the aeroplane lateral wall	☐ Checked	☐ Not Applicable
Flight Deck, if left unattended	☐ Checked	☐ Not Applicable
Between 5% and 10% of life-jacket pouches	☐ Checked	☐ Not Applicable
Aircraft service panels and service hatches	☐ Checked	☐ Not Applicable
Aircraft hold, unless it's sealed	☐ Checked	☐ Not Applicable
Items transported in the hold	☐ Checked	☐ Not Applicable
Wheel wells	☐ Checked	☐ Not Applicable
All crew rest areas	☐ Checked	☐ Not Applicable
Catering supplies	☐ Checked	☐ Not Applicable
Aircraft Registration	Flight Number	Origin of the previous flight
Date: _____		
Signature: _____		
PRINTED NAME: _____		
NETJETS		
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10.3.2 Protection of Aeroplanes

Aeroplanes are not only vulnerable to the introduction of prohibited articles and devices by passengers but also by other persons while the aeroplane is on the ground and either parked or being prepared for service / flight.

The first line of defence against unauthorised access to the aeroplane is the integrity of the landside / airside boundary. However, the prevention of unauthorised access to an aeroplane within this boundary also depends on security measures taken in the immediate proximity of the aeroplane and in the general apron / ramp area.

NTA will work with local FBO and Maintenance Organisations to ensure adequate security for aeroplanes parked on ramps and in hangars.

During the aeroplane security search / check, for an aeroplane in service, access shall be controlled from the start of the aeroplane security search / check until departure, in order to maintain the integrity of the search.

Regardless of where an aeroplane is parked at an airport, each of its external doors shall be protected against unauthorised access by:

- a. ensuring that persons seeking to gain unauthorised access are challenged promptly; or
- b. having the external door closed;
- c. having electronic means which will immediately detect unauthorised access.

It shall not apply to an aeroplane parked in a hangar that is locked or otherwise protected from unauthorised access.

Each aeroplane in service shall be placed under sufficient surveillance to detect unauthorised access.

Access to an aeroplane not in service, where external doors are closed and the aeroplane is in a part other than a critical part, shall be controlled as follows:

1. External doors shall be locked where only persons with an operational need shall be able to unlock these doors (see note); or
2. Security Seals should be applied to aeroplane panels / doors, as per [Security Seals](#).
3. Be monitored, ensuring that unauthorised access to the aeroplane is immediately detected.

In addition, when all staff are not screened for access into security restricted areas, each aeroplane should be visited at least once every 30 minutes by a foot or mobile patrol or placed under surveillance sufficient to detect unauthorised access and security seals checked.

An aeroplane shall, wherever possible, be parked away from perimeter fences or other easily penetrable barriers and in well-illuminated areas.

NOTE

Under circumstances where ad-hoc maintenance is required at stations where there is no NetJets maintenance centres with keys for the aeroplanes, and there is the need to leave the aeroplane unlocked for maintenance access, the following must be achieved:

1. Notify the Security Department through Lis-Security@netjets.com that an ad-hoc maintenance service is to be completed and there is the need for the crew to leave the aeroplane unlocked;
2. Security tamper evident seals must be applied by the crew, when leaving the aeroplane, as the primary mean of protecting the aeroplane from unauthorized access, according with the instructions under [Security Seals](#);
3. A security search before the initiation of any flight operations need to be performed according to instructions under [Searching and Checking the Aeroplane](#).

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10.3.3 Control of Access to Aeroplanes

To prevent unauthorised access, an aeroplane preparing to fly or in post-flight should not be left unattended. Aeroplane crew, maintenance, aircraft restocking, cleaning personnel or any other person servicing an aeroplane shall identify any person approaching or boarding the aeroplane and confirm their need to board the aeroplane; should there be any doubt or suspicion, airfield security staff shall be summoned.

Where greater security is considered necessary, security staff should be positioned to control access to the aeroplane or the general vicinity of the aeroplane and be in charge, overall, of the aeroplane's security. Unrecognised or unauthorised persons trying to access the aeroplane shall be challenged.

Tamper-evident security seals should be placed across panels / doors as per [Security Seals](#) making any unauthorised opening evident to persons carrying out predeparture inspections.

If possible for overnight parking, the best illuminated parking area should be selected.

All doors shall be confirmed closed, locked and security sealed prior to leaving the aeroplane, as per [Security Seals](#).

Airports will be reviewed by the Security department for a level of security threat. The handler may also be contacted to review what level of extra security is available, e.g. security guards, perimeter fences, etc. The final decision on the level of security or ultimately whether the flight can still be operated to a location, from a security perspective, will be taken by the Security Manager, who will be consulted prior to confirmation of any flight by Owner Services.

At airports where it is possible that unauthorised persons have gained access to an NTA aeroplane, additional action will be taken before the aeroplane is prepared for further service.

As a general rule, pre-flight check will include:

- An inspection of the exterior of the aeroplane, paying particular attention to wheel bays and technical areas;
- The details provisioned under [Searching and Checking the Aeroplane](#);
- The monitoring of aeroplane maintenance and servicing operations;
- The prevention of unauthorised access to the aeroplane;
- The screening of all passengers, cabin baggage and hold baggage;
- The supervision of the loading of all hold baggage, if applicable, and catering supplies;
- The security procedures for flight and cabin crew including their baggage;
- The review of security communications for the purpose of analysing any intelligence and threat information for its impact upon the flight.

In addition to the normal pre-flight inspections, the crew member doing the pre-flight shall pay special attention to any suspicious or unusual details such as tampering with the cabin or baggage doors, inspection panels, emergency hatches, etc.

10.3.3.1 Security Seals

Before leaving the aeroplane unattended, the aeroplane shall be sealed by the flight crew using red security seals assigned individually to these crew members according to specific instructions stated below with NetJets numbered / controlled security seals provided.

After sealing the aeroplane, the Flight Crew (PIC or SIC) shall pass on the security seals numbers by electronic means using 'WAYPOINT- Security Seals app'. Whenever an NTA aeroplane needs to be accessed by maintenance, aircraft restocking, cleaning personnel, or any other persons authorised by NTA, when leaving the aeroplane, it's doors must be closed, locked, and sealed. After all work orders conducted by the authorised person shall apply the blue security seals (located in the Aircraft Documents Folder) on designated locations per fleet (See below – Placement of Seals) and send the information on the seals applied to NTA maintenance / restock responsible department. If the aircraft is delivered directly to the Flight Crew, the application of the blue security seal is not required.

Seals shall not be removed from one NTA aircraft and applied to another.

The current or next crew scheduled on that aeroplane, prior to accessing the aircraft shall inspect the seals and seal numbers for signs of tampering and confirm these were the last serial seal numbers used, prior entering the aeroplane, using 'WAYPOINT- Security Seals app'.

This information is available 24/7 in 'WAYPOINT- Security Seals app'.

Placement of Security Seals

BD-700-1A10 & BD-700-1A11:

- Main cabin door (Lower right side of the main door);
- Cargo / Baggage compartment doors;
- Fuel service panel door;
- Aft equipment compartment door.

DA-2EASy:

- Main cabin door (Lower right side of the main door);
- Cargo / Baggage compartment doors.

BD-100:

- Main cabin door (Lower right side of the main door);
- Cargo / Baggage compartment doors.

CL650:

- Main cabin door (Lower right side of the main door);
- Cargo / Baggage compartment doors;
- Aft Equipment bay Door.

CE-560XLS:

- Main cabin door (Lower right side of the main door);
- Cargo / Baggage compartment doors.

CE-680A:

- Main cabin door (Lower right side of the main door);
- Cargo / Baggage compartment doors.

EMB-505:

- Main cabin door (Lower right side of the main door);
- Cargo / Baggage compartment doors.

In the event of Security Seals being tampered with, broken or suspicion of unlawful interference, the aeroplane Commander shall contact the Security department who will implement the appropriate security procedures. The crew will carry out a thorough aeroplane search, using as a guide the specific aeroplane checklist, as per OM-B.

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10.3.3.2 Specific Aeroplane Threat

When there is intelligence or other information which indicates that a specific NTA aeroplane may be subject to an act of unlawful interference, NTA in conjunction with the competent Authorities will coordinate all aspects deemed necessary to enable them to respond and implement the additional security measures considered necessary to counter the threat.

NTA will request security guards to conduct apron and aeroplane parking area surveillance and all apron workers briefed to be vigilant and report any suspicious activity immediately.

Before the arrival of an aeroplane under increased threat, a search should be made of the allotted parking area and its surroundings for any unauthorised persons and potential explosive devices which may be concealed within vehicles and ground equipment.

Where a clear and specific risk exists of an attack to a particular aeroplane, the NTA Head of Security will request such aeroplane to be escorted during its journey between the runway and the security parking position by mobile security patrols. The areas immediately adjacent to the runways and taxiways being used by the aeroplane are to be inspected prior to the movement of the aeroplane to ensure that no risk exists in the area. Depending on the nature of the threat, the inspection of the approach and take-off paths outside the airport perimeter may also be required.

Additional security measures are provided for specific flights considered under increased threat. This may include parking the aeroplane in a designated security parking area such as an isolated aeroplane parking position and individual guarding of such aeroplane during the time the aeroplane is on the ground. It may also include supplementary security screening of all passengers and luggage.

10.3.3.3 Operations at Remote Locations

When operating to remote locations with a known security risk, a risk assessment shall be conducted by the NTA Security department. Such flights should not depart before the commander is briefed about the assessment made either by phone, email, fax, or hardcopy.

That should cover country overview, city and airport area, as deemed necessary, and will include sudden changes in political, criminal, environmental, religious, commercial, intelligence, and media reports.

When applicable, crew members may be requested to complete a self-survey for specific destinations or event-driven security survey or incidents.

10.3.3.4 Security Threat Level Index

In order to ensure the effective application of preventative security measures, the level of threat shall be continually reviewed in response to domestic and international events.

NTA operations threat level index at any given location of operations is achieved by a systematic and ongoing process of intelligence collection, collation and evaluation. This is essential in identifying changes in trends, existing and potential threats and other significant developments.

The threat at some risk locations may be lowered by the application of additional mitigating measures, such as individual guarding of the aeroplane during the time the aeroplane is on the ground and crew's security escort.

NTA AVIATION SECURITY THREAT ASSESSMENT LEVEL INDEX

INSIGNIFICANT	Terrorism and or Security Insurgency are unlikely – NTA Operations Approved
LOW	Terrorism and or Security Insurgency possible but not likely – NTA Operations Approved with routine protective security measures.
MODERATE	Terrorism and or Security Insurgency it's a possibility – NTA Operations & Overnight Authorised – Additional and sustainable protective security measures reflecting the broad nature the threat combined with specific business and geographical vulnerabilities and judgements on acceptable risk. <u>Aviation Security Briefing required.</u>
SUBSTANTIAL	Terrorism and or Security Insurgency are highly likely – NTA Operations Approved with no Overnight Authorised – Additional and sustainable protective security measures reflecting the broad nature the threat combined with specific business and geographical vulnerabilities and judgements on acceptable risk. <u>Aviation Security Briefing required.</u>
SEVERE	Terrorism and or Security Insurgency is expected imminently. - NTA Operations Prohibited

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10.4 Passenger and Cabin Baggage

10.4.1 General Provisions

Third parties providing services to NTA are responsible for performing all NTA passengers and cabin baggage screening such as airports, handling agents / FBOs. NTA ensures through the aeroplane Commander and NTA representatives (Handling Agents / FBOs), that baggage is not placed on board an aeroplane unless the following measures are taken: Third countries where the security standards applied are recognised as equivalent to the common basic standards as regards passengers and cabin baggage are listed in Attachment 4-B of the Implementing Regulation.

Passengers and their cabin baggage arriving from a Member State where the aeroplane was in transit after having arrived from a third country not listed in Attachment 4-B of the Implementing Regulation shall be considered as passengers and cabin baggage arriving from a third country, unless there is a confirmation that these passengers and their cabin baggage were screened in that Member State.

References to third countries in this chapter and where applicable in Commission Implementing Decision C (2015) 8005 include other countries and territories to which, in accordance with Article 355 of the Treaty on the Functioning of the European Union, Title VI of Part Three of that Treaty does not apply.

For the purpose of this chapter:

- a. 'liquids, aerosols and gels' (LAGs) shall include pastes, lotions, liquid / solid mixtures, and the contents of pressurised containers, such as toothpaste, hair gel, drinks, soups, syrups, perfume, shaving foam, and other items with similar consistencies;
- b. 'security tamper-evident bag' (STEB) is a bag that conforms to the recommended security control guidelines of the International Civil Aviation Organisation;
- c. 'liquid explosive detection systems (LEDS) equipment' is a piece of equipment capable of detecting threat materials that meets the provisions laid down in a separate Annex to the Commission Implementing Decision.

10.4.2 Screening of Passengers and Cabin Baggage

Third parties providing services to NTA are responsible for performing all NTA passengers and cabin baggage screening such as airports, handling agents / FBOs. NTA ensures through the aeroplane Commander and NTA representatives (Handling Agents / FBOs), that baggage is not placed on board an aeroplane unless the following measures are taken:

- Passengers, or any cabin baggage of a passenger are not permitted to enter or remain on board any aeroplane unless the passenger or cabin baggage in question has been screened at a Domestic Airport. Passengers remaining on board as permitted under [Searching and Checking the Aeroplane](#) of this chapter must, along with their cabin baggage, be subjected to a hand search on board the aeroplane.
- Transfer and transit passengers and their cabin baggage arriving on an aeroplane which has been diverted to an airport which is not its original destination, as a result of a medical, technical, or weather-related emergency, may be subjected to screening.

NTA must notify ANAC of any such emergency within 24 hours of the diversion, if passengers and their cabin baggage are not screened.

10.4.2.1 Screening of Passengers

All departing passengers and supernumeraries (e.g. navigator or in-flight services representatives) on flights where mandatory screening is required and / or increased security threat (e.g. originating and transfer passengers, unless previously screened) shall be screened to prevent prohibited articles from being introduced into the security restricted areas and on board an aeroplane. This may include passengers in wheelchairs, or with orthopaedic or implanted electronic devices.

For passengers with reduced mobility, the nature of the passenger's disability shall be taken into account when choosing the method of screening. If a wheelchair, crutch, stick, or stretcher is being used by the passenger, it shall, if possible, be screened as cabin baggage.

Where a passenger is screened by a hand search, and that passenger is wearing Religious or Cultural Headgear, that Religious or Cultural Headgear may be screened by means of ETD equipment in conjunction with HHMD equipment as an alternative to screening by a hand search.

Transplant organs, human tissue, human blood and cells requiring emergency transportation for medical reasons are exempt from screening subject to the nominated courier providing appropriate certification from a known organisation demonstrating the legitimacy of the items and sufficient identification to establish their identity as a medical courier. The courier and their personal baggage are not exempt from screening.

Appropriate authorities may select at random categories of persons, flights or destinations, as deemed necessary, that shall be subject to special screening procedures or exempted from screening.

Any person who refuses to undergo screening before boarding or entering an aeroplane, where mandatory screening is required, must be denied boarding, and not allowed to pass the point of search. Additionally, any person denied passage for refusal to be processed, or some other security reason, should be referred to the local police authority.

Before screening, outer wear shall be taken off and shall be screened as cabin baggage, unless the concept of operations of equipment allows for outer wear to be kept on. The screener may request the passenger to undertake further divesting as appropriate.

Passengers will be screened by at least one of the following methods:

- a. hand search;
- b. walk-through metal detection equipment (WTMD);
- c. explosive detection dogs (EDD);
- d. explosive trace detection (ETD) equipment;
- e. security scanners which do not use ionising radiation;
- f. ETD equipment combined with hand held metal detection (HHMD) equipment;
- g. shoe metal detection (SMD) equipment;
- h. shoe explosive detection (SED) equipment.

Where the screener cannot determine whether or not the passenger is carrying prohibited articles, the passenger will be denied access to security restricted areas or rescreened to the screener's satisfaction.

Where a live animal is permitted to be carried in the cabin of an aeroplane, it shall be screened either as a passenger or as cabin baggage.

Passengers shall be entitled to opt out from a security scanner. In this case the passenger shall be screened by an alternative screening method including at least a hand search in accordance with the Implementing Regulation.

When a hand search is performed, it shall be carried out so as to reasonably ensure that the person is not carrying prohibited articles.

When WTMD equipment alarms, the cause of the alarm shall be resolved.

Hand-held metal detection (HHMD) equipment may only be used as a supplementary means of screening. It shall not replace the requirements of a hand search.

Explosive detection dogs, ETD equipment, SMD equipment and SED equipment may only be used as a supplementary means of screening. Explosive trace detection (ETD) equipment in combination with hand held metal detection (HHMD) equipment may only be used in cases where the screener considers a hand search of a given part of the person to be inefficient and/or undesirable.

10.4.2.2 Screening of Cabin Baggage

The cabin baggage of all departing passengers and supernumeraries (e.g. navigator or in-flight services representatives) where mandatory screening is required and / or increased security threat, i.e. originating and transfer passengers shall be screened prior to being allowed into security restricted areas and on board an aeroplane.

Any prohibited articles shall be removed from the passenger's possession or the passenger denied access into the security restricted area or the aeroplane, as appropriate.

Before screening, portable computers and other large electrical items shall be removed from cabin baggage and shall be screened separately, unless the cabin baggage is to be screened with Explosive Detection Systems (EDS) equipment meeting standard C2 or higher.

Cabin baggage shall be screened by at least one of the following methods:

- a. a hand search;
- b. x-ray equipment;
- c. explosive detection systems (EDS) equipment;
- d. automated prohibited items detection (APID) software in combination with point (c);
- e. explosive detection dogs in combination with point (a);
- f. ETD equipment.

Where the screener cannot determine whether or not the cabin baggage contains any prohibited articles, it shall be rejected or rescreened to the screener's satisfaction.

A hand search of cabin baggage shall consist of a manual check of the baggage, including its contents, as to reasonably ensure that it does not contain prohibited articles.

Where x-ray equipment is used, each image shall be viewed by the screener.

Where EDS equipment is used, each image shall be viewed by the screener or analysed by automated prohibited items detection (APID) software.

Where APID software is used, all alarms as referred above shall be resolved to the satisfaction of the screener so as to reasonably ensure that no prohibited articles are carried into the SRA or on board an aircraft. Where EDS equipment is used, all alarms as referred to above shall be resolved by screening the baggage again using an additional screening method.

Where EDS equipment has been installed before 1 July 2023 and is used without APID software, all alarms as referred to in point 12.4.1.3 shall be resolved to the satisfaction of the screener so as to reasonably ensure that no prohibited articles are carried into the SRA or on board an aircraft. When the identity of an article is unclear, the alarms shall be resolved by screening the baggage again using an additional screening method.

Where x-ray or EDS equipment is used, any item whose density impairs the ability of the screener to analyse the contents of the cabin baggage shall be taken out of the baggage. The bag shall be screened again and the item shall be screened separately as cabin baggage.

Any bag that is found to contain a large electrical item shall be screened again with the item no longer in the bag and the electrical item screened separately, unless the cabin baggage was screened with EDS equipment meeting standard C2 or higher.

Explosive detection dogs and explosive trace detection (ETD) equipment may only be used as a supplementary means of screening.

Where APID software is used in combination with EDS equipment meeting either of standards C1, C1+, C2 or C2+, the operator or entity using equipment shall ensure that the procedures are in accordance with the concept of operations of these standards as regards screening of large electronic items and screening of LAGs.

When conducting a hand search of cabin baggage, screeners must reconcile each bag to be searched with its owner and assess their behaviour and reaction to the search process. Passengers will be denied unsupervised access to the contents of cabin baggage that has not yet been searched to the screener's satisfaction.

Unless the alarm has been generated automatically and the screener is satisfied that no prohibited articles are present, where X-ray or EDS equipment is used, all alarms are to be resolved as follows:

- a. The cabin baggage must be rescreened by means of a hand search; or
- b. In the case where an item of cabin baggage is thought to contain an explosive device or incendiary device the screener must:
 - i. Ensure that the passenger is not able to access, interfere with, or touch the item;
 - ii. Ensure that the image of the item produced by the X-ray or EDS equipment can be viewed; and
 - iii. Consult a supervisor who must agree further action in relation to the item and the passenger.

Screening of liquids, aerosols, and gels (LAGs)

LAGs carried by passengers may be exempted from screening with EDS equipment upon entry to the SRA if the LAGs are in individual containers with a capacity not greater than 100 milliliters or equivalent in one transparent resealable plastic bag of a capacity not exceeding 1 litre, whereby the contents of the plastic bag fit comfortably and the bag is completely closed.

Before screening, LAGs shall be removed from cabin baggage and shall be screened separately from other items of cabin baggage, unless the equipment used for the screening of cabin baggage is also capable of screening multiple closed LAGs containers inside baggage.

Where LAGs have been removed from cabin baggage, the passenger shall present:

- a. all LAGs in individual containers with a capacity not greater than 100 milliliters or equivalent in one transparent resealable plastic bag of a capacity not exceeding 1 litre, whereby the contents of the plastic bag fit comfortably and the bag is completely closed; and

- b. all other LAGs, including STEBs containing LAGs.

100% of LAGs permitted to be carried into the security restricted area must be screened by LEDS equipment unless they are:

- a. exempt from screening with LEDS equipment pursuant to provision 4.1.3.1 of the Annex to the Implementing Decision; or
- b. are of a quantity no greater than 100ml or equivalent and either:
 - i. they were obtained at an airport or on board an aeroplane and are sealed in a STEB inside which is displayed satisfactory proof of purchase; or
 - ii. they are required for medical purposes or a special dietary requirement, including baby food, to be used during the Trip.

LAGs carried into a security restricted area or on board an aeroplane by persons other than passengers must not be exempted from screening.

LAGs to be used during the Trip, that are required for dietary or medical purposes and that do not fall within the exemption at provision 4.1.3.1(a) of the Annex to the Implementing Regulation, must not be carried into a security restricted area or into the cabin of an aeroplane unless accompanied by documentary proof of authenticity. Where the screener is not satisfied as to the authenticity of such LAGs, they must not be taken beyond the point of screening.

LAGs comprising baby food to be used during the Trip, and that do not fall within the exemption at point 4.1.3.1(a) of the Annex to the Implementing Regulation, must not be taken beyond the point of screening if the passenger is not accompanied by an infant.

The requirement that baby food must not be taken beyond the point of screening if the passenger is not accompanied by an infant does not apply to expressed breast milk.

The expressed breast milk referred above must not be frozen and must be held in a container or more than one container where no container has a capacity exceeding two liters.

10.4.3 Protection of Passengers and Cabin Baggage

In the event of detected or suspected mixing of screened departing passengers and unscreened persons, the following actions shall be taken:

- a. those parts where mixing was detected or suspected shall be cleared and then a search shall be carried out to reasonably ensure that no prohibited articles have been introduced to those parts; and
- b. departing passengers and their cabin baggage present in areas where mixing was detected or suspected shall be screened again.

The above shall not apply to screened departing passengers mixing with:

- a. crew members arriving from a third country departing on the same aeroplane and who have not left the critical part; and / or
- b. transit passengers departing on the same aeroplane.

The above shall not apply if mixing took place due to factors beyond the passengers' control and shall be considered to be met for an aeroplane that is subjected to an aeroplane security search.

Screened departing passengers shall not be mixed with arriving passengers who may not have been screened. Where these passengers cannot be physically separated, then the security objective shall be achieved by the application of compensatory measures in accordance with the assessment of the risk by the competent authority.

10.4.4 Potentially Unruly and Disruptive Passengers

Abusive, aggressive, or violent behaviour which could endanger the safety of the aeroplane, passengers, and crew is not acceptable. Refer to [Procedures for Dealing with Unruly \(Abusive\) Passengers](#) for additional information.

Disruptive passenger will be defined as a passenger who fails to respect the rules of conduct at an airport or on board an aircraft or to follow the instructions of the airport staff or crew members and, thereby, disturbs the good order and discipline at an airport or on board the aircraft.

Unruly passengers are defined as persons who commit, on board a civil aircraft, from the moment when the aircraft door is closed prior to take-off to the moment when it is reopened after landing, an act of:

- a. assault, intimidation, menace, or wilful recklessness which endangers good order or the safety of property or persons;
- b. assault, intimidation, menace, or interference with a crew member in the performance of duties or which lessens the ability to perform duties;
- c. wilful recklessness or damage to an aircraft, its equipment, or attendant structures and equipment such as to endanger good order and the safety of the aircraft or its occupants;
- d. communication of information which is known to be false, thereby endangering the safety of an aircraft in flight; and
- e. disobedience of lawful commands or instructions for safe, orderly, or efficient operations.

In many instances, passengers behave irrationally and do not calculate the consequences of their actions.

Wherever possible, the crew must attempt to control a situation and prevent it from escalating to an extent that the safety and security of the aeroplane is endangered.

Passengers may need to be advised that their behaviour is unacceptable and advised of the consequences should they choose not to moderate it.

The Commander should be able to assess and deal effectively with disruptive / unruly behaviour on board. If the situation cannot be controlled and, or whenever the Commander assesses that a situation will in any way affect the safety of the aircraft, any passenger, and cabin crew, the following should be considered:

- a. Request police / local authority response by meeting the aircraft on arrival;
- b. Divert the aeroplane and land.

In extreme cases, the Company may consider legal actions.

The Commander and / or cabin crew shall submit a written report after the flight in situations as those described above. This report must be addressed to the Security Manager. The Security Manager will then follow the directives as defined by the National Civil Aviation Authority Regulation no. 50/2005.

10.4.5 Transport of Deportees or Persons in Custody

Due to the nature of NTA operations, the transport of deportees or persons in custody does not apply to NTA operations.

10.4.5.1 Restrain Kits

Whenever the aircraft commander considers that a passenger is acting in such a way as to endanger or is likely to endanger the safety of the aircraft, other passengers, property or good order and discipline on board, they may take such appropriate measures, including coercion, against such persons as they deem necessary.

All Netjets aircraft are equipped with bags with restrain kits. These Kits are located in the cockpit of all aircraft. The decision to use them rests solely with the commander. The use of restrain kits is considered in the PNSAC, the Tokyo Convention, ICAO Annex 6, Decree-Law 71 / 84 of February 27 and Commission Decisions.

After the incident and as soon as possible, the aircraft commander must inform Operational Dispatch of their decision and action taken. Operational Dispatch must inform the Head of Security and Security department.

The report for unruly passengers - 'Unruly Passenger Incident Report' is the basis for a formal complaint against the passenger. A copy of this report must be sent to the national authority, and another filed upon arrival of the flight.

On arrival at the destination airport, the commander of the aircraft must notify their intention to disembark a passenger and hand them over to the airport security authorities.

In Portugal, unruly passengers must be handed over to the competent authority at the airport of disembarkation, accompanied by the respective occurrence reports signed by the captain of the aircraft, so that the appropriate criminal or administrative proceedings can be initiated, as applicable, under the terms of the law.

The Security department is responsible for submitting reports of unruly passengers to ANSAC.

Kit of retaining clamps:

- 2 cuffs;
- 1 cuff cutter;
- 1 Instructions for use.

Follow the instructions for use:

1. Once ready to secure the individual, put the cuff on your wrist and approach the unruly passenger from the rear;
2. Grab the unruly passenger's hand. As you pull their hand through the cable tie;
3. Repeat the same procedure with the other hand;
4. Once the cable tie is on, pull tight;
5. Remember to check for tightness regularly. Ensure that they're not causing injury;
6. You should only use the cutter in an emergency or if it is too tight. If it's too tight, handcuff it first before cutting the tight ones.

10.4.6 Prohibited Articles

Passengers shall not be permitted to carry into security restricted areas or on board an aeroplane the articles listed in Attachment 4-C of the implementing regulation.

NTA ensures that passengers are informed of the prohibited articles listed in Attachment 4-C via the NTA Operations Bulletin and the individual flight customer copy.

Example of information on prohibited articles available to the passenger:

ADVISORY NOTES

Aviation Security - Prohibited Articles – Cabin Baggage

In accordance with the relevant implementing regulation, passengers are not permitted to carry into security restricted areas or on board an aircraft the articles listed below:

- guns, firearms and other devices that discharge projectiles — devices capable, or appearing capable, of being used to cause serious injury by discharging a projectile, including firearms of all types, such as pistols, revolvers, rifles, shotguns, toy guns, replicas and imitation firearms capable of being mistaken for real weapons, component parts of firearms, excluding telescopic sights, compressed air and CO2 guns, such as pistols, pellet guns, rifles and ball bearing guns, signal flare pistols and starter pistols, bows, cross bows and arrows, harpoon guns and spear guns, slingshots and catapults;
- stunning devices — devices designed specifically to stun or immobilise, including devices for shocking, such as stun guns, tasers and stun batons, animal stunners and animal killers, disabling and incapacitating chemicals, gases and sprays, such as mace, pepper sprays, capicum sprays, tear gas, acid sprays and animal repellent sprays;
- objects with a sharp point or sharp edge — objects with a sharp point or sharp edge capable of being used to cause serious injury, including items designed for chopping, such as axes, hatchets and cleavers, ice axes and ice picks, razor blades, box cutters, knives with blades of more than 6 cm, scissors with blades of more than 6 cm as measured from the fulcrum, martial arts equipment with a sharp point or sharp edge, swords and sabres;
- workmen's tools — tools capable of being used either to cause serious injury or to threaten the safety of aircraft, including crowbars, drills and drill bits, including cordless portable power drills, tools with a blade or a shaft of more than 6 cm capable of use as a weapon, such as screwdrivers and chisels, saws, including cordless portable power saws, blowtorches, bolt guns and nail guns;
- blunt instruments — objects capable of being used to cause serious injury when used to hit, including: baseball and softball bats, clubs and batons, such as billy clubs, blackjacks and night sticks, martial arts equipment;
- explosives and incendiary substances and devices — explosives and incendiary substances and devices capable, or appearing capable, of being used to cause serious injury or to pose a threat to the safety of aircraft, including blasting caps, detonators and fuses, replica or imitation explosive devices, mines, grenades and other explosive military stores, fireworks and other pyrotechnics, smoke-generating canisters and smoke-generating cartridges, ammunition, dynamite, gunpowder and plastic explosives.

An exemption to the above may be granted on condition that Portuguese National Civil Aviation Authority has approved the article for carriage, in accordance with the NetJets Security Programme; and NetJets has been informed about the passenger and the article that the passenger is carrying prior to passengers boarding the aircraft and the applicable safety rules are complied with. These articles shall then be placed in secure conditions on board aircraft.

Aviation Security - Prohibited Articles – Hold Baggage

In accordance with the relevant implementing regulation, passengers are not permitted to carry in their hold baggage the articles listed below:

- explosives and incendiary substances and devices — explosives and incendiary substances and devices capable of being used to cause serious injury or to pose a threat to the safety of aircraft, including ammunition, blasting caps, detonators and fuses, mines, grenades and other explosive military stores, fireworks and other pyrotechnics, smoke-generating canisters and smoke-generating cartridges, dynamite, gunpowder and plastic explosives.

An exemption to the above may be granted on condition that the appropriate authority has national rules permitting carriage of the article and the applicable safety rules are complied with.

Smart Baggage

Smart Baggage is a bag (rucksack, suitcase or similar) that contains a battery/power bank. This device is used to recharge personal electronic devices (PEDs) i.e. smart phones, tablets, laptops etc. or provides motorized electric propulsion to the suitcase. If the battery/power bank cannot be readily detached from Smart Baggage by the customer, the Smart Baggage will not be accepted on the flight.

Baggage Weight – Hold Baggage

To calculate aircraft loading, NetJets uses an industry standard of 13kg (29lbs) for each passenger's hold baggage. If you plan to take more than this, please advise your Owners Services team in advance.

Operating Company

This flight will be operated by NetJets Transportes Aéreos S.A. – AOC PT-01/02

10.4.7 NTA Security Procedures

Passengers are encouraged to arrive at the airport prior to departure with enough time to clear airport security.

A crew member is to meet the passengers at the terminal at all times, clearly introduce themselves, and confirm their identification with the Company Crew Badge. They should also establish the identification of the passengers prior to escorting them to the aeroplane. The crew shall have a complete passenger manifest prior to departure.

10.4.8 Identification of Passengers

Prior to departure, crews are required to check the ID document of all passengers and ensure they match the document on file used for customs and immigration purposes (e.g. passport or ID card). Within Portuguese territory it is mandatory to present either a valid national ID card or valid passport. For other countries other than Portugal, their national identification rules apply. Where possible this identification shall be actioned using the Waypoint ID Reader functionality.

If the IJET manifest is not complete and ID details are missing, such as name, date of birth, Id number, passport number, and nationality, these details must be gathered by the crew and passed to Customer Services in Lisbon through Flight Dispatch, either electronically or by phone, prior to aeroplane departure. Those details must be entered in the IJET reservation by Customer Services and after details cleared with no security locks/advisories in IJET, Flight Dispatch will release the flight. If any security lock is preventing a flight to be released, security shall be contacted to resolve the security lock.

Crews are required to confirm each passenger ID on each flight and this responsibility cannot be delegated.

Any change, passenger loads or names, to the passenger manifest initially given for a flight shall be reported to Flight Dispatch and confirmed before departure that there is no security warning or lock in the flight and that this can be released. In the case of additional passengers, a further release is required from Flight Dispatch and IJET manifest shall be updated in advance of the flight release.

10.4.9 Baggage Security

The passengers shall positively identify all baggage before it is placed on the aeroplane. No unaccompanied baggage including personal items, parcels, courier items, postal mail, diplomatic mail, or company stores will be allowed on the aeroplane unless authorised by the Security department.

After landing, the crew shall ensure that all passenger baggage are removed and loaded on the correct vehicle. If there is any doubt about where the bags are going, a crew member should stay with the bags until they are sure they are being loaded on to the correct transport.

After all passengers have left the aeroplane, a crew member shall complete an interior walk-through to ensure no passenger bags or any other items (e.g. portable phones, laptop, tablets, etc.) have been left on the aeroplane.

Each crew member is responsible for their own crew luggage. Luggage which has been left unattended shall be rechecked by the respective crew member prior to being loaded on board the aeroplane.

10.4.10 Loss of Crew Items

If Crew identification, uniform, or aeroplane keys are lost or stolen, the affected crew member shall report the loss immediately to the relevant Chief Pilot and to the Security department.

Attachment 4-C PASSENGERS AND CABIN BAGGAGE

List of Prohibited Articles

Without prejudice to applicable safety rules, passengers are not permitted to carry the following articles into security restricted areas and on board an aircraft:

- a. guns, firearms, and other devices that discharge projectiles, except as authorised in accordance with [Carriage of Weapons, War Munitions, and Sporting Weapons](#) - devices capable, or appearing capable of being used to cause serious injury by discharging a projectile, including:
 - firearms of all types, such as pistols, revolvers, rifles, shotguns;
 - toy guns, replicas, and imitation firearms capable of being mistaken for real weapons;
 - component parts of firearms, excluding telescopic sights;
 - compressed air and CO2 guns, such as pistols, pellet guns, rifles, and ball bearing guns;
 - signal flare pistols and starter pistols;
 - bows, cross bows, and arrows;
 - harpoon guns and spear guns;
 - slingshots and catapults.
- b. stunning devices - devices designed specifically to stun or immobilise, including:
 - devices for shocking, such as stun guns, tasers, and stun batons;
 - animal stunners and animal killers;
 - disabling and incapacitating chemicals, gases, and sprays, such as mace, pepper sprays;
 - capsicum sprays, tear gas, acid sprays, and animal repellent sprays.
- c. objects with a sharp point or sharp edge - objects with a sharp point or sharp edge capable of being used to cause serious injury, including:
 - items designed for chopping, such as axes, hatchets, and cleavers;
 - ice axes and ice picks;
 - razor blades;
 - box cutters;
 - knives with blades of more than 6 cm;
 - scissors with blades of more than 6 cm as measured from the fulcrum;
 - martial arts equipment with a sharp point or sharp edge;

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- swords and sabres.
- d. workmen's tools - tools capable of being used either to cause serious injury or to threaten the safety of the aircraft, including:
- crow bars;
 - drills and drill bits, including cordless portable power drills;
 - tools with a blade or a shaft of more than 6 cm capable of use as a weapon, such as screwdrivers and chisels;
 - saws, including cordless portable power saws;
 - blowtorches;
 - bolt guns and nail guns.
- e. blunt instruments - objects capable of being used to cause serious injury when used to hit, including:
- baseball and softball bats;
 - clubs and batons, such as billy clubs, blackjacks, and night sticks;
 - martial arts equipment.
- f. explosives and incendiary substances and devices - explosives and incendiary substances and devices capable, or appearing capable of being used to cause serious injury or to pose a threat to the safety of aircraft, including:
- ammunition;
 - blasting caps;
 - detonators and fuses;
 - replica or imitation explosive devices;
 - mines, grenades, and other explosive military stores;
 - fireworks and other pyrotechnics;
 - smoke-generating canisters and smoke-generating cartridges;
 - dynamite, gunpowder, and plastic explosives.

10.5 Hold Baggage

10.5.0 General Provisions

Third countries where the security standards applied are recognised as equivalent to the common basic standards as regards hold baggage are listed in Attachment 5-A of the Implementing Regulation.

Hold baggage arriving from a Member State where the aeroplane was in transit after having arrived from a third country not listed in Attachment 5-A of the Implementing Regulation shall be considered as hold baggage arriving from a third country, unless there is a confirmation that the hold baggage was screened in that Member State.

References to third countries in this Chapter and where applicable in the Commission Implementing Decision include other countries and territories to which, in accordance with Article 355 of the Treaty on the Functioning of the European Union, Title VI of Part Three of that Treaty does not apply.

Due to the nature of NTA operations off-airport or curbside check-in of hold baggage does not apply to NTA operations.

10.5.1 Screening of Hold Baggage

Third parties providing services to NTA are responsible for performing all NTA hold baggage screening such as airports, handling agents / FBOs. NTA ensures through the aeroplane Commander and NTA representatives (Handling Agents / FBOs), that hold baggage is not placed on board an aeroplane unless the following measures are taken:

NTA ensures via their own means or via the airport service providers/third parties/handling agents / FBOs the proper identification of each single hold baggage piece, before the baggage being submitted to screening, either as accompanied baggage or unaccompanied baggage.

10.5.1.1 Accompanied Hold Baggage

All items of accompanied hold baggage, both originating and transfer hold baggage, unless previously screened, shall be screened by one of the methods before being loaded onto an aeroplane:

- a. A hand search; or
- b. Conventional x-ray equipment; or
- c. Explosive detection systems (EDS) equipment; or
- d. Explosive trace detection (ETD) equipment; or
- e. Explosive detection dogs.

Where the screener cannot determine whether or not the hold baggage contains any prohibited articles, it shall be rejected or rescreened to the screener's satisfaction.

A hand search shall consist of a thorough manual check of the baggage, including all its contents, so as to reasonably ensure that it does not contain prohibited articles.

Where x-ray or EDS equipment is used, any item whose density impairs the ability of the screener to analyse the contents of the baggage shall result in it being subject to another means of screening.

Screening by explosive trace detection (ETD) equipment shall consist of the analysis of samples taken from both the inside and the outside of the baggage and from its contents. The contents may also be subjected to a hand search.

The following categories of hold baggage may be exempted from screening:

- a. Transfer hold baggage which has arrived at the airport on a Domestic Flight and which has been screened in accordance with the Relevant Security Measures provided that, in the case of unaccompanied hold baggage, evidence of screening is held by the air carrier at the transfer airport;
- b. Transit hold baggage;
- c. Baggage which has been screened as cabin baggage in accordance with the Relevant Security Measures before becoming hold baggage.

10.5.1.2 Unaccompanied Hold Baggage

The carriage of unaccompanied baggage or parcels must be authorised by the Security department.

All items of unaccompanied baggage, both originating and transfer hold baggage, shall be screened by one of the methods laid down above under [10.5.1.1](#), before being loaded onto an aeroplane.

Hold baggage that becomes unaccompanied baggage due to factors other than those referred to in the Implementing Regulation shall be removed from the aeroplane and rescreened before loading it again.

The reason that the baggage became unaccompanied shall be recorded before it is loaded onto an aeroplane, unless the security controls as referred to in point above are applied.

Where unaccompanied hold baggage is screened by a hand search, it shall be supplemented by the use of explosive trace detection (ETD) equipment.

Screened cabin baggage placed in the hold of an aeroplane during the boarding process due to space constraints on board the aeroplane shall continue to be considered as cabin baggage.

As an exception, hold baggage that becomes unaccompanied baggage may be exempted from re-screening provided that an appropriate risk assessment is carried out by NTA.

10.5.2 Protection of Hold Baggage

Hold baggage to be carried on an NTA aeroplane shall be protected from unauthorised interference from the point at which it is accepted into the care of NTA until departure of the aeroplane on which it is to be carried.

The following measures shall be taken in protecting hold baggage:

- a. Prior to being loaded, hold baggage shall be held in the baggage make-up area or other storage area of an airport to which only authorised persons may have access.
- b. Any person entering a baggage make-up or storage area without authorisation shall be challenged and escorted out of the area.
- c. Originating and transfer hold baggage shall not be left unattended on the ramp or plane side prior to being loaded on the aeroplane.
- d. Tail-to-tail transfer hold baggage shall not be left unattended on the ramp or plane side prior to being loaded.

- e. Access to lost-and-found offices in the terminal shall be restricted to prevent unlawful access to baggage and materials.

Access to hold baggage which has been screened by the airport operator or accepted into the care of NTA may, in addition to those passengers granted access under point 5.2.1 of the Annex to the Implementing Regulation, also be granted to the following persons, subject to the Security department approval justifying that access is required for a legitimate reason:

- a. employees or agents of NTA or the airport operator;
- b. NTA Crew members about to depart from the airport, provided that they are supervised to ensure that:
 - i. no prohibited articles as listed in Attachment 5-B of the Implementing Decision are introduced into the hold baggage; or
 - ii. no prohibited articles as listed in Attachment 4-C of the Implementing Regulation are removed from the hold baggage and introduced into the security restricted area or on board an aeroplane; and
 - iii. Compliance Authority Personnel.

Hold baggage that has not been protected from unauthorised interference shall be rescreened.

Hold baggage that is in a critical part shall be considered as protected from unauthorised interference.

Hold baggage that is in a part other than a critical part shall be considered as protected from unauthorised interference if:

- a. it is secured baggage; or
- b. it is not left unattended at airports where alternative provisions apply.

When any package (may include dangerous goods), that appears to be damaged or leaking, is found on an aeroplane or on the ramp, the Commander must inform the Airport Authorities and the package shall be removed from the aeroplane, if necessary, and inspected by trained personnel to ensure that it is in a proper condition for transport and that no damage or contamination has occurred to the aeroplane or its load. In case of dangerous goods the safe disposal of the items shall be processed by the Airport Authorities.

Each crew member is responsible for their own crew luggage. Luggage which has been left unattended shall be rechecked by the respective crew member prior to being loaded on board the aeroplane.

10.5.3 Baggage Reconciliation

Identification of hold baggage

NTA ensures, through the aeroplane Commander and NTA representatives (Handling Agents / FBOs), that baggage is not placed on board an aeroplane unless the following measures are taken:

- a. Only baggage belonging to the passenger on board shall be on the flight on which it is to be carried; and
- b. Prior to loading, hold baggage shall be held in an area of the airport to which only authorised persons have access; and

- c. All items of baggage taken into the custody of NTA for carriage in an aeroplane shall be identified as either accompanied or unaccompanied.

The passengers shall positively identify all baggage before it is placed on the aeroplane.

No unaccompanied baggage including personal items, parcels, courier items, postal mail, diplomatic mail, or company stores will be allowed on the aeroplane unless authorised by the Security department. After landing, the crew will ensure that all passenger baggage are removed and loaded on the correct vehicle. If there is any doubt about where the bags are going, a crew member should stay with the bags until they are sure they are being loaded on to the correct transport. After all passengers have left the aeroplane, a crew member shall complete an interior walk through to ensure no passenger bags or any other items, e.g. portable phones, laptop, tablets, etc., have been left on the aeroplane.

A Hold Baggage Manifest must be prepared in respect of each flight containing information, including:

- a. the flight number and date; and
- b. in respect of each item of hold baggage loaded for a flight:
 - i. the baggage tag number or other identifier, approved by the airport authorities, that directly links that item of baggage with the passenger recorded as having placed it in the custody of the air carrier;
 - ii. where the item of hold baggage has been identified by manual means, the flight number and date of the flight;
 - iii. whether the item of hold baggage is accompanied or unaccompanied hold baggage; and
 - iv. in respect of each item of unaccompanied hold baggage confirmation that it has been subjected to the appropriate security controls.

Where hold baggage has been identified by manual means, prior to signing the Hold Baggage Manifest, the appointed person must:

- a. Examine each entry on the Hold Baggage Manifest to ensure that it relates to the relevant flight and fully compliant with the requirements;
- b. Ensure that there is an indication that hold baggage have been recorded on the manifest; and
- c. Compare the number of bags recorded on the Hold Baggage Manifest with the number of hold baggage items registered in the associated baggage acceptance records.

The appointed person must sign the Hold Baggage Manifest prior to take-off of the aeroplane, provided the appointed person is satisfied that all hold baggage loaded onto the aeroplane has been subjected to the appropriate security controls and authorised for carriage on the flight.

The appointed person must:

- a. sign each separate page of the Hold Baggage Manifest; or
- b. create and sign a summary that shows separately:
 - i. the flight number and date of the flight;
 - ii. the number of accompanied hold bags authorised for carriage that have been identified by manual means;

- iii. the number of unaccompanied hold bags authorised for carriage that have been identified by manual means; and
- iv. a declaration confirming that all reasonable steps have been taken to ensure that hold baggage has been appropriately identified and recorded on the Hold Baggage Manifest and that unaccompanied hold baggage has been appropriately screened.

In circumstances where no unaccompanied hold baggage has been loaded on to the relevant flight, the Hold Baggage Manifest must include a statement to that effect.

No item of hold baggage can be placed on board the aeroplane after the manifest has been signed without the authority of the appointed person.

Where an item of unaccompanied hold baggage has been placed on board an aeroplane, a copy of the completed Hold Baggage Manifest must be made available to the Commander of the aeroplane.

The following records must be retained for each flight and remain accessible at the airport of departure for a period of seven days:

- a. the completed Hold Baggage Manifest; and
- b. the Departure Control System records, or other records that confirm the number of items of hold baggage checked in for a flight.

10.5.4 Prohibited Articles

NTA ensures that passengers are informed of the prohibited articles listed in Attachment 5-B through the NetJets Operations Bulletin and the individual flight customer copy.

Passengers may be permitted to carry ammunition in their hold baggage provided that carriage is in compliance with national regulations (The Air Navigation (Dangerous Goods) Regulations 2002 (SI 2002/2786) as amended by SI 2015/970.) enforced by the PT CAA and that approval has been granted by NTA.

An article listed in paragraphs (a) - (e) of Attachment 4-C to the Implementing Regulation, may be permitted to remain within accompanied hold baggage belonging to, and in the possession of, member of the Crew in a security restricted area, provided that:

- a. The crew member has arrived at the screening facility with other crew members who are intending to travel on the same flight;
- b. The airport operator is satisfied that the carriage of such an article appears to be reasonably required by the crew member; and
- c. The airport operator has received verbal assurance from the senior member of the crew, that the bag containing the article in question will be:
 - i. Directly supervised by a person other than the crew member whose bag contains the article and at all times the bag is in a security restricted area;
 - ii. Taken directly to the aeroplane by vehicle and placed in the hold; and
 - iii. Similarly supervised at the arrival destination.

10.5.4.1 Attachment 5-B - List of Prohibited Articles

Passengers are not permitted to carry the following articles in their hold baggage:

- explosives and incendiary substances and devices - explosives and incendiary substances and devices capable of being used to cause serious injury or to pose a threat to the safety of aircraft, including:
 - ammunition;
 - blasting caps;
 - detonators and fuses;
 - mines, grenades, and other explosive military stores;
 - fireworks and other pyrotechnics;
 - smoke-generating canisters and smoke-generating cartridges;
 - dynamite, gunpowder, and plastic explosives.

10.6 Cargo and Mail

Not applicable to NTA operations.

10.7 Air Carrier Mail and Materials

Before being loaded into the hold of an aeroplane, air carrier mail and air carrier materials must be subjected to the measures set out in [10.5.1 Screening of Hold Baggage](#).

Before being loaded into any part of an aeroplane other than the hold, air carrier mail and air carrier materials must be subjected to the measures set out in [10.4.2 Screening of Passengers and Cabin Baggage](#).

Spare parts for aeroplane transported as air carrier material are not exempted from security controls.

Due to the nature of NTA operations, NTA does not hold check-in counters at airports therefore the protection of air carrier materials used for passenger and baggage processing (i.e. to surplus ticketing or computers, unattended check-in computers and / or surplus ticketing destruction) does not apply to NTA operations. Airport / FBO serving NTA hold that responsibility.

10.8 In-flight Supplies & Cleaning

10.8.1 General Provisions

Unless otherwise stated, NTA or an NTA regulated supplier of in-flight supplies shall, as appropriate, implement the measures set out in this Chapter and the corresponding provisions in the Implementing Regulation and Implementing Decision.

NTA requires catering companies to ensure that catering, stores and supplies placed on board NTA aeroplanes do not contain any prohibited articles that could endanger the safety of the aeroplane.

10.8.2 Security Controls

Any in-flight supply received showing signs of being tampered with, or where there is reason to believe that it has not been protected from unauthorised interference from the time that controls were applied, must be screened.

10.8.2.1 Screening

All in-flight supplies must be screened. Before receiving in-flight supplies from a regulated supplier, NTA must obtain confirmation of the regulated supplier status of the originating supplier.

The security programme of the supplier must be validated by a person acting on behalf of the regulated supplier or air carrier who has been appointed and trained as a Security Manager in accordance with point 11.2.5 of the Annex to the Implementing Regulation.

Premises used for the preparation and / or storage of catering, stores, and supplies intended to be carried on board a NTA aeroplane should be secured and access should be controlled at all times; all doors, windows or any other means of access into the buildings should be secured and/or access controlled. Access control measures are applied at all times to prevent access by unauthorised persons into the catering premises.

Any vehicles which are used for the transportation of catering supplies, stores and supplies to and from the aeroplane from companies located outside the security restricted area should be locked or sealed.

The seals on the delivery vehicle, if applicable, should be checked against appropriate documentation (vehicle dispatch certificate) by the designated person. If there is any discrepancy with the accompanying documentation that cannot be resolved or any signs of interference with the seals, the load should be regarded as 'unknown' and should not be taken on board an aeroplane.

Where there is any reason to believe that in-flight supplies to which security controls have been applied have been tampered with or have not been protected from unauthorised interference from the time that those controls were applied, they shall be screened before being allowed into security restricted areas.

Prior to delivery of the catering at the aeroplane, the catering should be checked by the crew for any signs of interference and to ensure it contains no prohibited articles.

When screening in-flight supplies, the means or method employed shall take into consideration the nature of the supplies and shall be of a standard sufficient to reasonably ensure that no prohibited articles are concealed in the supplies.

The following means or method of screening, either individually or in combination, shall be applied:

- a. visual check;
- b. hand search;
- c. x-ray equipment;
- d. EDS equipment;
- e. ETD equipment in combination with point (a);
- f. explosive detection dogs in combination with point (a);
- g. EVD equipment applied in accordance with the relevant provisions contained in Attachment 6-J of the EU Regulation 2015/1998 and in combination with point (a).

Where the screener cannot determine whether or not the item contains any prohibited articles, it shall be rejected or rescreened to the screener's satisfaction.

A visual check shall consist of a thorough visual check of the supplies and shall only be allowed:

- a. in combination with other methods; or
- b. where all parts of the supplies can actually be seen, with or without aids; or
- c. where supplies are of a nature that makes the concealment of prohibited articles impossible; or
- d. where ETD or EDD is not available and the size, weight or nature of the in-flight supplies does not permit the use of x-ray equipment, EDS equipment or a hand search. In this case the visual check shall include a careful examination of the packaging for signs of tampering and where items are consolidated, each individual item shall be checked.

A hand search shall consist of a thorough manual check of the supplies.

Catering, stores, and supplies present a possible means for the introduction of weapons, explosives, and other prohibited articles on board an aeroplane. Due to specific characteristics of NTA type of operations, it is not always possible to be supplied by a regulated supplier and / or a known supplier.

10.8.3 NTA Cleaning, Stores, and Supplies

Entities and / or suppliers providing NTA aeroplane cleaning services should ensure that security measures are implemented to prevent the introduction of weapons, explosives, and other dangerous devices into aeroplane cleaning supplies intended for carriage on NTA flights. These measures should be detailed in the cleaning company security programme.

All deliveries of cleaning supplies to be used for NTA aeroplane cleaning should be broken down as appropriate prior to being used on board an aeroplane to ensure that they do not contain any prohibited articles, and then held in secure conditions thereafter.

Deliveries which cannot be broken down (i.e. inter alia blankets, duvets, napkins, towels, headsets, sky cots, etc.) should be sealed or made tamper-evident by the originator of the goods. If there is any sign of interference or tampering, a hand search of the products or items to ensure they do not contain any prohibited articles should be conducted.

Whenever cleaning services are required, the crew must check the airport identification cards of the aeroplane cleaning personnel.

After deliveries of cleaning supplies, the crew should check the supplies on a random basis to ensure that they do not contain any prohibited articles and, if sealed, that there are no signs of interference.

Any item of cleaning supplies showing evidence that it may have been subject to unauthorised interference should be regarded as 'unknown'. It should either be searched by hand to ensure that it does not contain any prohibited articles or it should not be carried on an aeroplane.

When receiving supplies from a company that is not a regulated supplier or known supplier, the supplies shall be 100% screened in accordance with the relevant Implementing Regulation.

Whenever an NTA aeroplane needs to be accessed by restocking or cleaning personnel, if authorised by NTA, when leaving the aeroplane, doors are to be closed and sealed. The aeroplane should not be left unattended.

■ For placement of Security Seals please refer to [10.3 Aeroplane Security](#).

10.9 Airport Supplies

Not applicable to NTA operations.

10.10 In-flight Security Measures

10.10.1 Flight Crew Compartment Security

At all times, including during a flight, no person other than the flight crew members assigned to a flight shall have admission to or be carried in the cockpit, unless that person is:

- a. An operating crew member;
- b. A representative of the competent or inspecting authority, if required to be there for the performance their official duties; or
- c. Permitted by and carried in accordance with the instructions contained in the OM-A.

The commander shall ensure that:

- a. admission to the flight crew compartment does not cause distraction or interference with the operation of the flight; and
- b. all persons carried in the flight crew compartment are made familiar with the relevant safety procedures.

The commander shall make the final decision regarding the admission to the flight crew compartment.

NTA aeroplanes do not fall under the requirements set under the implementing regulation (EU) 2019/1387 - ORO.SEC.100 Flight crew compartment security — aeroplane.

Please refer to section [8.3.12 Admission to Flight Crew Compartment \(FCC\)](#) for additional information.

10.11 Staff Recruitment and Training

10.11.1 National Aviation Security Training Programme

ANAC developed and implemented a National Civil Aviation Security Training Programme to enable aircrew and ground personnel to implement aviation security requirements and to respond to acts of unlawful interference with aviation.

10.11.2 NTA Security Training Programme

Taking into consideration the relevant EU and National regulations and standards practices, recognised and recommended by international organisations, NTA developed an Aviation Security Training Programme, requiring certification by the National Civil Aviation Security Authority (ANSAC).

This Security Training Programme was created for its implementation at a company level, as a regulatory and crucial contribution to the protection of civil aviation. In order to provide aviation security training in a practical way, the Security Training Programme is a fundamental part for the implementation of a security and safety mindset within civil aviation, disseminated through NetJets.

The training programme falls within the framework of compliance with the requirements of the National Training Programme in Civil Aviation for Security (PNFSAC) developed by the appropriate National Civil Aviation Security Authority (ANSAC).

The programme, despite the direct applicability of PNFSAC under the Regulation (EU) No. 300/2008 and its implementing acts, aims to regulate and adapt the provisions of the Regulation (EU) No. 2015/1998 of the European Commission.

10.11.3 Recruitment

The following personnel shall have successfully completed an enhanced background check:

- a. persons being recruited to implement, or to be responsible for the implementation of screening, access control or other security controls in a security restricted area;
- b. persons with general responsibility at national or local level for ensuring that a security programme and its implementation meet all legal provisions (security managers);
- c. instructors;
- d. EU aviation security validators.

The following personnel shall have successfully completed an enhanced or a standard background check:

- a. Persons being recruited to implement, or to be responsible for the implementation of, screening, access control or other security controls elsewhere than a security restricted area;
- b. Persons having unescorted access to air cargo and mail, air carrier mail and air carrier material, in-flight supplies and airport supplies to which the required security controls have been applied;
- c. Persons having administrator rights or unsupervised and unlimited access to critical information and communications technology systems and data used for civil aviation security purposes as described in 1.7.1 in accordance with the national aviation security programme, or having been otherwise identified in the risk assessment in accordance with 1.7.3 of the EU Regulation 2015/1998 .

Unless otherwise specified in the EU Regulation 2015/1998 Regulation, whether an enhanced or a standard background check has to be completed shall be determined by the appropriate authority in accordance with applicable national rules.

In accordance with the applicable rules of Union and national law, an enhanced background check shall at least:

- a. Establish the person's identity based on documentary evidence;
- b. Cover criminal records in all states of residence during at least the preceding 5 years;
- c. Cover employment, education, and any gaps during at least the preceding 5 years;

- d. Cover intelligence and any other relevant information available to the competent national authorities that they consider may be relevant to the suitability of a person to work in a function which requires an enhanced background check.

Subject to the second paragraph below, a person's identity must, for the purposes of a background check, be established on production of one of the following original documents:

- a. establish the person's identity on the basis of documentary evidence;
- b. cover criminal records in all states of residence during at least the preceding 5 years;
- c. cover employment, education and any gaps during at least the preceding 5 years;
- d. cover intelligence and any other relevant information available to the competent national authorities that they consider may be relevant to the suitability of a person to work in a function which requires an enhanced background check.

In accordance with the applicable rules of Union and national law, a standard background check shall:

- a. establish the person's identity on the basis of documentary evidence;
- b. cover criminal records in all states of residence during at least the preceding 5 years;
- c. cover employment, education and any gaps during at least the preceding 5 years.

Enhanced background checks shall be considered as failed if not all the elements specified in points above are completed satisfactorily or if at any point in time these elements do not provide the necessary levels of assurance as to the reliability of the individual.

Background checks shall be subject to the following:

- a. a mechanism for the ongoing review of the elements specified in points 11.1.3 and 11.1.4 of the Regulation EU 2015 / 1998 through the prompt notification to the competent authority, operator or issuing entity, as applicable, of any occurrence that may affect the reliability of the individual. Modalities for the notification, exchange of information and content thereof between the competent authorities, operators and entities, shall be established and monitored in accordance with the national law; or
- b. a repeat at regular intervals not exceeding 12 months for enhanced background checks, or three years for standard background checks. Written confirmation must be obtained from employers, educational establishments or other sources capable of verifying the information provided by the person for the purpose of the background check.

Where a background check reveals that the person was self-employed, either wholly, or in part, during any of the period covered by the check, written confirmation of the dates of the periods of self-employment must be obtained from independent accountants, solicitors, or from the relevant government department or agency. Recruitment records are stored electronically for the duration of the contract with NTA. A standard background check or points (a) to (c) of an enhanced background check shall be completed before the person undergoes initial security training involving access to information which is not publicly available due to its security sensitivity. Where applicable, point (d) of unenhanced background check shall be completed before a person is allowed to implement, or to be responsible for the implementation of, screening, access control or other security controls.

Whenever an enhanced background check is required, it shall be fully completed before the person undergoes the training referred to in point 11.4. of the Regulation EU 2015 / 1998.

10.11.4 Training

Taking into consideration the relevant EU and National regulations and standards practices, recognised and recommended by international organisations, NTA complies with the requirements of the Approved Security Training Programme levels 6,7, 8, 13, 14,19, 20 and 21 as per the National Civil Aviation Security Training Programme. The NTA aircrew is directly responsible to conduct security searches for all NTA flights.

10.11.5 Certification or Approval

Certification or approval records shall be kept for all persons certified or approved, respectively, for at least the duration of their contract.

10.11.6 Recurrent Training

Recurrent training, where this is required for the persons performing tasks listed, must be delivered to at least the frequencies defined under the Security Training Programme.

10.11.7 Qualifications of Instructors

Instructors shall fulfil all of the following requirements:

- a. the successful completion of a background check in accordance with points 11.1.3 and 11.1.5 of the implementing regulation;
- b. competency in instructional techniques;
- c. knowledge of the work environment in the relevant aviation security field;
- d. competency in the security elements to be taught.

Certification and, where appropriate, re-certification of instructors must be obtained in accordance with the requirements of the relevant approved Security Training Programme.

Certificated instructors must deliver training courses in accordance with the delivery conditions set out in the relevant approved Security Training Programme.

10.12 Carriage of Weapons, War Munitions, and Sporting Weapons

10.12.1 Transport of Weapons and Munitions of War

Aeroplane operated by NTA shall not carry weapons and munitions of war, unless as specified below in [10.12.3](#) and [10.12.4](#).

Weapons of war and munitions of war can only be carried provided an approval to do so has been granted by the Security Department, and all the states concerned before a flight. The transportation of such weapons in the hold luggage is only permitted if an authorised and fully certified staff has confirmed that they are unloaded. They must be carried in the aircraft in a place which is inaccessible to passengers during flight and, in the case of firearms, unloaded, except as specified [10.12.2](#).

10.12.2 Stowage Requirements for Munitions of War

In exceptional circumstances, weapons of war and munitions of war may be carried other than in an inaccessible place on the aircraft and may be loaded, provided an approval to do so has been granted by the Security Department and all the States concerned before a flight.

The Security Department will notify the commander before a flight if weapons of war or munitions of war are to be carried on the aeroplane.

10.12.3 Bodyguards

Some of NTA owners are individuals whose status makes it necessary for them to be accompanied by bodyguards. Passengers should not be permitted to carry into security restricted areas or on board an aircraft items listed in Appendix 4-C of Regulation 2015-1989 (LIST OF PROHIBITED ITEMS - Passengers and Cabin Baggage).

According to Regulation UE 2015/1998 and ANAC Security Instruction No. 30712, an exemption from the provisions of point 4.4.1 of UE 2015/1998 regulation may be granted if:

- a. The competent authority has authorised the carriage of the item in question; and
- b. The air carrier has been informed of the passenger and the item being carried prior to boarding the aircraft; and
- c. Applicable security standards are met.

Upon all approvals set in this chapter are granted, when boarding the aeroplane, bodyguards on escort duty must proceed / be instructed as follows:

- a. Before boarding, the weapon(s) and their ammunition must be handed over in the boarding area, in the presence of the police, who will escort them to the aircraft;
- b. The firearm(s) and their ammunition mentioned above should preferably be transported in an area inaccessible to passengers. If NTA transports them in the hold, they must be properly packaged to ensure their safety and prevent loss;
- c. After landing, the weapon(s) and their ammunition will be handed over to their owner, still in the restricted security area, when applicable, in the presence of a member of the airport police.

10.12.4 Accredited Government Close Protection Officers

According to ANAC instruction no. 30712, accredited government close protection officers, while in Portuguese territory, must proceed as follows:

- a. Before boarding the aeroplane, at the boarding gate, all guns and ammunitions shall be handed in the presence of the escorting police (Police escorting to the aeroplane);
- b. Guns and ammunition should preferably be transported in the cockpit. If elected to be transported in the hold compartment, these should be properly packed making sure they are secure and assuring no person can access the hold compartment;
- c. After landing, all guns and ammunition should be returned to their owner / user while still in the security restricted area when applicable, in the presence of an airport police officer.

When not in Portuguese territory, accredited government close protection officers may retain control of their weapons with authorisation from the NTA Security department. In such cases the Commander will be informed of the retention of the weapons by the Close Protection Officers and made aware of any special procedures.

10.12.5 Carriage of Sporting Weapons and Amunition

As the operator of fractionally-owned aeroplanes, NTA frequently carries aeroplane owners, family members, and / or their guests who travel specifically to recreational venues which involve the carriage of sporting weapons and ammunition. Additionally, many of the aeroplane

types operated by NTA do not have baggage compartments external to the aeroplane cabin. In such aeroplane, baggage is stored in specific areas designed for the purpose within the confines of the cabin. In order to accommodate the need by owners or their guests to transport sporting weapons and ammunition, and acknowledging that the risk associated with such carriage, under these circumstances, is unrelated to the comparable risks of such carriage in general public transport operations, aeroplanes operated by NTA may carry such items, but only under the following conditions:

1. The weapons shall be for sporting or competitive purposes only. Firearms used for personal protection, other than those carried by bodyguards or accredited government close protection officers, do not fall within these provisions.
2. The weapons and ammunition shall be the personal property of the passenger and properly licensed as necessary by the passenger's country of residence.
3. Weapons shall be unloaded and stored in proper gun-cases or bags. Baggage should be placed in a baggage compartment.
4. Ammunition shall be properly packed in the passenger's baggage and the baggage placed in a baggage compartment.
5. The first choice of baggage compartment shall be external or inaccessible to the passengers in flight. If such a compartment is not available, place the ammunition-carrying baggage in a separate compartment than the one in which the weapons are stored, if available.
6. Ammunition may only be carried when securely boxed in quantities of 5 Kg or less, and corresponding to the latest Technical Instructions for the Safe Transport of Dangerous Goods by Air (ICAO Doc 9284- AN/905), including the Supplement and any Addendum, approved and published by decision of the council of the ICAO.
7. Transportation shall be approved by the NTA Security department and all the states concerned before a flight.
8. All Dangerous Goods regulations are met (see Chapter 9 - Dangerous Goods and Weapons for further instructions).
9. What is described for the transport of arms and ammunition in the points above does not exempt the full compliance of the rules for the transport of arms and ammunition imposed by each State where the NTA is operating, namely with regard to the entry and exit control of arms and ammunition by the national authorities, from each State.

10.12.6 Notifications

NTA and its customers must observe all regulations applicable to the export, import, and transit of weapons and ammunition applicable in the country of departure, transit, and destination.

Whenever any type of firearm is carried on board an NTA aeroplane, a special services note is to be entered by Owner Services in the reservation.

Ground Services and / or International Planning, as applicable, is to inform the relevant Airports / FBOs of the transportation of the firearms.

Upon receiving the approval / denial from the relevant Airports / FBOs, the reservation is to be updated and Owner Services informed of the course of action.

The Security department will brief the Commander if a passenger wishes to transport any firearm or ammunition if this is not previously stated on the reservation flight details, and will provide the necessary clarification on how to handle this transportation.

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The Commander of an NTA flight must notify the Security department if a passenger wishes to transport any firearm or ammunitions, if this is not previously stated on the reservation flight details.

10.13 NTA Contingency Planning

10.13.1 Description of Plans to Deal with Contingencies

10.13.1.1 Aeroplane Hijack

This chapter establishes procedures which enable crew members to act in the most appropriate manner to minimise the consequences of acts of unlawful interference.

After assessing the situation, the commander should communicate the circumstances concerning the threat to an air traffic services unit. However, serious consideration should be given to the consequences which may occur as a result of any action being taken by air traffic services or the competent authority of a State, that is contrary to the requests of the commander.

The Tokyo Convention specifies the powers of the commander while in flight and authorises specific action to be taken by the commander in the event of an unlawful act. In addition, the commander shall have command over all crew members including any security guards. If the commander is unable to do so, another qualified member of the crew should assume command, in accordance with the sequence provided for in the NTA Operations Manual. All crew members under direction of the commander shall assist them in every way.

Full account should be taken of the possibility that a person committing an unlawful act aboard an aeroplane in flight may be in an unstable state of mind or extremely volatile or nervous. Crew members should therefore:

- a. Refrain from unnecessary conversation or actions which may irritate the individual.
- b. Exercise due care to ensure that the passengers do not become excited or, in an attempt to overcome the individual, resort to action which may jeopardise safety. One practicable measure to restrain passengers would be to instruct them to fasten their seat belts.
- c. Crew shall not serve any alcoholic beverages.

Of prime importance to the safe conduct of the flight is the security of the flight deck area. Once the unlawful seizure has commenced, an attempt should be made to dissuade the offender from accessing the flight deck area.

If plain language is permitted, the flight crew will give the status of flight to ATC and as much information as deemed helpful. It should be stated whether violence has taken place on board.

Switch transponder to code 7500, the international code to signal a hijack. If not under radar control, transmit in an air / ground message the call sign followed by 'Transponder code seven five zero zero'. ATC will acknowledge receipt of code 7500 by asking if this is intentional. The Commander need only to reply 'affirmative' and proper authorities will be notified.

Switch transponder code from 7500 to 7700 to signal 'situation appears desperate, need all possible assistance'. This will alert all concerned that intervention may be required upon landing. If not under radar control, verbally transmit call sign followed 'transponder seven seven zero zero'.

When changing from 7500 to 7700 wait until code 7500 is acknowledged by the ATC. ATC will also acknowledge receipt of code 7700.

If in spite of adherence to all security procedures, a suspect person should succeed in entering the flight deck, try to prevent the seat of an active crew member being occupied by the hijacker. Use careful but firm behaviour.

Unless physically threatened, remain on the flight deck at all times.

If at all possible establish discreet Air-Ground Communications by:

- The Commander using headsets if possible;
- Ensuring that the Commander speaker is OFF;
- Tuning no.2 transceiver to the emergency frequency 121.5 MHz or frequency specified by the ATC;
- The second in command (and third flight deck crew member if available) monitoring the no.1 transceiver on en-route ATC frequency;
- The second in command speaker being ON at all times to give the impression to the hijacker that they are monitoring all air ground communications. This procedure may allow the Commander to receive messages not monitored by the hijacker.

After landing, pull the cockpit voice recorder circuit breaker to deactivate the voice recorder. Retention of the voice recorder tape may be useful in identifying and convicting a hijacker.

Further reference by voice communications from ATS units about the situation should consider the effect such transmissions will have on escalating or aggravating the situation.

Following an act of unlawful interference on board an aeroplane, the Commander will report all details to the Security department.

The Head of Security will submit, without delay, a report of such an act to the designated local Authority and to the Portuguese Civil Aviation Authority, according to the relevant procedures.

In order to prevent hijacking and sabotage, a set of precautionary measures has been established which can be varied according to the actual threat situation by decision of NTA, in cooperation with the local police and a competent authority.

Confidential information pertaining to security measures in force can be obtained from the Head of Security.

The Commander may require stronger precautionary measures than generally prescribed, if the actual situations so warrant. Special impromptu measures, however, decided by the Commander should always be discussed with the Head of Security prior to enforcing them.

In principle, aeroplane doors shall not be opened after the aeroplane has left the parking position.

In the event a request to open any aeroplane door is received via radio outside the parking area, the aeroplane shall return to the parking position and the doors shall be opened in the presence of handling agent staff. If aeroplane doors are forcibly opened, the control tower shall immediately be informed and the aeroplane must return to the parking position for investigation.

The flight crew compartment door, if fitted, shall be capable of being locked from within the compartment in order to prevent unauthorised access. The door should be opened from the cabin with a key stowed at a specified place in the cabin only known to the cabin crew.

In the event of the company being informed of critical political situations by the authority of the State and where an aeroplane is parked outside during night stop, then special security measures shall be undertaken. The Security department will inform the crew accordingly.

Generally the aeroplane must be fully be locked and secure in accordance with the procedures outlined in NTA Operations Manual and the Aeroplane Flight Manual during night stops, and / or when the aeroplane is left unattended.

The Company has established training programmes for relevant personnel. They cover the prevention of acts of unlawful interference or seizure of aeroplanes and are detailed in the OM-D.

10.13.2 Bomb Threat

Bomb threats are normally anonymous telephone calls which may provide information indicating the intention to commit an act of terrorism, or sabotage, against an aeroplane or airport installation. Written warnings and / or warnings by fax are rare but should be assessed using the same techniques as applied to a telephoned warning.

Bomb threats are usually intended to cause nuisance and can arise in different forms. Some are sufficiently vague not to warrant any special measures, whereas others may provide significant amounts of detail, against which a threat may be assessed to determine its relative credibility.

Threat warnings may be received by airports and / or air carriers either directly from the people issuing the warnings or from intermediaries (for example, press agencies or the media). In either case, the recipient should attempt to obtain as much information as possible about the threat in order to facilitate assessment of the threat and identification of the person(s) issuing it.

Occasionally the Head of Security may receive threat warnings directly, based on intelligence from domestic or foreign intelligence agencies, Authorities, and Police.

Bomb threats can be made against an aeroplane in the following operational phases:

- On the ground, before boarding of passengers or crew;
- On the ground, but after boarding of passengers and crew and before take-off;
- After take-off and during climb phase;
- During the flight cruise phase;
- During the descent phase; or
- After landing, but before passengers and crew have disembarked.

Main Potential Targets

Typically, but not limited to, threat warnings may be directed against or via:

1. Aircraft
 - a. On the ground; or
 - b. In the air.
2. Airports & Airline Facilities
 - a. Installations, facilities, and buildings;
 - b. Maintenance, Ground Support, and in-flight supplies.

10.13.2.1 Phases of Response

This section details the different phases of the response to bomb threats as follows:

- Receipt of the bomb threat call 10.13.2.1.1;
- Report of the bomb threat call 10.13.2.1.2;
- Threat assessment 10.13.2.1.3;
- Action to be taken 10.13.2.1.4.

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10.13.2.1.1 Receipt of the Bomb Threat Call

Anyone working at NTA may receive such calls. Procedures shall, therefore, be as simple as possible and distributed as a short list of instructions which will allow any of these people, when confronted with this situation, to get a minimum of information which could be useful for further analysis of the threat.

During the call, the recipient should try to follow the guidelines given in the NTA Bomb Threat Checklist (below), also available in the NTA Security Programme, having in mind all important characteristics of the call.

These instructions must be known by every employee who may be concerned, and they must be subject to periodic reminders by the management.

BOMB THREAT CHECKLIST

ACTIONS TO BE TAKEN ON RECEIPT OF A BOMB THREAT

- LISTEN WITHOUT INTERRUPTING THE CALLER AND WITHOUT PROVIDING INFORMATION
- TAKE NOTE OF THE NUMBER SHOWING ON YOUR PHONE DISPLAY, IF VISIBLE
- IF AVAILABLE, SWITCH ON THE PHONE RECORDER OR VOICEMAIL
- USE ANY POSSIBLE MEANS TO LENGTHEN THE CONVERSATION AS MUCH AS POSSIBLE
- IF POSSIBLE, WITHOUT DISTURBING THE CALLER, CALL A SUPERVISOR OR A COLLEAGUE TO LISTEN TO THE CALL AS WELL WHILE THE CALLER IS ON THE LINE
- DO NOT DESTROY YOUR PERSONAL NOTES AFTER THE CALL IS OVER
- AFTER THE CALL IS OVER DO NOT HANGUP THE PHONE
- RECORD BELOW THE EXACT WORDING OF THE THREAT:

ASK THE FOLLOWING QUESTIONS

Where is the bomb right now?

When is it going to explode?

What does it look like?

What kind of bomb is it?

What will cause it to explode?

Did you place the bomb?

Why?

What is your name?

If the bomb has been placed on an aeroplane:

A/C Registration A/C Type Location

RECORD TIME CALL COMPLETED

DATE & TIME (AM or PM)	PHONE NUMBER	DURATION OF THE CALL
		minutes

SEX OF CALLER (Male or Female)	AGE OF CALLER (approx.)	NAME (If Possible)

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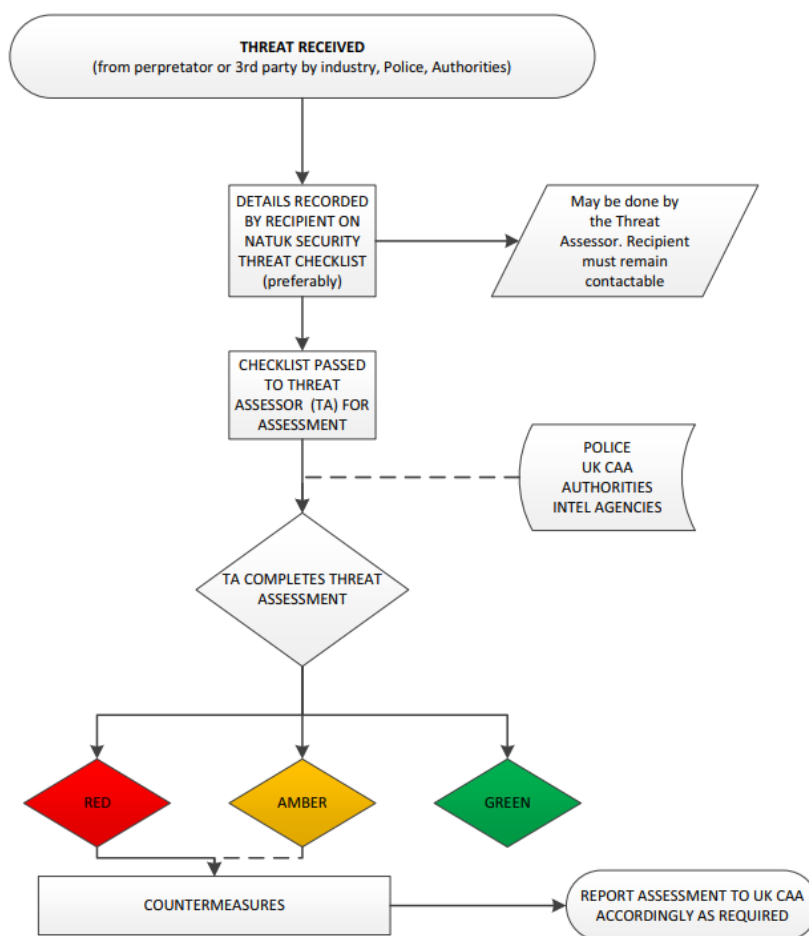
BOMB THREAT CHECKLIST		
TICK WHERE APPROPRIATE		
CALLER'S VOICE	THREAT LANGUAGE	BACKGROUND SOUNDS
☐ Calm	☐ Well-spoken	☐ House noises
☐ Crying	☐ Irrational	☐ Street noises
☐ Angry	☐ Taped message	☐ Animal noises
☐ Nasal	☐ Offensive	☐ Bar/Restaurant
☐ Slurred	☐ Incoherent	☐ Motor/Engines
☐ Clearing throat	☐ Message read by threat-maker	☐ Clear
☐ Excited	☐ Uneducated	☐ Voices
☐ Stutter	☐ Foul Language	☐ Static
☐ Disguised	☐ Intoxicated	☐ PA system
☐ Slow		☐ Booth
☐ Lisp		☐ Music
☐ Rapid		☐ Factory machinery
☐ Deep		☐ Office machinery
☐ Hoarse		☐ Other (specify) _____
☐ Laughter		
☐ Familiar (if so, whose voice did it sound like?) _____		
Did the caller seem familiarized with airports, airplanes, and/or aviation phraseology?		
☐ YES ☐ NO If YES please specify _____		
OTHER REMARKS		
Date: _____		
Signature: _____		
PRINT NAME: _____		
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10.13.2.1.2 Report of the Bomb Threat Call

The person who has received the call shall report the call as accurately as possible without delay to the immediate supervisor or manager, who will relay it to the Head of Security / Threat Assessor. For this purpose, the recipient shall detail the entire content of the call and, to such an extent as possible, provide all information which they are able to gather by conforming to the instructions for the receipt of bomb threat calls given in the Bomb Threat Checklist (above).

10.13.2.1.3 Threat Assessment

Threat Assessment Process



This threat assessment process is designed to lead the Head of Security / Threat Assessor to a logical assessment of the threat considering:

- Whether a specific target has been identified and / or;
- Whether specific details about the intended method of attack have been provided; these two elements should then be brought together in a single assessment of the level of detailed information the threat contains;
- Whether any of the above information can be and has been verified or discredited; and
- Whether there is any relevant background data that adds to or subtracts from the overall credibility; these two elements should then also be brought together to form an assessment of the credibility of the threat;

- e. Finally, the relevant security measures in place that would mitigate the chances of such an attack being carried out should be listed and an assessment should be made of their effectiveness or, conversely, any compliance or operational issues.

The Head of Security / Threat Assessor will first take all actions to initiate the threat assessment process.

Any potential threat received will be assessed, using the NTA Threat Assessment method utilising the Positive Target Identification (PTI) system, conducted by the Head of Security certified as Threat Assessor. In their absence, the Head of Security will appoint a certified Threat Assessor to replace them. Upon conducting the threat assessment, depending on the threat level categorised for the event, the Security Response Plan will be activated accordingly.

GREEN LEVEL

Assessed based on information to indicate strongly that the threat is a hoax e.g. part of a sequence of spurious warnings, where the predicted timing has already passed, nature of the call, etc.

AMBER LEVEL

Threat of doubtful credibility but where it is prudent to consider taking additional protective measures e.g. augmented security checks, especially if there are concerns about the effectiveness of existing countermeasures.

RED LEVEL

A credible and specific threat requiring immediate protective measures, e.g. Diversion if in flight or evacuation if on the ground./ Credible threat relating to a specific target, likely to involve danger.

Threat Classification

GREEN - A non-credible threat; no further action required.

AMBER - A threat of doubtful credibility but where it is prudent to consider taking additional protective measures e.g. augmented security checks, especially if there are concerns about the effectiveness of existing countermeasures.

RED - Security & Emergency plans activation - A credible and specific threat requiring immediate protective measures, e.g. diversion if in flight or evacuation if on the ground. Credible threat relating to a specific target, likely to involve danger.

10.13.2.1.4 Actions to be Taken

Bomb Threat In Flight In the event of a bomb threat in flight, responsibility for the decisions taken rests with the Commander.

Bomb Threat In Flight shall be considered that the aeroplane is taking-off or has taken off within the last five minutes, and shall only be contacted after passing 5000 ft.

In dealing with a bomb threat directed against an NTA aeroplane in flight, some or all of the following actions are to be taken led by the Head of Security in collaboration with the commander:

- response / decision, including whether reaction will be overt or covert;
- conduct of search procedures in flight;
- evaluation of possible effects of pressurisation or pressure equalisation;
- establishment of the order of search;
- arranging of search assignments for flight crew;
- announcement of the threat by the commander to the passengers;

- arranging for possible passenger participation in search;
- establishment of emergency procedures if a suspect explosive device is located; and
- determination of flight routing and landing and deplaning procedures.

If the aeroplane is taking-off or has taken-off within the last five minutes, the Head of Security or their deputy or Flight Dispatch will inform the aeroplane (only after passing 5000 ft) that a bomb threat call has been received which relates to that flight, using the following phraseology (for example):

- 'Fraction 1432 from (location), this is a security message, are you ready to copy?'
- 'The following message was received from (location) at hhmm GMT, quote, 'A bomb threat call has been received at hhmm which may relate to your flight, and which we believe is credible', unquote.'

If an aeroplane is taxiing to its take-off position, advise ATC to instruct the aeroplane to hold short of the take-off point due to security requirement.

Should a Bomb Threat be received in flight, actions to be taken by the commander will depend on the phase of flight when the warning is received, location of the aeroplane in relation to possible diversion airfields, built up areas and advice offered by the Head of Security or their deputy in association with the threat, as well as many other variables.

The commander may decide to divert to the nearest suitable airfield. In such circumstances, the commander should, unless already committed to a landing, consider avoiding over flight of densely populated areas and may be guided by the following:

- An emergency should be declared to ATC stating the nature of the emergency. The aeroplane should be diverted to the nearest airfield at which it can land safely, either civil or military. Identification of an appropriate airfield should be discussed with the Head of Security or the ATC.

Aeroplanes in flight should not be advised of any threat which has been assessed as GREEN. The only exception is where an aeroplane calls for an assessment of a threat received in flight. Such a threat might take the form of discovery of a message on board, or after a message has been received via ATC.

In those cases where the bomb threat assessment indicates that it is unlikely that there is a bomb on board the aeroplane, but where the Head of Security or the Commander wishes to be prudent - before committing, for example, to an extended flight across water - the aeroplane shall be diverted to an airfield at which it can land safely. Once it has landed, consideration should be given to taking the actions described above for threats against an aeroplane on the ground.

Suspect Explosive Device / Security Search

In case of a potential bomb threat, the immediate response and prime objective should be to land as soon as possible, taxi to a remote site and deplane.

However, under circumstances where the flight route is distant from a suitable airport and an immediate landing cannot be accomplished, and if instructed by the Head of Security and / or through the ATC an EOD (Explosive Ordnance Disposal Technician) a security search looking for suspect objects can be initiated taking into consideration the below details (follow OMB's Search Checklist where applicable as reference).

The commander will be instructed in the event that a security search is called for, and only on those circumstances a security search in-flight should be initiated.

Since suspect explosive devices can take many forms, it is impossible to provide clear procedures to deal with the situation. Elements to be considered may include:

- Time;
- Pressurisation - If direct control of the pressurisation is desired, then it will be necessary to switch the pressurisation to 'Manual' and use the Manual Toggle Valve' (Cherry Picker) for control;
- Aeroplane Altitude (Descend to max 10,000 ft);
- Proximity to populated areas.

The person responsible for the search teams should be briefed to look for unidentified objects or anything:

- that should not be there;
- that cannot be accounted for;
- that is out of place.

How to Search

The commander will designate a crew member to perform the search. The search should use the checklist below and the barrier search principle which is a thorough, systematic, and sectional search of the aeroplane starting at the front (i.e. flight deck) and working to aft., from the floor upwards, leaving behind a sterile area. As each section is searched, it should then be considered sterile and no items should be moved forward. When the final aft section is searched, the search is completed. If a suspect device is found, the search cannot be considered as completed and it should continue until 100% of the area is completed, potentially looking for secondary / additional devices.

Sections to be searched in the event that an in-flight Suspect Device Search is called for are:

1. Cockpit and Cockpit Compartments;
2. Galleys;
3. Passenger Cabin;
4. Toilets and Lavatories.

In general, a search should be conducted with extreme caution, using flashlights and mirrors when searching in confined or dark areas.

Areas to be searched should include all quick-latch or easy access doors. Equipment areas should be searched carefully as there is the possibility that a bomb may be hidden in, behind or under equipment.

Discovery of a Suspect Explosive Device in Flight – (Crew Actions) In the event that a suspect device is discovered, then the advice contained in [10.13.3 Discovery of a Suspect or Prohibited Article](#) should also be consulted - as a general rule a device should not be disturbed unless you are instructed otherwise.

Primary actions upon discovery of a suspect device are to:

- Communicate with ATC to seek advice on suspect device and authorisation to divert;
- Adjust altitude (10,000 ft or less);

- Maintain 1 PSI differential cabin pressure;
- Restrict manoeuvres to a minimum;
- Do not cut or disconnect any wires;
- Switch off all non-essential power in the cabin;
- If advised to do so by the EOD expert, move the suspect device in the attitude found to the LRBL.

If an immediate landing can be made, the item should be left in place, covered with polythene or other waterproof material, if available, and then packed around with pillows, blankets, coats, and other blast absorbent materials. The item itself must be kept dry (hence the waterproof material) but surrounding materials should be wet in order to reduce the risk of fire.

Searchers should not, as much as possible, touch suspicious objects but mark the location with a prominent object (but not touching the device) and withdraw from the immediate area to await further instructions. When a suspect device is discovered, the possibility of secondary devices should not be discounted as previously mentioned.

LRBL (Least Risk Bomb Location) - Location per Fleet

Phenom – Front Right Window nearest to the front bulkhead.

Challenger – MED - Main Entry Door.

Citation XL – One Window, Right or Left, Mid Forward Cabin.

Latitude – One Window, Right or Left, Mid Forward Cabin.

Falcon – RHS – Right Hand Side Window Mid Forward Cabin.

Global – MED - Main Entry Door.

LRBL (Least Risk Bomb Location) - Procedures

After choosing the primary location to place the suspect item (One window, mid-forward cabin, left or right), take the following steps:

- Place the suspect item as close to the window (LRBL) as possible;
- Cover the item with a waterproof material;
- Place a wet towel or blanket over the item;
- Pack over and around it with soft furnishings, such as blankets, cushions, etc.;
- Move any flammables, gas cylinders or metal objects away from the area;
- Move all passengers to the back of the aeroplane and land as soon as possible.

Bomb Threat On The Ground

When a bomb threat or warning is associated with an NTA aeroplane that is still on the ground, and once the warning has been assessed as credible, the NTA Head of Security in consultation with airport authorities will coordinate the following tasks:

- Disembark all passengers and crew with all cabin baggage;
- Move the aeroplane to a remote location such as an isolated parking position;
- Isolate and re-screen all passengers and their cabin baggage and hold them in a separate area until the crew members, hold baggage, and catering supplies have been inspected / screened, searched, and declared safe;

- Unload all hold baggage and require passengers to identify their baggage, which should then be screened or searched before it is reloaded (when applicable).
- Coordinate the search the aeroplane by appropriate authorities.

If the authorities are carrying out the search without crew involvement, then the suspect device security search checklist (if available under each OM-B) should be provided to them to assist with their search.

10.13.3 Discovery of a Suspect or Prohibited Article

When dealing with an incident of a suspect explosive device there are basic measures which if followed will help in protecting life and limiting damage to property. A general rule at NTA is that any suspect bag, package left unattended, or abandoned must only be opened by its owner or, if necessary, in the owner's absence, by an explosive disposal expert.

NTA policy regarding this issue is:

- always take any threat call seriously and handle with priority;
- regard any unidentified or abandoned package or bag as a potential suspect explosive device;
- immediately alert the Head of Security and, if necessary, the Airport Authorities or the FBO;
- evacuate the area in a direction away from and without approaching the suspect object;
- ensure the finder of the suspect explosive device is available for the arrival of the bomb disposal team in order to provide them with any useful information.

Do not:

- touch, manipulate, or move the suspect object;
- throw water or any other substance on the object;
- cover the object;
- remain in the immediate vicinity of the object nor allow others to do so;
- use radio equipment in the vicinity of the suspect device; or
- produce any sound, heat, or vibrations in the vicinity of the object.

In addition to the above guidelines, if a suspect explosive device is discovered on board an aeroplane, the Commander should direct that:

- the suspect object is not moved or opened;
- passengers should be moved as far away as possible, instructed to fasten their seat belts and keep their heads below the tops of the seat backs;
- portable oxygen, bottles of alcohol, and first aid kits should be removed from the vicinity. Fire extinguishers should be readily available.

10.13.3.1 Equipment Failure

Not applicable.

10.13.4 Enhanced Measures for an Increase in the Level of Threat

The security measures detailed in this chapter are provided to assist in determining the response to changes in a specific threat to NTA. These are intended to complement the additional measures within the context of contingency planning for emergencies, alert situations, and a general increase in the threat.

Measures are recommended for implementation at each one of three different levels of threat. These are:

GREEN ALERT - BASELINE MEASURES

In general terms this can be described as a normal threat situation where, whilst there is no intelligence to indicate that NTA has been targeted for attack, there is always the possibility of unlawful interference by groups and individuals about which there is no intelligence or from criminals, the mentally disturbed, disaffected employees, and the general public. At this level baseline measures need to be in place at all times to manage the possible risk.

AMBER ALERT - INTERMEDIATE MEASURES

Intelligence indicates the probability that a particular NTA aeroplane and / or a particular airport or airports have been targeted for attack. At this level baseline security measures should be enhanced to manage the increased risk.

RED ALERT - HIGH MEASURES

Intelligence indicates that an NTA aeroplane and / or airport premises and / or facilities have been specifically targeted for attack. At this level baseline security measures should be further enhanced to manage the increased risk.

The level of threat to civil aviation is determined by the appropriate authority as is the determination of countermeasures. The measures detailed herein are for guidance only suggesting the kind of enhanced measures appropriate authorities may like to consider at each level of threat. The determination of enhanced measures should depend on:

- the nature and degree of the threat to which a given operator or airport is subjected according to the threat assessment;
- the expected or anticipated duration of an enhanced threat situation;
- the layout of facilities and the organisation of all different parts of ground handling, at a given airport;
- security staff and equipment available;
- security standards which are currently applicable to flights and airports affected by an enhanced threat;
- the number of flights and passengers, the volume of baggage and cargo subject to enhanced threat procedures.

10.13.5 Security Sensitive Flights

Security measures may have to be provided for specific flights considered under increased threat. When an NTA flight is considered under increased threat, depending upon the nature of the threat, some of the security measures which should be considered for selective application to protect passengers, the public, and terminal facilities may include:

- Thorough aeroplane security searches before flight.

- Use runways or taxiways that cross roads to which the public has unrestricted access, the locations should be manned by security guards.
- Parking the aeroplane in a designated security parking area such as the isolated aeroplane parking position and individual guarding of such aeroplane during the time the aeroplane is on the ground.
- The size and frequency of apron and aeroplane parking area mobile security patrols should be increased and all apron workers briefed to be vigilant and report any suspicious activity immediately.
- Before the arrival of an aeroplane under increased threat, a search should be made of the allotted parking area and its surroundings for any unauthorised persons and potential explosive devices which may be concealed within vehicles and ground equipment.
- Where a specific risk exists of an attack on a particular aeroplane, it should be escorted during its journey between the runway and the security parking position by mobile security patrols. The areas immediately adjacent to the runways and taxiways being used by the aeroplane should be inspected prior to the movement of the aeroplane to ensure that no potential attackers are in the area. Depending on the nature of the threat the inspection of the approach and take-off paths outside the airport perimeter may also be required.
- Coordination with local policing authorities and emergency services to provide immediate assistance in the event of a security incident at or in the local area of the airport.
- Increasing the level of surveillance at NTA facilities.
- Provision of additional security awareness training to all staff.
- Depending on the nature of the sensitivity of a flight, the crew may be escorted from the airport to their hotel and back.

10.13.6 Transport of Inadmissible Passengers

NTA does not normally carry passengers who fall within this category. In the event, however, that circumstances at some time oblige the company to deal with such passengers, the following notes will provide guidance. The current National regulation concerning such events, must be fully complied.

Inadmissible Passengers (INADS)

Passengers who are refused admission to a country by authorities of such country, i.e. due to lack of a visa, expired passport, lack of funds, or suspected intent to illegally take up employment unless explicitly ordered otherwise by the authority refusing entry, an INAD shall be carried outbound again on an NTA flight to a country of their choice where there is no risk of being refused entry again.

10.13.7 Intoxicated Passengers

Especially in an emergency situation, an intoxicated passenger can be a danger to themselves or others.

Crew members shall monitor passengers for signs of intoxication during boarding. The Commander and, if required, the NJE Chief Pilot shall be informed in case passengers appear to be intoxicated and decide whether the passenger is refused carriage. In flight, no alcoholic beverages shall be served to passengers who have been deemed to be intoxicated.

Suggesting to a passenger that they may be under the influence of alcohol can potentially result in adverse passenger behaviour, in which case the Commander shall be notified. The procedures in this section are applicable in cases of undesirable passenger behaviour influenced by any psychoactive substance – not just alcohol.

10.14 Security Incident Reporting

This chapter identifies the types of security-related incidents that must be reported to the Security department or alternatively to Flight Dispatch via the 'Hot line' and sets out the means by which reports are to be made, in accordance with the ERP. It does not address the question of response or follow-up incidents.

10.14.1 Objectives

The objectives of the security-related incidents reporting procedures describe are to:

- Ensure that the correct people, departments, and agencies are provided with timely reports following incidents that are or may be detrimental to NetJets;
- Ensure that security incidents and losses are reported promptly and fully so that, where necessary, external agencies can be notified;
- Ensure that relevant incidents are reported promptly and correctly so that corrective actions, identifying root causes, to prevent a recurrence may be taken;
- Enable the security department to adjust the security measures necessary to meet the threat to NTA operations by analysing reported incidents.

10.14.2 Responsibilities

NTA shall investigate and report certain types of security-related incidents to ANAC appropriate authorities, or to the police. This may be verbal by phone or e-mail, if verbally notified this will be followed by a report in writing via e-mail or courier.

Head of Security: The Head of Security is responsible for investigating by whatever means are most appropriate and expedient with relation to the type of incident. Including, interviewing, and obtaining eye witness reports, security agency reports, and photo / video evidence. Report all security-related incidents to Senior Executives / Management via email, and also to ANAC and other relevant authorities. Unless local procedures require immediate reporting to the police, all reporting to agencies outside NTA is to be by, or after consultation, with the Head of Security.

Reports related with Potentially Unruly and or Disruptive passengers should follow ANAC Regulation No. 50/2005 directives.

Line Managers: Line Managers are responsible for establishing reporting procedures within their areas of responsibility that will ensure the requirements of paragraph [10.14.3 Reporting Security Incidents](#) below are met.

Staff and Agents: Every NTA employee has a responsibility to report any incident defined as 'reportable' (see [10.14.3.1](#)).

10.14.3 Reporting Security Incidents

Definition: 'Any security, asset protection or loss incident or event, whether of a personal or business nature, considered by the originator or observer to be sufficiently serious to prejudice the safety, security, profitability, or good name of NetJets'.

10.14.3.1 Reportable Incidents

Reportable incidents are categorised into two groups. Examples of the types of incident in each of these groups are in Annex A and B. The lists are not exhaustive. Where there are any doubts about the categorisation or reportability of an incident advice should be sought from the Head of Security.

ANAC and relevant authorities, if applicable, will be informed in writing of any of the security-related incidents reported on Annex A of [10.14.3.2](#).

10.14.3.2 Reporting Methods

Annex A: Incidents of the type listed shall be reported immediately to the Head of Security, and after the immediate contact is done, submit an Security report via the online security reporting system.

Annex B: Incidents of the type listed are to be reported and shall be submitted by the end of the working day or operational shifts in which the incident occurs via the online security reporting system or email to Lis-Security@netjets.com.

Written Reports: Initial reports made verbally are always to be submitted using the Security Report Form, available on the online security reporting system for adequate follow-up.

Annex A

- Declaration of War.
- Attack against airports, NTA aeroplanes, or premises.
- Bomb threats.
- Other threats (e.g. extortion / letter threats).
- Kidnapping of any NTA employee.
- Injury to passengers or NetJets staff as a result of a security incident.
- Any discovered breach of security processes, particularly discrepancies in passenger baggage reconciliation discovered after an aeroplane has departed.
- Loss from or of valuables, baggage, or diplomatic mail.
- Loss of notified firearms, including sporting weapons and ammunition.
- Discovery of un-notified firearms, including sporting weapons.
- Discovery of explosives, explosive devices, detonators, or incendiary devices.
- Security-related incidents involving unruly or disruptive passengers on board an aeroplane, whether in-flight or not. Unauthorised access to NTA aeroplane (e.g. tampering of security seals).
- Hostile or damaging media activity.
- Tests or inspections by Government bodies that are critical of NTA security and have a direct impact on the aeroplane operation.
- Security related incidents involving NTA owners when on board NetJets aeroplane, at FBOs, or being transported by an NTA third party service provider organised by NTA.
- Civil unrest in the Airport Networks where NTA operates.

- Suspicious or malicious damage to an aeroplane.
- Incidents including arrests implicating NTA contracted security personnel.

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Annex B

- Unauthorised access to areas controlled by NTA or its agents.
- Illegal access to other areas used by NTA.
- The taking into custody of an NTA employee whilst in the course of their business for an offence which, if proven, is liable to be punished by heavy fine or imprisonment.
- Threats against NTA staff.
- Loss from or of diplomatic mail, uniforms, ID documents, or security passes.
- Loss or misappropriation of NTA documents.
- Routine tests, Security audits or inspections conducted by national auditors or inspectors.
- Loss or misappropriation of NTA owners property, if travelling or being transported by NetJets.
- Security related incidents implicating NTA staff.
- Fire on NTA premises.
- Malicious damage to NTA premises or property.

10.14.3.3 Crime on Board

If a crime is committed on board during flight, the Commander is responsible for safeguarding the necessary evidence. This may include a search of clothes and belongings, or the arrest of suspects. In urgent cases they may arrange for a preliminary inquiry until officials can take over.

The Commander must notify the next airport in advance. After landing, they must report the incident to the ground staff for relay to the local competent police authorities. Upon return to home base, the Commander must forward a written report to the Head of Security, detailing events and actions taken. A copy shall be forwarded to the DO.

10.14.3.4 Security Reports Feedback, Follow-up, and Corrective Actions

Reports must be assessed and analysed to ascertain its accuracy, weaknesses, and value. The findings and subsequent recommendations should be addressed in workable and timely solutions. These findings and feedback of corrective actions should be communicated to the originator of the report via their email or in person.

10.15 Security Quality Assurance

Security Quality Assurance (SQA) is an auditing function that is conducted to ensure a means for monitoring the security and quality of services provided by external suppliers and subcontractors.

SQA refers to the planned and systematic activities implemented and followed by the Security Department in order to guarantee that quality and security requirements are being fulfilled. Being a systematic measurement, comparison with the standards, monitoring of processes and the associated feedback loop, confers prevention.

SQA activities i.e. audits and surveys, are performed by the Head of Security, or by a Security Quality Assurance team member delegated by the Head of Security to ensure that the functional areas meet regulatory requirements and established NTA security standards and procedures.

SQA activities results are presented to the Accountable Manager, by the Head of Security for periodic review and action.

10.15.1 Purpose

The performance of the SQA provides the Head of Security with unbiased information that can be used to:

- Identify areas of opportunity for continuous improvement to ensure zero security mishaps;
- Evaluate actual or potential risks to NTA operations;
- Provide input for business decisions, so that security concerns and costs be prevented or rectified;
- Provide visible commitment to security, safety and quality;
- Assure compliance according with the EU and UK implementing regulations, ICAO recommended practices, ECAC policy stated as Doc.30.

10.15.2 Security Quality Assurance Applicability

Security Quality Assurance defines a scope of auditing that encompasses all areas of the organisation that impact operational security. The SQA ensures a means of monitoring external suppliers that conduct operational functions for the NTA in terms of operational security (i.e. monitoring airports, FBOs, Hotels, Ground Transportation Vendors, etc.).

To achieve effective external monitoring, the following features are designed:

- Appointment of the Head of Security to oversee the program, as the nominated person for all Aviation Security matters;
- Operational independence of the SQA function;
- Head of Security responsibility to implement corrective actions & measures as defined in chapter 1 of the programme;
- Direct line of communication and reporting to the Accountable Manager.

10.15.3 Security Quality Control

The Security Quality Control (SQC) mechanism includes:

- Formal performance monitoring security audits. Security Audits are conducted on a yearly basis, or as required when performance issues are highlighted. The audit looks at compliance with legislation, regulations, operating rules and the NTA Security Programme.
- Security Surveys. These are conducted based on the NTA Security Quality Control Plan. Security Surveys are conducted when required, or to a maximum of 24 months of any previously surveyed location, to monitor compliance of security standards.
- Monitoring of processes and staff performance.

Further details on the SQA can be found under the Security Programme.

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Chapter 11 - Handling, Notifying and Reporting Accidents, Incidents and Occurrences and Using the CVR Recording

11.1 Post-Serious Incident / Accident Guidelines for Crew Members

1. If possible, attend to the immediate needs of the passengers, fellow crew members, and any other (injured) people.
2. Call +351 21 446 9090 (Hotline). NTA will assist with medical, legal, media, and personal support.
3. Secure the aeroplane and all its contents, if possible. An alternative is to handover the aeroplane to an appropriate authority, normally the police or officers from the accident investigation organisation.
4. Contact your families as soon as possible, but only tell them the following:
 - You are OK;
 - How to contact the crew;
 - They may be contacted by the media. They should refer all third-parties to NTA.
5. Review this checklist as a crew and take care of each other. Preferably stay together.
If this is not possible, coordinate whereabouts among the crew members.
6. Be aware that each crew member may be significantly affected by shock and symptoms may be delayed. Even if all crew members are uninjured, it is advisable to request medical assistance for all the crew. This will give each member time to recover mentally, as well as physically, away from media, etc.
7. Do not release any information except where needed to save life. If required, state 'I do not feel able to answer your questions at the moment'.
8. Do not hesitate to contact the hotline for further help or advice.

11.2 Definitions

Accident. An accident is an occurrence associated with the operation of an aeroplane which takes place between the time any person boards the aeroplane with the intention of flight until such time as all such persons have disembarked, in which:

1. A person is fatally or seriously injured as a result of:
 - being in the aeroplane; or
 - direct contact with any part of the aeroplane, including parts which have become detached from the aeroplane; or
 - direct exposure to jet blast, except when injuries are from natural causes, selfinflicted, or inflicted by other persons, or when the injuries are to stowaways hiding outside the areas normally available to the passengers and crew; or
2. The aeroplane sustains damage or structural failure which adversely affects the structural strength, performance or flight characteristics of the aeroplane; and would normally require major repair or replacement of the affected component, except for engine failure or damage, when the damage is limited to a single engine, (including its cowlings or accessories), to propellers, wing tips, antennas, probes, vanes, tyres, brakes, wheels,

fairings, panels, landing gear doors, windscreens, the aeroplane skin (such as small dents or puncture holes) or minor damages to main rotor blades, tail rotor blades, landing gear, and those resulting from hail or bird strike (including holes in the radome); or

3. The aeroplane is missing or is completely inaccessible. An aeroplane is considered missing when the official search has been terminated and the wreckage has not been located.

Fatal injury. Fatal injury means an injury which is sustained by a person in an accident and which results in their death within 30 days of the date of the accident.

Incident. An incident means an occurrence, other than an accident, associated with the operation of an aeroplane which affects or could affect the safety of operation.

Occurrence. An occurrence means an operational interruption, defect, fault or other irregular circumstance that has or may have influenced flight safety and that has not resulted in an accident or serious incident.

Serious incident. A serious incident means an incident involving circumstances indicating that there was a high probability of an accident and is associated with the operation of an aeroplane, which takes place between the time any person boards the aeroplane with the intention of flight until such time as all such persons have disembarked.

Serious injury. Serious injury means an injury which is sustained by a person in an accident and which involves one of the following:

- hospitalisation for more than 48 hours, commencing within 7 days from the date the injury was received;
- a fracture of any bone (except simple fractures of fingers, toes, or nose);
- lacerations which cause severe haemorrhage, nerve, muscle or tendon damage;
- injury to any internal organ;
- second or third degree burns, or any burns affecting more than 5% of the body surface.

11.3 Accident and Serious Incident - Handling and Notification

In order to meet the aims of providing immediate assistance to victims and threatened persons, and to minimise damage, it is of the highest importance that, whoever is the first to know of an accident or a serious incident (ATS or other authority, Commander, handling personnel, or others) shall notify the appropriate search and rescue services (i.e. rescue coordination centre and fire-fighting services / medical services) by the quickest means available as well as the Hotline (+351 21 446 9090) in order to activate NTA's Emergency Response Plan (ERP). The ERP Teams will provide any aid or information, requested or unsolicited, which may prove helpful in containing injuries and damages.

Regardless of action already taken by the authorities, NTA will endeavour to inform local authorities (i.e. police departments, accident investigation boards), and medical services (hospitals, ambulances, doctors).

NTA will be represented by the ERP's On-Site Team. Additionally, NTA may seek further support from the handling agent nearest to the place of occurrence. Notification should be sent to ANAC / GPIAAF in the shortest time and always within 6 hours of the occurrence time.

Contact details are:

Autoridade Nacional Da Aviação Civil (ANAC)

Rua B, Edifícios 4, 5 e 6

Aeroporto da Portela, n.º 4

1749-034 Lisboa

Portugal

Tel.: +351 21 284 22 26

Internet: www.anac.pt

Email: reportedeocorrencias@anac.pt

Gabinete de Prevenção e Investigação de Acidentes com Aeronaves e de Acidentes Ferroviários (GPIAAF) - Unidade da Aviação Civil

Praça Duque de Saldanha nº 31, 4º andar

1050-049 Lisboa

Portugal

Tel.: +351 212 739 230

Online: www.gpiaaf.gov.pt

Tel. (accident / incident reporting 24/7): +351 915 192 963

Email: occreport@gpiaaf.gov.pt / geral@gpiaaf.gov.pt

Upon being notified, all handling agents, and all other company units (i.e. crew scheduling, flight dispatch, maintenance and engineering) or any other persons holding documentation and other evidence relating to the flight, aeroplane / crew / passengers / load concerned shall seize and safeguard all such material.

NTA shall immediately report the accident to the appropriate local authorities. This immediate report shall, as far as possible, contain the following details:

- Name;
- Location and function of the reporting person;
- Date and time of the accident;
- Location of the accident;
- Phase of operation;
- Aeroplane type, registration, call-sign;
- Name of the operator, i.e. NTA;
- Name of the Commander;
- Purpose of the flight;
- Departure / destination aerodromes;
- Number of crew / passengers;
- Description of injury to persons or damage to property;
- Type of occurrence;
- Details of any dangerous goods carried.

As soon as possible, the Commander shall submit a safety report to the Safety department, who shall in turn immediately notify ANAC and GPIAAF.

Please refer to the Emergency Response Plan manual on the public drive for further details about the NetJets emergency response procedures.

11.4 Procedures for Verbal Notification

The commander is responsible to ensure that the following notifications are made:

- Notify ATS immediately, and thereby pilots of other aeroplanes, when immediate hazards are found to exist (ACAS RA, bird hazard, dangerous goods, failure of ground installations, extraordinary meteorological phenomena and other hazardous conditions);
- Notify ATS immediately, and thereby the appropriate aerodrome authority and pilots of other aeroplanes, and NTA, when their aeroplane, after suffering an incident / occurrence, constitutes a hazard to other aeroplanes, to vehicles, persons or property;
- Notify the appropriate local authority without delay in the event of any emergency situation that necessitated action in violation of local regulations and / or procedures;
- Notify the nearest authority, by the quickest available means, of any accident or serious incident resulting in injury, death, or substantial aeroplane damage.

11.5 Occurrence Reporting

- a. NetJets shall report to ANAC, and to any other organisation required to be informed, any accident, serious incident and occurrence as defined in Regulation (EU) No 996/2010 of the European Parliament and of the Council and Regulation (EU) No 376/2014.
- b. Without prejudice to point (a) NetJets shall report to ANAC and to the organisation responsible for the design of the aircraft any incident, malfunction, technical defect, exceeding of technical limitations or occurrence that would highlight inaccurate, incomplete or ambiguous information contained in the operational suitability data established in accordance with Regulation (EU) No 748/2012 or other irregular circumstance that has or may have endangered the safe operation of the aircraft and that has not resulted in an accident or serious incident.
- c. Without prejudice to point (a), NetJets shall report to ANAC, and to any other organisation required to be informed, any information security incident or vulnerability that may represent a significant risk to aviation safety as defined in requirement IS.I.OR.230 of Commission Implementing Regulation (EU) 2023/203.
- d. Without prejudice to Regulation (EU) No 996/2010 and Regulation (EU) No 376/2014, the reports referred in points (a) and (b) shall be made in a form and manner established by ANAC and shall contain all pertinent information about the conditions known to the operator.
- e. Reports shall be made as soon as practicable, but in any case within 72 hours of NetJets identifying the condition to which the report relates, unless exceptional circumstances prevent this.
- f. Where relevant, NetJets shall produce a follow-up report to provide details of actions it intends to take to prevent similar occurrences in the future, as soon as these actions have been identified. This report shall be produced in a form and manner established by ANAC.

11.6 Safety Reporting Forms

This section details the forms to be used for reporting all types of accident, incident and occurrence, instructions on how they are to be completed, the addresses to which they should be sent and the time allowed for this to be done.

NJE Safety Report

This report is available to all NetJets employees.

In accordance with NTA's Safety Policy this report shall be used by all NetJets employees to report any flight safety issue or event that may affect the safety of NetJets' customers, employees, or operations.

The report shall also be used by flight or cabin crew to report any occurrence subject to mandatory reporting in accordance with [11.7 Internal Safety Reporting Scheme](#).

Fatigue Report (FAT)

The Fatigue Report is an individual report and shall be completed by flight and cabin crew to report:

- a. Any situation where the crew member felt fatigued and consequently unfit to perform their duties; and / or
- b. Any situation where the crew member assess that there was the potential to feel fatigued during the duty day and consequently unfit to complete the allocated duty; and / or
- c. Any situation where a crew member felt, in hindsight, that their fatigue level adversely affected their performance; and / or
- d. Any safety occurrence that may be attributable wholly or partially to fatigue. (The technical / operational occurrence report continues to be reported through a Net- Jets Safety report, as applicable.)
- e. Any concern regarding potential fatigue contributing factors within NTA policies, procedures, etc.

The safety report forms are available for electronic completion using the safety reporting system. There is a link to the safety reporting system at the top of the Company intranet homepage. Crew members may also use the app available for download on their EFB.

Completion instructions are available in the guide in the NTA documentation folder on the company network drive and in the Safety collection on Comply365.

All safety reports after submission, based on the initial information available, are risk assessed using the Event Risk Classification (ERC) methodology. Safety issues are risk assessed using the Safety Issues Risk Assessment (SIRA) methodology. Detailed information regarding both of these risk assessment methodologies is provided in the Safety Management Handbook.

If the safety report is classified with a high risk, then immediate actions will be taken to address the issue.

The initial risk assessment may be revised once the analysis / investigation of the safety report is complete.

Unusual events which have been observed by the public (whether passengers or other persons) shall be reported, after landing, by the Commander via telephone to the DO or the NJE Chief Pilot. This will enable them to answer questions, if any, asked by the press and by any Authority. In addition, the Commander shall ensure that a safety report is submitted.

Safety reports shall be submitted to the Safety department either:

- Via the online safety reporting system;
- By Email: Lis-safety@netjets.com

Dangerous Goods Occurrence Report (DGOR)

A Safety NJE Safety report form is to be completed following a dangerous goods accident or serious incident. This form should also be used to report any occasion when undeclared or misdeclared dangerous goods are discovered in cargo, mail or unaccompanied baggage or when accompanied baggage contains dangerous goods which passengers or crew are not permitted to take on the aeroplane, within 72 hours, unless exceptional circumstances prevent this.

The Safety Report form, includes a DGOR section dedicated to capture required data. It should be duly completed, and sent as soon as possible, even if all the information is not available. Copies of all relevant documents and any photographs shall be attached to the report. Any further information, or any information not included in the initial report, should be sent as soon as possible. Providing it is safe to do so, all dangerous goods, packaging, documents, etc., relating to the occurrence shall be retained until after the initial report has been sent and the appropriate authorities have indicated whether or not these should continue to be retained.

DGOR Accident and Incident definitions (CAT.GEN.MPA.200(e)):

Dangerous goods accident: An occurrence associated with and related to the transport of dangerous goods by air which results in fatal or serious injury to a person or major property or environmental damage.

Dangerous goods incident: An occurrence other than a dangerous goods accident associated with and related to the transport of dangerous goods by air, not necessarily occurring on board an aeroplane, which results in injury to a person, property or environmental damage, fire, breakage, spillage, leakage of fluid or radiation or other evidence that the integrity of the packaging has not been maintained. Any occurrence relating to the transport of dangerous goods which seriously jeopardises an aeroplane or its occupants is also deemed to be a dangerous goods incident.

NOTE

A dangerous goods accident or incident may also constitute an aeroplane accident or incident as specified in ICAO Annex 13 — Aeroplane Accident.

NTA must report dangerous goods accidents and incidents to the appropriate authorities of the State of the Operator and the State in which the accident or incident occurred in accordance with the reporting requirements of those appropriate authorities.

NTA must report any occasion when undeclared or misdeclared dangerous goods are discovered in cargo or mail, to the appropriate authorities of the State of the Operator and the State in which this occurred. Must also report any occasion when dangerous goods that are not permitted are discovered by NTA (or is advised by the entity that discovers the dangerous goods) either in the baggage or on the person of passengers.

NOTE

This includes incidents involving dangerous goods that are not subject to all or part of the ICAO Technical Instructions through the application of an exception or of a special provision (e.g. an incident involving the short circuiting of a dry cell battery that is required to meet short-circuit prevention conditions in a special provision of 3;3).

In addition to the requirements of the ICAO Technical Instructions for the reporting of dangerous goods occurrences (above), ORO.GEN.160 requires that any incident which endangers or which, if not corrected, would endanger an aeroplane, its occupants or any other person is reported to the State of the Operator (ANAC). Dangerous goods occurrences are reportable under the Mandatory Occurrence Reporting Scheme in accordance with 11.7 Internal Safety Reporting Scheme.

The report shall also be used by flight or cabin crew to report any occurrence subject to mandatory reporting in accordance s 11.7 Internal Safety Reporting Scheme.

NOTE

Dangerous goods occurrences meeting the criteria of ORO.GEN.160 also meet the definition of a dangerous goods accident or incident (above), reportable in accordance with CAT.GEN.MPA.200(e). Accordingly, the report must be made to ANAC Safety Data within 72 hours (rather than 96), unless exceptional circumstances prevent this. Guidance to reports accidents or incidents with dangerous goods can be found on Chapter 11, Dangerous Goods Occurrence Report (DGOR).

A dangerous goods accident or dangerous goods incident not meeting the criteria of ORO.GEN.160 must be reported to the ANAC within 72 hours, unless exceptional circumstances prevent this. If necessary, a subsequent report shall be made as soon as possible giving all the details that were not known at the time the first report was sent. If a report has been made verbally, written confirmation shall be sent as soon as possible. Any type of accident or incident must be reported irrespective of whether the dangerous goods are in cargo, passengers' baggage, or crew baggage.

The first and any subsequent report shall be as precise as possible and contain such of the following data that are relevant:

- Date of the incident or accident or the finding of undeclared or misdeclared dangerous goods.
- Location, the flight number, and flight date.
- Description of the goods and the reference number of the air waybill, pouch, baggage tag, ticket, etc.
- Proper shipping name (including the technical name, if appropriate) and UN/ID number, when known.
- Class or division and any subsidiary hazard.
- Type of packaging, and the packaging specification marking on it.
- Quantity of dangerous goods.
- Name and address of the shipper, passenger, etc.
- Any other relevant details.

- Suspected cause of the incident or accident.
- Action taken.
- Any other reporting action taken.
- Name, title, address, and telephone number of the person making the report.

Copies of relevant documents and any photographs taken should be attached to a report.

NOTE

If safe to do so, the dangerous goods involved in the accident or incident should be held pending ANAC investigation.

11.7 Internal Safety Reporting Scheme

According with the Safety Policy, all employees are required to report any safety issue or event that may affect the safety of NetJets' customers, employees, or operations. In particular, the pilot in command, or, in cases where the pilot in command is unable to report the occurrence, any other crew member next in the chain of command of an aircraft shall report any occurrence included in the mandatory reporting list below.

The overall purpose of the internal safety reporting scheme is to use reported information to improve the level of the safety performance of NetJets and not to attribute blame.

The objectives of the scheme are to:

- a. enable an assessment to be made of the safety implications of each relevant incident and accident, including previous similar occurrences, so that any necessary action can be initiated; and
- b. ensure that knowledge of relevant incidents and accidents is disseminated, so that other persons and operators may learn from them.

The scheme is an essential part of the overall monitoring function and it is complementary to the normal day-to-day procedures and 'control' systems and is not intended to duplicate or supersede any of them. The scheme is a tool to identify those instances where routine procedures have failed.

All occurrence reports judged reportable by the person submitting the report shall be retained as the significance of such reports may only become obvious at a later date.

Significant risk-bearing incidents detected by FDM should, therefore, normally be the subject of mandatory occurrence reporting by the crew. If this is not the case, then the crew shall submit a retrospective report whenever informed by the Safety department of a mandatory reportable occurrence. Thereafter, the crew will have 72 hours to submit their retrospective report, which will then follow the normal process for reporting and analysing hazards, incidents and accidents.

Some events (such as ATS incidents, ACAS RAs, wildlife strikes, dangerous goods incidents or accidents, and unlawful interference) may require additional information and be subject to additional notification procedures.

The list below includes the mandatory reportable occurrences required by Regulation (EU) No 376/2014 and Regulation No 965/2012 which must be reported to the Safety department within 72 hours of occurrence. However, it is not exhaustive. Mandatory reportable

occurrences are those where the safety of the operation was or could have been endangered or which could have led to an unsafe condition. Any such occurrences should also be reported even if they do not appear on the list or whenever requested by the Safety department.

List of Mandatory Reportable Events

This list is structured in such a way that the pertinent occurrences are linked with categories of activities during which they are normally observed, according to experience, in order to facilitate the reporting of those occurrences. However, this list must not be understood as meaning that occurrences must not be reported in case they take place outside the category of activities to which they are linked in the list.

1. AIR OPERATIONS

1. Flight preparation

- a. Use of incorrect data or erroneous entries into equipment used for navigation or performance calculations which has or could have endangered the aircraft, its occupants or any other person.
- b. Carriage or attempted carriage of dangerous goods in contravention of applicable legislations including incorrect labelling, packaging and handling of dangerous goods.

2. Aircraft preparation

- a. Incorrect fuel type or contaminated fuel.
- b. Missing, incorrect or inadequate De-icing / Anti-icing treatment.

3. Take-off and landing

- a. Taxiway or runway excursion.
- b. Actual or potential taxiway or runway incursion.
- c. Final Approach and Take-off Area (FATO) incursion.
- d. Any rejected take-off.
- e. Inability to achieve required or expected performance during take-off, goaround or landing.
- f. Actual or attempted take-off, approach or landing with incorrect configuration setting.
- g. Tail, blade / wingtip or nacelle strike during take-off or landing.
- h. Approach continued against air operator stabilised approach criteria.
- i. Continuation of an instrument approach below published minimums with inadequate visual references.
- j. Go-around due to, or producing, a dangerous or potentially dangerous situation.
- k. Precautionary or forced landing.
- l. Short and long landing.
- m. Hard landing.

4. Any phase of flight

- a. Loss of control (including partial or temporary) regardless of the cause.

- b. Aircraft upset, exceeding normal pitch attitude, bank angle or airspeed inappropriate for the conditions.
- c. Level bust.
- d. Activation of any flight envelope protection, including stall warning, stick shaker, stick pusher and automatic protections.
- e. Unintentional deviation from intended or assigned track of the lowest of twice the required navigation performance or 10 nautical miles.
- f. Exceedance of aircraft flight manual limitation.
- g. Operation with incorrect altimeter setting.
- h. Jet blast or rotor and prop wash occurrences which have or could have endangered the aircraft, its occupants or any other person.
- i. Misinterpretation of automation mode or of any flight deck information provided to the flight crew which has or could have endangered the aircraft, its occupants or any other person.
- j. Loss of GPS signal in flight.

5. Other types of occurrences

- a. Unintentional release of cargo or other externally carried equipment.
- b. Loss of situational awareness (including environmental, mode and system awareness, spatial disorientation, and time horizon).
- c. Any occurrence where the human performance has directly contributed to or could have contributed to an accident or a serious incident.

2. TECHNICAL OCCURRENCES

1. Structure and systems

- a. Loss of any part of the aircraft structure in flight.
- b. Loss of a system.
- c. Loss of redundancy of a system.
- d. Leakage of any fluid which resulted in a fire hazard or possible hazardous contamination of aircraft structure, systems or equipment, or which has or could have endangered the aircraft, its occupants or any other person.
- e. Fuel system malfunctions or defects, which had an effect on fuel supply and / or distribution.
- f. Malfunction or defect of any indication system when this results in misleading indications to the crew.
- g. Abnormal functioning of flight controls such as asymmetric or stuck / jammed flight controls (for example: lift (flaps / slats), drag (spoilers), attitude control (ailerons, elevators, rudder) devices).

2. Propulsion (including engines) and auxiliary power units (APUs)

- a. Failure or significant malfunction of any part or controlling of a powerplant.

- b. Flameout, in-flight shutdown of any engine or APU when required (for example: ETOPS, MEL).
- c. Engine operating limitation exceedance, including overspeed or inability to control the speed of any high-speed rotating component (for example: APU, air starter, air cycle machine or air turbine motor).
- d. Failure or malfunction of any part of an engine, powerplant, APU or transmission resulting in any one or more of the following:
 - thrust-reversing system failing to operate as commanded;
 - inability to control power, thrust or rpm (revolutions per minute);
 - non-containment of components / debris.

3. INTERACTION WITH AIR NAVIGATION SERVICES (ANS) AND AIR TRAFFIC MANAGEMENT (ATM)

- a. Unsafe ATC clearance.
- b. Prolonged loss of communication with ATS or ATM Unit.
- c. Conflicting instructions from different ATS Units potentially leading to a loss of separation.
- d. Misinterpretation of radio-communication which has or could have endangered the aircraft, its occupants, or any other person.
- e. Intentional deviation from ATC instruction which has or could have endangered the aircraft, its occupants, or any other person.

4. EMERGENCIES AND OTHER CRITICAL SITUATIONS

- a. Any event leading to the declaration of an emergency ('Mayday' or 'PAN call').
- b. Any burning, melting, smoke, fumes, arcing, overheating, fire or explosion.
- c. Contaminated air in the cockpit or in the passenger compartment which has or could have endangered the aircraft, its occupants or any other person.
- d. Failure to apply the correct non-normal or emergency procedure by the flight or cabin crew to deal with an emergency.
- e. Use of any emergency equipment or non-normal procedure affecting in-flight or landing performance.
- f. Failure of any emergency or rescue system or equipment which has or could have endangered the aircraft, its occupants or any other person.
- g. Uncontrollable cabin pressure.
- h. Critically low fuel quantity (including 'MINIMUM FUEL' declarations) or fuel quantity at destination below required final reserve fuel.
- i. Any use of crew oxygen system by the crew.
- j. Incapacitation of any member of the flight or cabin crew that results in the reduction below the minimum certified crew complement.
- k. Crew fatigue impacting or potentially impacting their ability to perform safely their flight duties.

5. EXTERNAL ENVIRONMENT AND METEOROLOGY

- a. A collision or a near collision on the ground or in the air, with another aircraft, vehicle, terrain or obstacle.
- b. ACAS RA (Airborne Collision Avoidance System, Resolution Advisory).
- c. Activation of genuine ground collision system such as GPWS (Ground Proximity Warning System) / TAWS 'warning'.
- d. Wildlife strike including bird strike.
- e. FOD / debris.
- f. Unexpected encounter of poor runway surface conditions.
- g. Wake-turbulence encounters.
- h. Interference with the aircraft by firearms, fireworks, flying kites, laser illumination, high powered lights, lasers, Remotely Piloted Aircraft Systems, model aircraft or by similar means.
- i. A lightning strike which resulted in damage to the aircraft or loss or malfunction of any aircraft system.
- j. A hail encounter which resulted in damage to the aircraft or loss or malfunction of any aircraft system.
- k. Severe turbulence encounter or any encounter resulting in injury to occupants or deemed to require a 'turbulence check' of the aircraft.
- l. A significant wind shear or thunderstorm encounter which has or could have endangered the aircraft, its occupants or any other person.
- m. Icing encounter resulting in handling difficulties, damage to the aircraft or loss or malfunction of any aircraft system.
- n. Volcanic ash encounter.

6. SECURITY

- a. Bomb threat or hijack.
- b. Breach of security procedures.
- c. Difficulty in controlling intoxicated, violent or unruly passengers.
- d. Discovery of a stowaway.

7. RVSM

All recorded or communicated occurrences of height-keeping errors caused by malfunction of aircraft equipment or of operational nature, equal to or greater than:

- a. A total vertical error (TVE) of ± 90 m (± 300 ft);
- b. An altimetry system error (ASE) of ± 75 m (± 245 ft); and
- c. An assigned altitude deviation (AD) of ± 90 m (± 300 ft).

8. PERFORMANCE BASED NAVIGATION OPERATIONS

- a. A reportable event should be an event that adversely affects the safety of the operation and may be caused by actions or events external to the functioning of the aircraft navigation system.
- b. Technical defects and the exceedance of technical limitations, including:
 1. significant navigation errors attributed to incorrect data or a database coding error;
 2. unexpected deviations in lateral/vertical flight path not caused by flight crew input or erroneous operation of equipment;
 3. significant misleading information without a failure warning;
 4. total loss or multiple navigation equipment failure; and
 5. loss of integrity, e.g. RAIM function, whereas integrity was predicted to be available during preflight planning, are considered a reportable event.
- c. NetJets shall investigate if the occurrence is due to an improperly coded procedure or a navigation database error. NetJets shall initiate corrective actions for such an event.

9. PORTABLE ELECTRONIC DEVICES

Occurrences of suspected or confirmed interference that have potential safety implications should be reported to ANAC. Where possible, to assist follow-up and technical investigation, reports shall describe the offending device, identify the brand name and model number, its location in the aircraft at the time of the occurrence, interference symptoms and the results of actions taken by the crew.

The cooperation of the device owner shall be sought by obtaining contact details.

10. LOW VISIBILITY OPERATIONS (LVO)

- a. A reportable event should include:
 1. significant deviations from the flight path not caused by flight crew input;
 2. misleading information without flight deck alerts;
 3. loss of airborne navigation equipment functions necessary for the operation;
 4. loss of functions or facilities at the aerodrome necessary for the operation, including aerodrome operating procedures, ATC operation, navigation facilities, visual aids and electrical power supply;
 5. loss of other functions related to external infrastructure necessary for the operation; and
 6. any other event causing the approach or landing to be abandoned if occurring repeatedly.
- b. In addition to the normal reporting the reports shall be submitted to the aerodrome involved when relevant.

11. PBS CHARTER

The PBCS authorisation has a requirement for NTA to join and participate in the PBCS monitoring programme (Charter). The safety reporting system is to be used to ensure internal visibility and coordinated accountability of the charter requirements.

The following items shall be reported:

- Any operational safety concerns or incidents with the FANS 1/A system or PBCS.

12. DANGEROUS GOODS

- Dangerous goods found not to have been secured to prevent movement.
- Damage to packages of dangerous goods.
- NOTOC errors where dangerous goods have not been stowed in accordance with loading instructions.
- Failure to prepare electric wheelchairs in order to prevent accidental activation.
- Electric wheelchairs found not to have been stowed and secured correctly.
- Leak or spill of dangerous goods from passenger baggage or undeclared dangerous goods.

13. INFORMATION SECURITY

- Unauthorised system access;
- Corruption of flight data or unavailability of critical systems;
- Detection of malware;
- Attacks that degrade or interrupt air-ground communications, surveillance or links for operational data;
- Jamming or spoofing of radars, GNSS or VHF/UHF communication systems;
- Unauthorised access to employee or passenger personal data;
- Failure of ATC datalink, ADS-B, SWIM or other flight information services;
- Failure of flight planning systems;
- Unavailability of aircraft performance calculation systems;
- Any other occurrences listed in ANAC CIA 11/2025.

11.8 Crew Reporting Procedures

Crew members shall inform the Commander immediately of any incident that has endangered, or may have endangered, safety during the flight. Crew members shall ensure that the Commander is provided with all relevant information as soon as possible. Further information on crew coordination available in section [8.3.19 Crew Coordination](#).

11.9 Preservation, Production, and Use of Flight Recorder Recordings

Following an accident, a serious incident, or an occurrence identified by the investigating authority, crew members shall preserve FDR and CVR data by pulling the respective circuit breakers immediately after completion of the flight. They shall notify the NJE Chief Pilot or FOSM team, or their deputy, that the recorders have been disabled as soon as possible. Precautions must be taken to prevent the inadvertent reactivation of the recorders. The original recorded data shall be preserved for a period of 60 days unless otherwise directed by the investigating authority. The original recorded data shall be preserved for a period of 60 days unless otherwise directed by the investigating authority.

NTA shall make available any flight recorder recording that has been preserved, if so determined by ANAC.

Except for ensuring the CVR serviceability, CVR recordings shall not be disclosed or used for purposes other than for the investigation of an accident, a serious incident, or an occurrence identified by the investigating authority.

FDR recordings or data link recordings shall only be used for purposes other than for the investigation of an accident or an incident which is subject to mandatory reporting if such records are:

- Used by the operator for airworthiness or maintenance purposes only; or
- De-identified; or
- Disclosed under secure procedures.

Flight recorder recordings shall be protected against unauthorised use and may only be handled in accordance with the procedures described in the *Safety Management Handbook* and [NJMP 1.37](#).

FDM use shall not adversely affect the serviceability of equipment required for accident investigation.

For compliance with CAT.GEN.MPA.195, installation of flight recorders that take images of the flight crew compartment are not authorised.

Chapter 12 - Rules of the Air

The Rules of the Air are in accordance with relevant ICAO Annexes 2, 10 and 12. This chapter contains extracts from the respective ICAO Annexes and only highlights the most relevant and applicable portions. For further detail refer to ICAO Annex 2 (*Rules of the Air*), ICAO Annex 10 (*Aeronautical Telecommunications*) published in the Jeppesen Airway Manual in the Air Traffic Control section. Reference to the Emergency Section in the Jeppesen Airway Manual is also required as detailed in the paragraphs below.

For specific State differences from ICAO Annexes, refer to respective State Rules and Procedures, published in the Jeppesen Airway Manual - Air Traffic Control section.

NOTE

For ease of reference the numbering of the text in each sub paragraph is coincident with that of the ICAO documents published in the appropriate Jeppesen Airway Manual sections. Where the text conforms to the Standardised European Rules of the Air (SERA), reference is made to the respective article.

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12.1 General Rules

Also refer to ICAO Annex 2, *Rules of the Air*, Chapters 4 and 5 as published in the Jeppesen Airway Manual Air Traffic Control section.

2.1 Territorial Application of the Rule of the Air

For flight over the high seas, the rules specified in Annex 2 to the Chicago Convention shall apply without exception. For the purposes of continuity and seamless operation of air traffic services in particular within Functional Airspace Blocks, the provisions of Annex 11 to the Chicago Convention may be applied in airspace over high seas in a manner that is consistent with how those provisions are applied over the territory of the member States.

2.2 Compliance with the Rules of the Air

The operation of an aircraft either in flight or on the movement area of an aerodrome shall be in compliance with the general rules and, in addition, when in flight, either with:

- a. the visual flight rules; or
- b. the instrument flight rules.

3.1 Negligent or Reckless Operation of Aircraft

An aircraft shall not be operated in a negligent or reckless manner so as to endanger life or property of others.

3.2 Avoidance of Collision

3.2.1 Proximity

(SERA.3205)

An aircraft shall not be operated in such proximity to other aircraft as to create a collision hazard.

3.2.2 Right-of Way

(SERA.3210) Nothing in these rules shall relieve the pilot-in-command of an aircraft from the responsibility of taking such action, including collision avoidance manoeuvres based on resolution advisories provided by ACAS equipment, as will best avert collision.

Regardless of the type of flight or the class of airspace in which the aircraft is operating, vigilance for the purpose of detecting potential collisions shall be exercised on board an aircraft. This equally applies while operating on the movement area of an aerodrome.

- a. The aircraft that has the right-of-way shall maintain its heading and speed.
- b. An aircraft that is aware that the manoeuvrability of another aircraft is impaired shall give way to that aircraft.
- c. An aircraft that is obliged by the following rules to keep out of the way of another shall avoid passing over, under or in front of the other, unless it passes well clear and takes into account the effect of aircraft wake turbulence.
 1. Approaching head-on. When two aircraft are approaching head-on or approximately so and there is danger of collision, each shall alter its heading to the right.

2. Converging. When two aircraft are converging at approximately the same level, the aircraft that has the other on its right shall give way, except as follows:
 - i. power-driven heavier-than-air aircraft shall give way to airships, sailplanes and balloons;
 - ii. airships shall give way to sailplanes and balloons;
 - iii. sailplanes shall give way to balloons;
 - iv. power-driven aircraft shall give way to aircraft which are seen to be towing other aircraft or objects.
 3. Overtaking. An overtaking aircraft is an aircraft that approaches another from the rear on a line forming an angle of less than 70 degrees with the plane of symmetry of the latter, i.e. is in such a position with reference to the other aircraft that at night it should be unable to see either of the aircraft's left (port) or right (starboard) navigation lights. An aircraft that is being overtaken has the right-of-way and the overtaking aircraft, whether climbing, descending or in horizontal flight, shall keep out of the way of the other aircraft by altering its heading to the right, and no subsequent change in the relative positions of the two aircraft shall absolve the overtaking aircraft from this obligation until it is entirely past and clear.
 - i. Sailplanes overtaking. A sailplane overtaking another sailplane may alter its course to the right or to the left.
 4. Landing. An aircraft in flight, or operating on the ground or water, shall give way to aircraft landing or in the final stages of an approach to land.
 - i. When two or more heavier-than-air aircraft are approaching an aerodrome or an operating site for the purpose of landing, aircraft at the higher level shall give way to aircraft at the lower level, but the latter shall not take advantage of this rule to cut in front of another which is in the final stages of an approach to land, or to overtake that aircraft. Nevertheless, power-driven heavier-than-air aircraft shall give way to sailplanes.
 - ii. Emergency landing. An aircraft that is aware that another is compelled to land shall give way to that aircraft.
 5. Taking-off. An aircraft taxiing on the manoeuvring area of an aerodrome shall give way to aircraft taking off or about to take off.
- d. Surface movement of aircraft, persons and vehicles.
1. In case of danger of collision between two aircraft taxiing on the movement area of an aerodrome or equivalent part of an operating site, the following shall apply:
 - i. when two aircraft are approaching head on, or approximately so, each shall stop or where practicable alter its course to the right so as to keep well clear;
 - ii. when two aircraft are on a converging course, the one which has the other on its right shall give way;
 - iii. an aircraft which is being overtaken by another aircraft shall have the right-of-way and the overtaking aircraft shall keep well clear of the other aircraft.
 2. At a controlled aerodrome an aircraft taxiing on the manoeuvring area shall stop and hold at all runway-holding positions unless an explicit clearance to enter or cross the runway has been issued by the aerodrome control tower.

3. An aircraft taxiing on the manoeuvring area shall stop and hold at all lighted stop bars and may proceed further in accordance with (2) when the lights are switched off.

Unserviceable Stop Bars - Contingency Measures In situations where the lit stop bars cannot be turned off because of a technical problem, the airport operator may implement the following contingency measures:

- i. physically disconnecting the respective lit stop bar from its power supply;
- ii. physically obscuring the lights of the lit stop bar;
- iii. using a different route, until the malfunctioning system has been repaired.

In case of implementation of (a) or (b), a marshaller or a follow-me vehicle may need to be provided to lead the aircraft to cross the stop bar.

Moreover, in case of implementation of (b), care should be exercised to ensure correct implementation of the measure to avoid misunderstandings by the flight crew. In any case, the measures taken should not undermine the principle that a lit stop bar must not be crossed.

4. Movement of persons and vehicles at aerodromes:

- i. The movement of persons or vehicles, including towed aircraft, on the manoeuvring area of an aerodrome shall be controlled by the aerodrome control tower as necessary to avoid hazard to them or to aircraft landing, taxiing or taking off.
- ii. In conditions where low visibility procedures are in operation:
 - A. persons and vehicles operating on the manoeuvring area of an aerodrome shall be restricted to the essential minimum, and particular regard shall be given to the requirements to protect the ILS/MLS sensitive area(s) when Category II or Category III precision instrument operations are in progress;
 - B. subject to the provisions in (iii) the minimum separation between vehicles and taxiing aircraft shall be as specified by the Air Navigation Service Provider (ANSP) and approved by the competent authority taking into account the aids available;
 - C. when mixed ILS and MLS Category II or Category III precision instrument operations are taking place to the same runway continuously, the more restrictive ILS or MLS critical and sensitive areas shall be protected.
- iii. Emergency vehicles proceeding to the assistance of an aircraft in distress shall be afforded priority over all other surface movement traffic.
- iv. Subject to the provisions in (iii), vehicles on the manoeuvring area shall be required to comply with the following rules:
 - A. vehicles and vehicles towing aircraft shall give way to aircraft which are landing, taking off, taxiing or being towed;
 - B. vehicles shall give way to other vehicles towing aircraft;
 - C. vehicles shall give way to other vehicles in accordance with air traffic services unit instructions;
 - D. notwithstanding the provisions of (A), (B) and (C), vehicles and vehicles towing aircraft shall comply with instructions issued by the aerodrome control tower.

3.2.3 Lights to be Displayed by Aircraft

(SERA.3215)

Refer to [Display of Light](#).

3.2.4 Simulated Instrument Flights

(SERA.3220)

Not applicable.

3.2.5 Operation on and in the Vicinity of an Aerodrome

(SERA.3225)

An aircraft operated on or in the vicinity of an aerodrome shall:

- a. observe other aerodrome traffic for the purpose of avoiding collision;
- b. conform with or avoid the pattern of traffic formed by other aircraft in operation;
- c. except for balloons, make all turns to the left, when approaching for a landing and after taking off, unless otherwise indicated, or instructed by ATC;
- d. except for balloons, land and take off into the wind unless safety, the runway configuration, or air traffic considerations determine that a different direction is preferable.

3.3 Flight Plans

(SERA.4001) Refer to [ATS Flight Plan](#). For further detail refer to Jeppesen text pages, Air Traffic Control, ICAO Annex 2, Paragraph 3.3 Flight Plans.

3.4 Signals

(SERA.3301)

For depictions and interpretation of standard signals, refer to Appendix 1 to ICAO Annex 2: Jeppesen Text Pages, Air Traffic Control, ICAO Annex 2, Appendix 1.

- a. Upon observing or receiving any of the signals given in Appendix 1 to Annex 2, aircraft shall take such action as may be required by the interpretation of the signal given in that Appendix.
- b. The signals of Appendix 1 to Annex 2 shall, when used, have the meaning indicated therein. They shall be used only for the purpose indicated and no other signals likely to be confused with them shall be used.
- c. A signalman / marshaller shall be responsible for providing standard marshalling signals to aircraft in a clear and precise manner using the signals shown in Appendix 1 to Annex 2.
- d. Only persons trained, qualified and approved as required by the relevant Union or national legislation shall carry out the functions of a signalman / marshaller.
- e. The signalman / marshaller shall wear a distinctive fluorescent identification vest to allow the flight crew to identify that they are the person responsible for the marshalling operation.
- f. Daylight-fluorescent wands, table-tennis bats or gloves shall be used for all signalling by all participating ground staff during daylight hours. Illuminated wands shall be used at night or in low visibility.

3.5 Time

(SERA.3401)

- a. Coordinated Universal Time (UTC) shall be used and shall be expressed in hours and minutes and, when required, seconds of the 24-hour day beginning at midnight.
- b. A time check shall be obtained prior to operating a controlled flight and at such other times during the flight as may be necessary.
- c. Wherever time is utilised in the application of data link communications, it shall be accurate to within 1 second of UTC.
- d. Time in air traffic services.

Aerodrome control towers shall, prior to an aircraft taxiing for take-off, provide the pilot with the correct time, unless arrangements have been made for the pilot to obtain it from other sources. Air traffic services units shall, in addition, provide aircraft with the correct time on request. Time checks shall be given at least to the nearest minute.

3.6 Air Traffic Control

Refer to Jeppesen text pages, Air Traffic Control, ICAO Annex 2, Paragraph 3.6 Air Traffic Control Service for further detail.

3.6.1 ATC Clearances

(SERA.8015)

1. Air traffic control clearances shall be based solely on the requirements for providing air traffic control service.
2. Operation subject to clearance:
 1. An air traffic control clearance shall be obtained prior to operating a controlled flight, or a portion of a flight as a controlled flight. Such clearance shall be requested through the submission of a flight plan to an air traffic control unit.
 2. The pilot-in-command of an aircraft shall inform ATC if an air traffic control clearance is not satisfactory. In such cases, ATC will issue an amended clearance, if practicable.
 3. Whenever an aircraft has requested a clearance involving priority, a report explaining the necessity for such priority shall be submitted, if requested by the appropriate air traffic control unit.
 4. Potential reclearance in flight. If, prior to departure, it is anticipated that, depending on fuel endurance and subject to reclearance in flight, a decision may be taken to proceed to a revised destination aerodrome, the appropriate air traffic control units shall be so notified by the insertion in the flight plan of information concerning the revised route (where known) and the revised destination.
 5. An aircraft operated on a controlled aerodrome shall not taxi on the manoeuvring area without clearance from the aerodrome control tower and shall comply with any instructions given by that unit.
3. Clearances for transonic flight
 1. The air traffic control clearance relating to the transonic acceleration phase of a supersonic flight shall extend at least to the end of that phase.

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2. The air traffic control clearance relating to the deceleration and descent of an aircraft from supersonic cruise to subsonic flight shall seek to provide for uninterrupted descent at least during the transonic phase.

4. Contents of clearances

An air traffic control clearance shall indicate:

1. aircraft identification as shown in the flight plan;
2. clearance limit;
3. route of flight;
4. level(s) of flight for the entire route or part thereof and changes of levels if required;
5. any necessary instructions or information on other matters such as approach or departure manoeuvres, communications and the time of expiry of the clearance.

5. Read-back of clearances and safety-related information:

1. The flight crew shall read back to the air traffic controller safety-related parts of ATC clearances and instructions which are transmitted by voice. The following items shall always be read back:
 - i. ATC route clearances;
 - ii. clearances and instructions to enter, land on, take off from, hold short of, cross, taxi and backtrack on any runway; and
 - iii. runway-in-use, altimeter settings, SSR codes, newly assigned communication channels, level instructions, heading and speed instructions; and
 - iv. transition levels, whether issued by the controller or contained in ATIS broadcasts.
2. Other clearances or instructions, including conditional clearances and taxi instructions, shall be read back or acknowledged in a manner to clearly indicate that they have been understood and will be complied with.
3. The controller shall listen to the read-back to ascertain that the clearance or instruction has been correctly acknowledged by the flight crew and shall take immediate action to correct any discrepancies revealed by the read-back.
4. Voice read-back of CPDLC messages shall not be required, unless otherwise specified by the ANSP.

6. Coordination of clearances

1. An air traffic control clearance shall be coordinated between air traffic control units to cover the entire route of an aircraft or a specified portion thereof as described in provisions (2) to (6).
2. An aircraft shall be cleared for the entire route to the aerodrome of first intended landing:
 - i. when it has been possible, prior to departure, to coordinate the clearance between all the units under whose control the aircraft will come; or
 - ii. when there is reasonable assurance that prior coordination will be effected between those units under whose control the aircraft will subsequently come.

3. When coordination as in (2) has not been achieved or is not anticipated, the aircraft shall be cleared only to that point where coordination is reasonably assured; prior to reaching such point, or at such point, the aircraft shall receive further clearance, holding instructions being issued as appropriate.
4. When prescribed by the ATS unit, aircraft shall contact a downstream air traffic control unit, for the purpose of receiving a downstream clearance prior to the transfer of control point.
 - i. Aircraft shall maintain the necessary two-way communication with the current air traffic control unit whilst obtaining a downstream clearance.
 - ii. A clearance issued as a downstream clearance shall be clearly identifiable as such to the pilot.
 - iii. Unless coordinated, downstream clearances shall not affect the aircraft's original flight profile in any airspace, other than that of the air traffic control unit responsible for the delivery of the downstream clearance.
5. When an aircraft intends to depart from an aerodrome within a control area to enter another control area within a period of thirty minutes, or such other specific period of time as has been agreed between the area control centres concerned, coordination with the subsequent area control centre shall be effected prior to issuance of the departure clearance.
6. When an aircraft intends to leave a control area for flight outside controlled airspace, and will subsequently re-enter the same or another control area, a clearance from the point of departure to the aerodrome of first intended landing may be issued. Such clearance or revisions thereto shall apply only to those portions of the flight conducted within controlled airspace.

3.6.2 Adherence to Flight Plan

(SERA.8020)

- a. Except as provided for in (b) and (d) an aircraft shall adhere to the current flight plan or the applicable portion of a current flight plan submitted for a controlled flight unless a request for a change has been made and clearance obtained from the appropriate air traffic control unit, or unless an emergency situation arises which necessitates immediate action by the aircraft, in which event as soon as circumstances permit, after such emergency authority is exercised, the appropriate air traffic services unit shall be notified of the action taken and that this action has been taken under emergency authority.
 1. Unless otherwise authorised by the competent authority, or directed by the appropriate air traffic control unit, controlled flights shall, in so far as practicable:
 - i. when on an established ATS route, operate along the defined centre line of that route; or
 - ii. when on any other route, operate directly between the navigation facilities and / or points defining that route.
 2. Unless otherwise authorised by the competent authority, or directed by the appropriate air traffic control unit, an aircraft operating along an ATS route segment defined by reference to very high frequency omnidirectional radio ranges shall change over for its primary navigation guidance from the facility behind the aircraft to that ahead of it at, or as close as operationally feasible to, the changeover point, where established.
 3. Deviation from the requirements in (1) shall be notified to the appropriate ATS unit.

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- b. Inadvertent changes. In the event that a controlled flight inadvertently deviates from its current flight plan, the following action shall be taken:
1. Deviation from track: if the aircraft is off track, action shall be taken forthwith to adjust the heading of the aircraft to regain track as soon as practicable.
 2. Variation in true airspeed: if the average true airspeed at cruising level between reporting points varies or is expected to vary by plus or minus 5 per cent of the true airspeed, from that given in the flight plan, the appropriate air traffic services unit shall be so informed.
 3. Change in time estimate: if the time estimate for the next applicable reporting point, flight information region boundary or destination aerodrome, whichever comes first, is found to be in error in excess of 2 minutes from that notified to ATS, or such other period of time as is prescribed by the competent authority, a revised estimated time shall be notified as soon as possible to the ATS unit.
 4. Additionally, when an ADS-C agreement is in place, the air traffic services unit shall be informed automatically via data link whenever changes occur beyond the threshold values stipulated by the ADS-C event contract.
- c. Intended changes. Requests for flight plan changes shall include information as indicated hereunder:
1. Change of cruising level: aircraft identification; requested new cruising level and cruising speed at this level, revised time estimates (when applicable) at subsequent flight information region boundaries.
 2. Change of route:
 - i. Destination unchanged: aircraft identification; flight rules; description of new route of flight including related flight plan data beginning with the position from which requested change of route is to commence; revised time estimates; any other pertinent information.
 - ii. Destination changed: aircraft identification; flight rules; description of revised route of flight to revised destination aerodrome including related flight plan data, beginning with the position from which requested change of route is to commence; revised time estimates; alternate aerodrome(s); any other pertinent information.
- d. Weather deterioration below the VMC. When it becomes evident that flight in VMC in accordance with its current flight plan will not be practicable, a VFR flight operated as a controlled flight shall:
1. request an amended clearance enabling the aircraft to continue in VMC to destination or to an alternative aerodrome, or to leave the airspace within which an ATC clearance is required; or
 2. if no clearance in accordance with a) can be obtained, continue to operate in VMC and notify the appropriate ATC unit of the action being taken either to leave the airspace concerned or to land at the nearest suitable aerodrome; or
 3. if operated within a control zone, request authorisation to operate as a special VFR flight; or
 4. request clearance to operate in accordance with the instrument flight rules.

3.6.3 Position Reports

(SERA.8025)

- a. Unless exempted by the competent authority or by the appropriate air traffic services unit under conditions specified by that authority, a controlled flight shall report to the appropriate air traffic services unit, as soon as possible, the time and level of passing each designated compulsory reporting point, together with any other required information.

Position reports shall similarly be made in relation to additional points when requested by the appropriate air traffic services unit. In the absence of designated reporting points, position reports shall be made at intervals prescribed by the competent authority or specified by the appropriate air traffic services unit.

1. Controlled flights providing position information to the appropriate air traffic services unit via data link communications shall only provide voice position reports when requested.
 2. When a controlled flight has been exempted from the requirement to report at compulsory reporting points, pilots shall, unless automated position reporting is in effect, resume voice or CPDLC position reporting:
 - i. when so instructed;
 - ii. when advised that the ATS surveillance service has been terminated; or
 - iii. when advised that the ATS surveillance identification is lost.
- b. With due regard to requirements in SERA.14065 [Ch. 12 3.6.5.2] for communications changeover, the position report shall contain the following elements:
1. aircraft identification;
 2. position;
 3. time;
 4. speed, if assigned by ATC; and
 5. other elements as instructed by ATC.
- c. The elements described above shall be reported as described in the table below.

1	AIRCRAFT IDENTIFICATION	Report the aircraft radiotelephony call sign as per the flight plan.
2	POSITION	Report position in latitude (DD° or DD°MM followed by 'North' or 'South') and longitude (DDD° or DDD°MM followed by 'East' or 'West'), or as a significant point identified by a coded designator (2 to 5 characters), or as a significant point followed by magnetic bearing (NNN°) and distance in nautical miles from the point. Precede significant point with 'ABEAM', if applicable.

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3	TIME	Report time in hours and minutes UTC (HHMM) unless reporting time in minutes past the hour (MM) is prescribed on the basis of regional air navigation agreements. The time reported must be the actual time of the aircraft at the position and not the time of origination or transmission of the report. Time shall always be reported in hours and minutes UTC when issuing a special air-report.
4	FLIGHT LEVEL OR ALTITUDE	Report flight level (FL NNN) when on standard pressure altimeter setting. Report altitude in metres followed by 'METRES' or in feet followed by 'FEET' when on QNH. Report 'CLIMBING' (followed by the level) when climbing or 'DESCENDING' (followed by the level) when descending to a new level after passing the significant point.
5	NEXT POSITION AND ESTIMATED TIME OVER	Report the next reporting point and the estimated time over such reporting point, or report the estimated position that will be reached one hour later, according to the position reporting procedures in force. Use the data conventions specified in Item 2 for position. Report the estimated time over this position. Report time in hours and minutes UTC (HHMM) unless reporting time in minutes past the hour (MM) as prescribed by regional air navigation agreements.
6	ENSUING SIGNIFICANT POINT	Report the ensuing significant point following the 'next position and estimated time over'.

NOTE

Items 5 and 6 may be omitted unless required by the ATS unit, e.g. they are required in Oceanic and Remote Airspace.

3.6.4 Termination of Control

(SERA.8030)

A controlled flight shall, except when landing at a controlled aerodrome, advise the appropriate ATC unit as soon as it ceases to be subject to air traffic control service.

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3.6.5.1 Communications

(SERA.8035)

- a. An aircraft operated as a controlled flight shall maintain continuous air-ground voice communication watch on the appropriate communication channel of, and establish two-way communication as necessary with, the appropriate air traffic control unit, except as may be prescribed by the relevant ANSP in respect of aircraft forming part of aerodrome traffic at a controlled aerodrome.
 1. The requirement for an aircraft to maintain an air-ground voice communication watch shall remain in effect when CPDLC has been established.
- b. If a communication failure precludes compliance with point (a), the procedures for communication failures shall be followed, as specified in 3.6.5.3 (SERA.14083).
- c. In a HF environment, SELCAL or similar automatic signalling devices satisfy the requirement to maintain an air-ground voice communication watch.
- d. An aircraft may be permitted to communicate temporarily with a control unit other than the unit controlling the aircraft.

3.6.5.2 Procedures for air-ground voice communication channel changeover

(SERA.14065)

- a. Unless otherwise prescribed by the ANSP responsible for the provision of services and approved by the competent authority, the initial call to an air traffic services unit after a change of the air-ground voice communication channel shall contain the following elements:
 1. the designation of the ATS unit being called;
 2. call sign, immediately followed by the word "Heavy" or "Super" corresponding, as appropriate, to the wake turbulence category of the aircraft;
 3. level, including passing and cleared levels, if not maintaining the cleared level;
 4. speed, if assigned by ATC; and
 5. additional elements, as required by the ANSP responsible for the provision of services and approved by the competent authority.
- b. For aircraft being provided with aerodrome control service, the initial call shall contain:
 1. the designation of the ATS unit being called;
 2. position; and
 3. additional elements, as required by the ANSP responsible for the provision of services and approved by the competent authority.

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3.6.5.3 Communication Failure Procedures

- a. When an aircraft is unable to comply with 3.6.5.1 above, point (a), the flight crew shall attempt to establish contact on the previous channel used and, if not successful, on another channel appropriate to the route. If these attempts fail, the flight crew shall attempt to establish communication with:

1. the appropriate air traffic services unit;
2. other air traffic services units; or
3. other aircraft,

using all available means, including, inter alia, data link, satellite voice and mobile phones and, when successful, advise that contact on the assigned channel could not be established.

- b. If an expected communication from an aircraft is not received within a timeframe that suggests a communication failure, or if requested by other air traffic service units, the air traffic controller shall attempt to contact the aircraft on the assumed monitoring frequencies. If unsuccessful, the controller will:

1. Request assistance from other air traffic service units to call and relay messages to the aircraft.
2. Ask other aircraft on the route to try to establish communication and relay messages.
3. Notify the aircraft operator as soon as possible of the communication failure.

If attempts in b(i) and b.(ii) fail, blind transmissions of clearances shall not occur unless specifically requested by the originator. Other messages may be sent via blind transmission on the presumed frequencies. The provision of air traffic control service to flights operating in the airspace concerned shall be based on the premise that an aircraft experiencing communication failure complies with point (c).

- c. When an aircraft cannot comply with SERA.8035(a) and attempts to establish communication fail, the following procedures apply:

1. The crew should watch for visual signals at a controlled aerodrome.
2. The crew must set the transponder to Mode A Code 7600 and/or configure the ADS-B transmitter to indicate communication loss, following applicable procedures.
3. A VFR flight should continue in visual meteorological conditions, land at the nearest suitable aerodrome, and report arrival promptly.
4. An IFR flight must:
 - i. Maintain the last assigned speed and altitude for 20 minutes after a failed report or executing point 2. Adjust afterwards according to the flight plan (as amended, if applicable).
 - ii. If vectored by ATC (or directed to proceed offset using RNAV), continue to the specified limit (if applicable) and rejoin the last acknowledged route at the next significant point, adhering to minimum flight altitudes.
 - iii. Follow the last acknowledged route clearance to the appropriate navigation aid or fix of the destination aerodrome, holding as necessary to comply with c.4.iv below until descent begins.
 - iv. Start descent at or near the expected approach time or estimated time of arrival.

- v. Complete the instrument approach for the navigation aid or fix.
 - vi. Land, to the possible extent, within 30 minutes after the estimated arrival time or last expected approach time.
5. As an exception to 4., for standard instrument departure or arrival routes, follow specified radio communication failure procedures.
6. If an IFR flight encounters visual meteorological conditions and the pilot-in-command decides to continue to fly in visual meteorological conditions, the pilot shall set Mode A Code 7601, land at the nearest suitable aerodrome, and report arrival by the most expeditious means to the appropriate air traffic services unit. In the event of a communication failure while operating under IFR, NTA crews should attempt to continue under IFR and follow the appropriate procedures for communication failure. However, if continuing under IFR jeopardises the safety of the aircraft, crews may decide to follow the provisions of this paragraph whilst maintaining VMC.

In situations where multiple aircraft may be involved (even with functioning communications), the safety of operations is best assured when aircraft adhere to the last ATC clearance received and acknowledged, similar to the case of ground radio station failure.

For further information refer to Jeppesen Airway Manual, Emergency section, Emergency Data ICAO, Chapter 6.

3.7 Unlawful Interference

(SERA.11005)

- a. An aircraft which is being subjected to unlawful interference shall endeavour to set the transponder to Code 7500 and notify the appropriate ATS unit of, any significant circumstances associated therewith and any deviation from the current flight plan necessitated by the circumstances, in order to enable the ATS unit to give priority to the aircraft and to minimise conflict with other aircraft.
- b. If an aircraft is subjected to unlawful interference, the pilot-in-command shall attempt to land as soon as practicable at the nearest suitable aerodrome or at a dedicated aerodrome assigned by the competent authority unless considerations aboard the aircraft dictate otherwise.

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3.8 Interception

(SERA.11005)

- a. Except for intercept and escort service provided on request to an aircraft, interception of civil aircraft shall be governed by appropriate regulations and administrative directives issued by Member States in compliance with the Convention on International Civil Aviation, and in particular Article 3(d) under which ICAO Contracting States undertake, when issuing regulations for their State aircraft, to have due regard for the safety of navigation of civil aircraft.
- b. The pilot-in-command of a civil aircraft, when intercepted, shall:
 1. immediately follow the instructions given by the intercepting aircraft, interpreting and responding to visual signals in accordance with the specifications in Tables S11-1 and S11-2;
 2. notify, if possible, the appropriate air traffic services unit;
 3. attempt to establish radio-communication with the intercepting aircraft or with the appropriate intercept control unit, by making a general call on the emergency frequency 121,5 MHz, giving the identity of the intercepted aircraft and the nature of the flight; and if no contact has been established and if practicable, repeating this call on the emergency frequency 243 MHz;
 4. if equipped with SSR transponder, select Mode A, Code 7700, unless otherwise instructed by the appropriate air traffic services unit;
 5. if equipped with ADS-B or ADS-C, select the appropriate emergency functionality, if available, unless otherwise instructed by the appropriate air traffic services unit.
- c. If any instructions received by radio from any sources conflict with those given by the intercepting aircraft by visual signals, the intercepted aircraft shall request immediate clarification while continuing to comply with the visual instructions given by the intercepting aircraft.
- d. If any instructions received by radio from any sources conflict with those given by the intercepting aircraft by radio, the intercepted aircraft shall request immediate clarification while continuing to comply with the radio instructions given by the intercepting aircraft.
- e. If radio contact is established during interception but communication in a common language is not possible, attempts shall be made to convey instructions, acknowledgement of instructions and essential information by using the phrases and pronunciations in Table S11-3 and transmitting each phrase twice:
- f. As soon as an air traffic services unit learns that an aircraft is being intercepted in its area of responsibility, it shall take such of the following steps as are appropriate in the circumstances:
 1. attempt to establish two-way communication with the intercepted aircraft via any means available, including the emergency radio frequency 121,5 MHz, unless such communication already exists;
 2. inform the pilot of the intercepted aircraft of the interception;
 3. establish contact with the intercept control unit maintaining two-way communication with the intercepting aircraft and provide it with available information concerning the aircraft;
 4. relay messages between the intercepting aircraft or the intercept control unit and the intercepted aircraft, as necessary;

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5. in close coordination with the intercept control unit take all necessary steps to ensure the safety of the intercepted aircraft;
 6. inform air traffic services units serving adjacent flight information regions if it appears that the aircraft has strayed from such adjacent flight information regions.
- g. As soon as an air traffic services unit learns that an aircraft is being intercepted outside its area of responsibility, it shall take such of the following steps as are appropriate in the circumstances:
1. inform the air traffic services unit serving the airspace in which the interception is taking place, providing this unit with available information that will assist in identifying the aircraft and requesting it to take action in accordance with (f);
 2. relay messages between the intercepted aircraft and the appropriate air traffic services unit, the intercept control unit or the intercepting aircraft.

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Table S11-1

Signals initiated by intercepting aircraft and responses by intercepted aircraft

Series	INTERCEPTING Aircraft Signals	Meaning	INTERCEPTED Aircraft Responds	Meaning
1	DAY or NIGHT — Rocking aircraft and flashing navigational lights at irregular intervals (and landing lights in the case of a helicopter) from a position slightly above and ahead of, and normally to the left of, the intercepted aircraft (or to the right if the intercepted aircraft is a helicopter) and, after acknowledgement, a slow level turn, normally to the left (or to the right in the case of a helicopter) on the desired heading. <i>Note 1</i> Meteorological conditions or terrain may require the intercepting aircraft to reverse the positions and direction of turn given above in Series 1. <i>Note 2</i> If the intercepted aircraft is not able to keep pace with the intercepting aircraft, the latter is expected to fly a series of race-track patterns and to rock the aircraft each time it passes the intercepted aircraft.	You have been intercepted. Follow me.	DAY or NIGHT — Rocking aircraft, flashing navigational lights at irregular intervals and following.	Understood, will comply.
2	DAY or NIGHT — An abrupt breakaway manoeuvre from the intercepted aircraft consisting of a climbing turn of 90 degrees or more without crossing the line of flight of the intercepted aircraft.	You may proceed.	DAY or NIGHT — Rocking the aircraft.	Understood, will comply.
3	DAY or NIGHT — Lowering landing gear (if fitted), showing steady landing lights and overflying runway in use or, if the intercepted aircraft is a helicopter, overflying the helicopter landing area. In the case of helicopters, the intercepting helicopter makes a landing approach, coming to hover near to the landing area.	Land at this aerodrome.	DAY or NIGHT — Lowering landing gear, (if fitted), showing steady landing lights and following the intercepting aircraft and, if, after overflying the runway in use or helicopter landing area, landing is considered safe, proceeding to land.	Understood, will comply.

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Table S11-2

Signals initiated by intercepted aircraft and responses by intercepting aircraft

Series	INTERCEPTED Aircraft Signals	Meaning	INTERCEPTING Aircraft Responds	Meaning
4	DAY or NIGHT — Raising landing gear (if fitted) and flashing landing lights while passing over runway in use or helicopter landing area at a height exceeding 300 m (1 000 ft) but not exceeding 600 m (2 000 ft) (in the case of a helicopter, at a height exceeding 50 m (170 ft) but not exceeding 100 m (330 ft)) above the aerodrome level, and continuing to circle runway in use or helicopter landing area. If unable to flash landing lights, flash any other lights available.	Aerodrome you have designated is inadequate.	DAY or NIGHT — If it is desired that the intercepted aircraft follow the intercepting aircraft to an alternate aerodrome, the intercepting aircraft raises its landing gear (if fitted) and uses the Series 1 signals prescribed for intercepting aircraft. If it is decided to release the intercepted aircraft, the intercepting aircraft uses the Series 2 signals prescribed for intercepting aircraft.	Understood, follow me. Understood, you may proceed.
5	DAY or NIGHT — Regular switching on and off of all available lights but in such a manner as to be distinct from flashing lights.	Cannot comply.	DAY or NIGHT — Use Series 2 signals prescribed for intercepting aircraft.	Understood.
6	DAY or NIGHT — Irregular flashing of all available lights.	In distress.	DAY or NIGHT — Use Series 2 signals prescribed for intercepting aircraft.	Understood.

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Table S11-3

Phrases for use by INTERCEPTING aircraft			Phrases for use by INTERCEPTED aircraft		
Phrase	Pronunciation ⁽¹⁾	Meaning	Phrase	Pronunciation ⁽¹⁾	Meaning
CALL SIGN	<u>KOL</u> SA-IN	What is your call sign?	CALL SIGN (call sign) ⁽²⁾	<u>KOL</u> SA-IN (call sign)	My call sign is (call sign)
FOLLOW	FOL-LO	Follow me	WILCO	<u>VILL</u> -KO	Understood, will comply
DESCEND	DEE-SEND	Descend for landing			
			CAN NOT	<u>KANN</u> NOTT	Unable to comply
YOU LAND	YOU LAAND	Land at this aerodrome	REPEAT	REE-PEET	Repeat your instruction
			AM LOST	AM LOSST	Position unknown
PROCEED	PRO- <u>SEED</u>	You may proceed			
			MAYDAY	MAYDAY	I am in distress
			HIJACK ⁽³⁾	<u>HI-JACK</u>	I have been hijacked
			LAND (place name)	LAND (place name)	I request to land at (place name)
			DESCEND	DEE-SEND	I require descent

⁽¹⁾ In the second column, syllables to be emphasised are underlined.

⁽²⁾ The call sign required to be given is that used in radiotelephony communications with air traffic services units and corresponding to the aircraft identification in the flight plan.

⁽³⁾ Circumstances may not always permit, nor make desirable, the use of the phrase 'HIJACK'.

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12.2 Visual and Instrument Flight Rules

4. Visual Flight

4.1 Visual Meteorological Conditions

(SERA 5.001)

VMC visibility and distance from cloud minima are contained in Table S5-1.

Table S5-1

Table S5-1 (*)			
Altitude band	Airspace class	Flight visibility	Distance from cloud
At and above 3 050 m (10 000 ft) AMSL	A (**) B C D E F G	8 km	1 500 m horizontally 300 m (1 000 ft) vertically
Below 3 050 m (10 000 ft) AMSL and above 900 m (3 000 ft) AMSL, or above 300 m (1 000 ft) above terrain, whichever is the higher	A (**) B C D E F G	5 km	1 500 m horizontally 300 m (1 000 ft) vertically
At and below 900 m (3 000 ft) AMSL, or 300 m (1 000 ft) above terrain, whichever is the higher	A (**) B C D E	5 km	1 500 m horizontally 300 m (1 000 ft) vertically
	F G	5 km (***)	Clear of cloud and with the surface in sight

(*) When the height of the transition altitude is lower than 3 050 m (10 000 ft) AMSL, FL 100 shall be used in lieu of 10 000 ft.

(**) The VMC minima in Class A airspace are included for guidance to pilots and do not imply acceptance of VFR flights in Class A airspace.

(***) When so prescribed by the competent authority:

(a) flight visibilities reduced to not less than 1 500 m may be permitted for flights operating:

- (1) at speeds of 140 kts IAS or less to give adequate opportunity to observe other traffic or any obstacles in time to avoid collision; or
- (2) in circumstances in which the probability of encounters with other traffic would normally be low, e.g. in areas of low volume traffic and for aerial work at low levels;

(b) Helicopters may be permitted to operate in less than 1 500 m but not less than 800 m flight visibility, if manoeuvred at a speed that will give adequate opportunity to observe other traffic or any obstacles in time to avoid collision. Flight visibilities lower than 800 m may be permitted for special cases, such as medical flights, search and rescue operations and fire-fighting.

4.2 Visual Flight Rules

(SERA 5.005)

- a. Except when operating as a special VFR flight, VFR flights shall be conducted so that the aircraft is flown in conditions of visibility and distance from clouds equal to or greater than those specified in Table S5-1.
- b. Except when a special VFR clearance is obtained from an air traffic control unit, VFR flights shall not take off or land at an aerodrome within a control zone, or enter the aerodrome traffic zone or aerodrome traffic circuit when the reported meteorological conditions at that aerodrome are below the following minima:
 1. the ceiling is less than 450 m (1 500 ft); or
 2. the ground visibility is less than 5 km.

- c. When so prescribed by the competent authority, VFR flights at night may be permitted under the following conditions:
1. if leaving the vicinity of an aerodrome, a flight plan shall be submitted in accordance with SERA.4001(b)(6);
 2. flights shall establish and maintain two-way radio communication on the appropriate ATS communication channel, when available;
 3. the VMC visibility and distance from cloud minima as specified in Table S5-1 shall apply except that:
 - i. the ceiling shall not be less than 450 m (1 500 ft);
 - ii. the reduced flight visibility provisions specified in Table S5-1(a) and (b) shall not apply;
 - iii. in airspace classes B, C, D, E, F and G, at and below 900 m (3 000 ft) AMSL or 300 m (1 000 ft) above terrain, whichever is the higher, the pilot shall maintain continuous sight of the surface; and
 - iv. for mountainous terrain, higher VMC visibility and distance from cloud minima may be prescribed by the competent authority;
 4. except when necessary for take-off or landing, or except when specifically authorised by the competent authority, a VFR flight at night shall be flown at a level which is not below the minimum flight altitude established by the State whose territory is overflown, or, where no such minimum flight altitude has been established:
 - i. over high terrain or in mountainous areas, at a level which is at least 600 m (2 000 ft) above the highest obstacle located within 8 km of the estimated position of the aircraft;
 - ii. elsewhere than as specified in i), at a level which is at least 300 m (1 000 ft) above the highest obstacle located within 8 km of the estimated position of the aircraft.
- d. VFR flights shall not be operated:
1. at transonic and supersonic speeds unless authorised by the competent authority;
 2. above FL 195. Exceptions to this requirement are the following:
 - i. an airspace reservation has been established, where practical, by the Member States, in which VFR flights may be allowed; or
 - ii. airspace up to and including flight level 285, when VFR traffic in that airspace has been authorised by the responsible ATS unit in accordance with the authorisation procedures established by the Member States and published in the relevant aeronautical information publication.
- e. Authorisation for VFR flights to operate above FL 285 shall not be granted where a vertical separation minimum of 300 m (1 000 ft) is applied above FL 290.
- f. Except when necessary for take-off or landing, or except by permission from the competent authority, a VFR flight shall not be flown:
1. over the congested areas of cities, towns or settlements or over an open-air assembly of persons at a height less than 300 m (1 000 ft) above the highest obstacle within a radius of 600 m from the aircraft;

2. elsewhere than as specified in (1), at a height less than 150 m (500 ft) above the ground or water, or 150 m (500 ft) above the highest obstacle within a radius of 150 m (500 ft) from the aircraft.
- g. Except where otherwise indicated in air traffic control clearances or specified by the competent authority, VFR flights in level cruising flight when operated above 900 m (3 000 ft) from the ground or water, or a higher datum as specified by the competent authority, shall be conducted at a cruising level appropriate to the track.
- h. VFR flights shall comply with the provisions of Section 8:
 1. when operated within Classes B, C and D airspace;
 2. when forming part of aerodrome traffic at controlled aerodromes; or
 3. when operated as special VFR flights.
- i. A VFR flight operating within or into areas or along routes designated by the competent authority, in accordance with SERA.4001(b)(3) or (4), shall maintain continuous air-ground voice communication watch on the appropriate communication channel of, and report its position as necessary to, the air traffic services unit providing flight information service.
- j. An aircraft operated in accordance with the visual flight rules which wishes to change to compliance with the instrument flight rules shall:
 1. if a flight plan was submitted, communicate the necessary changes to be effected to its current flight plan; or
 2. as required by SERA.4001(b), submit a flight plan to the appropriate air traffic services unit as soon as practicable and obtain a clearance prior to proceeding IFR when in controlled airspace.

5. Instrument Flight Rules

(SERA.5015)

5.1 Rules Applicable to All IFR Flights

a. Aircraft equipment

Aircraft shall be equipped with suitable instruments and with navigation equipment appropriate to the route to be flown and in accordance with the applicable air operations legislation.

b. Minimum levels

Except when necessary for take-off or landing, or except when specifically authorised by the competent authority, an IFR flight shall be flown at a level which is not below the minimum flight altitude established by the State whose territory is overflown, or, where no such minimum flight altitude has been established:

1. over high terrain or in mountainous areas, at a level which is at least 600 m (2 000 ft) above the highest obstacle located within 8 km of the estimated position of the aircraft;
2. elsewhere than as specified in (1), at a level which is at least 300 m (1 000 ft) above the highest obstacle located within 8 km of the estimated position of the aircraft.

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c. Change from IFR flight to VFR flight

1. An aircraft electing to change the conduct of its flight from compliance with the instrument flight rules to compliance with the visual flight rules shall notify the appropriate air traffic services unit specifically that the IFR flight is cancelled and communicate thereto the changes to be made to its current flight plan.
2. When an aircraft operating under the instrument flight rules is flown in or encounters visual meteorological conditions it shall not cancel its IFR flight unless it is anticipated, and intended, that the flight will be continued for a reasonable period of time in uninterrupted visual meteorological conditions.
3. Change from IFR flight to VFR flight shall only be acceptable when a message initiated by the pilot-in-command containing the specific expression 'CANCELLING MY IFR FLIGHT', together with the changes, if any, to be made to the current flight plan, is received by an ATS unit. No invitation to change from IFR flight to VFR flight shall be made by ATS either directly or by inference.

5.2 Rules Applicable to IFR Flights within Controlled Airspace

(SERA.5020)

- a. IFR flights shall comply with the provisions of Section 8 when operated in controlled airspace.
- b. An IFR flight operating in cruising flight in controlled airspace shall be flown at a cruising level, or, if authorised by ATS unit to employ cruise climb techniques, between two levels or above a level, selected from the table of cruising levels (refer to Jeppesen chart), except that the correlation of levels to track prescribed therein shall not apply whenever otherwise indicated in air traffic control clearances or specified by the competent authority in aeronautical information publications.

5.3 Rules Applicable to IFR Flights Outside Controlled Airspace

(SERA.5025)

- a. Cruising levels

An IFR flight operating in level cruising flight outside of controlled airspace shall be flown at a cruising level appropriate to its track as specified (refer to Jeppesen chart), except when otherwise specified by the competent authority for flight at or below 900 m (3 000 ft) above mean sea level.

- b. Communications

An IFR flight operating outside controlled airspace but within or into areas, or along routes, designated by the competent authority in accordance with SERA.4001(b)(3) or (4) shall maintain an air-ground voice communication watch on the appropriate communication channel and establish two-way communication, as necessary, with the air traffic services unit providing flight information service.

- c. Position reports

An IFR flight operating outside controlled airspace and required by the competent authority to maintain an air-ground voice communication watch on the appropriate communication channel and establish two-way communication, as necessary, with the air traffic services unit providing flight information service, shall report position, as specified in SERA.8025 for controlled flights.

- d. Although an IFR flight operating in level cruising flight outside controlled airspace is to be flown at a cruising level appropriate to its track, as specified in the table of cruising levels, this does not preclude the use of cruise climb techniques.
- e. Aircraft electing to use the air traffic advisory service whilst operating under IFR within specified advisory airspace are expected to comply with the provisions of 'Air traffic Control Service', except that the flight plan and changes thereto are not subject to clearances and that two-way communication will be maintained with the unit providing the air traffic advisory service.

12.3 Communications and SSR Transponder

(SERA 6.005)

- a. Radio mandatory zone (RMZ)
 - 1. VFR flights operating in parts of Classes E, F or G airspace and IFR flights operating in parts of Classes F or G airspace designated as a radio mandatory zone (RMZ) by the competent authority shall maintain continuous air-ground voice communication watch and establish two-way communication, as necessary, on the appropriate communication channel, unless in compliance with alternative provisions prescribed for that particular airspace by the ANSP.
 - 2. Before entering a radio mandatory zone, an initial call containing the designation of the station being called, call sign, type of aeroplane, position, level, the intentions of the flight and other information as prescribed by the competent authority, shall be made by pilots on the appropriate communication channel.
- b. Transponder mandatory zone (TMZ)
 - 1. All flights operating in airspace designated by the competent authority as a transponder mandatory zone (TMZ) shall carry and operate SSR transponders capable of operating on Modes A and C or on Mode S, unless in compliance with alternative provisions prescribed for that particular airspace by the ANSP.
- c. Airspaces designated as radio mandatory zone and / or transponder mandatory zone shall be duly promulgated in the aeronautical information publications.

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12.4 Aeroplane Observations

12.4.1 Type of Aeroplane Observations

(SERA.12001)

The following aeroplane observations shall be made during any phase of the flight:

1. special aeroplane observations; and
2. other non-routine aeroplane observations.

12.4.2 Special Aeroplane Observations

(SERA.12005)

- a. Special observations shall be made and reported by all aeroplanes whenever the following conditions are encountered or observed:
 1. moderate or severe turbulence; or
 2. moderate or severe icing; or
 3. severe mountain wave; or
 4. thunderstorms, without hail, that are obscured, embedded, widespread or in squall lines; or
 5. thunderstorms, with hail, that are obscured, embedded, widespread or in squall lines; or
 6. heavy dust storm or heavy sandstorm; or
 7. volcanic ash cloud; or
 8. pre-eruption volcanic activity or a volcanic eruption.
- b. Competent authorities shall prescribe as necessary other conditions which shall be reported by all aeroplanes when encountered or observed.
- c. Flight crews shall compile the special air-reports as per [Reporting Procedures](#).

12.4.3 Other Non-routine Aeroplane Observations

(SERA.12010)

When other meteorological conditions not listed under SERA.12.4.2 (a), e.g. wind shear, are encountered and which, in the opinion of the pilot-in-command, may affect the safety or markedly affect the efficiency of other aeroplane operations, the pilot-in-command shall advise the appropriate air traffic services unit as soon as practicable.

12.4.4 Reporting of Aeroplane Observations by Voice Communication

(SERA.12015)

- a. Aeroplane observations shall be reported during flight at the time the observation is made or as soon thereafter as is practicable.
- b. Aeroplane observations shall be reported as air-reports and shall comply with the technical specifications in [Reporting Procedures](#).

12.4.5 Exchange of Air-reports

(SERA.12020)

- a. ATS units shall transmit, as soon as practicable, special and non-routine air-reports to:
 - other aeroplanes concerned;
 - the associated meteorological watch office (MWO); and
 - other ATS units concerned.
- b. Transmissions to aeroplanes shall be repeated at a frequency and continued for a period of time which shall be determined by the ATS unit concerned.

12.4.6 Reporting Procedures

- a. Special air-reports

The elements contained in special air-reports shall be:

- Message type designator;
- Aeroplane identification;
- Position or latitude and longitude;
- Time;
- Level or range of levels;
- Condition prompting the issuance of a special air-report, to be selected from the list presented in [Special Aeroplane Observations](#).

For detail reporting procedures including the formats of messages and the phraseologies refer to Jeppesen Airway Manual / General Airway Manual / Air Traffic Control / International Civil Aviation Organisation - Air Traffic Management / Appendix 1 - Instructions for Air-reporting by Voice Communications.

For reporting of Volcanic activity, refer to the Special Air-report of Volcanic Activity form (located below the Special Air-report in the Jeppesen Airway Manual described above).

Since a more detailed report is required after arrival (see point c.) it is recommended to refer to the Special Air-report of Volcanic Activity form to gather the other data that is not required to be reported by voice e.g. points 9-16 of the report form.

- b. Reporting of wind shear

When reporting aeroplane observations of wind shear encountered during the climb-out and approach phases of flight, the aeroplane type shall be included.

Where wind shear conditions in the climb-out or approach phases of flight were reported or forecast but not encountered, the pilot-in-command shall advise the appropriate air traffic services unit as soon as practicable unless the pilot-in-command is aware that the appropriate air traffic services unit has already been so advised by a preceding aeroplane.

- c. Post-flight reporting of volcanic activity

On arrival of a flight at an aerodrome, the completed report of volcanic activity shall be delivered by the aeroplane operator or a flight crew member, without delay, to the aerodrome meteorological office, or if such office is not easily accessible to arriving flight crew members, the completed form shall be dealt with in accordance with local arrangements made by the meteorological authority and the aeroplane operator.

NTA crew members shall comply with this by filing a NetJets Safety Report, in accordance with the mandatory reporting criteria - [Chapter 11 - Handling, Notifying and Reporting Accidents, Serious Incidents and Occurrences](#). The safety report will then be followed up by the Safety Department, including liaising with the applicable authorities and meteorological office.

12.5 Visual Signals Used to Warn an Unauthorised Aeroplane Flying in or About to Enter a Restricted, Prohibited or Danger Area

Also refer to ICAO Annex 2, Rules of the Air, Appendix 2 as published in the Jeppesen Airway Manual, Air Traffic Control section Appendix 1.

By day and by night, a series of projectiles discharged from the ground at intervals of 10 seconds, each showing, on bursting, red and green lights or stars will indicate to an unauthorised aeroplane that it is flying in or about to enter a restricted, prohibited or danger area, and that the aeroplane is to take such remedial action as may be necessary.

12.6 Procedures for Flight Crew Observing an Accident or Receiving a Distress Transmission

Refer to SEP Manual, Chapter 5.

12.7 Ground / Air Visual Codes for Use by Survivors, Description and Use of Signal Aids

Refer to SEP Manual, Chapter 5.

12.8 Distress and Urgency

These procedures are in accordance with ICAO Annex 2 and can be referred to in the Jeppesen Airway Manual in the Emergency Data section Chapter 2.5.

NOTE

None of the provisions in this section shall prevent the use, by an aeroplane in distress, of any means at its disposal to attract attention, make known its position and obtain help.

2.5.1 Distress Signals

2.5.1.1 The following signals, used either together or separately, mean that grave and imminent danger threatens, and immediate assistance is requested:

- a. A signal made by radiotelegraphy or by any other signalling method consisting of the group SOS (. . . - - - . . .) in the Morse Code);
- b. A radiotelephony distress signal consisting of the spoken word MAYDAY.

12.9 Air Traffic Control Service

12.9.1 Operation of Air Traffic Control Service - Visual Approach

(SERA 8005 (c))

1. Subject to the conditions described in point (2), clearance for an IFR flight to execute a visual approach may be requested by a flight crew or initiated by the air traffic controller. In the latter case, the concurrence of the flight crew should be required.
2. An IFR flight should only be cleared to execute a visual approach, provided the pilot can maintain visual reference to the terrain and:
 - a. the reported ceiling is at or above the level of the beginning of the initial approach segment for the aircraft so cleared; or
 - b. the pilot reports at the level of the beginning of the initial approach segment or at any time during the instrument approach procedure that the meteorological conditions are such that with reasonable assurance a visual approach and landing can be completed.

The pilot-in-command of the aircraft concerned is responsible for ensuring that the spacing from a preceding aircraft of a heavier wake turbulence category is acceptable. If it is determined that additional spacing is required, the flight crew should inform the ATC unit accordingly, stating their requirements.

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Chapter 13 - Aircraft Leasing

Due to commercial reasons, NTA may need to establish a wet or dry lease agreements. They shall comply with the regulations listed below and, if applicable, the respective AMCs and any other regulation referred there in:

- Commission Regulation (EU) No 965/2012
- Regulation (EC) No 1008/2008 (Operation Of Air Services)
- Commission Delegated Regulation (EU) 2020/723 (Acceptance Of Third-Country Certification Of Pilots)
- Commission Regulation (EU) No 748/2012 (Initial Airworthiness)
- Commission Regulation (EU) No 1321/2014 (Continuing Airworthiness) •
- Regulation (EU) 2018/1139 (Basic Regulation)
- Regulation (EC) No 2111/2005 (EU Ban List / Identity Of The Operating Air Carrier)
- Regulation (EC) No 785/2004 (Insurance Requirements For Air Carriers And Aircraft Operators)
- Commission Regulation (EU) No 452/2014 (Third Country Operators)
- Regulamento (INAC) 832/2010;.

NTA will also comply with any additional conditions and procedures published by ANAC in official channels (site, CIA or other).

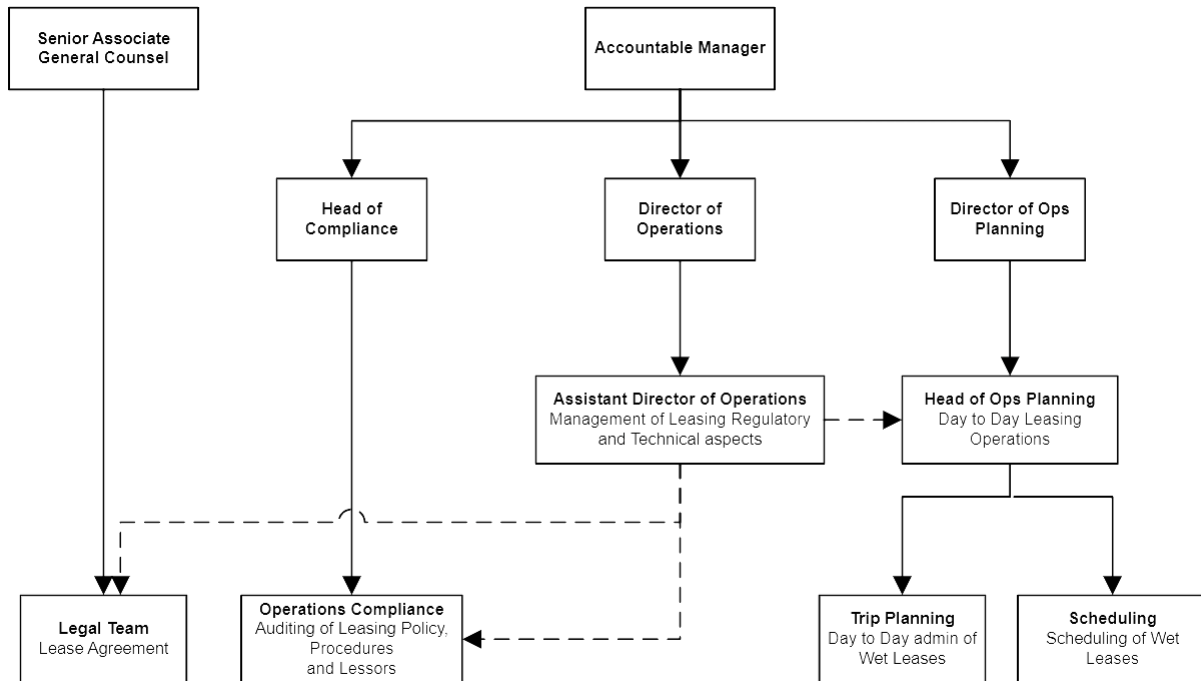
When leasing in or out aircraft from or to other operators, and without prejudice to Regulation (EC) No 1008/2008, any lease agreement established by NTA shall be subject to prior approval by ANAC. Any prerequisites part of this approval must be included in the lease agreement.

NTA shall not lease-in aircraft included in the list of operators subject to operational restrictions, registered in a State of which all operators under its oversight are subject to an operating ban or from an operator that is subject to an operating ban pursuant to the Regulation (EC) No 2111/2005.

When, for the purpose of a lease, the State of Registry transfers all or some of its functions and duties to the State of the Operator, an agreements under Article 83 bis of the Chicago Convention is required. All aircraft operating under such an agreement must carry a certified true copy of the Agreement Summary on board.

13.1 Management Responsibilities

The following is an organogram of the lines of responsibility with NTA with regard to leasing of aircraft. It includes those responsible for 'short term' and 'long term' leasing and those involved from a flight operations, continuing airworthiness and maintenance management, oversight and commercial perspective.



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13.2 Lease in

13.2.1 Dry Lease-In

NTA (lessee) is responsible for the continuing airworthiness of the aeroplane.

1. Aeroplane with a EU Registration

NTA shall ensure that:

- a. The aeroplane is included on the NTA AOC;
- b. The aeroplane Certificate of Airworthiness is in accordance with Regulation (EC) 748/2012;
- c. The aeroplane continued airworthiness is managed in accordance with Part-M.

2. Aircraft with a Non-EU Registration

NTA shall ensure that:

- a. An operational need has been identified that cannot be satisfied through leasing of an aeroplane registered in the EU;
- b. The duration of the dry lease-in does not exceed seven months in any 12 consecutive month period;
- c. Compliance with the applicable requirements of Regulation (EC) No 1321/2014 is ensured;
- d. The aeroplane is included on the NTA AOC;
- e. The aeroplane is equipped in accordance with Regulation (EU) No 965/2012AIR.

13.2.2 Wet Lease-In

General

The Operator (lessor) is responsible for continuing the airworthiness of the aeroplane.

NTA shall not wet lease-in an aeroplane that is included in the list of operators subject to operational restrictions, registered in a State of which all operators under its oversight are subject to an operational ban or from an operator that is subject to an operating ban pursuant to Regulation (EC) No 2111/2005.

1. Aircraft from an EU Operator

NTA shall ensure that:

- a. The operator has an AOC issued in accordance with Regulation (EU) No 965/2012; and
- b. The operator complies with Regulation (EU) 1321/2014.

2. Aircraft from a Non-EU Operator

NTA shall ensure that:

- a. The third country operator holds a valid AOC issued in accordance with ICAO Annex 6;

- b. The safety standards of the operator with regard to continuing airworthiness and air operations are equivalent to the applicable requirements established by Regulation (EU) No 965/2012 and Regulation (EU) 1321/2014; and
- c. The aeroplane has a standard Certificate of Airworthiness issued in accordance with ICAO Annex 8.

13.3 Lease-Out

13.3.1 Dry Lease-Out

General

NTA does not Dry Lease Out to any operator.

13.3.2 Wet Lease-Out

General

A wet lease-out is a contract between two air carriers in which an aeroplane (with crew) is operated under the AOC of the lessor. Therefore, the aeroplane is kept under NTA's AOC, but the commercial responsibility is transferred to the lessee.

As an exception to other leases, wet lease-out contracts do not require approval from ANAC, and only a notification is required.

NTA shall ensure that:

- a. Before the aeroplane is wet-leased out, ANAC is notified pursuant Regulamento (INAC) 832/2010 and Commission Regulation (EU) No 965/2012;
 - i. Additional information/documents, as required by the above regulations, shall be attached to the notification;
- b. The lessee must hold a valid AOC issued per Annex 6 to the Convention on International Civil Aviation;
- c. The lease contract contains a description of how the airworthiness of the aeroplane will be maintained.

Aircraft to a non-EU operator

As mandated by Commission Delegated Regulation (EU) 2020/723, in cases the lessee holds an AOC from a third country, FCM will need to obtain validation or exemption from the respective authority regarding their licenses and associated ratings, privileges or certificates. Exceptions may occur when in the presence of international agreements concluded between the Union and a third country per point (a) of Article 68(1) of Regulation (EU) 2018/1139. In this section, a third-country operator is one that requires a TCO authorisation to perform commercial air transport (CAT) operations into, within or out of an EU territory.

Technical Annex

Additional Technical Guidance

TA 1 Conversion of Units Units of Measurement

The metric system is used:

- Wind in M/S: $1 \text{ M/S} = 2 \text{ KTS (1,94)}$
- Speed in KM/H: $\text{KTS} = (\text{KM/H} / 2) + 10\%$
- Altitude in M: $\text{FT} = (\text{M} \times 3.28)$
- Distance in KM: $1 \text{ NM} = 1,85 \text{ KM}$
- Descent rate in M/S: $1 \text{ M/S} = 200 \text{ ft/min}$

Aeroplanes equipped with FMSs that have conversions pages / function, can obtain the majority of the conversions on the respective page(s).

- End of Additional Guidance -

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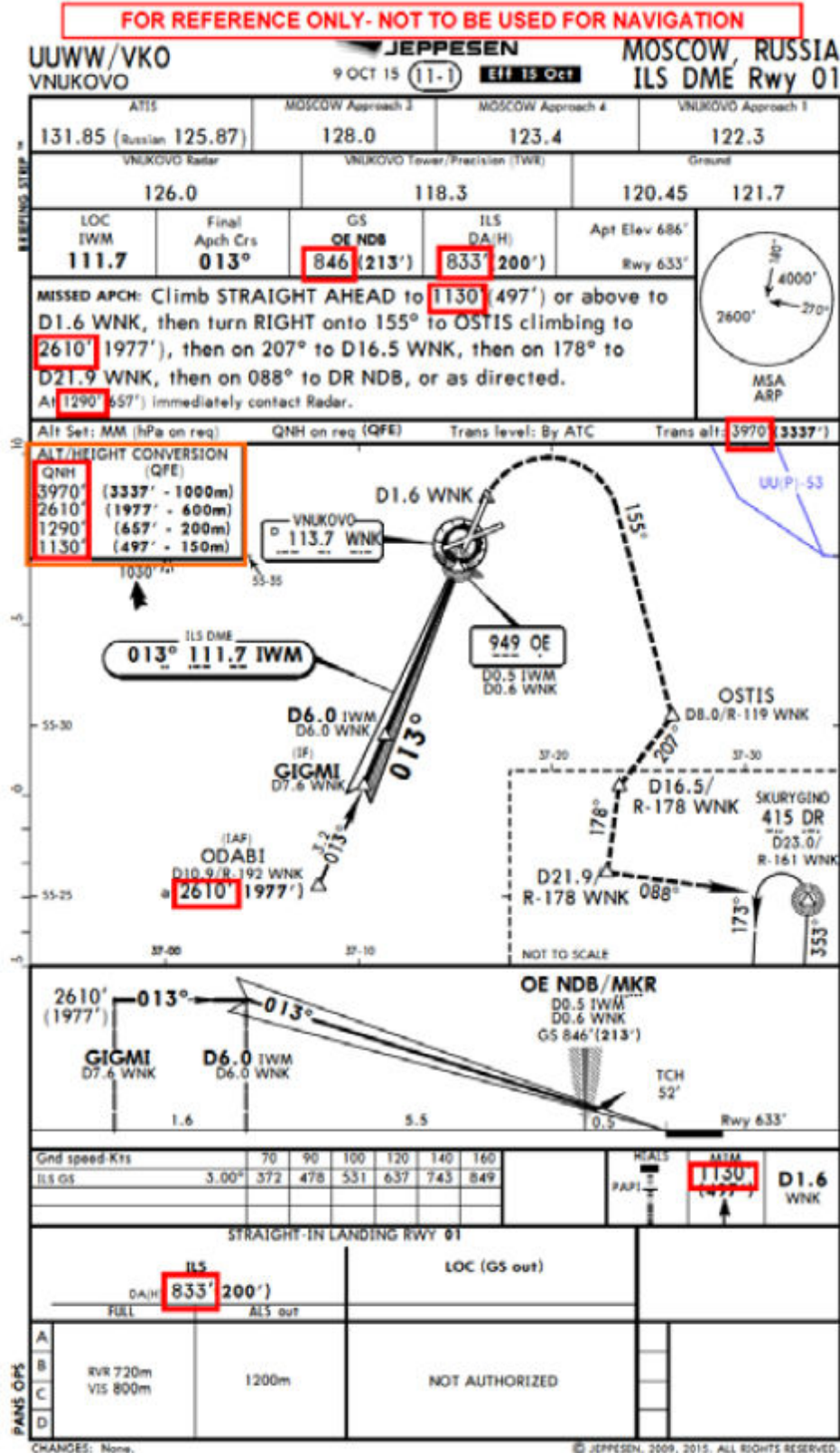
TA 2 Jeppesen Chart with Conversions

Below is an example of a Jeppesen Chart with the respective conversions:

- Values in brackets are QFE height in metres and / or feet.
- Values not in brackets are QNH altitudes in feet.

Converted altitudes in feet are boxed in **RED** in the chart below. Some charts may have a conversion table with the various altitude conversions (boxed in **orange**). Some may not and the conversion will be shown at the respective waypoint / fix.

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- End of Additional Guidance -

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TA 3 Meters to Feet Conversion Table**NOTE**

This table does not automatically convert QFE heights to QNH altitudes, it shall only be used when the altimeter setting datum matches the clearance datum, to convert metres to feet (e.g. altimeter is set to QFE and clearance is to a height based on QFE).

Aircraft types unable to use QFE, shall not use this table for operation in the metric / QFE environment and should instead refer to the aerodrome specific AHCC chart.

Meters	Feet
100	300
200	700
300	1000
400	1300
500	1600
600	2000
900	3000
1200	3900
1500	4900
1800	5900
2100	6900
2400	7900
2700	8900
3000	9800
3300	10800
3600	11800
3900	12800
4200	12800
4500	14800
4800	15700
5100	16700
5400	17700
5700	18700
6000	19700
6300	20700
6600	21700
6900	22600

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Meters	Feet
7200	23600
7500	24600
7800	25600
8100	26600
8600	28200
9100	29900
9600	31500
10100	33100
10600	34800
11100	36400
11600	38100
12100	39700
13100	43000
14100	46300
15100	49500
16100	52800

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TA 4 Millimetres Mercury to Hectopascals / Millibars Conversion

For example, 755 mm Mercury equals 1006.6 hPa (Mb).

Millimetres MERCURY to hPa or Mb (1 mm Mercury = 1.3332 hPa (Mb))										
MM	0	1	2	3	4	5	6	7	8	9
	HECTOPASCALS (or MILLIBARS)									
530	706.6	707.9	709.3	710.6	711.9	713.3	714.6	715.9	717.3	718.6
540	719.9	721.3	722.6	723.9	725.3	726.6	727.9	729.3	730.6	731.9
550	733.3	734.6	735.9	737.3	738.6	739.9	741.3	742.6	743.9	745.3
560	746.6	747.9	749.3	750.6	751.9	753.3	754.6	755.9	757.3	758.6
570	759.9	761.3	762.6	763.9	765.3	766.6	767.9	769.3	770.6	771.9
580	773.3	774.6	775.9	777.3	778.6	779.9	781.3	782.6	783.9	785.3
590	786.6	787.9	789.3	790.6	791.9	793.3	794.6	795.9	797.3	798.6
600	799.9	801.3	802.6	803.9	805.3	806.6	807.9	809.3	810.6	811.9
610	813.3	814.6	815.9	817.3	818.6	819.9	821.3	822.6	823.9	825.3
620	826.6	827.9	829.3	830.6	831.9	833.3	834.6	835.9	837.2	838.6
630	839.9	841.2	842.6	843.9	845.2	846.6	847.9	849.2	850.6	851.9
640	853.2	854.6	855.9	857.2	858.6	859.9	861.2	862.6	863.9	865.2
650	866.6	867.9	869.2	870.6	871.9	873.2	874.6	875.9	877.2	878.6
660	879.9	881.2	882.6	883.9	885.2	886.6	887.9	889.2	890.6	891.9
670	893.2	894.6	895.9	897.2	898.6	899.9	901.2	902.6	903.9	905.2
680	906.6	907.9	909.2	910.6	911.9	913.2	914.6	915.9	917.2	918.6
690	919.9	921.2	922.6	923.9	925.2	926.6	927.9	929.2	930.6	931.9
700	933.2	934.6	935.9	937.2	938.6	939.9	941.2	942.6	943.9	845.2
710	946.6	947.9	949.2	950.6	951.9	953.2	954.6	955.9	957.2	958.6
720	959.9	961.2	962.6	963.9	965.2	966.6	967.9	969.2	970.6	971.9
730	973.2	974.6	975.9	977.2	978.6	979.9	981.2	982.6	983.9	985.2
740	986.6	987.9	988.2	990.6	991.9	993.2	994.6	995.9	997.2	998.6
750	999.9	1001.2	1002.6	1003.9	1005.2	1006.6	1007.9	1009.2	1010.6	1011.9
760	1013.2	1014.6	1015.9	1017.2	1018.6	1019.9	1021.2	1022.6	1023.9	1025.2
770	1026.6	1027.9	1029.2	1030.6	1031.9	1033.2	1034.6	1035.9	1037.2	1038.6
780	1039.9	1041.2	1042.6	1043.9	1045.2	1046.6	1047.9	1049.2	1050.6	1051.9
790	1053.2	1054.6	1055.9	1057.2	1058.6	1059.9	1061.2	1062.6	1063.9	1065.2
800	1066.6	1067.9	1069.2	1070.6	1071.9	1073.2	1074.6	1075.9	1077.2	1078.6

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TA 5 QFE to QNH Altimeter Setting Conversion**Hectopascal / Millibar Equivalent for Airfield Elevation**

This table provides a correction factor in Hectopascals or Millibars for elevations on an airport. Application of the factoring this table will provide a QNH altimeter setting. To use the table, determine the correction factor by finding the elevation in hundreds of feet in the left column and reading across to the nearest ten feet. For example, 1220 feet equals 43.9. Add 43.9 to the QFE altimeter setting (hPa or Mb) and your altimeter should read the airport elevation.

Elev	-200	-190	-180	-170	-160	-150	-140	-130	-120	-110
	-7.3	-7	-6.6	-6.2	-5.9	-5.5	-5.1	-4.8	-4.4	-4
Elev	-100	-90	-80	-70	-60	-50	-40	-30	-20	-10
	-3.7	-3.3	-2.9	-2.6	-2.2	-1.8	-1.5	-1.1	-0.7	-0.4
Elev	+00	+10	+20	+30	+40	+50	+60	+70	+80	+90
0	0.0	0.4	0.7	1.1	1.5	1.8	2.2	2.6	2.9	3.3
100	3.7	4.0	4.4	4.8	5.1	5.5	5.8	6.2	6.6	6.9
200	7.3	7.7	8.0	8.4	8.8	9.1	9.5	9.8	10.2	10.6
300	10.9	11.3	11.7	12.0	12.4	12.8	13.1	13.5	13.8	14.2
400	14.6	14.9	15.3	15.6	16.0	16.4	16.7	17.1	17.5	17.8
500	18.2	18.5	18.9	19.3	19.6	20.0	20.3	20.7	21.1	21.4
600	21.8	22.1	22.5	22.9	23.2	23.6	23.9	24.3	24.7	25.0
700	25.4	25.7	26.1	26.4	26.8	27.2	27.5	27.9	28.2	28.6
800	29.0	29.3	29.7	30.0	30.4	30.7	31.1	31.5	31.8	32.2
900	32.5	32.9	33.2	33.6	33.9	34.3	34.7	35.0	35.4	35.7
1000	36.1	36.4	36.8	37.2	37.5	37.9	38.2	38.6	38.9	38.3
1100	39.6	40.0	40.3	40.7	41.1	41.4	41.8	42.1	42.5	42.8
1200	43.2	43.5	43.9	44.2	44.6	44.9	45.3	45.6	46.0	46.4
1300	46.7	47.1	47.4	47.8	48.1	48.5	48.8	49.2	49.5	49.9
1400	50.2	50.6	50.9	51.3	51.6	52.0	52.3	52.7	53.0	53.4
1500	53.7	54.1	54.4	54.8	55.1	55.5	55.8	56.2	56.5	56.9
1600	57.2	57.6	57.9	58.3	58.6	59.0	59.3	59.7	60.0	60.4
1700	60.7	61.1	61.4	61.8	62.1	62.5	62.8	63.2	63.5	63.8
1800	64.2	64.5	64.9	65.2	65.6	65.9	66.3	66.6	67.0	67.3
1900	67.7	68.0	68.4	68.7	69.0	69.4	69.7	70.1	70.4	70.8
2000	71.1	71.5	71.8	72.2	72.5	72.8	73.2	73.5	73.9	74.2
2100	74.6	74.9	75.3	75.6	75.9	76.3	76.6	77.0	77.3	77.7
2200	78.0	78.3	78.7	79.0	79.4	79.7	80.1	80.4	80.7	81.1
2300	81.4	81.8	82.1	82.5	82.8	83.1	83.5	83.8	84.2	84.5

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Elev	-200	-190	-180	-170	-160	-150	-140	-130	-120	-110
	-7.3	-7	-6.6	-6.2	-5.9	-5.5	-5.1	-4.8	-4.4	-4
Elev	-100	-90	-80	-70	-60	-50	-40	-30	-20	-10
	-3.7	-3.3	-2.9	-2.6	-2.2	-1.8	-1.5	-1.1	-0.7	-0.4
Elev	+00	+10	+20	+30	+40	+50	+60	+70	+80	+90
2400	84.8	85.2	85.5	85.9	86.2	86.6	86.9	87.2	87.6	87.9
2500	88.3	88.6	88.9	89.3	89.6	90.0	90.3	90.6	91.0	91.3
2600	91.6	92.0	92.3	92.7	93.0	93.3	93.7	94.0	94.4	94.7
2700	95.0	95.4	95.7	96.0	96.4	96.7	97.1	97.4	97.7	98.1
2800	98.4	98.7	99.1	99.4	99.8	100.1	100.4	100.8	101.1	101.4
2900	101.8	102.1	102.4	102.8	103.1	103.5	103.8	104.1	104.5	104.8
3000	105.1	105.5	105.8	106.1	106.5	106.8	107.1	107.5	107.8	108.1
3100	108.5	108.8	109.1	109.5	109.8	110.1	110.5	110.8	111.1	111.5
3200	111.8	112.1	112.5	112.8	113.1	113.5	113.8	114.1	114.5	114.8
3300	115.1	115.5	115.8	116.1	116.5	116.8	117.1	117.5	117.8	118.1
3400	118.5	118.8	119.1	119.4	119.8	120.1	120.4	120.8	121.1	121.4
3500	121.8	122.1	122.4	122.8	123.1	123.4	123.7	124.1	124.4	124.7
3600	125.1	125.4	125.7	126.0	126.4	126.7	127.0	127.4	127.7	128.0
3700	128.3	128.7	129.0	129.3	129.7	130.0	130.3	130.6	131.0	131.3
3800	131.6	131.9	132.2	132.6	132.9	133.3	133.6	133.9	134.2	134.5
3900	134.9	135.2	135.5	135.9	136.2	136.5	136.8	137.2	137.5	137.8
4000	138.1	138.5	138.8	139.1	139.4	139.8	140.1	140.4	140.7	141.1
4100	141.4	141.7	142.0	142.4	142.7	143.0	143.3	143.7	144.0	144.3
4200	144.6	145.0	145.3	145.6	145.9	146.2	146.6	146.9	147.2	147.5
4300	147.9	148.2	148.5	148.8	149.1	149.5	149.8	150.1	150.4	150.8
4400	151.1	151.4	151.7	152.0	152.4	152.7	153.0	153.3	153.6	154.0
4500	154.3	154.6	154.9	155.2	155.6	155.9	156.2	156.5	156.8	157.2
4600	157.5	157.8	158.1	158.4	158.8	159.1	159.4	159.7	160.0	160.4
4700	160.7	161.0	161.3	161.6	161.9	162.3	162.6	162.9	163.2	163.5
4800	163.8	164.2	164.5	164.8	165.1	165.4	165.8	166.1	166.4	166.7
4900	167.0	167.3	167.7	168.0	168.3	168.6	168.9	169.2	169.5	169.9
5000	170.2	170.5	170.8	171.1	171.4	171.8	172.1	172.4	172.7	173.0
5100	173.3	173.6	174.0	174.3	174.6	174.9	175.2	175.5	175.8	176.2
5200	176.5	176.8	177.1	177.4	177.7	178.0	178.3	178.7	179.0	179.3
5300	179.6	179.9	180.2	180.5	180.8	181.2	181.5	181.8	182.1	182.4

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Elev	-200	-190	-180	-170	-160	-150	-140	-130	-120	-110
	-7.3	-7	-6.6	-6.2	-5.9	-5.5	-5.1	-4.8	-4.4	-4
Elev	-100	-90	-80	-70	-60	-50	-40	-30	-20	-10
	-3.7	-3.3	-2.9	-2.6	-2.2	-1.8	-1.5	-1.1	-0.7	-0.4
Elev	+00	+10	+20	+30	+40	+50	+60	+70	+80	+90
5400	182.7	183.0	183.3	183.7	184.0	184.3	184.6	184.9	185.2	185.5
5500	185.8	186.1	186.5	186.8	187.1	187.4	187.7	188.0	188.3	188.6
5600	188.9	189.2	189.6	189.9	190.2	190.5	190.8	191.1	191.4	191.7
5700	192.0	192.3	192.6	193.0	193.3	193.6	193.9	194.2	194.5	194.8
5800	195.1	195.4	195.7	196.0	196.3	196.7	197.0	197.3	197.6	197.9
5900	198.2	198.5	198.8	199.1	199.4	199.7	200.0	200.3	200.6	200.9
6000	201.3	201.6	201.9	202.2	202.5	202.8	203.1	203.4	203.7	204.0
6100	204.3	204.6	204.9	205.2	205.5	205.8	206.1	206.4	206.7	207.1
6200	207.4	207.7	208.0	208.3	208.6	208.9	209.2	209.5	209.8	210.1
6300	210.4	210.7	211.0	211.3	211.6	211.9	212.2	212.5	212.8	213.1
6400	213.4	213.7	214.0	214.3	214.6	214.9	215.2	215.5	215.8	216.1
6500	216.4	216.7	217.0	217.3	217.6	217.9	218.2	218.5	218.8	219.1
6600	219.5	219.8	220.1	220.4	220.7	221.0	221.3	221.6	221.9	222.2
6700	222.5	222.7	223.0	223.3	223.6	223.9	224.2	224.5	224.8	225.1
6800	225.4	225.7	226.0	226.3	226.6	226.9	227.2	227.5	227.8	228.1
6900	228.4	228.7	229.0	229.3	229.6	229.9	230.2	230.5	230.8	231.1
7000	231.4	231.7	232.0	232.3	232.6	232.9	233.2	233.5	233.8	234.1
7100	234.4	234.7	235.0	235.2	235.5	235.8	236.1	236.4	236.7	237.0
7200	237.3	237.6	237.9	238.2	238.5	238.8	239.1	239.4	239.7	240.0
7300	240.3	240.6	240.8	241.1	241.4	241.7	242.0	242.3	242.6	242.9
7400	243.2	243.5	243.8	244.1	244.4	244.7	245.0	245.2	245.5	245.8
7500	246.1	246.4	246.7	247.0	247.3	247.6	247.9	248.2	248.5	248.8
7600	249.0	249.3	249.6	249.9	250.2	250.5	250.8	251.1	251.4	251.7
7700	252.0	252.2	252.5	252.8	253.1	253.4	253.7	254.0	254.3	254.6
7800	254.9	255.1	255.4	255.7	256.0	256.3	256.6	256.9	257.2	257.5
7900	257.7	258.0	258.3	258.6	258.9	259.2	259.5	259.8	260.1	260.3
8000	260.6	260.9	261.2	261.5	261.8	262.1	262.4	262.6	262.9	263.2
8100	263.5	263.8	264.1	264.4	264.6	264.9	265.2	265.5	265.8	266.1
8200	266.4	266.7	266.9	267.2	267.5	267.8	268.1	268.4	268.7	268.9
8300	269.2	269.5	269.8	270.1	270.4	270.6	270.9	271.2	271.5	271.8

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Elev	-200	-190	-180	-170	-160	-150	-140	-130	-120	-110
	-7.3	-7	-6.6	-6.2	-5.9	-5.5	-5.1	-4.8	-4.4	-4
Elev	-100	-90	-80	-70	-60	-50	-40	-30	-20	-10
	-3.7	-3.3	-2.9	-2.6	-2.2	-1.8	-1.5	-1.1	-0.7	-0.4
Elev	+00	+10	+20	+30	+40	+50	+60	+70	+80	+90
8400	272.1	272.4	272.6	272.9	273.2	273.5	273.8	274.1	274.3	274.6
8500	274.9	275.2	275.5	275.8	276.0	276.3	276.6	276.9	277.2	277.5
8600	277.7	278.0	278.3	278.6	278.9	279.1	279.4	279.7	280.0	280.3
8700	280.6	280.8	281.1	281.4	281.7	282.0	282.2	282.5	282.8	283.1
8800	283.4	283.6	283.9	284.2	284.5	284.8	285.1	285.3	285.6	285.9
8900	286.2	286.5	286.7	287.0	287.3	287.6	287.8	288.1	288.4	288.7
9000	289.0	289.2	289.5	289.8	290.1	290.4	290.6	290.9	291.2	291.5
9100	291.8	292.0	292.3	292.6	292.9	293.1	293.4	293.7	194.0	294.3
9200	294.5	294.8	295.1	295.4	295.6	295.9	296.2	296.5	296.7	297.0
9300	297.3	297.6	297.8	298.1	298.4	298.7	299.0	299.2	299.5	299.8
9400	300.1	300.3	300.6	300.9	301.2	301.4	301.7	302.0	302.3	302.5
9500	302.8	303.1	303.4	303.6	303.9	304.2	304.5	304.7	305.0	305.3
9600	305.5	305.8	306.1	306.4	306.6	306.9	307.2	307.5	307.7	308.0
9700	308.3	308.6	308.8	309.1	309.4	309.6	309.9	310.2	310.5	310.7
9800	311.0	311.3	311.6	311.8	212.1	312.4	312.6	312.9	313.2	313.5
9900	313.7	314.0	314.3	314.5	314.8	315.1	315.4	315.6	351.9	316.2

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TA 6 Guidance Material for Checking of Terrain Clearance - Altitude Corrections / Temperature Corrections

Which chart to use: LSZH has an elevation of 1417 ft, so the conservative chart to use is elevation 1000 ft. However, if a more accurate correction is required, interpolation is permitted.

LSZH ILS / LOC RUNWAY 14

Procedure altitude at Final Approach Fix (FAF)

MOCA at the FAF is 2490 ft and the FAF procedure altitude is 4000 ft.

Assuming an extreme temperature of -50°C and using the next highest altitude from the chart we can see that the correction to be used is 430 ft.

Temperature Correction Chart - Elevation 1000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	1200	10	20	30	40	50	60
300	1300	20	30	40	60	70	90
400	1400	20	40	60	80	100	120
500	1500	30	50	70	90	120	150
600	1600	30	60	80	110	140	170
700	1700	40	70	100	130	160	200
800	1800	40	70	110	150	190	230
900	1900	50	80	120	160	210	260
1000	2000	50	90	140	180	230	290
1500	2500	80	140	200	270	350	430

MOC altitude corrected for temperature is 2490 ft + 430 ft = 2920 ft.

So at the uncorrected 4000 ft there will be a clearance of 1080ft above the MOC, so there is no need to correct the FAF altitude when the temperature is -50°C or warmer.

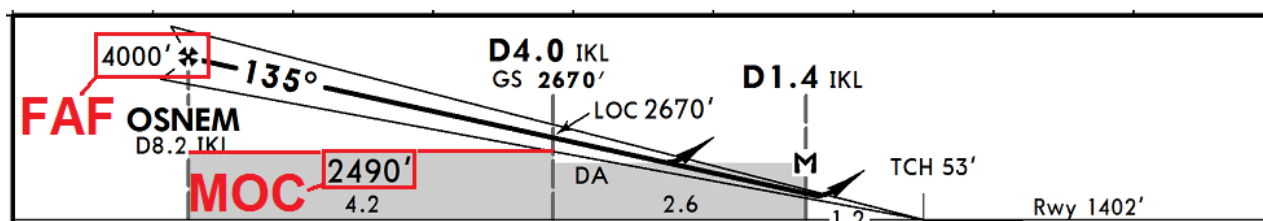
Procedure altitude at step-down fix & minima

Assuming temperature of -10°C.

The step-down fix at D4.0 will always have to be corrected when cold temperature corrections apply as it is inside the FAF, as will the minima.

Step-down-fix: The MOC altitude at the stepdown fix is 2490 ft so the correction will be 140 ft and the minimum altitude at the step-down-fix will be **2630 ft** corrected for temperature (if no MOC was published then the correction would be applied to the 2670 ft).

Minima: For all approach categories the minima is 1602 ft, being conservative and using the 1700 ft correction, the minima will be 1602 ft + 70 ft = **1672 ft** corrected for temperature.



LSZH AMIKI transition via segment ZH701 - TRA - D12.9 IKL - OSNEM - Procedural Approach

Assuming temperature of -10°C.

The IAF is at AMIKI and there is an IF at D12.9 IKL, any correction between the IAF AMIKI and the IF D 12.9 IKL only needs to be applied if it is greater than 200 ft.

Segments up until ZH701: the MSA is 4500 ft, the correction will be 400 ft so the corrected MSA is 4900 ft, as the minimum procedural altitude until ZH701 is 6000 ft (uncorrected), this gives 1100 ft clearance above the corrected MSA. **No temperature correction is required for this segment.**

Segment ZH701 to TRA: the minimum procedure altitude is 5000 ft, this segment enters the area where the MSA is 5700 ft, so then the obstacle clearance is being given by the procedure design rather than by being above MSA, therefore the procedure altitude of 5000 ft needs to be corrected. Since the correction is 400ft which is >200 ft, this correction needs to be applied. **The corrected minimum altitude for this segment is 5400 ft.**

Segment TRA to OSNEM: the minimum procedure altitude is 4000 ft which is below MSA, the correction is 300 ft, part of this segment (D12.9 IKL - OSNEM) is inside the IF so the correction needs to be applied. **The corrected minimum altitude for this segment is 4300 ft.**

NOTE

Note that OSNEM is also the FAF, which as shown previously does not itself require a correction, however in this case the correction required for the segment between the IF and the FAF means the FAF (OSNEM) will be crossed at the corrected 4300 ft.



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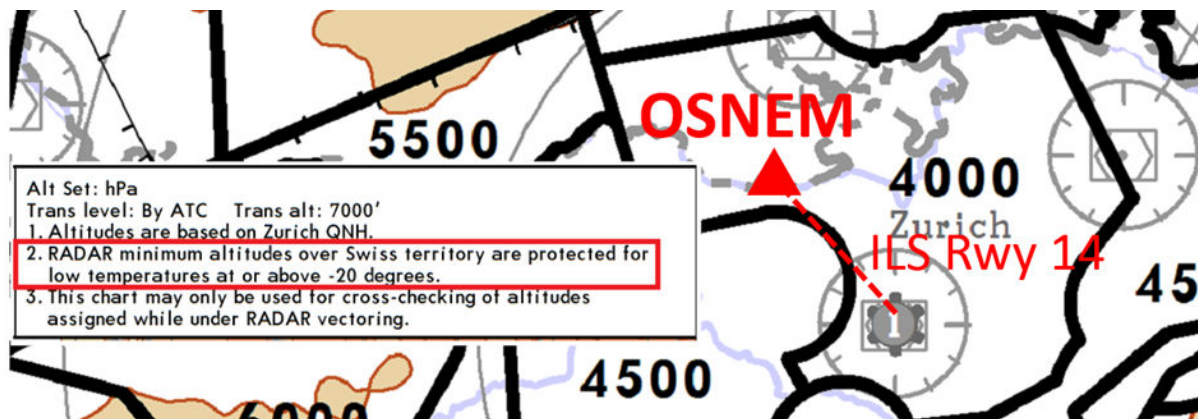
Radar Vectors for ILS Runway 14

The altitudes shown on the MRVA chart are protected down to -20°C, below that temperature ATC is still responsible for making cold temperature corrections to ensure terrain clearance whilst under radar vectors.

The FAF (OSNEM) is in an area where the MRVA is 4000 ft and as shown in the example LSZH ILS / LOC Runway 14 the FAF itself does not need to be corrected even down to - 50°C.

The ILS passes through an area where the MRVA is 4500 ft but at this point the aircraft will be within the LSZH ILS / LOC Runway 14 procedure, so the MRVA will not apply.

Therefore for Radar Vectored ILS to Runway 14 the only corrections required are the step-down-fix and the minima, neither of which need to be advised to ATC.



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TA 6.1 Temperature Correction Chart - Elevation Sea Level

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	200	20	20	30	40	50	60
300	300	20	30	50	60	80	90
400	400	30	40	60	80	100	120
500	500	30	50	70	100	120	150
600	600	40	60	90	120	150	180
700	700	40	70	100	140	170	210
800	800	50	80	120	150	190	240
900	900	50	90	130	170	220	270
1000	1000	60	100	140	190	240	300
1500	1500	90	150	210	280	360	450
2000	2000	200	200	300	400	500	600
3000	3000	200	300	500	600	800	900
4000	4000	300	400	600	800	1000	1200
5000	5000	300	500	800	1000	1300	1500
6000	6000	400	600	900	1200	1500	1800
7000	7000	400	700	1000	1400	1800	2200
8000	8000	500	800	1200	1600	2000	2500
9000	9000	600	900	1300	1800	2300	2800
10000	10000	600	1000	1500	2000	2500	3100
11000	11000	700	1100	1600	2200	2800	3400
12000	12000	700	1200	1800	2400	3000	3700
13000	13000	800	1300	1900	2600	3300	4100
14000	14000	900	1500	2100	2800	3600	4400
15000	15000	900	1600	2300	3000	3800	4700
16000	16000	1000	1700	2400	3200	4100	5100
17000	17000	1000	1800	2600	3400	4400	5400
18000	18000	1100	1900	2700	3600	4700	5800
19000	19000	1200	2000	2900	3900	4900	6100
20000	20000	1200	2100	3100	4100	5200	6400
25000	25000	1600	2700	3900	5200	6700	8200
30000	30000	1900	3300	4800	6400	8200	10100

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TA 6.2 Temperature Correction Chart - Elevation 1,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	1200	10	20	30	40	50	60
300	1300	20	30	40	60	70	90
400	1400	20	40	60	80	100	120
500	1500	30	50	70	90	120	150
600	1600	30	60	80	110	140	170
700	1700	40	70	100	130	160	200
800	1800	40	70	110	150	190	230
900	1900	50	80	120	160	210	260
1000	2000	50	90	140	180	230	290
1500	2500	80	140	200	270	350	430
2000	3000	100	200	300	400	500	600
3000	4000	200	300	400	600	700	900
4000	5000	200	400	600	800	1000	1200
5000	6000	300	500	700	900	1200	1500
6000	7000	300	600	800	1100	1400	1800
7000	8000	400	700	1000	1300	1700	2100
8000	9000	400	800	1100	1500	1900	2400
9000	10000	500	900	1300	1700	2200	2700
10000	11000	500	1000	1400	1900	2400	3000
11000	12000	600	1100	1500	2100	2700	3300
12000	13000	600	1100	1700	2300	2900	3600
13000	14000	700	1200	1800	2500	3200	3900
14000	15000	800	1300	2000	2700	3400	4200
15000	16000	800	1400	2100	2900	3700	4600
16000	17000	900	1500	2300	3100	3900	4900
17000	18000	900	1600	2400	3300	4200	5200
18000	19000	1000	1700	2600	3500	4500	5600
19000	20000	1000	1800	2700	3700	4700	5900
24000	25000	1300	2300	3500	4700	6100	7600
29000	30000	1600	2900	4300	5800	7500	9400

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TA 6.3 Temperature Correction Chart - Elevation 2,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	2200	10	20	30	40	50	60
300	2300	20	30	40	60	70	90
400	2400	20	40	50	70	90	110
500	2500	30	40	70	90	110	140
600	2600	30	50	80	110	140	170
700	2700	30	60	90	120	160	200
800	2800	40	70	100	140	180	220
900	2900	40	80	110	160	200	250
1000	3000	50	80	130	170	220	280
1500	3500	70	120	190	260	330	410
2000	4000	100	200	300	400	500	600
3000	5000	200	300	400	600	700	900
4000	6000	200	400	500	700	900	1200
5000	7000	300	500	700	900	1200	1400
6000	8000	300	500	800	1100	1400	1700
7000	9000	300	600	900	1300	1600	2000
8000	10000	400	700	1100	1400	1800	2300
9000	11000	400	800	1200	1600	2100	2600
10000	12000	500	900	1300	1800	2300	2900
11000	13000	500	1000	1400	2000	2600	3200
12000	14000	600	1000	1600	2200	2800	3500
13000	15000	600	1100	1700	2300	3000	3800
14000	16000	600	1200	1900	2500	3300	4100
15000	17000	700	1300	2000	2700	3500	4400
16000	18000	700	1400	2100	2900	3800	4700
17000	19000	800	1500	2300	3100	4000	5000
18000	20000	800	1600	2400	3300	4300	5400
23000	25000	1100	2000	3100	4300	5600	7000
28000	30000	1300	2500	3900	5300	6900	8700

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TA 6.4 Temperature Correction Chart - Elevation 3,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	3200	10	20	30	40	50	60
300	3300	20	30	40	50	70	80
400	3400	20	30	50	70	90	110
500	3500	20	40	60	80	110	140
600	3600	20	50	70	100	130	160
700	3700	30	60	80	120	150	190
800	3800	30	60	100	130	170	210
900	3900	30	70	110	150	190	240
1000	4000	40	80	120	160	210	270
1500	4500	50	100	180	240	320	400
2000	5000	100	200	300	400	500	600
3000	6000	100	300	400	500	700	800
4000	7000	200	300	500	700	900	1100
5000	8000	200	400	600	900	1100	1400
6000	9000	300	500	700	1000	1300	1700
7000	10000	300	600	900	1200	1500	1900
8000	11000	300	600	1000	1400	1800	2200
9000	12000	400	700	1100	1500	2000	2500
10000	13000	400	800	1200	1700	2200	2800
11000	14000	400	900	1400	1900	2400	3100
12000	15000	500	1000	1500	2100	2700	3400
13000	16000	500	1000	1600	2200	2900	3700
14000	17000	500	1100	1700	2400	3100	3900
15000	18000	600	1200	1900	2600	3400	4200
16000	19000	600	1300	2000	2800	3600	4500
17000	20000	600	1400	2100	2900	3900	4900
22000	25000	800	1800	2800	3900	5100	6400
27000	25000	1000	2200	3500	4900	6400	8100

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TA 6.5 Temperature Correction Chart - Elevation 4,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	4200	10	20	30	40	40	60
300	4300	10	20	40	50	60	80
400	4400	20	30	50	70	80	110
500	4500	20	40	60	80	100	130
600	4600	20	40	70	100	120	160
700	4700	20	50	80	110	140	180
800	4800	30	60	90	130	160	210
900	4900	30	60	100	140	180	230
1000	5000	30	70	110	160	200	260
1500	5500	40	100	160	230	300	380
2000	6000	100	200	300	400	500	600
3000	7000	100	200	400	500	700	800
4000	8000	200	300	500	700	900	1100
5000	9000	200	400	600	800	1100	1300
6000	10000	200	400	700	1000	1300	1600
7000	11000	200	500	800	1100	1500	1900
8000	12000	300	600	900	1300	1700	2100
9000	13000	300	600	1000	1500	1900	2400
10000	14000	300	700	1100	1600	2100	2700
11000	15000	300	800	1300	1800	2300	3000
12000	16000	400	900	1400	1900	2600	3200
13000	17000	400	900	1500	2100	2800	3500
14000	18000	400	1000	1600	2300	3000	3800
15000	19000	500	1100	1700	2400	3200	4100
16000	20000	500	1100	1800	2600	3500	4400
21000	25000	600	1500	2500	3500	4600	5900
26000	30000	800	1900	3100	4400	5800	7400

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TA 6.6 Temperature Correction Chart - Elevation 5,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	5200	10	20	20	30	40	50
300	5300	10	20	30	50	60	80
400	5400	10	30	40	60	80	100
500	5500	10	30	50	80	100	130
600	5600	20	40	60	90	120	150
700	5700	20	40	70	100	140	170
800	5800	20	50	80	120	160	200
900	5900	20	60	90	130	180	220
1000	6000	20	60	100	150	200	250
1500	6500	30	90	150	220	290	370
2000	7000	100	200	200	300	400	500
3000	8000	100	200	300	500	600	800
4000	9000	100	300	400	600	800	1000
5000	10000	100	300	500	800	1000	1300
6000	11000	200	400	600	900	1200	1500
7000	12000	200	500	700	1100	1400	1800
8000	13000	200	500	900	1200	1600	2100
9000	14000	200	600	1000	1400	1800	2300
10000	15000	200	600	1100	1500	2000	2600
11000	16000	300	700	1200	1700	2200	2800
12000	17000	300	800	1300	1800	2400	3100
13000	18000	300	800	1400	2000	2700	3400
14000	19000	300	900	1500	2100	2900	3700
15000	20000	300	900	1600	2300	3100	3900
20000	25000	400	1300	2200	3100	4200	5300
25000	30000	600	1600	2700	4000	5300	6800

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TA 6.7 Temperature Correction Chart - Elevation 6,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	6200	10	10	20	30	40	50
300	6300	10	20	30	40	60	70
400	6400	10	20	40	60	80	100
500	6500	10	30	50	70	100	120
600	6600	10	30	60	90	110	140
700	6700	10	40	70	100	130	170
800	6800	10	40	80	110	150	190
900	6900	20	50	90	130	170	210
1000	7000	20	50	90	140	190	240
1500	7500	20	80	140	210	280	360
2000	8000	100	100	200	300	400	500
3000	9000	100	200	300	500	600	800
4000	10000	100	200	400	600	800	1000
5000	11000	100	300	500	700	1000	1200
6000	12000	100	300	600	900	1200	1500
7000	13000	100	400	700	1100	1400	1700
8000	14000	100	500	800	1100	1500	2000
9000	15000	200	500	900	1300	1700	2200
10000	16000	200	600	1000	1400	1900	2500
11000	17000	200	600	1100	1600	2100	2700
12000	18000	200	700	1200	1700	2300	3000
13000	19000	200	700	1300	1900	2500	3200
14000	20000	200	800	1400	2000	2700	3500
19000	25000	300	1000	1900	2800	3800	4900
24000	30000	300	1300	2400	3600	4900	6300

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TA 6.8 Temperature Correction Chart - Elevation 7,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	7200	10	10	20	30	40	50
300	7300	10	20	30	40	60	70
400	7400	10	20	40	50	70	90
500	7500	10	30	50	70	90	120
600	7600	10	30	60	80	110	140
700	7700	10	30	70	90	130	160
800	7800	10	40	80	100	140	180
900	7900	10	40	90	120	160	210
1000	8000	10	50	90	130	180	230
1500	8500	10	70	140	190	260	340
2000	9000	100	100	200	300	400	500
3000	10000	100	200	300	400	600	700
4000	11000	100	200	400	600	700	1000
5000	12000	100	300	500	700	900	1200
6000	13000	100	300	500	800	1100	1400
7000	14000	100	300	600	900	1300	1700
8000	15000	100	400	700	1100	1500	1900
9000	16000	100	400	800	1200	1700	2100
10000	17000	100	500	900	1300	1800	2400
11000	18000	100	500	1000	1500	2000	2600
12000	19000	100	600	1100	1600	2200	2900
13000	20000	100	600	1200	1800	2400	3100
18000	25000	100	800	1600	2500	3400	4400
23000	30000	200	1100	2100	3200	4400	5700

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TA 6.9 Temperature Correction Chart - Elevation 8,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	8200	0	10	20	30	40	50
300	8300	0	20	30	40	50	70
400	8400	0	20	30	50	70	90
500	8500	0	20	40	60	90	110
600	8600	0	30	50	70	100	130
700	8700	0	30	60	90	120	150
800	8800	0	30	60	100	140	180
900	8900	0	40	70	110	150	200
1000	9000	0	40	80	120	170	220
1500	9500	0	60	120	180	250	330
2000	10000	0	100	200	300	400	500
3000	11000	0	200	300	400	500	700
4000	12000	0	200	300	500	700	900
5000	13000	0	200	400	600	900	1100
6000	14000	0	300	500	800	1000	1400
7000	15000	0	300	600	900	1200	1600
8000	16000	0	300	700	1000	1400	1800
9000	17000	0	400	700	1100	1600	2000
10000	18000	0	400	800	1300	1700	2300
11000	19000	0	400	900	1400	1900	2500
12000	20000	0	500	1000	1500	2100	2700
17000	25000	0	700	1400	2200	3000	4000
22000	30000	0	900	1800	2800	4000	5200

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TA 6.10 Temperature Correction Chart - Elevation 9,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	9200	0	10	20	30	40	50
300	9300	0	10	20	40	50	70
400	9400	0	20	30	50	70	90
500	9500	0	20	40	60	80	110
600	9600	0	20	40	70	100	130
700	9700	0	20	50	80	110	150
800	9800	0	30	60	90	130	170
900	9900	0	30	60	100	140	190
1000	10000	0	30	70	110	160	210
1500	10500	0	40	100	170	240	310
2000	11000	0	100	200	300	400	500
3000	12000	0	100	200	400	500	700
4000	13000	0	200	300	500	700	900
5000	14000	0	200	400	600	800	1100
6000	15000	0	200	500	700	1000	1300
7000	16000	0	200	500	800	1200	1500
8000	17000	0	300	600	900	1300	1700
9000	18000	0	300	700	1100	1500	2000
10000	19000	0	300	700	1200	1700	2200
11000	20000	0	400	800	1300	1800	2400
16000	25000	0	500	1200	1900	2700	3500
21000	30000	0	700	1500	2500	3600	4700

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TA 6.11 Temperature Correction Chart - Elevation 10,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	10200	0	10	20	20	30	40
300	10300	0	10	20	30	50	60
400	10400	0	10	30	40	60	80
500	10500	0	10	30	50	80	100
600	10600	0	20	40	60	90	120
700	10700	0	20	50	70	110	140
800	10800	0	20	50	80	120	160
900	10900	0	20	60	90	140	180
1000	11000	0	20	60	110	150	200
1500	11500	0	30	90	160	220	300
2000	12000	0	100	200	300	300	400
3000	13000	0	100	200	400	500	600
4000	14000	0	100	300	500	600	800
5000	15000	0	100	300	600	800	1000
6000	16000	0	200	400	700	900	1200
7000	17000	0	200	500	800	1100	1500
8000	18000	0	200	500	900	1200	1700
9000	19000	0	200	600	1000	1400	1900
10000	20000	0	200	700	1100	1600	2100
15000	25000	0	400	1000	1600	2400	3200
20000	30000	0	500	1300	2200	3200	4300

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TA 6.12 Temperature Correction Chart - Elevation 11,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	11200	0	10	20	20	30	40
300	11300	0	10	20	30	50	60
400	11400	0	10	30	40	60	80
500	11500	0	10	30	50	70	100
600	11600	0	10	40	60	90	120
700	11700	0	10	40	70	100	130
800	11800	0	10	50	80	110	150
900	11900	0	20	50	90	130	170
1000	12000	0	20	60	100	140	190
1500	12500	0	20	80	140	210	280
2000	13000	0	100	200	200	300	400
3000	14000	0	100	200	300	500	600
4000	15000	0	100	300	400	600	800
5000	16000	0	100	300	500	700	1000
6000	17000	0	100	400	600	900	1200
7000	18000	0	100	400	700	1000	1400
8000	19000	0	100	500	800	1200	1600
9000	20000	0	200	500	900	1300	1800
14000	25000	0	200	800	1400	2100	2800
19000	30000	0	300	1100	1900	2900	3900

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TA 6.13 Temperature Correction Chart - Elevation 12,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	12200	0	10	10	20	30	40
300	12300	0	10	20	30	40	60
400	12400	0	10	20	40	60	80
500	12500	0	10	30	50	70	90
600	12600	0	10	30	60	80	110
700	12700	0	10	30	60	90	130
800	12800	0	10	40	70	110	150
900	12900	0	10	40	80	120	160
1000	13000	0	10	50	90	130	180
1500	13500	0	10	70	130	200	270
2000	14000	0	100	100	200	300	400
3000	15000	0	100	200	300	400	600
4000	16000	0	100	200	400	600	800
5000	17000	0	100	300	500	700	900
6000	18000	0	100	300	600	800	1100
7000	19000	0	100	400	600	1000	1300
8000	20000	0	100	400	700	1100	1500
13000	25000	0	100	600	1200	1800	2500
18000	30000	0	100	900	1700	2500	3500

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TA 6.14 Temperature Correction Chart - Elevation 13,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	13200	0	0	10	20	30	40
300	13300	0	0	20	30	40	50
400	13400	0	0	20	40	50	70
500	13500	0	0	20	40	60	90
600	13600	0	0	30	50	80	110
700	13700	0	0	30	60	90	120
800	13800	0	0	30	70	100	140
900	13900	0	0	40	70	110	160
1000	14000	0	0	40	80	120	170
1500	14500	0	0	60	120	180	260
2000	15000	0	0	100	200	300	400
3000	16000	0	0	200	300	400	600
4000	17000	0	0	200	400	500	700
5000	18000	0	0	200	400	700	900
6000	19000	0	0	300	500	800	1100
7000	20000	0	0	300	600	900	1300
12000	25000	0	0	500	1000	1600	2200
17000	30000	0	0	700	1400	2200	3100

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TA 6.15 Temperature Correction Chart - Elevation 14,000 ft

Height AAL (FT)	Altitude (FT)	0°C	-10°C	-20°C	-30°C	-40°C	-50°C
200	14200	0	0	10	20	30	40
300	14300	0	0	10	30	40	50
400	14400	0	0	20	30	50	70
500	14500	0	0	20	40	60	80
600	14600	0	0	20	50	70	100
700	14700	0	0	20	50	80	120
800	14800	0	0	30	60	90	130
900	14900	0	0	30	70	100	150
1000	15000	0	0	30	70	120	160
1500	15500	0	0	50	110	170	240
2000	16000	0	0	100	200	300	400
3000	17000	0	0	100	300	400	500
4000	18000	0	0	200	300	500	700
5000	19000	0	0	200	400	600	900
6000	20000	0	0	200	500	700	1000
11000	25000	0	0	400	800	1300	1900
16000	30000	0	0	500	1200	1900	2700

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TA 7 TCAS / ACAS Guidance Material

TRAFFIC ADVISORY - TA

Traffic Advisories (TAs) indicate the approximate position relative to the subject aeroplane, either in azimuth only, or azimuth and altitude, of a nearby transponding aeroplane which may become a threat.

A TA is intended to alert the crew that a RA, requiring a change in flight path, may follow. A visual search should immediately be concentrated on that part of the sky where the TA indicates the conflicting traffic to be. If the potential threat cannot be seen and gives cause for concern, air traffic control assistance should be requested in deciding whether a change of flight path is required.

As per the 'see and avoid' concept, if a potential threat is seen, and considered to pose a definite risk of collision, the pilot should manoeuvre their aeroplane as necessary to avoid it, making sure that the area into which they are manoeuvring is clear. Flight crew shall be aware that the potential threat may not be the aircraft that triggered the TA.

Once clear of the potential threat, and any other subsequent conflicts, the pilot should resume their previously cleared flight path and advise ATC of any deviation from their clearance.

RESOLUTION ADVISORY - RA

Resolution Advisories (RA) recommend manoeuvres or manoeuvre restrictions in the vertical plane to resolve conflicts with an aeroplane transponding SSR Mode C. Whenever both aeroplanes are operating TCAS II in RA mode, TCAS II coordinates the RAs.

A RA is intended to advise pilots on the manoeuvre they shall carry out in order to achieve or maintain adequate separation from an established threat. After any RA manoeuvres crew should only rely on the 'Clear of traffic' to assess that they are clear of traffic. They shall not rely on their visual acquisition which might not be the one that has triggered the event.

The required manoeuvre shall be initiated immediately and crew members not involved in its execution shall ensure that the sky ahead is clear of other traffic and continue the visual search for the established threat.

Once the ACAS II indicates that adequate separation has been achieved, or ATC information shows that there is no longer a conflict, the aeroplane should be promptly returned to its intended flight path, and ATC informed.

Manoeuvres shall never be made in a direction opposite to that given in an RA.

VISUAL ACQUISITION LIMITATIONS

The visual assessment of traffic can be misleading: At high altitude, it is difficult to assess the range and heading of traffic as well as its relative height. At low altitude, the heavy aeroplane attitude at low speed makes it difficult to assess whether it is climbing or descending.

Visual acquisition does not provide any information about the intent of other traffic. The traffic in visual contact may not be the threat that triggers the RA. A visual manoeuvre relative to the wrong visual traffic may degrade the situation against the real threat.

SYSTEM LIMITATIONS ACAS II:

- Does not relieve the crew from maintaining an active lookout.
- Relies upon information received from transponder equipped aeroplanes.
- Is not capable of resolving with complete accuracy the bearing, heading or vertical rates of an intruding aeroplane; a tolerance of 30 degrees in bearing resolution is common. It's also crucial to change scales on the display in order to get the most precise bearing resolution.

Pilots shall be aware of the limitations of the particular equipment, as the full range of TAs and RAs may not be produced beyond the minimum and maximum altitudes specified for its operation.

Even if TAs and RAs are suspected of being nuisance or false advisories, they shall be treated as genuine. Having a visual acquisition shall not bring the alert stage down.

SYSTEM INHIBITIONS

When either a GPWS / EGPWS or wind shear warning is active, TCAS may automatically switch to the TA-only mode of operation with TCAS aural annunciations inhibited. TCAS remains in TA-only mode for 10 seconds after the GPWS / EGPWS or wind shear warning is removed.

Increase descent RAs are inhibited below 1450 ft AGL.

Descend RAs are inhibited below 1100 ft AGL.

All RAs are inhibited below 1000 ft AGL.

All ACAS aural annunciations are inhibited below 500 ft AGL.

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TA 8 General Guidelines for Operations in Turbulence, Windshear, and Near Thunderstorms

1. Fasten shoulder harness.
2. Consider switching cockpit lighting to high intensity if lightning may be encountered.
3. Turn off auto-throttle and manually set thrust to maintain target speed.
4. Fly the recommended turbulence speed according to AOM.
5. Switch on engine ignition and / or de-icing equipment according to AOM procedures.
6. Use autopilot as recommended in the AOM. If autopilot is off, leave stabiliser trim in level-flight trim setting (do not over control).

Altitude

At maximum cruise altitude, the margin between low speed and high speed buffet is small and any increase of g-loads, whether caused by manoeuvring or by turbulence, may lead to handling difficulties. Therefore, avoid selecting maximum cruise altitude.

Large altitude variations are possible in severe turbulence, which may have to be accepted in order to maintain the appropriate attitude.

Airspeed / Power Setting

Large speed fluctuations are to be expected; therefore:

1. Do not chase airspeed!
2. Maintain the recommended turbulence speed as target speed.
3. Set thrust as required and then do not change it unless required by large and / or persistent airspeed or altitude variations.
4. If subjected to a sudden onset of turbulence, do not slow the aeroplane down hurriedly.

Attitude

Endeavour to maintain a constant attitude and avoid over controlling. As mentioned above, altitude deviations may have to be accepted, but attitude is of primary importance.

Use of Autopilot and Flight Director

The Flight Director can effectively reduce workload since it will encourage smooth control inputs and is therefore recommended for use in turbulence. Autopilot should be used in accordance with the AOM.

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TA 9 Thunderstorms

Thunderstorms are classified as:

- Frontal thunderstorms;
- Air mass thunderstorms - Convective and Orographic.

Thunderstorms may also be associated with jetstreams, volcanic activity, mountain wave activity and marked temperature inversions - at the planning stage, such features can help predict where thunderstorm activity may be expected.

Hazards associated with thunderstorms include:

Icing, Turbulence, Windshear, Lightning, and Heavy Precipitation, hence the avoidance criteria in [Adverse and Potentially Hazardous Atmospheric Conditions](#).

Thunderstorms typically extend up to 25,000 ft vertically during winter time, often up to the tropopause during summer, and 10 to 20 km horizontally.

Weather radar should be used effectively in order to avoid flying close to thunderstorms, and to select the best path if faced with multiple returns.

Flight under a Cumulonimbus anvil should be avoided where possible as precipitation (including hail) may be encountered.

Thunderstorms can often be relatively short-lived (20-30 minutes), so delaying a take-off or landing during times of thunderstorm activity may be prudent to avoid encountering the hazards mentioned above.

Lightning

Lightning is not a direct danger to the crew and passengers as the electrical load will stay outside the aeroplane (Faraday Cage) but lightning strikes may disturb or damage the navigation system.

Lightning is associated with cumulonimbus / thunderstorm activity, so employing the advice offered on avoiding thunderstorms will help to avoid lightning. However, reports of aircraft suffering lightning strikes in clear air, 25 NM or more from the nearest storm, are common.

Although crew incapacitation due to lightning is unlikely (other than momentary flash blindness), direct effects from lightning current attachment to and flow through the aeroplane may include:

- Pit marks, often seen along the fuselage, or holes in the trailing edge of wing and tail tips.
- Melting of rivets.
- Puncturing of non-metallic structures.
- Puncturing and de-lamination of composites.
- Slight deformation of metal skins and structures.
- Welding or roughening of moveable hinges and bearings.
- Damage to parts that may conduct lightning current other than the airframe, such as bonding straps and pitot tubes.

The flash-induced electromagnetic field, and the momentary increase of the voltage and current at the aeroplane structure, may produce indirect effects including:

- Interruption of instruments and navigation equipment.
- Damage to electronic and electrical equipment.
- Tripping of circuit breakers.
- Loss of electrical power.
- Engine flame out.

An imminent lightning strike may be indicated to flight crews by the build-up of static discharge, which causes interference on ADF indicators or noise in communication receivers. Another indication is St. Elmo's fire, which is visible at night as small electrical discharges running across the windshield and sparking on the wings.

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TA 10 Icing**Icing Conditions in Flight**

Ice formation on the ground and its effects are explained in [De-icing and Anti-icing on the Ground](#).

Flight in icing conditions presents additional hazards, and pilots need to understand the effects that in-flight icing conditions can have on aeroplane performance and handling, as well as the use and limitations of aeroplane de-ice and anti-ice equipment.

Ice Formation

Small water droplets tend to follow the aerodynamic flow around the wing, and are swept away, but larger, heavier droplets tend to 'impinge' aft of the leading edge; the larger the droplet, the further aft it will impinge.

Furthermore, large droplets are relatively warm and as a result tend to run back and spread out in liquid state before freezing. This effect is known as 'flowback', and its effect depends directly upon the size and temperature of the water droplets, as well as the velocity of the aircraft and wing surface temperature. Impingement and flowback result in ice collecting not only on the leading edge, but also on the upper and lower surfaces of a wing, where there may be no ice protection.

Types of Ice

Rime Ice: associated with stratiform cloud, small water droplets and relatively cold environment (-15°C or colder). Some droplets impact the wing and tailplane leading edges with little or no flowback, freezing quickly into rime ice. Therefore, leading edge anti/deicing equipment is relatively effective against this type of ice.

Clear Ice: associated with cumuliform clouds, larger water droplets and relatively warm environment (-10°C or warmer). Droplets tend to impinge further aft and flow back before freezing to form sheet ice on the upper and lower surfaces of the wing. These areas may not be protected by anti/de-icing equipment. Any significant deposit of clear ice can significantly degrade performance and can form on aerofoil surfaces with few visual cues to the pilot. The absence of indications of ice formation on the windshield posts and / or wipers, does not mean that clear ice is not forming. The terms 'freezing drizzle' or 'freezing rain' in a terminal forecast / report indicate a strong probability of conditions for clear ice.

Mixed Ice: is a combination of clear and rime ice, with all the hazards of both. It forms when the temperature is in the range -10°C to -15°C, and the proportional make-up varies with the weather system.

Freezing drizzle or rain will cause ice build-up on or beyond the anti and / or de-icing surfaces and will lead to a degradation of lift, increase in drag and aeroplane mass which could exceed the control limits / engine thrust of the aeroplane or lead to uncontrollable asymmetric flight control forces. Hence why flight into known freezing drizzle or rain of any intensity is prohibited.

Similarly, severe icing (by its definition) can exceed the capabilities of the anti and / or de-icing systems of the aeroplane to protect the aeroplane from the build-up of ice. Flying with the autopilot and / or auto-throttle engaged when in severe icing conditions can mask the effects of ice build-up, including trim changes and increased engine thrust. Therefore, it is recommended that flight in anticipated areas of significant icing is done with the auto-throttle and / or autopilot disengaged if workload permits.

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The AFM/AOM shall always be consulted for aeroplane type-specific information but the following sections offer general guidance for flight in icing conditions.

1. De-ice and / or anti-ice equipment shall be turned on prior to entering areas with the risk of icing.
2. Strictly observe any minimum speeds associated with icing conditions - stalling speeds with contaminated wings are higher than normal.
3. Winter jet-streams and storm centres are more intense and clear air turbulence is more common. Frequently check weather conditions for destinations and alternates as ceilings and visibilities can be very low. Particular care shall be taken when there are reports of freezing rain or drizzle. There is the possibility of longer holding times in these conditions when landing weather is at or below minima.
4. Make sure that the arrival briefing includes landing conditions, braking, use of reverse thrust, flaps and spoilers as well as ground handling / taxiing techniques (see also [De-icing and Anti-icing on the Ground](#)).

Exposure to Severe Icing

Severe icing is often associated with supercooled large droplets (i.e. freezing drizzle or rain).

The most effective means of identifying severe icing conditions are cues that can be seen, felt or heard. This includes visual inspection of aeroplane surfaces, i.e. wings or windscreen. At temperatures near freezing it may be possible to detect large droplets splashing or splattering upon impact with the windscreen.

When exposed to severe icing, perform the following actions:

- Disengage the autopilot and hand-fly the aeroplane. The autopilot may mask important cues or may self-disconnect and present unusual attitudes or control conditions;
- Advise air traffic control and promptly exit the condition, using control inputs that are as smooth and small as possible;
- Change heading, altitude or both to find an area that is warmer than freezing, substantially colder than the current ambient temperature, or free of clouds;
- When severe icing conditions exist, reporting may assist other crews in maintaining vigilance. Submit a pilot report of the observed icing conditions. It is important not to understate the conditions or effects of the icing observed.

Control Anomalies in Icing Conditions

The formation of ice on the aeroplane structure could create a situation from which the pilot might have difficulty recovering and, in some instances, may not be able to recover at all. For this reason, the following paragraphs describe two particular types of control upsets likely to be encountered during flight in severe icing conditions, both of which highlight the importance of exiting the environment as soon as possible.

Roll upset: an uncommanded and uncontrolled roll phenomenon which may occur as a consequence of, or prior to, a wing stall due to anomalous forces that cause the ailerons to deflect, or because the ailerons have lost effectiveness. Deflection of ailerons or loss of aileron effectiveness may be caused by ice accumulation in a sensitive area of the wing aft of the de-icing boots. It occurs under unusual circumstances associated with supercooled large droplets and, rarely, normal cloud droplets in a very narrow temperature range near freezing. Roll upset can result from severe icing conditions even without the usual symptoms of ice accumulation or a perceived aerodynamic stall.

It is important to review the AFM / AOM for aeroplane type-specific information. Also, pilots should check any available icing related bulletins from the aeroplane manufacturer.

Tailplane (empennage) stall: It is another significant hazard of airframe icing. Sharp edged surfaces are more susceptible to collecting ice than large blunt ones. For this reason, the tailplane may begin accumulating ice before the wings and can accumulate faster. There have been reports of ice on the tailplane without any visible ice on the wing. This can occur if the tailplane has not been or cannot be de-iced.

A tailplane stall occurs when, as with the wing, the critical angle of attack is exceeded. Since the horizontal stabiliser counters the natural nose down tendency caused by the centre of lift of the main wing, the aeroplane will react by pitching nose down, sometimes uncontrollably, when the tailplane is stalled. Application of flaps can aggravate or initiate the stall. Use caution when applying flaps during approach if there is the possibility of icing on the tailplane.

Perhaps the most important characteristic of a tailplane stall is the relatively high airspeed at the onset and, if it occurs, the suddenness and magnitude of the nose down pitch. A stall is more likely to occur when the flaps are approaching the fully extended position, after nose down pitch and airspeed changes following flap extension, or during flight through wind gusts.

WARNING

Once a tailplane stall is encountered, the stall condition tends to worsen with increased airspeed and possibly may worsen with increased power settings at the same flap setting. Airspeed, at any flap setting, in excess of the aeroplane manufacturer's recommendations for the flight and environmental conditions, accompanied by uncleared ice contamination on the tailplane, may result in a tailplane stall and uncommanded pitch down from which a recovery may not be possible. A tailplane stall may occur at speeds less than the flaps extension speed.

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TA 11 Turbulence

Turbulence is defined as a disturbed, irregular flow of air with embedded irregular whirls or eddies and waves. An aeroplane in turbulent flow is subjected to irregular and random motions while more or less maintaining its intended flight path.

Types of Turbulence

Turbulence may be one of the following types:

- Convective turbulence;
- Orographic turbulence;
- Clear air turbulence;
- Wake turbulence.

Convective Turbulence

Convective turbulence is caused by thermal instability, often associated with thunderstorm activity, and can cause extreme air motion with vertical speeds up to 6000 ft/min.

Orographic Turbulence

Orographic turbulence is caused by terrain, with mountain waves on the lee side of a mountain typically producing lenticular and rotor clouds. The strongest turbulence may be found in rotor clouds.

Clear Air Turbulence (CAT)

Clear air turbulence is of special significance, since its presence cannot be detected before it is encountered. It is caused by large wind shears with rapid changes of wind direction, horizontally and / or vertically. Abrupt changes of wind direction in a sharp trough line may cause considerable turbulence and a change of flight level will normally alleviate the problem.

CAT may also be expected on the upper side of a sloping tropopause. Further large horizontal and vertical shears of wind speed in the transition zone between cold and warm air masses, as well as at the tropopause associated with jetstreams may cause severe CAT. These areas of turbulence are normally shallow, narrow and extended patches which move with the wind.

Wake Turbulence

All precautionary measures are covered at [Wake Turbulence](#).

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TA 12 Windshear

Windshear is defined as any rapid change in wind direction and / or speed along the flight path of an aeroplane.

Windshear Mechanism

Windshear, with or without turbulence, alters the lift force acting on an aeroplane, resulting in a significant sinking or rising motion. Therefore, windshear may be categorised as:

- Increasing performance shear caused by increasing headwind / decreasing tailwind component or vertical updrafts;
- Decreasing performance shear caused by decreasing headwind / increasing tailwind component or vertical down-drafts.

Conditions

Conditions for potentially hazardous windshears are:

- Convective conditions (thunderstorms, rain / snow showers);
- Frontal systems;
- Jet streams;
- Strong or gusty surface winds;
- Temperature inversion;
- Mountain waves;
- Sea breeze circulations.

Recognition

There is only limited time for windshear recognition and action, typically 5 to 15 seconds. Several factors can impede windshear recognition:

- Marginal weather conditions;
- High crew work load conditions;
- Illusion of normality.

(During the initial part of the windshear encounter, everything may appear normal. Even severe windshear onset may not provide dramatic early indications to the flight crew).

Downbursts

The term 'downburst' describes a severe downward rush of air and its outburst of damaging winds on or near the ground. The effects can last for 5 mins and affect a comparatively small area (microburst), or up to 20 mins and impact a much larger area (macroburst).

Downbursts can occur wherever convective weather conditions exist. Approximately 5 percent of all thunderstorms produce microbursts. Downdrafts associated with microbursts are typically only a few hundred to 1000m across. When the downburst hits the ground, it spreads out horizontally and may form one or more horizontal vortex rings around the downdraft up to 2000 ft AGL. This can cause very powerful updrafts and roll forces in conjunction with wind speed changes up to 45 kt. Depending on the movement and the height

of the base of the parent cloud, microbursts may occur as stationary or moving, surface or mid-air, wet or dry. Microbursts have occurred in relatively dry conditions of light rain or precipitation that evaporates before reaching the earth's surface.

Frontal Windshear

Frontal wind shear is possible in both cold and warm fronts but exists in a different relative location in each type of front. Because the cold front boundary slopes back behind the frontal surface, the wind shear line also slopes back. The frontal boundary of the warm front slopes upward ahead of the surface front and so does the wind shear. Significant wind shears can be expected if there is a large difference in surface temperature across the front or if the front is moving rapidly (> 30 kt).

Wind Speed Changes

Large wind speed changes near the ground can be found in many meteorological situations, including frontal conditions. Terrain irregularities or buildings which interrupt the wind flow can produce significant wind shears close to the ground.

Sea Breeze Circulations

At night during the spring and summer months strong jet winds can occur close to the ground, when radiation cooling reaches maximum and a moderate to strong pressure gradient exists.

Lake and sea breeze windshear is caused by the differential heating rate of land and water under conditions of strong solar radiation. When the offshore component of the prevailing surface wind is not too large and the temperature difference is big enough warm air rising over the land will be substituted with colder air from the sea. It begins as a light breeze in the mid-morning and increases through the day to 15 kt up to a height of 1200 ft, and often moves up to 50 km inland. With moist and unstable air over the land, a sea breeze front with characteristics of thunderstorms may occur.

At night the sea breeze will change to a land breeze because the radiation cooling from the land surface occurs more rapidly than over the sea. The temperature differences at night are smaller than during the day and therefore the land breeze is not as strong as the sea breeze. Normally only moderate windshear is caused by a land or sea breeze, unless a sea breeze front develops, which may encourage microbursts and the associated windshear hazard. Flight crews shall be aware of sea breeze windshear, especially on flights to islands in summer time where the runways are often close to the shore. Generally airfields near the coast will be more affected than those inland.

Conclusion

The actions to be taken by the flight crews can be summed up as follows:

- Evaluation of the weather situation;
- Avoidance of known windshear;
- Being prepared to employ standard recovery techniques if subject to windshear.

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TA 13 Jetstreams

Jetstreams are narrow bands of wind, generally westerly, with extremely high speeds (up to 300 kt), usually occurring near the tropopause. Jetstreams can be several thousand miles long, and have a width of several miles.

Clear Air Turbulence (CAT)

In mid-latitudes there is a common area for CAT around the jet stream, above and below the jet core and to the polar side.

To avoid or to leave the areas of CAT the following procedures should be considered:

- When flying parallel with the jetstream, changing altitude by up to 1000 ft;
- When flying perpendicular to the jetstream, from the warm to the cold side, descend 1000 ft; from the cold to the warm side, climb 1000 ft;
- If the temperature is changing in the CAT area the flight should be continued on course as the CAT area may be crossed in a short time;
- If the temperature remains constant the course should be altered in order to leave the CAT area.

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TA 14 Volcanic Ash

Volcanic ash may extend for several hundred miles, and eruptions may send ash plumes up to 40,000 ft. Do not rely on weather-radar to detect ash clouds or volcanic dust as they cannot be detected by such systems.

Conditions within Volcanic Ash Clouds

If the aeroplane enters an ash cloud the flight crew can expect:

- Smoke;
- Dust or acrid odours similar to electric sparks in the cockpit;
- At night, heavy static discharges around the windshield;
- A decrease in engine rpm;
- A decrease in fuel flow;
- An increase in EGT;
- Engine surge or overheat;
- Engine flame-out;
- Unreliable airspeed indication;
- Pressurisation and electrical systems may be affected;
- At night, St. Elmo's fire or other static discharges accompanied by a bright orange glow in the engine inlets.

Procedures

The following procedures are applicable unless manufacturer's guidance states otherwise.

When encountering ash cloud the following procedures are recommended:

1. Fly on the upwind side, turn on continuous ignition and declare an emergency.
2. Do not climb in order to overfly the ash cloud.
3. Reduce power to idle to provide additional engine stall margin and lower turbine temperature.
4. Try to escape the ash cloud by descending and flying a 180° turn (if terrain clearance permits).
5. Monitor attitude versus airspeed, keep as many engines running as possible.
6. Restart any engine which was shut down due to EGT/ITT exceeding the limits. If an engine fails to restart, repeated attempts should be made immediately to avoid solidification of molten ash on the turbine blades.
7. Turn on all accessory air bleeds including all air conditioning packs, nacelles, and wing anti-ice. This will provide an additional engine stall margin by reducing engine pressure.

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Additional Procedures

For additional procedures, the AOM / Emergency checklist of the respective type of aeroplane shall be checked.

Following an encounter with an ash cloud a landing at the next suitable airport is recommended. If the visibility through the windshield is impaired, a runway with maximum lighting facilities (e.g. CAT III capability) should be considered. Flight crews are requested to make special air-reports when volcanic eruption is observed or when volcanic ash cloud is observed or encountered.

Information on volcanic activities, including pre-emptive activity, volcanic eruption and volcanic ash clouds, is transmitted to aeroplanes operating on routes that could be affected. This information is also published by NOTAMS and SIGMETS up to a 12 hour validity period and, if necessary, recommendations on re-routings will be included.

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TA 15 Heavy Precipitation

Heavy precipitation may occur as rain showers, snow showers and hail. The greatest hazards to flight are reduced visibility and the risk of icing in combination with low temperature.

On the ground, contaminated runways may influence landing performance and crosswind limitations, as well as presenting a risk of aquaplaning. The special procedures of the AOM of the respective aeroplane shall be followed.

Partial loss of orientation may occur after change over from instruments to visual flying during the approach, especially in snow showers and blowing snow. In falling or blowing snow, landing lights should be used with caution as the reflected light may actually reduce the effective visibility, and even cause false impression of drift during the flare and roll-out.

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TA 16 Sand Storms

A sandstorm (SS) or dust storm (DS) occurs when strong dry winds are blowing over the desert raising and carrying along clouds of sand or dust. Such strong winds are usually the result of convective currents created by intense heating of the ground and are often strong enough to move dunes. The clouds are often dense enough to obscure the sun and reduce the visibility almost to zero.

A SS normally carries sand no higher than 10 m, while a DS can reach FL 150 or higher.

Sandstorm (SS) and / or dust storms (DS) should be avoided whenever possible. Considerable damage can be done to an aeroplane's windscreen, leading edges and engine compressor blades by the abrasive action of the sand particles.

If a landing in a SS or DS is unavoidable, the following risks shall be considered:

- clogging of pitots;
- reduction of engine performance;
- visual illusions after changing over from instrument to visual flying;
- landing lights should be used with care as reflected light may reduce the visibility and cause disorientation;
- reverse thrust may considerably reduce visibility;
- sand or dust covering the airport may reduce the ability to assess the RWY dimensions;
- the brake efficiency may be affected.

On the ground, aeroplanes should ideally be kept under cover if sand storms or dust storms are forecast or in progress. Alternatively, all engine blanks and cockpit covers should be fitted, as well as the blanks and covers for the various system and instrument intakes and probes. These should be carefully removed before flight to ensure that accumulations of dust are not deposited in the orifices which the covers are designed to protect.

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TA 17 Mountain Waves

Mountain waves and downslope windshear are caused by a significant airflow crossing a mountain range, together with special atmospheric conditions. The strong vertical and horizontal wind shears, so-called rotor turbulence, represent a danger at low heights as well as the strong downslope wind at the lee side of the mountains. Frequently, a second rotor will form up to 100 NM from the lee side of the mountain.

Flight crews should be aware of the potential hazard at airports within the flow regime of the wave. Depending on moisture content of the air, lenticular (lens shaped) clouds may be present.

Approaching a Mountain Range

When approaching a mountain range from the upwind side, there will usually be a smooth updraft. Therefore, it is not quite as dangerous an area as the lee of the range. From the leeward side consider increasing altitude, because downdrafts can exceed the climb capability of the aeroplane; turbulence can also be expected.

Manoeuvring

If severe turbulence is encountered, reduce to manoeuvring speed and fly away from the turbulent area. Windshear recovery techniques may need to be employed if faced with a significant downdraught.

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TA 18 Significant Temperature Inversions

Strong inversions in pre-cold-front situations may be associated with strong low altitude jet winds immediately above the ground. However, the main negative performance factor is caused by the decrease in engine power resulting from the temperature rise as the aircraft climbs through an inversion.

There may also be associated reduced visibility.

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TA 19 Wake Vortex Turbulence

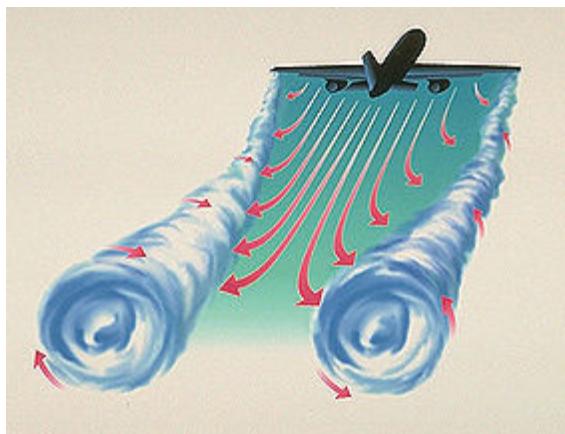
Additional details on wake turbulence and vortex generation.

Wake vortex turbulence is defined as turbulence which is generated by the passage of an aircraft in flight. It will be generated from the point when the nose landing gear of an aircraft leaves the ground on take-off and will cease to be generated when the nose landing gear touches the ground during landing.

The turbulence which an aircraft encounters when flying into the wake of another aircraft may be of such severity as to make control difficult. For instance, the wake from large jet aircraft can impose rolling moments exceeding the roll control power of some aircraft. Furthermore, turbulence generated within the vortices can damage aircraft components and equipment if it is encountered at close range.

Vortex Generation

Lift is generated by the creation of a pressure differential over the wing surfaces. The lowest pressure occurs over the upper wing surface and the highest pressure under the wing. The pressure differential triggers the roll-up of the airflow aft of the wing resulting in swirling air masses trailing downstream of the wing tips. After the roll-up is completed, the wake consists of two counter rotating cylindrical vortices.



Vortex Strength

The strength of the vortex is governed by the weight, speed, and shape of the wing of the generating aircraft. The vortex characteristics of a given aircraft will be changed by extension of flaps or other high-lift devices, or by change in speed. However, the main factor determining the strength of the vortex is aircraft weight. For a given aircraft type the greatest vortex strength occurs when it is:

HEAVY - CLEAN - SLOW

Vortex Behaviour

Trailing vortices have certain behavioural characteristics which can help pilots visualise probable wake location and thereby take precautions to avoid it. Since trailing vortices are a by-product of wing lift, they are generated from the moment an aircraft leaves the ground until touchdown on landing. Prior to take-off or touchdown pilots are therefore advised to note the rotation or touchdown point of the preceding aircraft.

The rate at which the vortices decay depends on the atmospheric conditions at the time: they are most persistent in calm, stable air. Under such conditions, considerable turbulence may be encountered several miles behind the generating aircraft.

The mutual interaction of a pair of vortices induces a downward movement, the rate of sink being something of the order of 500 ft/min, which continues until they begin to dissipate at about 1000 feet below the generating aircraft's flight path. In addition, vortices are carried by the prevailing wind. Pilots are therefore recommended to fly at or above the generating aircraft's flight path and to avoid the area behind and below it.

When near the ground (at a height approximately equal to half the span of the generating aircraft) the vortices begin to move horizontally away from each other. On a runway where there is a light cross wind, the upwind vortex may remain more or less stationary over the runway. Similarly, a tailwind can move the vortices of the preceding aircraft into the touchdown zone.

Table 1: Example list of aircraft types assigned to RECAT-EU categories (Category names and letters are indicative)

'SUPER HEAVY'	'UPPER HEAVY'	'LOWER HEAVY'	'UPPER MEDIUM'	'LOWER MEDIUM'	'LIGHT'
'CAT-A'	'CAT-B'	'CAT-C'	'CAT-D'	'CAT-E'	'CAT-F'
A388	A332	A306	A318	AT43	FA10
A124	A333	A30B	A319	AT45	FA20
(...)	A343	A310	A320	AT72	D328
	A345	B703	A321	B712	E120
	A346	B752	AN12	B732	BE40
	A359	B753	B736	B733	BE45
	B744	B762	B737	B734	H25B
	B748	B763	B738	B735	JS32
	B772	B764	B739	CL60	JS41
	B773	B783	C130	CRJ1	LJ35
	B77L	C135	IL18	CRJ2	LJ60
	B77W	DC10	MD81	CRJ7	SF34
	B788	DC85	MD82	CRJ9	P180
	IL96	IL76	MD83	DH8D	C650
	(...)	MD11	MD87	E135	C525
		TU22	MD88	E145	C180
		TU95	MD90	E170	C152
		(...)	T204	E175	(...)
			TU16	E190	
			(...)	E195	
				F70	
				F100	
				GLF4	
				RJ85	

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'SUPER HEAVY'	'UPPER HEAVY'	'LOWER HEAVY'	'UPPER MEDIUM'	'LOWER MEDIUM'	'LIGHT'
'CAT-A'	'CAT-B'	'CAT-C'	'CAT-D'	'CAT-E'	'CAT-F'
				RJ1H	
				(...)	

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TA 20 Navigation Aids**ILS**

Except where topographical features dictate, or operational requirements permit a reduction of the specified limits, the ILS localiser coverage sector extends from the localiser antenna to distances of:

- 25 NM within $\pm 10^\circ$ of the front course; and
- 17 NM within $\pm 35^\circ$ of the front course.

Presence of an ILS identifier does not confirm the integrity of a glideslope signal and, consequently, a GS check in accordance with the procedures shall be performed.

VOR Course Errors

When tuning manually a VOR station, care shall be taken to select the correct frequency. If by mistake the adjacent channel, i.e. a frequency offset by 50 KHZ, is selected, the following phenomena may occur:

- The 'off' flag disappears;
- Station identification may be received, though probably distorted;
- Course information, usually close to radial 180 or 360, may be shown which does not alter even when the course of the aeroplane is changed.

Whenever a radial is indicated, which is unlikely for the course flown, and / or when the VOR identification signal is distorted, the VOR frequency selected shall be cross-checked against the published frequency.

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TA 21 Hot and High Operations

At high density altitude (HDA) airfields, not only is the aeroplane performance restricted in terms of approach / climb performance and take-off performance but the aeroplane energy levels during an approach are high and as such are very similar to the energy of an aeroplane under the conditions of a 'rushed approach' to an airfield in ISA conditions.

This is due to a higher TAS, and thus groundspeed, in hot and high conditions compared to ISA conditions.

Examples: Phoenix, USA, the temperature can rise to 45 degrees (ISA + 32). The elevation of the airfield is 1135 ft and the density altitude is 5000 ft. An approach IAS of 120 kts equates to a groundspeed in still air of 132 kts.

Appreciation of the environmental conditions and the impact on performance is the key to avoid a poorly managed approach.

Density Altitude

To calculate Density Altitude a rule of thumb is:

'Density Altitude = Airfield Altitude + 1000 ft per 8 degrees above ISA'

Awareness of the Density Altitude

'Hot and high' conditions can be anticipated and therefore covered in approach briefing. Most of the problems arise because the TAS increases with increasing density altitude (refer to table). Therefore, a comparison of IAS and TAS will give a good indication of the problems ahead.

Horizontal Profile

The turning radius will increase as the TAS increases. It may be necessary to reduce speed by the increment in TAS (due to high density altitude) to achieve the horizontal profile.

Vertical Profile

When flying a 3° profile the rule of thumb is rate of descent = 5 x groundspeed. Therefore, if the TAS is higher it follows that the V/S will be higher by an equivalent amount.

Energy Management

Approaches are normally based on 'gates' that are fixed distance (or height, it amounts to the same thing) for touchdown. With a higher TAS this distance will be covered a lot more quickly, leaving less time to slow down. Allowing a minimum of 50% extra distance would seem to be a good rule of thumb. However, it could be more for very high density altitudes. This should be increased further if the factor (e.g. tailwind, non-normal configuration, high approach speeds due to weight, etc.) are also present.

Example: A speed of 250 kts is required below 10,000 ft. On a continuous descent approach one would aim to pass the 250/10,000 gate at approximately 35 nm from the runway. However, at a HDA airfield 250 kts IAS can give a TAS / Groundspeed of 330 kts. A lot more energy to lose and a potential to get high and fast if not recognised and acted upon early. The higher TAS also increases the radii of turn. Adherence to the published procedures will grant obstacle clearance. Strict adherence to procedure design speeds is mandatory. This is particularly important where the airfield is situated in a valley, like Samedan. Crews are to slow down and configure early where possible.

Braking

The kinetic energy will be higher on touchdown and this has to be dissipated by the brakes. The use of reverse thrust and an appropriate runway turn-off will keep brake temperatures down.

Other Factors

Consider the appropriateness of tankering into and out of HDA airfields and the effect on performance.

Check landing distance. Plan to use most of the runway on a long roll out. Use reverse thrust if available and minimum braking to avoid hot brakes.

Summary

1. 'Hot and high' conditions shall be anticipated and briefed. Comparing IAS to TAS will give a good indication of likely problems.
2. Turning radius will increase. Adherence to procedure design airspeed is mandatory.
3. The ability of the aeroplane to follow a descent profile is not necessarily an indication of how well it will slow up.
4. It could take up to 50% further to decelerate (more if the density altitude is very high or if other factors are present). The 'gates' should be moved back to more appropriate position to allow more time and distance to slow up.
5. Touchdown speeds will be higher. Choose appropriate braking.

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TA 22 PBN**RNP Specifications
RNP 10**

RNP 10 implementation is for the oceanic and remote phases of flight where ground-based navigation aids do not exist, except possibly at isolated locations. Hence aeroplane navigation will need to be based on a long-range navigation capability of acceptable performance using inertial navigation and / or global positioning systems.

RNP 10 supports a longitudinal separation minima of 80 NM (RNAV distance) or 10 minutes using the Mach Number Technique. This may be reduced to 50 NM when RNP 10 is supported by either direct VHF contact or for PBCS-authorized aircraft when RNP 10 is supported by RCP 240 / RSP 180.

At least two FMS and at least two independent LRNS are required for operations in oceanic and remote areas.

In all cases, where navigation relies only on inertial systems, a usage limit of 6.2 hours (without updating) is set from the time the inertial system is placed into the navigation mode.

RNP 4

RNP4 is an oceanic and remote area navigation specification to support 30 NM track spacing (Lateral and Longitudinal) with ADS-C and CPDLC required. To meet this more accurate navigation requirement, two independent LRNS are required for which GNSS sensors are mandatory. If GNSS is used as a stand-alone LRNS, an integrity check is foreseen (fault detection and exclusion). Additional aeroplane requirements include two long range communication systems (LRCSS) in order to operate in RNP4 designated airspace.

The separation minima may be reduced for PBCS-authorized aircraft; when RNP 4 is supported by RCP 240 / RSP 180, separation minima of 23 NM lateral and 30 NM or 5 minutes longitudinal may be applied.

During operations in airspace or on routes designated as RNP 4, the lateral total system error must be within ± 4 NM for at least 95% of the total flight time.

The appropriate AIP that is in Jeppesen Airway Manual (En-route Section) shall be consulted to assess coverage of HF and SATCOM. The additional requirements will include use of automatic dependent surveillance (ADS) and / or CPDLC when the separation standard is 30 NM lateral and / or longitudinal.

The on-board navigation data must be current and include appropriate procedures.

Flight crew shall not modify waypoints that have been retrieved from the database. User-defined data may be entered and used.

For parallel offset operations en-route in RNP 4, transitions to and from the offset track shall maintain an intercept angle of no more than 45° unless specified otherwise by ATC.

Additional requirements are specified in other documents, such as national operating rules, aeronautical information publications (AIPs) and the Regional Supplementary Procedures (Doc 7030). These are found in the Jeppesen Airway Manual.

RNP 2

RNP 2 is primarily intended for a diverse set of en-route applications, particularly in geographic areas with little or no ground NAVAID infrastructure, limited or no ATS surveillance, and low to medium density traffic. Use of RNP 2 in continental applications

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requires a lower continuity requirement than used in oceanic/remote applications. The RNP 2 specification is based upon GNSS and therefore GNSS is required. RNP 2 shall not be used in areas of known GNSS signal interference.

The Lateral separation minima for RNP 2 is 50 NM, reduced to 23 NM with RCP 240 / RSP 180, and further reduced to 18 NM with direct controller-pilot VHF voice.

During operations in airspace or on routes designated as RNP 2, the lateral total system error must be within ± 2 NM for at least 95% of the total flight time.

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RNP 1

RNP 1 is a terminal airspace specification which may also be applied in the en-route flight phase. RNP 1 may be thought of as P-RNAV (RNAV 1) plus a requirement for GNSS as the primary navigation sensor. It is designed to connect en-route airspace structure to instrument approaches (STARs and runway transitions), and for departures (SIDs).

As in the case of P-RNAV, RNP 1 terminates at the FAF. During operations in airspace or on routes designated as RNP 1, the lateral total system error must be within ± 1 NM for at least 95% of the total flight time.

RNP APCH

The ICAO term to describe RNAV (GNSS) approaches, the charting and avionics will retain the existing labels as a legacy to existing operations. However, the approach operations conform to RNP principles. Refers to any approach which relies solely on RNAV guidance. Such approaches require approval of the operator and crew training. The default RNP value for an RNP APCH is 0.3 nm.

RNP APCH requires an approach and landing using instruments for navigation guidance based on an instrument approach procedure. There are two methods for executing instrument approach operations:

- A two-dimensional (2D) instrument approach operation, using lateral navigation guidance only; and
- A three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance.

NOTE

Lateral and vertical guidance refers to the guidance provided either by:

- A ground-based radio navigation aid; or
- Computer-generated navigation data from ground-based, space-based, self-contained navigation aids, or a combination of these.

Instrument approaches are designed as either 2D or 3D procedures. A 2D procedure utilises a Minimum Descent Altitude and, by design, any vertical guidance is advisory; for example, NDB, VOR, or RNP APCH with LNAV (only) minima. Conversely, a 3D procedure utilises a Decision Altitude and includes electronically-generated vertical guidance; for example, ILS, or RNP APCH with VNAV minima.

It is important to recognise that the design of an instrument approach procedure does not stipulate how it is to be flown. For example, all the 2D approach procedures listed above may be flown using 3D operational techniques (fleet specific) - as overlay or RNP approaches. Moreover, NTA operational policy requires approaches to be flown using the CDFA technique to a Decision Altitude.

RNAV

An acronym derived from 'Area Navigation'. A method of navigating which permits aircraft operation on any desired flight path using self-contained or ground-based aids, or any combination thereof. Also refers to terminology to define Minima associated with Final Approaches flown without reference to ground-based NAVAIDs.

RNAV 5 (B-RNAV)

RNAV 5 (B-RNAV) is mandatory as the primary means of navigation in en route airspace, including designated feeder routes (SIDs & STARs) in and out of notified Terminal Areas. Implementation RNAV5 in designated airspace and regional procedures are located in the Jeppesen Airway Manual in the ATC Section and in the notes section of the Jepp FDPro en-route charts.

RNAV5 defines RNAV operations which satisfy a required track- keeping accuracy of ± 5 NM for at least 95% of the flight time.

This level of navigation accuracy is comparable with that which can be achieved by conventional navigation techniques on airways defined by VOR/DME, when VORs are less than 100 NM apart. In general terms, RNAV equipment operates by automatically determining aeroplane position from one, or a combination, of the following together with the means to establish and follow a desired path:

- DME/DME;
- VOR/DME;
- INS/IRS;
- GPS/GNSS (with limitations if it is the only available sensor).

The required equipment and limitations are stated in the respective AFM and MEL.

This Standard is applicable to all aeroplanes under IFR operating in appropriately designated and / or notified airspace. RNAV5 implementation will be reflected in the Jeppesen Airway Manual sections: Air Traffic Control / Air Traffic Control Data.

RNAV 2

The RNAV 2 specification is applicable to all ATS routes, including routes in the en-route domain, SIDs and STARs. It also applies to IAPs up to the FAF.

RNAV 2 is primarily developed for RNAV operations in a radar environment (for SIDs, radar coverage is expected prior to the first RNAV course change). However, RNAV 2 may be used in a non-radar environment or below minimum vectoring altitude if the implementing State ensures appropriate system safety and accounts for lack of on-board performance monitoring and alerting.

RNAV 2 procedures are in use worldwide and different requirements with regards to the RNAV system performance are established. E.g. some procedures require DME/DME as a minimum and do not accept VOR/DME.

RNAV 2 operations are based upon the use of RNAV equipment that automatically determines aeroplanes position in the horizontal plane using inputs from the following types of positioning sensor:

- Aeroplane Position Determined by DME/DME; or
- Aeroplane Position Determined by Global Navigation Satellite System (GNSS); or
- Aeroplane Position Determined by DME/DME/IRU.

RNAV 1 (P-RNAV)

Initial application is in the Terminal Area and RNAV 1 track keeping equates to cross track accuracy of RNP 1 (lateral better than +/- 1 NM for 95% of the flight time). RNAV 1 procedures are designed to a common set of design principles specific to RNAV equipped aeroplanes.

RNAV 1 procedures are in use worldwide and different requirements with regards to the RNAV system performance are established. E.g. some procedures require DME/DME as a minimum and do not accept VOR/DME.

RNAV 1 operations are based upon the use of RNAV equipment that automatically determines aeroplanes position in the horizontal plane using inputs from the following types of positioning sensor:

- Aeroplane Position Determined by DME/DME; or
- Aeroplane Position Determined by VOR/DME; or
- Aeroplane Position Determined by Global Navigation Satellite System (GNSS); or
- Aeroplane Position Determined by INS or IRS with Auto Updating.

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TA 23 SID/STAR Navigation Specifications

Navigation specifications / functionalities applicable to SIDs and STARs are:

- RNAV-1 (P-RNAV);
- RNAV-2;
- RNP-1;
- RNAV-5 (B-RNAV) - usually only used for STARs but some states may differ from ICAO and use RNAV-5 for SIDs.

Sensors normally applicable to SIDs and STARs are:

- GNSS (GPS);
- DME/DME;
- IRU.

Conventional navigation aids may also be required for certain segments of the SID/STAR (e.g. initial portion of a SID).

If applicable the chart will state which sensors (or combination) and/or conventional navigation aids are required and which are not permitted to be used; OM-B / AFM procedures shall be complied with to meet these requirements.

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TA 24 En-route Navigation Specifications

Navigation specifications applicable to en-route continental (generally airspace with ground-based surveillance available, e.g. radar):

- RNAV-1 (P-RNAV);
- RNAV-2;
- RNP-2;
- RNAV-5 (B-RNAV) (VOR/DME, DME/DME, IRU, GNSS).

Navigation specifications applicable to en-route Oceanic / Remote (generally airspace without ground-based surveillance, e.g. procedural separation):

- RNAV-10 (RNP-10);
- RNP-4;
- RNP-2.

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TA 25 Identification of PBN Approach Procedures

In 2015 ICAO published Circular 336, which is a new protocol for the naming of RNAV final approach procedures. In accordance with the general move away from the older term 'RNAV' to the newer concept of Performance Based Navigation, a final approach designed to be flown using RNAV techniques will, forthwith, be referred to as an RNP Approach. However, since states have until 2022 to comply with the requirements of the Circular, and there is some doubt about its global implementation, it is likely that pilots will encounter a variety of terms for RNP Approaches in the interim.

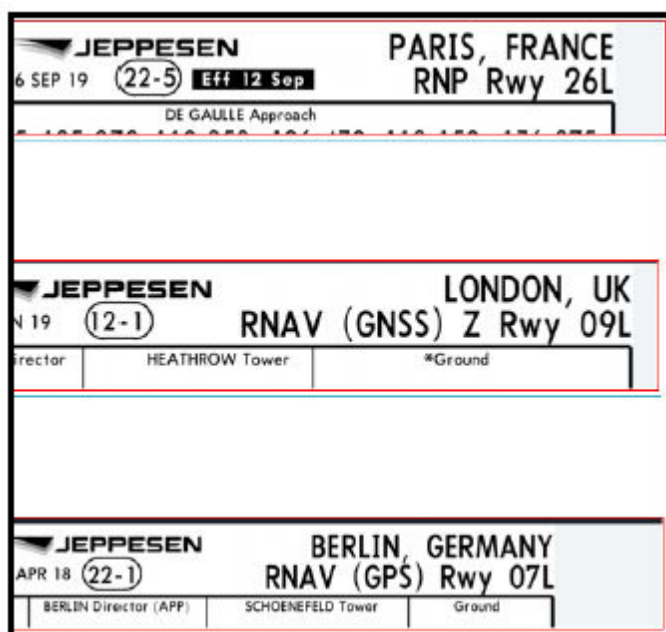
RNP APCHs are charted as RNP RWY XX, RNAV (GNSS) RWY XX, or RNAV (GPS) RWY XX. A minima line is provided on the approach chart for each of the available types of RNP APCH procedures:

- Non-precision approach (NPA) - lateral navigation (LNAV) or localiser performance (LP) minima line;
- APV BARO - LNAV/VNAV (vertical navigation) minima line; and
- APV SBAS - localiser performance with vertical guidance (LPV) minima line.

NOTE

RNP AR Approaches charted as RNP RWY XX (AR) or RNAV (RNP) with a note stating Authorisation Required or Special Aircraft and Aircrew Authorisation required. **These are NOT approved and shall not be conducted.**

For example, charts titles are:



Example of a typical RNP approach minima box:

PV, LNAV/VNAV, MAP at DA							
NAV, MAP at RW26L							
Standard				STRAIGHT-IN LANDING RWY 26L			
LPV CAT I				LNAV/VNAV		LNAV CDFA	
DA(H) 517' (200')				DA(H) A: 647' (330')		DA/MDA(H) A: 740' (423') C: 770' (453')	
				BC: 657' (340')		B: 750' (433') D: 790' (473')	
				D: 667' (350')			
		TDZ or CL out	ALS out	ALS out		ALS out	
A B C D	RVR 550m	RVR 550m	RVR 1200m	RVR 800m	RVR 1500m	RVR 1200m	RVR 1500m
						RVR 1300m	RVR 1500m
						RVR 1400m	RVR 2100m
				RVR 900m	RVR 1600m	RVR 1500m	RVR 2200m
RVR 750m when a Flight Director or Autopilot or HUD to DA is not used.							

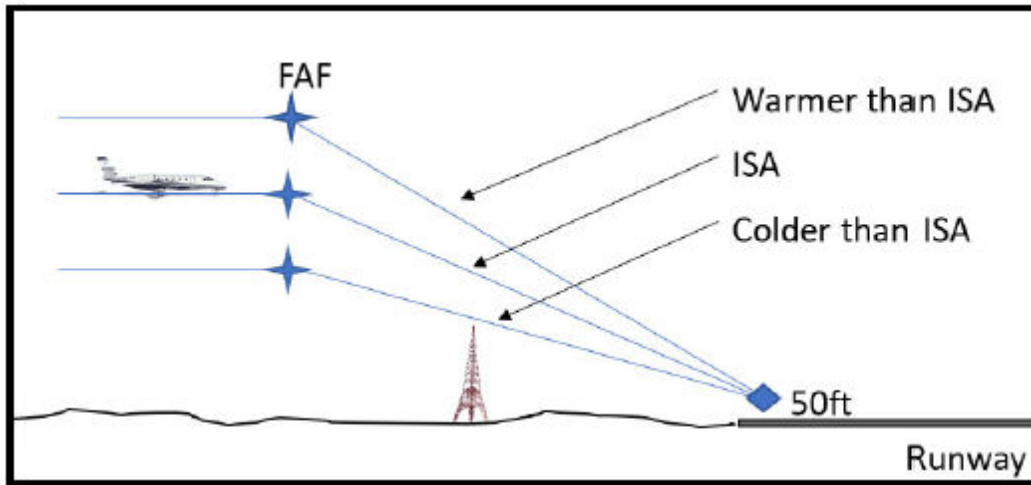
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TA 26 Temperature Effect

The VNAV path coincides with the published angle only when the temperature is ISA (standard). It will be steeper in warmer temperatures and shallower in colder temperatures.



RNAV Final Approach Procedures which have LNAV/VNAV minima stated will usually have a published minimum temperature stated on the chart, below which the procedure must not be flown to LNAV/VNAV minima. If this is the case, the approach may be flown to temperature corrected LNAV minima or LPV minima. Visual alignment with either the PAPI or VASI will ensure obstacle clearance. The altimeter error will be greater near the FAF and smaller near the runway.

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TA 27 Receiver Autonomous Integrity Monitoring (RAIM)

This is a technique whereby a GNSS receiver / processor determines the integrity of the GNSS navigation signals using only GNSS signals or GNSS signals augmented with altitude. This determination is achieved by a consistency check among redundant pseudorange measurements. At least one satellite in addition to those required for navigation has to be in view for the receiver to perform the RAIM function. Four satellites will give a navigation solution but five are needed for the equipment to conduct RAIM and six or more will allow the equipment to lock out a rogue satellite. The aircraft turning, or terrain masking during approach, can block the signals from the satellites and cause short duration RAIM outages.

RAIM detects faults with redundant GNSS-range measurements. That is, when more satellites are available than needed to produce a position fix, the extra range measurements should all be consistent with the computed position. A range measurement which differs significantly from the expected value (an outlier) may indicate a fault of the associated satellite or another signal integrity problem (e.g. ionospheric dispersion).

NTA dispatch procedures perform a RAIM check for every flight. Flight crew may assume that RAIM prediction is satisfactory unless information is received to the contrary. RAIM prediction will be provided via the OFP. This is valid from 1 hour prior to and 1 hour after the destination ETA on the OFP.

For those GNSS systems relying on RAIM and necessitating a check of its availability for RNP APCH, the flight crew shall perform a new RAIM availability check if ETA is more than 1 hour different from the ETA used during the pre-flight planning.

For aeroplanes that have a functioning SBAS receiver, the RAIM check is not required. If SBAS (EGNOS, WAAS, etc.) is lost, RAIM prediction must be performed. For aeroplanes navigating with SBAS receivers, GNSS RAIM availability shall be checked in areas where the SBAS signal is unavailable.

If RAIM Check is not available before the flight, the availability of RAIM shall be confirmed using the latest NOTAMs and information from the US Coastguard. Alternatively, EUROCONTROL's AUGUR website (<https://augur.eurocontrol.int/>) can provide a prediction for the availability of RAIM.

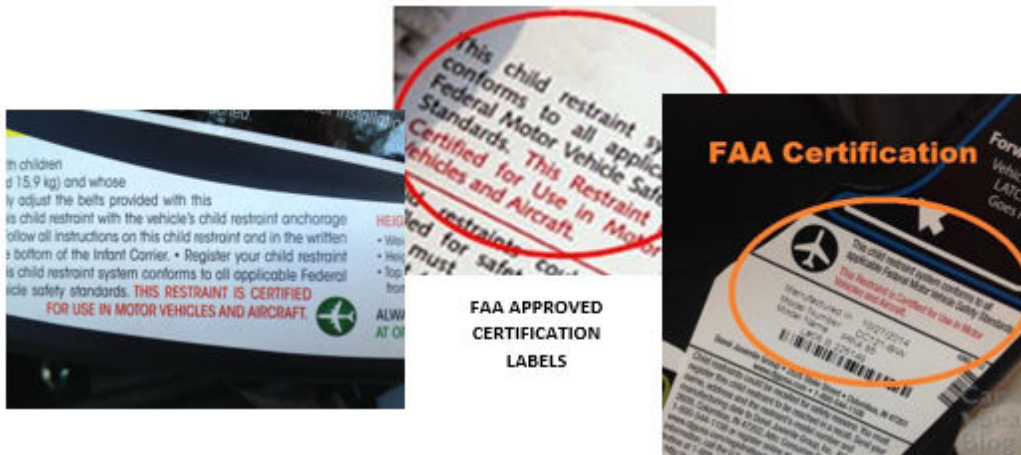
In the event that a RAIM outage of 5 minutes or more is predicted, crew may not fly any part of an RNP APCH during that outage time - from the Initial Approach Fix to the end of the Missed Approach.

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TA 28 CRDS: Examples of Approved CRD Labelling**NOTE**

These images illustrate labelling on aviation approved CRD, but are not exhaustive.

**EU CERTIFICATION LABEL****TRANSPORT CANADA CERTIFICATION LABEL****FAA APPROVED
CERTIFICATION
LABELS****AUSTRALIA & NEW ZEALAND
APPROVED CERTIFICATION LABELS**

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TA 29 Operations in Class E Airspace

Many European regional airports, notably in Germany and Switzerland, make extensive use of class E airspace. Due to the nature of the ATC environment, operating to and from these airports presents the crew with a unique set of challenges. It should be noted that:

- In class E airspace, whilst both IFR and VFR flight are permitted, ATC does not provide separation between IFR and VFR traffic;
- IFR traffic shares responsibility for separation from uncontrolled VFR traffic with that traffic;
- VFR traffic is not required to have an ATC clearance, nor be fitted with a transponder.

There is a risk that when operating in Class E airspace, ATC may have no knowledge of conflicting VFR traffic. Collision avoidance is the responsibility of the crew, even when under IFR. Separation is based on the see and avoid principle.

To facilitate the visual acquisition of flights which are not separated, a speed limitation of 250 KTS is imposed by the airspace structure for both VFR and IFR flights. NTA recommends that speed is reduced below 200 KTS to aid acquisition and avoidance of VFR traffic.

Class	Type of Flight	Separation provided	Separation provided	Speed limitation	Radio communication requirement	Subject to an ATC clearance
E	IFR	IFR from IFR	Air traffic control service and, as far as practical traffic information about VFR flights	250 KTS IAS below 10000 ft AMSL	Continuous two-way	Yes
	VFR	Nil	Traffic information as far as practical	250KTS IAS below 10000 ft AMSL	No	No

The risk of a TCAS or airprox event in Class E airspace may be mitigated by considering:

- Maintain an effective visual lookout to see and avoid VFR traffic, regardless of the flight rules;
- Optimising the use of automation to allow more spare capacity to scan for other traffic.

Operational recommendations

The following operational recommendations should be considered to mitigate the risk of a TCAS or airprox event when operating in Class E airspace:

- During the pre-flight briefing, identify Class E airspace and attribute airprox threat to VFR traffic. Remember that VFR traffic is not required to have an ATC clearance, nor be fitted with a transponder.
- Maximise the opportunity to maintain an effective lookout for VFR traffic, by briefing early and making maximum use of automation;

- Consider task allocation to maximise effective lookout for VFR traffic;
- Display all exterior lights, including pulse lights (if fitted);
- Reduce speed. Just as for VFR flight, consider a maximum speed of 200kts;
- Consider deploying first stage of flap to lower the nose, maximising visual lookout;
- Request VFR traffic information and if necessary, delay your arrival or departure to deconflict;
- Minimise visual approaches - they require additional attention and increase flight time in proximity to VFR traffic;
- Minimise the time spent in Class E airspace by, for example, expediting the climb or descent.

See and avoid

See and avoid is the last line of defence to avoid a TCAS or airprox event in class E airspace.

See and avoid requires:

- Effective visual scanning;
 - Brief and prepare early;
 - Task allocation - consider one pilot responsible for the visual scan;
 - Effective scanning is accomplished by a series of short, regularly-spaced eye movements that bring successive areas of the sky into the central visual field. Each movement should not exceed 10°, and each area should be observed for at least one second to enable detection.
- Good situational awareness:
 - Collecting information about the operating environment from radio transmissions by other aircraft and ground stations;
 - Considering crew cooperation by optimising workload management and task allocation.
- Effective flight path management by optimal use of automated systems.
 - Higher levels of automation enable extra opportunity for the visual lookout task;
 - Relieve the PF from routine handling tasks and thus allows time and resources to enhance their situational awareness of the threat environment.

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TA 30 Contaminated & Wet Runway Guidance Material**Deriving RWYCC using the RSD and braking action**

It is possible that some states / aerodromes may not publish it is therefore necessary to define how a runway surface description without an RWYCC may be related to RWYCC performance data.

The RWYCC to be used shall be the most restrictive (i.e. the lowest number) which results from entering the table below using the:

1. Runway surface conditions description;
2. Reported Braking Action.

E.g., if the report gives: -20°C, Compacted snow, and Braking Action Medium, then the lowest number RWYCC is produced by the Medium Braking action so the RWYCC to be used is 3.

RUNWAY CONDITION CODE (RCC)	RUNWAY SURFACE CONDITION DESCRIPTION	REPORTED BREAKING ACTION
6	• Dry	-
5	• Frost • Wet (includes damp and 1/8 in (3 mm) depth or less of water) Frost 1/8 in (3 mm) depth or less of: • Slush • Dry snow • Wet snow	Good
4	-15°C and Colder OAT: • Compacted snow	Good to Medium [1]
3	• Wet ('slippery when wet' runway) • Dry snow or wet snow (any depth) over compacted snow Greater than 1/8 in (3 mm) depth of: • Dry snow • Wet snow Warmer than -15°C OAT: • Compact snow	Medium [1]
2	Greater than 1/8 in (3 mm) depth of: • Water • Slush	Medium [1] to Poor
1	• Ice	Poor

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RUNWAY CONDITION CODE (RCC)	RUNWAY SURFACE CONDITION DESCRIPTION	REPORTED BREAKING ACTION
0	• Wet ice • Water on top of compact snow • Dry snow or wet snow over ice	Nil

Wet runway coverage

A runway will have 'Wet' in the Runway Surface Descriptor (RSD) field when the Wet coverage exceeds 10%, with up to 25% being reported as 25%.

Aeroplane performance is only affected when the coverage exceeds 25% in at least one third, so where the coverage is <25% in all thirds and there is no significant degradation in braking action, then the (RWYCC 6) will be issued for all thirds and the runway can be considered 'Dry' for performance purposes.

So, a runway where 10% to 25% is wet may be reported as 6/6/6 25/25/25 NR/NR/NR WET/WET/WET, and since all thirds are RWYCC '6' the runway can be considered dry for performance calculations despite having 'Wet' as the RSD.

As soon as the coverage exceeds 25% and/or the braking action is degraded in any third, a RWYCC '5' will be issued for that third and the Runway must now be considered 'Wet' for performance purposes¹.

Contaminated runway coverage

Contamination is reported when the coverage exceeds 10%. Runway contamination affects aeroplane performance only when the coverage exceeds 25% in at least one third.

Coverage from 10% to 25% is reported as 25% so when 25% is reported, coverage may be significantly less than 25% and is provided only for situational awareness.

With a contaminant coverage of 25% or less, and a RWYCC of 5, performance calculations may then assume a wet runway.

Coverage more than 25% in any third, combined with the required depth if a loose contaminant (see below), will result in a RWYCC <5 being issued for that third and the entire runway will be considered to be contaminated.

In other words, a runway will be considered contaminated^[1] if one of the thirds has contaminant coverage in excess of 25% at the required depth if applicable and has a RWYCC of <5.

Loose contaminant depth

According to the Runway Condition Assessment Matrix (RCAM), RwyCC – 5 which is equivalent to a 'Wet' runway includes:

Up to and including 3mm depth of:

- Water
- Slush

¹ If only the last third is affected then it may be possible to shorten the runway as described in [8.1.2.4.5 CONTAMINATED RUNWAY LIMITATIONS](#).

- Dry Snow
- Wet Snow

Therefore, when a runway is affected by one of these contaminants, it will only be considered contaminated when the depth exceeds 3mm and has the required coverage (see above)^[1].

NOTE

For solid contaminant such as compacted snow or ice, there is no depth requirement to be considered contaminated, only coverage.

RwyCC Downgrade

When the contamination coverage and/or depth do not technically meet the criteria for the runway to be considered contaminated, BUT the runway inspection team assess they will sufficiently impact stopping performance as if the runway were contaminated, the RWYCC may include a downgrade.

This means that the RWYCC issued will be lower than would normally be expected when entering into the RCAM table using the reported type, depth, and coverage of contaminant.

For example: A runway covered to 25% with 4mm of Wet Snow is not technically contaminated as the coverage does not exceed 25%, so this would normally result in a RWYCC of 5. However, if the coverage is concentrated around the area where touchdown will occur, and braking will take place the runway inspection team may issue a downgrade to RWYCC 4 or even RWYCC 3.

Therefore, when a runway is issued with a RWYCC of less than 5, it is considered contaminated.

When calculating landing performance for aircraft which only have 'legacy performance' available and a downgraded RWYCC has been issued, the appropriate 'ALT RWYCC X' selection should be used in place of the depth and type of contaminant^[1].

RwyCC Upgrade

A RWYCC upgrade is where the RWYCC number is higher than would be expected when entering into the RCAM table with the contaminant type (e.g. when a runway is contaminated with Ice which would normally result in a RWYCC 1 but it is issued a RWYCC 3).

Upgrading a primary RWYCC 5, 4, 3 or 2 determined from the observed contaminant type is not allowed.

A RWYCC 1 or 0 can, in exceptional cases, be upgraded to a maximum of 3 even when the contaminant that has caused this primary classification has not been removed. It may have been treated with sand or gravel, or simply provide exceptional friction due to its inherent characteristics, as assessed by trained aerodrome personnel.

When calculating landing performance for aircraft which only have 'legacy performance' available and an upgraded RWYCC has been issued, the appropriate 'ALT RWYCC X' selection should be used in place of the depth and type of contaminant.

Additional Supporting Data**Table 6. Data supporting PDCS 'ALT RWYCC X' selection referred to in [8.1.2.4.1](#)**

RWYCC	PDCS Selection	Data used by aircraft type					
		Phenom 300	Citation XLS	Citation Latitude	Challenger 350	Challenger 650	Global 5000/6000
4	ALT RWYCC 4	COMPACT SNOW	COMPACT SNOW	COMPACT SNOW	COMPACT SNOW	COMPACT SNOW	COMPACT SNOW
3	ALT RWYCC 3	WET SNOW 1/8 in	SNOW 1 in	WET SNOW 1/8 in	DRY SNOW	LOOSE SNOW	LOOSE SNOW
2	ALT RWYCC 2	SLUSH 1/8 in	SLUSH COVERED 1/8 in	WET SNOW 1/8 in	DRY SNOW	SLUSH / STANDING WATER	STANDING WATER
1	ALT RWYCC 1	NetJets operations not permitted 8.1.2.4.5					

- End of Additional Guidance -[Return to Main Content](#)

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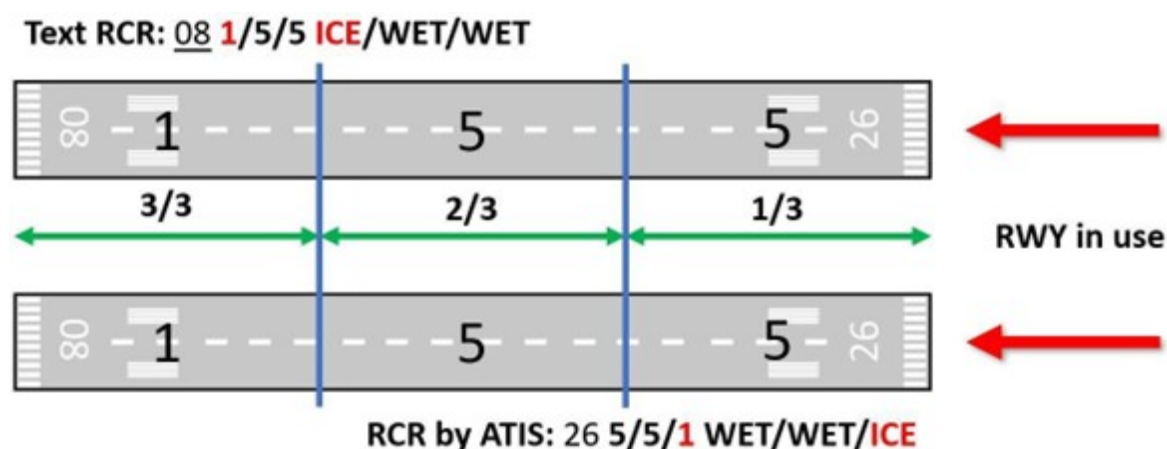
TA 31 Operations Using Only the First 2 / 3 of The Runway**Example 1 - Departing EGHH RWY 26**

Runway Condition Report: EGHH 12280700 08 1/5/5 50/100/100 NR/NR/NR ICE/WET/WET

CAUTION

A text RCR will always be in the direction of the lowest runway number, regardless of the runway in use.

If the RCR is given by ATIS then it will be in the direction of the runway in use. **Therefore, always verify the direction of the report you are using, particularly if using this procedure to shorten the runway!**

Figure 1 - Runway Condition Report (RCR) exemplified**Example 1:**

In this example the runway in use in EGHH is RWY26, however the text report is for RWY08. So, when assessing RWY26 the reports for the thirds of the runway will need to be re-ordered in the reverse.

The text report is for the lowest RWY number: **08 1/5/5 ICE/WET/WET** so re-ordered for RWY26 it will be **26 5/5/1 Wet/Wet/Ice**.

If we need to use the worst third of the runway when making the performance assessment, the RWYCC of 1 (RSD of Ice, last third of the RWY in use) means operation is prohibited. Therefore, you need to check if the runway length required for departure is within the first two thirds of the overall runway length.

To check whether operations are possible in the first two thirds of the runway, shorten the runway using the iPreFlight Genesis function as follows:

1. Input the departure data, applicable to the first two-thirds (in this case, the runway is 'Wet'), select the runway to be used and then select the applicable 'TORA' to enter the 'shorten' menu.

Figure 2 – Example 1 EGHH RWY 26

PERFORMANCE CRITERIA

Runway Condition	WET RWY	Bleeds	BLO OPEN
Flap	6 DEG	Miscellaneous	TRs, APU OFF
		Inoperative	None

RUNWAYS Planned Takeoff Weight: lb

RUNWAY	EQP	MAX TO WGT	TORA	NOTES	INTXN
☐ 08	Straight-Out	-- / Tc	7451 ft		SHOW
☒ 26	Straight-Out	98687 lb / -0	6647 ft		SHOW
☐ 26EXT	Straight-Out	98717 lb / -0	7254 ft	VIEW	

2. Select 'Shorten Runway' and 'Liftoff End', then calculate one third of the ASDA (in this case 2282 ft), enter this figure in the 'Shorten Runway Length By' box and click 'Apply'.

Figure 3 – Example 1 EGHH RWY 26 – Runway Shortening

Runway Shortening - EGHH 26

Apply Shortening To:

Shorten Runway Length By:

New TORA : 4365 ft TODA : 4365 ft ASDA : 4365 ft

NOTAM ID (optional)

NOTES

RWY 26 TEMP DATA FOR USE WHEN 2282 ft CLOSED AT LIFTOFF END

3. This will reduce the TORA, TODA and ASDA to the most limiting minus one third of the ASDA (in this case the TORA 6647 ft – 2282 ft = 4365 ft).

NOTE

The runway is now annotated with '**Shortened**' and the limiting take-off mass is reduced to 81142 lbs which is the limit to take-off considering only the first two thirds of the runway.

Figure 4 – Example 1 EGHH RWY 26 – Runway Shortened

PERFORMANCE CRITERIA					
Runway Condition	WET RWY		Bleeds	BLD OPEN	
Flap	6 DEG		Miscellaneous	TRs, APU OFF	
			Inoperative	None	

RUNWAYS					
Planned Takeoff Weight:		80000 lb	Add Obstacles		Runway Details
RUNWAY	EOP	MAX TO WGT	TORA	NOTES	INTXN
08	Straight-Out	-- / TW	7451 ft		SHOW
26	Straight-Out	98687 lb / -0	6647 ft		SHOW
26TSR Shortened	Straight-Out	81142 lb / FL	TORA 4365 ft TODA 4365 ft ASDA 4365 ft	VIEW	
26EXT	Straight-Out	98717 lb / -0	7254 ft	VIEW	

Example 2 - Landing EGHH RWY 26

Runway Condition Report: EGHH 12280700 08 1/3/3 75/100/100 NR/NR/NR ICE/COMPACT SNOW/COMPACT SNOW

In this example the report is for RWY08 **08 1/3/3 Ice/Compact Snow/Compact Snow**.

As we are using RWY26 the thirds of the runway will need to be re-ordered in the reverse **26 3/3/1 Compact Snow/Compact Snow/Ice**.

If we need to use the worst third of the runway when making the performance assessment, the RWYCC of 1 (RSD of Ice, last third of the RWY in use) means operation is prohibited. Therefore, you need to check if the LDR is within the first two thirds of the runway.

To check whether operations are possible in the first two thirds of the runway, shorten the runway using the iPreFlight function as follows:

1. Input the arrival data, applicable to the first two thirds (in this case the runway is RWYCC '3' or RSD 'Compact Snow' depending on the options available and whether you are making a 'Dispatch' or 'FLDTA' assessment) select the runway to be used and then select the applicable 'LDA' to enter the 'shorten' menu.

Figure 5 – Example 2 EGHH RWY 08

PERFORMANCE CRITERIA

Runway Condition	COMPACT SNOW	Bleeds	BLD OPEN
Flap	30 DEG	Miscellaneous	TRs, APU OFF, DISP STD LD 60%
		Inoperative	None

Landing Possible On Any Runway

Planned Landing Weight: lb

Missed Approach Gradient Available: 13.87% or 842 FT/NM ⓘ Runway Details

RUNWAY	MAX LD WGT ⓘ	LDA	LAND DIST	NOTES	REQD MISS APP CLIMB ⓘ
☐ 08	78600 lb / ST	6030 ft	4150 ft		Specify
☒ 26	78600 lb / ST	6463 ft	4150 ft		Specify
☐ 26EXT	78600 lb / ST	6463 ft	4150 ft	VIEW	Specify

2. Select '**Shorten Runway**' and 'Liftoff End', then calculate one third of the LDA (in this case 2155 ft), enter this figure in the 'Shorten Runway Length By' box and click 'Apply'.

Figure 6 – Example 2 EGHH RWY 08 – Runway Shortening

Close Runway Shortening - EGHH 26

Shorten Runway
Shorten Declared Distances

Cancel
Apply

Apply Shortening To: Arrival End Liftoff End

Shorten Runway Length By: 2155 FT M

New LDA : 4308 ft

NOTAM ID (optional)

NOTES

RWY 26 TEMP DATA FOR USE WHEN 2155 ft CLOSED AT LIFTOFF END

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3. This will reduce the LDA by one-third. Note the runway is now annotated with 'Shortened' and the limiting landing mass is reduced to 58011 lbs, which is the limit for landing considering only the first two-thirds of the runway.

Figure 7 – Example 2 EGHH RWY 08 – Runway Shortened

PERFORMANCE CRITERIA

Runway Condition	COMPACT SNOW	Bleeds	BLD OPEN
Flap	30 DEG	Miscellaneous	TRs, APU OFF, DISP STD LD 60%
		Inoperative	None

Landing Possible On Any Runway

RUNWAYS

Planned Landing Weight: lb

Missed Approach Gradient Available: 13.87% or 842 FT/NM ⓘ Runway Details

RUNWAY	MAX LD WGT ⓘ	LDA	LAND DIST	NOTES	REQD MISS APP CLIMB ⓘ
☐ 08	78600 lb / ST	6030 ft	4150 ft		Specify
☐ 26	78600 lb / ST	6463 ft	4150 ft		Specify
☒ 26TSR Shortened	58011 lb / FL	4308 ft	4150 ft	VIEW	Specify
☐ 26EXT	78600 lb / ST	6463 ft	4150 ft	VIEW	Specify

- End of Additional Guidance -

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TA 32 Verifying obstacle database in iPreFlight

iPreFlight incorporates two different obstacle sources:

1. **Terrain + Obstacles on the airport ICAO chart:** loaded directly to iPreFlight after downloading the Airfield database.



These obstacles are automatically considered on take-off performance calculations made by Dispatch and on iPreFlight. We advise crew members to regularly download the Airfield database to ensure these obstacles are current.

2. **NOTAM obstacles:** Our OpsAerodata team manually reviews these obstacles and loads them into a separate APG database. They are automatically available to Dispatch but are not on the iPreFlight database.

A list of any NOTAM obstacles considered during the dispatch analysis is included in the performance flight release. The potentially relevant obstacles are those that have been declared to be 'in corridor.' (see the attached flight release out of KMIA).

The 'in corridor' obstacles must be added using the Add Obstacles functionality in iPreFlight to be accounted for in a take-off performance analysis. Failing to add these obstacles may result in iPreFlight giving an incorrect higher permitted MTOM when compared to the flight release documentation.

NOTE

Just adding the 'in corridor' obstacles to iPreFlight is only applicable where the 'DP' used on iPreFlight matches the 'DP' used on the dispatch analysis which lists the obstacles. This is because the 'DP' defines the corridor. If a different DP is used, then all NOTAM'd obstacles must be added for the iPreFlight calculation to be correct.

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NOTE

If there are no obstacles listed in the airport NOTAMs and none listed on the dispatch release then iPreFlight can be used for any DP without adding any obstacles.

Departure- KMIA - RWY 08L GRVD	ATOW 92684 Flap 6 DEG		Wind 00000		Elevation 9 ft				
	Rwy Cond DRY		ALT 1023		Slope 0.0% Uphill				
	Configuration: BLD OPEN; TRs; APU OFF;								
	Temp	T/O Dist	EPR	V1	Vr	V2	VFTO	MATOW (Limit)	FRA AEO(AGL) OEI(MSL)
	36	6331	1.52	128	128	138	188	93400 (-O)	1509
	30	6147	1.53	127	127	138	188	94258 (-O)	1509
	24 (Target)	6049	1.53	127	127	138	188	94310 (-O)	1509
	18	5944	1.53	127	127	138	188	94364 (-O)	1509
	Departure Notes								
	Takeoffs from:		08L		08L-K2				
TORA/TODA/ASDA		8600/8600/8600		6975/6975/6975					
MATOW		94310		93435					
Notes:		NOTAM 05/282 has been accounted for RWY 08L out of corridor NOTAM 05/233 has been accounted for RWY 08L out of corridor NOTAM 05/231 has been accounted for RWY 08L out of corridor NOTAM 07/184 has been accounted for RWY 08L out of corridor NOTAM 07/183 has been accounted for RWY 08L out of corridor NOTAM 01/185 has been accounted for RWY 08L out of corridor NOTAM 01/184 has been accounted for RWY 08L out of corridor NOTAM 04/093 has been accounted for RWY 08L out of corridor NOTAM 04/092 has been accounted for RWY 08L out of corridor NOTAM 04/091 has been accounted for RWY 08L out of corridor NOTAM 04/090 has been accounted for RWY 08L out of corridor NOTAM 05/444 has been accounted for RWY 08L in corridor NOTAM 06/504 has been accounted for RWY 08L out of corridor NOTAM 06/503 has been accounted for RWY 08L out of corridor NOTAM 4/6874 has been accounted for RWY 08L out of corridor NOTAM 08/276 has been accounted for RWY 08L out of corridor NOTAM 08/275 has been accounted for RWY 08L out of corridor NOTAM 4/5880 has been accounted for RWY 08L out of corridor							
PDCS Code References ST - Structural Limit FL - Field Length Limit -O - Obstacle Limit TS - Tire Speed Limit BE - Brake Energy Limit MC - Minimum Control Speed Limit FP - TO Thrust Time Limit LW - Landing Weight Limit CL - Climb Limit SP - Climb Speed Limit									

Tail No: _____ Date: _____
Request-Leg No: _____ Regulation: EU-OPS

PERFORMANCE DATA

NOTAM 08/496 has been accounted for RWY 08L out of corridor
NOTAM 09/128 has been accounted for RWY 08L out of corridor
NOTAM 09/127 has been accounted for RWY 08L out of corridor
NOTAM 09/399 has been accounted for RWY 08L out of corridor
NOTAM 09/439 has been accounted for RWY 08L out of corridor
NOTAM 09/527 has been accounted for RWY 08L in corridor
NOTAM 09/526 has been accounted for RWY 08L in corridor
NOTAM 11/135 has been accounted for RWY 08L out of corridor
NOTAM 11/134 has been accounted for RWY 08L out of corridor
NOTAM 11/223 has been accounted for RWY 08L out of corridor
NOTAM 11/222 has been accounted for RWY 08L out of corridor
NOTAM 11/221 has been accounted for RWY 08L out of corridor
NOTAM 11/220 has been accounted for RWY 08L out of corridor
NOTAM 12/134 has been accounted for RWY 08L out of corridor
NOTAM 12/133 has been accounted for RWY 08L out of corridor
NOTAM 12/132 has been accounted for RWY 08L out of corridor
NOTAM 12/131 has been accounted for RWY 08L out of corridor
NOTAM 4/1123 has been accounted for RWY 08L out of corridor
NOTAM 5/0237 has been accounted for RWY 08L out of corridor

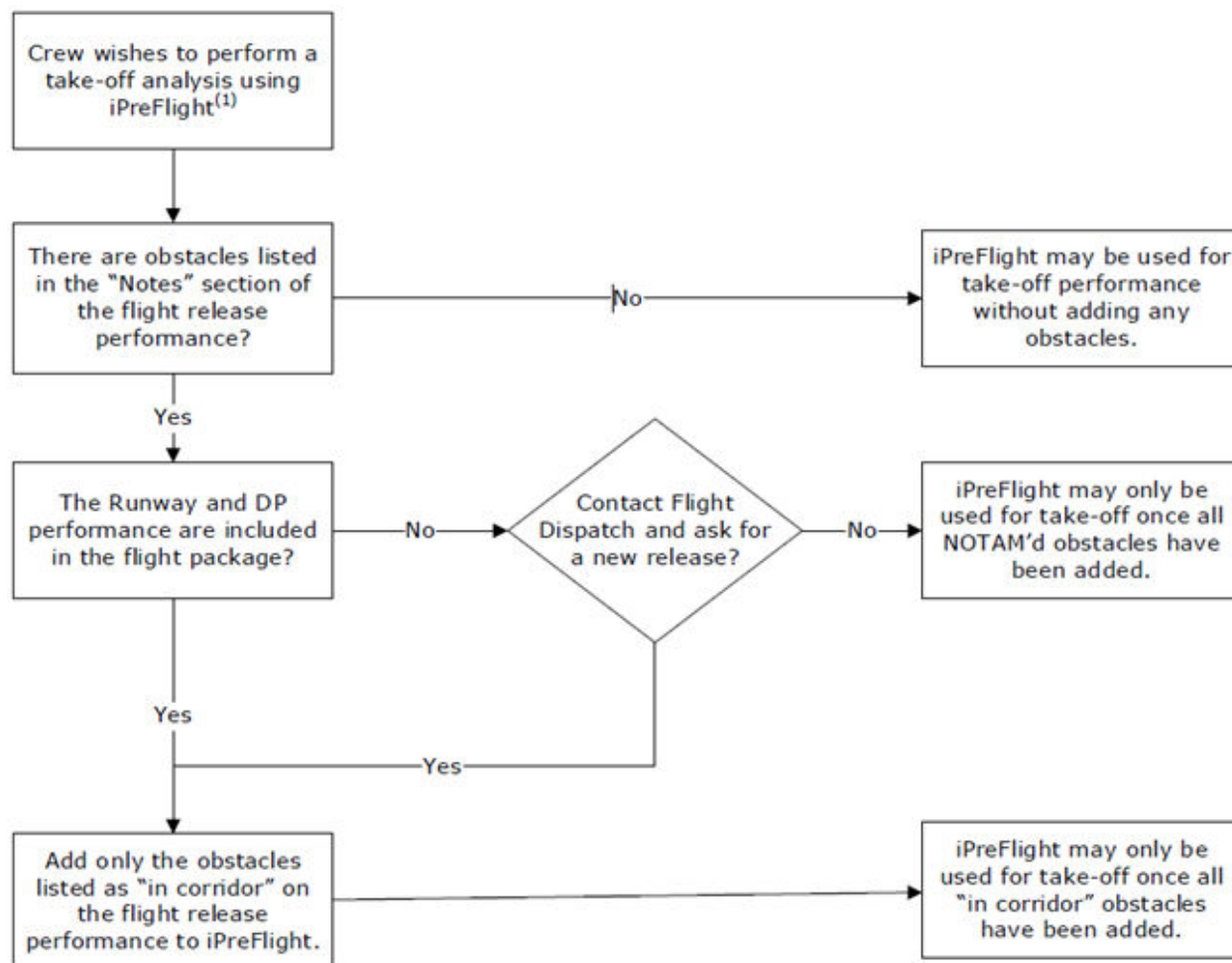
Engine Out Procedure(s)

Fly Runway Heading

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The process to decide on the need to use the Add Obstacle functionality on iPreFlight:

(A)



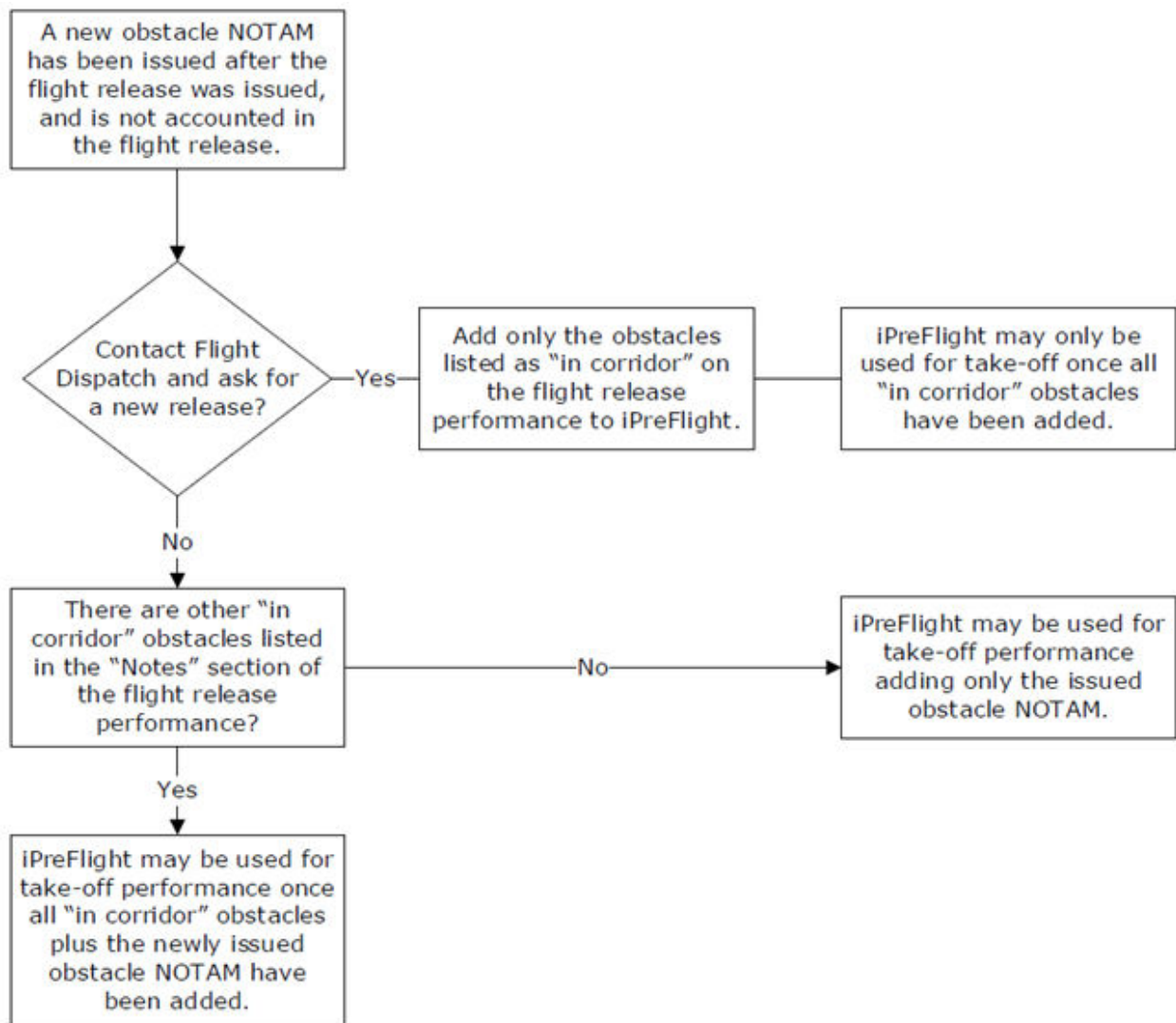
NOTE

Performing a take-off analysis using iPreFlight may be due to any number of reasons, including departure:

- from an intersection not listed on the flight release
- using a different configuration than shown on the flight release
- using a different runway from that shown in the flight release
- with conditions that differ from those show on the flight release

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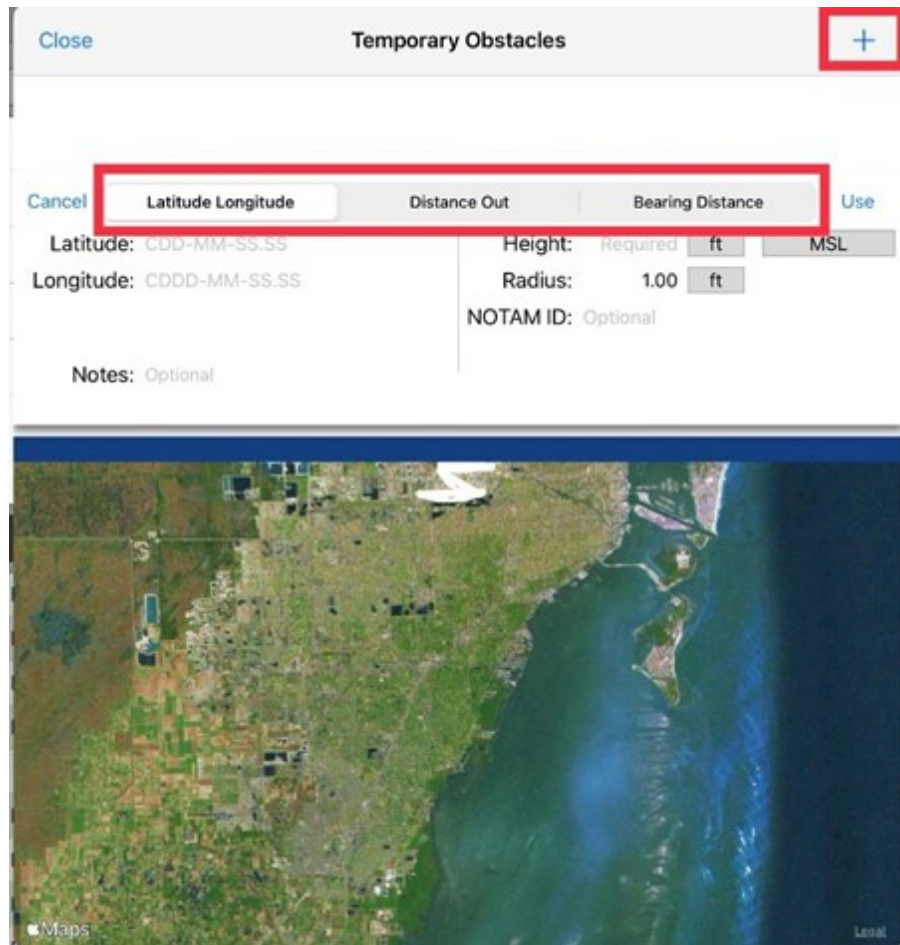
(B)

**How to use the Add Obstacle functionality:**

1. On the Runway Analysis screen, input the airport and runway and click **Add Obstacles**.



2. A Temporary Obstacles window will open. Click the + sign to start adding obstacles. Depending on the NOTAM text, you can opt for a Latitude/Longitude, Distance Out or Bearing Distance configuration.



3. Input the obstacle details. You can set the NOTAM ID equal to the NOTAM number for easier reference. Use a standard radius of 72 ft if no radius is provided on the NOTAM text. Height is actually referring to obstacle elevation (MSL). When finished, click **Use**.

MIA OBST CRANE (ASN UNKNOWN) 254808N0801123W (4.35NM E
APCH END RWY 26L) 610FT (602FT AGL) FLAGGED 2406030800-
2512300100

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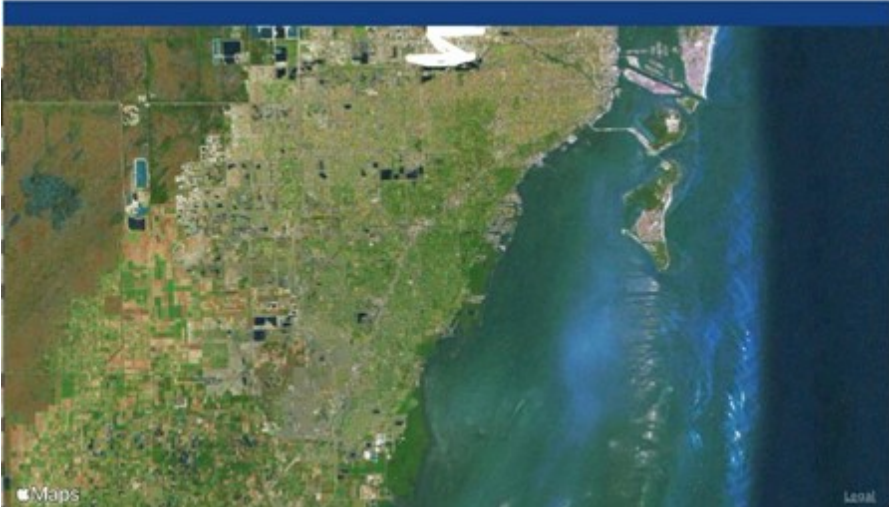
Close Temporary Obstacles +

Cancel Latitude Longitude Distance Out Bearing Distance Use

Latitude: N25-48-08
Longitude: W080-11-23

Height: 610 ft MSL
Radius: 72 ft
NOTAM ID: 05/444

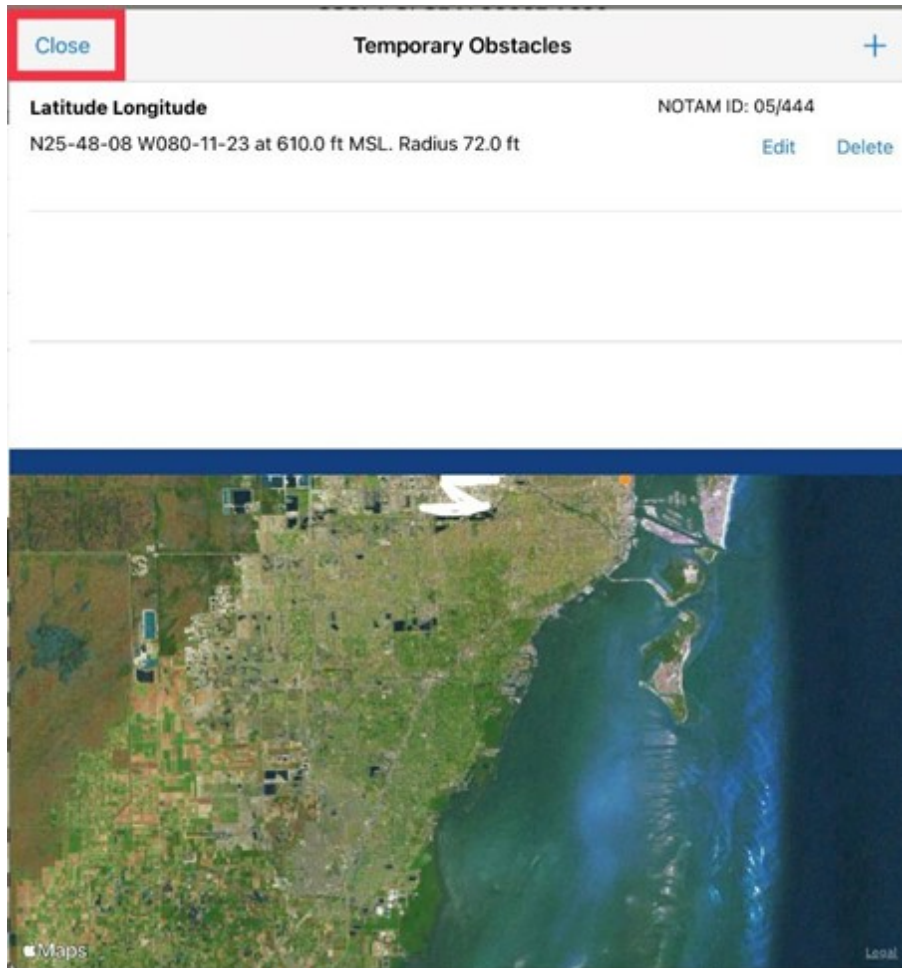
Notes: Optional



The image shows a mobile application interface for adding temporary obstacles. At the top is a header bar with 'Close' on the left, 'Temporary Obstacles' in the center, and a '+' icon on the right. Below the header is a form with several sections. The first section has a 'Cancel' button on the left and a 'Use' button on the right, which is highlighted with a red rectangle. Between these buttons are three tabs: 'Latitude Longitude', 'Distance Out', and 'Bearing Distance'. The 'Latitude Longitude' tab is selected. Below the tabs, the form contains fields for 'Latitude: N25-48-08', 'Longitude: W080-11-23', 'Height: 610 ft MSL', 'Radius: 72 ft', and 'NOTAM ID: 05/444'. There is also a 'Notes: Optional' field. At the bottom of the form is a satellite map view showing a coastal area with a white aircraft icon.

- When finished adding obstacles, click **Close**.

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5. The **Add Obstacle** button will now be highlighted, indicating the analysis is considering manually added obstacles.



Please be aware that iPreFlight will not permanently store these obstacles. If you exit the Runway Analysis screen, the obstacles you added will be removed.

TA 33 Correct application of FLDTA vs Dispatch Criteria

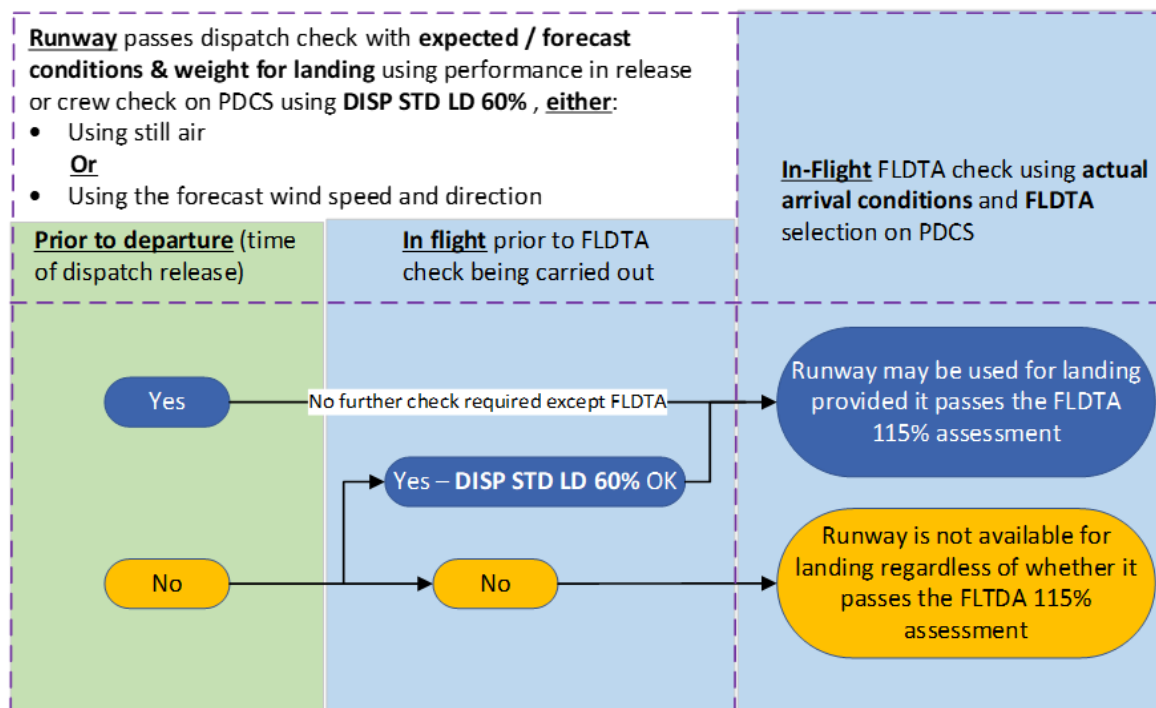
This annex aims to clarify the correct application of OM-A 8.3.28.1 (b), which regulates whether a runway will be available for landing using the FLDTA 115% check. The requirements laid out below are established in CAT.OP.MPA.303, CAT.POL.A.230 and CAT.POL.A.235 of Regulation (EU) No 965/2012 (as amended) (Air Operations).

RELEVANT PROCEDURES
In-flight Determination of Landing Distance at Time of Arrival (OM-A 8.3.28.1(b))
No approach to land shall be continued unless both: - The landing distance available (LDA) on the target runway is at least 115% of the landing distance required (LDTA) at estimated landing time, based on performance data for landing distance assessment. - The aerodrome and intended runway have previously been assessed and found to meet the dispatch criteria.
Landing performance requirements for dispatch (OM-A 8.1.2.4.4)
Paragraphs (C) and (D) establish the requirements for dispatch. These must be met, assuming: - The 'most favourable runway in still air', and - The 'most likely to be assigned runway with the forecast winds', unless one alternate aerodrome is assigned, where only a. needs to be satisfied. Note that this scenario is a simplification – full detail, including criteria for dry, wet and contaminated runways, see OM-A. Due to the above-mentioned requirements, an aircraft may be dispatched to a specific aerodrome without any assessment of the runway designated for landing, thus necessitating the inclusion of the requirements in OM-A 8.3.28.1(b).

Practical examples and a graphical representation of the procedure are included next. Those aim to highlight the following:

- When running an in-flight assessment, if a runway has already passed the dispatch assessment, only the FLDTA assessment is needed (even in cases when those planning criteria are no longer met in-flight)
- Even if the dispatch criteria for the likely runway fail at the planning stage, and under certain conditions, the flight may still be dispatched. That runway must then pass the dispatch assessment in-flight to be used, and conservative management is recommended. No landing is permitted if the dispatch criteria are not met.

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Safety and practical relevance of an accurate FLDTA check

The FLDTA check acts as the final line of defence regarding landing performance, particularly when conditions diverge from those at the time of dispatch. It is essential that all variables are entered accurately to represent the actual landing conditions, namely:

- If you don't plan to use maximum reverse thrust, select 'No TRs'.
- If you plan to cross the threshold at a speed greater than V_{REF} then select ' $V_{REF}+5$ ', ' $V_{REF}+10$ ' or other, as appropriate.

It must also be noted that all performance calculations assume maximum braking for the duration of the landing roll.

If it becomes apparent during the approach that the conditions for the FLDTA check will not be satisfied (for instance, the threshold will be crossed at a greater speed), then a go-around must be conducted, and the FLDTA assessment must be repeated using the updated conditions. Therefore, one can avoid such an outcome by selecting a conservative set of conditions for the FLDTA check (for example, if anticipating crossing the threshold at $V_{REF}+5$, then use $V_{REF}+10$ for the FLDTA check). Example #2 illustrates the importance of an accurate and conservative FLDTA check when conditions have changed from those expected at dispatch.

EXAMPLE #1

A Citation XLS is required in LSZA (Lugano)
 Forecast: TAF LSZA XXXXXXZ 0X09/0X18 VRB01KT CAVOK PROB40 TEMPO XX10/XX15 9999 FEW050
 Planned landing mass: 14900 lbs

EXAMPLE #1

Dispatch assessment:

- **Runway 01 (LDA 4282 ft)** - maximum landing mass is 14951 lbs for still air landing within 60% of LDA ✓
- **Runway 19 (LDA 3724 ft)** - is not long enough to support landing within 60% of LDA at any weight ✗

In-flight assessment:

During the flight, the wind changed to 190/05 kt, and the crew saved 550 lbs of fuel, resulting in an expected landing mass of 15500 lbs.

Runway 01 (LDA 4282 ft) ✓

1. FLDTA max landing mass is 18700 lbs ✓
2. Dispatch criteria could be met at the time of dispatch ✓

Runway 19 (LDA 3724 ft) ✗

1. FLDTA max landing mass is 17727 lbs ✓
2. Dispatch criteria could NOT be met ✗

Conclusion:

Landing on Runway 01 is permitted as both criteria are met.

Landing on Runway 19 is not permitted since it was not possible to previously meet the dispatch criteria, regardless of the result of the FLDTA assessment.

NOTE

The in-flight expected landing mass exceeds that permitted by dispatch criteria (due to the fuel savings en-route). However, the aircraft can still land on runway 01 since the dispatch requirement was satisfied at the time of dispatch.

EXAMPLE #2

A Falcon 2000EX is required in LPCS (Cascais)
Forecast: TAF LPCS XXXXXXZ 0X09/0X18 VRB01KT 8000 SCT020 OVC030
Planned landing mass: 27500 lbs

EXAMPLE #2

Dispatch assessment:

- **Runway 35 (LDA 4593 ft)** - maximum landing mass is 28231 lbs for still air landing within 60% of LDA ✓
- **Runway 17 (LDA 3970 ft)** - maximum landing mass is 22375 lbs for still air landing within 60% of LDA ✗

NOTE

Since there is no rain in the forecast, the dispatch assessment is done using DRY criteria.

In-flight assessment:

During the flight, rain started to fall, and when the FLDTA check was carried out, the runway was wet (RwyCC 5).

Runway 35 (LDA 4593 ft) ✓

(See Remark in Conclusion)

1. FLDTA max landing mass (WET - RWYCC 5 / TRs / $V_{REF}+0$) is 34040 lbs ✓
2. Dispatch criteria could be met at the time of dispatch (for DRY, as forecasted) ✓

Runway 17 (LDA 3970 ft) ✗

1. FLDTA max landing mass is (RWYCC 5 / TRs / $V_{REF}+0$) 27553 lbs ✓
2. Dispatch criteria for DRY could **not** be met (Max 22375 lbs DRY 60%) ✗

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EXAMPLE #2

Conclusion:

Landing on runway 35 is permitted as both criteria are met (see Note).

Landing on runway 17 is not permitted since it was not possible to previously meet the dispatch criteria, regardless of the result of the FLDTA assessment.

Remark:

Although the FLDTA assessment passed with the ideal conditions given above (RWYCC 5 / TRs / $V_{REF}+0$), it is important to understand that this result is only valid if the presumptions are strictly met ($V_{REF}+0$ at the threshold and TRs used to their full extent).

If, in practice, the crew believes those conditions are impractical and the threshold would be crossed at $V_{REF}+5$ and only idle reverse thrust will be used, then it is **essential** that the FLDTA check assessment reflects this assumption (by selecting RWYCC 5 / No TRs / $V_{REF}+5$). With these assumptions, the FLDTA maximum landing mass for runway 35 is 25319 lbs, well below the planned landing mass, so the FLDTA assessment would **not** allow landing in runway 35 with wet conditions and the new assumptions.

With the slightly different scenario where the TRs are used to their fullest extent, but the threshold is crossed at $V_{REF}+10$ (selection RWYCC5 / TRs / $V_{REF}+10$), then the FLDTA maximum landing mass for runway 35 is 28136 lbs, only 636 lbs above the planned landing mass. Although landing on runway 35 is permitted, it should be noted that the LDR is 86% of the LDA, and the safety margin is only 14% of the runway length.

These more practical scenarios demonstrate how any deviation from the conditions used for the FLDTA assessment could severely increase the risk of an overrun.

EXAMPLE #3

A Challenger 350 is required in LSZA (Lugano)
Forecast: TAF LSZA XXXXXXZ 0X09/0X18 19015KT 5000 OVC035
Planned landing mass: 29000 lbs

LSZA Weather Minima is:

- Runway 01 – Ceiling 5100 ft / Visibility 8 km
- Runway 19 – Ceiling 3100 ft / Visibility 5 km

From the above, landing in runway 19 is required.

Dispatch assessment:

- **Runway 01 (LDA 4282 ft)** - maximum landing mass is 31098 lbs for still air landing within 60% of LDA ✓
- **Runway 19 (LDA 3724 ft)** - maximum landing mass is 29325 lbs for landing with the forecast wind of 190/15KT within 60% of LDA ✓

(Note that the maximum landing mass for runway 19 is only 325 lbs above planned landing mass)

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EXAMPLE #3

In-flight assessment:

During the flight, the weather remains as forecasted except the wind decreases to 190/05KT. Since runway 19 already passed the dispatch check at the time of dispatch using the forecast wind of 190/15, it is available with only an FLDTA check using the actual wind of 190/05.

Runway 01 (LDA 4282 ft) ✓

1. FLDTA max landing mass (wind 190/05) is 34150 lbs ✓
2. Dispatch criteria was met at the time of dispatch (still air) ✓

Runway 19 (LDA 3724 ft) ✓

1. FLDTA max landing mass (wind 190/05) is 34150 lbs ✓
2. Dispatch criteria was met at the time of dispatch (wind 190/15) ✓

Conclusion:

Landing on Runway 19 is permitted as both criteria are met. Note that the dispatch assessment would not pass with a reduced headwind of 190/05. Regardless, because it had previously passed at the time of dispatch with the forecasted wind, only an FLDTA assessment is required.

EXAMPLE #4

A Citation Latitude is required in LSZA (Lugano)
Forecast: TAF LSZA XXXXXXZ 0X09/0X18 VRB01KT 5000 RA OVC035
Planned landing mass: 25000 lbs

LSZA Weather Minima is:

- Runway 01 – Ceiling 5100 ft / Visibility 8 km
- Runway 19 – Ceiling 3100 ft / Visibility 5 km

From the above, landing in runway 19 is required.

Dispatch assessment:

- **Runway 01 (LDA 4282 ft)** - maximum landing mass is 25233 lbs for still air (WET) ✓
- **Runway 19 (LDA 3724 ft)** - maximum landing mass is 19892 lbs for still air (WET) ✗

NOTE

Since RA (Rain) is forecasted, the dispatch assessment must be done using WET criteria.

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EXAMPLE #4

In-flight assessment:

Near top of descent, a weather check shows conditions are better than the forecast available for dispatch. No rain is present or expected, and the runways are declared dry. Therefore, we can reasonably assume the runway will remain dry upon arrival and re-evaluate the dispatch assessment using DRY data.

A new dispatch performance assessment is carried out using DRY conditions:

- **Runway 01 (LDA 4282 ft)** - maximum landing mass is 27575 lbs for still air landing DRY within 60% of LDA ✓
- **Runway 19 (LDA 3724 ft)** - maximum landing mass is 25238 lbs for still air landing DRY within 60% of LDA ✓

Subsequently, the FLDTA check is completed as follows:

Runway 01 (LDA 4282 ft) ✓

1. FLDTA max landing mass is 27575 lbs ✓
2. Dispatch criteria were met using in-flight re-dispatch (DRY) ✓

Runway 19 (LDA 3724 ft) ✓

1. FLDTA max landing mass is 27575 lbs ✓
2. Dispatch criteria were met using in-flight re-dispatch (DRY) ✓

Conclusion:

The assessment performed using the improved conditions allows the use of runway 19 for landing, but it is necessary to validate both criteria in-flight

EXAMPLE #5

A Citation XLS is required in LSZA (Lugano)
Forecast: TAF LSZA XXXXXXZ 0X09/0X18 28024KT 5000 OVC035
Planned landing mass: 14900 lbs

LSZA Weather Minima is:

- Runway 01 – Ceiling 5100 ft / Visibility 8 km
- Runway 19 – Ceiling 3100 ft / Visibility 5 km

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EXAMPLE #5

Dispatch assessment:

- **Runway 01 (LDA 4282 ft)** - maximum landing mass is 14951 lbs for still air landing within 60% of LDA ✓
- **Runway 19 (LDA 3724 ft)** - is not long enough to support landing within 60% of LDA at any weight ✗

Note that the forecasted ceiling is below the minimum for runway 01. From OM-A 8.1.2.4.4, and as long as an adequate alternate is selected, the flight may be dispatched with performance meeting only the 'most favourable runway in still air'. In this case, the flight passes the dispatch assessment notwithstanding the weather forecast 'not permitting' a landing on the runway which meets the dispatch performance requirements.

In-flight assessment:

In flight, the wind remains as forecasted, 28024KT (24kt crosswind on runway 19), and the Ceiling remains at 3500 ft. Therefore, runway 19 is the only possibility for landing due to the weather minima.

Runway 19 (LDA 3724 ft) ✗

- FLDTA max landing mass for runway 19 is 17288 lbs ✓
- Dispatch criteria for Runway 19 could not be met at the time of dispatch or the current time ✗

Conclusion:

Landing on Runway 19 is **NOT** permitted in this scenario.

Scenario drill-down:

The aircraft could be dispatched (relying on the performance for runway 01), even knowing it would likely only be able to land on runway 19. If the requirement for the runway to pass the dispatch criteria was not enforced, and only FLDTA was performed, the landing scenario would represent a significant reduction in safety margins, as highlighted in comparison with the scenario where the aircraft lands in runway 19 with the criteria for Fractional Ops, as described next.

Assessment with Fractional Ops (80%) rules results in a maximum landing mass for LSZA runway 19 in the given conditions of 15431 lbs (more than 500 lbs above the planned landing mass).

However, for the flight to be operated under Fractional Ops (80%) rules, the crosswind limit is set at 15kt (per the Fractional Ops Manual). This restriction mitigates the risk presented by the reduced safety margin (1.25 vs 1.67), which means the flight would not have been possible under Fractional Ops.

In direct comparison, under the 'safer' commercial 60% rules, the crosswind mitigation does not apply, and the flight would be permitted to land if no dispatch assessment was necessary for the landing runway.

This possible discrepancy emphasizes the significance of verifying the dispatch criteria for the landing runway.

TA 34 Runway Status Lights (RWSL)

Runway Status Lights (RWSL) are an automatic, advisory runway safety system designed to assist pilots in maintaining situational awareness and reducing the risk of runway incursions and runway-related incidents. Using real-time surface surveillance data, RWSL systems illuminate in pavement red lights to indicate that a runway, intersection, or departure path is temporarily unsafe due to conflicting traffic. RWSL operates independently of ATC clearances and does not provide clearance or routing instructions.

In general:

- Do not enter, cross, or depart from a runway when RWSL red lights are illuminated (including RIL, THL, or REL).
- If RWSL illuminate after receiving an ATC clearance, stop or reject the manoeuvre as appropriate and clarify with ATC.
- Treat extinguished lights as advisory only; normal ATC procedures and clearances remain mandatory.
- RWSL indications must always be respected as an additional safety barrier, but they do not replace ATC instructions or standard operating procedures.

Further technical details, operational logic, and pilot guidance are available at:

- Jeppesen Publications (FD Pro): *North America* □ *North America Airway Manual* □ *State Rules and Procedures – United States* □ *Section 1. Airport Lightning Aids*
- Skybrary (<https://skybrary.aero/articles/runway-status-lights-rwsl>)
- EUROCONTROL (<https://www.eurocontrol.int/system/runway-status-lights>)

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